How the Brain Learns

A Systems-Level Introduction to Neuroscience for Engineers and Technologists

Braun Brelin



Santiago Ramón y Cajal (1852–1934), pioneer of modern neuroscience.

“Every man can, if he so desires, become the sculptor of his own brain.”

— Santiago Ramón y Cajal, Reglas y consejos sobre investigación científica (preface to the second edition, 1898)

Preface

This book is written for readers who are already technically literate but who have never been formally trained in neuroscience. If you are a software engineer, physicist, mathematician, data scientist, or practitioner in a neighboring technical field, you already possess many of the intellectual tools required to understand the brain. What you may lack is not ability, but orientation.

The central aim of this book is to replace metaphor with mechanism. Wherever possible, explanations are grounded in physical processes that occur in living tissue: the movement of ions, the opening and closing of channels, the release of chemicals, and the slow structural changes that follow experience.

Throughout this book, the brain is treated as an adaptive biological control system. It is not a computer in the conventional sense, nor is it a passive information-processing device. Its primary objective is viability: remaining functional under uncertainty.

This perspective has important consequences. Learning is not instantaneous because physical change is slow and metabolically expensive. Forgetting is not a failure of memory, but a necessary feature of a system that must remain flexible.

The book is organized cumulatively, in six parts. Part I lays the foundations: the physics and chemistry that any living system must obey (Chapter 0, which readers comfortable with both may skim), what kind of system the brain is, and how its cells signal. Part II describes the machinery of learning: the chemical signals that decide when learning happens, the circuits in which it happens, and the plasticity that stores it. Part III builds upward to perception, emotion, decision, and habit. Part IV holds the brain up against machine learning, asking what artificial intelligence gets right and wrong and what it could borrow. Part V draws the threads together: whether one principle lies beneath the mechanisms, why the brain is built the way it is, what its design asks of schools, workplaces, and technology, and finally a single model of how the brain learns. Part VI is a case study: an attempt to build a brain from the book's principles, and what its failures teach.

This is not a book about neuroscience as a collection of facts. It is a book about neuroscience as a way of explaining behavior.

If, by the end of this book, you can explain why the brain uses spikes rather than continuous signals and why learning must be regulated, you will have acquired a coherent mental model of how the human brain works.

How to Use This Book

This book is designed to be read sequentially. Unlike reference manuals or survey texts, its chapters build upon one another in a deliberate order. Later discussions rely on mechanisms, constraints, and distinctions introduced earlier, often implicitly. Skipping ahead may create the impression of understanding while quietly weakening the causal chain that gives the subject coherence.

Neuroscience spans multiple levels of explanation, from molecular processes to behavior and experience. This book moves between those levels repeatedly, but it does so in a controlled way. When a concept is introduced at a higher level, it is eventually anchored to mechanisms at lower levels.

Technically trained readers often bring powerful habits of abstraction to new domains. These habits are an asset, but they can also become a liability if applied too early. Biological systems are constrained in ways that engineered systems are not.

Each chapter is written to be read twice. The first pass should be uninterrupted and relatively quick, with the goal of understanding the narrative arc. The second pass should be slower and more critical, with attention paid to how explanations are grounded in physical processes.

A companion volume, *Student Notes and Labs*, accompanies this text. For each of Chapters 1 to 16 it provides a summary of the chapter's argument and a set of laboratory exercises, most of which can be worked by hand or with a few lines of code; printed check values let you confirm your work as you go. An instructor's answer key gives worked solutions and model answers.

The book does not assume memorization as a primary learning strategy. Terminology is introduced sparingly and reused consistently.

Finally, this book rewards patience. Biological intelligence is not optimized for transparency, and many of its solutions appear indirect when viewed through an engineering lens.

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A Note on Levels of Explanation

One of the greatest sources of confusion in neuroscience, and in discussions about the brain more generally, comes from a failure to distinguish between different levels of explanation. Intelligent readers often assume that disagreement reflects uncertainty or lack of rigor, when in fact it reflects the fact that different questions are being answered at different levels.

At the most basic level, the brain is a physical system governed by the laws of physics and chemistry. Ions move according to electrochemical gradients. Proteins change shape. Chemical reactions occur with finite speed and nonzero error rates.

At the next level are cells (neurons and glial cells) and their immediate interactions. Here we describe how neurons integrate inputs, generate spikes, and modify their connections in response to activity.

Above the cellular level lies the level of circuits and systems. At this scale, we consider populations of neurons and the patterns of activity they sustain.

At the highest level commonly discussed are cognitive and behavioral descriptions. Here we use terms such as perception, memory, attention, emotion, and decision-making.

Problems arise when levels are mixed without acknowledgment. Explaining a behavioral phenomenon by invoking a neurotransmitter without specifying the circuit in which it operates conflates molecular and systems-level explanations.

This book moves deliberately between levels of explanation, but it does so with care.

Understanding levels of explanation is not a preliminary technicality; it is a foundational skill.

Part I: Foundations

Chapter 0

Foundations: The Physical and Chemical Basis of Life

> *"What an organism feeds upon is negative entropy. Or, to put it less paradoxically, the essential thing in metabolism is that the organism succeeds in freeing itself from all the entropy it cannot help producing while alive."*
>
> *— Erwin Schrödinger, What Is Life? (1944)*

Neurons are cells before they are anything else, and the brain's capabilities and limits follow from the physics and chemistry of the matter it is made of. This chapter supplies that background: what a cell is, what energy and entropy demand of it, how charge and diffusion behave in salt water, what proteins and the other large molecules do, and how one cell becomes a nervous system. Readers who have studied biology can skim it. Readers who have not will find here every term the later chapters assume, defined once and then used without apology.

What a Cell Is

Every living thing on Earth is made of cells. A bacterium is one cell; an adult human body is on the order of 30 trillion of them, the classic estimate being 37 trillion and a recent recount giving about 36 trillion for a reference man and 28 trillion for a reference woman.[1] Across that range of scale, all cells share the features that define being alive.[2]

A cell is a compartment bounded by a membrane. Inside the membrane is a crowded chemical environment where thousands of reactions run at once; outside is whatever the world happens to be. The membrane separates the controlled interior from the uncontrolled exterior, and without that boundary there is no chemistry a cell could call its own. Figure 0.1 shows the standard inventory of an animal cell: a nucleus that holds the DNA, mitochondria that produce most of the cell's usable energy, a system of internal membranes that makes and sorts proteins, and a skeleton of filaments that gives the cell its shape. A neuron has every one of these parts.

Cells are not passive containers. They maintain their internal environment, resist degradation, harvest energy from their surroundings, respond to signals, and reproduce. Each of these activities is work, and a cell that stops doing the work dies.

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[1] Bianconi, E., Piovesan, A., Facchin, F., Beraudi, A., Casadei, R., Frabetti, F., Vitale, L., Pelleri, M. C., Tassani, S., Piva, F., Perez-Amodio, S., Strippoli, P., & Canaider, S. (2013). An estimation of the number of cells in the human body. *Annals of Human Biology*, 40(6), 463–471; Hatton, I. A., Galbraith, E. D., Merleau, N. S. C., Miettinen, T. P., Smith, B. M., & Shander, J. A. (2023). The human cell count and size distribution. *Proceedings of the National Academy of Sciences*, 120(39), e2303077120. Bianconi and colleagues gave the 37 trillion estimate; Hatton and colleagues the 36 trillion and 28 trillion figures.

[2] Alberts, B., Heald, R., Johnson, A., Morgan, D., Raff, M., Roberts, K., & Walter, P. (2022). *Molecular Biology of the Cell* (7th ed.). W. W. Norton. Chapter 1.

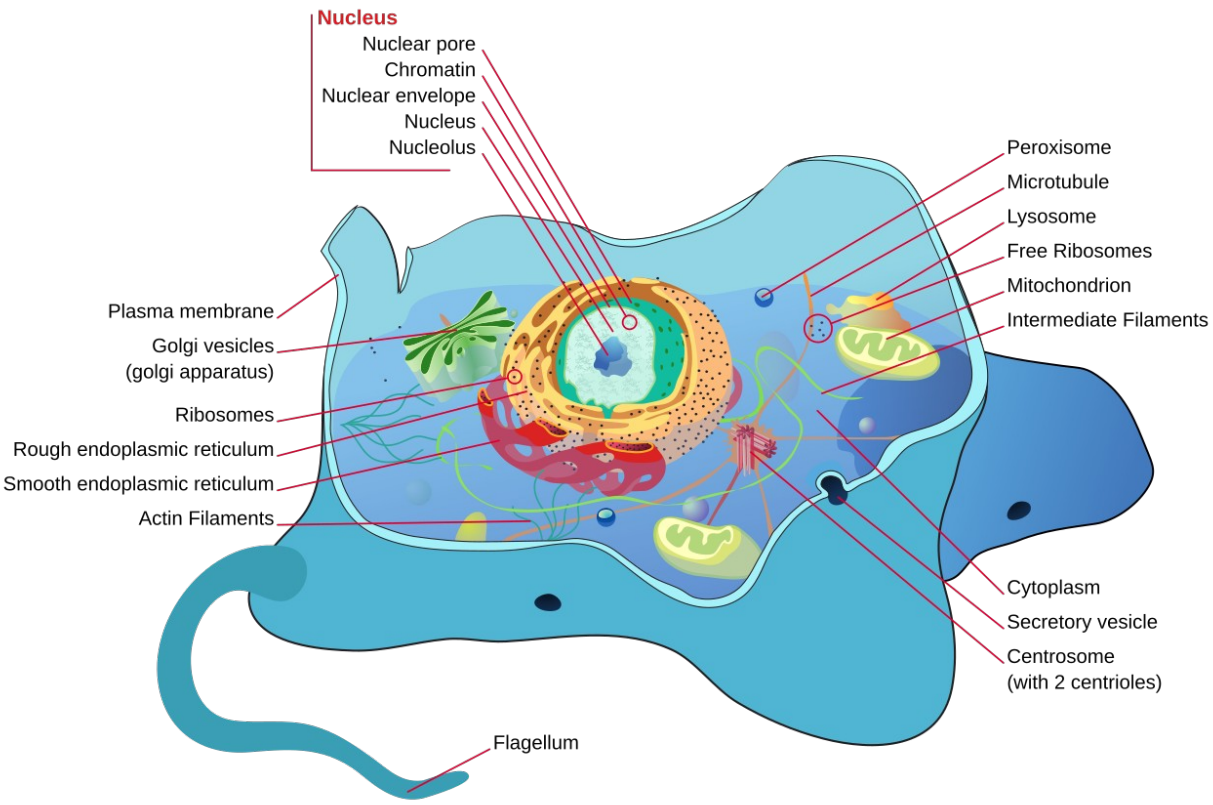


Figure 0.1. The parts of an animal cell. A membrane bounds the cell; the nucleus holds the DNA; mitochondria supply most of the cell's usable energy; the endoplasmic reticulum and Golgi apparatus make and sort proteins; filaments give the cell its shape. Every one of these parts is present in a neuron.

Cells come in two kinds. Prokaryotes, the bacteria and archaea, are small and structurally simple: they have no nucleus, and their DNA lies loose in the cell interior. Despite that simplicity they dominated the planet for more than two billion years and remain its most numerous organisms.[3] Eukaryotes, which include plants, animals, fungi, and the single-celled protists, are larger and more elaborate. Their DNA is enclosed in a membrane-bound nucleus, and their interior is divided into organelles, specialized compartments that each perform a particular job. All neurons are eukaryotic cells, and the distinction matters because eukaryotic machinery is what made multicellular organisms, specialized cell types, and eventually nervous systems possible.[4]

Energy and Order

Cells obey the same physical laws as everything else. There is no vital force and no exemption for living matter, and the laws that matter most for a neuron are the two laws of thermodynamics.[5]

[3] Whitman, W. B., Coleman, D. C., & Wiebe, W. J. (1998). Prokaryotes: The unseen majority. *Proceedings of the National Academy of Sciences*, 95(12), 6578–6583.

[4] Alberts, B., Heald, R., Johnson, A., Morgan, D., Raff, M., Roberts, K., & Walter, P. (2022). *Molecular Biology of the Cell* (7th ed.). W. W. Norton. Chapter 1 gives the prokaryote and eukaryote comparison.

[5] Helmholtz, H. von (1847). *Über die Erhaltung der Kraft*. Berlin: Reimer; Clausius, R. (1865). Ueber verschiedene für die Anwendung bequeme Formen der Hauptgleichungen der mechanischen Wärmetheorie. *Annalen der Physik und Chemie*, 125, 353–400. Helmholtz gave energy conservation its general form, after earlier statements by Julius Robert Mayer and James Prescott Joule, with physiology among his examples; Clausius named entropy and stated the second law in the form used here.

The first law says that energy is neither created nor destroyed, only transformed. When a cell breaks down glucose, the chemical energy in the sugar becomes other forms: some is captured as useful work, the rest leaves as heat, and the total is unchanged. The second law says that in any spontaneous process the total entropy of the universe increases. Entropy is a measure of disorder, or more precisely of the number of microscopic arrangements compatible with what we can see, and left alone, ordered things become disordered. A cell deprived of fuel does not pause; it dissolves.

It follows that maintaining the organized structure of a cell requires constant work. Figure 0.2 shows the arrangement. The cell takes in molecules of low entropy and high free energy (glucose and oxygen), uses them to keep its own structure in order, and exports entropy to its surroundings as heat and as small, disordered waste molecules such as carbon dioxide and water. The cell's own entropy stays steady, the entropy of the surroundings rises by more than the cell's would otherwise have risen, and the second law is satisfied. Schrödinger's phrase for this, in the epigraph, is that the organism feeds on negative entropy; he added in a note to the same chapter that, writing for physicists alone, he would have said it feeds on free energy, the portion of energy that is available to do work.[6]

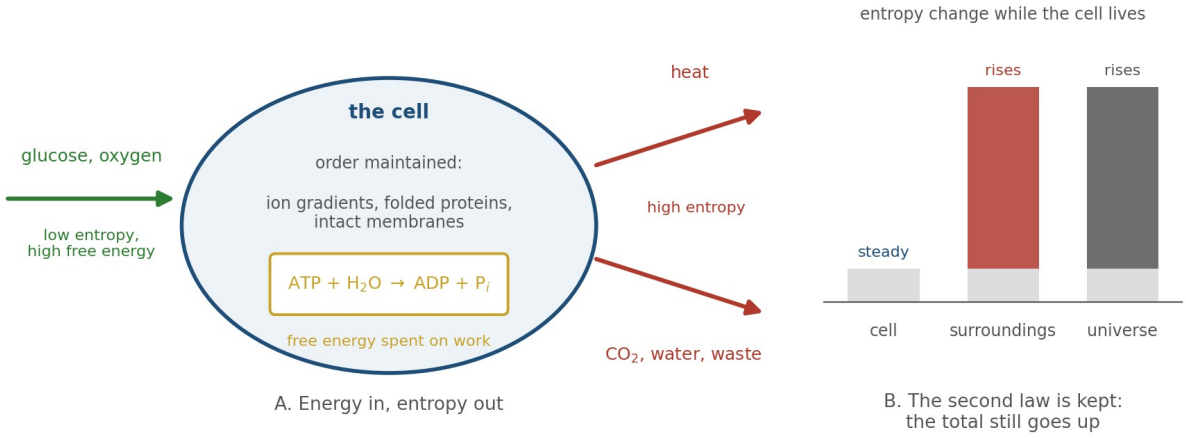


Figure 0.2. How a cell keeps its order. (A) Glucose and oxygen enter with low entropy and high free energy; the cell spends that free energy to maintain ion gradients, folded proteins, and intact membranes, and exports entropy as heat and waste. (B) The cell's entropy holds steady while the entropy of its surroundings, and so of the universe, rises. An illustrative diagram.

The currency in which cells spend free energy is a small molecule called adenosine triphosphate, or ATP. Mitochondria make most of it by oxidizing the breakdown products of glucose, and nearly every energy-requiring process in a cell, from pumping ions to building proteins, is paid for by splitting ATP into adenosine diphosphate and phosphate. It is tempting to say that the energy is "stored in the phosphate bond" and released when the bond breaks, but that is a misconception: breaking any bond costs energy. ATP hydrolysis releases free energy because the products are more stable than the reactant, through relief of the repulsion among the closely packed negative charges, better solvation of the products by water, and resonance stabilization of the released phosphate.[7] The distinction is worth keeping in mind,

[6] Schrödinger, E. (1944). *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press. Chapter 6, "Order, Disorder and Entropy," and its appended note, where Schrödinger says that for physicists he would have let the discussion turn on free energy.

because the brain spends much of its ATP on one task, keeping ions on the correct sides of neuronal membranes.

Charge and Gradients

Neurons are electrical devices, but their electricity is not the electricity in a wire. In a wire, electrons move through a metal; in a neuron, ions move through water and across membranes, which is far slower and is why the brain's signals look nothing like a circuit board's. Understanding the difference means understanding electrochemistry.

Ions are atoms or molecules that carry an electric charge because they have lost or gained electrons. The ions that matter most for a neuron are sodium, potassium, and calcium, all positively charged (calcium carries two charges), and chloride, which is negative.[8] When ions are distributed unequally across a barrier, two tendencies act on them. One is the concentration gradient: molecules spread from where they are concentrated to where they are scarce, because spreading out is the more probable arrangement, which is the second law again in different clothes. The other is the electrical gradient: opposite charges attract, like charges repel, and a positive ion is drawn toward a negatively charged region. The sum of the two is the electrochemical gradient, and it is what every neural signal runs on.

A membrane with unequal ions on its two sides is a charged battery, and neurons discharge it to signal.[9] Figure 0.3 shows how the battery is charged. A protein in the membrane, the sodium-potassium pump, uses the energy of one ATP to push three sodium ions out of the cell and pull two potassium ions in, so that potassium ends up concentrated inside and sodium outside.[10] A second class of membrane protein, the ion channel, forms a pore that lets one kind of ion through and no other, and opens and closes in response to voltage or to a chemical. When a channel opens, ions run down their gradient and the membrane voltage changes. The pumps charge the battery slowly and continuously; the channels discharge it in milliseconds. Chapter 2 builds the neuron's signal from exactly these two parts.

[7] Nelson, D. L., & Cox, M. M. (2021). *Lehninger Principles of Biochemistry* (8th ed.). W. H. Freeman. Chapter 13, on the free energy of ATP hydrolysis and why "high-energy bond" is a misnomer.

[8] Hille, B. (2001). *Ion Channels of Excitable Membranes* (3rd ed.). Sinauer Associates. Chapter 1 for the ions of excitable membranes and their typical concentrations.

[9] Nernst, W. (1889). Die elektromotorische Wirksamkeit der Ionen. *Zeitschrift für physikalische Chemie*, 4, 129–181; Goldman, D. E. (1943). Potential, impedance, and rectification in membranes. *Journal of General Physiology*, 27(1), 37–60. Nernst gave the equilibrium potential for a single ion; Goldman the voltage set by several ions with different permeabilities.

[10] Skou, J. C. (1957). The influence of some cations on an adenosine triphosphatase from peripheral nerves. *Biochimica et Biophysica Acta*, 23, 394–401; Hille, B. (2001). *Ion Channels of Excitable Membranes* (3rd ed.). Sinauer Associates. Skou discovered the sodium-potassium pump; Hille gives its three-to-two stoichiometry and the distinction between pumps and channels.

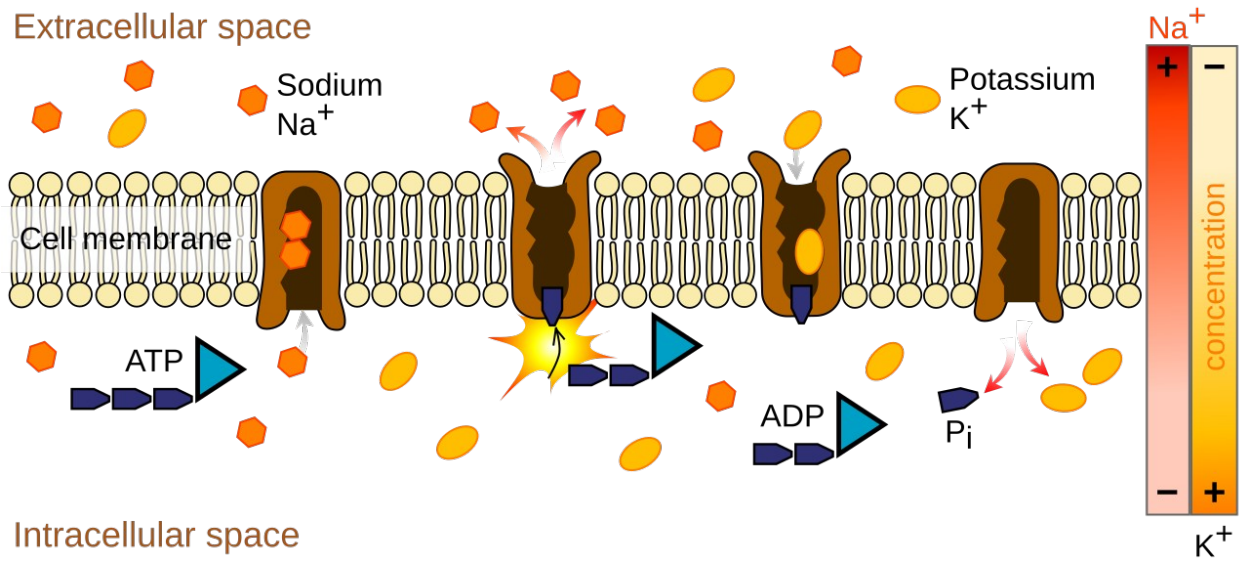


Figure 0.3. The sodium-potassium pump charges the membrane. Each cycle spends one ATP to move three sodium ions out and two potassium ions in, so the cell ends with potassium concentrated inside and sodium outside. The bars at the right show the two gradients the pump maintains.

Diffusion

Molecules in solution are in constant random motion because of thermal energy. That motion carries substances from regions of high concentration to regions of low concentration, a process called diffusion.[11] Diffusion is fundamental to cell biology: neurotransmitters diffuse across synaptic gaps, oxygen diffuses from blood into tissue, and waste diffuses away from cells. No energy is spent on it; the molecules simply wander, and wandering evens things out.[12]

Diffusion has one property that shapes the whole design of the nervous system. The time a diffusing molecule takes to cover a distance grows with the square of the distance.[13] A small molecule such as a neurotransmitter crosses the twenty-nanometer gap of a synapse in a fraction of a microsecond and drifts across a micrometer-sized dendritic spine in under a millisecond, but the same molecule would take years to diffuse the length of a meter-long axon. This is why neurons evolved a different mechanism for long-distance signaling, the electrical impulse that Chapter 2 describes, and why chemistry is reserved for the short hop across the synapse.

Molecules and Bonds

Atoms join into molecules by sharing or transferring electrons, and the forces that hold them together are chemical bonds.[14] The point that matters for a neuron is that bonds come in two strengths. Covalent bonds, in which two atoms share a pair of electrons, are strong and stable; they hold the atoms of a

[11] Fick, A. (1855). Ueber Diffusion. *Annalen der Physik und Chemie*, 94, 59–86; Einstein, A. (1905). Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen. *Annalen der Physik*, 322(8), 549–560. Fick's law describes the flux; Einstein derived it from random molecular motion.

[12] Berg, H. C. (1993). *Random Walks in Biology* (expanded ed.). Princeton University Press.

[13] Berg, H. C. (1993). *Random Walks in Biology* (expanded ed.). Princeton University Press. Chapter 1. For a small molecule in water the diffusion coefficient is of the order of one billionth of a square meter per second, which gives roughly a fifth of a microsecond for 20 nanometers, half a millisecond for a micrometer, and about fifteen years for a meter.

[14] Pauling, L. (1960). *The Nature of the Chemical Bond* (3rd ed.). Cornell University Press.

water molecule together and form the backbone of every protein and every strand of DNA. The weak interactions are the interesting ones. Ionic attractions between opposite charges, and hydrogen bonds, which form when a hydrogen atom bonded to oxygen or nitrogen is drawn toward another oxygen or nitrogen, are each a small fraction of the strength of a covalent bond, and they can be made and broken at body temperature. That is what lets a protein fold into a shape, change its shape when something binds to it, and change back. An ion channel opens because weak bonds rearrange; the two strands of DNA separate for copying because hydrogen bonds let go.[15] Figure 0.4 shows the three kinds.

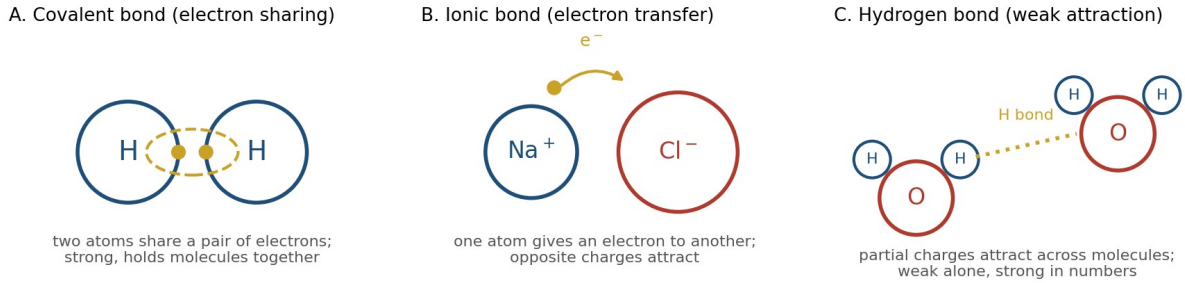


Figure 0.4. Three kinds of chemical bond. (A) A covalent bond shares a pair of electrons and is strong. (B) An ionic bond is the attraction between a positive and a negative ion. (C) A hydrogen bond is a weak attraction between partial charges on neighboring molecules; many together shape proteins and hold the two strands of DNA.

The Molecules of Life

Four classes of large molecules do most of the work in a cell, and Figure 0.5 sketches them. Proteins are the machines. They catalyze reactions, form structural elements, transport materials, send and receive signals, and perform mechanical work. A protein is a chain of amino acids folded into a specific three-dimensional shape, and the shape determines the function: Christian Anfinsen showed in the 1960s that a protein unfolded in the test tube refolds into its working shape on its own, so the sequence of amino acids contains the instructions for the shape.[16] Ion channels are proteins. Neurotransmitter receptors are proteins. The machinery that releases neurotransmitter from vesicles, the small membrane-bound packets in which a neuron stores it, consists of proteins.

Nucleic acids store and transmit the genetic information. DNA holds the instructions for building proteins, and RNA carries those instructions from the DNA to the protein-building machinery.[17] Carbohydrates, the sugars, store energy and form structural materials; glucose is the brain's primary fuel, and the brain, at about two percent of body mass, consumes about a fifth of the body's glucose-derived energy at rest.[18] Lipids, the fats, form membranes, store energy, and act as signaling molecules. The membrane around every cell and every organelle is a double layer of lipids with water-loving heads

[15] Watson, J. D., & Crick, F. H. C. (1953). Molecular structure of nucleic acids: A structure for deoxyribose nucleic acid. *Nature*, 171, 737–738; Nelson, D. L., & Cox, M. M. (2021). *Lehninger Principles of Biochemistry* (8th ed.). W. H. Freeman.

[16] Anfinsen, C. B. (1973). Principles that govern the folding of protein chains. *Science*, 181(4096), 223–230. Anfinsen's Nobel lecture, summarizing the refolding experiments on ribonuclease.

[17] Crick, F. (1970). Central dogma of molecular biology. *Nature*, 227, 561–563.

[18] Mergenthaler, P., Lindauer, U., Dienel, G. A., & Meisel, A. (2013). Sugar for the brain: The role of glucose in physiological and pathological brain function. *Trends in Neurosciences*, 36(10), 587–597.

facing outward and water-fearing tails facing inward, a structure first inferred in 1925 from the observation that the lipid extracted from red blood cells covers twice the cells' surface area.[19]

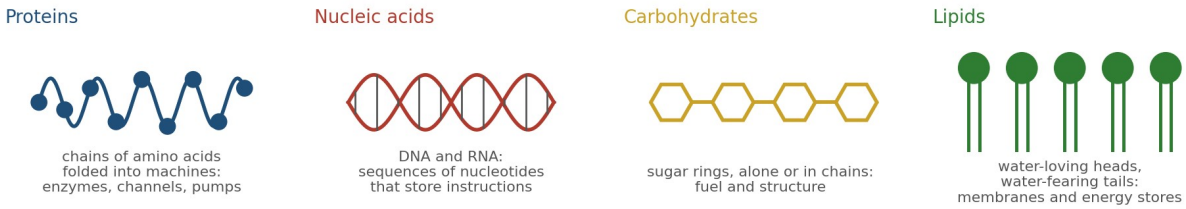


Figure 0.5. The four kinds of large molecule. Proteins are chains of amino acids folded into machines; nucleic acids are sequences of nucleotides that store instructions; carbohydrates are sugars used for fuel and structure; lipids have water-loving heads and water-fearing tails and form membranes.

Enzymes

Most of the reactions in a cell would run far too slowly to support life without help. Enzymes are proteins that accelerate specific reactions, by factors of a million in ordinary cases and by up to a hundred thousand trillion in the most extreme case measured, an enzyme called OMP decarboxylase whose reaction would otherwise take 78 million years to run halfway.[20] An enzyme works by holding the reacting molecules together in the right orientation and chemical surroundings. It is not used up; a single enzyme molecule can process thousands of molecules per second.[21] And it is specific: each enzyme typically catalyzes one reaction or a small family of related ones, the fit between enzyme and substrate being, in Emil Fischer's image of 1894, that of a lock and key.[22] Specificity is what lets a cell decide which reactions happen, when, and where. Neural signaling depends on enzymes at every step: synthesizing neurotransmitters, breaking them down, making and consuming ATP, and modifying ion channels.

How These Laws Were Discovered

In 1665, Robert Hooke looked at a sliver of cork through a microscope and saw small compartments he called cells, from the Latin *cella*, a small room. He was looking at the walls of dead plant cells, but the name stuck, and his drawing, reproduced in Figure 0.6, is the first published description of a cell.[23] Over the next 170 years microscopes improved and observations accumulated, until in 1838 and 1839 Matthias

[19] Gorter, E., & Grendel, F. (1925). On bimolecular layers of lipoids on the chromocytes of the blood. *Journal of Experimental Medicine*, 41(4), 439–443; Singer, S. J., & Nicolson, G. L. (1972). The fluid mosaic model of the structure of cell membranes. *Science*, 175(4023), 720–731.

[20] Radzicka, A., & Wolfenden, R. (1995). A proficient enzyme. *Science*, 267(5194), 90–93. Orotidine 5'-phosphate decarboxylase accelerates its reaction by a factor of about 10^17; the uncatalyzed half-life is 78 million years.

[21] Nelson, D. L., & Cox, M. M. (2021). *Lehninger Principles of Biochemistry* (8th ed.). W. H. Freeman. Chapter 6, on enzyme kinetics and turnover numbers.

[22] Fischer, E. (1894). Einfluss der Configuration auf die Wirkung der Enzyme. *Berichte der deutschen chemischen Gesellschaft*, 27(3), 2985–2993. The "lock and key" account of specificity.

[23] Hooke, R. (1665). *Micrographia: or Some Physiological Descriptions of Minute Bodies Made by Magnifying Glasses*. London: J. Martyn and J. Allestry, for the Royal Society. Observation XVIII, "Of the Schematisme or Texture of Cork."

Schleiden and Theodor Schwann proposed the cell theory: all living organisms are composed of cells, and the cell is the basic unit of life.[24] Rudolf Virchow later popularized the principle, first stated by François-Vincent Raspail, that every cell arises from a pre-existing cell.[25] Whatever the apparent diversity of life, trees, mushrooms, fish, and humans share a common structural basis.

[24] Schleiden, M. J. (1838). Beiträge zur Phytogenesis. *Archiv für Anatomie, Physiologie und wissenschaftliche Medicin*, 5, 137–176; Schwann, T. (1839). *Mikroskopische Untersuchungen über die Uebereinstimmung in der Structur und dem Wachsthum der Thiere und Pflanzen*. Berlin: Sander. English translation by H. Smith, *Microscopical Researches into the Accordance in the Structure and Growth of Animals and Plants* (Sydenham Society, 1847).

[25] Virchow, R. (1858). *Die Cellularpathologie in ihrer Begründung auf physiologische und pathologische Gewebelehre*. Berlin: Hirschwald. The maxim *omnis cellula e cellula*, which Virchow popularized from 1855, had been stated by Raspail in 1825.

Schem: XI.

Fig: 1.

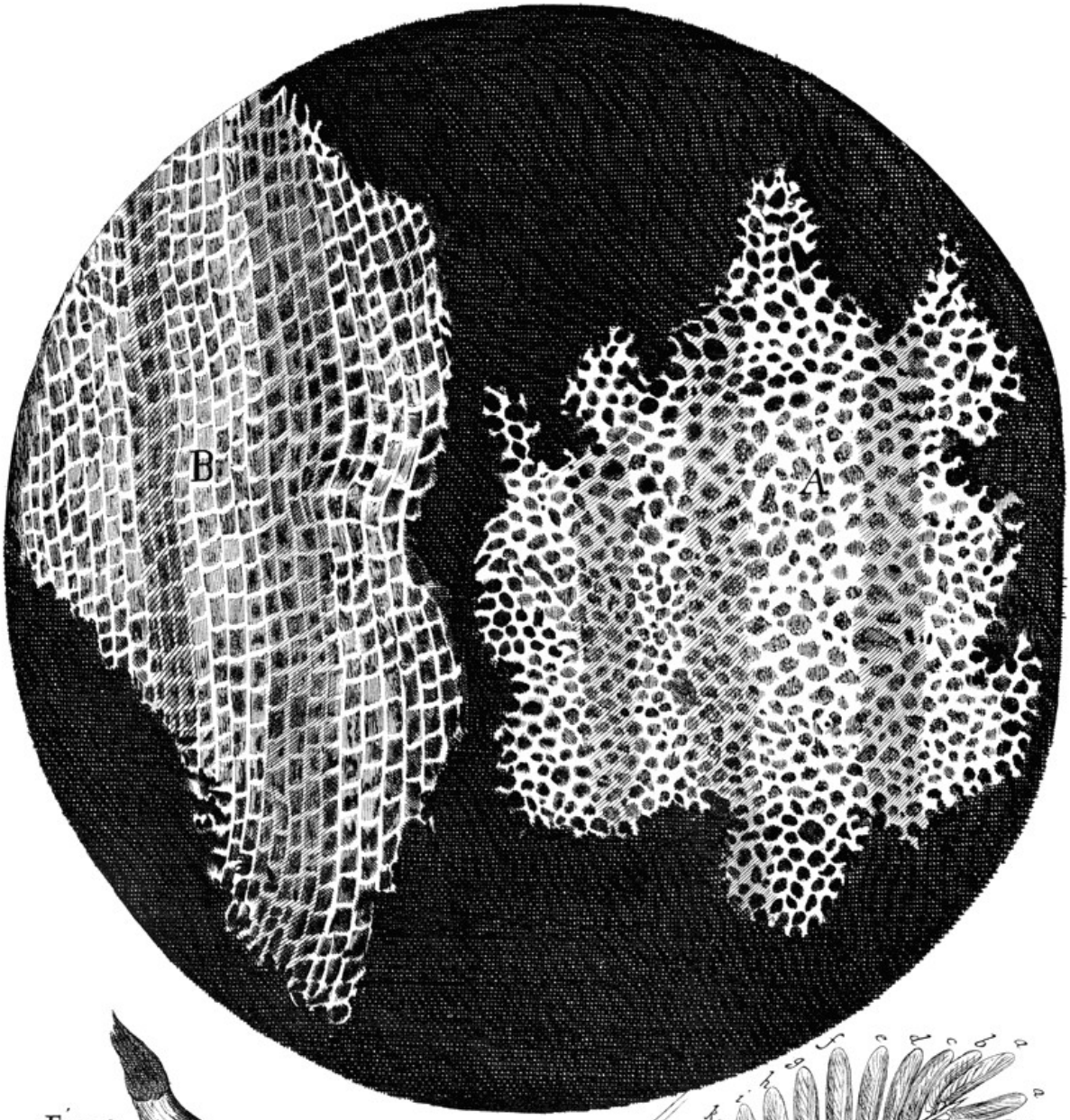


Fig: 2.

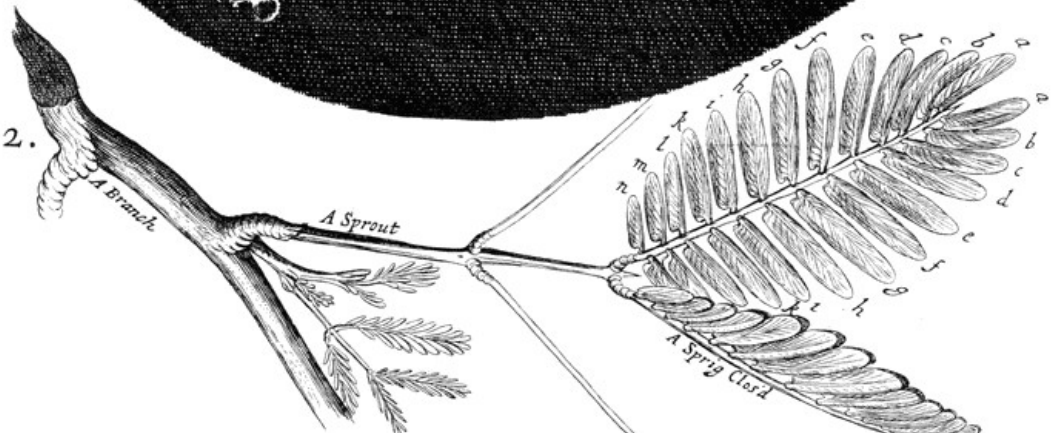


Figure 0.6. Hooke's drawing of cork (1665). The first published description of cells, from Micrographia; what Hooke saw were the walls of dead plant cells.

The laws of thermodynamics emerged from a practical concern with steam engines. Sadi Carnot in 1824 asked how much work a given quantity of heat could yield; Rudolf Clausius, by 1865, had given the answer its general form and had coined the word entropy.[26] Eduard Buchner's demonstration in 1897 that yeast extract with no living cells in it still ferments sugar was the turning point for biology, because it showed that the chemistry of life could be taken out of the cell and studied as chemistry.[27] Vitalism, the belief that life needs a special non-physical force, did not die at once; Hans Driesch defended it into the twentieth century. But by the middle of that century, when Schrödinger wrote the book quoted at the head of this chapter, the laws of physics had been applied to living matter without finding any exemption, and vitalism had no scientific case left.[28]

Around 1780, Luigi Galvani found that electrical stimulation made a frog's leg twitch, the first demonstration that electricity has a role in living tissue.[29] His contemporary Alessandro Volta, arguing with him about what the experiment meant, invented the battery.[30] In the century and a half that followed, scientists worked out how cells use ions and gradients, and the culmination came in 1952, when Alan Hodgkin and Andrew Huxley published a mathematical model of the action potential in the squid giant axon, an axon so large (up to a millimeter across) that electrodes could be threaded inside it. Figure 0.7 shows one impulse computed from their equations: the sodium conductance rises first and fast and drives the voltage up, the potassium conductance rises later and more slowly and drives it back down, and the brief undershoot at the end is the potassium channels still open. Hodgkin and Huxley received the Nobel Prize for this work in 1963.[31]

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[26] Carnot, S. (1824). *Réflexions sur la puissance motrice du feu et sur les machines propres à développer cette puissance*. Paris: Bachelier; Clausius, R. (1865). Ueber verschiedene für die Anwendung bequeme Formen der Hauptgleichungen der mechanischen Wärmetheorie. *Annalen der Physik und Chemie*, 125, 353–400.

[27] Buchner, E. (1897). Alkoholische Gährung ohne Hefezellen. *Berichte der deutschen chemischen Gesellschaft*, 30(1), 117–124.

[28] Schrödinger, E. (1944). *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press. Schrödinger's book is the classic mid-century statement that living matter obeys the known laws of physics.

[29] Galvani, L. (1791). *De viribus electricitatis in motu musculari commentarius*. Bologna: Ex Typographia Instituti Scientiarum. Galvani's experiments began around 1780; the full account was published in 1791.

[30] Volta, A. (1800). On the electricity excited by the mere contact of conducting substances of different kinds. *Philosophical Transactions of the Royal Society of London*, 90, 403–431.

[31] Hodgkin, A. L., & Huxley, A. F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *Journal of Physiology*, 117(4), 500–544; Hille, B. (2001). *Ion Channels of Excitable Membranes* (3rd ed.). Sinauer Associates. The 1963 Nobel Prize in Physiology or Medicine was shared with John Eccles. Hille, Chapter 2, describes the squid giant axon preparation.

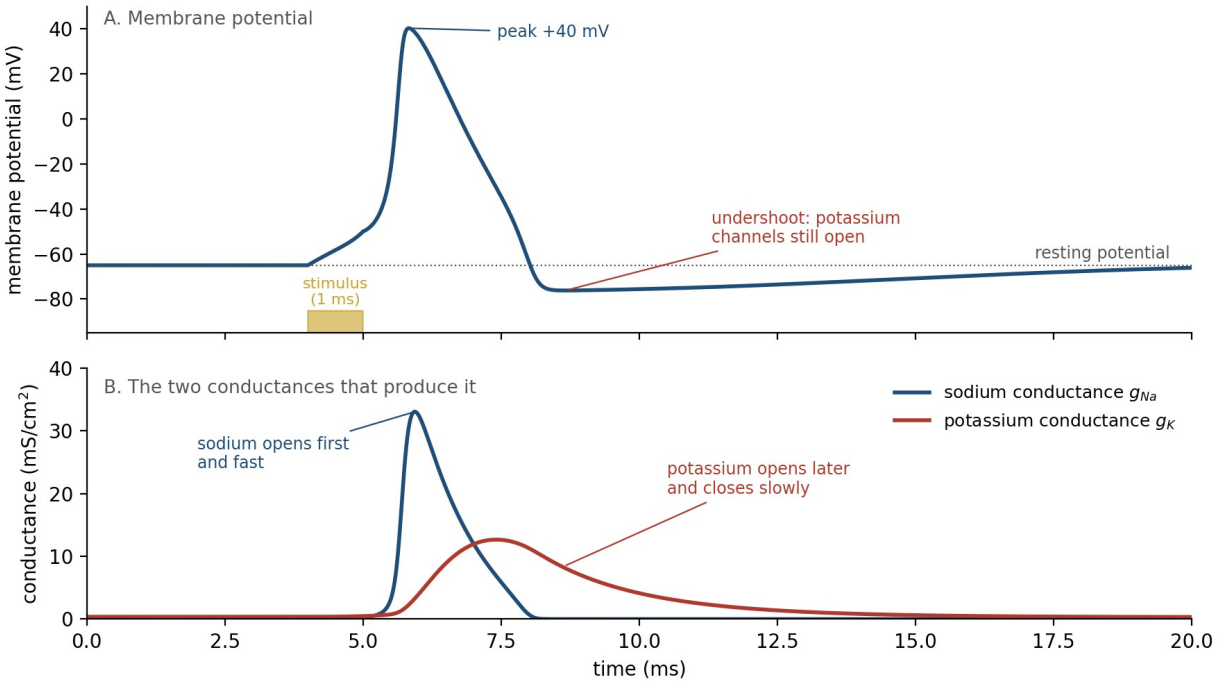


Figure 0.7. One action potential from the Hodgkin and Huxley equations (a simulation). (A) The membrane potential of a model squid giant axon after a one-millisecond stimulus: a rise to about +40 mV, a fall, and a brief undershoot. (B) The sodium conductance opens first and fast; the potassium conductance opens later and closes slowly. Both panels are computed from the 1952 model, not recorded.

The twentieth century then brought the structure of DNA in 1953, the genetic code in the 1960s,[32] the mechanisms of protein synthesis, and the details of the metabolic pathways. That work continues as structural biology, proteomics, and systems biology.

From One Cell to a Brain

A single cell is remarkable, but a brain is not a single cell. The human brain contains approximately 86 billion neurons and a similar number of non-neuronal cells, most of them glia, the support cells that insulate axons, regulate the chemistry around synapses, and clear debris, all organized into precise structures.[33] How does that organization arise?

Every human begins as one fertilized egg. Through repeated division, differentiation, and migration, that one cell becomes tens of trillions organized into tissues, organs, and systems, and Figure 0.8 shows the first weeks of the process.[34] The sequence is not random. Cells follow genetic programs inherited from their parent cells that specify when to divide, when to differentiate into a specialized type, where

[32] Watson, J. D., & Crick, F. H. C. (1953). Molecular structure of nucleic acids: A structure for deoxyribose nucleic acid. *Nature*, 171, 737–738; Nirenberg, M. W., & Matthaei, J. H. (1961). The dependence of cell-free protein synthesis in E. coli upon naturally occurring or synthetic polyribonucleotides. *Proceedings of the National Academy of Sciences*, 47(10), 1588–1602; Crick, F. H. C., Barnett, L., Brenner, S., & Watts-Tobin, R. J. (1961). General nature of the genetic code for proteins. *Nature*, 192, 1227–1232.

[33] Azevedo, F. A. C., Carvalho, L. R. B., Grinberg, L. T., Farfel, J. M., Ferretti, R. E. L., Leite, R. E. P., Jacob Filho, W., Lent, R., & Herculano-Houzel, S. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5), 532–541; Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (Eds.). (2021). *Principles of Neural Science* (6th ed.). McGraw-Hill. Kandel and colleagues, Chapter 7, for the functions of glia.

[34] Barresi, M. J. F., & Gilbert, S. F. (2019). *Developmental Biology* (12th ed.). Sinauer Associates.

to migrate, and how to connect. Development is a cascade in which early events constrain later possibilities: a cell that has committed to the nervous system cannot later become a liver cell, and a neuron that has migrated to the cortex cannot relocate to the spinal cord. Conrad Waddington's image for this was a ball rolling down a landscape of branching valleys, each fork closing off the valleys not taken.[35] This is why early development is so critical and why errors in it propagate forward.

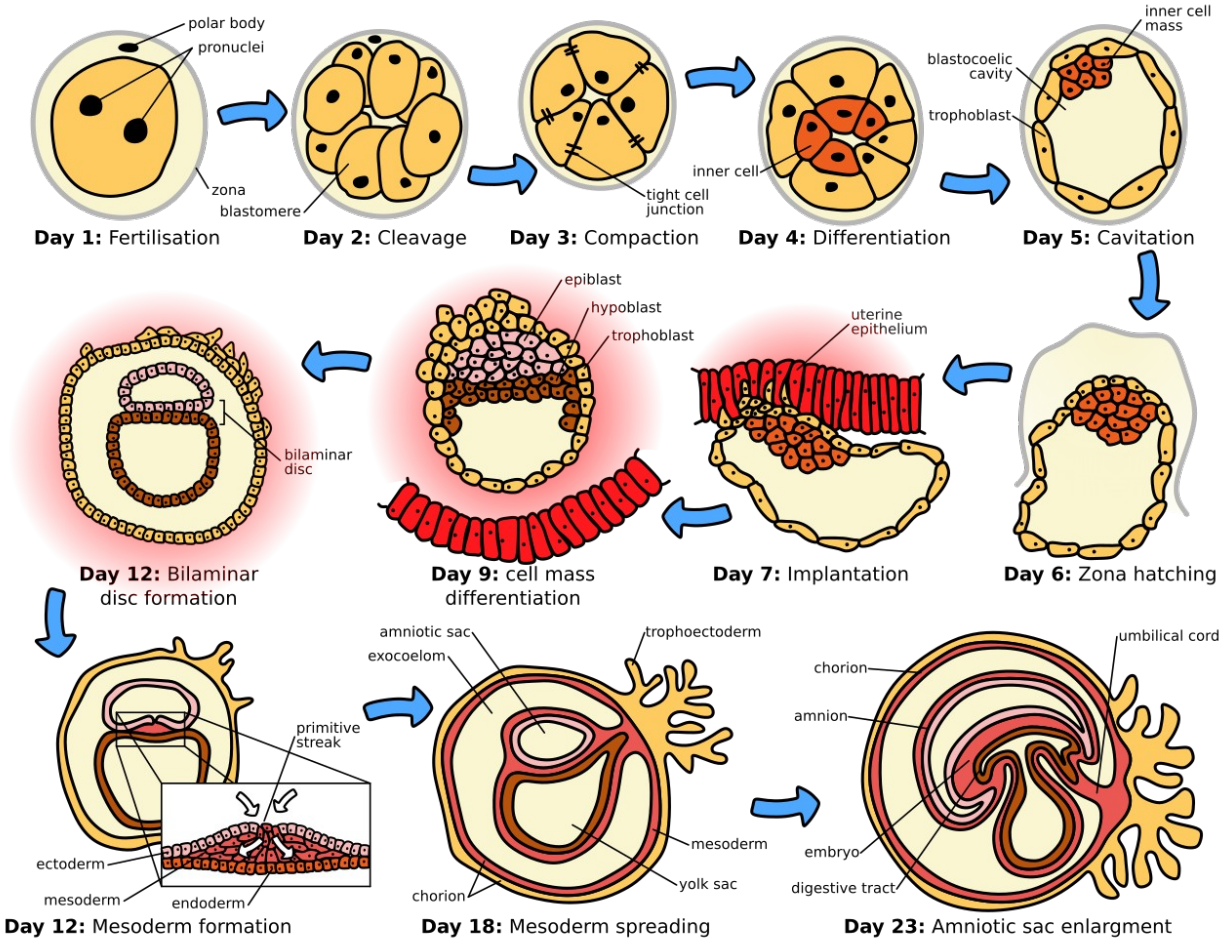


Figure 0.8. The first weeks of a human embryo. From the fertilized egg (top left), repeated division produces a ball of cells, which implants in the uterine wall and organizes into the layers from which every tissue, including the nervous system, will form.

Cells do not develop in isolation. They constantly send and receive chemical signals that coordinate their behavior, and biologists sort the signals by how far they travel. Autocrine signals act on the cell that produced them. Paracrine signals act on nearby cells. Endocrine signals, the hormones, travel through the bloodstream to act on distant cells. Synaptic signaling is treated as a category of its own: a neuron releases a chemical across a gap of about twenty nanometers onto one particular target cell, which makes it a chemical signal with the addressing precision of a wire.[36] These signals let cells coordinate

[35] Waddington, C. H. (1957). *The Strategy of the Genes*. London: Allen & Unwin. Waddington's "epigenetic landscape" is the standard image of progressively restricted cell fate.

[36] Alberts, B., Heald, R., Johnson, A., Morgan, D., Raff, M., Roberts, K., & Walter, P. (2022). *Molecular Biology of the Cell* (7th ed.). W. W. Norton. Chapter 15 classifies contact-dependent, paracrine, synaptic, and endocrine signaling as separate modes.

during development, maintain tissue organization in adults, and respond to changing conditions. The nervous system is, in one sense, an elaboration of cell signaling specialized for speed and precision.

Self-Organization

Complex organization can emerge from simple rules followed by many agents without any agent knowing the plan. This principle, called self-organization, operates throughout biology; Alan Turing showed in 1952 that two chemicals reacting and diffusing at different rates can produce stripes and spots from a uniform start.[37]

The wiring of the visual system shows how the brain uses it, and Figure 0.9 shows the two stages. Axons from the retina grow to their targets in the thalamus and the superior colliculus (not directly to the cortex), and they arrive in rough order: an axon from the temporal side of the retina stops near the front of the target, and an axon from the nasal side runs to the back. The ordering is chemical. Roger Sperry proposed in 1963 that growing axons and their targets carry matching chemical labels, and the labels turned out to be graded: the target expresses a molecule, ephrin-A, in a gradient from front to back, temporal axons carry more of its receptor than nasal axons do, and the more receptor an axon carries the sooner the gradient repels it and stops it.[38] The result is a coarse map, with each axon's terminal arbor spread over a wide region. The second stage sharpens it. Before the eyes open, the retina generates spontaneous waves of activity that sweep across it, so that neighboring retinal cells fire together and distant ones do not; after birth, vision supplies the same correlations. Connections between cells that fire together are strengthened and the rest are pruned, by the rule the book returns to in Chapter 5, Donald Hebb's rule that neurons which fire together wire together, and each arbor shrinks to one position.[39] Blocking the retinal waves leaves the coarse map in place but prevents the sharpening.[40]

[37] Turing, A. M. (1952). The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society B*, 237(641), 37–72; Camazine, S., Deneubourg, J.-L., Franks, N. R., Sneyd, J., Theraulaz, G., & Bonabeau, E. (2001). *Self-Organization in Biological Systems*. Princeton University Press.

[38] Sperry, R. W. (1963). Chemoaffinity in the orderly growth of nerve fiber patterns and connections. *Proceedings of the National Academy of Sciences*, 50(4), 703–710; Cang, J., & Feldheim, D. A. (2013). Developmental mechanisms of topographic map formation and alignment. *Annual Review of Neuroscience*, 36, 51–77. Sperry's chemoaffinity hypothesis; Cang and Feldheim review the Eph receptor and ephrin gradients that implement it.

[39] Hebb, D. O. (1949). *The Organization of Behavior: A Neuropsychological Theory*. New York: Wiley; Katz, L. C., & Shatz, C. J. (1996). Synaptic activity and the construction of cortical circuits. *Science*, 274(5290), 1133–1138. "Fire together, wire together" paraphrases Hebb's postulate; Katz and Shatz review spontaneous retinal waves and activity-dependent refinement.

[40] Cang, J., & Feldheim, D. A. (2013). Developmental mechanisms of topographic map formation and alignment. *Annual Review of Neuroscience*, 36, 51–77. Mice lacking normal retinal waves form a coarse map whose arbors fail to refine.

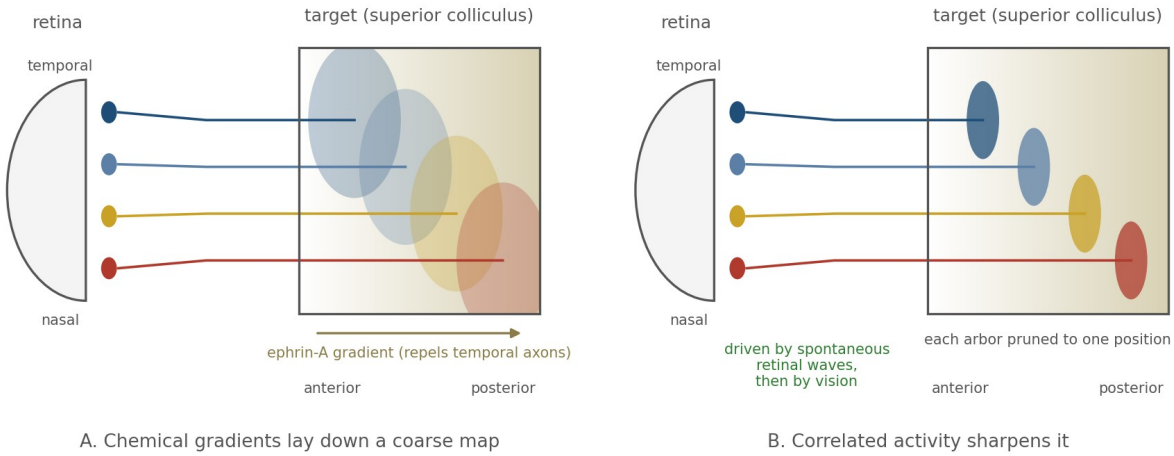


Figure 0.9. Two stages in the wiring of the visual map. (A) Axons from the temporal (top) to nasal (bottom) retina terminate in order along the target because a gradient of ephrin-A repels temporal axons sooner; each arbor is still broad. (B) Correlated activity, from spontaneous retinal waves and then from vision, prunes each arbor to a single position. An illustrative diagram.

No central controller directs either stage. No blueprint specifies where each connection goes. A gradient and a local rule, applied by a million axons at once, produce a map, and the same combination of chemical cues and activity-dependent refinement builds circuits throughout the brain.[41] The result is structure that looks designed but was never designed. It evolved, and then it developed.

Why This Foundation Matters

Everything that follows depends on what this chapter has covered. Neurons are cells. They obey thermodynamics and electrochemistry. They are built from proteins, lipids, and the other molecules of life, they develop according to genetic programs, and they wire themselves up by self-organization. When later chapters describe an action potential, they are describing a physical event governed by ion gradients and channels. When they describe learning, they are describing physical changes in synapses governed by molecular biology. When they describe memory, they are describing physical traces left in neural tissue.

How such mechanisms relate to subjective experience, the felt quality of seeing red or tasting coffee, is a question this book does not try to answer.[42] Neuroscience has explained specific functions of specific circuits in some detail, how the retina encodes an image, for instance, and it has not explained that.[43] This book stays with what can be explained: the physical mechanisms of learning, memory, perception, and action, which are remarkable enough to warrant careful study on their own. The brain is not a metaphor. It is matter, and matter is enough.

[41] Tessier-Lavigne, M., & Goodman, C. S. (1996). The molecular biology of axon guidance. *Science*, 274(5290), 1123–1133; Katz, L. C., & Shatz, C. J. (1996). Synaptic activity and the construction of cortical circuits. *Science*, 274(5290), 1133–1138.

[42] Levine, J. (1983). Materialism and qualia: The explanatory gap. *Pacific Philosophical Quarterly*, 64(4), 354–361; Chalmers, D. J. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3), 200–219.

[43] Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (Eds.). (2021). *Principles of Neural Science* (6th ed.). McGraw-Hill; Purves, D., Augustine, G. J., Fitzpatrick, D., Hall, W. C., LaMantia, A.-S., Mooney, R. D., Platt, M. L., & White, L. E. (Eds.). (2018). *Neuroscience* (6th ed.). Sinauer Associates.

Next Chapter

Chapter 1: What the Brain Is (and Is Not)

Figure and Epigraph Sources

Figure Attribution

Figures 0.2, 0.4, 0.5, 0.7, and 0.9 were drawn for this book. The traces in Figure 0.7 are a simulation computed from the Hodgkin and Huxley (1952) equations with squid giant axon parameters at 6.3 degrees Celsius, not recorded data. Figures 0.2 and 0.9 are illustrative diagrams.

Figure 0.1: “Animal cell structure en.svg” by Mariana Ruiz Villarreal (LadyofHats), public domain, via Wikimedia Commons. Figure 0.3: “Scheme sodium-potassium pump-en.svg” by Mariana Ruiz Villarreal (LadyofHats), public domain, via Wikimedia Commons. Figure 0.6: Robert Hooke, *Micrographia* (1665), Observation XVIII, public domain. Figure 0.8: “HumanEmbryogenesis.svg” by Wikimedia Commons user Zephyris, CC BY-SA 3.0, via Wikimedia Commons.

Epigraph

Erwin Schrödinger, *What Is Life? The Physical Aspect of the Living Cell* (Cambridge University Press, 1944), chapter 6, “Order, Disorder and Entropy.”

Chapter 1

What the Brain Is (and Is Not)

> *"The Brain is wider than the Sky."*
>
> *— Emily Dickinson, The Brain is wider than the Sky (c. 1862)*

What This Chapter Is For

This book is about the brain, but it is not written primarily for neuroscientists. It is written for readers who already know how to think carefully about complex systems, engineers, programmers, physicists, and mathematicians among them, but who have not yet learned how biological intelligence resists many of the assumptions that work so well in their own fields. Chapter 0 supplied the cell biology and physics that the rest of the book assumes, and readers who already have that background can skip it; this chapter supplies the frame.

Neuroscience presents a peculiar challenge. Unlike astronomy or particle physics, its object of study is not remote. We experience the brain continuously, from the inside, and this intimacy creates confidence without understanding. We feel that we know what thinking is, what memory is, what perception is, simply because we experience them. Yet introspection is a poor guide to mechanism. In a series of experiments in the 1970s, Richard Nisbett and Timothy Wilson showed that people asked why they had made a choice gave fluent, confident reasons that had nothing to do with the factors that actually drove it: shoppers asked to pick the best of four identical pairs of stockings laid out in a row preferred the rightmost pair by about four to one, and explained the preference by its knit, its sheerness, anything but its position.[44] The brain does not reveal its workings by reflecting on them.

For technically trained readers, this difficulty is compounded by metaphor. Since the 1940s, when Warren McCulloch and Walter Pitts showed that networks of idealized neurons could implement any function of propositional logic, and John von Neumann borrowed their vocabulary for the first description of a stored-program computer, the language of computation has been the dominant way of speaking about minds.[45] We talk about inputs and outputs, representations and processing, memory and storage. These terms are not wrong, but they are dangerously incomplete. They import assumptions about precision, modularity, and design that do not hold for biological systems.

If you approach the brain as if it were a poorly documented piece of software, you will find yourself constantly asking why it is so badly engineered. Why is learning slow? Why is memory unreliable? Why

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[44] Nisbett, R. E., & Wilson, T. D. (1977). Telling more than we can know: Verbal reports on mental processes. *Psychological Review*, 84(3), 231–259.

[45] McCulloch, W. S., & Pitts, W. (1943). A logical calculus of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics*, 5(4), 115–133; von Neumann, J. (1945). *First Draft of a Report on the EDVAC*. Moore School of Electrical Engineering, University of Pennsylvania. Reprinted in *IEEE Annals of the History of Computing*, 15(4), 27–75 (1993); Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, 59(236), 433–460; McCarthy, J., Minsky, M. L., Rochester, N., & Shannon, C. E. (1955). A proposal for the Dartmouth Summer Research Project on Artificial Intelligence. Reprinted in *AI Magazine*, 27(4), 12–14 (2006); von Neumann, J. (1958). *The Computer and the Brain*. Yale University Press. McCulloch and Pitts are the start of the vocabulary; von Neumann's EDVAC report borrows their idealized neurons by name; the Dartmouth proposal of 1955 is the start of artificial intelligence as a field; von Neumann's posthumous lectures are the most careful early statement of the comparison.

are perceptual judgments biased? Why does the system forget, hallucinate, and mispredict? The answer is not that evolution did a bad job. It is that the brain was never trying to solve the problem you think it was solving, and this chapter exists to reframe the problem. It does so by saying what the brain is not (a digital computer), what it is (a physical system with a history, built to keep a body alive), and what follows from that for noise, redundancy, and the meaning of optimization. Along the way it introduces a few of the numbers and experiments on which the rest of the book depends.

The Brain Is Not a Digital Computer

The comparison between brains and computers is so natural that it often escapes notice. We say the brain "processes information" as casually as we say the heart "pumps blood." The analogy feels descriptive rather than metaphorical, and this ease is precisely what makes it dangerous.

Digital computers are artifacts of design, and the design is easy to draw. Figure 1.1 shows the architecture von Neumann described in 1945 for the EDVAC, the plan that almost every computer since has followed. A central processing unit, made of a control unit and an arithmetic and logic unit, fetches instructions and data from a separate memory over a single channel, operates on them, and writes results back; input and output devices hang off the sides. Three features of the drawing matter here. Memory is distinct from computation. Control is centralized, in a unit that decides what happens next. And time is discretized: a clock sets when each step occurs, and the components are engineered so that the signal at every step is unambiguously a zero or a one.[46] Noise, in such a machine, is an enemy to be suppressed, though von Neumann himself knew that the brain did not work that way. In lectures given in 1952 he asked how reliable computation could be built from unreliable components and showed that it could, by redundancy and majority voting, which is one reason the contrast drawn in this section is a contrast of emphasis rather than of kind.[47]

[46] von Neumann, J. (1945). *First Draft of a Report on the EDVAC*. Moore School of Electrical Engineering, University of Pennsylvania. Reprinted in *IEEE Annals of the History of Computing*, 15(4), 27–75 (1993).

[47] von Neumann, J. (1956). Probabilistic logics and the synthesis of reliable organisms from unreliable components. In C. E. Shannon & J. McCarthy (Eds.), *Automata Studies* (pp. 43–98). Princeton University Press. The text is based on lectures given at the California Institute of Technology in January 1952.

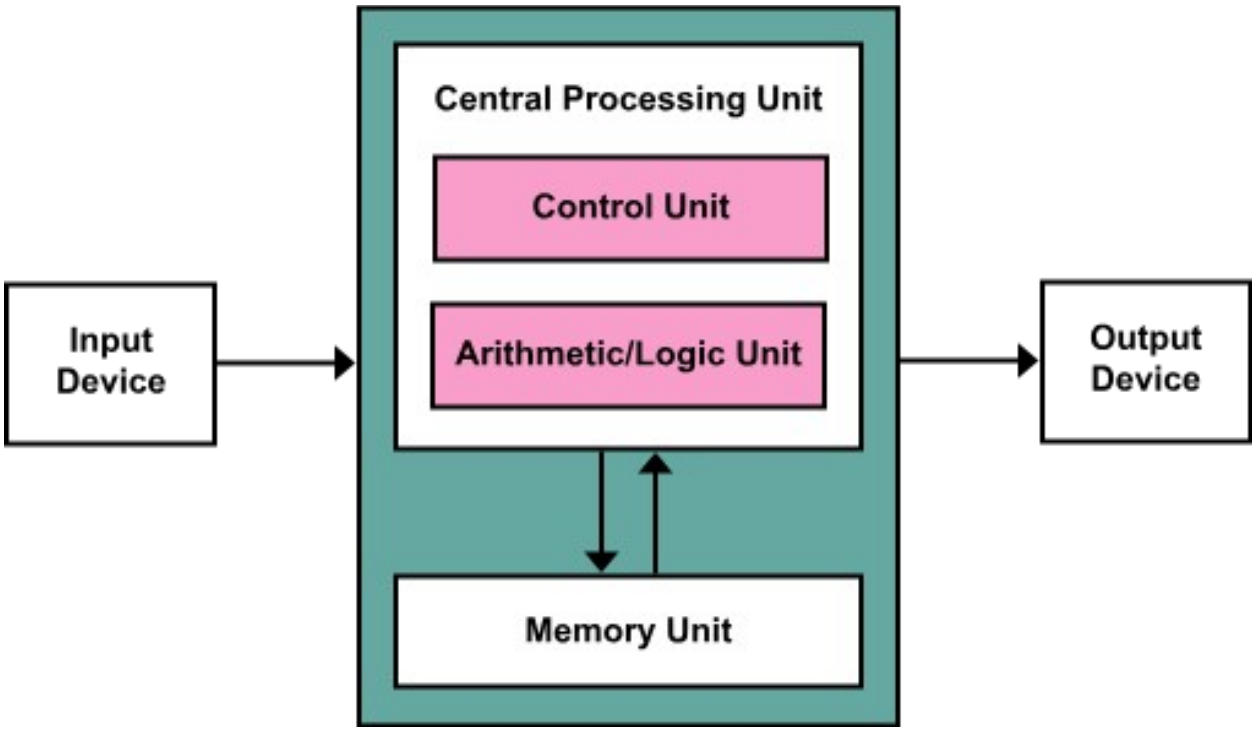


Figure 1.1. Classical von Neumann architecture. Note the single pair of arrows between the memory unit and the central processing unit: every instruction and every datum crosses that channel, one step per tick of a clock the diagram leaves implicit. Nothing in the brain corresponds to that channel or that clock.

Brains are none of these things. A human brain contains about 86 billion neurons, the electrically excitable cells that carry its signals, together with a roughly equal number of non-neuronal cells, and the neurons are joined by well over a hundred trillion synapses, the junctions at which one neuron influences another.[48] There is no instruction stream to tell these cells what to do, no central controller, and no boundary between storage and processing: the same synapses that carry a signal are the places where the system's memory is written, and they change as a result of use.[49] Signals are continuous in time, variable in amplitude, and context dependent. Neurons operate asynchronously, with no clock, and the system rewires itself while it is running.

The difference shows most clearly in time. A neuron's action potential, the brief electrical pulse by which it signals, lasts about a millisecond. The chemical step at a synapse, in which the arriving pulse releases a transmitter molecule that diffuses across a gap and opens channels on the receiving cell, adds about half a millisecond more; Bernard Katz and Ricardo Miledi measured that delay at the junction between a frog's nerve and muscle in 1965.[50] The fastest axons in the human body, the fibers along which neurons send their pulses, conduct at roughly 100 meters per second, and most are much slower. A transistor, by contrast, switches in picoseconds, and a processor clock ticks several billion times a

[48] Azevedo, F. A. C., Carvalho, L. R. B., Grinberg, L. T., Farfel, J. M., Ferretti, R. E. L., Leite, R. E. P., Jacob Filho, W., Lent, R., & Herculano-Houzel, S. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5), 532–541. The counts are 86.1 billion neurons and 84.6 billion non-neuronal cells, with only 19 percent of the neurons in the cerebral cortex.

[49] Kandel, E. R. (2001). The molecular biology of memory storage: A dialogue between genes and synapses. *Science*, 294(5544), 1030–1038.

[50] Katz, B., & Miledi, R. (1965). The measurement of synaptic delay, and the time course of acetylcholine release at the neuromuscular junction. *Proceedings of the Royal Society of London B*, 161(985), 483–495.

second. Measured step for step against a processor's clock, the brain is about a million times slower than a laptop. Yet Simon Thorpe and his colleagues showed in 1996 that a person can decide whether a photograph flashed for 20 milliseconds contains an animal, and that the brain's electrical response to the two categories diverges within about 150 milliseconds of the picture appearing.[51] At a few milliseconds per step, 150 milliseconds allows perhaps a hundred sequential steps; Jerome Feldman and Dana Ballard called this the hundred-step constraint, and it rules out any account of perception that resembles a long serial program.[52] Whatever the brain is doing, it is doing it with enormous breadth and very little depth: billions of slow, unreliable components working at once.

These differences are not superficial. They reflect fundamentally different goals. Computers are designed to produce correct answers to well-specified problems. Brains evolved to keep organisms alive in uncertain, hostile environments, where a fast, approximately right answer is worth more than a slow, exact one, and where precision is valuable only insofar as it supports survival.[53] This distinction explains why many computational metaphors break down under scrutiny. When we look for symbols, registers, or clean data structures in the brain, we are often disappointed. When we look instead for dynamics, feedback, and constraint-driven adaptation, the system begins to make sense.

A more honest statement is not that the brain is a computer, but that both brains and computers are physical systems capable of computation under radically different constraints. David Marr's distinction between what a system computes, the procedure it uses, and the hardware that runs the procedure is the standard way of keeping those levels of analysis apart (a different axis from the levels of biological organization in the front matter), and it cuts both ways: the brain may compute something a computer can also compute while sharing nothing with it at the level of procedure or hardware.[54] Conflating the two hides exactly the differences that matter.

The Brain Is a Physical System

At the most basic level, the brain is matter. It occupies space, has mass, consumes energy, and obeys the same physical laws as any other biological tissue. Its signals arise from the movement of ions, charged atoms such as sodium and potassium, across the membranes that enclose its cells, and its components are proteins that change shape, channels that open and close, and molecules that diffuse through crowded cellular environments. Every thought corresponds to physical change, and the clearest demonstration that the physical account is the whole account came in 1952.

Alan Hodgkin and Andrew Huxley worked on the giant axon of the squid, a nerve fiber about half a millimeter across, large enough to thread a wire down its inside. Holding the voltage across the membrane fixed and measuring the current that flowed, they separated it into a sodium current and a potassium current, each switched on and off by the voltage itself with its own time course. They described the two with four differential equations and then asked the equations what a nerve impulse should look like. The equations produced the impulse: its shape, its threshold, and its conduction velocity, which Huxley computed by hand on a mechanical calculator at 18.8 meters per second against a

[51] Thorpe, S., Fize, D., & Marlot, C. (1996). Speed of processing in the human visual system. *Nature*, 381(6582), 520–522.

[52] Feldman, J. A., & Ballard, D. H. (1982). Connectionist models and their properties. *Cognitive Science*, 6(3), 205–254.

[53] Sterling, P., & Laughlin, S. (2015). *Principles of Neural Design*. MIT Press.

[54] Churchland, P. S., & Sejnowski, T. J. (1992). *The Computational Brain*. MIT Press; Marr, D. (1982). *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. W. H. Freeman. Marr's three levels (computational, algorithmic, implementational) are the standard way of keeping the two apart.

measured 21.2.[55] Nothing was left over. The impulse was not a signal carried by the membrane; it was the membrane's physics, and that is the sense in which this book treats every mental event. Figure 1.2 shows the organ in which those events occur, a midline magnetic resonance image of a human head, with the folded sheet of the cerebral cortex above, the cerebellum tucked beneath it at the back, and the brainstem descending toward the spinal cord.

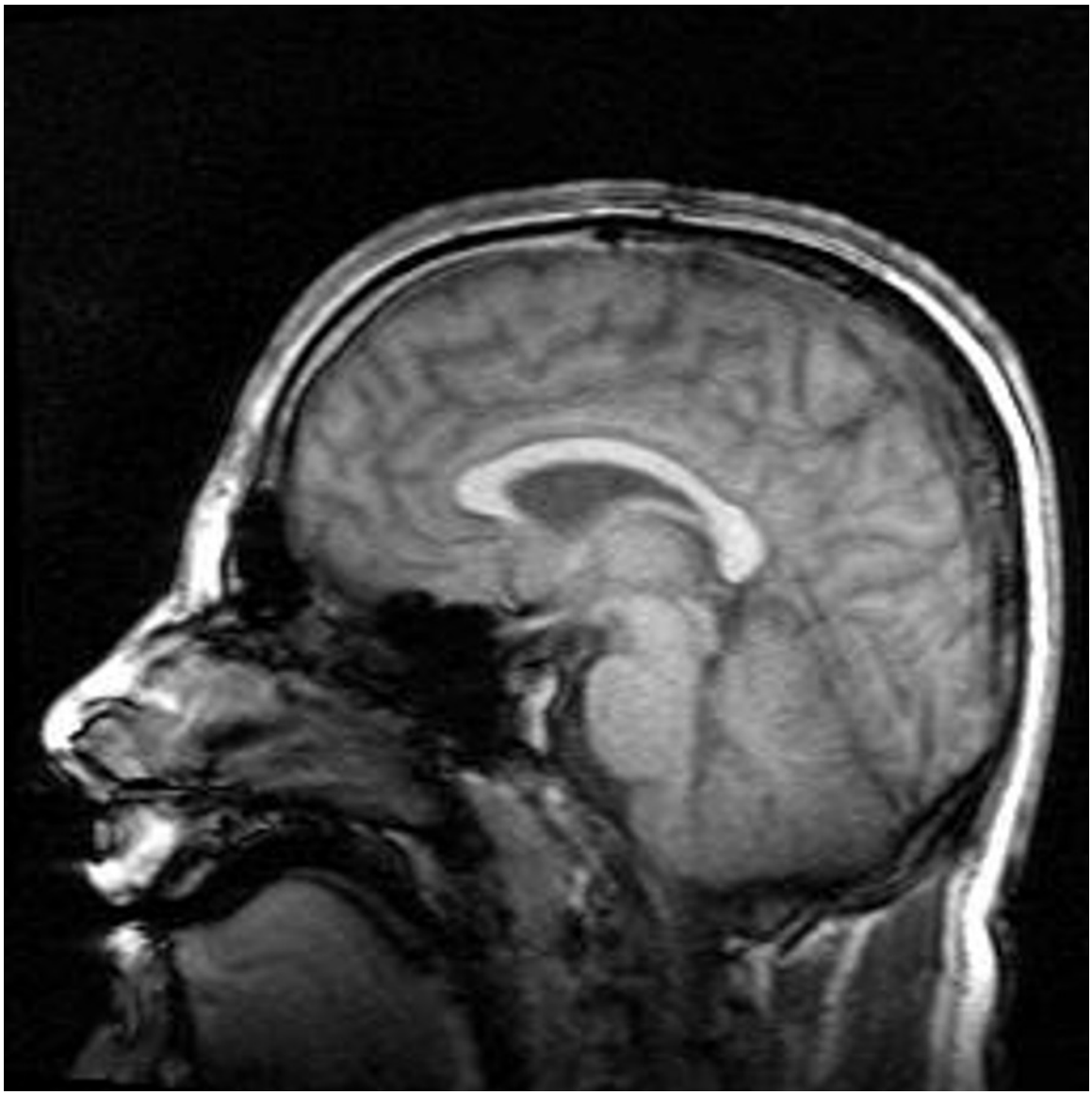


Figure 1.2. Midline MRI of a human head. The folded cerebral cortex, the cerebellum beneath it, and the brainstem are visible. None of this layout was specified in advance; Chapter 0 described how tissues like these develop.

This fact is easy to state and difficult to absorb. Many accounts of cognition implicitly treat the physical substrate as incidental, as if the real processes occurred at an abstract level divorced from biology. In reality, the properties of cognition are inseparable from the properties of the tissue that generates it.

[55] Hodgkin, A. L., & Huxley, A. F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *Journal of Physiology*, 117(4), 500–544. The calculated velocity of 18.8 meters per second and the observed 21.2 meters per second, both at 18.5 degrees Celsius, are reported in the paper's section on the propagated action potential.

Physical processes take time, consume energy, and are noisy, and they leave traces. Memory persists because structural change persists. Fatigue, intoxication, and disease affect cognition because they affect metabolism, signaling, and cellular integrity.

The energy budget is the constraint that shapes everything else. The adult human brain is about two percent of body mass and uses about twenty percent of the body's resting energy, which works out to roughly 20 watts: the power of a dim light bulb, running a system with 86 billion processors.[56] David Attwell and Simon Laughlin traced where that energy goes in the gray matter of the rodent brain, the tissue that holds the cell bodies and synapses. Nearly half, by their estimate 47 percent, is spent on action potentials, a further 34 percent on the effects of the main excitatory transmitter, glutamate, on the receiving cells, 13 percent on maintaining the resting voltage of cells that are not signaling at all, and the small remainder on recycling transmitter and other steps.[57] Signaling is expensive, and the budget cannot cover much of it: Peter Lennie calculated that the cortex can afford to have only about one neuron in a hundred firing at any moment.[58] The brain is therefore forced into sparse codes, low average firing rates, and wiring plans that minimize the length of axons, and many of the design choices the later chapters describe, from the inhibition that keeps cortical firing sparse in Chapter 4 to the use of prediction to suppress expected signals in Chapter 6, are what a 20-watt budget looks like from the inside.

Even the brain's apparent inefficiencies make sense in this light. Synapses fail, neurons die, and signals degrade, and rather than eliminating these problems, evolution built systems that tolerate them. Feedback compensates for unreliability, and plasticity, the capacity of neurons and synapses to change their structure and strength with use, compensates for damage: after a stroke destroys part of the motor cortex, the region of cortex that drives movement, the representation of the hand can re-form in the surviving tissue around the lesion, provided the hand is used.[59] What looks sloppy from the standpoint of a designer is often robust from the standpoint of an organism. To understand the brain, we must abandon the idea that biology is merely an imperfect implementation of abstract computation. Biology is the source of the computation.

Why Learning Is Slow

The first question a technical reader asks about the brain is usually why it learns so slowly. A network on a computer can be retrained in an afternoon; a person takes years to learn a language and a decade to master an instrument. The answer has two parts, and they are different in kind.

The first is that lasting change is molecular, and molecules take time. The earliest steps of synaptic plasticity are fast: a second of intense stimulation can strengthen a hippocampal synapse within a minute or two, by inserting receptor proteins that are already waiting near the membrane.[60] But a

[56] Mink, J. W., Blumenschine, R. J., & Adams, D. B. (1981). Ratio of central nervous system to body metabolism in vertebrates: Its constancy and functional basis. *American Journal of Physiology*, 241(3), R203–R212; Sterling, P., & Laughlin, S. (2015). *Principles of Neural Design*. MIT Press. Twenty percent of a resting metabolic rate of roughly 100 watts is about 20 watts.

[57] Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*, 21(10), 1133–1145.

[58] Lennie, P. (2003). The cost of cortical computation. *Current Biology*, 13(6), 493–497.

[59] Nudo, R. J. (2013). Recovery after brain injury: Mechanisms and principles. *Frontiers in Human Neuroscience*, 7, 887.

[60] Malinow, R., & Malenka, R. C. (2002). AMPA receptor trafficking and synaptic plasticity. *Annual Review of Neuroscience*, 25, 103–126.

change that lasts more than a few hours requires the cell to synthesize new proteins and to build or enlarge the physical structures of the synapse, and that takes hours; blocking protein synthesis leaves the early change intact and erases the late one. Eric Kandel's work on the sea slug Aplysia showed the same division at the level of a single reflex: a single shock to the tail produces a heightened withdrawal reflex that lasts minutes and needs no new protein, while five shocks spaced over time produce one that lasts days to weeks and depends on genes being switched on in the nucleus.[61] Learning that is meant to persist is slow because it is construction.

The second reason is that fast change is dangerous. A memory system that rewrote itself completely on every experience would be unable to keep anything, because each new pattern would overwrite the old. When Michael McCloskey and Neal Cohen trained an artificial network on one set of arithmetic facts and then on a second set, the second erased the first, a failure they named catastrophic interference.[62] James McClelland, Bruce McNaughton, and Randall O'Reilly argued in 1995 that the brain avoids this by dividing the labor: the hippocampus, a structure deep in the temporal lobe, learns in a single exposure, and then replays what it has learned to the cortex slowly, interleaved with everything the cortex already knows, so that the slow learner is never forced to choose between the new memory and the old.[63] Slow learning in the cortex is a requirement of stability, not a limitation of the hardware, and fast learning is quarantined where it can do no harm. Chapter 5 shows how both halves of this answer work in the synapse.

Time, Change, and the Absence of Reset

One of the most underappreciated differences between brains and machines is that biological systems have no reset button. A computer can be restarted, its memory cleared, its parameters reinitialized. A brain cannot be returned to an earlier state without destroying the system itself. Every experience leaves some trace, however small. Development, learning, aging, and injury accumulate, and the brain you have today is the product of a continuous, irreversible history.

The cleanest demonstration is also one of the oldest in systems neuroscience. In 1963 Torsten Wiesel and David Hubel sutured one eyelid of newborn kittens closed for the first months of life and then recorded from neurons in the visual cortex. Almost every neuron answered only to the eye that had stayed open; the closed eye, though its retina and optic nerve were intact, had lost its cortical territory.[64] Later experiments showed that the loss was largely permanent, and that the same closure in an adult cat did nothing. Experience during a sensitive period, a window of early life in which a circuit is unusually open to being shaped, had set the initial conditions for all later experience, and there was no way back to the state before it. Eric Knudsen's review of sensitive periods finds the same logic in the owl's auditory map,

[61] Kandel, E. R. (2001). The molecular biology of memory storage: A dialogue between genes and synapses. *Science*, 294(5544), 1030–1038.

[62] McCloskey, M., & Cohen, N. J. (1989). Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24, 109–165.

[63] McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. (1995). Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, 102(3), 419–457.

[64] Wiesel, T. N., & Hubel, D. H. (1963). Single-cell responses in striate cortex of kittens deprived of vision in one eye. *Journal of Neurophysiology*, 26(6), 1003–1017.

in birdsong, and in human language: early experience has outsized effects not because it is special in some abstract sense, but because it shapes the starting point from which everything later is learned.[65]

The same irreversibility operates over evolutionary time. Engineers often look for optimal solutions; biological systems settle for workable ones, and once a solution supports survival it tends to persist, even if better ones might exist in principle. Evolution, as François Jacob put it, is a tinkerer, not an engineer: it cannot redesign from scratch, only modify what is already there.[66] The recurrent laryngeal nerve is the standard illustration. In fish, the nerve runs a short, direct course from the brain, past the heart, to the gills. As the neck lengthened in land vertebrates, the nerve stayed hooked under an artery near the heart, so in a human it descends into the chest and loops back up to the larynx, and in a giraffe the round trip is several meters for a destination a short distance from where the nerve began.[67] No designer would draw that path; a tinkerer who could only lengthen what was already there had no choice. Stephen Jay Gould and Richard Lewontin warned that the same history means not every feature of an organism is an adaptation at all; some are byproducts of the way other features were built.[68] Historical contingency is what it means to be a living system.

The Brain as an Adaptive Control System

If the brain is not a computer, what kind of system is it? A useful framing is to view it as an adaptive control system embedded in a body. What it controls is not the world but a set of internal variables that must stay within bounds for the organism to live: core temperature, blood sugar, blood oxygen, water balance, and, at a longer range, distance from predators and proximity to food and mates. Walter Cannon named the maintenance of these variables homeostasis, and W. Ross Ashby made the idea precise: an organism's behavior is whatever keeps its essential variables inside their limits, and learning is the process by which the controller changes its own organization when the old behavior no longer does so.[69] The brain's role is to regulate behavior in an environment that is uncertain, dynamic, and only partially observable, and it does this by continuously coupling perception and action, adjusting its internal organization based on the consequences of its own behavior. The division of labor is simple to state. The brain generates behavior, and the world corrects it.

From this perspective, cognition is not a detached process that observes the world and computes answers. It is an activity inseparable from action. Perception is an active process of inference shaped by prior experience and biological need; Hermann von Helmholtz said so in 1867, calling perception an unconscious inference from inadequate sensory evidence, and the contemporary view of the brain as a prediction machine that issues guesses and listens for its errors is a direct descendant of his.[70] Learning is the mechanism by which the control system adapts. Memory is the persistence of those adaptations.

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[65] Knudsen, E. I. (2004). Sensitive periods in the development of the brain and behavior. *Journal of Cognitive Neuroscience*, 16(8), 1412–1425.

[66] Jacob, F. (1977). Evolution and tinkering. *Science*, 196(4295), 1161–1166.

[67] Wedel, M. J. (2012). A monument of inefficiency: The presumed course of the recurrent laryngeal nerve in sauropod dinosaurs. *Acta Palaeontologica Polonica*, 57(2), 251–256.

[68] Gould, S. J., & Lewontin, R. C. (1979). The spandrels of San Marco and the Panglossian paradigm: A critique of the adaptationist programme. *Proceedings of the Royal Society of London B*, 205(1161), 581–598.

[69] Cannon, W. B. (1932). *The Wisdom of the Body*. W. W. Norton; Ashby, W. R. (1952). *Design for a Brain*. Chapman & Hall; Powers, W. T. (1973). *Behavior: The Control of Perception*. Aldine.

[70] Helmholtz, H. von (1867). *Handbuch der physiologischen Optik*. Leipzig: Voss. English edition: *Treatise on Physiological Optics*, vol. III, trans. J. P. C. Southall (Optical Society of America, 1925); Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204.

Chapter 6 takes up the inference, and Chapter 4 the generation and correction of movement; here the point is only that none of these is a separate faculty bolted onto a controller. They are what controlling a body in an uncertain world requires.

The objective of this system is therefore not truth in the abstract but viability. The brain must keep the organism alive long enough to reproduce, and Ashby's essential variables, or in Humberto Maturana and Francisco Varela's later phrase the continued self-production of the living system, are what it serves.[71] Accuracy is a means, and viability is the end. When precision conflicts with speed, energy efficiency, or robustness, precision often loses. One contemporary and contested attempt to formalize this idea, Karl Friston's free-energy principle, treats the organism as minimizing the long-run surprise of its own sensory states; whether that formulation adds anything to Ashby's is disputed, but the core claim, that the system's job is to stay within bounds, is not.[72]

This framing helps explain why human cognition is full of heuristics, biases, and shortcuts. Herbert Simon pointed out in 1956 that an organism with limited time and computation cannot optimize and must instead settle for options that are good enough, a strategy he called satisficing; Amos Tversky and Daniel Kahneman documented the systematic errors that fast heuristics produce; and Gerd Gigerenzer and Henry Brighton showed that in uncertain environments, where the data are few and noisy, simple rules that ignore most of the available information often predict better than models that use all of it.[73] These are trade-offs that favor survival under uncertainty, and the next section examines the most basic of them.

Noise, Degeneracy, and the Myth of Optimization

Engineers are trained to suppress noise, and they use redundancy deliberately and well: error-correcting memory, mirrored disks, and the triple-redundant flight computers of an airliner all protect a system by duplicating a component and comparing the copies. Biological systems are built from parts far noisier than any engineer would accept, and their redundancy is of a different kind.

The noise first. Christopher Allen and Charles Stevens recorded from single synapses in slices of rodent hippocampus in 1994, stimulating one axon and watching for the response in the cell it contacted, and found that more than half of the arriving impulses evoked no response at all. They ruled out failures of the stimulating electrode, failures of conduction along the axon, and the artificially cool temperature of the bath, and concluded that the synapse itself simply fails to release transmitter on most trials.[74] A cortical neuron shows the same unreliability at the level of its output. Michael Shadlen and William Newsome analyzed neurons in the visual cortex of monkeys watching identical repeated stimuli and found that the number of spikes fired on each repeat varied so widely that the variance was about as large as the mean: the same stimulus never produces the same spike train twice.[75] Noise enters at every

[71] Ashby, W. R. (1952). *Design for a Brain*. Chapman & Hall; Maturana, H. R., & Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living*. D. Reidel.

[72] Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.

[73] Simon, H. A. (1956). Rational choice and the structure of the environment. *Psychological Review*, 63(2), 129–138; Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157), 1124–1131; Gigerenzer, G., & Brighton, H. (2009). Homo heuristicus: Why biased minds make better inferences. *Topics in Cognitive Science*, 1(1), 107–143.

[74] Allen, C., & Stevens, C. F. (1994). An evaluation of causes for unreliability of synaptic transmission. *Proceedings of the National Academy of Sciences*, 91(22), 10380–10383.

[75] Shadlen, M. N., & Newsome, W. T. (1998). The variable discharge of cortical neurons: Implications for connectivity, computation, and information coding. *Journal of Neuroscience*, 18(10), 3870–3896.

stage, from the random opening of individual ion channels to the random arrival of photons at the retina, and its sources are well enough characterized that they can be budgeted.[76]

Rather than eliminating this variability, the brain exploits it. Populations of neurons average over the failures of their members, so that a signal carried by a hundred unreliable synapses is reliable even when each one fails more often than not. Feedback stabilizes the dynamics of the whole, and the components regulate themselves: neurons monitor their own average activity over a day or more and scale their synapses up or down to keep it within a working range, a mechanism Gina Turrigiano and Sacha Nelson called homeostatic plasticity, so that a circuit whose components drift is held at the operating point where it is most sensitive.[77] Chapter 4 shows how inhibition prevents runaway excitation on a faster timescale, and Chapter 5 shows why slow cortical learning is itself a stabilizing device.

The redundancy is where the contrast with engineering is sharpest. An engineer's redundancy is duplication: three identical computers running the same program. The brain's redundancy is what Gerald Edelman and Joseph Gally called degeneracy, the ability of structurally different elements to perform the same function.[78] Many different combinations of neurons in a population can produce the same movement; many different synaptic configurations can store the same memory; many different genes can be deleted without changing the organism, because other pathways cover the loss. Degenerate systems are robust in a way duplicated ones are not. A duplicated system fails when all its copies meet the same unanticipated condition, while a degenerate one has, in effect, many partially independent answers to every problem, and degeneracy is also the raw material of evolution, since a change to one element is tolerated while the others keep the function alive. Distributed representations prevent single points of failure for the same reason. The recovery after stroke described earlier is degeneracy at work: the surviving tissue was never a spare copy of the lost tissue, but it could be taught to do part of its job.

Optimization in biology does not mean maximizing performance on a benchmark. It means maintaining function across a wide range of conditions with limited resources, and the resource that binds hardest is the 20 watts. Peter Sterling and Simon Laughlin argue that, judged by that standard, the nervous system is a near-optimal design: it sends information at the lowest rate that will do, uses the cheapest signal that will carry it, and builds its wiring to the minimum length that connects what must be connected.[79] By the standard that matters to an organism, the brain is not inefficient at all. Nor does the brain optimize nothing: it has no single global objective, but Chapters 3 and 6 show many of its circuits minimizing a local error or cost.

Why This Matters Before We Study Neurons

This chapter has stayed above the level of cells on purpose. That is not because the details are unimportant, but because they are easily misinterpreted without the right conceptual frame.

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[76] Faisal, A. A., Selen, L. P. J., & Wolpert, D. M. (2008). Noise in the nervous system. *Nature Reviews Neuroscience*, 9(4), 292–303.

[77] Turrigiano, G. G., & Nelson, S. B. (2004). Homeostatic plasticity in the developing nervous system. *Nature Reviews Neuroscience*, 5(2), 97–107.

[78] Edelman, G. M., & Gally, J. A. (2001). Degeneracy and complexity in biological systems. *Proceedings of the National Academy of Sciences*, 98(24), 13763–13768.

[79] Sterling, P., & Laughlin, S. (2015). *Principles of Neural Design*. MIT Press.

Figure 1.3 shows what the frame is for. It is Santiago Ramón y Cajal's 1899 drawing of a single Purkinje cell from the human cerebellum, made from tissue prepared with the Golgi stain, a silver method devised by Camillo Golgi that, for reasons still not understood, blackens only about one cell in a hundred and leaves the rest invisible, so that one neuron stands out entire against its unstained neighbors. Drawings like this established that the nervous system is made of discrete, branching cells rather than a continuous net, and the branching is the point. A Purkinje cell's dendritic tree, the branched input region of the neuron that fills the upper part of the drawing, receives on the order of 200,000 inputs, and its single axon, the line leaving the base of the cell body, is the only output of the cerebellar cortex. A reader who looks at the drawing with this chapter's frame sees a component shaped by its constraints: a cell whose geometry is a wiring solution, whose inputs are individually unreliable, whose output is a slow train of millisecond pulses, and whose synapses are among the sites at which the cerebellum learns to correct movements before their errors are felt. Chapter 4 explains how.

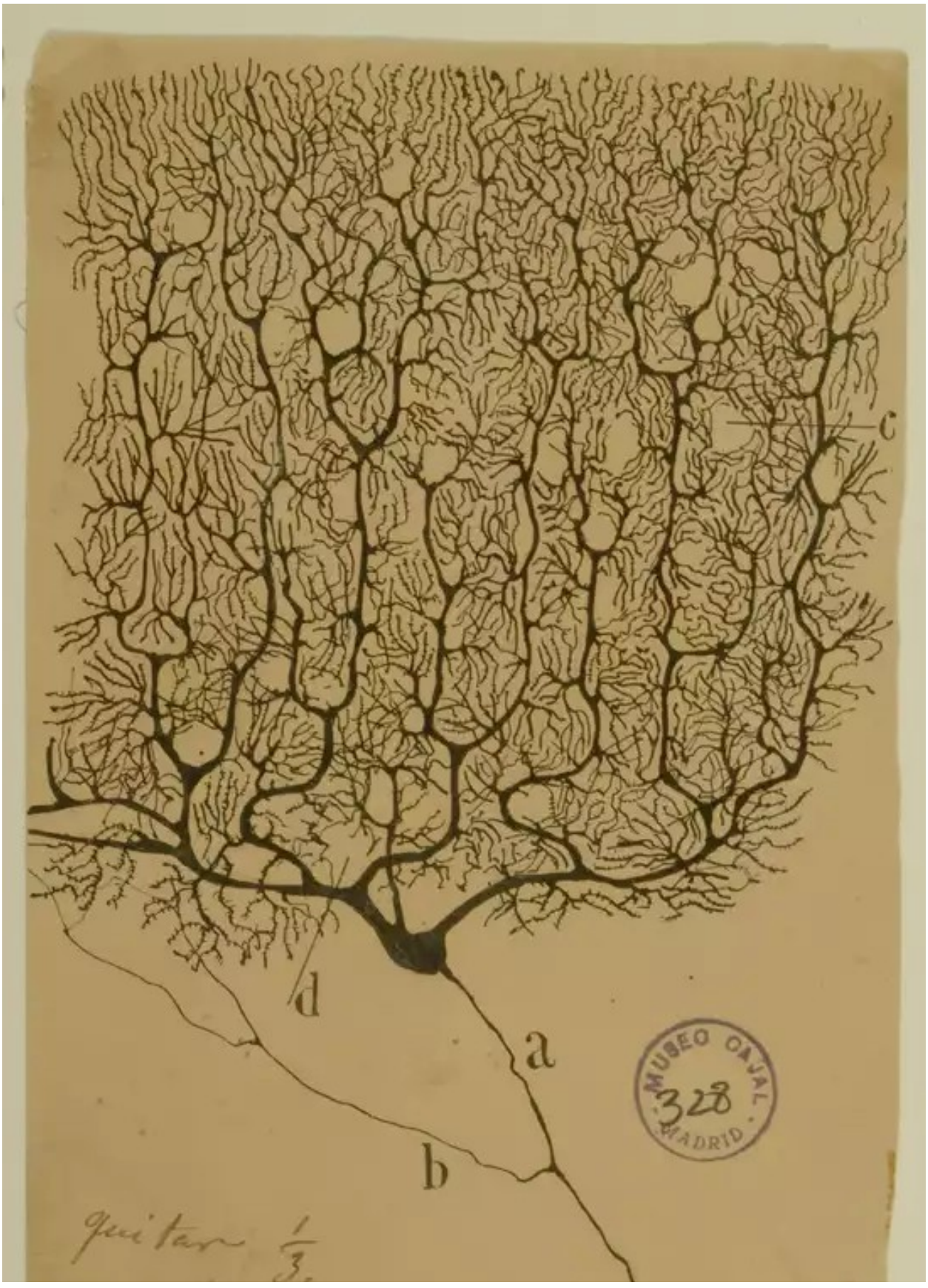

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Figure 1.3. Cajal's 1899 drawing of a Purkinje cell from the human cerebellum, Golgi stain. The flat, densely branched dendritic tree fills the upper part of the drawing; the single axon leaves the base of the cell body. Drawings like this established neurons as discrete, branching cells.

When we turn to neurons in the next chapter, we will encounter cells that signal electrically, communicate chemically, and change physically in response to activity. Without the framing established here, these mechanisms can seem arbitrary or unnecessarily complex. With it, they become intelligible as solutions shaped by constraint, history, and function. If there is one idea to carry forward, it is this: the brain is not an engineered solution to a clean problem. It is an evolved solution to a messy one, and understanding it requires patience, humility, and a willingness to revise intuitions forged in other domains. Neuroscience begins not with facts but with learning how to see.

Implications for Artificial Intelligence

The artificial neural networks that now dominate machine learning were named for the brain, and the comparison drawn in this chapter bears on them in several ways.

First, energy. A brain runs on about 20 watts, and that budget, not any abstract preference for elegance, is what forced it into sparse activity, low firing rates, local wiring, and prediction. Training a large model consumes energy that is measured in megawatt-hours, and running it is not cheap either. As power becomes the binding constraint on artificial systems, the brain's economies, which Attwell, Laughlin, and Lennie have made quantitative, are a design guide rather than a curiosity.

Second, noise. Artificial networks are built on deterministic arithmetic and made insensitive to small perturbations, when they are, only as an afterthought. The brain is built from components that fail more often than they succeed and gets reliability from populations, feedback, and degeneracy. Techniques that inject noise during training, or that spread a representation across many units so that no single one is essential, are rediscoveries of this strategy, and the brain suggests they could be pushed much further.

Third, the absence of reset. A network can be reinitialized and retrained from scratch, and most are; the brain cannot, and so it had to solve the problem of learning continually without erasing what it already knew. The catastrophic interference that McCloskey and Cohen demonstrated is still unsolved in general for artificial networks, and the brain's answer, a fast learner that replays into a slow one, was arrived at independently in machine learning in the form of experience replay. Chapter 5 tells that story.

Fourth, control. Most artificial systems are trained to map inputs to outputs and are then frozen. The brain is a controller that generates behavior, observes the consequences, and corrects, and it is never frozen. Reinforcement learning is the branch of machine learning that takes this seriously, and the agents it produces face exactly the problems this chapter has described: a world that is only partially observable, feedback that is delayed and noisy, and no clean separation between the part that acts and the part that learns. The brain is an existence proof that those problems can be solved by a system with no central controller at all, and Chapter 4 describes the machinery that makes a central controller unnecessary.

Finally, a caution. Nothing in this chapter says that the brain's solutions are the only ones, or that an artificial system must copy them. The brain is a tinkered solution under constraints that silicon does not share: it cannot duplicate a component exactly, it cannot run a clock, and it cannot be reset. Some of what it does is a response to those constraints and would be foolish to imitate. The point of the

comparison is to know which is which, and that requires understanding the brain on its own terms, which is what the rest of this book attempts.

The Central Lesson

The brain is not a digital computer that happens to be built from cells. It is a physical system, made of slow, noisy, energy-limited components, with an irreversible history written into its structure at every scale, whose job is not to compute correct answers but to keep a body alive by generating behavior and letting the world correct it. Its slowness, its unreliability, and its biases are what that job looks like when it is done within a budget of 20 watts by a tinkerer who could never start over. Every mechanism the following chapters describe should be read as a solution to that problem, not to the one an engineer would have posed.

Next Chapter

Chapter 2: Neurons: The Brain's Signaling Cells

Figure and Epigraph Sources

Figure Attribution

Figure 1.1: "Von Neumann Architecture" diagram by Wikimedia Commons user Kapooht, CC BY-SA 3.0, via Wikimedia Commons.

Figure 1.2: "Mri brain side view" (midsagittal magnetic resonance image of a human head) by Oliver Stollmann, free-use attribution license, via Wikimedia Commons.

Figure 1.3: Santiago Ramón y Cajal, drawing of a Purkinje cell from the human cerebellum, Golgi stain (1899), ink on paper, Instituto Cajal (CSIC), Madrid; public domain.

Epigraph

Emily Dickinson, "The Brain—is wider than the Sky—" (c. 1862), poem 632 in *The Poems of Emily Dickinson*, ed. T. H. Johnson (Harvard University Press, 1955); Fr598 in R. W. Franklin's edition (1998). The epigraph gives the first line without Dickinson's dashes.

Chapter 2

Neurons: The Brain's Signaling Cells

> *"Like the entomologist in pursuit of brightly coloured butterflies, my attention hunted, in the flower garden of the gray matter, cells with delicate and elegant forms, the mysterious butterflies of the soul, the beating of whose wings may some day—who knows?—clarify the secret of mental life."*
>
> *— Santiago Ramón y Cajal, Recollections of My Life (1937)*

Matter That Signals

The previous chapter argued that the brain is not a digital computer but a physical, biological system shaped by evolutionary constraint. That argument now has to be paid for in detail. If cognition arises from matter, what kind of matter can support perception, memory, and action, and what do its physical properties allow and forbid?

The answer begins with a single kind of cell. At the end of the nineteenth century, Santiago Ramón y Cajal stained slices of brain with Camillo Golgi's silver method (Chapter 1) and drew what he saw: discrete, branching cells, each bounded by its own membrane, touching but not fused with their neighbors. Wilhelm Waldeyer gave the cell its name, the neuron, in 1891, and the claim that the nervous system is built from such units, the neuron doctrine, rests chiefly on Cajal's drawings.[80] The cells Cajal called the butterflies of the soul were pyramidal neurons of the cerebral cortex, and the hope in the epigraph above, that their study would one day explain the mind, is the program this book follows.

Neurons are living cells, governed by the same physical and chemical laws as every other biological system; they consume energy, respond to stress, and eventually fail, and yet, organized in vast networks, they give rise to perception, memory, and thought. This chapter follows a signal through one neuron, from the chemical event at a synapse to the electrical impulse that leaves along the axon, and asks at each step what the physics costs and what it buys. Readers who want the underlying cell biology in more detail will find it in Chapter 0.

A Neuron Is a Cell

That statement is easy to say and easy to forget. Neurons are not logic gates. They do not execute instructions, store variables, or manipulate symbols; those descriptions are metaphors imposed at higher levels of analysis, useful for some purposes and misleading for others.[81] At the level of mechanism, a neuron integrates physical inputs and generates physical outputs according to biophysical rules, and it has no place to hold a variable apart from its own structure.

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[80] Waldeyer, W. (1891). Ueber einige neuere Forschungen im Gebiete der Anatomie des Centralnervensystems. *Deutsche medicinische Wochenschrift*, 17, 1213–1218, 1244–1246, 1267–1269, 1287–1289, 1331–1332, 1352–1356; Ramón y Cajal, S. (1909–1911). *Histologie du système nerveux de l'homme et des vertébrés* (2 vols., trans. L. Azoulay). Paris: Maloine; Shepherd, G. M. (1991). *Foundations of the Neuron Doctrine*. Oxford University Press. The word "neuron" was introduced by Waldeyer in 1891; the doctrine that the nervous system is built of discrete cells rests chiefly on Cajal's histology.

[81] Marr, D. (1982). *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. W. H. Freeman.

Like all cells, a neuron is bounded by a membrane, a film of fatty molecules about five nanometers thick that water and charged particles cannot easily cross.[82] On either side of the membrane is a salt solution, but not the same one. Inside, potassium ions are concentrated and sodium ions are scarce; outside, the reverse. The difference is maintained by a protein embedded in the membrane, the sodium-potassium pump, which on each cycle pushes three sodium ions out and brings two potassium ions in. Each cycle is paid for with one molecule of ATP (adenosine triphosphate), the small molecule that cells use as an energy currency, spending it to drive reactions that would not run on their own. Because the membrane at rest lets potassium leak out more easily than sodium leaks in, the unequal distribution leaves the inside of the cell about 70 millivolts negative relative to the outside. This is the resting potential, and every signal the neuron produces is a brief departure from it.

Figure 2.1 shows the arrangement: two salt solutions, a thin membrane, pumps that burn ATP without pause, and channels, pores in the membrane that open and close, through which ions cross when the cell signals. The cost of keeping this order is large. By Attwell and Laughlin's estimate in Chapter 1, most of the energy budget goes to pumping back the ions that signaling lets through.[83]

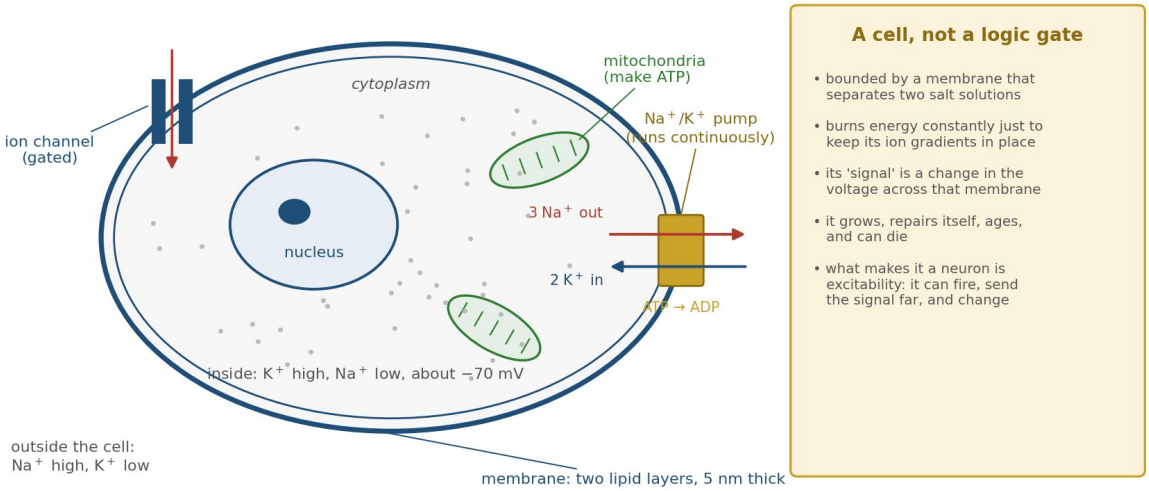


Figure 2.1. A neuron is a living cell. A membrane separates two salt solutions; pumps burn ATP continuously to keep sodium out and potassium in, and gated channels let ions cross when the cell signals. Nothing here is a logic gate.

What distinguishes neurons from most other cells is not excitability alone, which they share with muscle and some gland cells, but a combination: the ability to generate electrical signals, to transmit them over long distances, to influence particular other cells, and to modify those influences over time.[84]

[82] Alberts, B., Johnson, A., Lewis, J., Morgan, D., Raff, M., Roberts, K., & Walter, P. (2015). *Molecular Biology of the Cell* (6th ed.). Garland Science. The pump moves three sodium ions out and two potassium ions in for each ATP molecule it consumes.

[83] Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*, 21(10), 1133–1145.

[84] Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (Eds.). (2021). *Principles of Neural Science* (6th ed.). McGraw-Hill.

The Polarized Structure of Neurons

Despite their diversity, most neurons share a common structural plan, and the plan constrains how information can flow. At one end are the dendrites, branching structures that receive signals from other neurons. At the center is the cell body, or soma, which holds the nucleus and the cell's metabolic machinery. Extending from the soma is the axon, a long, thin projection that carries the neuron's output, a brief all-or-none electrical impulse called the action potential, or spike, sometimes for a millimeter and sometimes for a meter. In long-range axons the fiber is wrapped in myelin, an insulating sheath of fatty membrane laid down in many layers by neighboring support cells and interrupted at regular intervals by short bare gaps; most local axons in the cortex are unmyelinated. At its far end the axon branches into terminals that contact other cells.

The inputs are numerous. Megías and colleagues reconstructed pyramidal neurons from the CA1 region of the rat hippocampus and counted about 30,000 excitatory synapses and 1,700 inhibitory ones on a single cell, and a typical neuron in the mouse cortex carries several thousand.[85] The Purkinje cell of the cerebellum, whose drawing closed Chapter 1, receives on the order of 200,000. Against those thousands of inputs stands one axon. It may branch to reach thousands of targets, but it carries a single signal, the same to every branch. Cajal called this arrangement the law of dynamic polarization: signals enter through the dendrites and soma and leave through the axon, in one direction.[86]

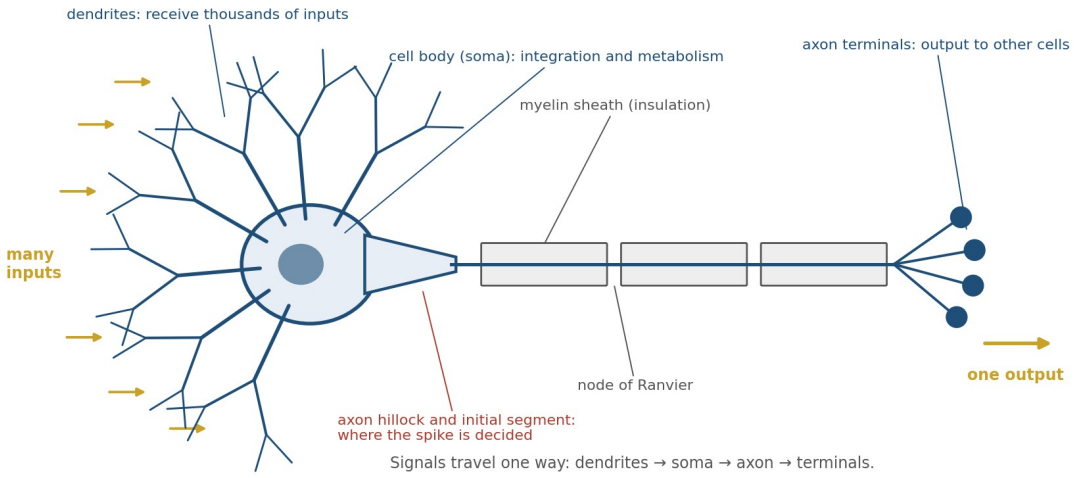


Figure 2.2. The polarized plan of a neuron. Thousands of inputs arrive on the dendrites and soma; a single axon, insulated by myelin in long-range axons, carries the one output, decided at the initial segment, to the terminals. Signals travel one way.

Figure 2.2 shows the plan. The many-to-one shape is the first constraint on what a neuron can compute: whatever happens on the dendrites must eventually be reduced to a single decision, made at a single place, about whether and when to fire.

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[85] Megías, M., Emri, Z., Freund, T. F., & Gulyás, A. I. (2001). Total number and distribution of inhibitory and excitatory synapses on hippocampal CA1 pyramidal cells. *Neuroscience*, 102(3), 527–540; Braitenberg, V., & Schüz, A. (1998). *Cortex: Statistics and Geometry of Neuronal Connectivity* (2nd ed.). Springer. A rat hippocampal CA1 pyramidal cell carries roughly 30,000 excitatory and 1,700 inhibitory synapses.

[86] Ramón y Cajal, S. (1909–1911). *Histologie du système nerveux de l'homme et des vertébrés* (2 vols., trans. L. Azoulay). Paris: Maloine. Cajal's "law of dynamic polarization."

The Physical Form of Inputs to a Neuron

Neurons receive their inputs at specialized junctions that Charles Sherrington named synapses in 1897, before anyone knew how they worked.[87] The answer came from a frog's heart. In 1921 Otto Loewi stimulated the vagus nerve of one isolated heart, which slowed its beat, then transferred the fluid that had bathed it to a second heart, which slowed too. Something released by the nerve had carried the message.[88] That something was a neurotransmitter, and most synapses in the brain work the same way.

At a chemical synapse, an electrical impulse arriving at the axon terminal opens channels that let calcium ions in. The calcium triggers small membrane-bound packets of neurotransmitter, called vesicles, to fuse with the terminal membrane and release their contents into the synaptic cleft, a gap about twenty nanometers wide between the terminal and the receiving cell. The transmitter diffuses across the cleft in a fraction of a millisecond and binds receptors on the membrane of the receiving neuron, and the receptors open ion channels.[89] Figure 2.3 shows the sequence, including the clearing of transmitter from the cleft by reuptake into cells, enzymatic breakdown, and diffusion away.

[87] Foster, M., & Sherrington, C. S. (1897). *A Textbook of Physiology, Part III: The Central Nervous System* (7th ed.). Macmillan. The term was coined by Sherrington.

[88] Loewi, O. (1921). Über humorale Übertragbarkeit der Herznervenwirkung. *Pflügers Archiv für die gesamte Physiologie*, 189, 239–242.

[89] Katz, B. (1969). *The Release of Neural Transmitter Substances*. Liverpool University Press; Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (Eds.). (2021). *Principles of Neural Science* (6th ed.). McGraw-Hill. The synaptic cleft is about 20 nanometers wide.

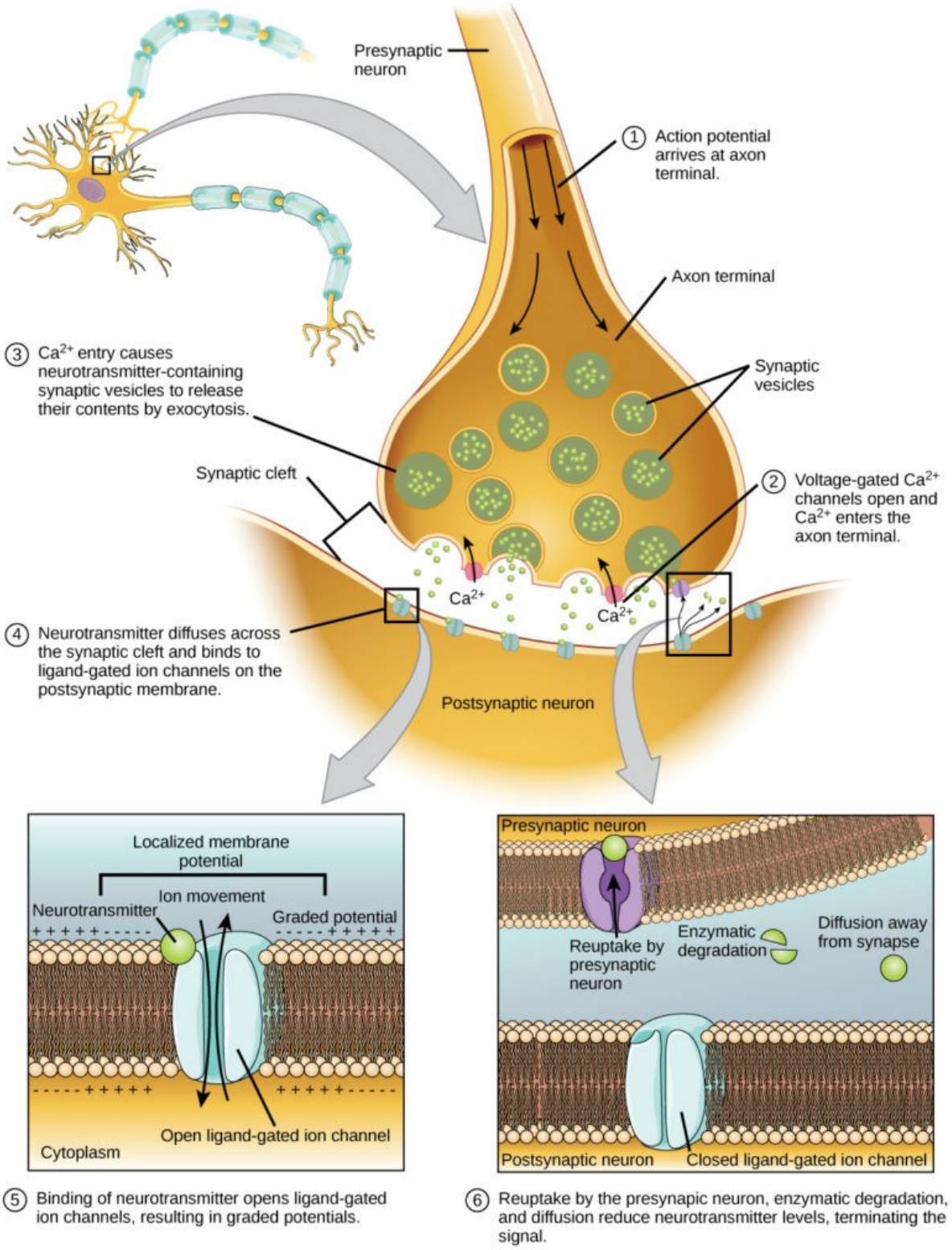


Figure 2.3. A chemical synapse. An action potential arriving at the terminal opens calcium channels; calcium triggers vesicles to release neurotransmitter into the cleft; the transmitter binds receptors on the postsynaptic membrane and produces a small graded potential, then is cleared by reuptake, breakdown, and diffusion.

The point to hold on to is that the immediate input to a neuron is not an electrical spike but a small, local chemical event that produces a graded electrical change, larger or smaller depending on how much transmitter was released and how many receptors received it.[90] This is what makes the synapse adjustable, and it is why learning, as later sections show, lives mostly at synapses.

Not every synapse is chemical. Some neurons, especially the inhibitory interneurons of the cortex and hippocampus, are also joined by electrical synapses, or gap junctions: protein channels that connect the interiors of two cells directly, so that current flows from one to the other with no transmitter and almost no delay. Barry Connors and Michael Long reviewed the evidence that such coupling is widespread in the mammalian brain, and mice lacking the main neuronal gap-junction protein, connexin 36, show weakened gamma oscillations, the fast rhythm that Chapter 4 describes.[91] Electrical synapses synchronize; chemical synapses do almost everything else.

Quanta, Failures, and Noise

The chemical event at a synapse is not a smooth flow but a set of discrete packets, and it is unreliable. Both facts were discovered by Bernard Katz and his colleagues at the junction between nerve and muscle in the frog, the largest and most accessible synapse there is.

In 1952 Paul Fatt and Katz, recording with a fine electrode inside a muscle fiber, saw small depolarizations of about half a millivolt that appeared at random, with no nerve impulse to trigger them. They had the same time course as the full response to a nerve impulse, only much smaller, and curare, which blocks the muscle's receptors for the transmitter acetylcholine, shrank them.[92] Two years later José del Castillo and Katz lowered the calcium and raised the magnesium in the bath until the nerve impulse released very little transmitter and found that the response to each impulse now fluctuated in steps: sometimes nothing, sometimes one unit the size of the spontaneous event, sometimes two or three. The counts followed the statistics of independent random events. They concluded that transmitter is released in fixed packets, quanta, each almost certainly the contents of one vesicle, and that each impulse releases a random number of them.[93]

The same randomness governs synapses in the brain, where an impulse often releases one vesicle or none. Chapter 1 described Christopher Allen and Charles Stevens's finding that at single hippocampal synapses more than half of the arriving impulses evoked no response at all. In pairs of connected cortical pyramidal neurons recorded by Henry Markram and his colleagues, the amplitude of the response to a single presynaptic impulse varied from trial to trial so much that its standard deviation was on average about half its mean.[94] A. Aldo Faisal, Luc Selen, and Daniel Wolpert surveyed the sources

[90] Fatt, P., & Katz, B. (1951). An analysis of the end-plate potential recorded with an intra-cellular electrode. *Journal of Physiology*, 115(3), 320–370.

[91] Connors, B. W., & Long, M. A. (2004). Electrical synapses in the mammalian brain. *Annual Review of Neuroscience*, 27, 393–418; Hormuzdi, S. G., Pais, I., LeBeau, F. E. N., Towers, S. K., Rozov, A., Buhl, E. H., Whittington, M. A., & Monyer, H. (2001). Impaired electrical signaling disrupts gamma frequency oscillations in connexin 36-deficient mice. *Neuron*, 31(3), 487–495.

[92] Fatt, P., & Katz, B. (1952). Spontaneous subthreshold activity at motor nerve endings. *Journal of Physiology*, 117(1), 109–128.

[93] del Castillo, J., & Katz, B. (1954). Quantal components of the end-plate potential. *Journal of Physiology*, 124(3), 560–573; Katz, B. (1969). *The Release of Neural Transmitter Substances*. Liverpool University Press.

[94] Allen, C., & Stevens, C. F. (1994). An evaluation of causes for unreliability of synaptic transmission. *Proceedings of the National Academy of Sciences*, 91(22), 10380–10383; Markram, H., Lübke, J., Frotscher, M., Roth, A., & Sakmann, B. (1997). Physiology and anatomy of synaptic connections between thick tufted pyramidal neurones in the developing rat neocortex. *Journal of Physiology*, 500(2), 409–440. In Markram and colleagues' recordings the coefficient of variation of unitary EPSP amplitude averaged 0.52.

of such noise in the nervous system, from the random opening of single ion channels to the random number of vesicles released, and concluded that noise is present at every stage and limits what any single element can do reliably.[95] The neuron's input is therefore not a clean number but a sample from a distribution, and the cell must make its decisions from many such samples at once.

Excitatory and Inhibitory Postsynaptic Potentials

Not all synaptic inputs push a neuron toward firing; some do the opposite. When transmitter binds receptors on a dendrite or the soma, it changes the membrane's permeability to particular ions, and the direction of the change determines whether the synapse excites or inhibits.

An excitatory postsynaptic potential (EPSP) slightly depolarizes the membrane, making the inside less negative and moving it closer to the threshold at which an impulse is triggered. EPSPs are produced mostly by glutamate acting on receptors whose channels admit positive ions without much selectivity; sodium flowing in outweighs potassium flowing out, and the net inward current depolarizes the cell.[96] An inhibitory postsynaptic potential (IPSP) moves the membrane away from threshold or clamps it near rest, usually by opening channels for chloride or potassium ions, and makes firing less likely.[97] Figure 2.4 shows both, and what happens when they combine.

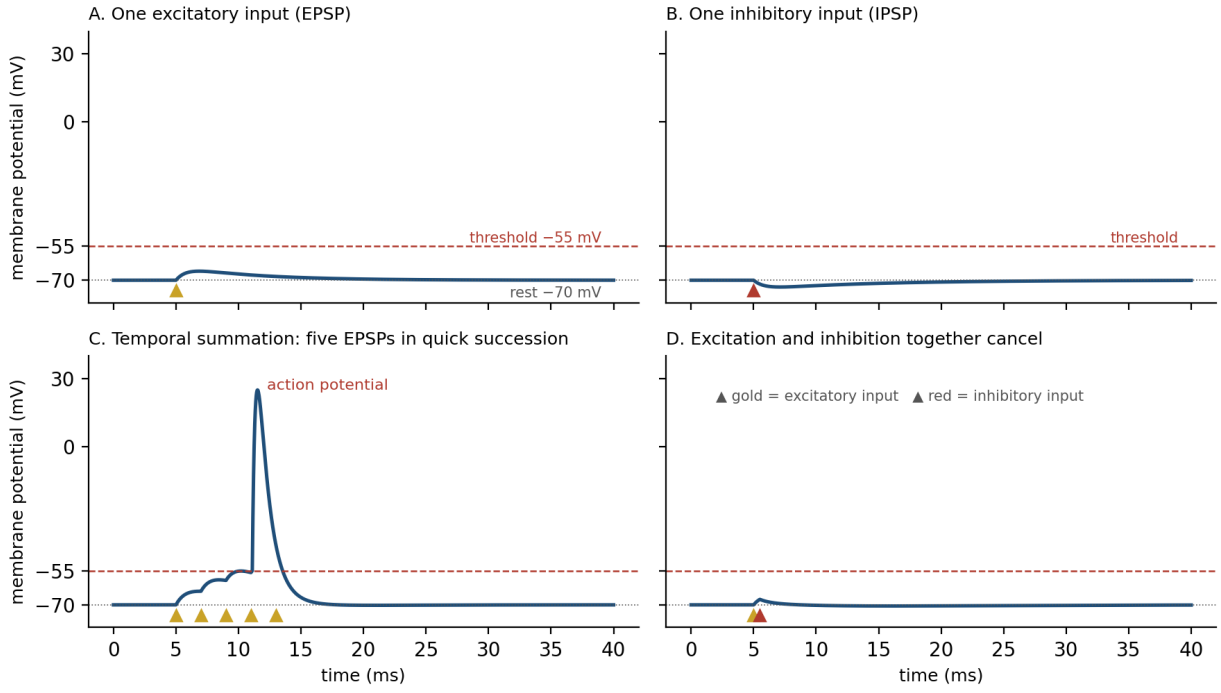


Figure 2.4. Excitatory and inhibitory postsynaptic potentials. A: one EPSP depolarizes the membrane. B: one IPSP hyperpolarizes it. C: five EPSPs in quick succession sum to threshold and trigger an action potential. D: an EPSP and an IPSP arriving together nearly cancel. Traces are simulated, not recorded, and the EPSPs are exaggerated for clarity: real ones are a fraction of a millivolt to a few millivolts, and tens are needed to reach threshold.

[95] Faisal, A. A., Selen, L. P. J., & Wolpert, D. M. (2008). Noise in the nervous system. *Nature Reviews Neuroscience*, 9(4), 292–303.

[96] Eccles, J. C. (1964). *The Physiology of Synapses*. Springer.

[97] Coombs, J. S., Eccles, J. C., & Fatt, P. (1955). The specific ionic conductances and the ionic movements across the motoneuronal membrane that produce the inhibitory post-synaptic potential. *Journal of Physiology*, 130(2), 326–374.

Real synaptic potentials are small. Markram and his colleagues measured the response at the soma of one cortical pyramidal neuron to a single impulse in another and found an average of 1.3 millivolts, ranging from about 0.15 to 5.5 millivolts across connections; Sen Song and his colleagues, recording from hundreds of such pairs, found that most connections were weak and a few strong, with a long-tailed distribution of sizes.[98] A few millivolts is the exception, a fraction of a millivolt the rule. EPSPs and IPSPs are graded, local, and short-lived, and they fade over tens of milliseconds as charge leaks back across the membrane. The neuron's firing reflects the balance of excitation and inhibition summed across its dendritic tree and soma over time.[99]

A worked example shows what that summing means. Suppose the membrane rests at −70 millivolts and the threshold for firing is about −55, fifteen millivolts higher. Suppose each excitatory input adds half a millivolt at the soma, and, as a first approximation, that inputs add linearly. Thirty inputs arriving within a couple of milliseconds then sum to fifteen millivolts and fire the cell. Now spread the same thirty inputs evenly across a tenth of a second. Each EPSP decays with a time constant, the time for its size to fall to about a third, which for illustration we take to be twenty milliseconds, so the sum never builds: the average depolarization is the rate of arrival, three inputs every ten milliseconds, times the size of each, times the time constant, or about three millivolts, far below threshold. The same thirty inputs fire the cell when they arrive together and leave it silent when they straggle. Add ten IPSPs of one millivolt each arriving with the burst, and the thirty excitatory inputs are no longer enough; about fifty are needed. Thirty synapses are a tenth of a percent of the 30,000 on a CA1 pyramidal cell. What drives a neuron is not how many of its inputs are active over a long period but how many are active together, and against how much inhibition.

Dendrites Are Not Passive Cables

The synaptic potentials spread from where they arise, along the dendrites and into the soma. The baseline description of this spread was worked out by Wilfrid Rall, who in 1959 treated a dendrite as an electrical cable: a thin conducting core surrounded by a leaky insulator. In a passive cable, a voltage change shrinks as it travels and is smeared out in time, because charge leaks across the membrane along the way and the membrane's capacitance slows every change.[100] A synapse at the tip of a long dendrite therefore has much less effect at the soma than the same synapse near it; in a pyramidal neuron a distal input can shrink many times over before it arrives.[101] Some neurons partly compensate by making their distal synapses stronger, and for small, scattered inputs the passive account is a good approximation of what the dendrite does.[102]

But dendrites are not passive cables. Their membranes carry voltage-gated channels for sodium, calcium, and potassium, channels that open when the membrane depolarizes, and they carry NMDA receptors, a type of glutamate receptor whose current grows as the membrane depolarizes because depolarization expels a magnesium ion that plugs the channel at rest (Chapter 5 tells that story). With

[98] Markram, H., Lübke, J., Frotscher, M., Roth, A., & Sakmann, B. (1997). Physiology and anatomy of synaptic connections between thick tufted pyramidal neurones in the developing rat neocortex. *Journal of Physiology*, 500(2), 409–440; Song, S., Sjöström, P. J., Reigl, M., Nelson, S. B., & Chklovskii, D. B. (2005). Highly nonrandom features of synaptic connectivity in local cortical circuits. *PLoS Biology*, 3(3), e68.

[99] Magee, J. C. (2000). Dendritic integration of excitatory synaptic input. *Nature Reviews Neuroscience*, 1(3), 181–190.

[100] Rall, W. (1959). Branching dendritic trees and motoneuron membrane resistivity. *Experimental Neurology*, 1(5), 491–527.

[101] London, M., & Häusser, M. (2005). Dendritic computation. *Annual Review of Neuroscience*, 28, 503–532.

[102] Magee, J. C. (2000). Dendritic integration of excitatory synaptic input. *Nature Reviews Neuroscience*, 1(3), 181–190.

these channels a dendrite can generate its own regenerative events. Jackie Schiller and her colleagues showed in 2000 that when enough neighboring synapses on a thin basal dendrite of a cortical pyramidal neuron are activated together, the branch fires a local NMDA spike, a plateau of depolarization lasting tens of milliseconds and confined to that branch.[103] Matthew Larkum, Julius Zhu, and Bert Sakmann found that the long apical dendrite of a layer 5 pyramidal neuron has its own zone near the top that fires broad calcium spikes, which drive the axon to fire bursts.[104] A dendrite, on this evidence, is not a wire that delivers its inputs to the soma but a set of semi-independent compartments, each able to detect when its own inputs coincide and to amplify them.

These dendritic spikes matter for learning. Nace Golding, Nathan Staff, and Nelson Spruston showed that synapses on the far dendrites of hippocampal CA1 neurons, which the cell's own output impulse barely reaches, can be strengthened when cooperative input triggers a local dendritic spike, even when the neuron does not fire at all.[105] The integration that the earlier sections described is real, but it is done by a structure with active parts.

The Axon Initial Segment: Where Spikes Begin

All this graded activity converges on one place. Just beyond the soma lies the axon initial segment, the first few tens of micrometers of the axon, and it is packed with voltage-gated sodium channels at a far higher density than the soma or dendrites. Maarten Kole and his colleagues estimated that in cortical pyramidal neurons the sodium-channel density at the initial segment is about fifty times that in the nearby dendrites, and their models could account for where and how the neuron initiates impulses only with such a density.[106] When the summed depolarization arriving there reaches threshold, about fifteen millivolts above rest, the initial segment fires an action potential. That the impulse starts in the axon rather than the soma was inferred by John Eccles's group from recordings in spinal motor neurons in 1957, and shown directly by Lucy Palmer and Greg Stuart, who used voltage-sensitive dyes to watch the impulse arise in cortical pyramidal neurons and located its origin about 35 micrometers down the axon from the point where it leaves the soma.[107] Figure 2.5 shows the arrangement.

[103] Schiller, J., Major, G., Koester, H. J., & Schiller, Y. (2000). NMDA spikes in basal dendrites of cortical pyramidal neurons. *Nature*, 404(6775), 285–289; London, M., & Häusser, M. (2005). Dendritic computation. *Annual Review of Neuroscience*, 28, 503–532.

[104] Larkum, M. E., Zhu, J. J., & Sakmann, B. (1999). A new cellular mechanism for coupling inputs arriving at different cortical layers. *Nature*, 398(6725), 338–341.

[105] Golding, N. L., Staff, N. P., & Spruston, N. (2002). Dendritic spikes as a mechanism for cooperative long-term potentiation. *Nature*, 418(6895), 326–331.

[106] Kole, M. H. P., Ilschner, S. U., Kampa, B. M., Williams, S. R., Ruben, P. C., & Stuart, G. J. (2008). Action potential generation requires a high sodium channel density in the axon initial segment. *Nature Neuroscience*, 11(2), 178–186.

[107] Coombs, J. S., Curtis, D. R., & Eccles, J. C. (1957). The generation of impulses in motoneurones. *Journal of Physiology*, 139(2), 232–249; Palmer, L. M., & Stuart, G. J. (2006). Site of action potential initiation in layer 5 pyramidal neurons. *Journal of Neuroscience*, 26(6), 1854–1863.

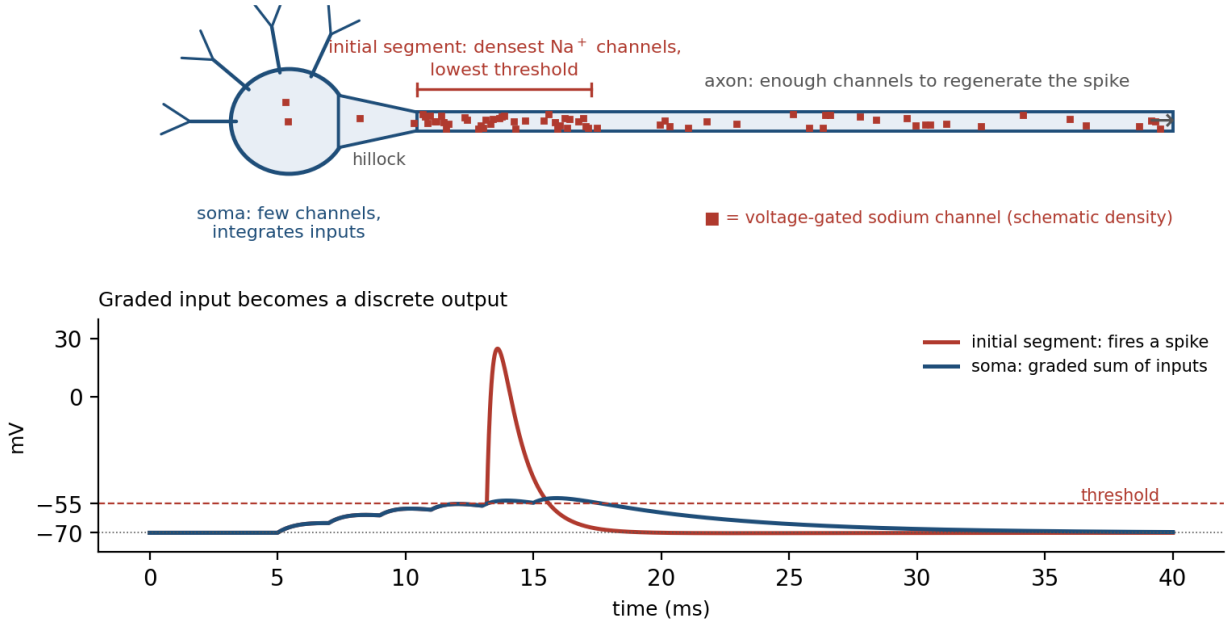


Figure 2.5. The axon initial segment. Top: voltage-gated sodium channels are sparse in the soma and densest in the first stretch of axon, which therefore has the lowest threshold. Bottom: the soma's graded sum of inputs (blue) crosses threshold there and the initial segment fires a stereotyped spike (red). Simulated traces.

Once initiated, the action potential does not only travel forward. It invades the soma at full amplitude, which is why every electrode in a neuron's cell body records the familiar spike, and from the soma it spreads back into the dendrites. Greg Stuart and Bert Sakmann recorded simultaneously from the soma and from the apical dendrite of the same layer 5 pyramidal neurons in 1994. They found voltage-gated sodium channels in patches of dendritic membrane and showed that the spike appears first in the axon and then actively propagates back into the dendritic tree, carried by the dendrites' own channels rather than merely leaking into them. This is the back-propagating action potential.[108] Back-propagation weakens with distance and can fail in the finest distal branches, and it is regulated by the dendrite's potassium channels and by inhibition, but over much of the dendritic tree it reports to every synapse that the neuron has just fired.[109]

That report is what allows a synapse to know whether its input helped. Chapter 5 shows that the back-propagating spike, arriving at a synapse whose NMDA receptors have just bound glutamate, expels their magnesium block and admits the calcium that triggers lasting change. It also couples inputs: Larkum and his colleagues found that one back-propagating spike, coinciding within a few milliseconds with weak input to the far apical dendrite, could trigger a dendritic calcium spike and a burst of output.[110]

The division of labor is therefore subtler than a simple picture of passive input and active output. Dendrites integrate, mostly linearly for weak input and nonlinearly when inputs cluster; the soma, with

[108] Stuart, G. J., & Sakmann, B. (1994). Active propagation of somatic action potentials into neocortical pyramidal cell dendrites. *Nature*, 367, 69–72.

[109] Stuart, G., Spruston, N., Sakmann, B., & Häusser, M. (1997). Action potential initiation and backpropagation in neurons of the mammalian CNS. *Trends in Neurosciences*, 20(3), 125–131; Golding, N. L., Staff, N. P., & Spruston, N. (2002). Dendritic spikes as a mechanism for cooperative long-term potentiation. *Nature*, 418(6895), 326–331.

[110] Larkum, M. E., Zhu, J. J., & Sakmann, B. (1999). A new cellular mechanism for coupling inputs arriving at different cortical layers. *Nature*, 398(6725), 338–341.

a lower density of sodium channels, relays rather than decides; the initial segment decides; and the axon transmits. Evolution separated integration, decision, and transmission into distinct regions while letting the decision travel back to every place where it can be used for learning.

How the Action Potential Propagates

In 1939 Alan Hodgkin and Andrew Huxley threaded a fine electrode down the inside of a squid's giant axon and measured, for the first time, the voltage across a nerve membrane during an impulse. The inside swung from its negative resting value not merely to zero, as the leading theory of the day predicted, but well past it, to tens of millivolts positive.[111]

Chapter 1 described how, by 1952, Hodgkin and Huxley had separated the current into its sodium and potassium components and written equations that reproduced the impulse. The mechanism the equations describe is this. When the membrane depolarizes past threshold, voltage-gated sodium channels open within a fraction of a millisecond, sodium ions rush in down their concentration gradient, and the inflow depolarizes the membrane further, which opens more sodium channels. This positive feedback drives the inside to about +40 millivolts. Within a millisecond the sodium channels close of their own accord, a process called inactivation, while slower voltage-gated potassium channels open and let potassium flow out, returning the membrane to rest and briefly below it. For a millisecond or two afterward the inactivated sodium channels cannot reopen, and the membrane cannot fire again; this refractory period limits a neuron to a few hundred impulses per second at most.[112] The ions exchanged are a tiny fraction of those in the cell, and the pumps restore the balance later, at the cost of ATP.

Propagation follows from the same mechanism. Sodium entering at the active patch of membrane spreads inside the axon and depolarizes the patch ahead of it to threshold, which then fires, injecting its own current further along. The patch behind is refractory and cannot fire again, so the impulse travels in one direction only. Each patch regenerates the impulse from its own ion gradients, so the impulse arrives at the far end of a meter-long axon as large as it started. No particle travels the length of the axon; what travels is a pattern of channel openings, each triggered by the one before.

[111] Hodgkin, A. L., & Huxley, A. F. (1939). Action potentials recorded from inside a nerve fibre. *Nature*, 144, 710–711.

[112] Hodgkin, A. L., & Huxley, A. F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *Journal of Physiology*, 117(4), 500–544; Hodgkin, A. L. (1964). *The Conduction of the Nervous Impulse*. Liverpool University Press; Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (Eds.). (2021). *Principles of Neural Science* (6th ed.). McGraw-Hill.

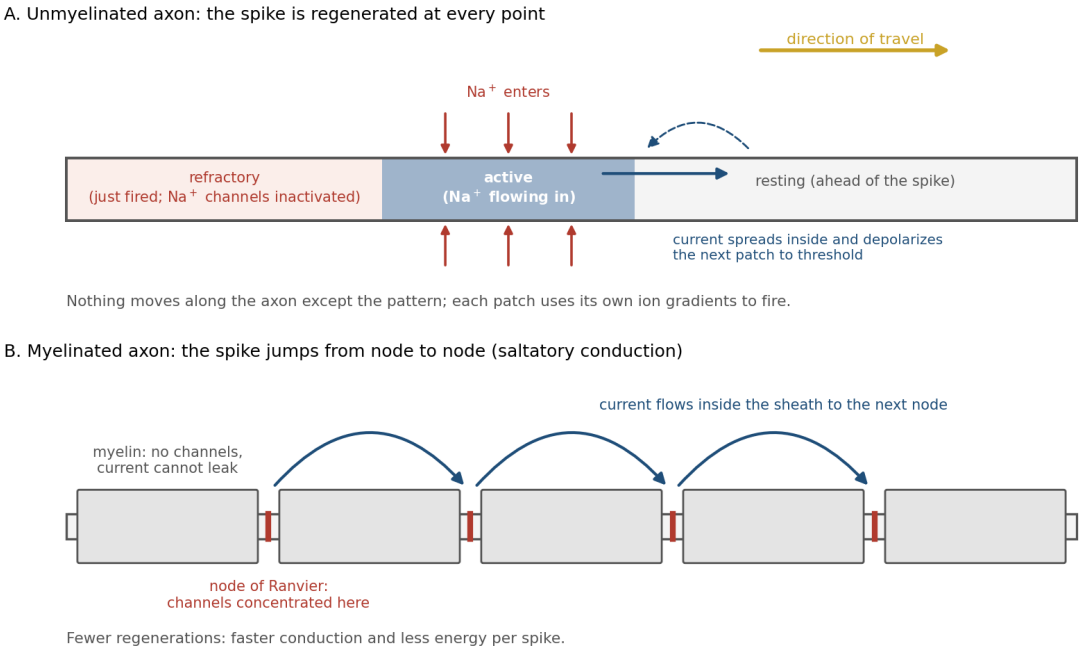


Figure 2.6. How the action potential travels. A: in a bare axon, sodium entering the active patch spreads inside and brings the next patch to threshold, while the patch behind is refractory; the spike is regenerated at every point. B: in a myelinated axon, channels are confined to the nodes and the spike jumps from node to node (saltatory conduction).

Figure 2.6 shows both forms of propagation. In a bare axon the impulse creeps, at speeds of the order of a meter per second in thin fibers. Myelin changes this. Because the insulating sheath prevents current from leaking out and reduces the charge needed to change the voltage, the current from one bare gap, or node, can bring the next node, up to a millimeter or so away, to threshold, and sodium channels are concentrated at the nodes. The impulse jumps from node to node, a mode of conduction called saltatory, from the Latin for leaping. Andrew Huxley and Robert Stämpfli demonstrated it in 1949 by measuring the currents around single frog nerve fibers along their length and showing that excitation passes from node to node, skipping the insulated stretches between.[113] The fastest myelinated axons conduct at around 100 meters per second, against about a meter per second for thin unmyelinated fibers, and at a lower metabolic cost, since only the nodes exchange ions.[114]

Rate, Timing, and the All-or-None Spike

If every action potential is the same size, how does a neuron say more or less? Edgar Adrian answered the question in 1926 with Yngve Zotterman. Working with a small muscle of the frog, they cut away tissue until the nerve carried impulses from a single stretch receptor, then hung weights from the muscle. Each impulse was the same size whatever the weight. A heavier weight produced impulses at a higher frequency, and a sustained weight produced a frequency that declined over seconds as the receptor adapted.[115] The message was carried by the number of impulses per second, not by their size.

[113] Huxley, A. F., & Stämpfli, R. (1949). Evidence for saltatory conduction in peripheral myelinated nerve fibres. *Journal of Physiology*, 108(3), 315–339.

[114] Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (Eds.). (2021). *Principles of Neural Science* (6th ed.). McGraw-Hill.

[115] Adrian, E. D., & Zotterman, Y. (1926). The impulses produced by sensory nerve-endings: Part II. The response of a single end-organ. *Journal of Physiology*, 61(2), 151–171; Adrian, E. D. (1928). *The Basis of Sensation: The Action of the Sense Organs*.

Adrian generalized the finding across sense organs, and firing rate has been the default currency of neural coding ever since. Figure 2.7A shows the principle: a stronger stimulus produces more spikes, not larger ones.

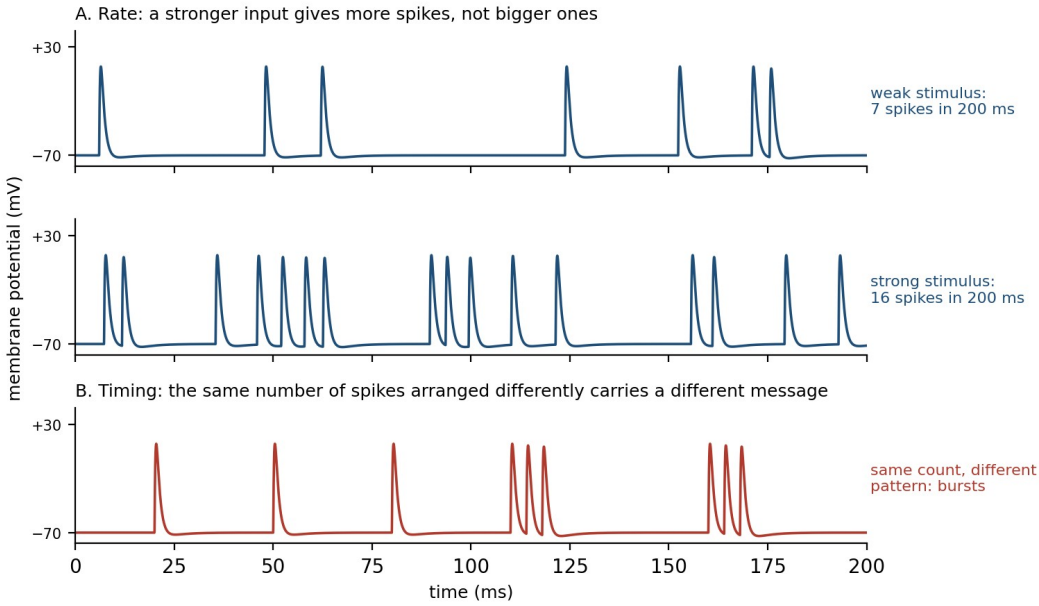


Figure 2.7. Spikes are all the same size. A: a stronger stimulus produces more spikes, not larger ones. B: the same number of spikes arranged as bursts carries a different message. Information lies in rate and timing. Simulated traces.

The all-or-none rule needs one qualification. Action potentials are stereotyped, and their presence rather than their size is the message, but small variations in their shape do modulate what they release. In hippocampal mossy fibers, Jörg Geiger and Peter Jonas found that the impulse in the axon terminal broadens up to threefold during high-frequency firing, letting more calcium in and releasing more transmitter. Henrik Alle and Geiger, and separately Yousheng Shu, David McCormick, and their colleagues in cortex, showed that subthreshold voltage changes at the soma spread hundreds of micrometers down the axon and change how much transmitter each subsequent spike releases.[116] The axon carries a mainly digital signal with an analog component riding on it.

Rate is also not the only code. Figure 2.7B shows the same number of spikes arranged as bursts, which a downstream neuron with the summing properties described above would read differently from evenly spaced spikes. Zachary Mainen and Terrence Sejnowski showed in 1995 that cortical neurons can time their spikes precisely: driven by a constant current, a neuron in a brain slice fired at slightly different times on each trial, but driven by a fluctuating current resembling real synaptic input, it fired at times reproducible to within a millisecond from trial to trial.[117] Speed argues the same way. In Simon Thorpe's experiment of Chapter 1, brain signals distinguishing photographs with and without an animal appeared

Christophers.

[116] Geiger, J. R. P., & Jonas, P. (2000). Dynamic control of presynaptic Ca2+ inflow by fast-inactivating K+ channels in hippocampal mossy fiber boutons. *Neuron*, 28(3), 927–939; Alle, H., & Geiger, J. R. P. (2006). Combined analog and action potential coding in hippocampal mossy fibers. *Science*, 311(5765), 1290–1293; Shu, Y., Hasenstaub, A., Duque, A., Yu, Y., & McCormick, D. A. (2006). Modulation of intracortical synaptic potentials by presynaptic somatic membrane potential. *Nature*, 441(7094), 761–765.

[117] Mainen, Z. F., & Sejnowski, T. J. (1995). Reliability of spike timing in neocortical neurons. *Science*, 268(5216), 1503–1506.

about 150 milliseconds after a twenty-millisecond flash.[118] If that signal passes through something like ten stages of processing between the eye and the decision, each stage has roughly fifteen milliseconds, time enough for a neuron firing at typical rates to emit one spike or none. A code that required counting spikes over long windows could not work at that speed. Information is carried by the rate of firing, by the timing of spikes, and by which neurons fire together, and how the brain weighs these in each circuit is an active research question.[119]

Plasticity: Learning as Physical Change

Synapses strengthen or weaken with use. Donald Hebb proposed in 1949 that when one cell repeatedly takes part in firing another, some growth process or metabolic change makes the connection more effective, and in 1973 Timothy Bliss and Terje Lømo found such a change in the hippocampus of anesthetized rabbits: brief high-frequency trains of impulses in the input pathway left the response of the receiving cells enlarged for hours afterward, a phenomenon now called long-term potentiation, or LTP.[120]

What changes is physical. A strengthened synapse can release more transmitter per impulse, can have more receptors in its membrane, and can sit on a larger spine, the small protrusion from the dendrite that carries most excitatory synapses in the cortex. Masanori Matsuzaki, Haruo Kasai, and their colleagues released glutamate onto single spines with flashes of light and showed that the stimulation that potentiates a synapse also enlarges its spine, and that in small spines the enlargement persists for more than an hour.[121] Figure 2.8 shows the before and after: more docked vesicles, a larger spine head, more receptors, and an EPSP that more than doubles.

[118] Thorpe, S., Fize, D., & Marlot, C. (1996). Speed of processing in the human visual system. *Nature*, 381(6582), 520–522.

[119] Rieke, F., Warland, D., de Ruyter van Steveninck, R., & Bialek, W. (1997). *Spikes: Exploring the Neural Code*. MIT Press.

[120] Hebb, D. O. (1949). *The Organization of Behavior: A Neuropsychological Theory*. Wiley; Bliss, T. V. P., & Lømo, T. (1973). Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit following stimulation of the perforant path. *Journal of Physiology*, 232(2), 331–356.

[121] Matsuzaki, M., Honkura, N., Ellis-Davies, G. C. R., & Kasai, H. (2004). Structural basis of long-term potentiation in single dendritic spines. *Nature*, 429, 761–766.

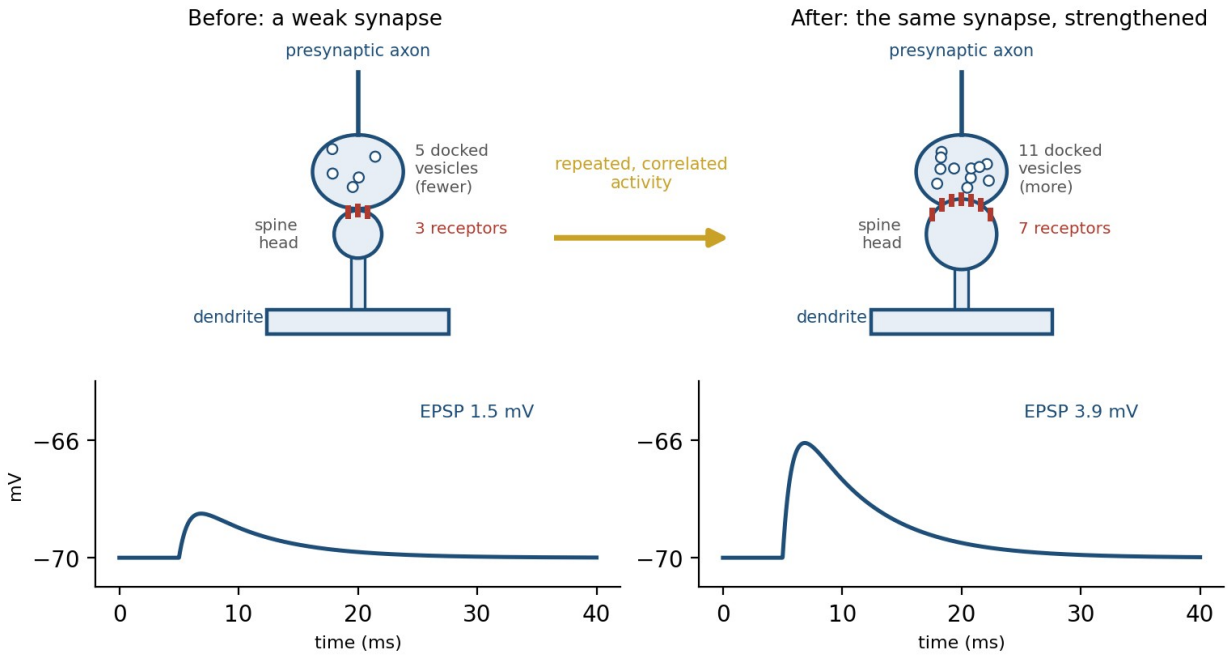


Figure 2.8. Learning as physical change. Before: a weak synapse with fewer docked vesicles and few receptors gives a small EPSP. After repeated correlated activity, the same synapse holds more docked vesicles, a larger spine head, and more receptors, and the EPSP more than doubles. Simulated traces.

This is the sense in which learning is physical. Storage in the brain is physical and distributed, held in the strengths and structure of many synapses, and there is no separate memory in which records sit apart from the machinery that uses them. The synapse that encodes a past experience is the same synapse that transmits the next impulse, so every use of the memory runs through the stored change and may alter it. Chapter 5 develops this account, from the molecular steps of potentiation to the reasons it makes memory reconstructive.

Why Neurons Look Inefficient, and Why They Are Not

Judged as switching elements, neurons are slow, noisy, and expensive. An action potential lasts about a millisecond; a transistor in a modern processor switches in roughly a hundred picoseconds, ten million times faster. A synapse fails more often than it succeeds at many connections, and the same stimulus never produces exactly the same spikes twice. Faisal, John White, and Laughlin calculated that axons thinner than about a tenth of a micrometer would be unusable, because the random opening of single sodium channels would fire spontaneous impulses often enough to drown the real ones; the thinnest axons found in brains sit just above that limit, so noise sets a floor on how small the wiring can be.[122] And the brain is about two percent of the body's mass but uses about twenty percent of its resting energy, most of it, by Attwell and Laughlin's estimate, on signaling itself: their budget predicts that a cortex paying these costs can afford to have no more than about fifteen percent of its neurons active at the same time, and Peter Lennie's later calculation, described in Chapter 1, puts the affordable fraction nearer one in a hundred.[123] Figure 2.9 summarizes the three costs.

[122] Faisal, A. A., White, J. A., & Laughlin, S. B. (2005). Ion-channel noise places limits on the miniaturization of the brain's wiring. *Current Biology*, 15(12), 1143–1149.

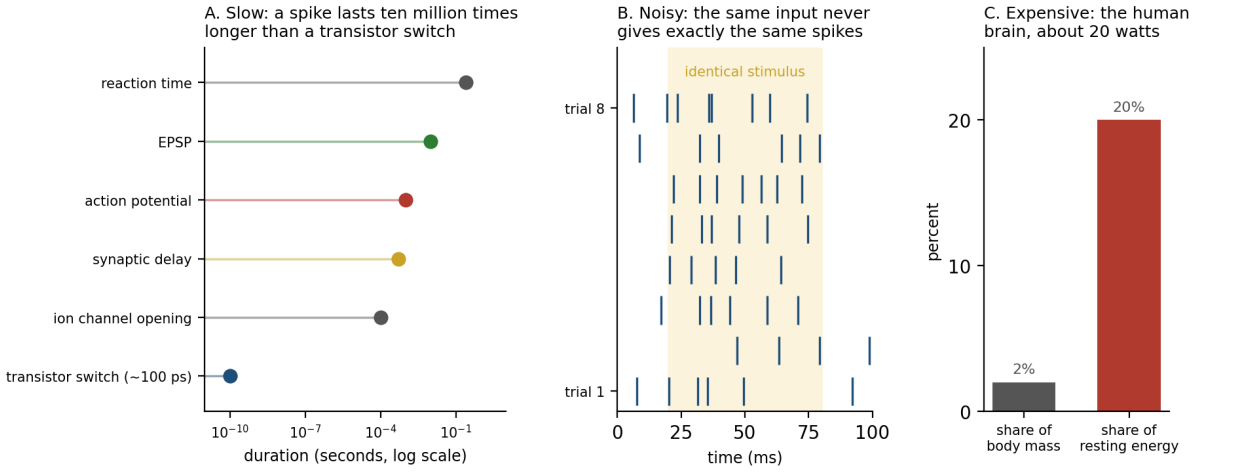


Figure 2.9. Slow, noisy, and expensive. A: a spike lasts about a millisecond, ten million times longer than a transistor switch of about 100 picoseconds. B: the same stimulus gives a different spike pattern on every trial (simulated raster). C: the brain is about 2 percent of body mass but uses about 20 percent of resting energy.

The slowness and noise are tolerable because the brain computes with populations rather than single cells. Thirty unreliable synapses that must coincide to fire a neuron, as in the worked example above, amount to a vote, and a vote among many noisy voters is reliable even when each voter is not. Slowness is offset by parallelism: billions of neurons each doing a little at once. The expense buys something no transistor has, a signaling element that can rewire itself while it works. Peter Sterling and Simon Laughlin argue, as Chapter 1 noted, that under these constraints neural design is close to efficient, with neurons firing sparsely because every spike is paid for.[124] What looks like inefficiency is the shape of a good solution to a different problem from the one a computer solves.

A Neuron in Context

The account so far has treated a neuron's response as fixed by its inputs and its synapses, but the same neuron, with the same wiring and the same input, can respond quite differently depending on the chemical setting it is in. Figure 2.10 shows the same neuron answering the same train of input spikes with three spikes in one chemical setting and eleven in another; the difference comes from substances such as dopamine, acetylcholine, and norepinephrine that are released widely and change how neurons fire and how their synapses learn.

[123] Raichle, M. E., & Gusnard, D. A. (2002). Appraising the brain's energy budget. *Proceedings of the National Academy of Sciences*, 99(16), 10237–10239; Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*, 21(10), 1133–1145; Lennie, P. (2003). The cost of cortical computation. *Current Biology*, 13(6), 493–497.

[124] Sterling, P., & Laughlin, S. (2015). *Principles of Neural Design*. MIT Press.

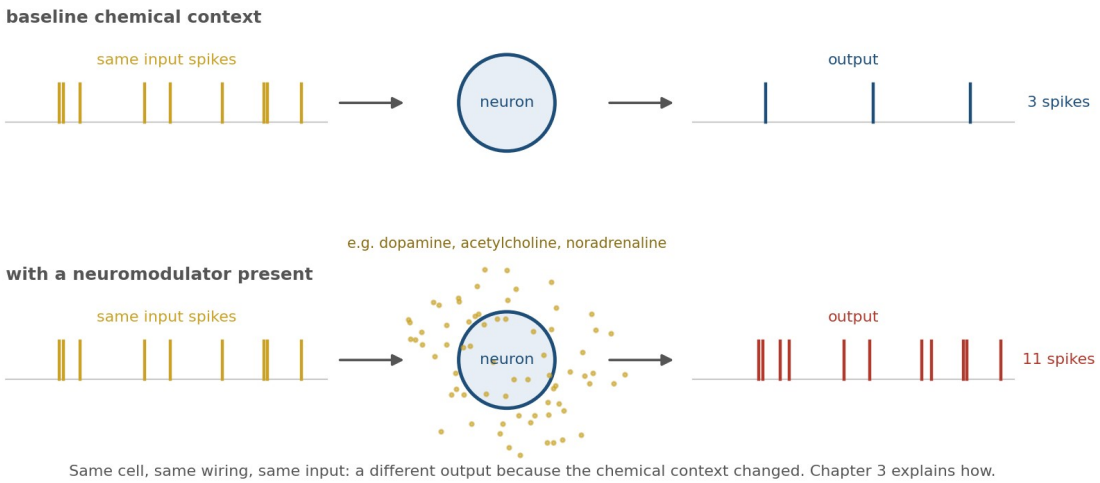


Figure 2.10. The same input, a different output. With no neuromodulator present the neuron responds to a train of input spikes with three spikes; with a neuromodulator present it responds with eleven. The wiring has not changed. Chapter 3 explains how chemical context does this. Schematic, not data.

Those substances are the neuromodulators, and they are the subject of the next chapter. They decide much of when the physical changes described above are made and kept, and they turn one circuit into many.

Implications for Artificial Intelligence

The artificial neuron began as a deliberate simplification of the one described in this chapter. Warren McCulloch and Walter Pitts proposed in 1943 a unit that sums its weighted inputs and fires if the sum exceeds a threshold, and showed that networks of such units could compute any logical function.[125] The units of modern deep networks are its descendants: a weighted sum followed by a smooth nonlinearity. Several of this chapter's findings bear on how far that simplification reaches.

First, a biological neuron is not a single unit but a small network. Panayiota Poirazi, Bartlett Mel, and a colleague found that the firing of a detailed model of a hippocampal pyramidal neuron was well predicted by a two-layer network, in which each thin dendritic branch acts as a separate nonlinear subunit and the soma sums their outputs. David Beniaguev, Idan Segev, and Michael London went further and trained deep networks to reproduce, millisecond by millisecond, the output of a detailed model of a cortical pyramidal neuron; they needed a network five to eight layers deep, and when they removed the model's NMDA receptors a network with a single hidden layer sufficed.[126] Albert Gidon, Matthew Larkum, and their colleagues found that dendrites of human cortical neurons produce a kind of calcium spike whose amplitude is largest for inputs just above threshold and smaller for stronger ones, which lets a single neuron compute exclusive-or, a function that a single artificial unit cannot

[125] McCulloch, W. S., & Pitts, W. (1943). A logical calculus of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics*, 5(4), 115–133.

[126] Poirazi, P., Brannon, T., & Mel, B. W. (2003). Pyramidal neuron as two-layer neural network. *Neuron*, 37(6), 989–999; Beniaguev, D., Segev, I., & London, M. (2021). Single cortical neurons as deep artificial neural networks. *Neuron*, 109(17), 2727–2739.

compute.[127] If one cortical neuron is worth a small deep network, comparing brains and models by counting units understates what the brain's units do.

Second, the brain communicates with sparse, all-or-none events whose cost scales with how many are sent, while a conventional artificial network computes every unit on every step. As hardware becomes power-limited, the energy figures above are a reason to take event-driven, spiking computation seriously.

Third, biological components are unreliable, and the system is robust anyway. Dropout, a training technique that randomly silences units, makes artificial networks generalize better.[128] Synaptic failure may serve the brain in a related way, by keeping any one pathway from becoming indispensable, though this is an analogy, not an established equivalence.

Finally, the brain and artificial networks both store what they learn in connection weights, so the difference is not where knowledge is kept but how it changes. In an artificial network, weights are changed by a separate training procedure and are fixed when the network is used, unless it is trained online. In the brain, synapses are adjusted during use, by rules that depend on signals available at the synapse itself, among them the back-propagating spike that tells it whether the neuron fired. How such local rules add up to learning that serves the whole organism is a central question for both fields.

The Central Lesson

A neuron is a living cell that turns many small, noisy, chemical inputs into one all-or-none electrical output, and changes its own connections in the process. Its dendrites integrate actively rather than passively, its axon initial segment makes the decision, its spike travels forward to other cells and back to its own synapses, and every step is paid for in energy and limited by noise. These physical facts are not implementation details beneath the level of computation; they define what a neuron can compute, how fast, at what cost, and how it learns.

Next Chapter

Chapter 3: Neuromodulation: How Chemical Context Shapes Learning and Behavior

Figure and Epigraph Sources

Figure Attribution

Figures 2.1, 2.2, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, and 2.10 were drawn for this book. The voltage traces in Figures 2.4, 2.5, 2.7, and 2.8 and the spike raster in Figure 2.9 are illustrative simulations, not recorded data; the EPSP amplitudes in Figure 2.4 are exaggerated for clarity. Figure 2.10 is a schematic.

Figure 2.3: the chemical synapse figure from *Anatomy and Physiology* by OpenStax (Rice University), licensed CC BY 4.0, via openstax.org.

[127] Gidon, A., Zolnik, T. A., Fidzinski, P., Bolduan, F., Papoutsi, A., Poirazi, P., Holtkamp, M., Vida, I., & Larkum, M. E. (2020). Dendritic action potentials and computation in human layer 2/3 cortical neurons. *Science*, 367(6473), 83–87.

[128] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15, 1929–1958.

Epigraph

Santiago Ramón y Cajal, *Recollections of My Life*, trans. E. Horne Craigie with the assistance of Juan Cano (American Philosophical Society, 1937; reprinted MIT Press, 1989), p. 363 of the MIT Press edition. Cajal's autobiography *Recuerdos de mi vida* was published in Spanish in 1901–1917.

Part II: The Machinery of Learning

Chapter 3

Neuromodulation: How Chemical Context Shapes Learning and Behavior

> *"The transmission of the effects of nerve impulses, by the release of chemical agents, first became an experimental reality in 1921."*
>
> *— Henry Dale, Nobel Lecture (1936)*

Why Neurons Alone Are Not Enough

Chapter 2 built a neuron. It receives inputs on its dendrites, integrates them, and when the membrane at the axon initial segment crosses threshold it fires an action potential that travels down the axon and crosses synapses onto other cells. That account is accurate, and it explains how signals move through the brain in milliseconds. It leaves out something that any account of learning needs: a reason why the same neuron, carrying the same signal, should change its connections on one occasion and not on another.

If neural function were set entirely by fast electrical signaling through fixed wiring, the brain would be a complex but rigid machine. The same input would reliably produce the same output, and whatever learning occurred would occur whenever neurons fired, with no regard for whether the moment deserved it. That is not what organisms do. A sound can be ignored at one moment and trigger rapid learning at the next. A mistake can be shrugged off today and permanently reshape behavior tomorrow. Even when the same neurons fire in similar patterns, learning sometimes happens and sometimes does not, and the difference lies in the state of the animal rather than in the pattern.

Electrical signaling explains how neurons communicate. It does not explain when learning happens, what is worth learning, or why the organism's internal state matters so much. Answering those questions requires looking at the chemical environment in which neural activity unfolds, and that environment is set by a small number of cell groups whose job is not to carry detailed content but to change how everything else handles it.[129] This is neuromodulation, and it is the subject of this chapter.

Fast Signals and Slow Control

Neural communication runs on more than one timescale. At the fastest, an action potential crosses a synapse in under a millisecond, and the glutamate or GABA it releases acts on receptors a few tens of nanometers away and is cleared within milliseconds. Figure 3.1A shows this arrangement: one sender, one receiver, a cleft about 20 nanometers wide. Fast transmission of this kind carries perception, reflexes, and moment-to-moment decisions.

Learning cannot run at that speed, and should not. If every fluctuation of activity produced a lasting change, the brain would overwrite what it knew as fast as it acquired it; this is the stability-plasticity dilemma, and Chapter 5 returns to it.[130] A learning system needs a way to let some experiences leave

[129] Marder, E. (2012). Neuromodulation of neuronal circuits: Back to the future. *Neuron*, 76(1), 1–11; Dayan, P. (2012). Twenty-five lessons from computational neuromodulation. *Neuron*, 76(1), 240–256.

[130] Carpenter, G. A., & Grossberg, S. (1988). The ART of adaptive pattern recognition by a self-organizing neural network. *Computer*, 21(3), 77–88; Abraham, W. C., & Robins, A. (2005). Memory retention: The synaptic stability versus plasticity

traces while most do not, and the decision has to be made on the basis of something slower and broader than the activity of any one synapse.

Neuromodulators are that something. Dopamine, norepinephrine, serotonin, and acetylcholine are released by small groups of neurons whose axons branch across large territories of the brain, and much of their release is not into a synaptic cleft but from swellings along the axon, called varicosities, into the extracellular space (Figure 3.1B). From there the molecule diffuses over micrometers and reaches receptors on many cells and many synapses at once; this is called volume transmission, in contrast to the point-to-point transmission of Figure 3.1A.[131] The signals themselves can be fast. A dopamine neuron's burst in response to a surprise lasts about a tenth of a second, and locus coeruleus neurons answer a salient stimulus with a similar latency. What is slow is the action. Most neuromodulator receptors are not ion channels but G-protein-coupled receptors, which work by setting off chains of chemical reactions inside the cell, and their effects on excitability and synaptic strength unfold over seconds to minutes and sometimes longer.[132]

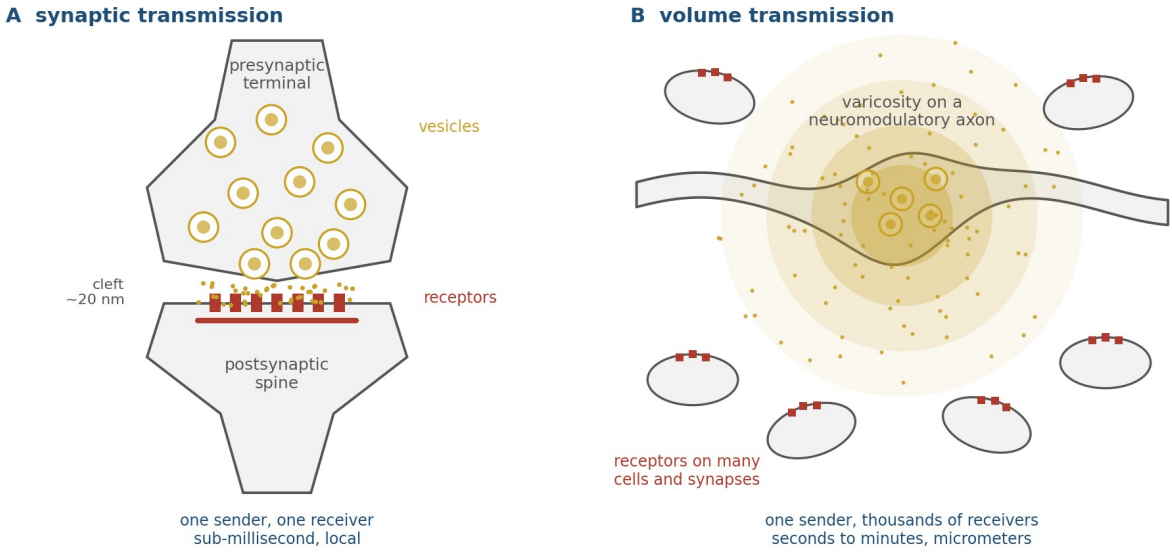


Figure 3.1. Synaptic transmission (A) versus volume transmission (B). A fast transmitter crosses a cleft about 20 nanometers wide to receptors on one partner cell. A neuromodulator released from an axonal varicosity diffuses through the extracellular space and reaches receptors on many cells and synapses, with effects that last seconds to minutes.

The consequence is that a neuromodulator does not carry detailed sensory content. It changes how circuits respond to the content they are already receiving: it alters excitability, biases the competition between pathways, and decides whether, how strongly, and in which direction synapses are permitted to change.[133] Figure 3.2 shows the principle at a single synapse. The same three presynaptic spikes

dilemma. *Trends in Neurosciences*, 28(2), 73–78.

[131] Fuxe, K., Dahlström, A. B., Jonsson, G., Marcellino, D., Guescini, M., Dam, M., Manger, P., & Agnati, L. (2010). The discovery of central monoamine neurons gave volume transmission to the wired brain. *Progress in Neurobiology*, 90(2), 82–100.

[132] Nadim, F., & Bucher, D. (2014). Neuromodulation of neurons and synapses. *Current Opinion in Neurobiology*, 29, 48–56; Schultz, W. (1998). Predictive reward signal of dopamine neurons. *Journal of Neurophysiology*, 80(1), 1–27; Sara, S. J. (2009). The locus coeruleus and noradrenergic modulation of cognition. *Nature Reviews Neuroscience*, 10(3), 211–223. The receptor-level cascades are slow; the phasic dopamine and locus coeruleus responses that trigger them are not.

[133] Bargmann, C. I. (2012). Beyond the connectome: How neuromodulators shape neural circuits. *BioEssays*, 34(6), 458–465.

produce a small response when the modulator level is low and a larger one when it is high, and at the high level the synapse is also opened to lasting change.

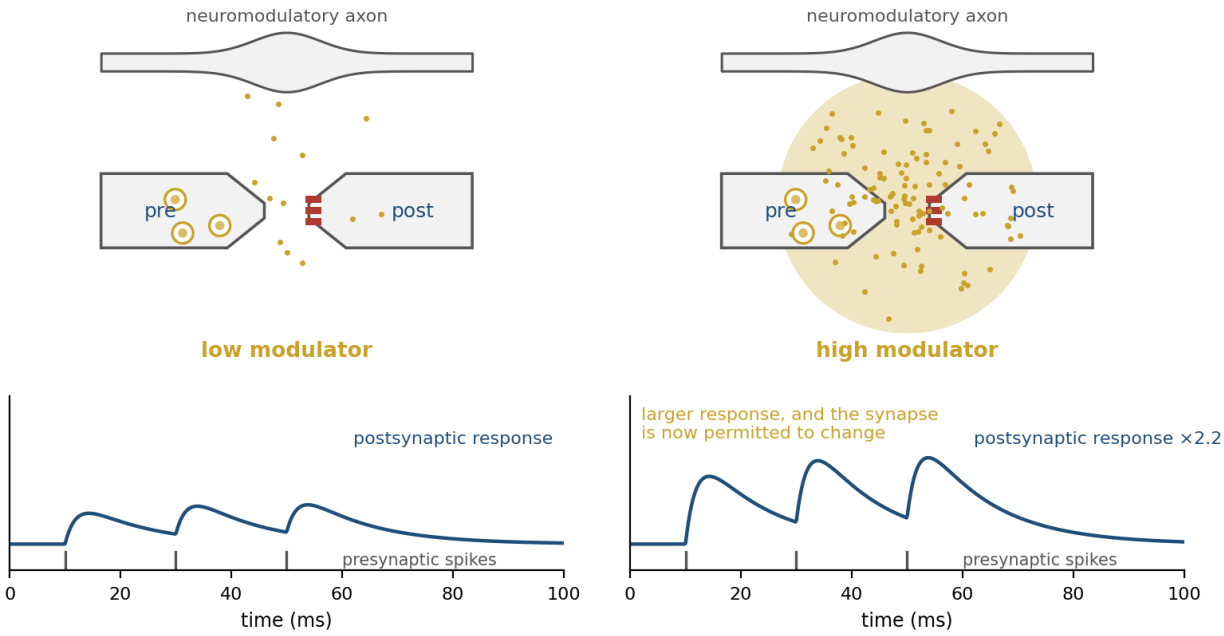


Figure 3.2. Neuromodulators create chemical context at synapses. The same synapse receiving the same three presynaptic spikes gives a small response when the modulator level is low (left) and a larger response when it is high (right); the traces are illustrative. High modulator also opens the synapse to lasting change.

It is worth being precise about what "opened to lasting change" means, because the rest of the book depends on it. Synaptic plasticity, the lasting change in the strength of a connection, can be induced by presynaptic and postsynaptic activity alone: the potentiation and depression that Chapter 5 describes are produced in brain slices by patterns of stimulation with no neuromodulator added. What neuromodulators do is gate and scale much of that plasticity in the intact brain, and decide whether it is consolidated. Kittens whose visual cortex was depleted of both norepinephrine and acetylcholine failed to show the rewiring that normally follows closing one eye. Adult rats in which a tone was paired with stimulation of the cholinergic nucleus basalis, or of the dopaminergic ventral tegmental area, expanded the area of auditory cortex devoted to that tone, something the tone alone never does.[134] Hebbian plasticity is the mechanism; neuromodulation is, for much of it, the permission.

Why Neuromodulatory Systems Are Diffuse

The neurons that release these modulators are few, and they sit far from the regions they control. In 1964 Annica Dahlström and Kjell Fuxe, using a method that made monoamines fluoresce, mapped the dopamine, norepinephrine, and serotonin cell groups of the rat brainstem; the cholinergic groups of the

[134] Bear, M. F., & Singer, W. (1986). Modulation of visual cortical plasticity by acetylcholine and noradrenaline. *Nature*, 320(6058), 172–176; Kilgard, M. P., & Merzenich, M. M. (1998). Cortical map reorganization enabled by nucleus basalis activity. *Science*, 279(5357), 1714–1718; Bao, S., Chan, V. T., & Merzenich, M. M. (2001). Cortical remodelling induced by activity of ventral tegmental dopamine neurons. *Nature*, 412(6842), 79–83.

basal forebrain and brainstem were charted two decades later by Marsel Mesulam and colleagues.[135] Figure 3.3 shows the four systems on a schematic midsagittal view. Dopamine neurons sit in the ventral tegmental area and substantia nigra of the midbrain and project to the striatum, the input stage of the basal ganglia described in Chapter 4, and to the prefrontal cortex. Norepinephrine comes almost entirely from the locus coeruleus, a nucleus in the pons that sends axons to nearly the whole forebrain and to the cerebellum. Serotonin comes from the raphe nuclei along the midline of the brainstem. Acetylcholine comes from the basal forebrain and from two nuclei in the brainstem that reach the thalamus.

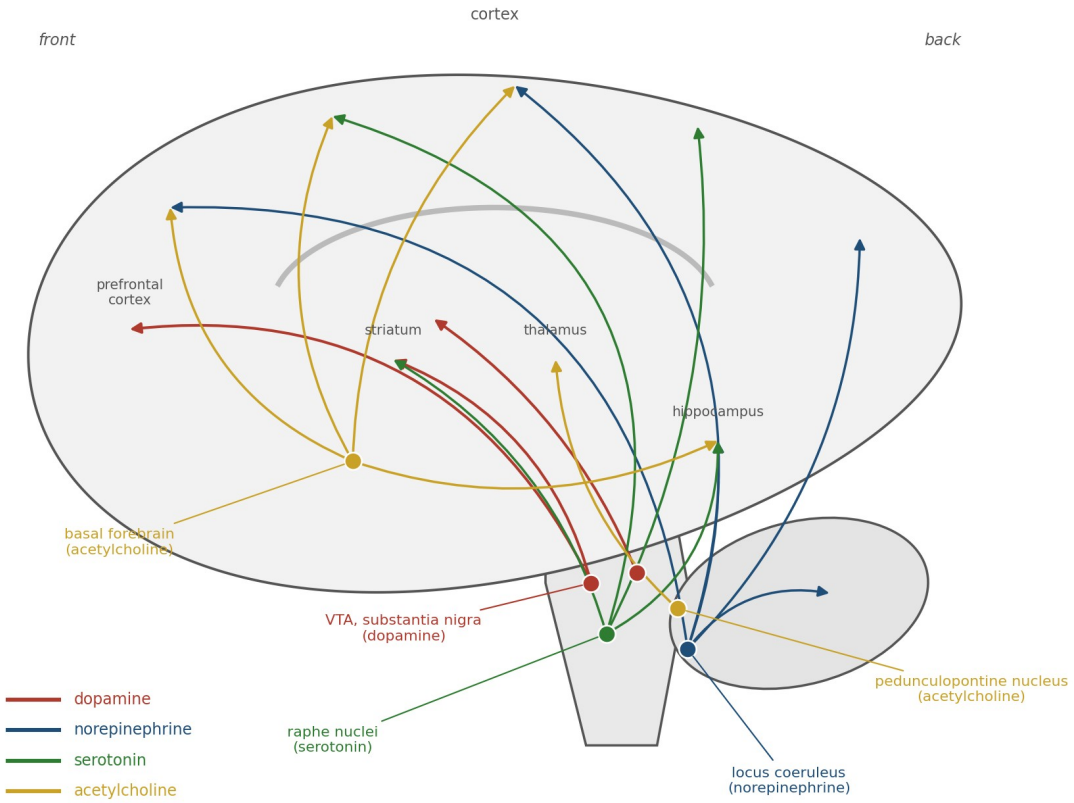


Figure 3.3. Major neuromodulatory systems in the brain, on a schematic midsagittal view with the front at the left. Small nuclei in the brainstem and basal forebrain send dopamine, norepinephrine, serotonin, and acetylcholine across cortex, striatum, thalamus, hippocampus, and cerebellum.

A word on naming. Neuroscientists label neurons by the transmitter they release, with the suffix "-ergic," from the Greek *ergon*, work: dopaminergic neurons release dopamine, noradrenergic neurons release norepinephrine, serotonergic neurons release serotonin, and cholinergic neurons release acetylcholine. The label says what chemical influence a cell exerts, not what it represents.

The numbers are striking. The human nucleus basalis of Meynert, the main cholinergic source for the neocortex, holds roughly 200,000 cholinergic neurons per hemisphere, and the whole basal forebrain

[135] Dahlström, A., & Fuxe, K. (1964). Evidence for the existence of monoamine-containing neurons in the central nervous system. I. Demonstration of monoamines in the cell bodies of brain stem neurons. *Acta Physiologica Scandinavica*, 62, Supplement 232, 1–55; Mesulam, M.-M., Mufson, E. J., Wainer, B. H., & Levey, A. I. (1983). Central cholinergic pathways in the rat: An overview based on an alternative nomenclature (Ch1–Ch6). *Neuroscience*, 10(4), 1185–1201.

cholinergic system amounts to a few hundred thousand cells; the cortex they innervate contains about 16 billion neurons.[136] Figure 3.4 shows that system on its own, and the section on acetylcholine below returns to what it does. A single neuromodulatory neuron compensates with the size of its axon. When Wakoto Matsuda and colleagues filled individual dopamine neurons of the rat substantia nigra and reconstructed their axons, each neuron's arbor in the striatum was about half a meter in total length and covered on average 2.7 percent of the striatum's volume, up to about 6 percent in the largest; estimates of the number of release sites on one such neuron run to several hundred thousand in the rat and more than a million in the human.[137] One cell, in other words, can speak to hundreds of thousands of synapses at once.

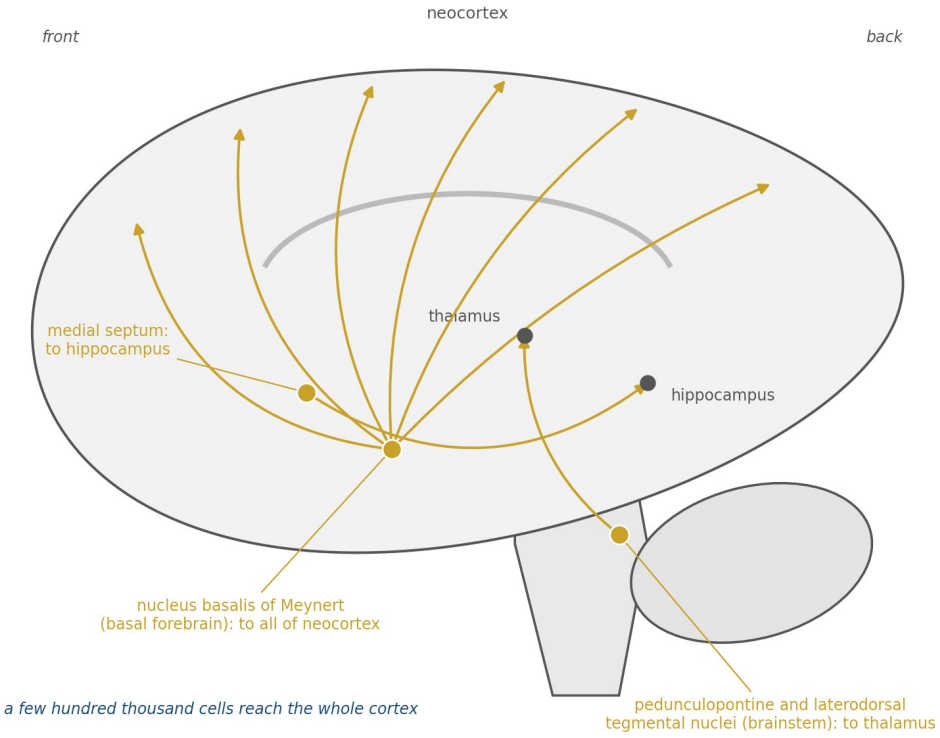


Figure 3.4. The cholinergic system. The nucleus basalis of Meynert reaches all of neocortex, the medial septum reaches the hippocampus, and the pedunculopontine and laterodorsal tegmental nuclei of the brainstem reach the thalamus. A few hundred thousand cells innervate a cortex of about 16 billion neurons.

[136] Mesulam, M.-M., & Geula, C. (1988). Nucleus basalis (Ch4) and cortical cholinergic innervation in the human brain: Observations based on the distribution of acetylcholinesterase and choline acetyltransferase. *Journal of Comparative Neurology*, 275(2), 216–240; Azevedo, F. A. C., Carvalho, L. R. B., Grinberg, L. T., Farfel, J. M., Ferretti, R. E. L., Leite, R. E. P., Jacob Filho, W., Lent, R., & Herculano-Houzel, S. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5), 532–541. Mesulam and Geula counted about 210,000 cholinergic neurons in the human nucleus basalis per hemisphere; Azevedo and colleagues put the human cerebral cortex at about 16 billion neurons.

[137] Matsuda, W., Furuta, T., Nakamura, K. C., Hioki, H., Fujiyama, F., Arai, R., & Kaneko, T. (2009). Single nigrostriatal dopaminergic neurons form widely spread and highly dense axonal arborizations in the neostriatum. *Journal of Neuroscience*, 29(2), 444–453; Bolam, J. P., & Pissadaki, E. K. (2012). Living on the edge with too many mouths to feed: Why dopamine neurons die. *Movement Disorders*, 27(12), 1478–1483. Matsuda and colleagues report arbors covering on average 2.7 percent of the rat neostriatum (range about 0.5 to 5.7 percent); Bolam and Pissadaki estimate several hundred thousand release sites per rat neuron and more than a million per human neuron.

Why broadcast a signal so widely rather than deliver it to the circuits that need it? Because what these systems carry is context, not instruction. They report whether the organism is safe or threatened, whether events are going better or worse than expected, and whether the environment is stable or has just changed, and those are facts that every circuit needs at the same time.[138] A broadcast lets a few hundred thousand cells coordinate the behavior of billions without a central controller deciding who should hear.[139]

Dopamine: Schultz's Monkeys

Dopamine is often described as the brain's reward or pleasure chemical, and the description is wrong in an instructive way.[140] What corrected it was a series of experiments by Wolfram Schultz and colleagues, beginning in the 1980s and summarized in a 1998 review, in which fine microelectrodes recorded the activity of single dopamine neurons in the midbrain of macaque monkeys while the monkeys performed simple tasks for drops of fruit juice.[141]

The neurons fired at a slow, steady rate, a few spikes per second, most of the time. When a drop of juice arrived unannounced, they produced a brief burst, starting within about a tenth of a second and lasting about as long. So far this looked like a reward signal. The monkeys were then trained with a cue, a light or a sound that preceded the juice by a second or so. Over days, as the monkeys learned that the cue predicted the juice, the burst moved. It now came at the cue, and the juice itself, when it arrived as predicted, evoked nothing at all. Finally, on trials in which the cue appeared but the juice was withheld, the neurons did something a reward detector cannot do: at the moment the juice should have arrived, they fell silent, dipping below their baseline rate for a few hundred milliseconds. The dip was timed to the expected delivery, which showed that the neurons carried a prediction of when, not merely whether.[142] Figure 3.5 shows the three conditions as simulated rasters and histograms.

[138] Yu, A. J., & Dayan, P. (2005). Uncertainty, neuromodulation, and attention. *Neuron*, 46(4), 681–692; Doya, K. (2002). Metalearning and neuromodulation. *Neural Networks*, 15(4–6), 495–506; Dayan, P. (2012). Twenty-five lessons from computational neuromodulation. *Neuron*, 76(1), 240–256.

[139] Lee, S.-H., & Dan, Y. (2012). Neuromodulation of brain states. *Neuron*, 76(1), 209–222.

[140] Berridge, K. C., & Robinson, T. E. (1998). What is the role of dopamine in reward: Hedonic impact, reward learning, or incentive salience? *Brain Research Reviews*, 28(3), 309–369; Wise, R. A. (2004). Dopamine, learning and motivation. *Nature Reviews Neuroscience*, 5(6), 483–494.

[141] Schultz, W., Apicella, P., & Ljungberg, T. (1993). Responses of monkey dopamine neurons to reward and conditioned stimuli during successive steps of learning a delayed response task. *Journal of Neuroscience*, 13(3), 900–913; Schultz, W. (1998). Predictive reward signal of dopamine neurons. *Journal of Neurophysiology*, 80(1), 1–27.

[142] Hollerman, J. R., & Schultz, W. (1998). Dopamine neurons report an error in the temporal prediction of reward during learning. *Nature Neuroscience*, 1(4), 304–309; Schultz, W. (1998). Predictive reward signal of dopamine neurons. *Journal of Neurophysiology*, 80(1), 1–27.

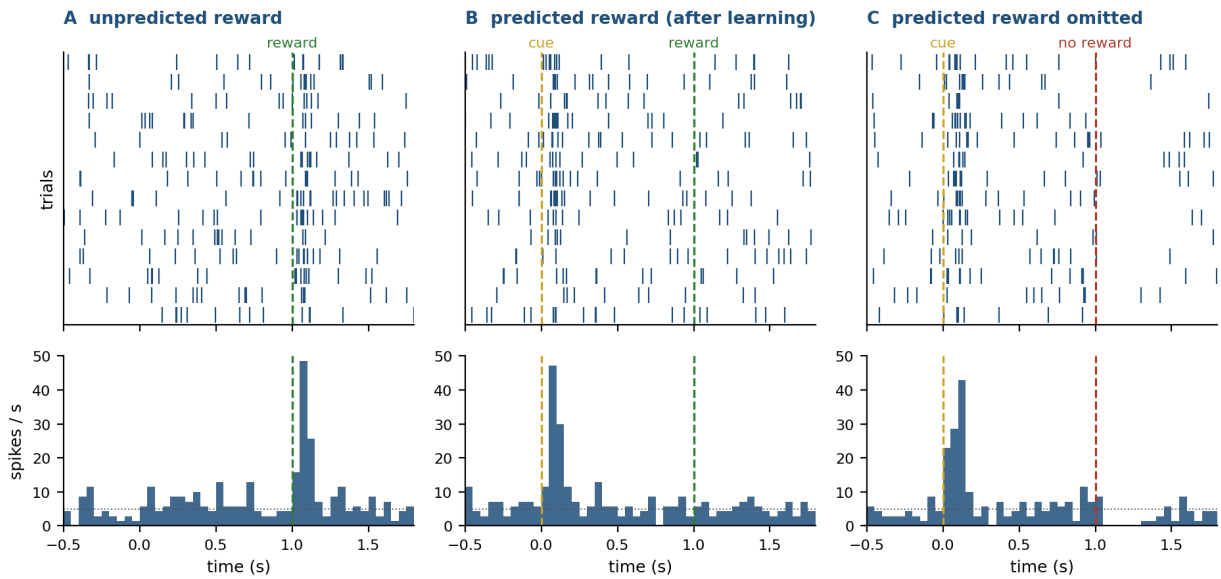


Figure 3.5. Dopamine neurons signal a reward prediction error. Simulated spike rasters and firing-rate histograms in the three conditions of Schultz's experiments. An unpredicted reward evokes a burst (A). After learning, the burst moves to the cue and the predicted reward evokes nothing (B). When a predicted reward is withheld, firing dips below baseline at the moment the reward was due (C).

The neurons were not reporting reward. They were reporting the difference between the reward the monkey received and the reward it had predicted. An unpredicted reward is a positive difference: a burst. A fully predicted reward is no difference: silence. A predicted reward that fails to arrive is a negative difference: a dip. This quantity is called a reward prediction error, and later work showed that the size of the burst scales with the size of the error, so that the signal is quantitative rather than merely present or absent.[143]

The interpretation that made these recordings famous came from computer science. In 1988 Richard Sutton had described an algorithm for learning to predict future reward called temporal-difference learning. At each moment the learner holds an estimate of how much reward is still to come. When the next moment arrives it compares the reward actually received, plus its new estimate of what remains, against the estimate it held a moment before; the difference between these two successive predictions, the temporal difference, is the error it uses to correct the earlier one. Over repeated trials the correction propagates backward in time, from the reward to whatever reliably preceded it.[144] In 1996 Read Montague, Peter Dayan, and Terrence Sejnowski observed that Schultz's recordings matched this error term in detail, and in 1997 Schultz, Dayan, and Montague laid the two side by side: the burst that migrates from the juice to the cue, and the dip when the predicted juice is missing, are what a temporal-

[143] Bayer, H. M., & Glimcher, P. W. (2005). Midbrain dopamine neurons encode a quantitative reward prediction error signal. *Neuron*, 47(1), 129–141; Eshel, N., Bukwich, M., Rao, V., Hemmelder, V., Tian, J., & Uchida, N. (2015). Arithmetic and local circuitry underlying dopamine prediction errors. *Nature*, 525(7568), 243–246.

[144] Sutton, R. S. (1988). Learning to predict by the methods of temporal differences. *Machine Learning*, 3(1), 9–44; Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press; Rescorla, R. A., & Wagner, A. R. (1972). A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. In A. H. Black & W. F. Prokasy (Eds.), *Classical Conditioning II: Current Research and Theory* (pp. 64–99). Appleton-Century-Crofts. The Rescorla-Wagner model is the trial-level ancestor of the temporal-difference rule.

difference error looks like as learning proceeds.[145] Figure 3.6 shows the migration in a simulation of the algorithm. Within a trial the error starts at the reward and, over trials, moves back to the cue, so that the error at the reward falls to zero as the error at the cue rises.

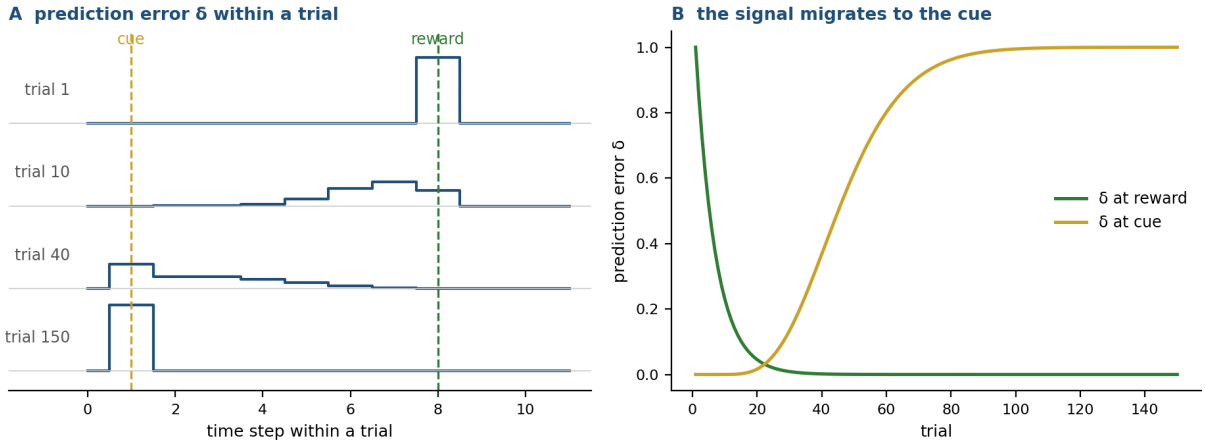


Figure 3.6. Dopamine signals shift to predictive cues after learning. In a simulation of temporal-difference learning, the prediction error within a trial starts at the reward and, over trials, migrates backward to the cue (A), so the error at the reward falls to zero while the error at the cue rises to the full reward value in this undiscounted simulation (B).

A correlation between a neuron's firing and an algorithm's variable is not proof that the neuron causes learning. The causal test came later. When dopamine neurons in rats were driven with light, by optogenetics, at the moment a reward would normally have produced no error because it was already predicted, the rats learned about a cue they would otherwise have ignored, as if an error had occurred.[146]

Two cautions belong here. First, the reward-prediction-error account describes the phasic response of a large population of midbrain dopamine neurons in tasks with appetitive outcomes; it is not a description of all dopamine neurons or of everything dopamine does. Some dopamine neurons respond to aversive events as well as rewarding ones, some to novelty or salience, and recordings from dopamine neurons and axons in mice have found signals tied to movement, position, and other variables; whether the slow ramps of dopamine seen in the striatum are prediction errors or something else is an active debate.[147] The right statement is that a large population of dopamine neurons signals a reward prediction error, and that the population is heterogeneous in ways still being worked out. Second, the error in question is a single number about value; Chapter 6 introduces a different, high-dimensional prediction error about sensory content, and the two should not be confused.

[145] Montague, P. R., Dayan, P., & Sejnowski, T. J. (1996). A framework for mesencephalic dopamine systems based on predictive Hebbian learning. *Journal of Neuroscience*, 16(5), 1936–1947; Schultz, W., Dayan, P., & Montague, P. R. (1997). A neural substrate of prediction and reward. *Science*, 275(5306), 1593–1599.

[146] Steinberg, E. E., Keiflin, R., Boivin, J. R., Witten, I. B., Deisseroth, K., & Janak, P. H. (2013). A causal link between prediction errors, dopamine neurons and learning. *Nature Neuroscience*, 16(7), 966–973.

[147] Matsumoto, M., & Hikosaka, O. (2009). Two types of dopamine neuron distinctly convey positive and negative motivational signals. *Nature*, 459(7248), 837–841; Bromberg-Martin, E. S., Matsumoto, M., & Hikosaka, O. (2010). Dopamine in motivational control: Rewarding, aversive, and alerting. *Neuron*, 68(5), 815–834; Howe, M. W., & Dombeck, D. A. (2016). Rapid signalling in distinct dopaminergic axons during locomotion and reward. *Nature*, 535(7613), 505–510; Engelhard, B., Finkelstein, J., Cox, J., Fleming, W., Jang, H. J., Ornelas, S., Koay, S. A., Thiberge, S. Y., Daw, N. D., Tank, D. W., & Witten, I. B. (2019). Specialized coding of sensory, motor and cognitive variables in VTA dopamine neurons. *Nature*, 570(7762), 509–513; Jeong, H., Taylor, A., Floeder, J. R., Lohmann, M., Mihalas, S., Wu, B., Zhou, M., Burke, D. A., & Namboodiri, V. M. K. (2022). Mesolimbic dopamine release conveys causal associations. *Science*, 378(6626), eabq6740.

What the dopamine signal does not do is say which synapses should change. It is broadcast to the entire striatum and much of the cortex at once, and it carries one scalar. It does say when, and in which direction: bursts drive potentiation at synapses onto striatal neurons that carry one class of dopamine receptor, while dips favor the opposite change at synapses carrying the other, and the burst must arrive within a narrow window after the synapse was active, a point taken up below.[148]

Credit Assignment

Underneath all learning lies a question that Marvin Minsky named in 1961: when an action leads to success or failure, which of the many internal components that contributed deserve the credit or the blame? This is the credit assignment problem.[149]

Artificial neural networks solve it with care. The architecture is known, the objective is explicit, and the error at the output can be computed by comparison with a correct answer. The backpropagation algorithm then carries that error backward through the network, layer by layer, and assigns to every weight an update proportional to its exact contribution to the error.[150] It is precise, and it requires a complete, labeled, differentiable description of the system in which it runs.

The brain has no known way to do this exactly, and whether it approximates it is one of the field's open questions. Francis Crick raised the objection in 1989: nothing in the cortex looked like a mechanism for sending each synapse its share of a global error.[151] Since then the objection has weakened without disappearing. Random feedback weights turn out to be enough for a network to learn, so the backward pathway need not mirror the forward one; bursts of spikes in cortical pyramidal cells, driven by inputs to their apical dendrites, can carry an error signal separate from the signal the cell forwards; and Timothy Lillicrap and colleagues argued in 2020 that the brain may well implement something functionally like backpropagation, with the feedback connections that cortical anatomy supplies in abundance carrying the errors.[152] Chapters 6, 10, and 11 return to the question. The point for this chapter is narrower: whatever the brain does about credit assignment, it is not the clean procedure of the artificial case, and a large part of what it does instead is visible in the neuromodulatory systems.

Consider what makes the problem hard. The brain receives no labels. It is never told the correct answer, only whether things went well or badly, and that news arrives late, after a delay of seconds or minutes during which millions of synapses have been active for unrelated reasons. Circuits are recurrent and overlapping rather than neatly layered. Neurons take part in many functions at once. By the time an

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[148] Shen, W., Flajolet, M., Greengard, P., & Surmeier, D. J. (2008). Dichotomous dopaminergic control of striatal synaptic plasticity. *Science*, 321(5890), 848–851; Reynolds, J. N. J., Hyland, B. I., & Wickens, J. R. (2001). A cellular mechanism of reward-related learning. *Nature*, 413(6851), 67–70; Yagishita, S., Hayashi-Takagi, A., Ellis-Davies, G. C. R., Urakubo, H., Ishii, S., & Kasai, H. (2014). A critical time window for dopamine actions on the structural plasticity of dendritic spines. *Science*, 345(6204), 1616–1620. Shen and colleagues showed that D1-receptor and D2-receptor striatal neurons respond to dopamine with opposite directions of plasticity.

[149] Minsky, M. (1961). Steps toward artificial intelligence. *Proceedings of the IRE*, 49(1), 8–30.

[150] Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323(6088), 533–536.

[151] Crick, F. (1989). The recent excitement about neural networks. *Nature*, 337(6203), 129–132.

[152] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276; Payeur, A., Guerguiev, J., Zenke, F., Richards, B. A., & Naud, R. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24(7), 1010–1019; Lillicrap, T. P., Santoro, A., Marris, L., Akerman, C. J., & Hinton, G. (2020). Backpropagation and the brain. *Nature Reviews Neuroscience*, 21(6), 335–346.

outcome is known, the detailed pattern of activity that produced it has been overwritten by everything that happened since, a difficulty Eugene Izhikevich called the distal reward problem.[153] In many cases the question "which synapse caused this outcome?" has no single answer, because thousands contributed and none was decisive.

Learning Under Uncertainty

A system that cannot identify exact causes must learn under uncertainty. The brain does not know which synapse mattered or which action was decisive. What it can know is that something important just happened, and the neuromodulatory systems are built to say so.

The approximate solution works in two steps. When an outcome is better or worse than expected, dopamine broadcasts a signal that marks the recent past as significant. Synapses that were active shortly before the signal, and only those, become eligible for modification.[154] No individual update is guaranteed to be correct; a synapse that was active by coincidence is strengthened along with the ones that mattered. But the updates are small, and they accumulate. Over many trials, patterns of activity that tend to precede good outcomes are strengthened more often than they are weakened, and patterns that precede bad ones the reverse, which is Edward Thorndike's law of effect from 1911, now with a mechanism.[155] The system acquires biases, statistical tendencies toward what has worked. Bias here does not mean error. It means the gradual shaping of probability under uncertainty, and it is what learning looks like when the learner cannot see the causal chain.

It is tempting to summarize this by saying that the brain, unlike an artificial network, does not optimize a loss function. That is true in one sense and false in another, and the book needs both. No single global objective is handed to the brain and minimized by gradient descent; there is no external label, no fixed architecture, and no one error. But many circuits are well described as minimizing a local error or cost. The temporal-difference error of Figure 3.6 is such a quantity, and the dopamine system is, to a good approximation, the machinery that reduces it. Chapter 6 describes cortex as reducing a sensory prediction error, and shows the brain combining cues in a way that matches a computed optimum. The accurate statement is that the brain has many local objectives and no global one, and that neuromodulators are among the signals that define them.

Eligibility Traces and the Three-Factor Rule

The two-step solution requires something at the synapse that remembers recent activity long enough for the modulatory signal to arrive. That something is called an eligibility trace. The term comes from reinforcement learning, where it names a decaying record of which states and actions were recently visited; Harry Klopf proposed a synaptic version in 1982, before there was evidence for one.[156]

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[153] Izhikevich, E. M. (2007). Solving the distal reward problem through linkage of STDP and dopamine signaling. *Cerebral Cortex*, 17(10), 2443–2452.

[154] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85; Gerstner, W., Lehmann, M., Liakoni, V., Corneil, D., & Brea, J. (2018). Eligibility traces and plasticity on behavioral time scales: Experimental support of neoHebbian three-factor learning rules. *Frontiers in Neural Circuits*, 12, 53.

[155] Thorndike, E. L. (1911). *Animal Intelligence: Experimental Studies*. Macmillan; Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.

[156] Klopf, A. H. (1982). *The Hedonistic Neuron: A Theory of Memory, Learning, and Intelligence*. Hemisphere; Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.

The evidence now exists. When a synapse is active, with the presynaptic neuron firing shortly before the postsynaptic one, biochemical processes put it into a transient, modifiable state. If a neuromodulatory signal arrives while that state persists, the synapse changes; if not, the state fades and nothing lasting happens. The windows have been measured. In 2014 Sho Yagishita and colleagues, working in slices of mouse striatum, activated a single dendritic spine with a flash of light that released glutamate at it while the neuron fired, then released dopamine at varying delays by stimulating dopamine axons with light. The spine enlarged, which is the structural signature of potentiation, only when dopamine arrived between about 0.3 and 2 seconds after the activity; earlier or later, nothing.[157] In hippocampal slices the window is longer. Zuzanna Brzosko, Wolfram Schultz, and Ole Paulsen found that a pairing protocol that normally produces depression was converted into potentiation if dopamine was applied within about a minute afterward, but not if the delay was ten minutes, and recordings in living rats have found windows of similar length at synapses from cortex onto the striatum.[158] Eligibility traces in the brain, then, last seconds to minutes, which is the timescale on which outcomes follow actions. Figure 3.7 shows the logic: a spike pair sets a trace that decays, a dopamine pulse that arrives during the decay changes the weight, and the same pulse arriving after the trace has gone changes nothing.

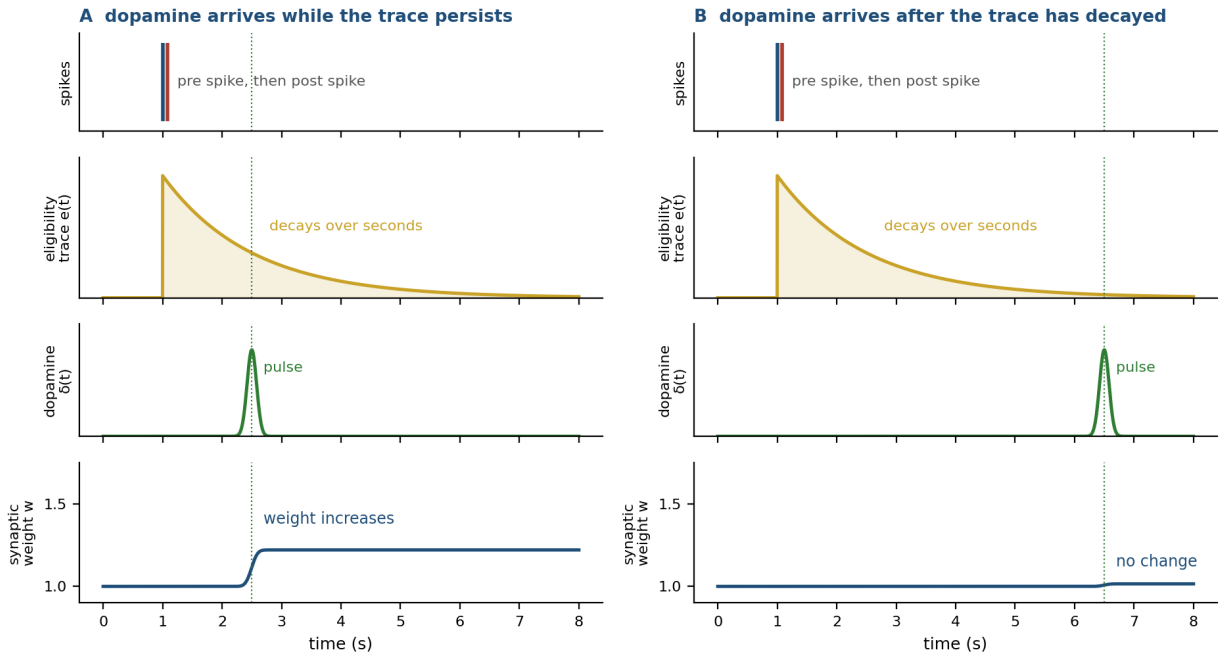


Figure 3.7. Eligibility traces bridge the gap between activity and the learning signal. In this illustrative simulation a pre-then-post spike pair sets a trace that decays over seconds. A dopamine pulse that arrives while the trace persists changes the synaptic weight (A). The same pulse arriving after the trace has decayed changes nothing (B).

[157] Yagishita, S., Hayashi-Takagi, A., Ellis-Davies, G. C. R., Urakubo, H., Ishii, S., & Kasai, H. (2014). A critical time window for dopamine actions on the structural plasticity of dendritic spines. *Science*, 345(6204), 1616–1620.

[158] Brzosko, Z., Schultz, W., & Paulsen, O. (2015). Retroactive modulation of spike timing-dependent plasticity by dopamine. *eLife*, 4, e09685; Fisher, S. D., Robertson, P. B., Black, M. J., Redgrave, P., Sagar, M. A., Abraham, W. C., & Reynolds, J. N. J. (2017). Reinforcement determines the timing dependence of corticostriatal synaptic plasticity in vivo. *Nature Communications*, 8, 334; He, K., Huertas, M., Hong, S. Z., Tie, X., Hell, J. W., Shouval, H., & Kirkwood, A. (2015). Distinct eligibility traces for LTP and LTD in cortical synapses. *Neuron*, 88(3), 528–538. Measured windows run from about a second for dopamine action on striatal spines to about a minute in hippocampal slices; He and colleagues found separate traces for potentiation and depression in cortex, each lasting seconds.

Put together, this is a three-factor learning rule. A change that lasts requires three things to coincide within the lifetime of the trace: the presynaptic neuron was active, the postsynaptic neuron fired, and a neuromodulatory signal arrived.[159] The first two factors are Hebb's, and Chapter 5 takes them apart; the third is the subject of this chapter. Figure 3.8 writes the rule out. The eligibility trace rises with each spike pairing by an amount set by the relative timing of the two spikes (a rule Chapter 5 explains), decays with a time constant of seconds, and is multiplied by the dopamine signal; the weight moves only while both the trace and the signal are present, and the sign of the pairing and the sign of the signal together set the direction.

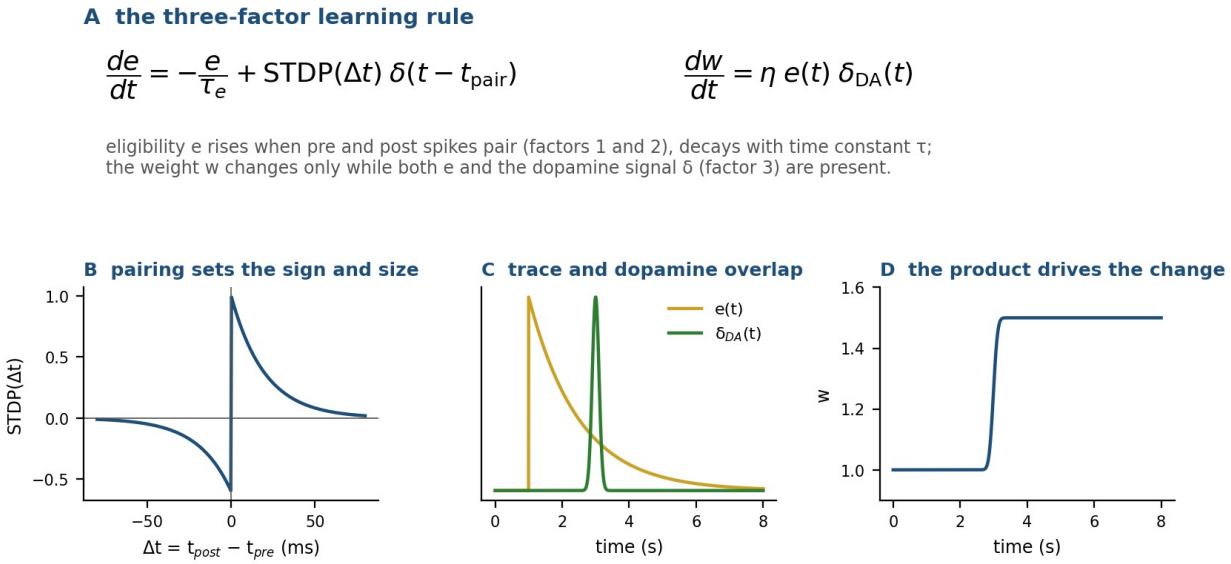


Figure 3.8. The three-factor rule written out (A). The eligibility trace e rises with each spike pairing according to the spike-timing window (B), decays, and is multiplied by the dopamine signal (C); the weight changes only while both are present (D). The curves are illustrative.

Eligibility traces let learning bridge the gap between action and outcome without any knowledge of the causal chain, and they are the reason a scalar broadcast can produce selective change: the signal reaches every synapse, but only the recently active ones are listening. What physically happens when a trace is captured, the receptors inserted, the spine enlarged, the proteins synthesized, is the subject of Chapter 5, which also describes synaptic tagging and capture, a slower relative of the trace, in which a tag set at a synapse lets it collect proteins whose synthesis a modulatory signal triggers up to an hour or two later.[160]

[159] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85; Gerstner, W., Lehmann, M., Liakoni, V., Corneil, D., & Brea, J. (2018). Eligibility traces and plasticity on behavioral time scales: Experimental support of neoHebbian three-factor learning rules. *Frontiers in Neural Circuits*, 12, 53.

[160] Kandel, E. R. (2001). The molecular biology of memory storage: A dialogue between genes and synapses. *Science*, 294(5544), 1030–1038; Malenka, R. C., & Bear, M. F. (2004). LTP and LTD: An embarrassment of riches. *Neuron*, 44(1), 5–21; Frey, U., & Morris, R. G. M. (1997). Synaptic tagging and long-term potentiation. *Nature*, 385, 533–536.

Norepinephrine, Serotonin, and Acetylcholine

Dopamine is the best understood of the four systems because reward is the easiest context to manipulate in an experiment. The other three are at least as important, and each has been tied to a context of its own.

Norepinephrine, from the locus coeruleus, is the signal of surprise and arousal. Locus coeruleus neurons fire a brief burst at a novel or salient stimulus, and their tonic rate tracks the animal's state from drowsiness through alert engagement to agitation. Gary Aston-Jones and Jonathan Cohen proposed that the system sets the gain of cortical circuits in one of two modes: a phasic mode, in which bursts sharpen responses to task-relevant events, and a tonic mode, in which a high baseline makes the animal responsive to everything and disengaged from the current task, which is the right setting when the current task has stopped paying.[161] In the computational account of Angela Yu and Peter Dayan, norepinephrine reports unexpected uncertainty, the signal that the world has changed and the current model should be distrusted, while acetylcholine reports the expected uncertainty of a world that is noisy but stable.[162] Chapter 6 builds on this distinction when it treats attention as the weighting of evidence by its reliability.

Serotonin, from the raphe nuclei, is the least settled. It has been tied to patience, to aversion, to behavioral inhibition, and to the regulation of plasticity itself, and the reviews that try to unify these accounts are candid that no unification has yet succeeded.[163] One concrete result stands out. Katsuhiko Miyazaki, Kayoko Miyazaki, and Kenji Doya recorded from serotonin neurons of the dorsal raphe while rats waited at a port for a reward that came only after a delay. The neurons raised their firing throughout the wait and dropped it when the rat gave up and left.[164] On one account, then, serotonin promotes the stability that lets an animal hold a course of action against the pull of an immediate alternative, which is roughly the opposite of what dopamine does when it marks a new option as better than expected.

Acetylcholine sets the balance between taking in new information and using old. Its anatomy, shown in Figure 3.4, comes in three parts: the nucleus basalis of Meynert in the basal forebrain sends axons to all of the neocortex, the medial septum reaches the hippocampus, and the pedunculopontine and laterodorsal tegmental nuclei of the brainstem reach the thalamus. Michael Hasselmo and colleagues showed that acetylcholine selectively suppresses transmission at the recurrent synapses by which cortical and hippocampal circuits complete stored patterns, while leaving the inputs that carry new information intact. High acetylcholine, as during attentive waking, therefore favors encoding: the circuit is driven by what is arriving and is prevented from being captured by what it already knows. Low acetylcholine, as during slow-wave sleep, the deep stage of sleep, favors recall and the replay of stored patterns that Chapter 5 connects to consolidation.[165] The map-expansion experiment described earlier, in

[161] Aston-Jones, G., & Cohen, J. D. (2005). An integrative theory of locus coeruleus–norepinephrine function: Adaptive gain and optimal performance. *Annual Review of Neuroscience*, 28, 403–450; Sara, S. J. (2009). The locus coeruleus and noradrenergic modulation of cognition. *Nature Reviews Neuroscience*, 10(3), 211–223.

[162] Yu, A. J., & Dayan, P. (2005). Uncertainty, neuromodulation, and attention. *Neuron*, 46(4), 681–692.

[163] Dayan, P., & Huys, Q. J. M. (2009). Serotonin in affective control. *Annual Review of Neuroscience*, 32, 95–126; Cools, R., Roberts, A. C., & Robbins, T. W. (2008). Serotoninergic regulation of emotional and behavioural control processes. *Trends in Cognitive Sciences*, 12(1), 31–40.

[164] Miyazaki, K., Miyazaki, K. W., & Doya, K. (2011). Activation of dorsal raphe serotonin neurons underlies waiting for delayed rewards. *Journal of Neuroscience*, 31(2), 469–479.

which a tone paired with nucleus basalis stimulation claimed a larger share of auditory cortex, is the same system acting as permission for cortical plasticity.[166]

Together these systems regulate not what the brain thinks but how it thinks: how strongly to respond, how much to trust the current model, whether to persist or to switch, and whether to write or to read.

Clinical Evidence

The clearest evidence that neuromodulatory tone shapes learning and behavior comes from the clinic, where it has been altered deliberately in humans for more than sixty years.

Parkinson's disease is the founding case. In 1960 Herbert Ehringer and Oleh Hornykiewicz measured dopamine in postmortem brains and found it almost absent from the striatum of patients who had died with the disease, the first time a neurological disorder had been tied to the loss of a specific transmitter.[167] Arvid Carlsson had shown three years earlier that animals immobilized by reserpine, a drug that empties the brain's stores of monoamines, recovered when given L-DOPA, the precursor from which neurons make dopamine; in 1967 George Cotzias gave patients L-DOPA by mouth in gradually increasing doses and reported, in the sober prose of a medical journal, that rigidity and slowness of movement were in some patients nearly abolished.[168] L-DOPA remains the standard treatment. Its effect on learning, as distinct from movement, was shown by Michael Frank, Lauren Seeberger, and Randall O'Reilly in 2004. Patients chose repeatedly between pairs of symbols that paid off with different probabilities. Off medication, with striatal dopamine low, they were better at learning which symbol to avoid than which to choose; on medication, the reverse. The same patients, the same task, and a shift in dopamine tone changed whether they learned more from good outcomes or from bad.[169] This is what the three-factor rule predicts if bursts and dips carry different lessons and medication raises the floor under the dips.

Dopamine can also be reduced. The antipsychotic drugs used for schizophrenia block dopamine receptors, and in 1976 Ian Creese, David Burt, and Solomon Snyder showed that the clinical dose of each drug was predicted by how tightly it bound the dopamine receptor that would later be called D2; the drugs' parkinsonian side effects, slowed movement and tremor, are dopamine loss produced by pharmacology.[170] Acetylcholine has a parallel story. In 1982 Peter Whitehouse and colleagues found that the nucleus basalis of patients who had died with Alzheimer's disease had lost most of its neurons, and the cholinergic hypothesis of the memory loss of dementia followed. The drugs built on it, donepezil, galantamine, and rivastigmine, inhibit the enzyme that breaks acetylcholine down and so raise its level

[165] Hasselmo, M. E., & Bower, J. M. (1993). Acetylcholine and memory. *Trends in Neurosciences*, 16(6), 218–222; Hasselmo, M. E. (2006). The role of acetylcholine in learning and memory. *Current Opinion in Neurobiology*, 16(6), 710–715.

[166] Kilgard, M. P., & Merzenich, M. M. (1998). Cortical map reorganization enabled by nucleus basalis activity. *Science*, 279(5357), 1714–1718.

[167] Ehringer, H., & Hornykiewicz, O. (1960). Verteilung von Noradrenalin und Dopamin (3-Hydroxytyramin) im Gehirn des Menschen und ihr Verhalten bei Erkrankungen des extrapyramidalen Systems. *Klinische Wochenschrift*, 38, 1236–1239.

[168] Carlsson, A., Lindqvist, M., & Magnusson, T. (1957). 3,4-Dihydroxyphenylalanine and 5-hydroxytryptophan as reserpine antagonists. *Nature*, 180(4596), 1200; Cotzias, G. C., Van Woert, M. H., & Schiffer, L. M. (1967). Aromatic amino acids and modification of parkinsonism. *New England Journal of Medicine*, 276(7), 374–379.

[169] Frank, M. J., Seeberger, L. C., & O'Reilly, R. C. (2004). By carrot or by stick: Cognitive reinforcement learning in parkinsonism. *Science*, 306(5703), 1940–1943.

[170] Creese, I., Burt, D. R., & Snyder, S. H. (1976). Dopamine receptor binding predicts clinical and pharmacological potencies of antischizophrenic drugs. *Science*, 192(4238), 481–483.

at every cholinergic synapse. A Cochrane review of the trials found that they help: a modest but consistent improvement in cognitive scores at six months, which is what one would expect from raising the tone of a system that permits encoding without restoring the neurons that were lost.[171]

Deep brain stimulation belongs in this section with a caveat attached. In 1991 Alim-Louis Benabid and colleagues reported that chronic electrical stimulation of a thalamic nucleus through an implanted electrode suppressed tremor, and in 1998 Patricia Limousin and colleagues extended the method to the subthalamic nucleus in advanced Parkinson's disease, where it relieves rigidity and slowness and lets medication be reduced.[172] Figure 3.9 shows the hardware: a lead through a burr hole in the skull, its tip in a nucleus a few millimeters across, connected under the skin to a pulse generator below the collarbone. The procedure is often called neuromodulation, and in the clinical sense of the word it is. In the sense used in this chapter it is not. The stimulation is electrical, it acts on the activity of a circuit rather than on the level of any transmitter, and how it works, whether by silencing the target, by driving it in a regular rhythm that the rest of the circuit can ignore, or by disrupting a pathological oscillation, is still debated.[173] It is strong evidence that the basal ganglia circuit of Chapter 4 can be pushed from outside. It is not direct evidence about neuromodulator tone, and it is placed here so that the two senses of the word are not confused.

[171] Whitehouse, P. J., Price, D. L., Struble, R. G., Clark, A. W., Coyle, J. T., & DeLong, M. R. (1982). Alzheimer's disease and senile dementia: Loss of neurons in the basal forebrain. *Science*, 215(4537), 1237–1239; Birks, J. S. (2006). Cholinesterase inhibitors for Alzheimer's disease. *Cochrane Database of Systematic Reviews*, (1), CD005593.

[172] Benabid, A. L., Pollak, P., Gervason, C., Hoffmann, D., Gao, D. M., Hommel, M., Perret, J. E., & de Rougemont, J. (1991). Long-term suppression of tremor by chronic stimulation of the ventral intermediate thalamic nucleus. *The Lancet*, 337(8738), 403–406; Limousin, P., Krack, P., Pollak, P., Benazzouz, A., Ardouin, C., Hoffmann, D., & Benabid, A.-L. (1998). Electrical stimulation of the subthalamic nucleus in advanced Parkinson's disease. *New England Journal of Medicine*, 339(16), 1105–1111.

[173] Lozano, A. M., Lipsman, N., Bergman, H., Brown, P., Chabardes, S., Chang, J. W., Matthews, K., McIntyre, C. C., Schlaepfer, T. E., Schulder, M., Temel, Y., Volkmann, J., & Krauss, J. K. (2019). Deep brain stimulation: Current challenges and future directions. *Nature Reviews Neurology*, 15(3), 148–160.

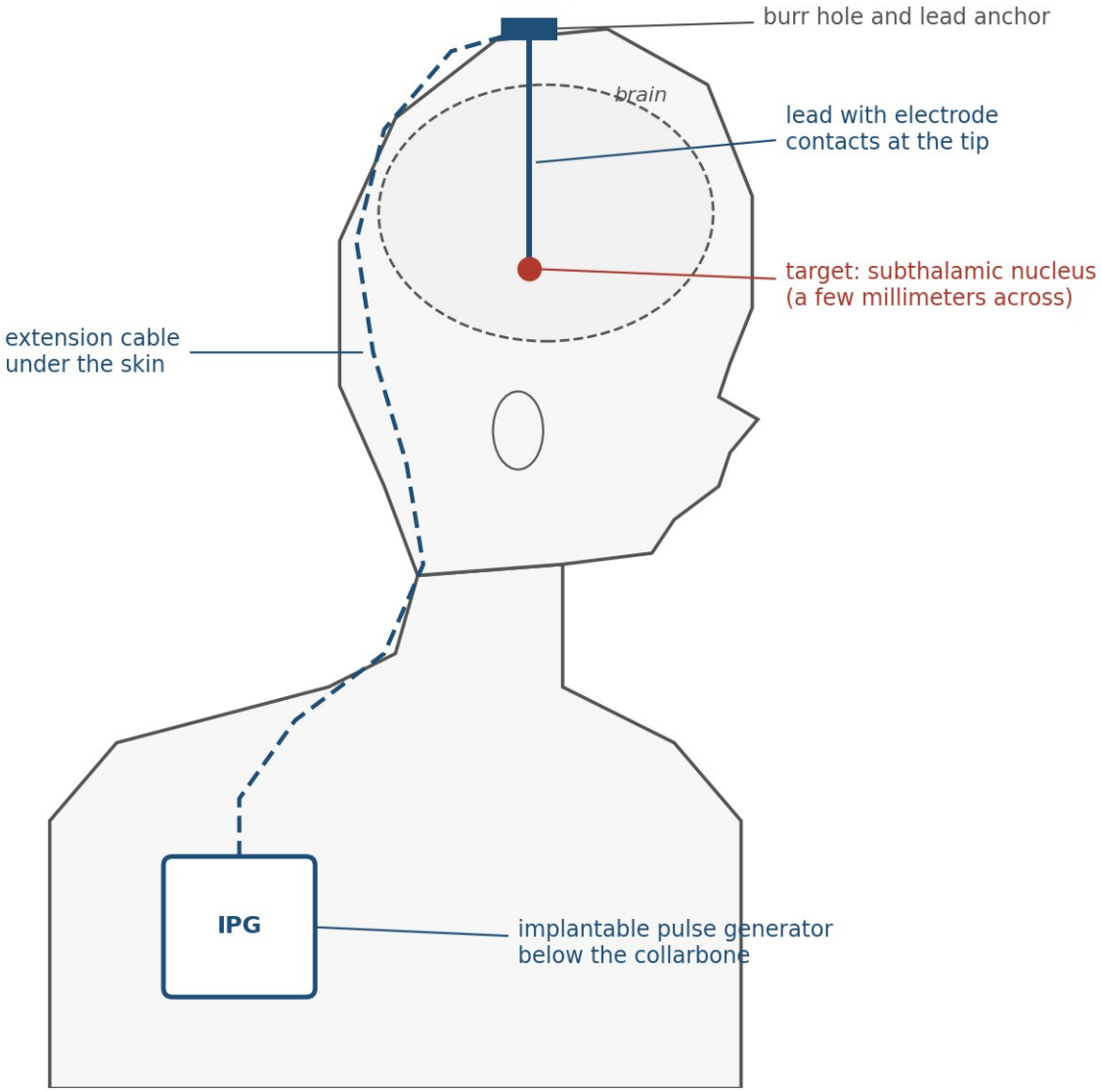


Figure 3.9. Deep brain stimulation hardware in place. A lead enters through a burr hole in the skull and ends in the subthalamic nucleus; an extension cable under the skin connects it to a pulse generator implanted below the collarbone. The intervention is electrical and acts on circuit activity, not on neuromodulator levels.

Implications for Artificial Intelligence

Modern artificial neural networks are extraordinarily capable within the assumptions they were built on. They learn well when the objective is explicit, feedback is immediate, and the architecture is fixed, and under those conditions backpropagation gives each weight an exact gradient, which is possible

precisely because the conditions hold. Biological intelligence evolved under none of them. Real organisms learn where goals are ambiguous, feedback is delayed, and the causal structure of outcomes is hidden. The brain's mechanisms reflect that: a global modulatory signal, local eligibility traces, and incremental biasing of behavior over time rather than an exact gradient. Several implications follow.

First, the brain separates activity from learning, and does so from the inside. In an artificial system the separation is imposed from outside by a switch. During training every forward pass is followed by a weight update; in deployment the weights are frozen and no forward pass changes anything. The brain has no such switch. It runs and learns continuously, and the decision about whether a given moment of activity is allowed to change the wiring is made internally, by the neuromodulatory context, on a timescale of seconds. No change lasts unless the context says it should. A system that must keep learning after deployment needs something of this kind.

Second, learning rules themselves can be state-dependent. Neuromodulators change how the same activity is interpreted: what counts as a surprise, how large an update is warranted, whether to encode or to recall. Kenji Doya proposed in 2002 that the four systems correspond to the four quantities a reinforcement learner has to set, the error signal, the learning rate, the exploration rate, and the discount on future reward, and recent work has trained networks whose plasticity is gated by a learned modulatory signal, or has found that a recurrent network trained on many tasks develops within its activity a fast learning system that resembles the one dopamine trains in prefrontal cortex.[174]

Third, approximate credit assignment can be enough. A network trained with random feedback weights learns almost as well as one with the exact backward pathway, which means precise attribution is not a requirement for useful adaptation.[175] The brain appears to go further still, broadcasting a single number to an entire region, and it learns well enough to survive.

Finally, neuromodulation reframes the objective. The brain reduces its many local errors under a constraint that an artificial system rarely faces: it must remain viable while it learns.[176] Success is defined operationally, by survival and by what works, and the neuromodulatory systems are the brain's running estimate of how that is going. As artificial systems move out of curated datasets and into open-ended interaction with the world, these principles may prove more important than further refinements of gradient descent.

The Central Lesson

Learning is gated, broadcast, and biased. The brain has no known way to trace an outcome exactly back to the synapses that caused it, and it does not need one. Instead a few hundred thousand neuromodulatory neurons broadcast a running summary of how things are going, whether events are better or worse than expected, whether the world has changed, whether to take in or to recall, and that summary decides whether the activity of the moment is permitted to leave a lasting trace. The signal is a

[174] Doya, K. (2002). Metalearning and neuromodulation. *Neural Networks*, 15(4–6), 495–506; Miconi, T., Rawal, A., Clune, J., & Stanley, K. O. (2019). Backpropamine: Training self-modifying neural networks with differentiable neuromodulated plasticity. In *International Conference on Learning Representations*; Wang, J. X., Kurth-Nelson, Z., Kumaran, D., Tirumala, D., Soyer, H., Leibo, J. Z., Hassabis, D., & Botvinick, M. (2018). Prefrontal cortex as a meta-reinforcement learning system. *Nature Neuroscience*, 21(6), 860–868.

[175] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276.

[176] Ashby, W. R. (1952). *Design for a Brain*. Chapman & Hall.

scalar that reaches everything, and it becomes selective only because synapses hold a short-lived record of their own recent activity. The changes it produces are small and sometimes wrong, but they accumulate, and over time they bias neural activity and behavior toward the patterns that have tended to succeed.

Next Chapter

Chapter 4: Circuits, Networks, and Coordinated Behavior

Figure and Epigraph Sources

Figure Attribution

Figures 3.1 to 3.9 were drawn for this book. The response traces in Figure 3.2 and the spike rasters, histograms, and curves in Figures 3.5 to 3.8 are illustrative simulations, not recorded data; Figure 3.6 is an undiscounted temporal-difference simulation. The brain outlines in Figures 3.3, 3.4, and 3.9 are schematic, nuclei are placed approximately, and the hardware in Figure 3.9 is generic rather than any manufacturer's design.

Epigraph

Henry H. Dale, "Some Recent Extensions of the Chemical Transmission of the Effects of Nerve Impulses," Nobel Lecture, Stockholm, 12 December 1936, opening sentence; reprinted in *Nobel Lectures, Physiology or Medicine 1922–1941* (Amsterdam: Elsevier, 1965), and online at nobelprize.org. Dale shared the 1936 Nobel Prize in Physiology or Medicine with Otto Loewi, whose 1921 experiments on the frog heart (Chapter 2) first showed that a nerve acts on its target by releasing a chemical substance.

Chapter 4

Circuits, Networks, and Coordinated Behavior

> *"Swiftly the head mass becomes an enchanted loom where millions of flashing shuttles weave a dissolving pattern, always a meaningful pattern though never an abiding one; a shifting harmony of subpatterns."*
>
> *— Charles Sherrington, Man on His Nature (1940)*

From Cells to Circuits

The previous chapters built a neuron and then surrounded it with chemical context. A neuron integrates inputs, fires when a threshold is crossed, and changes physically with use, and neuromodulators decide how much of that change is kept. That account is complete at the level of the cell. It explains almost nothing about behavior.

A single neuron cannot walk, breathe, decide, or remember. It cannot even compute a difference between two inputs in a way that outlasts the inputs themselves. Every capacity that we recognize as behavior belongs to groups of neurons connected in particular ways. The unit of function in the nervous system is not the neuron. It is the circuit.

This creates a problem of description. The human brain contains about 86 billion neurons and, in the neocortex alone, about 164 trillion synapses.[177] No account at the level of individual connections is possible, and none would be illuminating if it were. What makes circuits intelligible is that the nervous system reuses a small number of wiring patterns, at every scale, in every region. Learn the patterns and the regions become variations on a theme.

This chapter is about those patterns, the ways they are stacked into larger structures, and the rhythms that coordinate them. It ends where Chapter 1 began: with a system that produces coordinated behavior without anyone in charge.

The Simplest Circuit

Sherrington's reflex arc remains the clearest place to start. A sensory neuron in a muscle spindle detects stretch. Its axon enters the spinal cord and synapses directly on a motor neuron, which contracts the same muscle. Tap the tendon below the kneecap: the quadriceps stretches and the leg kicks. Two neurons, one synapse, no brain required.[178]

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[177] Azevedo, F. A. C., Carvalho, L. R. B., Grinberg, L. T., Farfel, J. M., Ferretti, R. E. L., Leite, R. E. P., Jacob Filho, W., Lent, R., & Herculano-Houzel, S. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5), 532–541; Pakkenberg, B., Pelvig, D., Marner, L., Bundgaard, M. J., Gundersen, H. J. G., Nyengaard, J. R., & Regeur, L. (2003). Aging and the human neocortex. *Experimental Gerontology*, 38(1–2), 95–99; Tang, Y., Nyengaard, J. R., De Groot, D. M. G., & Gundersen, H. J. G. (2001). Total regional and global number of synapses in the human brain neocortex. *Synapse*, 41(3), 258–273. Azevedo and colleagues count the neurons; Tang and colleagues count about 164 trillion synapses in the neocortex, and Pakkenberg and colleagues give a similar figure of about 150 trillion for young adults.

[178] Sherrington, C. S. (1906). *The Integrative Action of the Nervous System*. Charles Scribner's Sons; Liddell, E. G. T., & Sherrington, C. S. (1924). Reflexes in response to stretch (myotatic reflexes). *Proceedings of the Royal Society B*, 96, 212–242.

Even this minimal circuit contains a second element that is easy to overlook. The same sensory axon also excites an inhibitory interneuron, which suppresses the motor neurons of the opposing muscles. The quadriceps contracts while the hamstrings are actively prevented from resisting. This is reciprocal inhibition, and it is the first lesson of circuit organization: useful behavior requires turning things off as much as turning them on.[179]

Reflexes are not primitive leftovers. They are the fastest and most reliable components of the motor system, and nearly everything the brain does with movement is built by modulating them.

A Small Vocabulary of Motifs

Across the nervous system, a handful of wiring motifs recur. Figure 4.1 shows six of them, beginning with the reflex arc just described. Each buys the organism something specific.

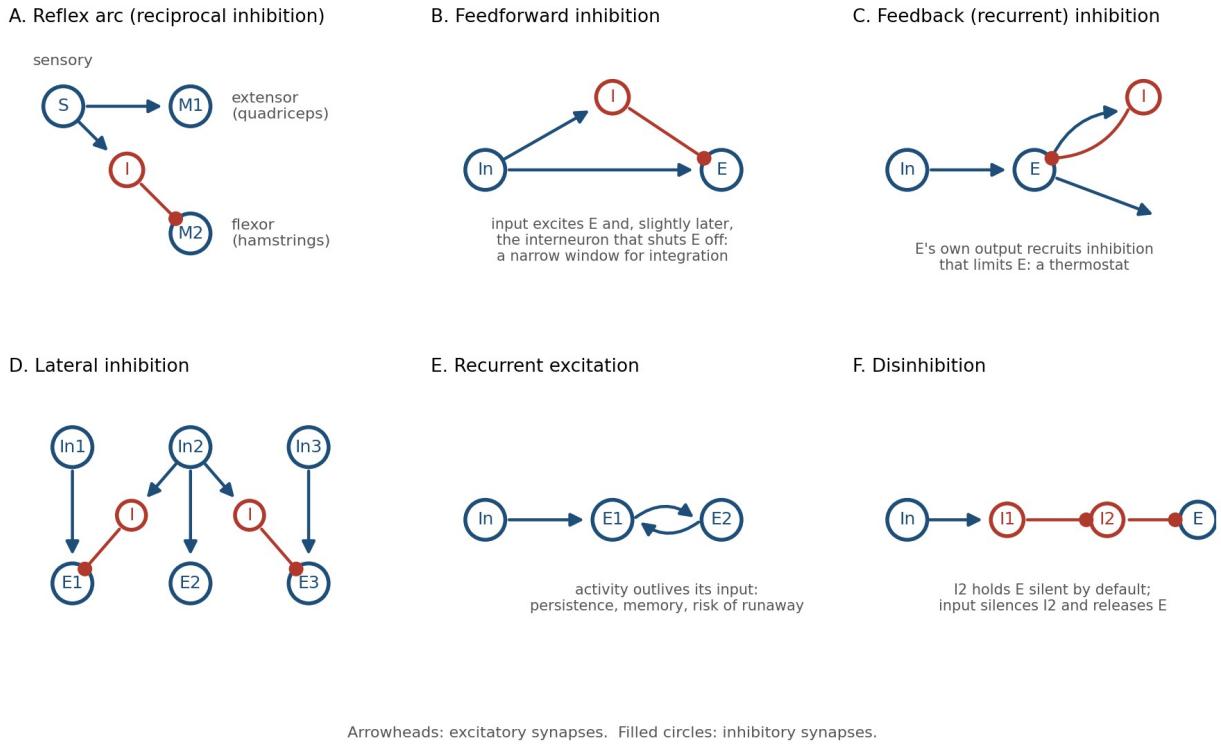


Figure 4.1. Six circuit motifs that recur throughout the nervous system. Arrowheads mark excitatory synapses and filled circles inhibitory ones; panel A is the knee-jerk arc, in which the stretched extensor is excited and its flexor antagonist is inhibited.

Feedforward excitation is the trivial case: a signal passes from one neuron to the next. Chains of it carry information from the senses toward the muscles. On its own it transmits but does not shape.

Feedforward inhibition adds a branch. The input excites a target neuron and, through an interneuron, inhibits the same target a millisecond or two later. The effect is a narrow window: the target can respond only to inputs that arrive close together in time. Inputs that straggle are canceled. This motif

[179] Sherrington, C. S. (1906). *The Integrative Action of the Nervous System*. Charles Scribner's Sons. Sherrington introduced the term reciprocal innervation in a series of notes to the Royal Society from 1893 onward and summarized the principle in the 1906 lectures.

turns a sluggish integrator into a coincidence detector, and it appears wherever timing precision matters, from the cerebellum to the auditory brainstem.[180]

Feedback inhibition closes a loop. The target's own output recruits an interneuron that inhibits the target. The harder the population fires, the harder it is suppressed. This is a thermostat, and it is the primary reason cortical activity does not explode.[181]

Lateral inhibition lets neighbors suppress one another. The neuron receiving the strongest input wins and silences its competitors. The consequences include sharpened contrast, winner-take-all selection, and the ability to represent a stimulus by which neuron is active rather than by how active everything is.[182]

Recurrent excitation lets activity feed back on itself. A pattern of firing can be sustained after its input has gone, which is the basis of working memory, the holding of information in mind for seconds, and of pattern completion, described below.[183] It is also the most dangerous motif in the brain. Without inhibition to contain it, recurrent excitation is a seizure.

Disinhibition is inhibition of an inhibitor. A target is held silent by default; an input that suppresses the inhibitor releases the target. This sounds like a roundabout way to excite something, but it has a property direct excitation lacks: the default state is off, so nothing happens unless the circuit is explicitly opened. The basal ganglia, discussed below, are built almost entirely from it.[184]

Inhibition Is Not the Opposite of Excitation

Engineers encountering these motifs tend to file inhibition under "negative weights" and move on. That misses most of what inhibition does.

Roughly one in five cortical neurons is inhibitory, and those neurons come in dozens of types, each targeting a different part of the excitatory cell: the dendrites, the soma, the axon initial segment, or other interneurons.[185] Because they target different compartments, they do different jobs. Inhibition at the dendrite vetoes particular inputs. Inhibition at the soma vetoes output. Inhibition at the axon initial segment controls exactly when a spike may leave.

Most importantly, excitation and inhibition are balanced. In an awake cortex, every excitatory input to a neuron is tracked by a nearly equal inhibitory input arriving a few milliseconds later.[186] The neuron

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[180] Pouille, F., & Scanziani, M. (2001). Enforcement of temporal fidelity in pyramidal cells by somatic feed-forward inhibition. *Science*, 293(5532), 1159–1163.

[181] Isaacson, J. S., & Scanziani, M. (2011). How inhibition shapes cortical activity. *Neuron*, 72(2), 231–243.

[182] Hartline, H. K., Wagner, H. G., & Ratliff, F. (1956). Inhibition in the eye of *Limulus*. *Journal of General Physiology*, 39(5), 651–673. Lateral inhibition was first measured in the compound eye of the horseshoe crab.

[183] Goldman-Rakic, P. S. (1995). Cellular basis of working memory. *Neuron*, 14(3), 477–485; Wang, X.-J. (2001). Synaptic reverberation underlying mnemonic persistent activity. *Trends in Neurosciences*, 24(8), 455–463.

[184] Letzkus, J. J., Wolff, S. B. E., Meyer, E. M. M., Tovote, P., Courtin, J., Herry, C., & Lüthi, A. (2011). A disinhibitory microcircuit for associative fear learning in the auditory cortex. *Nature*, 480, 331–335; Pi, H.-J., Hangya, B., Kvitsiani, D., Sanders, J. I., Huang, Z. J., & Kepecs, A. (2013). Cortical interneurons that specialize in disinhibitory control. *Nature*, 503, 521–524. These papers describe disinhibitory microcircuits in cortex; the basal ganglia case is cited below.

[185] Markram, H., Toledo-Rodriguez, M., Wang, Y., Gupta, A., Silberberg, G., & Wu, C. (2004). Interneurons of the neocortical inhibitory system. *Nature Reviews Neuroscience*, 5(10), 793–807; Tremblay, R., Lee, S., & Rudy, B. (2016). GABAergic interneurons in the neocortex: From cellular properties to circuits. *Neuron*, 91(2), 260–292.

lives close to threshold, held there by two large forces that almost cancel. Small changes in either one move it across the line.

Figure 4.2 shows why this arrangement is worth its cost. It is a simulation of a recurrently connected excitatory population, the same network three times, with only the strength of inhibition changed. With no inhibition the population does not respond to a brief input and then stop; the input tips it into a runaway that ends only when every cell is firing flat out, and it stays there after the input is gone, a seizure in miniature. Add inhibition that tracks the excitation and the same population produces a brief, bounded response that ends when the input ends. Add too much and the response is muted. The balanced regime is expensive, because two large currents are spent to produce a small net effect, and it is the only regime in which the circuit is both sensitive and stable.[187]

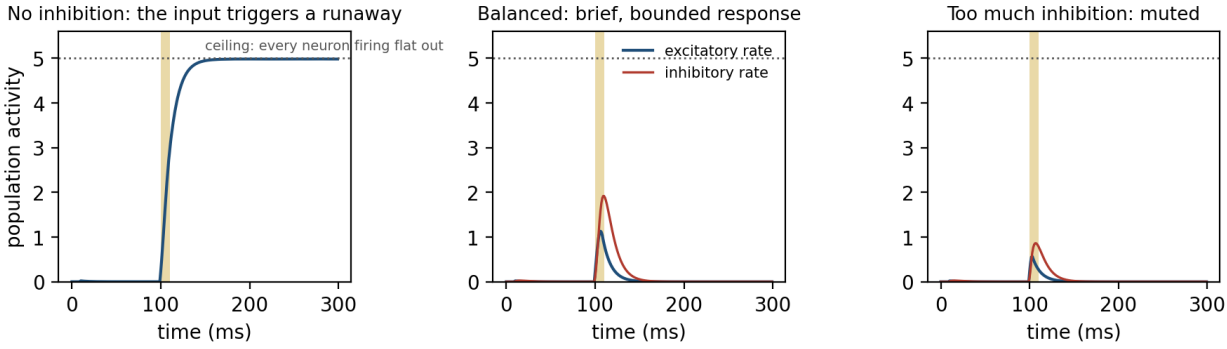


Figure 4.2. A simulated excitatory population with three levels of inhibition. The yellow band is a brief input: with no inhibition it triggers a runaway to the ceiling that outlasts the input, with balanced inhibition the response is brief and proportionate, and with excessive inhibition the input barely registers.

Balance is also what makes the circuit fast. A neuron held near threshold by opposing forces can be flipped by a single well-timed input. A neuron resting far below threshold must be dragged up first. Inhibition is not the brake on the system. It is what makes the system responsive.[188]

Computation Before the Brain

The retina shows how much can be done with these motifs before any signal reaches the cortex. Figure 4.3 is Cajal's drawing of it. Light enters at the left, crosses several layers of neurons, and reaches the photoreceptors at the right, so the signal those cells generate travels back through the layers before it leaves through the axons of the ganglion cells, which together form the optic nerve.

[186] Wehr, M., & Zador, A. M. (2003). Balanced inhibition underlies tuning and sharpens spike timing in auditory cortex. *Nature*, 426, 442–446; Okun, M., & Lampl, I. (2008). Instantaneous correlation of excitation and inhibition during ongoing and sensory-evoked activities. *Nature Neuroscience*, 11(5), 535–537.

[187] van Vreeswijk, C., & Sompolinsky, H. (1996). Chaos in neuronal networks with balanced excitatory and inhibitory activity. *Science*, 274(5293), 1724–1726; Shadlen, M. N., & Newsome, W. T. (1998). The variable discharge of cortical neurons: Implications for connectivity, computation, and information coding. *Journal of Neuroscience*, 18(10), 3870–3896.

[188] Isaacson, J. S., & Scanziani, M. (2011). How inhibition shapes cortical activity. *Neuron*, 72(2), 231–243.

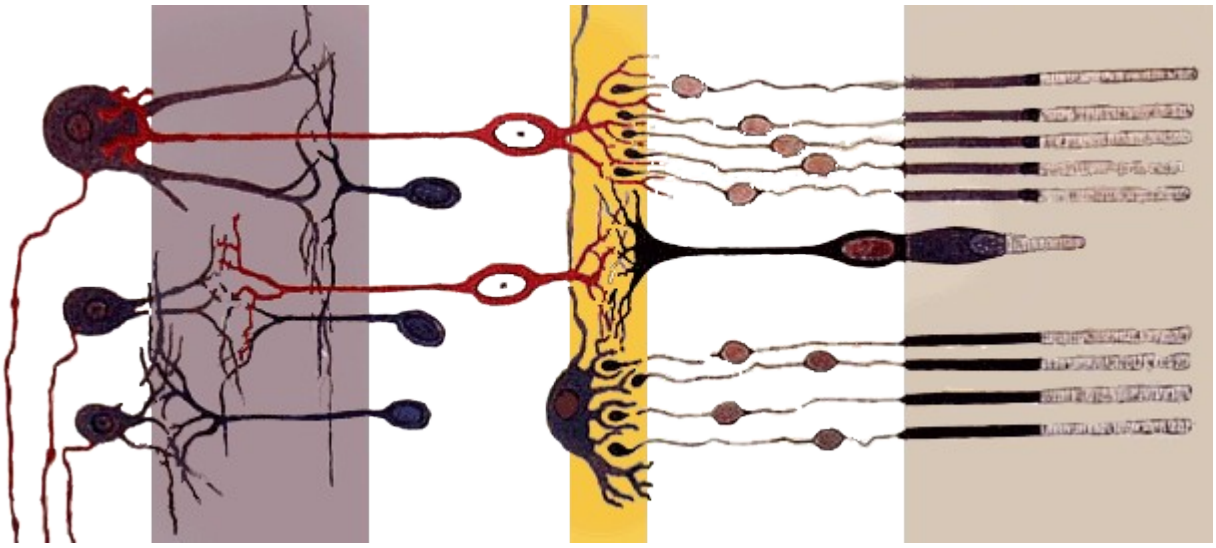


Figure 4.3. Santiago Ramón y Cajal's drawing of the vertebrate retina. Light enters from the left and crosses several layers of neurons before reaching the photoreceptors at the right; the processed signal leaves through the ganglion cell axons at the far left.

Light falls on photoreceptors, but the signal that leaves the eye has already passed through three layers of circuitry. Horizontal cells implement lateral inhibition across the photoreceptor layer, so a uniformly lit region produces little output while an edge produces a strong one. Amacrine cells add feedforward and feedback inhibition that make ganglion cells sensitive to change in time as well as space. By the time the optic nerve leaves the eye, the image has been converted into a map of edges and changes. Uniform brightness, which carries little information, has been largely discarded.[189]

Lateral inhibition can be seen directly. Figure 4.4 shows a staircase of uniform gray steps; each step looks lighter along its border with the darker neighbor and darker along its border with the lighter one, although the luminance within every step is constant. The fringes are Mach bands, described by the physicist Ernst Mach in 1865, and the lower panel shows where they come from: a cell just on the bright side of an edge is inhibited by fewer bright neighbors than a cell in the middle of the step, so it fires more, and a cell just on the dark side is inhibited by more, so it fires less. The bands are the signature of a circuit that trades a faithful copy of the input for an informative one, under a strict budget of axons.[190]

[189] Masland, R. H. (2012). The neuronal organization of the retina. *Neuron*, 76(2), 266–280; Wässle, H. (2004). Parallel processing in the mammalian retina. *Nature Reviews Neuroscience*, 5(10), 747–757.

[190] Ratliff, F. (1965). *Mach Bands: Quantitative Studies on Neural Networks in the Retina*. Holden-Day. Mach described the bands in 1865; Ratliff's monograph connects them quantitatively to lateral inhibition.

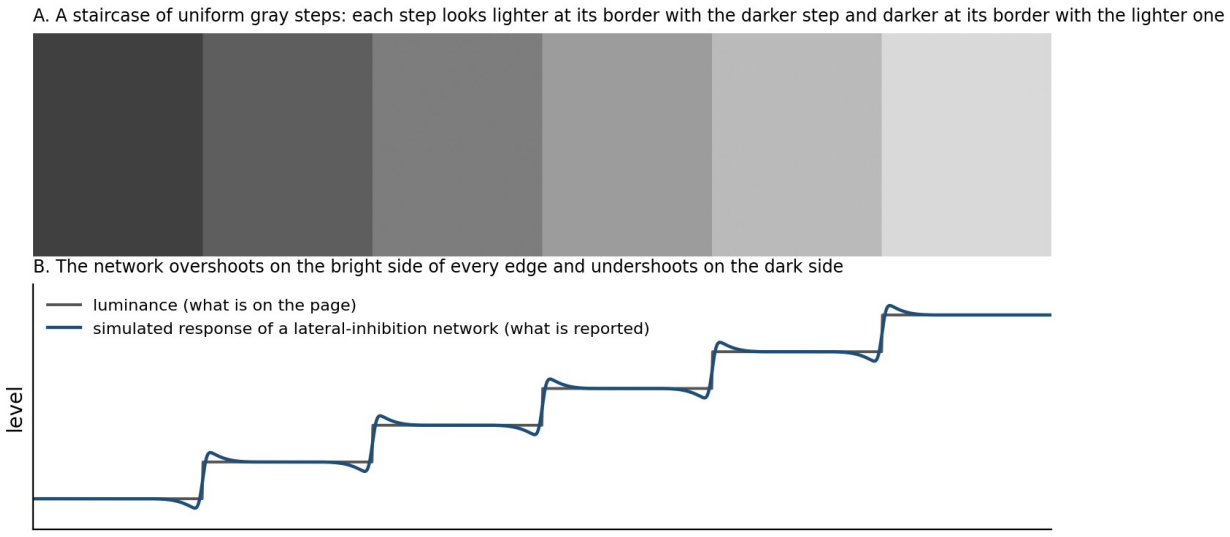


Figure 4.4. Mach bands, lateral inhibition made visible. The staircase is made of uniform gray steps, yet each step looks lighter at its border with the darker step and darker at its border with the lighter one; the lower panel compares the true luminance with the illustrative response of a lateral-inhibition network, which overshoots on the bright side of every edge and undershoots on the dark side.

Circuits That Generate Their Own Rhythm

Some circuits do not wait for input at all.

Walking, swimming, breathing, chewing, and the peristalsis of the gut are rhythmic, and the rhythm is generated by small networks in the spinal cord and brainstem called central pattern generators.[191] The surprising fact, established in the early twentieth century and confirmed many times since, is that these networks produce their rhythm without any rhythmic input. A spinal cord isolated from the brain and from sensory feedback will still generate the alternating pattern of locomotion when given a steady chemical or electrical drive.[192]

The minimal pattern generator is the half-center oscillator in Figure 4.5. Two populations inhibit each other, so only one can be active at a time. Each population also fatigues while active, through adaptation in its neurons or depression at its synapses. The active side weakens, its grip on the other side loosens, the other side escapes and takes over, and the cycle repeats. Mutual inhibition plus fatigue produces alternation from a constant input.[193]

[191] Marder, E., & Bucher, D. (2001). Central pattern generators and the control of rhythmic movements. *Current Biology*, 11(23), R986–R996; Smith, J. C., Ellenberger, H. H., Ballanyi, K., Richter, D. W., & Feldman, J. L. (1991). Pre-Bötzinger complex: A brainstem region that may generate respiratory rhythm in mammals. *Science*, 254(5032), 726–729. The respiratory rhythm generator in the brainstem is the pre-Bötzinger complex, identified by Smith and colleagues.

[192] Brown, T. G. (1911). The intrinsic factors in the act of progression in the mammal. *Proceedings of the Royal Society B*, 84(572), 308–319; Grillner, S., & Zangger, P. (1979). On the central generation of locomotion in the low spinal cat. *Experimental Brain Research*, 34(2), 241–261.

[193] Brown, T. G. (1914). On the nature of the fundamental activity of the nervous centres; together with an analysis of the conditioning of rhythmic activity in progression, and a theory of the evolution of function in the nervous system. *Journal of Physiology*, 48(1), 18–46. Brown's 1914 paper proposed the half-center model.

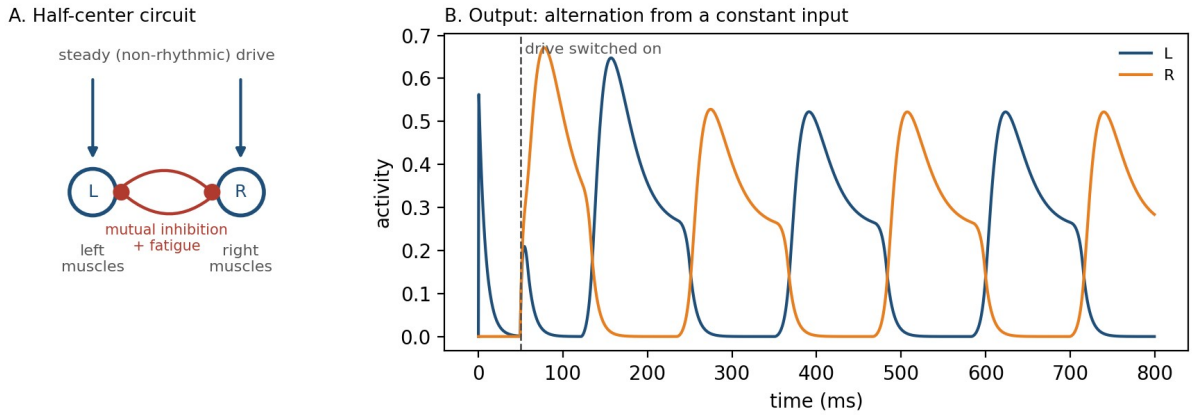


Figure 4.5. A half-center oscillator. Two populations inhibit each other and each fatigues while active; given a constant drive they alternate, as the simulated output trace on the right shows, because the active side tires, the suppressed side escapes, and the cycle repeats.

Sensory feedback does not drive the rhythm; it tunes it. Signals from the limbs adjust timing and amplitude so that a walking pattern survives a stone underfoot. The division of labor is the one Chapter 1 stated: “The brain generates behavior, and the world corrects it.”[194]

The Cortical Microcircuit

The cerebral cortex is a sheet a few millimeters thick, folded to fit the skull. Most of it is neocortex, the six-layered kind (older cortical regions such as the hippocampus have three layers), and across the whole neocortical sheet the same six-layered circuit is repeated with only local variations. Figure 4.6 shows it as Cajal drew it. Two of the panels use the Nissl stain, which marks only cell bodies, and the third uses the Golgi stain, which fills a few whole neurons, dendrites and axons included; in all three the layers appear as bands of differing cell density.

[194] Rossignol, S., Dubuc, R., & Gossard, J.-P. (2006). Dynamic sensorimotor interactions in locomotion. *Physiological Reviews*, 86(1), 89–154.

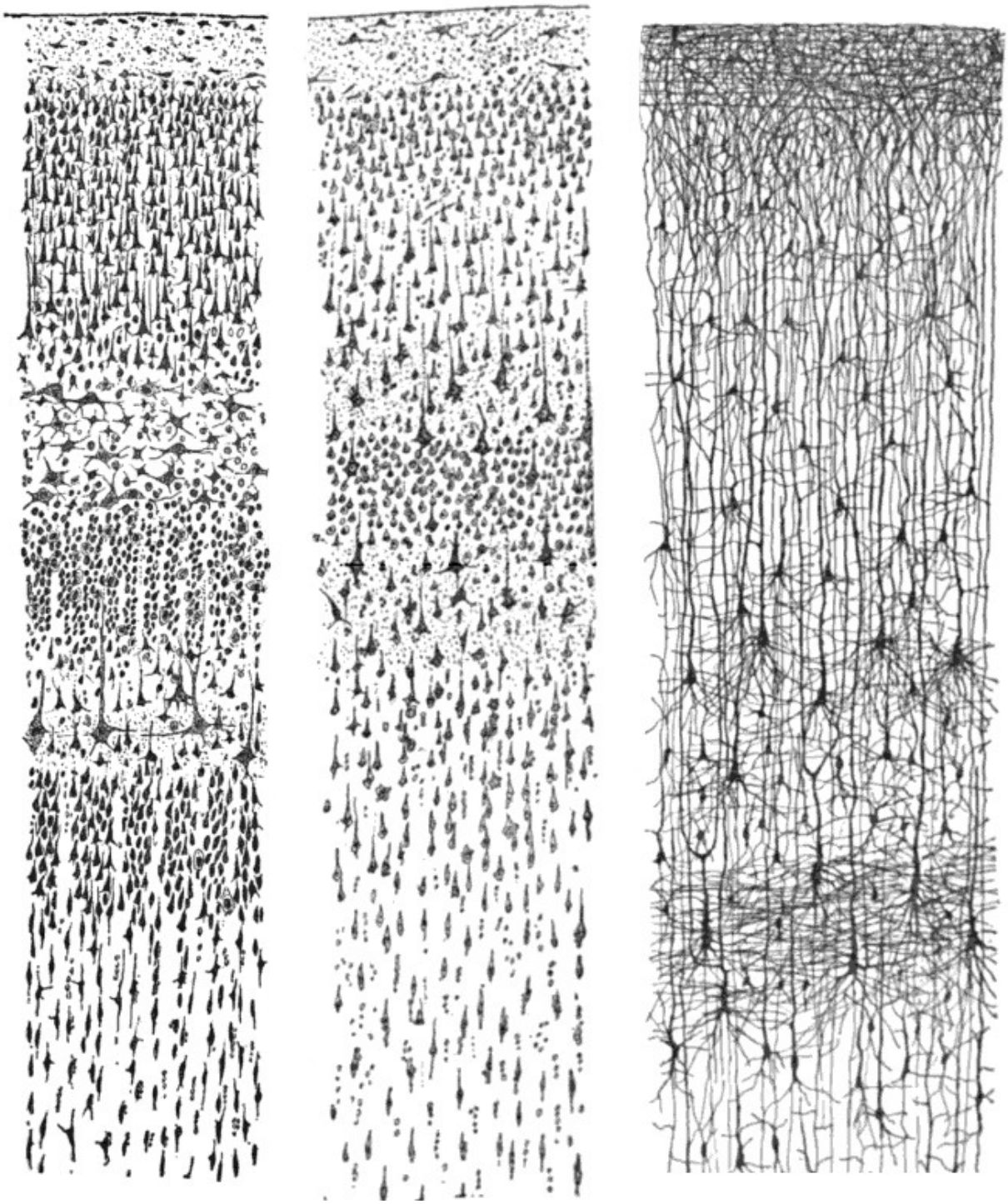


Figure 4.6. Three of Cajal's drawings of the cerebral cortex. Nissl stains show cell bodies in adult human visual cortex (left) and motor cortex (center), and a Golgi stain shows whole neurons in the cortex of a one-and-a-half-month-old infant (right); the six layers are visible as bands of differing cell density.

Input from the thalamus, the cortex's gateway for the senses, arrives mainly in layer 4. Layer 4 neurons project upward to layers 2 and 3, where most local processing and most connections to other cortical areas occur. Layers 2 and 3 project down to layer 5, which sends the cortex's output to the brainstem,

spinal cord, and basal ganglia. Layer 6 projects back to the thalamus, closing a loop that lets the cortex modulate its own input.[195]

Within this stack, every motif from Figure 4.1 is present. Feedforward inhibition sharpens the timing of thalamic input. Recurrent excitation among layer 2 and 3 neurons sustains and amplifies patterns. Feedback and lateral inhibition keep the amplification bounded. The whole arrangement has been called the canonical microcircuit, and it is approximately the same whether the cortex in question is processing vision, sound, touch, language, or plans.[196]

Vertically, the neurons stacked above one another through all six layers share their inputs, forming columns like those in Figure 4.7. A column in visual cortex responds to one orientation at one location; a column in somatosensory cortex responds to one whisker.[197] The column is not a rigid unit, and its boundaries blur in many species, but the principle is robust: the cortex is a tiled array of nearly identical circuits, differing mostly in what they are connected to.[198]

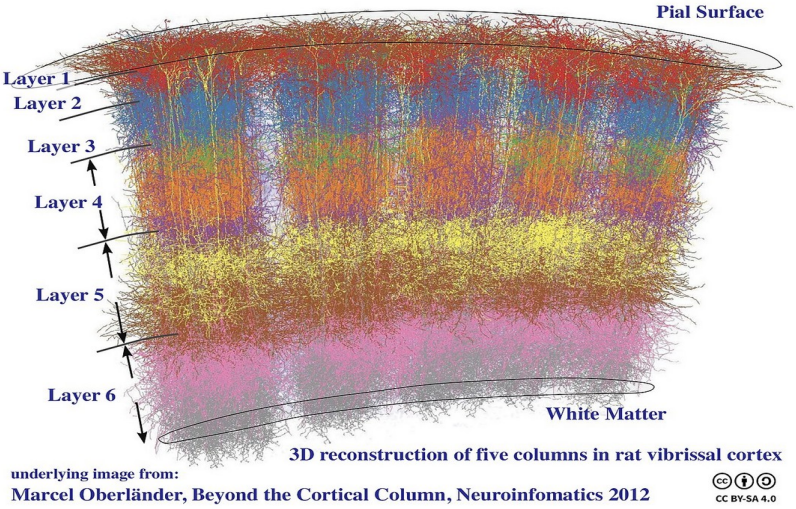


Figure 4.7. A three-dimensional reconstruction of five neighboring columns in rat somatosensory cortex, each corresponding to one whisker. The six layers run horizontally; the columns run vertically through all of them.

This is the strongest hint in neuroanatomy that cortical computation is general. The same circuit learns to see or to hear depending on where its input comes from, which is why, in a famous series of experiments in ferrets, retinal axons rerouted during development into the auditory pathway produced an auditory cortex that could see: the animals, tested as adults, responded to light through it.[199]

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[195] Douglas, R. J., & Martin, K. A. C. (2004). Neuronal circuits of the neocortex. *Annual Review of Neuroscience*, 27, 419–451; Harris, K. D., & Shepherd, G. M. G. (2015). The neocortical circuit: Themes and variations. *Nature Neuroscience*, 18(2), 170–181.

[196] Douglas, R. J., Martin, K. A. C., & Whitteridge, D. (1989). A canonical microcircuit for neocortex. *Neural Computation*, 1(4), 480–488; Douglas, R. J., & Martin, K. A. C. (2004). Neuronal circuits of the neocortex. *Annual Review of Neuroscience*, 27, 419–451.

[197] Mountcastle, V. B. (1957). Modality and topographic properties of single neurons of cat's somatic sensory cortex. *Journal of Neurophysiology*, 20(4), 408–434; Hubel, D. H., & Wiesel, T. N. (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *Journal of Physiology*, 160(1), 106–154.

[198] Horton, J. C., & Adams, D. L. (2005). The cortical column: A structure without a function. *Philosophical Transactions of the Royal Society B*, 360(1456), 837–862.

[199] Sur, M., Garraghty, P. E., & Roe, A. W. (1988). Experimentally induced visual projections into auditory thalamus and cortex. *Science*, 242(4884), 1437–1441; von Melchner, L., Pallas, S. L., & Sur, M. (2000). Visual behaviour mediated by retinal projections

Two Loops for Action

The cortex proposes. Two subcortical systems dispose, in different ways.

The basal ganglia solve the problem of selection. At any moment the cortex represents many candidate actions in parallel, and the body can perform only one. The basal ganglia are the disinhibition motif scaled up to an organ. Their output nuclei fire continuously, holding the thalamus and brainstem motor circuits under tonic inhibition, so that by default no action is released. A cortical pattern that excites the striatum, the input stage, inhibits the inhibitors covering one action, and that action alone is freed. Figure 4.8 draws this direct pathway; a second, indirect pathway through the striatum raises the inhibition on competing actions, and a hyperdirect path from cortex to the output nuclei can stop everything at once.[200]

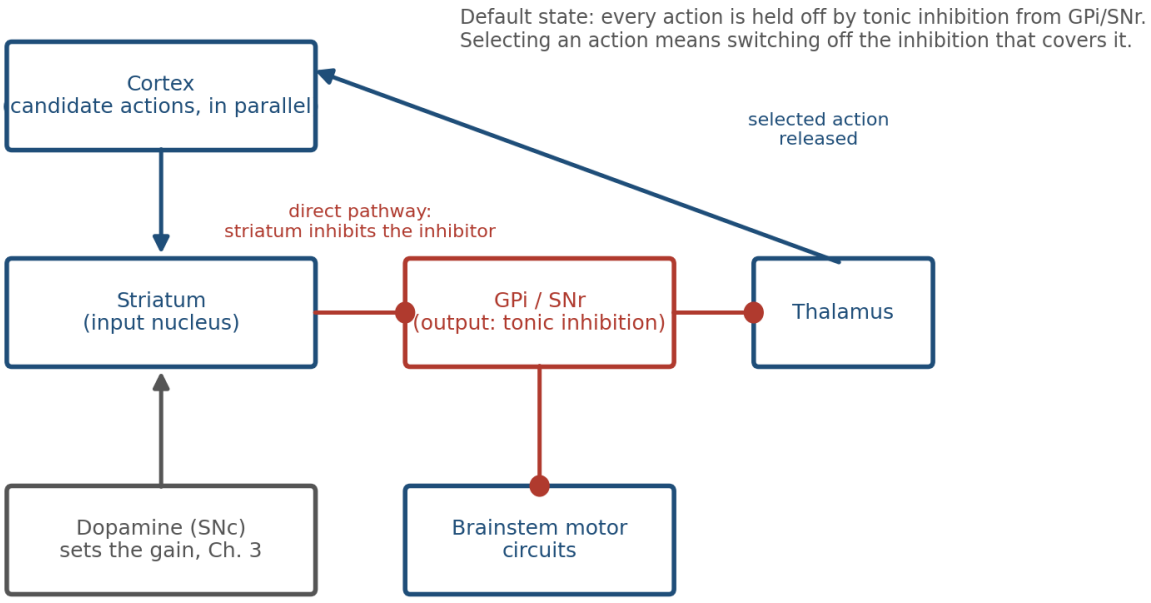


Figure 4.8. The direct pathway through the basal ganglia. Output nuclei inhibit motor targets by default; the striatum selects an action by inhibiting the inhibition that covers it, and dopamine sets how readily the striatum does this.

The design has two features that matter. The first is that it can fail in either direction, and the clinic shows both. Parkinson's disease is the loss of the dopamine neurons of the substantia nigra that bias the striatum toward release, and the result is too little movement: slowness, rigidity, and difficulty starting. Huntington's disease destroys the striatal neurons of the indirect pathway that hold competing actions off, and the result is chorea, a stream of unwanted movements; a lesion of the subthalamic nucleus, a station on that same path, produces hemiballismus, violent flinging of the limbs on one side. The classical model of Albin, Young, and Penney and of DeLong is built on exactly this contrast between

directed to the auditory pathway. *Nature*, 404, 871–876.

[200] Chevalier, G., & Deniau, J. M. (1990). Disinhibition as a basic process in the expression of striatal functions. *Trends in Neurosciences*, 13(7), 277–280; Mink, J. W. (1996). The basal ganglia: Focused selection and inhibition of competing motor programs. *Progress in Neurobiology*, 50(4), 381–425; Redgrave, P., Prescott, T. J., & Gurney, K. (1999). The basal ganglia: A vertebrate solution to the selection problem? *Neuroscience*, 89(4), 1009–1023.

hypokinetic and hyperkinetic disorders. The circuit does not fail safe: it fails toward too little when the release signal is lost and toward too much when the brakes are lost.[201] The second feature is that because dopamine acts on the striatum, the reward prediction error of Chapter 3, carried by a large population of midbrain dopamine neurons, arrives at exactly the synapses that decide which action is released. The basal ganglia are where "learning should happen now" meets "this is what was chosen." Habits are the result.[202]

The cerebellum solves a different problem: precision. It holds about 69 of the brain's 86 billion neurons, more than the rest of the brain combined, arranged in a circuit of striking regularity.[203] Each Purkinje cell, the large neuron that is the sole output of the cerebellar cortex, receives about two hundred thousand weak inputs from parallel fibers, carrying the current state of the body and the current motor command, and a single powerful input from one climbing fiber, which fires when a movement goes wrong.[204] Figure 4.9 lays out the circuit.

[201] Alexander, G. E., DeLong, M. R., & Strick, P. L. (1986). Parallel organization of functionally segregated circuits linking basal ganglia and cortex. *Annual Review of Neuroscience*, 9, 357–381; Albin, R. L., Young, A. B., & Penney, J. B. (1989). The functional anatomy of basal ganglia disorders. *Trends in Neurosciences*, 12(10), 366–375; DeLong, M. R. (1990). Primate models of movement disorders of basal ganglia origin. *Trends in Neurosciences*, 13(7), 281–285. The Albin, Young, and Penney model attributes hypokinetic disorders to overactive output nuclei and hyperkinetic ones, chorea and hemiballismus among them, to underactive output nuclei after loss of the indirect pathway or of the subthalamic nucleus.

[202] Schultz, W., Dayan, P., & Montague, P. R. (1997). A neural substrate of prediction and reward. *Science*, 275(5306), 1593–1599; Reynolds, J. N. J., Hyland, B. I., & Wickens, J. R. (2001). A cellular mechanism of reward-related learning. *Nature*, 413, 67–70; Graybiel, A. M. (2008). Habits, rituals, and the evaluative brain. *Annual Review of Neuroscience*, 31, 359–387; Yin, H. H., & Knowlton, B. J. (2006). The role of the basal ganglia in habit formation. *Nature Reviews Neuroscience*, 7(6), 464–476.

[203] Azevedo, F. A. C., Carvalho, L. R. B., Grinberg, L. T., Farfel, J. M., Ferretti, R. E. L., Leite, R. E. P., Jacob Filho, W., Lent, R., & Herculano-Houzel, S. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5), 532–541; Eccles, J. C., Ito, M., & Szentágothai, J. (1967). *The Cerebellum as a Neuronal Machine*. Springer. Azevedo and colleagues count roughly 69 of the brain's 86 billion neurons in the cerebellum.

[204] Napper, R. M. A., & Harvey, R. J. (1988). Number of parallel fiber synapses on an individual Purkinje cell in the cerebellum of the rat. *Journal of Comparative Neurology*, 274(2), 168–177; Eccles, J. C., Ito, M., & Szentágothai, J. (1967). *The Cerebellum as a Neuronal Machine*. Springer.

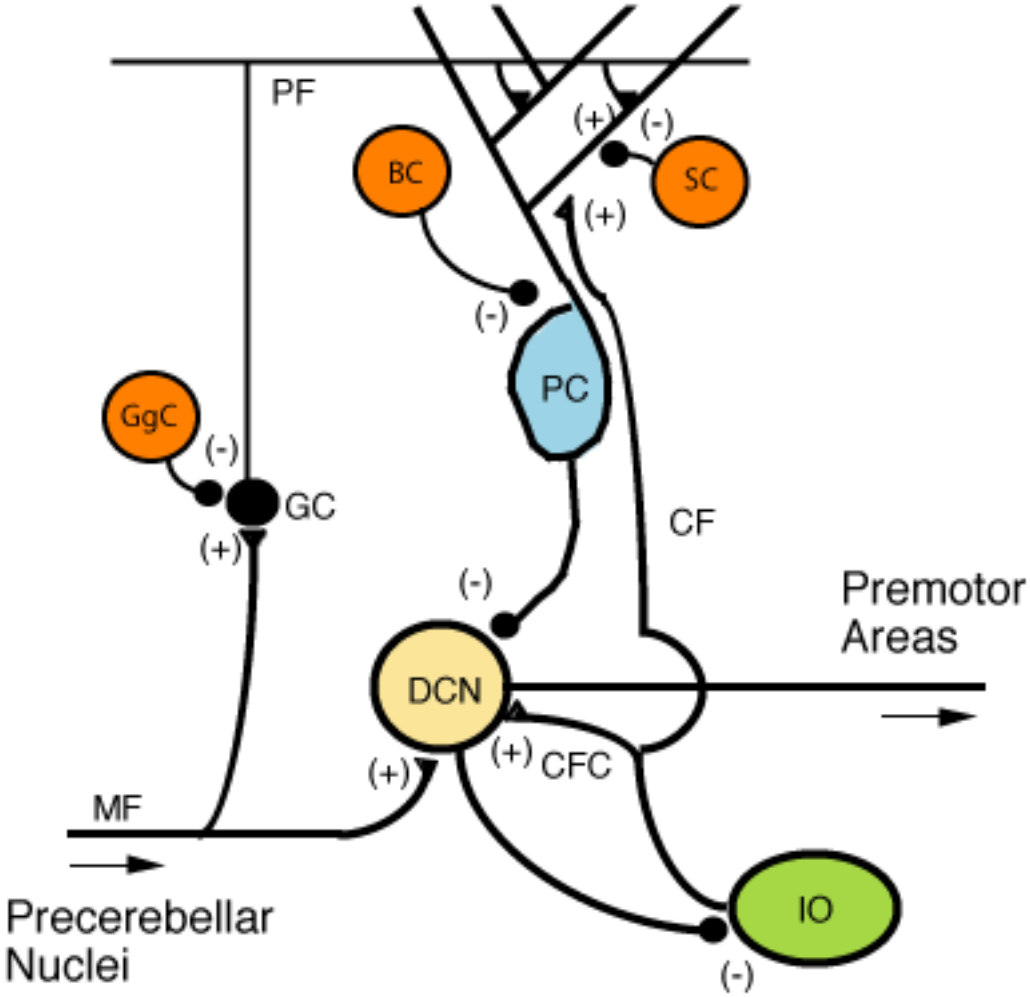


Figure 4.9. The cerebellar microcircuit. Mossy fibers (MF) and parallel fibers (PF) carry context to the Purkinje cell (PC); a single climbing fiber (CF) from the inferior olive (IO) carries an error signal. Purkinje cells inhibit the deep cerebellar nuclei (DCN), the cerebellum's output.

In the classical theory of Marr, Albus, and Ito, the climbing fiber is a teaching signal. When it fires, the parallel fiber synapses that were recently active are weakened, which is an eligibility trace mechanism of the kind Chapter 3 described, implemented in a circuit so regular that it was one of the first to be modeled mathematically; Ito demonstrated the weakening, cerebellar long-term depression, in 1982.[205] The full picture is richer. Mice in which this depression was blocked still learned several cerebellar tasks, including the adjustment of the reflex that keeps the eyes on a target while the head turns and the conditioning of an eyeblink to a tone, so plasticity at other sites, at the Purkinje cell's other synapses, among the interneurons of the molecular layer, and in the deep nuclei, must share the work; and climbing fibers have since been found to fire not only for errors but for expected reward and for learned

[205] Marr, D. (1969). A theory of cerebellar cortex. *Journal of Physiology*, 202(2), 437–470; Albus, J. S. (1971). A theory of cerebellar function. *Mathematical Biosciences*, 10(1–2), 25–61; Ito, M., Sakurai, M., & Tongroach, P. (1982). Climbing fibre induced depression of both mossy fibre responsiveness and glutamate sensitivity of cerebellar Purkinje cells. *Journal of Physiology*, 324, 113–134; Ito, M. (2001). Cerebellar long-term depression: Characterization, signal transduction, and functional roles. *Physiological Reviews*, 81(3), 1143–1195.

predictions of what a movement will produce.[206] What survives is the division of labor: a massive context input, a sparse instructive input, and synapses that change when the two coincide. Over repetitions, the cerebellum learns to predict the sensory consequences of a command and to issue corrections before errors occur.[207] Reaching, balance, eye movements, and the timing of speech all depend on it. Its lesions do not abolish movement; they make movement clumsy.[208]

The Hippocampal Loop

The hippocampus is the circuit most closely identified with memory, and its anatomy, shown in Figure 4.10, is a chain of the motifs already described.[209]

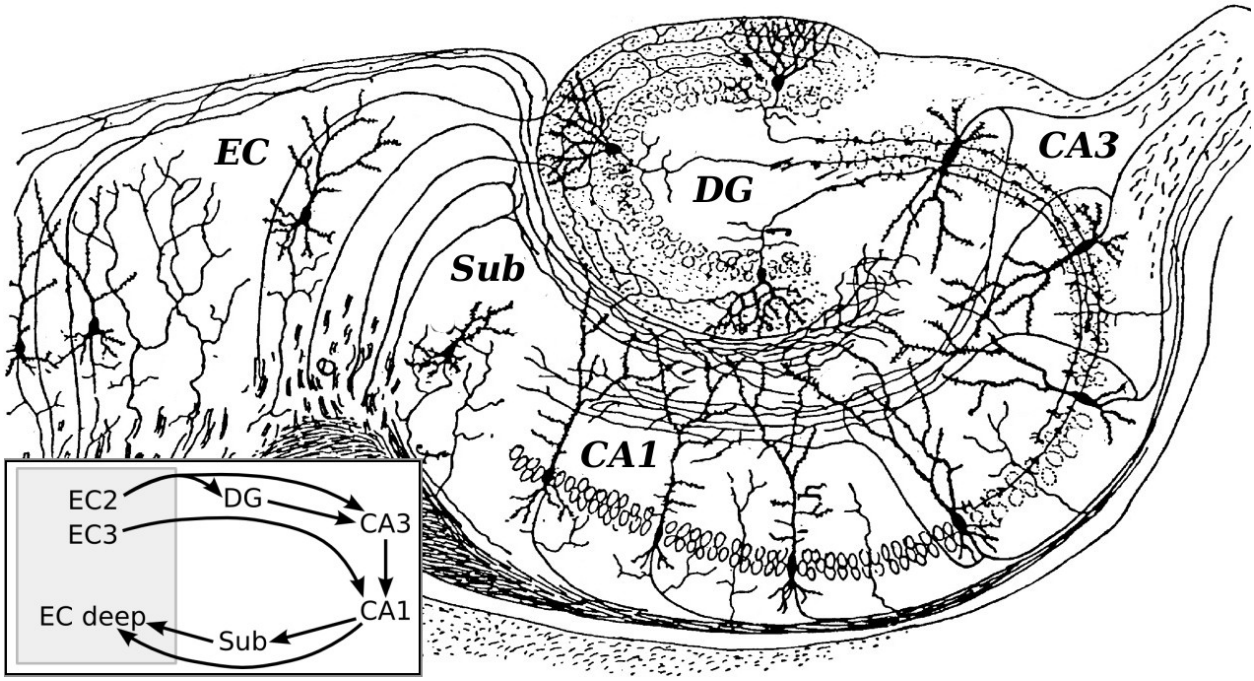


Figure 4.10. Cajal's drawing of the hippocampal formation, with a modern summary of its main pathway: entorhinal cortex (EC) to dentate gyrus (DG) to CA3 to CA1 and back to entorhinal cortex.

Cortical input enters through the entorhinal cortex and passes to the dentate gyrus, whose sparse, strongly inhibited granule cells make similar inputs less similar, a process called pattern separation.[210] The dentate projects to CA3, whose pyramidal cells are densely connected to one another: recurrent

[206] Schonewille, M., Gao, Z., Boele, H.-J., Vinueza Veloz, M. F., Amerika, W. E., Šimek, A. A. M., De Jeu, M. T., Steinberg, J. P., Takamiya, K., Hoebeek, F. E., Linden, D. J., Huganir, R. L., & De Zeeuw, C. I. (2011). Reevaluating the role of LTD in cerebellar motor learning. *Neuron*, 70(1), 43–50; Heffley, W., Song, E. Y., Xu, Z., Taylor, B. N., Hughes, M. A., McKinney, A., Joshua, M., & Hull, C. (2018). Coordinated cerebellar climbing fiber activity signals learned sensorimotor predictions. *Nature Neuroscience*, 21(10), 1431–1441; Kostadinov, D., Beau, M., Blanco-Pozo, M., & Häusser, M. (2019). Predictive and reactive reward signals conveyed by climbing fiber inputs to cerebellar Purkinje cells. *Nature Neuroscience*, 22(6), 950–962. Schonewille and colleagues blocked parallel fiber long-term depression in three mouse lines and found vestibulo-ocular reflex adaptation and eyeblink conditioning intact.

[207] Wolpert, D. M., Miall, R. C., & Kawato, M. (1998). Internal models in the cerebellum. *Trends in Cognitive Sciences*, 2(9), 338–347.

[208] Holmes, G. (1939). The cerebellum of man. *Brain*, 62(1), 1–30.

[209] Andersen, P., Bliss, T. V. P., & Skrede, K. K. (1971). Lamellar organization of hippocampal excitatory pathways. *Experimental Brain Research*, 13(2), 222–238; Amaral, D. G., & Witter, M. P. (1989). The three-dimensional organization of the hippocampal formation: A review of anatomical data. *Neuroscience*, 31(3), 571–591.

excitation on a large scale. A partial cue that activates some CA3 cells can regenerate the whole pattern, which is pattern completion.[211] CA3 projects to CA1, which on one account compares the retrieved pattern with the current input, delivered to CA1 directly from the entorhinal cortex, and returns the result to the cortex; the comparator role is a hypothesis supported by modeling and slice physiology rather than an established function.[212]

Separation, storage, completion, and comparison, in that order, each implemented by a motif from Figure 4.1. The circuit is described here only in outline; Chapter 5 returns to how its synapses change.

Rhythms

If circuits are the brain's structure, rhythms are its timing.

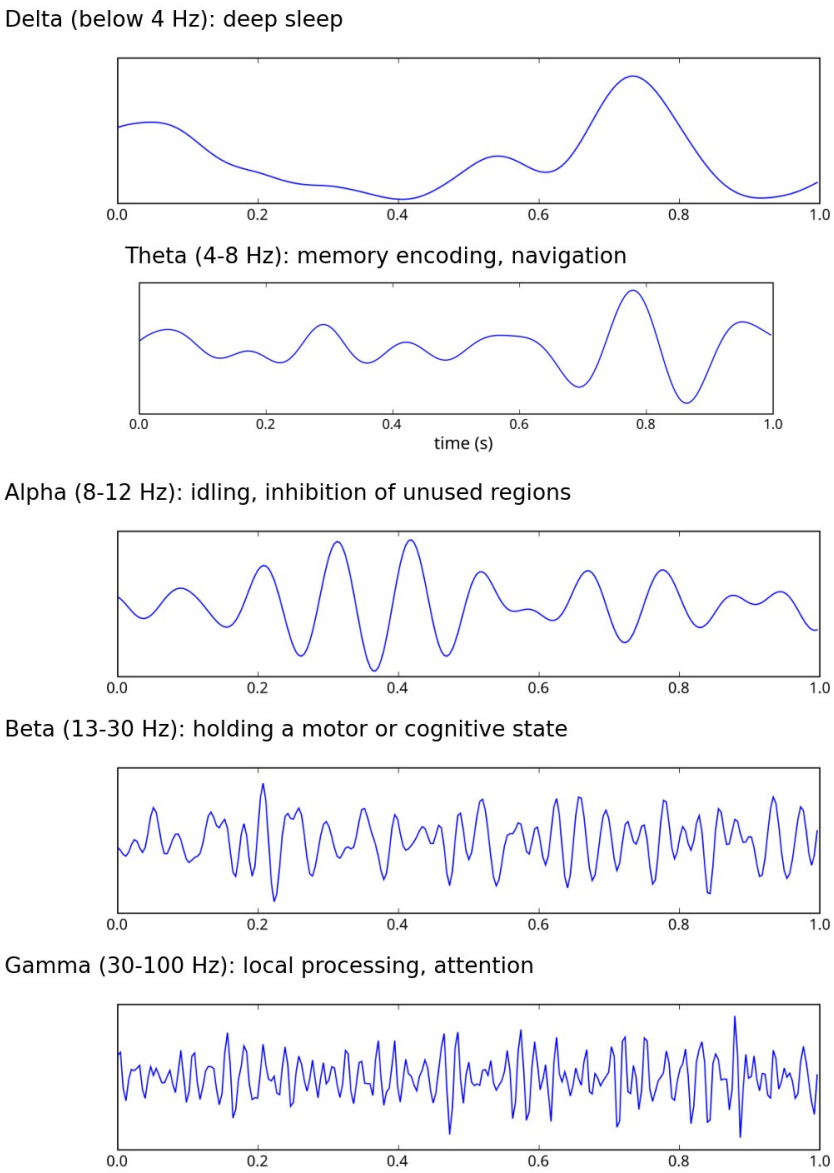
Record the voltage at the scalp, the electroencephalogram or EEG, and you see oscillations, conventionally sorted into bands named by Greek letters: delta or slow waves (below 4 Hz) during deep sleep, theta (4 to 8 Hz) during exploration and memory tasks, alpha (8 to 12 Hz) over idle cortex, beta (13 to 30 Hz) while a posture or a plan is held, and gamma (30 to 100 Hz) wherever local processing is intense. Figure 4.11 shows one second of each.[213]

[210] Leutgeb, J. K., Leutgeb, S., Moser, M.-B., & Moser, E. I. (2007). Pattern separation in the dentate gyrus and CA3 of the hippocampus. *Science*, 315(5814), 961–966.

[211] Marr, D. (1971). Simple memory: A theory for archicortex. *Philosophical Transactions of the Royal Society B*, 262(841), 23–81; Treves, A., & Rolls, E. T. (1994). Computational analysis of the role of the hippocampus in memory. *Hippocampus*, 4(3), 374–391; Nakazawa, K., Quirk, M. C., Chitwood, R. A., Watanabe, M., Yeckel, M. F., Sun, L. D., Kato, A., Carr, C. A., Johnston, D., Wilson, M. A., & Tonegawa, S. (2002). Requirement for hippocampal CA3 NMDA receptors in associative memory recall. *Science*, 297(5579), 211–218.

[212] Hasselmo, M. E., & Schnell, E. (1994). Laminar selectivity of the cholinergic suppression of synaptic transmission in rat hippocampal region CA1: Computational modeling and brain slice physiology. *Journal of Neuroscience*, 14(6), 3898–3914; Lisman, J. E., & Grace, A. A. (2005). The hippocampal-VTA loop: Controlling the entry of information into long-term memory. *Neuron*, 46(5), 703–713.

[213] Buzsáki, G. (2006). *Rhythms of the Brain*. Oxford University Press; Buzsáki, G., & Draguhn, A. (2004). Neuronal oscillations in cortical networks. *Science*, 304(5679), 1926–1929.



One second of scalp EEG filtered into five frequency bands. Traces: Hugo Gamboa, Wikimedia Commons, CC BY-SA 3.0.

Figure 4.11. One second of scalp EEG filtered into five frequency bands. Each band is associated with a characteristic behavioral state, though the associations are statistical rather than exclusive.

Oscillations are the collective behavior of circuits built from the motifs above. Feedback inhibition is intrinsically rhythmic: excitation recruits inhibition, inhibition silences excitation, inhibition decays, excitation recovers. A population of excitatory cells coupled to a population of fast interneurons oscillates in the gamma range for exactly this reason, and a network of interneurons inhibiting one another can do the same on its own.[214]

[214] Whittington, M. A., Traub, R. D., & Jefferys, J. G. R. (1995). Synchronized oscillations in interneuron networks driven by metabotropic glutamate receptor activation. *Nature*, 373, 612–615; Buzsáki, G., & Wang, X.-J. (2012). Mechanisms of gamma oscillations. *Annual Review of Neuroscience*, 35, 203–225; Cardin, J. A., Carlén, M., Meletis, K., Knoblich, U., Zhang, F., Deisseroth, K., Tsai, L.-H., & Moore, C. I. (2009). Driving fast-spiking cells induces gamma rhythm and controls sensory responses. *Nature*, 459, 663–667. Whittington and colleagues describe rhythm generated among interneurons alone; Cardin and colleagues, and Buzsáki and Wang, the loop between excitatory cells and interneurons.

Rhythms do real work. A neuron is most excitable at one phase of the oscillation and least at the opposite phase, so a shared rhythm creates windows during which inputs are effective. It has been proposed that this is how the brain routes information through fixed wiring: two regions whose oscillations are aligned so that one's spikes arrive in the other's excitable phase would communicate well, two regions out of alignment would communicate poorly, and changing the phase relation would change the effective wiring without changing a synapse. This is the communication-through-coherence hypothesis, and it is the brain's candidate answer to a problem every large network has.[215] The evidence is mixed. Coherence between areas does rise with attention, as the hypothesis predicts, but modeling and recordings from visual cortex show that coherence between two areas can arise simply from their connectivity and the sender's oscillatory power, with no phase-dependent gating at all, and the frequency of gamma shifts with the contrast and size of a stimulus, which makes it an unreliable clock for aligning distant areas.[216] Whether rhythm routes information or merely reflects it is an open question.

Rhythms also coordinate learning. In the hippocampus, place cells, neurons that each fire when the animal is at one particular location, fire at progressively earlier phases of the theta cycle as the animal crosses that location, compressing a sequence of seconds into a sequence of tens of milliseconds. That compressed sequence falls within the window in which synapses can change. The rhythm converts experience into a format plasticity can use.[217]

The Architecture of the Whole

Step back from any single region and the brain is a network of regions, which its anatomists now call the connectome.[218] Its wiring is neither random nor fully regular, and the compromise it strikes, shown in Figure 4.12, is the same one found in power grids, social networks, and the internet.

[215] Fries, P. (2005). A mechanism for cognitive dynamics: Neuronal communication through neuronal coherence. *Trends in Cognitive Sciences*, 9(10), 474–480; Fries, P. (2015). Rhythms for cognition: Communication through coherence. *Neuron*, 88(1), 220–235.

[216] Schneider, M., Broggini, A. C., Dann, B., Tzanou, A., Uran, C., Sheshadri, S., Scherberger, H., & Vinck, M. (2021). A mechanism for inter-areal coherence through communication based on connectivity and oscillatory power. *Neuron*, 109(24), 4050–4067; Ray, S., & Maunsell, J. H. R. (2015). Do gamma oscillations play a role in cerebral cortex? *Trends in Cognitive Sciences*, 19(2), 78–85.

[217] O'Keefe, J., & Dostrovsky, J. (1971). The hippocampus as a spatial map: Preliminary evidence from unit activity in the freely-moving rat. *Brain Research*, 34(1), 171–175; O'Keefe, J., & Recce, M. L. (1993). Phase relationship between hippocampal place units and the EEG theta rhythm. *Hippocampus*, 3(3), 317–330; Skaggs, W. E., McNaughton, B. L., Wilson, M. A., & Barnes, C. A. (1996). Theta phase precession in hippocampal neuronal populations and the compression of temporal sequences. *Hippocampus*, 6(2), 149–172.

[218] Sporns, O., Tononi, G., & Kötter, R. (2005). The human connectome: A structural description of the human brain. *PLoS Computational Biology*, 1(4), e42; Bullmore, E., & Sporns, O. (2009). Complex brain networks: Graph theoretical analysis of structural and functional systems. *Nature Reviews Neuroscience*, 10(3), 186–198. The word connectome was introduced by Sporns, Tononi, and Kötter.

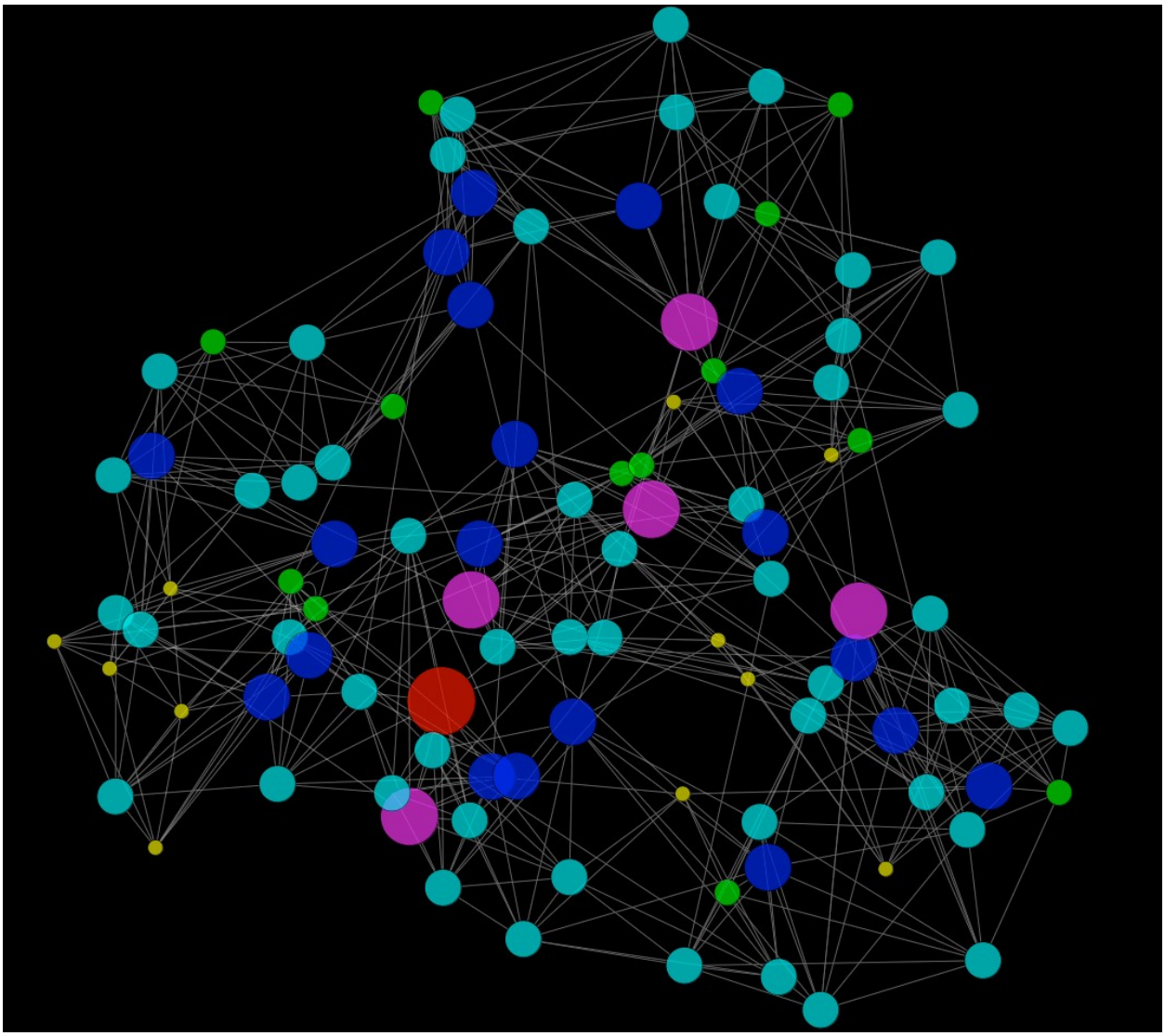


Figure 4.12. A small-world network: most connections are local, a few are long-range. The result is dense local clustering with short paths between any two nodes.

Most connections are short, linking neighbors into dense clusters. A minority are long, linking clusters to one another. The result is a small-world network: any two regions are only a few connections apart, yet almost all wiring is local and cheap. The same structure has been shown neuron by neuron in the nematode worm, the one animal whose complete wiring diagram was known when the idea was proposed; at the level of single neurons the mammalian brain is an extrapolation. Long axons are metabolically expensive, occupy volume, and slow signals, so the brain uses as few of them as it can while keeping the path length between regions short.[219] Figure 4.13 shows those long axons in a living human brain, reconstructed by tracing the direction in which water diffuses along the fiber bundles.

[219] Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of ‘small-world’ networks. *Nature*, 393, 440–442; Bassett, D. S., & Bullmore, E. (2006). Small-world brain networks. *The Neuroscientist*, 12(6), 512–523; Chklovskii, D. B., Schikorski, T., & Stevens, C. F. (2002). Wiring optimization in cortical circuits. *Neuron*, 34(3), 341–347; Bullmore, E., & Sporns, O. (2012). The economy of brain network organization. *Nature Reviews Neuroscience*, 13(5), 336–349.

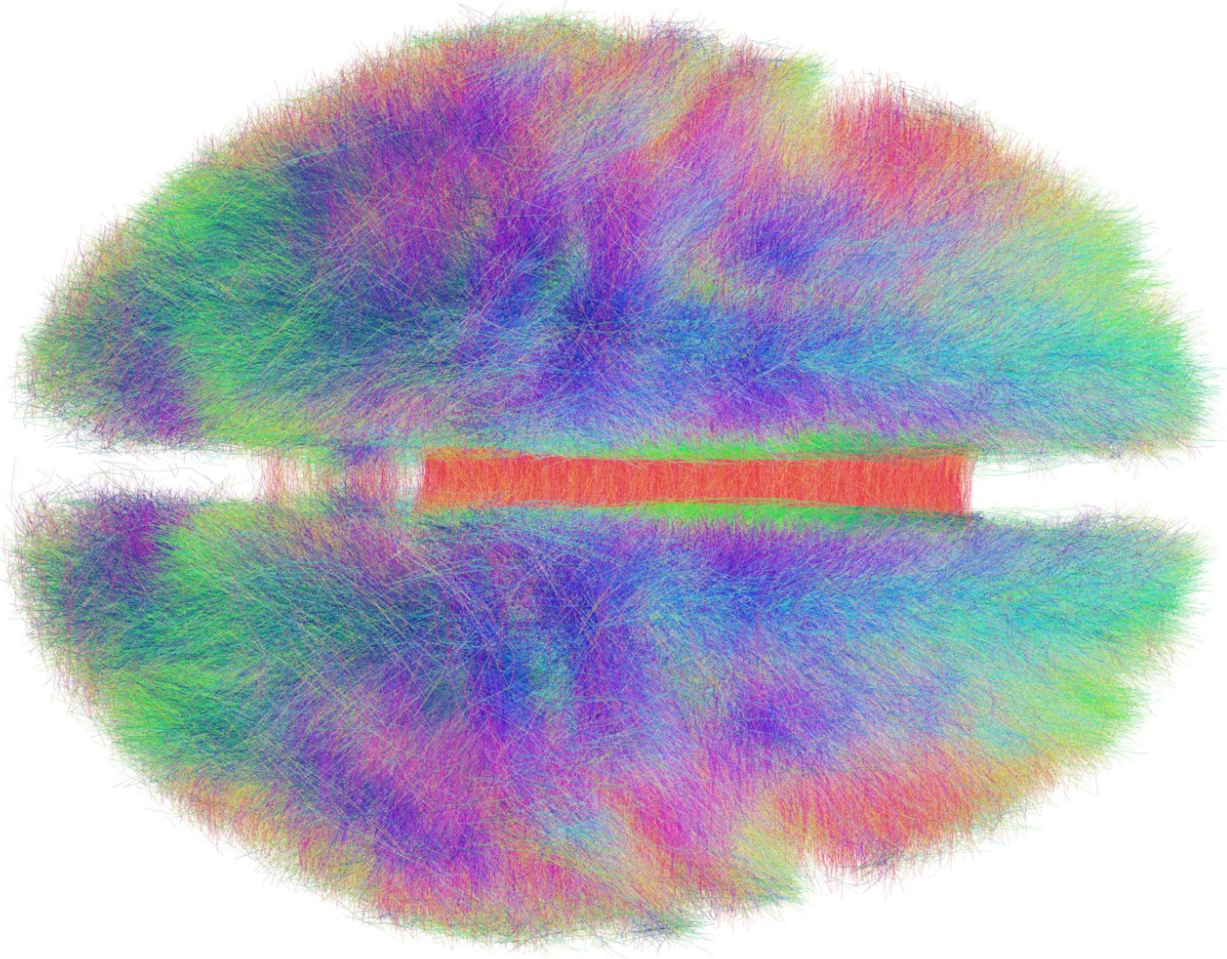


Figure 4.13. Long-range fiber tracts of the human brain reconstructed from diffusion MRI, viewed from above. Colors indicate fiber direction.

A few regions are hubs with far more long-range connections than average, and these hubs are disproportionately connected to one another, forming a core through which most long-distance traffic passes.[220] The arrangement is efficient and fragile in the same way an airline network is: it makes the system fast, and it makes damage to a hub far more consequential than damage elsewhere. Several neurological and psychiatric conditions, schizophrenia and Alzheimer's disease among them, are now described, at least in part, as disorders of the hubs: across 26 disorders, the sites of gray matter loss fell disproportionately on hub regions.[221]

Coordination Without a Conductor

Chapter 1 claimed that the brain has no central controller. This chapter has described the machinery that makes a controller unnecessary.

[220] van den Heuvel, M. P., & Sporns, O. (2011). Rich-club organization of the human connectome. *Journal of Neuroscience*, 31(44), 15775–15786.

[221] Crossley, N. A., Mechelli, A., Scott, J., Carletti, F., Fox, P. T., McGuire, P., & Bullmore, E. T. (2014). The hubs of the human connectome are generally implicated in the anatomy of brain disorders. *Brain*, 137(8), 2382–2395.

Reflexes handle the fastest demands locally. Pattern generators produce rhythmic behavior on their own and accept correction from the body. Balanced inhibition keeps every circuit sensitive without letting it run away. The basal ganglia release one action at a time from a default of silence. The cerebellum corrects movements before errors are felt. Rhythms may decide which regions can talk to which, moment by moment. A small-world wiring plan keeps all of this within a few connections of everything else.

No component sees the whole. Each solves a local problem using the same small vocabulary of motifs, and the behavior of the animal is what those local solutions add up to. The pattern is meaningful, as Sherrington said, and never abiding: it is re-created from moment to moment by circuits that were never designed to produce it.[222]

Implications for Artificial Intelligence

Artificial neural networks borrowed the neuron and left the circuit behind. Several of this chapter's lessons bear directly on how that might be repaired.

First, inhibition is not a negative weight. In most artificial networks a weight can be positive or negative, so the same unit excites some targets and inhibits others, and nothing enforces balance; a biological neuron is one or the other, a rule known as Dale's principle. Biological circuits maintain excitation and inhibition as separate populations and keep them in near-cancellation, which buys speed, stability, and sensitivity at once. Networks built to respect the division are a largely unexplored design space, and the first attempts to train them show that, with the learning rule adjusted, they can match unconstrained networks.[223]

Second, selection by disinhibition is a different control strategy from choosing the largest output. The basal ganglia hold all actions off by default and release one. The default of silence makes "do nothing" a first-class action, which is often what an agent in an uncertain environment should do, and it gives the circuit two recognizable failure modes, too little and too much, rather than a diffuse degradation.

Third, rhythm may be a routing mechanism. Fixed weights cannot change which parts of a network talk to each other, so artificial systems add attention mechanisms to do it (Chapter 6 returns to attention).[224] The brain, on the coherence hypothesis, does it with phase. Oscillatory gating offers a way to reconfigure effective connectivity without changing parameters, at low cost.

Fourth, the brain's wiring budget shapes its architecture. Small-world organization, local clustering, and sparse long-range connections are consequences of paying for every axon. Artificial networks pay far less per connection than an axon costs, in parameters, memory, and arithmetic rather than in volume and metabolism, and they are correspondingly dense; the literature on pruning and sparsity exists because even that smaller cost adds up.

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[222] Sherrington, C. S. (1940). *Man on His Nature*. Cambridge University Press.

[223] Cornford, J., Kalajdzievski, D., Leite, M., Lamarquette, A., Kullmann, D. M., & Richards, B. A. (2021). Learning to live with Dale's principle: ANNs with separate excitatory and inhibitory units. *International Conference on Learning Representations (ICLR)*.

[224] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.

Finally, the cortex suggests that general computation can come from a single repeated circuit with different inputs. The question of what that circuit is doing, in a form abstract enough to transfer, remains open. It is one of the most important open problems in both neuroscience and machine learning.

The Central Lesson

Behavior belongs to circuits, not cells. The nervous system builds its circuits from a small vocabulary of motifs in which inhibition does most of the shaping, stacks them into regions that each solve one kind of problem, and coordinates the regions through rhythm and a wiring plan that keeps everything close. No part is in charge. Coordinated behavior is what the parts produce together.

Next Chapter

Chapter 5: Plasticity and Memory: How Experience Becomes Structure

Figure and Epigraph Sources

Figure Attribution

Figures 4.1, 4.2, 4.4, 4.5, and 4.8 were drawn for this book. Figure 4.2, the lower panel of Figure 4.4, and the output trace in Figure 4.5 are illustrative simulations of simple rate models, not recordings.

Figure 4.3: Santiago Ramón y Cajal, structure of the mammalian retina (1900), public domain, via Wikimedia Commons.

Figure 4.6: Santiago Ramón y Cajal, drawings of the cerebral cortex (1899), public domain, via Wikimedia Commons (“Cajal cortex drawings”): Nissl-stained adult human visual cortex, Nissl-stained adult human motor cortex, and Golgi-stained cortex of a one-and-a-half-month-old infant.

Figure 4.7: “Cortical Columns,” underlying image from Marcel Oberländer, Beyond the Cortical Column, Neuroinformatics 2012; composite by Wikimedia Commons user Bodysurfinyon, CC BY-SA 4.0.

Figure 4.9: “CerebCircuit” by Wikipedia user Nrets, CC BY-SA 3.0, via Wikimedia Commons.

Figure 4.10: Santiago Ramón y Cajal, drawing of the hippocampus (1911), public domain, with circuit inset added by Wikimedia Commons contributors.

Figure 4.11: EEG band traces (“Eeg delta”, “Eeg theta”, “Eeg alpha”, “Eeg beta”, “Eeg gamma”) by Hugo Gamboa, CC BY-SA 3.0, via Wikimedia Commons; composite drawn for this book.

Figure 4.12: “Watts-Strogatz small-world model, 100 nodes” by Wikimedia Commons user Keiono, CC BY-SA 3.0.

Figure 4.13: “The Human Connectome” by Wikimedia Commons user Andreashorn, CC BY-SA 4.0.

Epigraph

Charles Sherrington, *Man on His Nature* (Cambridge University Press, 1940), the Gifford Lectures of 1937–38.

Chapter 5

Plasticity and Memory: How Experience Becomes Structure

> *"When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased."*
>
> *— Donald Hebb, The Organization of Behavior (1949)*

What Kind of Storage Memory Is

Every chapter since the primer of Chapter 0 has ended at the same door. Chapter 1 said that the brain has no reset button because every experience leaves a trace. Chapter 2 said that a neuron cannot hold a variable, so learning must be physical change. Chapter 3 said that neuromodulators decide whether much of that change is kept. Chapter 4 said that the hippocampus separates, stores, completes, and compares, and deferred the question of how its synapses change. This chapter walks through the door.

The first thing to leave behind is the computer's vocabulary of storage. A computer writes a value to an address and reads it back unchanged. The brain does store, in the plain sense that experience changes its tissue and the change persists, but the storage has none of those properties. There is no address: a memory is spread across many synapses in many regions, and no one of them holds it. There is no write operation that leaves the rest of the machinery untouched, and no read that leaves the record as it found it. What there is instead is a system whose physical structure is altered by use, so that the next time similar activity arrives it unfolds differently. Storage in the brain is physical, distributed, and reconstructive, and the rest of this chapter is an account of what each of those three words means. Memory is not something the brain has. It is something the brain's tissue has become.

This reframing is not a philosophical nicety. It predicts most of what follows: why learning is slow, why memories blur into one another, why recalling a memory can change it, why sleep matters, and why the brain spends energy to forget.

What Can Change

Hebb's postulate, quoted above, is the founding statement of the field, and it is deliberately vague about mechanism: "some growth process or metabolic change."[225] Three quarters of a century later the list of things that can change is long. The important ones fall into four groups.

The strength of an existing synapse can change. On the presynaptic side, the probability that an action potential releases neurotransmitter can rise or fall. On the postsynaptic side, the number of receptors in the membrane can be increased or reduced within minutes, by inserting or removing receptor proteins from a reserve just beneath the surface. A synapse that doubles its receptors roughly doubles its effect.[226]

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[225] Hebb, D. O. (1949). *The Organization of Behavior: A Neuropsychological Theory*. Wiley. The postulate appears on p. 62.

[226] Malinow, R., & Malenka, R. C. (2002). AMPA receptor trafficking and synaptic plasticity. *Annual Review of Neuroscience*, 25, 103–126; Malenka, R. C., & Bear, M. F. (2004). LTP and LTD: An embarrassment of riches. *Neuron*, 44(1), 5–21.

The structure of the synapse can change. Most excitatory synapses on cortical neurons sit on dendritic spines, small protrusions a micrometer long. Spines grow, shrink, appear, and vanish. A spine that enlarges carries a stronger synapse; a spine that disappears takes its synapse with it. Imaging the same neurons over weeks shows a population of spines in constant slow turnover, with a stable core that persists for the life of the animal and a changeable fringe that reflects recent experience.[227]

The neuron's own excitability can change. The density and type of ion channels in the membrane are adjusted by activity, so that the same synaptic input produces more or fewer spikes. This is plasticity of the cell rather than of the connection, and it is less specific but often faster.[228]

Finally, new synapses can form between neurons that were not connected, and in a few regions, chiefly the dentate gyrus of the hippocampus, new neurons are born throughout life, at least in rodents.[229] Whether the adult human dentate gyrus does the same is disputed. One 2018 study of postmortem tissue found new neurons undetectable after childhood, while two others, in 2018 and 2019, found them persisting into old age, and the disagreement turns on how the tissue was fixed and which molecular markers count as proof of a young neuron.[230] These are the slowest changes and the ones most clearly tied to structural growth.

All four share a property that distinguishes them from writing to memory: they are modifications of machinery that is also doing something else. The synapse that is strengthened is still transmitting. The spine that grows is still part of a dendrite that integrates thousands of inputs. There is no separate medium.

Hebb's Rule and Its Problem

Hebb's postulate says that cells that fire together wire together. The strength of a synapse should increase when the presynaptic and postsynaptic neurons are active at the same time.

This rule is attractive because it is local. Each synapse needs only information available at the synapse itself, the activity of the two cells it joins, and no global error signal. It also has an immediate problem, which the first computational models exposed in the 1970s and early 1980s: it only goes up.[231] A synapse strengthened by correlated activity makes that correlation more likely, which strengthens it further.

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[227] Grutzendler, J., Kasthuri, N., & Gan, W.-B. (2002). Long-term dendritic spine stability in the adult cortex. *Nature*, 420, 812–816; Holtmaat, A., & Svoboda, K. (2009). Experience-dependent structural synaptic plasticity in the mammalian brain. *Nature Reviews Neuroscience*, 10(9), 647–658; Yang, G., Pan, F., & Gan, W.-B. (2009). Stably maintained dendritic spines are associated with lifelong memories. *Nature*, 462, 920–924.

[228] Zhang, W., & Linden, D. J. (2003). The other side of the engram: Experience-driven changes in neuronal intrinsic excitability. *Nature Reviews Neuroscience*, 4(11), 885–900.

[229] Altman, J., & Das, G. D. (1965). Autoradiographic and histological evidence of postnatal hippocampal neurogenesis in rats. *Journal of Comparative Neurology*, 124(3), 319–335; Eriksson, P. S., Perfilieva, E., Björk-Eriksson, T., Alborn, A.-M., Nordborg, C., Peterson, D. A., & Gage, F. H. (1998). Neurogenesis in the adult human hippocampus. *Nature Medicine*, 4(11), 1313–1317.

[230] Sorrells, S. F., Paredes, M. F., Cebrian-Silla, A., Sandoval, K., Qi, D., Kelley, K. W., James, D., Mayer, S., Chang, J., Auguste, K. I., Chang, E. F., Gutierrez, A. J., Kriegstein, A. R., Mathern, G. W., Oldham, M. C., Huang, E. J., Garcia-Verdugo, J. M., Yang, Z., & Alvarez-Buylla, A. (2018). Human hippocampal neurogenesis drops sharply in children to undetectable levels in adults. *Nature*, 555, 377–381.; Boldrini, M., Fulmore, C. A., Tartt, A. N., Simeon, L. R., Pavlova, I., Poposka, V., Rosoklija, G. B., Stankov, A., Arango, V., Dwork, A. J., Hen, R., & Mann, J. J. (2018). Human hippocampal neurogenesis persists throughout aging. *Cell Stem Cell*, 22(4), 589–599.; Moreno-Jiménez, E. P., Flor-García, M., Terreros-Roncal, J., Rábano, A., Cafini, F., Pallas-Bazarra, N., Ávila, J., & Llorens-Martín, M. (2019). Adult hippocampal neurogenesis is abundant in neurologically healthy subjects and drops sharply in patients with Alzheimer's disease. *Nature Medicine*, 25(4), 554–560. Sorrells and colleagues found no young neurons in adult dentate gyrus; Boldrini and colleagues and Moreno-Jiménez and colleagues, using differently fixed tissue, found them into the ninth decade.

Left alone, Hebbian learning drives every active synapse to its ceiling and erases whatever selectivity it began with.

A working learning rule therefore needs three things Hebb did not specify: a way to weaken synapses, a bound on how strong they can become, and a criterion for which correlations count. The biology supplies all three, and the rest of this chapter is largely an account of how.

The Coincidence Detector

The mechanism that implements Hebb's "fire together," for most forms of LTP, is a single receptor type, the NMDA receptor.

Glutamate, the brain's main excitatory transmitter, acts on two kinds of receptor, each named for the laboratory chemical that activates it selectively: the AMPA receptor and the NMDA receptor. Most fast excitatory transmission uses AMPA receptors: glutamate binds, the channel opens, sodium enters, the membrane depolarizes. The NMDA receptor also binds glutamate and also conducts, but at the resting membrane potential its pore is plugged by a magnesium ion from outside the cell (Figure 5.1).[232] Glutamate alone opens nothing. The plug is expelled only when the membrane is already depolarized, by many simultaneous AMPA inputs or by an action potential propagating back from the soma into the dendrite, the back-propagating spike of Chapter 2.[233]

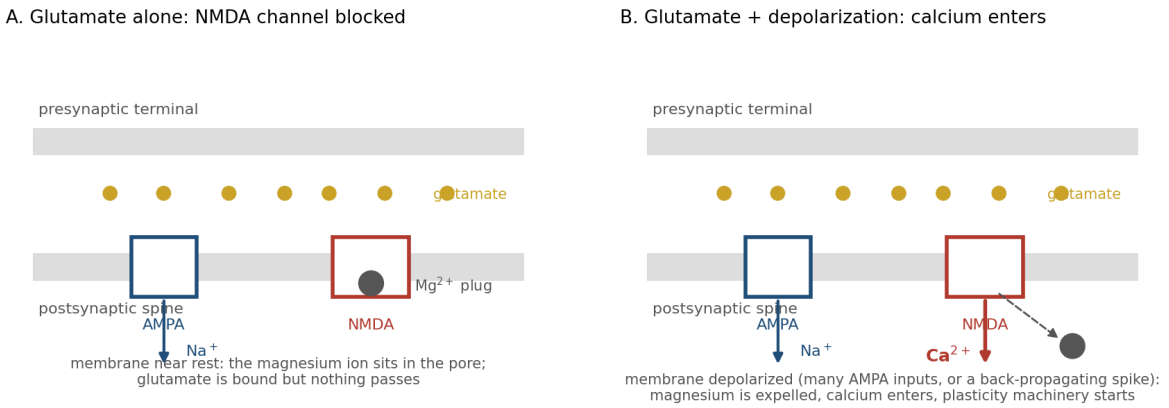


Figure 5.1. The NMDA receptor as a coincidence detector. In A, glutamate is bound but the magnesium plug holds; in B, depolarization has expelled the plug and calcium enters, so the calcium reports that presynaptic and postsynaptic activity coincided.

So the NMDA receptor conducts only when two conditions hold at once: the presynaptic neuron has released glutamate, and the postsynaptic neuron is depolarized. It is a molecular AND gate for the two

[231] von der Malsburg, C. (1973). Self-organization of orientation sensitive cells in the striate cortex. *Kybernetik*, 14(2), 85–100; Oja, E. (1982). A simplified neuron model as a principal component analyzer. *Journal of Mathematical Biology*, 15(3), 267–273. Oja's 1982 rule is the standard remedy: a normalizing term that bounds the total weight.

[232] Nowak, L., Bregestovski, P., Ascher, P., Herbet, A., & Prochiantz, A. (1984). Magnesium gates glutamate-activated channels in mouse central neurones. *Nature*, 307, 462–465; Mayer, M. L., Westbrook, G. L., & Guthrie, P. B. (1984). Voltage-dependent block by Mg2+ of NMDA responses in spinal cord neurones. *Nature*, 309, 261–263.

[233] Stuart, G. J., & Sakmann, B. (1994). Active propagation of somatic action potentials into neocortical pyramidal cell dendrites. *Nature*, 367, 69–72.

halves of Hebb's rule. What passes through it when it opens is calcium, and calcium is the signal that sets the plasticity machinery in motion.[234]

LTP and LTD

The two outcomes of the enzyme cascade that calcium sets off are long-term potentiation, a lasting increase in synaptic strength, and long-term depression, a lasting decrease. LTP was first measured in the hippocampus in 1973; LTD was established in the cerebellum in 1982 and convincingly in the hippocampus only in 1992, two decades after LTP. Both are now known throughout the brain.[235]

What decides the direction is the calcium signal itself. A large, brief rise in calcium, the kind produced by strong input coinciding with postsynaptic spikes, activates kinases, which add phosphate groups to proteins and drive extra AMPA receptors into the synapse. A smaller, more prolonged rise, the kind produced by weak or poorly timed input, preferentially activates phosphatases, which remove phosphate groups and pull receptors out. The same ion in different amounts produces opposite results (Figure 5.2A).[236]

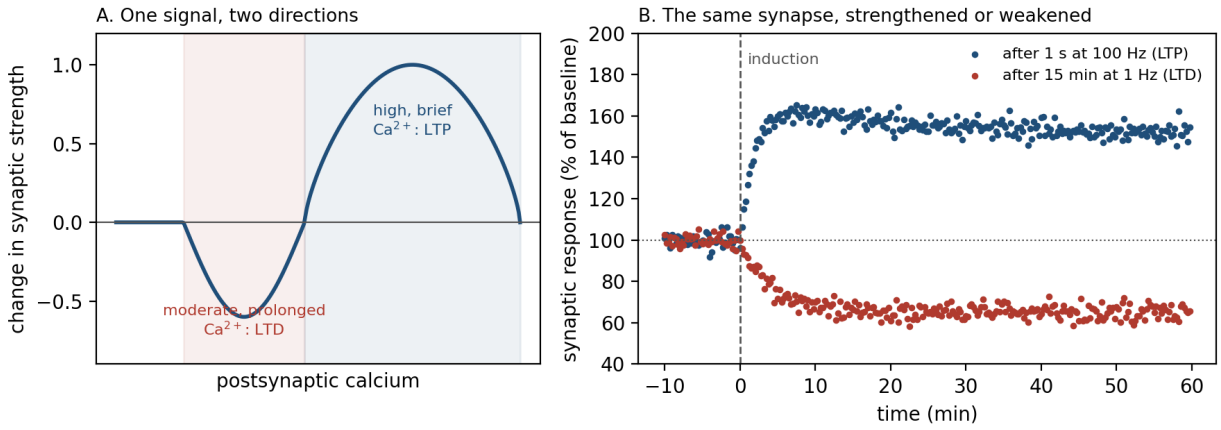


Figure 5.2. One signal, two directions. A: postsynaptic calcium determines the direction of change; modest, prolonged rises weaken a synapse, large brief rises strengthen it. B: simulated time course of a synapse's response after brief high-frequency stimulation (LTP) and after prolonged low-frequency stimulation (LTD).

[234] Lynch, G., Larson, J., Kelso, S., Barrionuevo, G., & Schottler, F. (1983). Intracellular injections of EGTA block induction of hippocampal long-term potentiation. *Nature*, 305, 719–721; Bliss, T. V. P., & Collingridge, G. L. (1993). A synaptic model of memory: Long-term potentiation in the hippocampus. *Nature*, 361, 31–39.

[235] Bliss, T. V. P., & Lømo, T. (1973). Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit following stimulation of the perforant path. *Journal of Physiology*, 232(2), 331–356; Ito, M., Sakurai, M., & Tongroach, P. (1982). Climbing fibre induced depression of both mossy fibre responsiveness and glutamate sensitivity of cerebellar Purkinje cells. *Journal of Physiology*, 324, 113–134; Dudek, S. M., & Bear, M. F. (1992). Homosynaptic long-term depression in area CA1 of hippocampus and effects of N-methyl-D-aspartate receptor blockade. *Proceedings of the National Academy of Sciences*, 89(10), 4363–4367; Mulkey, R. M., & Malenka, R. C. (1992). Mechanisms underlying induction of homosynaptic long-term depression in area CA1 of the hippocampus. *Neuron*, 9(5), 967–975. Long-term depression was first established in the cerebellum (Ito, Sakurai, and Tongroach, 1982); the hippocampal form was convincingly shown in 1992, two decades after LTP.

[236] Lisman, J. (1989). A mechanism for the Hebb and the anti-Hebb processes underlying learning and memory. *Proceedings of the National Academy of Sciences*, 86(23), 9574–9578; Artola, A., & Singer, W. (1993). Long-term depression of excitatory synaptic transmission and its relationship to long-term potentiation. *Trends in Neurosciences*, 16(11), 480–487; Malenka, R. C., & Bear, M. F. (2004). LTP and LTD: An embarrassment of riches. *Neuron*, 44(1), 5–21.

Experimentally, a second of stimulation at 100 Hz potentiates a synapse within minutes and the potentiation persists for hours in a slice of brain tissue kept alive in a dish and for weeks or longer in a living animal. Fifteen minutes at 1 Hz depresses the same synapse (Figure 5.2B).[237] Neither effect, in its induction and early phase, requires anything from outside the synapse and its immediate neighborhood; whether the change lasts is a separate question, to which the chapter returns. Hebb's growth process is real, and so is its opposite.

LTP has two phases. Early LTP, lasting an hour or two, uses proteins already present in the spine. Late LTP requires new protein synthesis and gene expression in the nucleus, and it is accompanied by visible enlargement of the spine. Block protein synthesis within an hour of learning and the memory forms but does not last.[238] This is the first of the chapter's two consolidation clocks.

Timing Is Causation

Hebb's phrase "takes part in firing" contains a subtlety that the rate-based LTP experiments missed. For cell A to take part in firing cell B, A must fire first.

Experiments in the late 1990s, pairing single presynaptic and postsynaptic spikes at controlled intervals, showed that the synapse respects this order. If the presynaptic spike precedes the postsynaptic spike by a few milliseconds to a few tens of milliseconds, the synapse strengthens. If the order is reversed, it weakens. The effect falls off exponentially with the interval and is negligible beyond about 50 ms in either direction (Figure 5.3).[239]

[237] Dudek, S. M., & Bear, M. F. (1992). Homosynaptic long-term depression in area CA1 of hippocampus and effects of N-methyl-D-aspartate receptor blockade. *Proceedings of the National Academy of Sciences*, 89(10), 4363–4367; Abraham, W. C., Logan, B., Greenwood, J. M., & Dragunow, M. (2002). Induction and experience-dependent consolidation of stable long-term potentiation lasting months in the hippocampus. *Journal of Neuroscience*, 22(21), 9626–9634. The standard low-frequency protocol is 900 pulses at 1 Hz; Abraham and colleagues followed potentiation in freely moving rats for months.

[238] Flexner, J. B., Flexner, L. B., & Stellar, E. (1963). Memory in mice as affected by intracerebral puromycin. *Science*, 141(3575), 57–59; Frey, U., Krug, M., Reymann, K. G., & Matthies, H. (1988). Anisomycin, an inhibitor of protein synthesis, blocks late phases of LTP phenomena in the hippocampal CA1 region in vitro. *Brain Research*, 452(1–2), 57–65; Kandel, E. R. (2001). The molecular biology of memory storage: A dialogue between genes and synapses. *Science*, 294(5544), 1030–1038; Matsuzaki, M., Honkura, N., Ellis-Davies, G. C. R., & Kasai, H. (2004). Structural basis of long-term potentiation in single dendritic spines. *Nature*, 429, 761–766.

[239] Markram, H., Lübke, J., Frotscher, M., & Sakmann, B. (1997). Regulation of synaptic efficacy by coincidence of postsynaptic APs and EPSPs. *Science*, 275(5297), 213–215; Magee, J. C., & Johnston, D. (1997). A synaptically controlled, associative signal for Hebbian plasticity in hippocampal neurons. *Science*, 275(5297), 209–213; Bi, G.-Q., & Poo, M.-M. (1998). Synaptic modifications in cultured hippocampal neurons: Dependence on spike timing, synaptic strength, and postsynaptic cell type. *Journal of Neuroscience*, 18(24), 10464–10472.

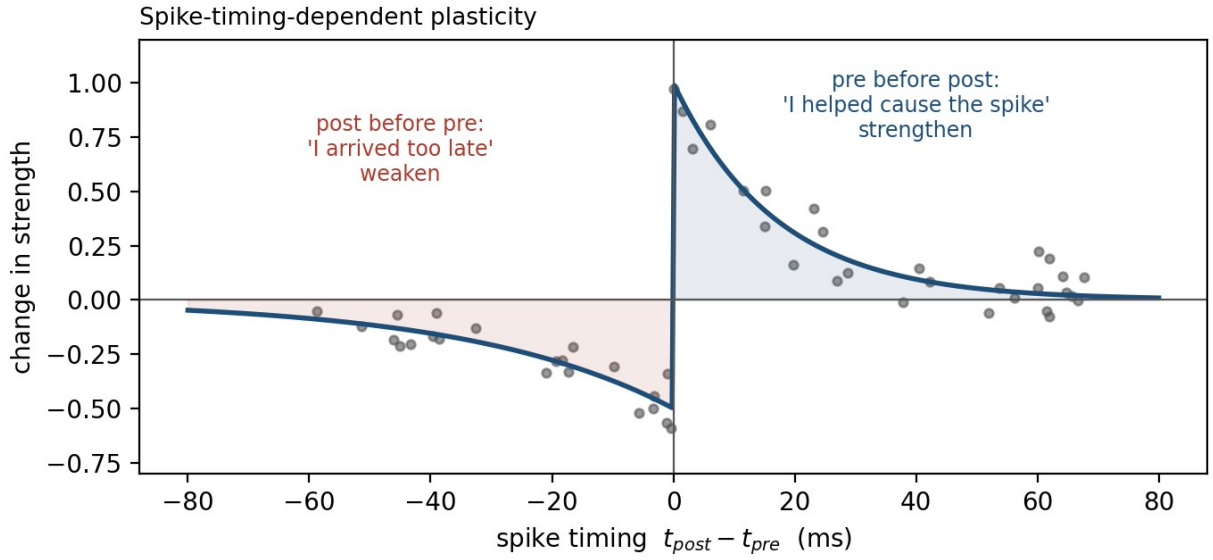


Figure 5.3. Spike-timing-dependent plasticity. A presynaptic spike shortly before a postsynaptic spike strengthens the synapse; the reverse order weakens it. Dots: simulated single-pairing measurements.

This is spike-timing-dependent plasticity, and its logic is causal. A synapse whose input arrived just before the neuron fired may have helped cause the spike, and it is rewarded. A synapse whose input arrived just after the spike could not have contributed, and it is penalized. The window is set by the kinetics of the NMDA receptor: a back-propagating spike that arrives while glutamate is still bound expels the magnesium and admits a large calcium pulse. A spike that arrives before the glutamate finds the receptor closed; the glutamate that follows then meets a membrane that has already repolarized and produces only a small, slow calcium signal.[240]

The timing rule does two things rate-based Hebbian learning cannot. It strengthens chains of neurons that fire in sequence, which is how a circuit can learn an ordered pattern. And it provides the weakening that Hebb's rule lacked, applied precisely to the connections that fire out of order.[241]

Keeping the Whole Neuron Stable

Local rules, even with weakening built in, do not by themselves keep a neuron's total input within a usable range. A neuron whose synapses have all been potentiated fires to everything; one whose synapses have all been depressed fires to nothing. Two further mechanisms, operating on the whole cell rather than the single synapse, prevent this.

The first adjusts the rule itself. The calcium threshold that separates LTD from LTP is not fixed; it slides with the neuron's recent average activity. After a period of intense firing the threshold rises, so that the same input now produces depression where before it produced potentiation. After a quiet period it falls.

[240] Caporale, N., & Dan, Y. (2008). Spike timing-dependent plasticity: A Hebbian learning rule. *Annual Review of Neuroscience*, 31, 25–46; Feldman, D. E. (2012). The spike-timing dependence of plasticity. *Neuron*, 75(4), 556–571.

[241] Gerstner, W., Kempter, R., van Hemmen, J. L., & Wagner, H. (1996). A neuronal learning rule for sub-millisecond temporal coding. *Nature*, 383, 76–81; Song, S., Miller, K. D., & Abbott, L. F. (2000). Competitive Hebbian learning through spike-timing-dependent synaptic plasticity. *Nature Neuroscience*, 3(9), 919–926.

This sliding threshold, proposed by Bienenstock, Cooper, and Munro in 1982, makes the neuron self-correcting: the more it fires, the harder it becomes to strengthen its inputs (Figure 5.4A).[242]

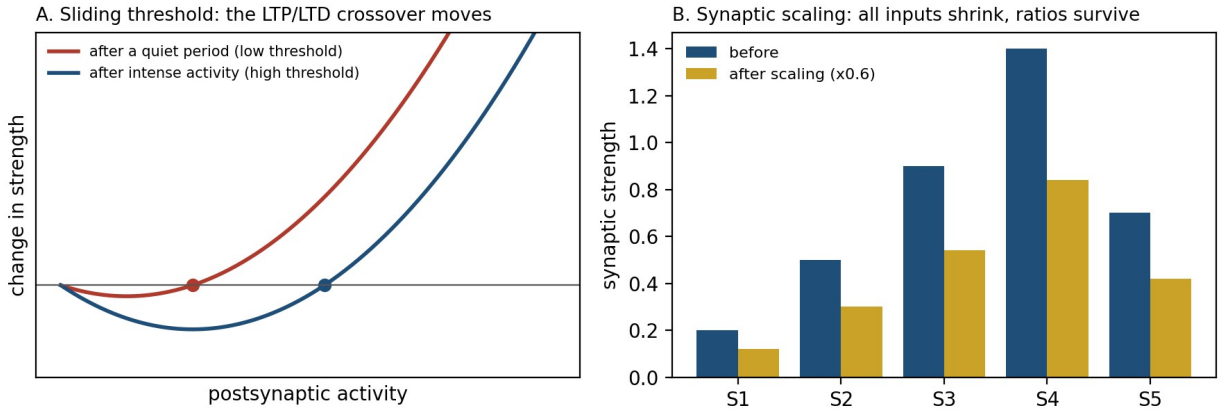


Figure 5.4. Two whole-neuron stabilizers. A: the crossover point between depression and potentiation slides with the neuron's recent activity. B: synaptic scaling multiplies every input by the same factor, preserving the pattern of relative strengths that learning established.

The second adjusts all the synapses at once. If a neuron's firing stays too high for hours, every excitatory synapse on it is scaled down by the same factor; if too low, all are scaled up. Because the scaling is multiplicative, the ratios between synapses are preserved (Figure 5.4B), and the ratios are where the learned information lives. This synaptic scaling, the form of homeostatic plasticity that Chapter 1 introduced, is slow, taking a day or two, and it is the cellular counterpart of the excitation-inhibition balance of Chapter 4.[243]

Inhibitory synapses are plastic too, and their plasticity is largely homeostatic: inhibition onto a neuron grows in step with the excitation it receives.[244] Between the sliding threshold, scaling, and inhibitory plasticity, the brain solves the stability problem that a pure Hebbian rule creates, and it does so without a global supervisor.

The Third Factor

Chapter 3 described the three-factor learning rule: a synapse changes lastingly when it was recently active, when its neuron fired, and when a neuromodulatory signal arrives saying that something significant happened. The mechanisms in this chapter supply the first two factors, and the slice experiments above show that the first two alone can induce potentiation and depression. What the third

[242] Bienenstock, E. L., Cooper, L. N., & Munro, P. W. (1982). Theory for the development of neuron selectivity: Orientation specificity and binocular interaction in visual cortex. *Journal of Neuroscience*, 2(1), 32–48; Kirkwood, A., Rioult, M. G., & Bear, M. F. (1996). Experience-dependent modification of synaptic plasticity in visual cortex. *Nature*, 381, 526–528.

[243] Turrigiano, G. G., Leslie, K. R., Desai, N. S., Rutherford, L. C., & Nelson, S. B. (1998). Activity-dependent scaling of quantal amplitude in neocortical neurons. *Nature*, 391, 892–896; Turrigiano, G. G. (2008). The self-tuning neuron: Synaptic scaling of excitatory synapses. *Cell*, 135(3), 422–435.

[244] Vogels, T. P., Sprekeler, H., Zenke, F., Clopath, C., & Gerstner, W. (2011). Inhibitory plasticity balances excitation and inhibition in sensory pathways and memory networks. *Science*, 334(6062), 1569–1573; Xue, M., Atallah, B. V., & Scanziani, M. (2014). Equalizing excitation–inhibition ratios across visual cortical neurons. *Nature*, 511, 596–600.

factor decides is whether, how strongly, and for how long the change is kept, and it plugs into the machinery in a specific place.[245]

Early LTP sets a tag at the synapse, a transient biochemical mark lasting an hour or two. Late LTP requires proteins that are made in the cell body and shipped to every dendrite. Only tagged synapses capture them. A weak stimulus that sets a tag but does not trigger protein synthesis produces LTP that fades, unless, within the tag's lifetime, something else triggers synthesis. Dopamine, acting on the cell, is one of the things that can.[246] The result is that a synapse activated during an unremarkable event can be consolidated retroactively by a reward that arrives an hour later. This is synaptic tagging and capture, a slower, molecular relative of Chapter 3's eligibility trace that works over hours rather than seconds.[247]

From Synapse to Engram

A memory is not a synapse. It is a pattern of strengthened connections among a set of neurons (Figure 5.5), and the set is sparse: a few to a few tens of percent of the cells in a region take part in encoding any particular event, from about 2 to 6 percent in the dentate gyrus of the hippocampus to 10 to 20 percent in the lateral amygdala.[248]

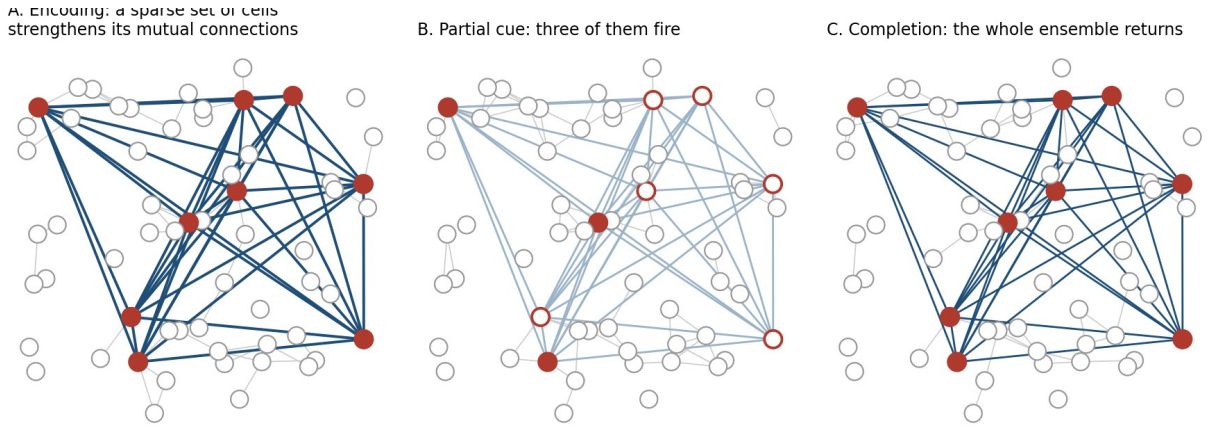


Figure 5.5. A memory as a sparse ensemble. Encoding strengthens the connections among a small set of cells (A); a partial cue activates a few of them (B); the strengthened recurrent connections complete the pattern (C).

[245] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85; Gerstner, W., Lehmann, M., Liakoni, V., Corneil, D., & Brea, J. (2018). Eligibility traces and plasticity on behavioral time scales: Experimental support of neoHebbian three-factor learning rules. *Frontiers in Neural Circuits*, 12, 53.

[246] Frey, U., Schroeder, H., & Matthies, H. (1990). Dopaminergic antagonists prevent long-term maintenance of posttetanic LTP in the CA1 region of rat hippocampal slices. *Brain Research*, 522(1), 69–75; Wang, S.-H., Redondo, R. L., & Morris, R. G. M. (2010). Relevance of synaptic tagging and capture to the persistence of long-term potentiation and everyday spatial memory. *Proceedings of the National Academy of Sciences*, 107(45), 19537–19542.

[247] Frey, U., & Morris, R. G. M. (1997). Synaptic tagging and long-term potentiation. *Nature*, 385, 533–536; Redondo, R. L., & Morris, R. G. M. (2011). Making memories last: The synaptic tagging and capture hypothesis. *Nature Reviews Neuroscience*, 12(1), 17–30; Lisman, J., Grace, A. A., & Duzel, E. (2011). A neoHebbian framework for episodic memory; role of dopamine-dependent late LTP. *Trends in Neurosciences*, 34(10), 536–547.

[248] Han, J.-H., Kushner, S. A., Yiu, A. P., Cole, C. J., Matynia, A., Brown, R. A., Neve, R. L., Guzowski, J. F., Silva, A. J., & Josselyn, S. A. (2007). Neuronal competition and selection during memory formation. *Science*, 316(5823), 457–460; Reijmers, L. G., Perkins, B. L., Matsuo, N., & Mayford, M. (2007). Localization of a stable neural correlate of associative memory. *Science*, 317(5842), 1230–1233; Josselyn, S. A., & Tonegawa, S. (2020). Memory engrams: Recalling the past and imagining the future. *Science*, 367(6473), eaaw4325. The engram-size figures by region are summarized in Josselyn and Tonegawa (2020).

The word for this set is the engram, a term coined in 1904 for a hypothetical physical trace of memory.[249] For a century the engram was an inference. In the last twenty years it has become an experimental object. Neurons that are active during a learning event can be labeled genetically and later made to fire with light, a technique called optogenetics. Reactivating them, in the absence of any cue, produces the behavior the animal learned, such as freezing in a chamber where it was once shocked, a memory that depends on the hippocampus and the amygdala, the two structures of the medial temporal lobe shown in Figure 5.6. Silencing them prevents recall even when the cue is present. And activating the engram of one place while an animal is being shocked in another creates a memory of a fear that never happened: the animal later freezes in the first place.[250]

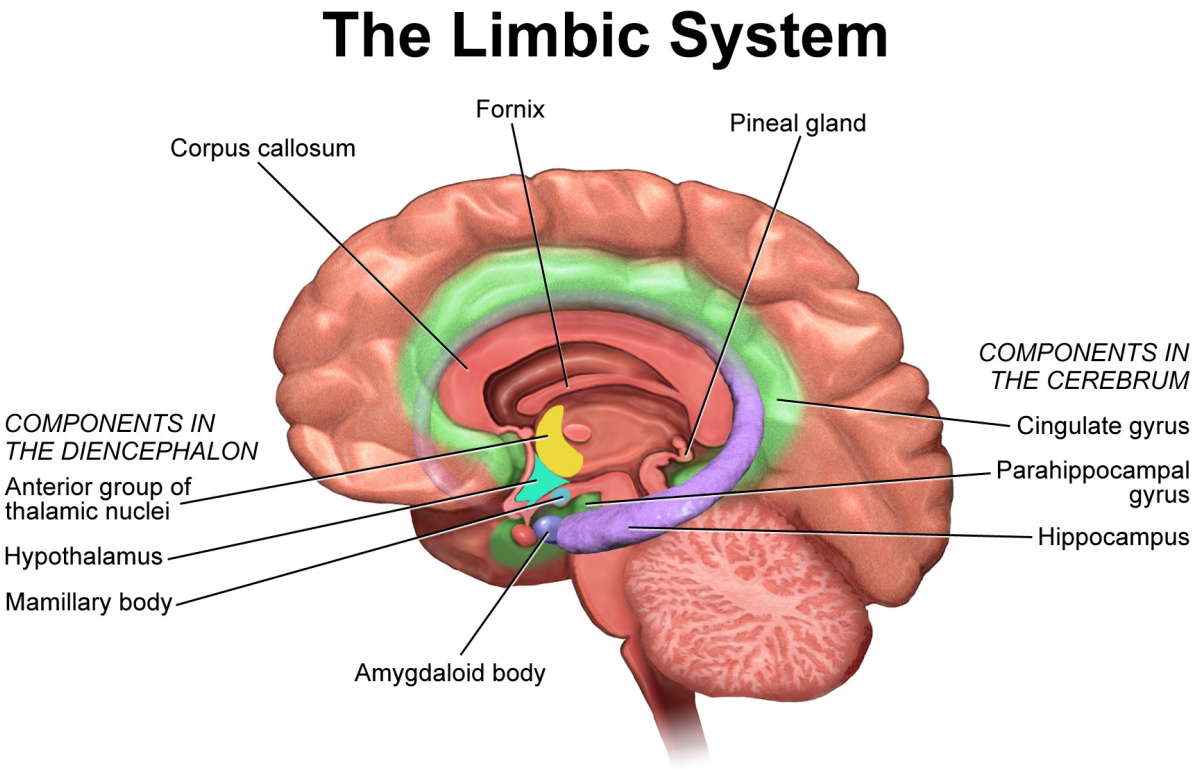


Figure 5.6. The hippocampus and amygdala in the medial temporal lobe, with the structures around them. The hippocampus encodes episodes; the amygdala attaches emotional weight.

[249] Semon, R. (1904). *Die Mneme als erhaltendes Prinzip im Wechsel des organischen Geschehens*. Leipzig: Engelmann; Josselyn, S. A., Köhler, S., & Frankland, P. W. (2015). Finding the engram. *Nature Reviews Neuroscience*, 16(9), 521–534.

[250] Liu, X., Ramirez, S., Pang, P. T., Puryear, C. B., Govindarajan, A., Deisseroth, K., & Tonegawa, S. (2012). Optogenetic stimulation of a hippocampal engram activates fear memory recall. *Nature*, 484, 381–385; Han, J.-H., Kushner, S. A., Yiu, A. P., Hsiang, H.-L., Buch, T., Waisman, A., Bontempi, B., Neve, R. L., Frankland, P. W., & Josselyn, S. A. (2009). Selective erasure of a fear memory. *Science*, 323(5920), 1492–1496; Denny, C. A., Kheirbek, M. A., Alba, E. L., Tanaka, K. F., Brachman, R. A., Laughman, K. B., Tomm, N. K., Turi, G. F., Losonczy, A., & Hen, R. (2014). Hippocampal memory traces are differentially modulated by experience, time, and adult neurogenesis. *Neuron*, 83(1), 189–201; Ramirez, S., Liu, X., Lin, P.-A., Suh, J., Pignatelli, M., Redondo, R. L., Ryan, T. J., & Tonegawa, S. (2013). Creating a false memory in the hippocampus. *Science*, 341(6144), 387–391; Tonegawa, S., Liu, X., Ramirez, S., & Redondo, R. (2015). Memory engram cells have come of age. *Neuron*, 87(5), 918–931.

These experiments establish that a specific, identifiable set of cells and connections is both necessary and sufficient for a specific memory.[251] They also show the mechanism of recall that Chapter 4 called pattern completion. The engram cells are connected to one another more strongly than to their neighbors, as in Figure 5.5. A cue that activates a few of them drives the rest through those connections, and the whole pattern returns. Recall is a dynamical event, not a lookup.[252]

Two Clocks

A memory that has been encoded is not yet durable. It becomes durable through two processes on very different timescales (Figure 5.7A).

Synaptic consolidation is the hour-scale transition from early to late LTP described above: protein synthesis, spine growth, and the structural stabilization of the changed synapses. During this window the memory can be blocked by drugs, by a second competing experience, or by a blow to the head. After it, the memory survives these insults.[253]

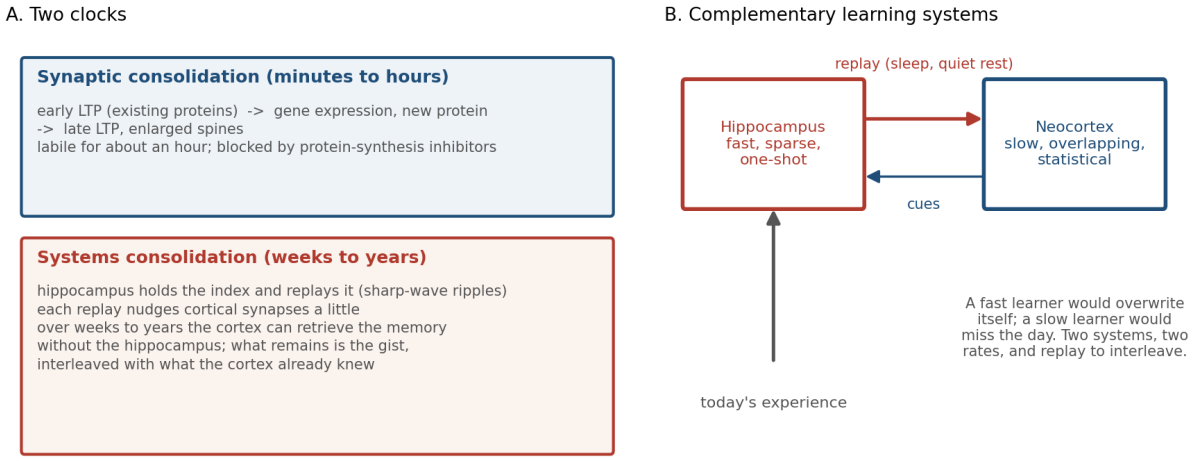


Figure 5.7. A: synaptic consolidation stabilizes a memory within hours; systems consolidation moves it over weeks and years. B: complementary learning systems: a fast hippocampal learner captures the day, a slow cortical learner integrates it, and replay carries one to the other.

Systems consolidation is the transfer, over weeks to years, of the memory from dependence on the hippocampus to independence from it. Patients with hippocampal damage lose the ability to form new episodic memories, memories of particular events, and lose recent memories, but retain old ones, with the boundary falling years before the injury; in the case of the patient H.M., described below, the gap

[251] Josselyn, S. A., & Tonegawa, S. (2020). Memory engrams: Recalling the past and imagining the future. *Science*, 367(6473), eaaw4325.

[252] Marr, D. (1971). Simple memory: A theory for archicortex. *Philosophical Transactions of the Royal Society B*, 262(841), 23–81; Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8), 2554–2558; Nakazawa, K., Quirk, M. C., Chitwood, R. A., Watanabe, M., Yeckel, M. F., Sun, L. D., Kato, A., Carr, C. A., Johnston, D., Wilson, M. A., & Tonegawa, S. (2002). Requirement for hippocampal CA3 NMDA receptors in associative memory recall. *Science*, 297(5579), 211–218.

[253] Müller, G. E., & Pilzecker, A. (1900). Experimentelle Beiträge zur Lehre vom Gedächtnis. *Zeitschrift für Psychologie und Physiologie der Sinnesorgane*, Ergänzungsband 1, 1–300; McGaugh, J. L. (2000). Memory: A century of consolidation. *Science*, 287(5451), 248–251.

covered roughly the eleven years before his surgery.[254] The hippocampus, on the standard account, holds a rapidly formed index that binds together the cortical regions that represented the experience. Over time, repeated reactivation strengthens the direct cortical connections until the index is no longer needed.[255] The standard account is contested. Multiple-trace theory, and its successor, trace-transformation theory, hold that what becomes independent of the hippocampus is a schematic, semanticized version of the memory, while the detailed episode that can be re-experienced remains hippocampal for as long as it exists at all. The evidence is that H.M. and patients like him could recount the facts of their remote past but not relive particular episodes from it, and that in rats memories that stay rich in contextual detail remain sensitive to hippocampal lesions at any age.[256]

The theory that explains why the brain would bother with two systems, called complementary learning systems, was set out in 1995 by McClelland, McNaughton, and O'Reilly, and it is the most important idea in this chapter for readers from machine learning. A single network that learns fast enough to capture a single experience will overwrite what it already knows; this is catastrophic interference, and it is a mathematical property of distributed representations, not a biological quirk.[257] A network that learns slowly enough to avoid interference cannot capture a single experience at all. The solution is two networks: a fast, sparse hippocampal learner that stores each day's events with minimal overlap, and a slow cortical learner that extracts their statistical structure. The fast one trains the slow one gradually, by replaying its contents interleaved with the cortex's existing knowledge (Figure 5.7B).[258]

Sleep

The replay is real. It happens during slow-wave sleep, the deep stage of sleep in which the cortex produces large, slow waves, and also during quiet rest while the animal is awake.[259]

Place cells, the hippocampal neurons of Chapter 4 that each fire when the animal is at one particular location, fire in sequence as a rat runs along a track. During the slow-wave sleep that follows, the same cells fire in the same sequence, compressed roughly twentyfold, during brief high-frequency events

[254] Scoville, W. B., & Milner, B. (1957). Loss of recent memory after bilateral hippocampal lesions. *Journal of Neurology, Neurosurgery & Psychiatry*, 20(1), 11–21; Squire, L. R., & Alvarez, P. (1995). Retrograde amnesia and memory consolidation: A neurobiological perspective. *Current Opinion in Neurobiology*, 5(2), 169–177; Frankland, P. W., & Bontempi, B. (2005). The organization of recent and remote memories. *Nature Reviews Neuroscience*, 6(2), 119–130; Corkin, S. (2002). What's new with the amnesic patient H.M.? *Nature Reviews Neuroscience*, 3(2), 153–160. Corkin (2002) puts H.M.'s retrograde amnesia at about eleven years.

[255] Teyler, T. J., & DiScenna, P. (1986). The hippocampal memory indexing theory. *Behavioral Neuroscience*, 100(2), 147–154; Kitamura, T., Ogawa, S. K., Roy, D. S., Okuyama, T., Morrissey, M. D., Smith, L. M., Redondo, R. L., & Tonegawa, S. (2017). Engrams and circuits crucial for systems consolidation of a memory. *Science*, 356(6333), 73–78.

[256] Nadel, L., & Moscovitch, M. (1997). Memory consolidation, retrograde amnesia and the hippocampal complex. *Current Opinion in Neurobiology*, 7(2), 217–227.; Winocur, G., & Moscovitch, M. (2011). Memory transformation and systems consolidation. *Journal of the International Neuropsychological Society*, 17(5), 766–780.; Steinvorth, S., Levine, B., & Corkin, S. (2005). Medial temporal lobe structures are needed to re-experience remote autobiographical memories: Evidence from H.M. and W.R. *Neuropsychologia*, 43(4), 479–496. Steinvorth, Levine, and Corkin report that H.M.'s remote autobiographical memories were semantic rather than episodic.

[257] McCloskey, M., & Cohen, N. J. (1989). Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24, 109–165; French, R. M. (1999). Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4), 128–135.

[258] McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. (1995). Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, 102(3), 419–457; Kumaran, D., Hassabis, D., & McClelland, J. L. (2016). What learning systems do intelligent agents need? Complementary learning systems theory updated. *Trends in Cognitive Sciences*, 20(7), 512–534.

[259] Rasch, B., & Born, J. (2013). About sleep's role in memory. *Physiological Reviews*, 93(2), 681–766.

called sharp-wave ripples.[260] The same sequences also run, forward and in reverse, during ripples in the pauses between runs, and silencing those waking ripples impairs the learning of a spatial task.[261] The ripples drive cortical activity.[262] Disrupt them selectively and the animal's memory of the track is impaired; prolong them and it improves.[263] Cue a memory during sleep, with an odor or a sound that was present during learning, and that memory is preferentially strengthened.[264] Figure 5.8 shows where in a night these events fall: the slow-wave stages, which dominate the first half, carry the ripples, while the rapid-eye-movement (REM) stages, which lengthen toward morning, do something different.

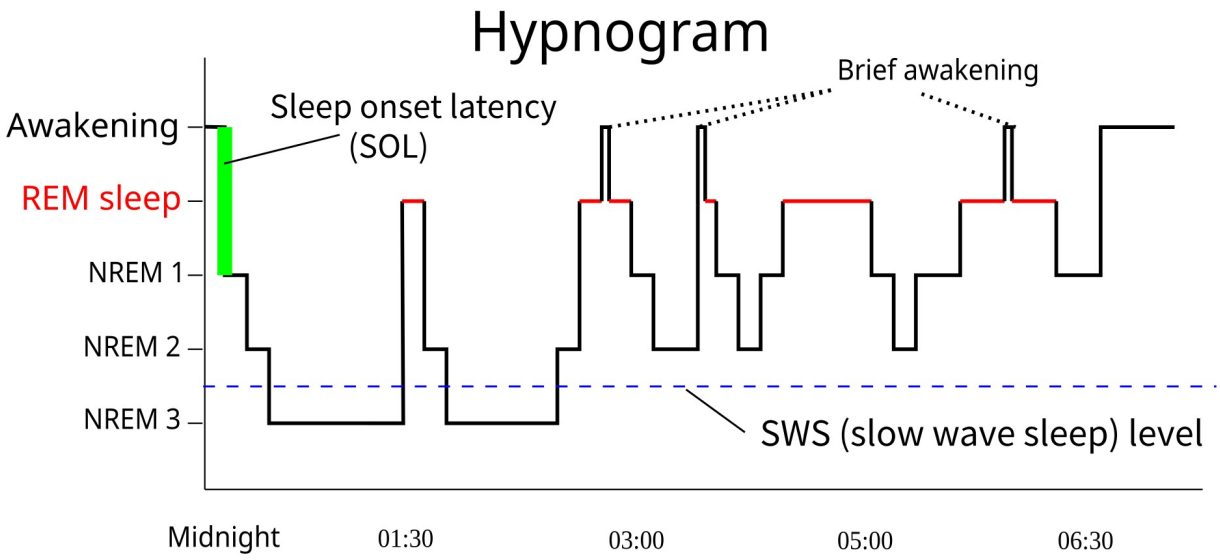


Figure 5.8. A night's sleep, with its alternation of slow-wave and REM stages. Hippocampal replay and sharp-wave ripples are concentrated in slow-wave sleep.

Sleep also appears to do a complementary job: scaling down. Synapses strengthened during the day make the cortex progressively more excitable and more expensive to run. Across a night of slow-wave sleep the average synaptic strength falls, by the homeostatic mechanisms of the previous section. The

[260] Wilson, M. A., & McNaughton, B. L. (1994). Reactivation of hippocampal ensemble memories during sleep. *Science*, 265(5172), 676–679; Lee, A. K., & Wilson, M. A. (2002). Memory of sequential experience in the hippocampus during slow wave sleep. *Neuron*, 36(6), 1183–1194; Buzsáki, G. (2015). Hippocampal sharp wave-ripple: A cognitive biomarker for episodic memory and planning. *Hippocampus*, 25(10), 1073–1188.

[261] Foster, D. J., & Wilson, M. A. (2006). Reverse replay of behavioural sequences in hippocampal place cells during the awake state. *Nature*, 440, 680–683.; Jadhav, S. P., Kemere, C., German, P. W., & Frank, L. M. (2012). Awake hippocampal sharp-wave ripples support spatial memory. *Science*, 336(6087), 1454–1458.

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[263] Girardeau, G., Benchenane, K., Wiener, S. I., Buzsáki, G., & Zugaro, M. B. (2009). Selective suppression of hippocampal ripples impairs spatial memory. *Nature Neuroscience*, 12(10), 1222–1223; Ego-Stengel, V., & Wilson, M. A. (2010). Disruption of ripple-associated hippocampal activity during rest impairs spatial learning in the rat. *Hippocampus*, 20(1), 1–10; Fernández-Ruiz, A., Oliva, A., Fermino de Oliveira, E., Rocha-Almeida, F., Tingley, D., & Buzsáki, G. (2019). Long-duration hippocampal sharp wave ripples improve memory. *Science*, 364(6445), 1082–1086.

[264] Rasch, B., Büchel, C., Gais, S., & Born, J. (2007). Odor cues during slow-wave sleep prompt declarative memory consolidation. *Science*, 315(5817), 1426–1429; Rudoy, J. D., Voss, J. L., Westerberg, C. E., & Paller, K. A. (2009). Strengthening individual memories by reactivating them during sleep. *Science*, 326(5956), 1079.

net effect is to preserve the ratios that encode what was learned while returning the total to a sustainable baseline. This synaptic homeostasis hypothesis is supported by electron-microscope measurements showing cortical synapses about 18 percent smaller after sleep than after waking, but it is contested: sleep also builds, and in mice the REM stages that follow motor learning both prune newly formed spines and strengthen the ones that survive, so the night's net effect is a reweighting rather than a uniform discount.[265] On either view sleep is not a pause in learning but its second half.[266]

Recall Changes the Record

Because recall is reactivation of the engram, and reactivation reopens the window of synaptic plasticity, recalling a memory makes it labile again. For some hours after retrieval, a consolidated memory can be strengthened, weakened, or altered, and must be reconsolidated to persist. Block protein synthesis in that window and the memory, which was stable the day before, is lost.[267]

This reconsolidation is the mechanism behind two things that look like faults from the storage point of view. Memories drift: each recall is an opportunity for the present context to be blended into the trace, and eyewitnesses who have told their story many times are confident and wrong in ways that match what they have been told since.[268] And memories can, perhaps, be revised. In rats, a fear memory reactivated and then extinguished within the reconsolidation window was weakened more durably than one extinguished outside it, and a 2010 study reported the same in people. The human result has not held up well: a preregistered replication with a larger sample found no persistent attenuation, other laboratories have reported both successes and failures, and so whether reconsolidation can be exploited in patients, as several approaches to trauma assume, is unresolved.[269]

A record that changes when read is a bad database. It is a good adaptive system, because the point of a memory is not fidelity to the past but usefulness in the future, and the future keeps changing.

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[265] Li, W., Ma, L., Yang, G., & Gan, W.-B. (2017). REM sleep selectively prunes and maintains new synapses in development and learning. *Nature Neuroscience*, 20(3), 427–437.; Frank, M. G., & Cantera, R. (2014). Sleep, clocks, and synaptic plasticity. *Trends in Neurosciences*, 37(9), 491–501. The 18 percent figure is from de Vivo et al. (2017), cited below.

[266] Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: From synaptic and cellular homeostasis to memory consolidation and integration. *Neuron*, 81(1), 12–34; de Vivo, L., Bellesi, M., Marshall, W., Bushong, E. A., Ellisman, M. H., Tononi, G., & Cirelli, C. (2017). Ultrastructural evidence for synaptic scaling across the wake/sleep cycle. *Science*, 355(6324), 507–510; Diering, G. H., Nirujogi, R. S., Roth, R. H., Worley, P. F., Pandey, A., & Huganir, R. L. (2017). Homer1a drives homeostatic scaling-down of excitatory synapses during sleep. *Science*, 355(6324), 511–515.

[267] Misanin, J. R., Miller, R. R., & Lewis, D. J. (1968). Retrograde amnesia produced by electroconvulsive shock after reactivation of a consolidated memory trace. *Science*, 160(3827), 554–555; Nader, K., Schafe, G. E., & LeDoux, J. E. (2000). Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. *Nature*, 406, 722–726.

[268] Loftus, E. F., & Palmer, J. C. (1974). Reconstruction of automobile destruction: An example of the interaction between language and memory. *Journal of Verbal Learning and Verbal Behavior*, 13(5), 585–589; Loftus, E. F. (2005). Planting misinformation in the human mind: A 30-year investigation of the malleability of memory. *Learning & Memory*, 12(4), 361–366.

[269] Monfils, M.-H., Cowansage, K. K., Klann, E., & LeDoux, J. E. (2009). Extinction-reconsolidation boundaries: Key to persistent attenuation of fear memories. *Science*, 324(5929), 951–955; Schiller, D., Monfils, M.-H., Raio, C. M., Johnson, D. C., LeDoux, J. E., & Phelps, E. A. (2010). Preventing the return of fear in humans using reconsolidation update mechanisms. *Nature*, 463, 49–53; Brunet, A., Orr, S. P., Tremblay, J., Robertson, K., Nader, K., & Pitman, R. K. (2008). Effect of post-retrieval propranolol on psychophysiologic responding during subsequent script-driven traumatic imagery in post-traumatic stress disorder. *Journal of Psychiatric Research*, 42(6), 503–506; Chalkia, A., Schroyens, N., Leng, L., Vanhasbroeck, N., Zenses, A.-K., Van Oudenhove, L., & Beckers, T. (2020). No persistent attenuation of fear memories in humans: A registered replication of the reactivation-extinction effect. *Cortex*, 129, 496–509. Chalkia and colleagues' registered replication of Schiller et al. (2010) found no persistent attenuation of fear.

Forgetting Is a Mechanism

The default fate of a memory is to be lost. Ebbinghaus measured the forgetting curve in 1885 on himself and lists of nonsense syllables; it has been replicated for every kind of material since.[270] Retention drops steeply in the first hours and days and then more slowly, a shape that later analyses found to be close to a power law and much the same across kinds of material and laboratories.[271] Meaning changes how fast the material is learned in the first place, as Ebbinghaus found when he compared his syllables with stanzas of Byron, far more than it changes the shape of the loss.

Three things slow the curve. Retrieval slows it: a memory that is actively recalled, rather than re-read, decays more slowly afterward, and the effect grows when the retrievals are spaced out (Figure 5.9).[272] Sleep slows it, through the consolidation above. And emotional significance slows it, through the third factor.[273]

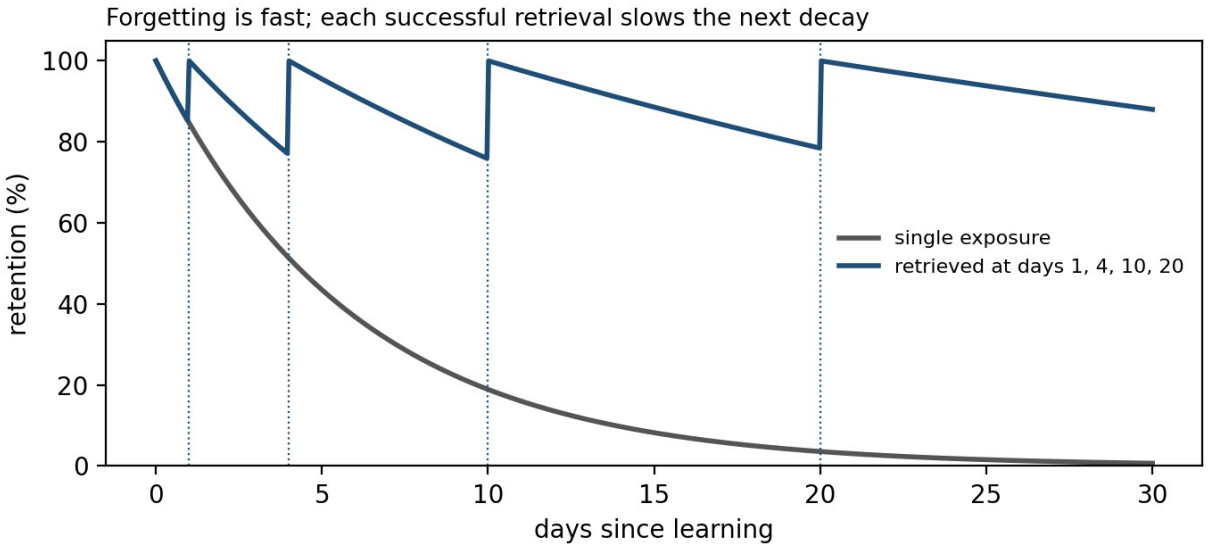


Figure 5.9. Forgetting and spaced retrieval. Illustrative curves of retention after a single exposure (gray) and with retrieval at increasing intervals (blue); each successful retrieval slows the subsequent decay.

Forgetting is also actively produced. In rodents, where adult neurogenesis is not in doubt, the new neurons of the dentate gyrus rewire the circuits they join; increasing neurogenesis in mice accelerates the loss of recent memories, and suppressing it preserves them.[274] Small signaling proteins in the spine

[270] Ebbinghaus, H. (1885). *Über das Gedächtnis: Untersuchungen zur experimentellen Psychologie*. Leipzig: Duncker & Humblot. English edition: *Memory: A Contribution to Experimental Psychology*, trans. H. A. Ruger & C. E. Bussenius (Teachers College, Columbia University, 1913); Murre, J. M. J., & Dros, J. (2015). Replication and analysis of Ebbinghaus' forgetting curve. *PLoS ONE*, 10(7), e0120644.

[271] Wixted, J. T., & Ebbesen, E. B. (1991). On the form of forgetting. *Psychological Science*, 2(6), 409–415.

[272] Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3), 249–255; Karpicke, J. D., & Roediger, H. L. (2008). The critical importance of retrieval for learning. *Science*, 319(5865), 966–968; Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. *Psychological Bulletin*, 132(3), 354–380.

[273] McGaugh, J. L. (2004). The amygdala modulates the consolidation of memories of emotionally arousing experiences. *Annual Review of Neuroscience*, 27, 1–28.

[274] Akers, K. G., Martinez-Canabal, A., Restivo, L., Yiu, A. P., De Cristofaro, A., Hsiang, H.-L., Wheeler, A. L., Guskjolen, A., Niibori, Y., Shoji, H., Ohira, K., Richards, B. A., Miyakawa, T., Josselyn, S. A., & Frankland, P. W. (2014). Hippocampal neurogenesis

actively shrink synapses that are not being used.[275] These mechanisms cost energy, which is the strongest evidence that forgetting is not decay but function. A system that must remain flexible cannot afford to keep everything, and what it keeps must be generalized, which means the specifics must go.[276]

Kinds of Memory, Kinds of Circuit

Psychology distinguishes many kinds of memory: episodic and semantic, procedural, emotional, working. The distinction is real, but it is not a hierarchy of storage tiers. Each kind is a different circuit using a different physical mechanism (Figure 5.10).[277]

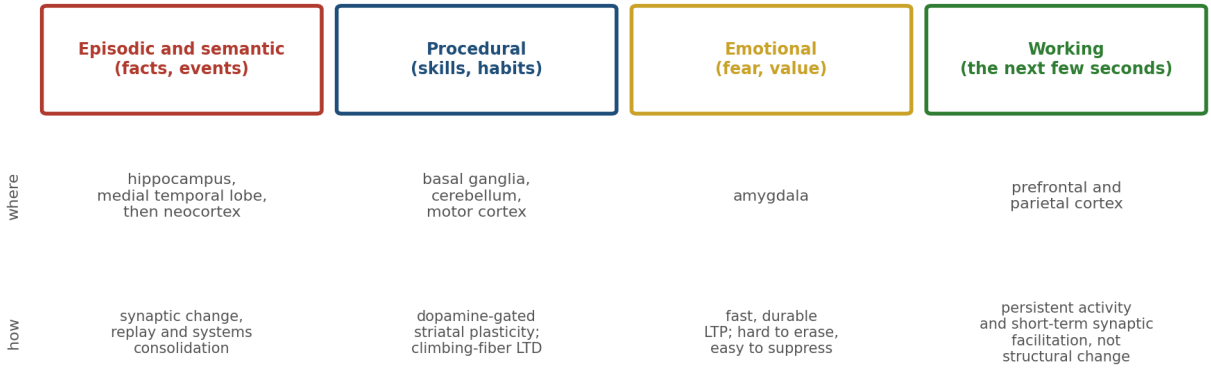


Figure 5.10. Kinds of memory are kinds of circuit. Each row is a different region and a different physical mechanism, not a different level of one storage system.

Episodic and semantic memory, the memory for events and facts, depend on the hippocampus and its systems consolidation into cortex. Procedural memory, for skills and habits, lives in the basal ganglia and cerebellum of Chapter 4, where dopamine-gated striatal plasticity and climbing-fiber-driven cerebellar LTD change what the body does without any episode being recalled.[278]

Emotional memory depends on the amygdala, where LTP is fast, durable, and notoriously hard to erase, which is why fear is easily learned and rarely unlearned, only suppressed.[279] Working memory, the few seconds of holding a number in mind, involves no lasting synaptic change at all. It is carried by persistent activity in prefrontal and parietal circuits, sustained by the recurrent excitation of Chapter 4,

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regulates forgetting during adulthood and infancy. *Science*, 344(6184), 598–602.

[275] Shuai, Y., Lu, B., Hu, Y., Wang, L., Sun, K., & Zhong, Y. (2010). Forgetting is regulated through Rac activity in Drosophila. *Cell*, 140(4), 579–589; Davis, R. L., & Zhong, Y. (2017). The biology of forgetting: A perspective. *Neuron*, 95(3), 490–503.

[276] Hardt, O., Nader, K., & Nadel, L. (2013). Decay happens: The role of active forgetting in memory. *Trends in Cognitive Sciences*, 17(3), 111–120; Richards, B. A., & Frankland, P. W. (2017). The persistence and transience of memory. *Neuron*, 94(6), 1071–1084.

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[278] Yin, H. H., & Knowlton, B. J. (2006). The role of the basal ganglia in habit formation. *Nature Reviews Neuroscience*, 7(6), 464–476; Ito, M. (2001). Cerebellar long-term depression: Characterization, signal transduction, and functional roles. *Physiological Reviews*, 81(3), 1143–1195.

[279] Rogan, M. T., Stäubli, U. V., & LeDoux, J. E. (1997). Fear conditioning induces associative long-term potentiation in the amygdala. *Nature*, 390, 604–607; LeDoux, J. E. (2000). Emotion circuits in the brain. *Annual Review of Neuroscience*, 23, 155–184; Bouton, M. E. (2004). Context and behavioral processes in extinction. *Learning & Memory*, 11(5), 485–494.

and, on a more recent view in which the activity need not be continuous, by short-term facilitation of synapses that lasts seconds and leaves no structural trace.[280]

A patient can lose one of these without the others. The amnesic patient H.M., who could form no new episodic memories after the front part of both hippocampi and the medial temporal cortex around them were removed, learned to trace a star in a mirror as well as anyone, improving day by day while denying each day that he had ever seen the task. His skill was in a circuit his lesion had not touched.[281]

Why Learning Is Slow

Chapter 1 promised that learning is slow because molecular change is slow. This chapter has shown the detail. Early LTP forms in minutes and lasts an hour or two; late LTP takes hours and new protein; systems consolidation takes weeks; a skill takes thousands of repetitions because striatal and cerebellar synapses change by small increments gated by a scarce teaching signal.

But the chapter has also shown that the slowness is not only a limitation of the hardware. It is a design requirement. The stability-plasticity dilemma, the fact that a system cannot both change quickly and retain what it knows, is solved by learning slowly in the cortex and letting the hippocampus take the speed.[282] Fast learning is quarantined to a structure built for it, and everything else changes at a pace that protects what is already there. Slowness is the price of not overwriting yourself, which, for a system without a reset button, is the price of staying coherent.

Implications for Artificial Intelligence

Memory is where biological and artificial learning differ most, and where the biology has the most to offer.

First, catastrophic interference is the central unsolved problem of continual learning in artificial networks, and the brain's solution is architectural: two learners with different rates, and replay to interleave them. Experience replay in reinforcement learning, in which an agent stores its past transitions and retrains on them in shuffled order, was invented in 1992, two years before hippocampal replay was first recorded, and the two were recognized as analogs only afterward: a convergence, not a borrowing. Systems that train a fast episodic store and distill it gradually into slow weights are the closest artificial analog of consolidation.[283]

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[280] Funahashi, S., Bruce, C. J., & Goldman-Rakic, P. S. (1989). Mnemonic coding of visual space in the monkey's dorsolateral prefrontal cortex. *Journal of Neurophysiology*, 61(2), 331–349; Goldman-Rakic, P. S. (1995). Cellular basis of working memory. *Neuron*, 14(3), 477–485; Mongillo, G., Barak, O., & Tsodyks, M. (2008). Synaptic theory of working memory. *Science*, 319(5869), 1543–1546; Stokes, M. G. (2015). 'Activity-silent' working memory in prefrontal cortex: A dynamic coding framework. *Trends in Cognitive Sciences*, 19(7), 394–405.

[281] Scoville, W. B., & Milner, B. (1957). Loss of recent memory after bilateral hippocampal lesions. *Journal of Neurology, Neurosurgery & Psychiatry*, 20(1), 11–21; Milner, B. (1962). Les troubles de la mémoire accompagnant des lésions hippocampiques bilatérales. In P. Passouant (Ed.), *Physiologie de l'hippocampe* (pp. 257–272). Paris: Centre National de la Recherche Scientifique; Milner, B., Corkin, S., & Teuber, H.-L. (1968). Further analysis of the hippocampal amnesic syndrome: 14-year follow-up study of H.M. *Neuropsychologia*, 6(3), 215–234; Corkin, S. (2002). What's new with the amnesic patient H.M.? *Nature Reviews Neuroscience*, 3(2), 153–160; Corkin, S., Amaral, D. G., González, R. G., Johnson, K. A., & Hyman, B. T. (1997). H.M.'s medial temporal lobe lesion: Findings from magnetic resonance imaging. *Journal of Neuroscience*, 17(10), 3964–3979. The MRI study found the posterior hippocampus partly spared.

[282] Grossberg, S. (1980). How does a brain build a cognitive code? *Psychological Review*, 87(1), 1–51; Abraham, W. C., & Robins, A. (2005). Memory retention: The synaptic stability versus plasticity dilemma. *Trends in Neurosciences*, 28(2), 73–78.

Second, local rules can be enough. Backpropagation needs a global error signal delivered to every weight. LTP, LTD, and timing-dependent plasticity need only what is present at the synapse plus a broadcast modulator. Whether local three-factor rules can match gradient descent on hard problems is open, but the brain is an existence proof that some biologically available rule can learn language, motor skill, and science.[284]

Third, stability must be engineered. Pure Hebbian learning runs away, and the brain's answer is a sliding threshold, synaptic scaling, and inhibitory plasticity. Normalization layers and weight decay are rough analogs; a homeostatic mechanism that adapts to each unit's own history is closer to the biology and less explored.[285]

Fourth, forgetting is worth building in. A model that stores everything generalizes poorly and grows without bound. The brain spends energy to remove specifics and keep gist. Machine unlearning, pruning, and the regularizers of continual learning are crude equivalents, and few deployed systems forget deliberately.

Finally, a memory that changes when read has a use. Artificial retrieval systems return the stored record unchanged. Biological recall reconsolidates, blending the record with the present. That costs fidelity and buys relevance. For agents that must act in a changing world, the trade is worth considering.

The Central Lesson

Experience becomes structure. A coincidence detector at the synapse turns correlated activity into calcium, calcium into receptor traffic and spine growth, and spine growth into a lasting change in what the circuit does next. Whole-cell mechanisms keep the process stable, neuromodulators decide when it counts, and two consolidation clocks turn a day's traces into durable knowledge without overwriting the knowledge that was there. Recall reopens the trace, and forgetting prunes it. What is stored is physical, distributed, and reconstructive: a change in the tissue, spread over many synapses, that is rebuilt at every recall rather than read.

Next Chapter

Chapter 6: Perception as Inference: How the Brain Sees What It Expects

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[283] Lin, L.-J. (1992). Self-improving reactive agents based on reinforcement learning, planning and teaching. *Machine Learning*, 8(3–4), 293–321; Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S., & Hassabis, D. (2015). Human-level control through deep reinforcement learning. *Nature*, 518, 529–533; Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., & Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526; Parisi, G. I., Kemker, R., Part, J. L., Kanan, C., & Wermter, S. (2019). Continual lifelong learning with neural networks: A review. *Neural Networks*, 113, 54–71.

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[285] Ioffe, S., & Szegedy, C. (2015). Batch normalization: Accelerating deep network training by reducing internal covariate shift. *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 448–456; Krogh, A., & Hertz, J. A. (1992). A simple weight decay can improve generalization. *Advances in Neural Information Processing Systems*, 4, 950–957.

Figure and Epigraph Sources

Figure Attribution

Figures 5.1, 5.2, 5.3, 5.4, 5.5, 5.7, 5.9, and 5.10 were drawn for this book. The traces in Figure 5.2B, the dots in Figure 5.3, and the curves in Figure 5.9 are illustrative simulations, not data.

Figure 5.6: “Limbic System” by BruceBlaus (Blausen.com staff, “Medical gallery of Blausen Medical 2014,” WikiJournal of Medicine), CC BY 3.0, via Wikimedia Commons.

Figure 5.8: “Sleep Hypnogram” by Wikimedia Commons user RazerM, CC BY-SA 3.0; composited on a white background.

Epigraph

Donald O. Hebb, *The Organization of Behavior: A Neuropsychological Theory* (Wiley, 1949), p. 62.

Part III: Mind and Behavior

Chapter 6

Perception as Inference: How the Brain Sees What It Expects

> *"Such objects are always imagined as being present in the field of vision as would have to be there in order to produce the same impression on the nervous mechanism, the eyes being used under ordinary normal conditions."*
>
> *— Hermann von Helmholtz, Treatise on Physiological Optics (1867)*

The Problem Perception Solves

Chapter 1 said, almost in passing, that perception is an active process of inference shaped by prior experience. Every chapter since has built the machinery that makes the sentence true: neurons that integrate evidence, circuits that amplify and compare, neuromodulators that set the context, and synapses that turn experience into structure. This chapter uses that machinery to answer a question that sounds too simple to need one. When you open your eyes, why do you see a world rather than a pattern of light?

The light that reaches the retina is not a world. It is a two-dimensional projection, blurred by the optics of the eye, sampled by a mosaic of receptors, and interrupted three times a second by eye movements. Worse, the projection is ambiguous in a way that no amount of resolution can fix. A trapezoid on the retina can be a trapezoid face-on, a rectangle receding from the viewer, or a painted shape on a billboard; the retina cannot distinguish them, because they all produce the same image (Figure 6.1). A patch of gray can be a dark surface in bright light or a bright surface in shadow. A sound that grows louder can be a source approaching or a source turning toward you. Mathematically this is an inverse problem, and inverse problems of this kind are ill-posed: the evidence does not determine the answer.

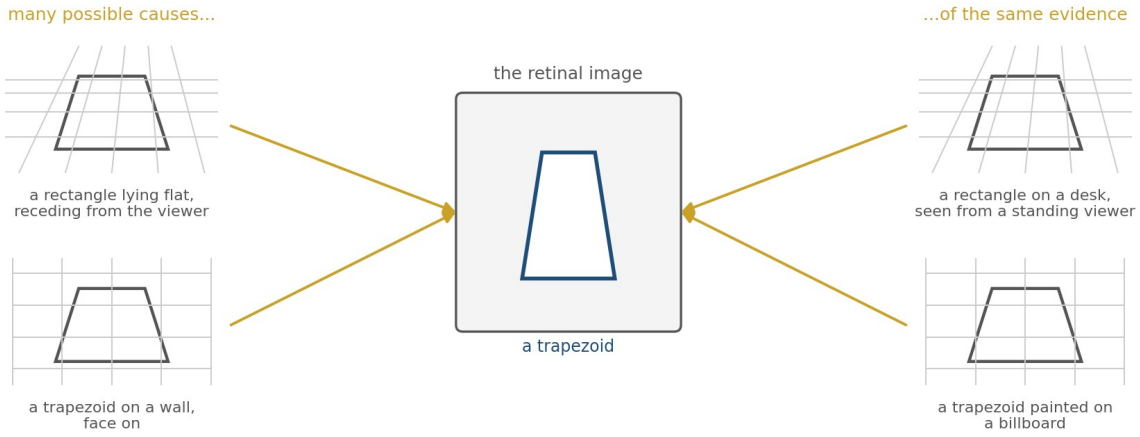


Figure 6.1. The inverse problem. Many arrangements of surfaces in the world produce the same image on the retina. The evidence alone cannot say which one is present.

Helmholtz saw this clearly in the 1860s and drew the only possible conclusion. If the evidence does not determine the answer and yet we perceive an answer, something must be supplying the rest. He called it

unconscious inference: the perceptual system settles on the state of the world that, under ordinary conditions, would most probably have produced the sensations it is receiving.[286] The judgment is not made by reasoning and cannot be overruled by it, which is why an illusion persists after you know it is an illusion. A century and a half later the idea has been made precise, provisionally mapped onto cortical anatomy, and measured with electrodes, and this chapter is an account of that work. The thesis can be stated in one sentence. Perception is not the reading of a signal but the construction of a hypothesis, built from what the senses report and from what the brain already expects, and weighted by how much each can be trusted.

The Senses Deliver Evidence, Not Pictures

Begin with what the senses actually provide, because it is less than intuition suggests.

The human retina has about 92 million rods and about 4.6 million cones, but only about a million ganglion cell axons leave each eye.[287] The retina therefore compresses by roughly a hundred to one before any signal reaches the brain, and Chapter 4 described what the compression keeps: contrast rather than brightness, change rather than steady state, edges rather than surfaces. Estimates of the information rate of the optic nerve are on the order of ten megabits per second, comparable to a single compressed video stream.[288]

That capacity is not spread evenly. Fine resolution is confined to the fovea, a patch of densely packed cones whose region of highest acuity covers only one to two degrees of the visual field, roughly the width of a thumbnail held at arm's length. Outside the fovea acuity collapses, and at fifteen degrees to one side there is a region with no receptors at all, where the optic nerve exits. This blind spot is about six degrees across, a dozen times the width of the full moon, and you have never seen it, because the brain fills it with whatever surrounds it.[289]

Nor does the eye hold still. It jumps about three times a second in rapid movements called saccades, resting for a few hundred milliseconds between jumps. During each jump the image sweeps across the retina at hundreds of degrees per second and vision is suppressed.[290] What arrives in the brain is not a stable, wide, high-resolution picture. It is a sequence of small, sharp, overlapping samples, each a fraction of a second long, separated by gaps, with a hole in each one. The stable scene you experience is not among the inputs. It is an output.

The same holds for the other senses. The cochlea delivers a running spectrum, not words. Touch delivers the deformation of skin, not the shape of the object. In every case the sense organ reports local physical quantities, and the rest is inference.

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[286] Helmholtz, H. von (1867). *Handbuch der physiologischen Optik*. Leipzig: Voss. English edition: *Treatise on Physiological Optics*, vol. III, trans. J. P. C. Southall (Optical Society of America, 1925). The doctrine of unconscious inference is set out in §26 of vol. III.

[287] Curcio, C. A., Sloan, K. R., Kalina, R. E., & Hendrickson, A. E. (1990). Human photoreceptor topography. *Journal of Comparative Neurology*, 292(4), 497–523; Jonas, J. B., Schmidt, A. M., Müller-Bergh, J. A., Schlötzer-Schrehardt, U. M., & Naumann, G. O. H. (1992). Human optic nerve fiber count and optic disc size. *Investigative Ophthalmology & Visual Science*, 33(6), 2012–2018.

[288] Koch, K., McLean, J., Segev, R., Freed, M. A., Berry, M. J., Balasubramanian, V., & Sterling, P. (2006). How much the eye tells the brain. *Current Biology*, 16(14), 1428–1434.

[289] Ramachandran, V. S. (1992). Blind spots. *Scientific American*, 266(5), 86–91.

[290] Matin, E. (1974). Saccadic suppression: A review and an analysis. *Psychological Bulletin*, 81(12), 899–917; Rayner, K. (1998). Eye movements in reading and information processing: 20 years of research. *Psychological Bulletin*, 124(3), 372–422.

Bayes in One Paragraph

The mathematics of inference under uncertainty was worked out in the eighteenth century and is short enough to state in full. Suppose the brain must estimate some quantity, say the direction of a sound. Before the sound arrives there is a prior: a distribution over directions that encodes what is usually true, built from a lifetime of sounds. When the sound arrives there is a likelihood: for each possible direction, how probable are these particular sensations? Bayes's rule says that the posterior, the distribution the brain should act on, is proportional to the prior multiplied by the likelihood.[291] The prior says what to expect; the evidence says what was observed; the posterior is the compromise, and it leans toward whichever of the two is more precise.

Figure 6.2 shows the consequence for the simplest case, where both are bell curves. When the evidence is sharp, the posterior sits almost on the evidence, and the prior barely matters. When the evidence is noisy, the posterior sits near the prior, and the senses barely matter. Neither extreme is a failure. A system that always trusted its senses would be captured by every flicker of noise; a system that always trusted its prior would be blind. The optimal weighting is set by relative precision, the inverse of a distribution's variance (the narrower the curve, the more precise), and it changes from moment to moment as the quality of the evidence changes.

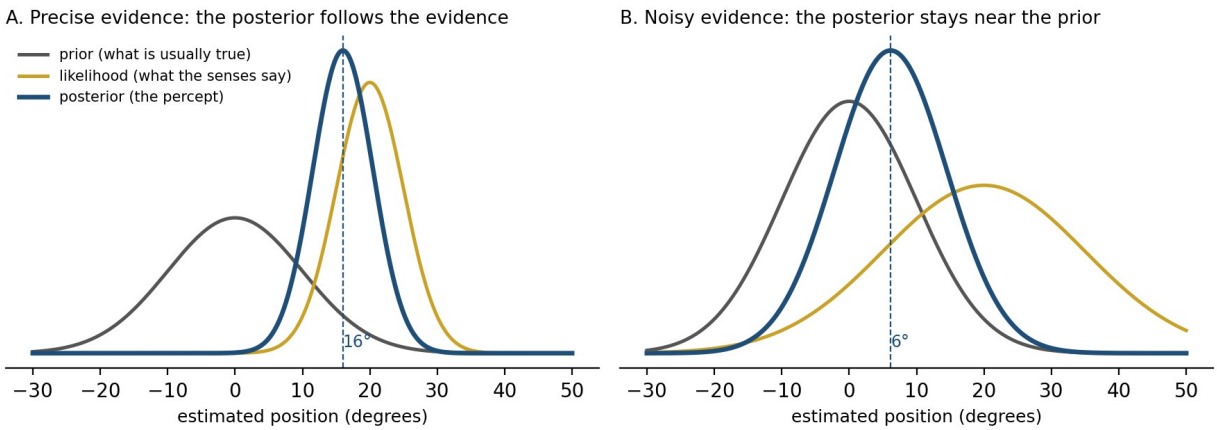


Figure 6.2. Combining a prior with sensory evidence. A: precise evidence pulls the posterior toward itself. B: the same prior with noisy evidence; the posterior stays near the prior. The weighting is set by the relative precisions.

This is the sense in which perception is inference. The brain does not compute Bayes's rule symbolically any more than a thrown ball solves differential equations. But the system behaves, to a good approximation, as the rule says it should, and the rest of this chapter is about the machinery that produces that behavior and the places where the approximation shows.

Where do the priors come from? From Chapter 5. A prior is learned structure: the accumulated statistical regularities of a lifetime of input, written into synaptic weights by the plasticity mechanisms already described. The light-from-above prior is the clearest example. Light has come from above for the whole of evolutionary history and for the whole of each individual life, and the visual system assumes it. A disc shaded light on top is seen as a bump, a disc shaded light on the bottom as a dent, and rotating

[291] Bayes, T. (1763). An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London*, 53, 370–418; Knill, D. C., & Pouget, A. (2004). The Bayesian brain: The role of uncertainty in neural coding and computation. *Trends in Neurosciences*, 27(12), 712–719.

the image swaps them (Figure 6.3B).[292] No one is taught this, and chicks reared from hatching under light from below still behave as if light came from above, so the assumption is at least partly in place before experience. But it is not fixed. The prior that human adults actually hold is light from above and a little to the left, and in 2004 Adams, Graf, and Ernst shifted it: after observers had touched shaded discs whose felt shape agreed with a light source rotated away from the usual direction, their judgments of bumps and dents moved toward the new direction, and the shift carried over to a lightness task that had never been trained.[293] A prior is in the weights, and the weights can be rewritten.

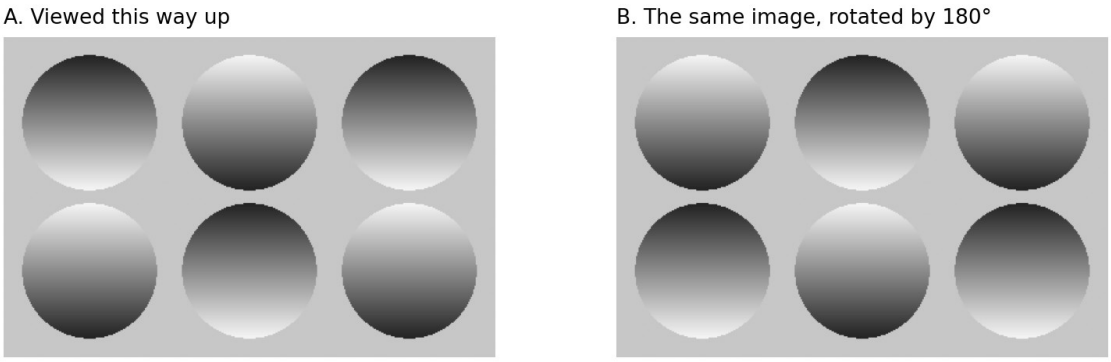


Figure 6.3. The light-from-above prior. A: discs shaded lighter on top are seen as bumps and the others as dents. B: the same image rotated by 180 degrees; every disc reverses. The shading is identical in both panels; only the assumption about where the light is differs.

The Hierarchy

Perception is carried out by a hierarchy of cortical areas, and the visual hierarchy is the best understood (Figure 6.4). Retinal ganglion cells project to the lateral geniculate nucleus of the thalamus, the LGN, and the LGN projects to the primary visual cortex, V1, at the back of the brain. From V1 the signal passes to V2, V4, and the inferotemporal cortex, IT, along what is called the ventral stream, and in parallel to areas specialized for motion and location along the dorsal stream.[294]

[292] Ramachandran, V. S. (1988). Perception of shape from shading. *Nature*, 331, 163–166.

[293] Hershberger, W. (1970). Attached-shadow orientation perceived as depth by chickens reared in an environment illuminated from below. *Journal of Comparative and Physiological Psychology*, 73(3), 407–411; Sun, J., & Perona, P. (1998). Where is the sun? *Nature Neuroscience*, 1(3), 183–184; Adams, W. J., Graf, E. W., & Ernst, M. O. (2004). Experience can change the 'light-from-above' prior. *Nature Neuroscience*, 7(10), 1057–1058.

[294] Ungerleider, L. G., & Mishkin, M. (1982). Two cortical visual systems. In D. J. Ingle, M. A. Goodale, & R. J. W. Mansfield (Eds.), *Analysis of Visual Behavior* (pp. 549–586). MIT Press; Felleman, D. J., & Van Essen, D. C. (1991). Distributed hierarchical processing in the primate cerebral cortex. *Cerebral Cortex*, 1(1), 1–47.

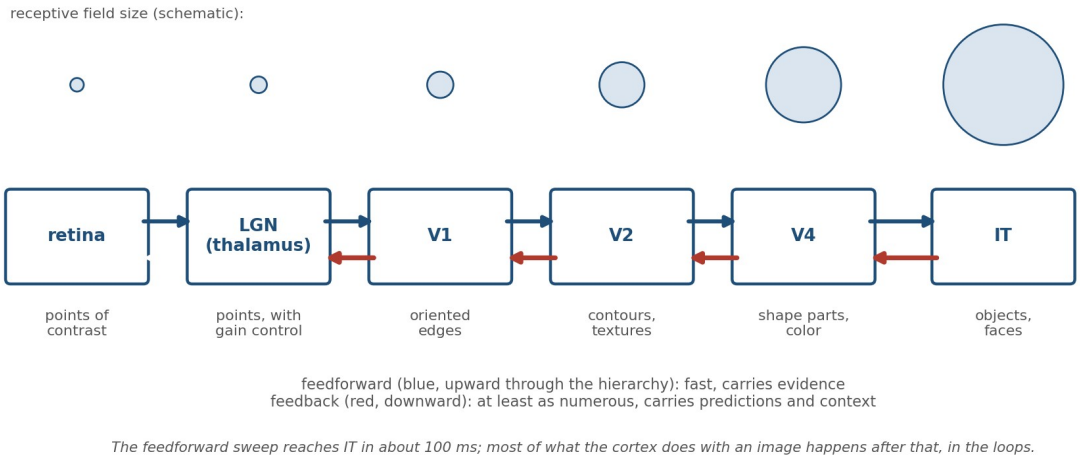


Figure 6.4. The ventral visual pathway. Receptive fields grow and preferences become more abstract at each stage. Feedforward connections carry evidence upward; feedback connections, at least as numerous, carry predictions and context downward. In mammals there is essentially no feedback to the retina.

Two things change at each stage. The receptive field, the region of the visual field that can influence a neuron, grows, from a fraction of a degree in V1 to most of the visual field in IT. And the preferred stimulus becomes more abstract. A V1 neuron responds to an oriented edge at one place.[295] A V4 neuron responds to a curved contour or a color in a larger region. An IT neuron responds to an object, a face, or a hand, more or less regardless of where it appears, how large it is, or how it is lit.[296] The hierarchy builds invariance: the same thing is recognized as the same thing across the enormous variation in images that it can produce.

This feedforward sweep is fast. An image flashed for a few tens of milliseconds produces a selective response in IT about a hundred milliseconds later, and a human can categorize a scene as containing an animal or not in about a hundred and fifty milliseconds.[297] Convolutional neural networks were modeled on this sweep: Fukushima's neocognitron of 1980 stacked alternating layers of feature detectors and pooling units in explicit imitation of Hubel and Wiesel's simple and complex cells, and every later convolutional network keeps that plan.[298] The best of them predict the responses of IT neurons to novel images better than any model built by hand.[299] This is the part of perception that artificial systems have reproduced.

But the sweep is not the whole hierarchy, nor even most of it. Feedback connections, running from each area back down to the areas below, are at least as numerous as the feedforward ones. In the most

[295] Hubel, D. H., & Wiesel, T. N. (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *Journal of Physiology*, 160(1), 106–154.

[296] Gross, C. G., Rocha-Miranda, C. E., & Bender, D. B. (1972). Visual properties of neurons in inferotemporal cortex of the macaque. *Journal of Neurophysiology*, 35(1), 96–111.

[297] Thorpe, S., Fize, D., & Marlot, C. (1996). Speed of processing in the human visual system. *Nature*, 381, 520–522; Hung, C. P., Kreiman, G., Poggio, T., & DiCarlo, J. J. (2005). Fast readout of object identity from macaque inferior temporal cortex. *Science*, 310(5749), 863–866.

[298] Fukushima, K. (1980). Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position. *Biological Cybernetics*, 36(4), 193–202.

[299] Yamins, D. L. K., Hong, H., Cadieu, C. F., Solomon, E. A., Seibert, D., & DiCarlo, J. J. (2014). Performance-optimized hierarchical models predict neural responses in higher visual cortex. *Proceedings of the National Academy of Sciences*, 111(23), 8619–8624.

complete count of macaque visual pathways to date, Markov and colleagues found more feedback pathways than feedforward pathways; a count of pathways is not a count of axons or synapses, though, and whether feedback outnumbers feedforward at every stage is not established.[300] At the thalamus it plainly does. The LGN, the thalamic relay that the retina feeds, receives only about a tenth of its synapses from the retina; the rest come from the cortex and the brainstem.[301] Anatomically, the visual system is not a pipeline from eye to cortex. It is a loop, and the downward half is at least as large as the upward half. Any account of perception has to say what the downward half is for.

Predictive Coding

The account that has organized the field for the last three decades is called predictive coding, and it is a direct implementation of Helmholtz's idea in the anatomy of Figure 6.4.[302]

Each level of the hierarchy holds a hypothesis about what the level below is receiving. It sends that hypothesis downward as a prediction. The level below compares the prediction with what it actually has and sends upward only the difference, the prediction error. The level above uses the error to revise its hypothesis, sends a revised prediction, and the cycle repeats until the error is small. What ascends the hierarchy is not the image but the part of the image that was not expected (Figure 6.5A).

This is prediction error in a different sense from Chapter 3, and the two should not be confused. There, a single number, broadcast by a large population of dopamine neurons, said how much better or worse than expected an outcome was: a scalar about value, the quantity that temporal-difference learning computes. Here, a whole pattern of errors about content, one for every feature at every level, passes between cortical layers and never leaves the sensory hierarchy. Readers who know temporal-difference learning will recognize the family resemblance, and theorists have tried to unify the two, but they are different signals carried by different cells, and the sensory one has no reward in it.

[300] Markov, N. T., Vezoli, J., Chameau, P., Falchier, A., Quilodran, R., Huissoud, C., Lamy, C., Misery, P., Giroud, P., Ullman, S., Barone, P., Dehay, C., Knoblauch, K., & Kennedy, H. (2014). Anatomy of hierarchy: Feedforward and feedback pathways in macaque visual cortex. *Journal of Comparative Neurology*, 522(1), 225–259.

[301] Sherman, S. M., & Guillery, R. W. (2006). *Exploring the Thalamus and Its Role in Cortical Function* (2nd ed.). MIT Press.

[302] Mumford, D. (1992). On the computational architecture of the neocortex. II. The role of cortico-cortical loops. *Biological Cybernetics*, 66(3), 241–251; Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1), 79–87; Friston, K. (2005). A theory of cortical responses. *Philosophical Transactions of the Royal Society B*, 360(1456), 815–836.

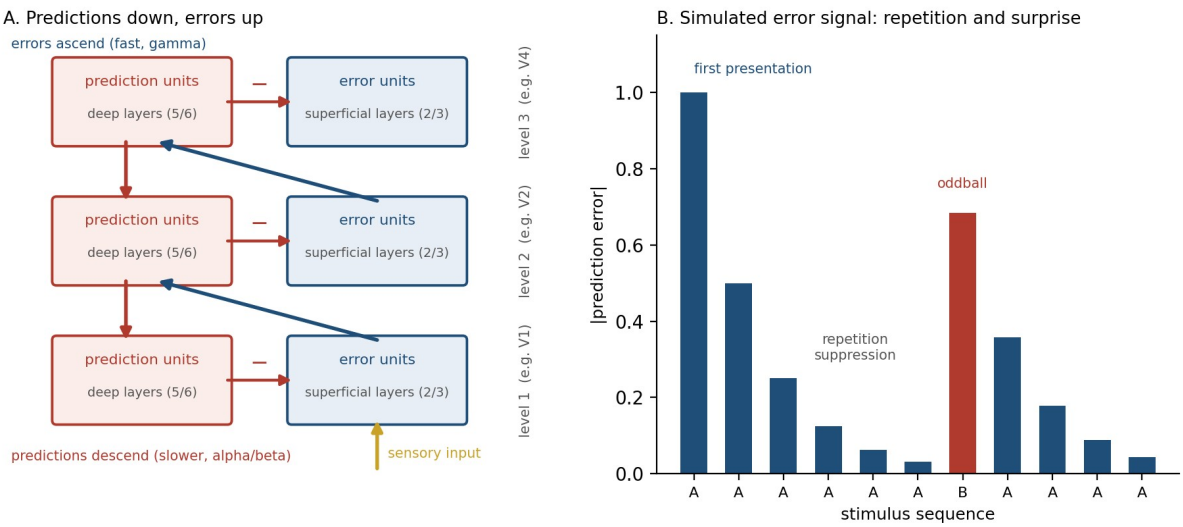


Figure 6.5. Predictive coding. A: each level sends predictions down and receives errors up; the proposed mapping onto the canonical microcircuit places error units in the superficial layers and prediction units in the deep layers. B: a simulated error signal for a tone repeated six times, replaced once, and repeated again. Error falls with repetition and spikes at the change.

The scheme has three consequences that look surprising until the scheme is understood. The first is that an expected stimulus produces less cortical activity than an unexpected one. If the prediction is good, the error is small, and the error is what is transmitted. The second is that the absence of an expected stimulus produces activity: a prediction with nothing to cancel it is itself an error. The third is that what a neuron responds to depends on context, because the prediction arriving from above depends on everything the higher level knows.

All three are observed, though the first needs care. A stimulus repeated several times evokes a smaller response each time, in single neurons and in whole-brain imaging; this is called repetition suppression, and it is one of the most reproducible phenomena in sensory cortex (Figure 6.5B).[303] On its own, though, it is weak evidence for prediction, because a fatigued neuron or a population sharpening its tuning would show the same decline, and the review that catalogs these models declines to choose among them. The decisive test holds repetition fixed and varies expectation. In 2008 Summerfield and colleagues showed pairs of faces in blocks where a repeat was likely and blocks where it was rare; the same repeated face suppressed the fusiform face area more when the repeat was expected, so expectation, not repetition itself, was setting the response, and this expectation suppression has since been separated in time from repetition suppression in human auditory cortex.[304]

[303] Grill-Spector, K., Henson, R., & Martin, A. (2006). Repetition and the brain: Neural models of stimulus-specific effects. *Trends in Cognitive Sciences*, 10(1), 14–23. The review compares fatigue, sharpening, and facilitation models of repetition suppression without endorsing one.

[304] Summerfield, C., Trittschuh, E. H., Monti, J. M., Mesulam, M.-M., & Egner, T. (2008). Neural repetition suppression reflects fulfilled perceptual expectations. *Nature Neuroscience*, 11(9), 1004–1006; Todorovic, A., & de Lange, F. P. (2012). Repetition suppression and expectation suppression are dissociable in time in early auditory evoked fields. *Journal of Neuroscience*, 32(39), 13389–13395.

A tone omitted from a regular sequence evokes a response at the moment it should have occurred.[305] The deviant tone in a sequence of standards evokes a distinctive electrical signature at the scalp, the mismatch negativity, about one to two hundred milliseconds after it arrives; it is detectable, though attenuated, in sleeping adults, it is present in newborn infants, and it reflects a prediction being violated rather than anything about the tone itself.[306]

Neurons in V1, finally, respond to a line most strongly when it ends inside their receptive field, a long-known puzzle called end-stopping, which the first predictive coding model of the visual cortex, published in 1999, reproduced without being designed to: a line that continues is predictable from its surroundings, and a line that stops is not.[307]

The scheme also fits the microcircuit of Chapter 4 in a specific way. The superficial layers of cortex, which project forward, carry activity dominated by fast gamma rhythms; the deep layers, which project backward, carry activity dominated by the slower alpha and beta rhythms. Predictive coding proposes that the superficial layers hold the error units and the deep layers the prediction units, so that errors ascend on gamma and predictions descend on alpha and beta. Recordings from several areas at once support the direction of this traffic.[308] It is a model, and the mapping is not settled, but it is the kind of model that could be wrong in detail, which is the kind worth having.

Illusions Are Correct Answers

If perception is inference, illusions are what inference looks like when the prior is right about the world and wrong about the picture. They are the most direct evidence we have for which priors the visual system holds.

In Adelson's checker-shadow figure (Figure 6.6), the squares marked A and B are printed in exactly the same gray. They do not look it, and they will not look it after you have measured them. The visual system has inferred a cylinder, a light source, and a shadow, and it has discounted the shadow before reporting the surface color of B. That is the correct computation. In a real scene, a surface in shadow that reflects as much light as a surface in the open really is lighter, and a visual system that reported raw luminance would misjudge every surface that fell in shade. The figure defeats the system by presenting a shadow with nothing behind it.[309]

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[305] Wacongne, C., Labyt, E., van Wassenhove, V., Bekinschtein, T., Naccache, L., & Dehaene, S. (2011). Evidence for a hierarchy of predictions and prediction errors in human cortex. *Proceedings of the National Academy of Sciences*, 108(51), 20754–20759.

[306] Näätänen, R., Gaillard, A. W. K., & Mäntysalo, S. (1978). Early selective-attention effect on evoked potential reinterpreted. *Acta Psychologica*, 42(4), 313–329; Näätänen, R., Paavilainen, P., Rinne, T., & Alho, K. (2007). The mismatch negativity (MMN) in basic research of central auditory processing: A review. *Clinical Neurophysiology*, 118(12), 2544–2590.

[307] Hubel, D. H., & Wiesel, T. N. (1965). Receptive fields and functional architecture in two nonstriate visual areas (18 and 19) of the cat. *Journal of Neurophysiology*, 28(2), 229–289; Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1), 79–87. End-stopped ("hypercomplex") cells were first described by Hubel and Wiesel.

[308] Bastos, A. M., Usrey, W. M., Adams, R. A., Mangun, G. R., Fries, P., & Friston, K. J. (2012). Canonical microcircuits for predictive coding. *Neuron*, 76(4), 695–711; van Kerkoerle, T., Self, M. W., Dagnino, B., Gariel-Mathis, M.-A., Poort, J., van der Togt, C., & Roelfsema, P. R. (2014). Alpha and gamma oscillations characterize feedback and feedforward processing in monkey visual cortex. *Proceedings of the National Academy of Sciences*, 111(40), 14332–14341; Bastos, A. M., Vezoli, J., Bosman, C. A., Schoffelen, J.-M., Oostenveld, R., Dowdall, J. R., De Weerd, P., Kennedy, H., & Fries, P. (2015). Visual areas exert feedforward and feedback influences through distinct frequency channels. *Neuron*, 85(2), 390–401.

[309] Adelson, E. H. (2000). Lightness perception and lightness illusions. In M. S. Gazzaniga (Ed.), *The New Cognitive Neurosciences* (2nd ed., pp. 339–351). MIT Press.

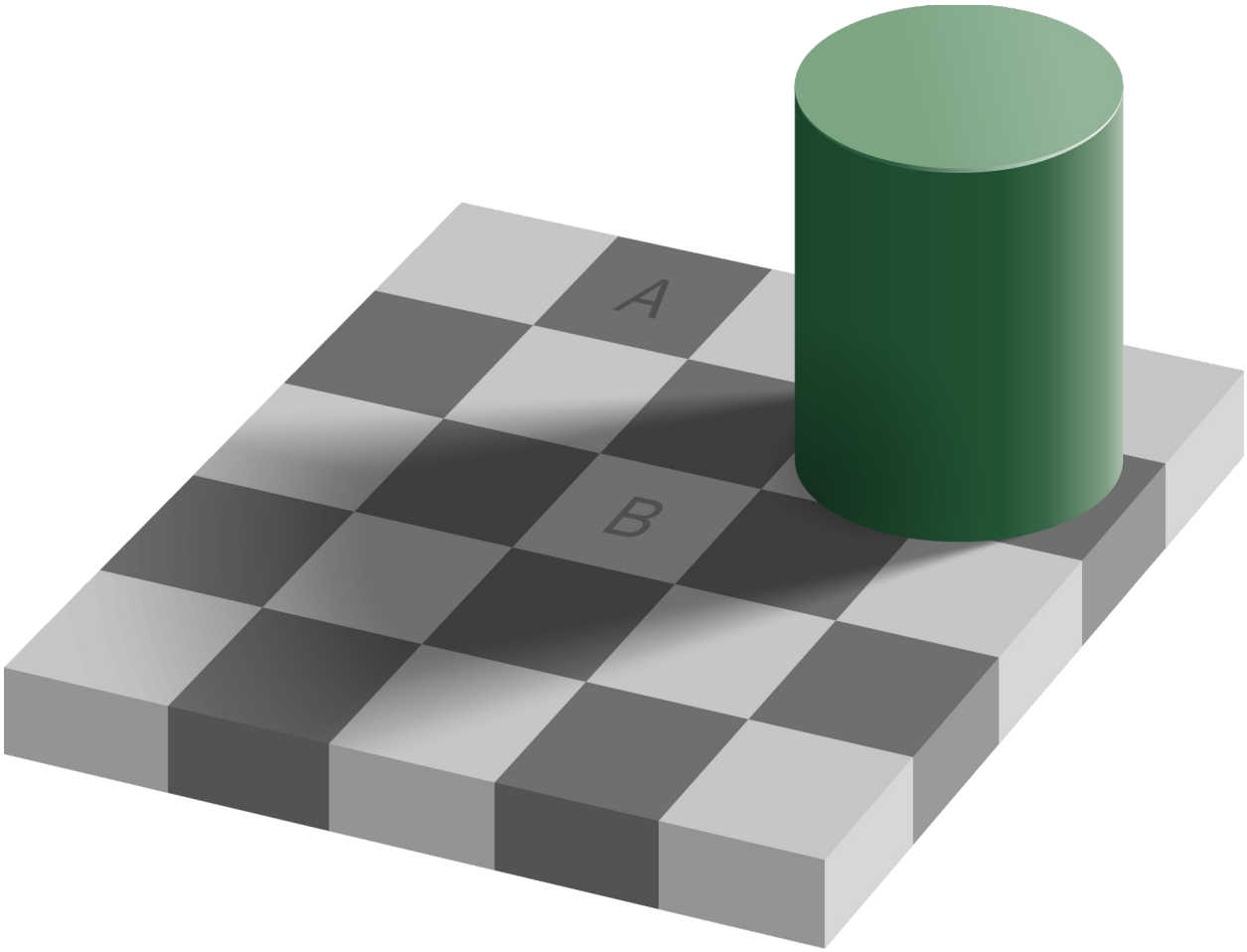


Figure 6.6. Adelson's checker-shadow illusion. The squares marked A and B are the same shade of gray. The visual system infers a shadow and discounts it, which is the right inference for a real scene.

The Kanizsa triangle (Figure 6.7A) contains no triangle. Three discs with wedges removed and three corner fragments are arranged so that the most probable cause of the image is a white triangle lying on top of three black discs and an outlined triangle. The visual system reports that cause, complete with edges that are not in the image and a surface that looks whiter than the identical white around it. Neurons in V2 respond to these illusory contours as if they were real, which, under the hypothesis, they are.[310]

[310] Kanizsa, G. (1955). Margini quasi-percettivi in campi con stimolazione omogenea. *Rivista di Psicologia*, 49(1), 7–30; von der Heydt, R., Peterhans, E., & Baumgartner, G. (1984). Illusory contours and cortical neuron responses. *Science*, 224(4654), 1260–1262.

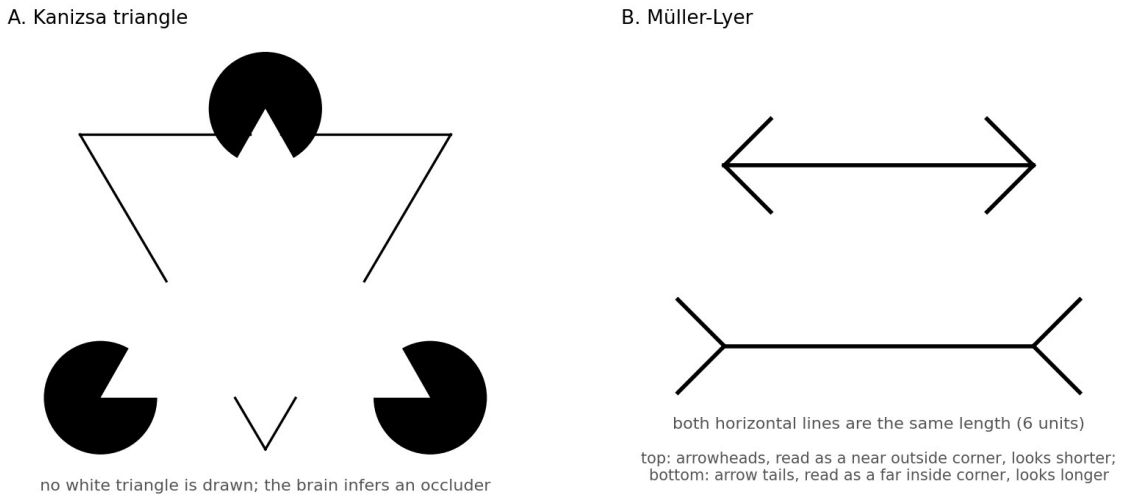


Figure 6.7. Two illusions as inferences. A: the Kanizsa triangle; the occluding white triangle is inferred, not drawn. B: the Müller-Lyer figure; the two horizontal lines are the same length but are read as corners at different depths.

In the Müller-Lyer figure (Figure 6.7B) two equal lines look unequal. One explanation, due to Richard Gregory, is that the fins are read as the edges of corners in a world of right-angled rooms and buildings, what Gregory called a carpentered world: arrowheads as the near corner of a building, arrow tails as the far corner of a room. A far corner that subtends the same angle as a near one must be longer, and the system scales accordingly.[311] The explanation is debated, and that is the point. Each illusion is a probe, and arguments about illusions are arguments about which prior the system is using.

The hollow-mask illusion is the strongest of all. A face mask viewed from the inside, concave, is seen as convex, a normal face, even when it is slowly rotating and stereo vision and shading both report the truth. The prior that faces are convex is so strong that it overrides the evidence completely. The illusion weakens at close range, where stereo evidence becomes precise enough to win, and it is measurably weaker in people with schizophrenia, an observation to which the chapter returns.[312]

What all of these show is that the output of the perceptual system is not the image and not the world but the posterior, and that the posterior can be moved by manipulating either the evidence or the prior.

Attention as Precision

Bayes's rule weights evidence by its precision, and the brain cannot know the precision of a signal without estimating it. That estimate has to be represented somewhere and has to change what the circuit does. The proposal that follows from predictive coding is that this is what attention is: the gain applied to prediction errors, raised where the evidence is expected to be reliable and lowered where it is not.[313]

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[311] Müller-Lyer, F. C. (1889). Optische Urteilstäuschungen. *Archiv für Anatomie und Physiologie, Physiologische Abteilung*, Supplement, 263–270; Gregory, R. L. (1963). Distortion of visual space as inappropriate constancy scaling. *Nature*, 199, 678–680.

[312] Gregory, R. L. (1970). *The Intelligent Eye*. Weidenfeld & Nicolson; Dima, D., Roiser, J. P., Dietrich, D. E., Bonnemann, C., Lanfermann, H., Emrich, H. M., & Dillo, W. (2009). Understanding why patients with schizophrenia do not perceive the hollow-mask illusion using dynamic causal modelling. *NeuroImage*, 46(4), 1180–1186.

[313] Feldman, H., & Friston, K. J. (2010). Attention, uncertainty, and free-energy. *Frontiers in Human Neuroscience*, 4, 215.

The proposal unifies a great deal. Attention to a location increases the response of neurons whose receptive fields cover it and decreases the response of neighbors, which is the biased competition of Chapter 4's lateral inhibition with a top-down bias setting the winner.[314] Attention increases the gamma synchrony of the attended population, which, through the oscillatory gating described in Chapter 4, lets its spikes arrive during the receptive phase of the next area and so increases their weight.[315] And attention makes the attended evidence count for more in the posterior, which is what raising its precision means.

Precision also has a chemistry, and it is the chemistry of Chapter 3. Acetylcholine, released from the basal forebrain into cortex, increases the gain of feedforward sensory input relative to recurrent and feedback input; a cholinergic cortex trusts its senses more and its expectations less.[316] One influential account assigns acetylcholine to expected uncertainty, the known unreliability of a familiar cue, and norepinephrine to unexpected uncertainty, the moment when the world changes and the current model should be discarded.[317] The noradrenergic burst that accompanies a surprise is, on this reading, a global instruction to lower the precision of every prediction at once.

The cost of precision weighting is that what is not attended is barely seen. In the best-known demonstration, observers counting passes in a basketball video fail to notice a person in a gorilla suit walking through the middle of the scene.[318] In change blindness, large alterations to a scene go unnoticed if they coincide with a blink or a cut.[319] These are not failures of the eye; the gorilla was on the retina for several seconds. They are the consequence of a system that transmits only what it has been told to find surprising.

Combining the Senses

The clearest quantitative test of the Bayesian account uses two senses at once. If the brain weights evidence by precision, then when vision and touch both report the width of an object, the combined estimate should lie between them, closer to the more reliable sense, and should be more precise than either alone. The weights can be computed exactly from the reliability of each sense measured separately.

In 2002 Marc Ernst and Martin Banks ran this experiment with a bar that observers could see and feel at the same time, degrading vision by adding noise to the display. With clear vision, the estimate followed the eyes. As the visual noise increased, the weight shifted toward touch, and at every noise level the observed weights matched the computed optimum (Figure 6.8).[320] The brain was not favoring vision

[314] Desimone, R., & Duncan, J. (1995). Neural mechanisms of selective visual attention. *Annual Review of Neuroscience*, 18, 193–222.

[315] Fries, P., Reynolds, J. H., Rorie, A. E., & Desimone, R. (2001). Modulation of oscillatory neuronal synchronization by selective visual attention. *Science*, 291(5508), 1560–1563; Fries, P. (2015). Rhythms for cognition: Communication through coherence. *Neuron*, 88(1), 220–235.

[316] Hasselmo, M. E. (2006). The role of acetylcholine in learning and memory. *Current Opinion in Neurobiology*, 16(6), 710–715.

[317] Yu, A. J., & Dayan, P. (2005). Uncertainty, neuromodulation, and attention. *Neuron*, 46(4), 681–692.

[318] Simons, D. J., & Chabris, C. F. (1999). Gorillas in our midst: Sustained inattentional blindness for dynamic events. *Perception*, 28(9), 1059–1074.

[319] Rensink, R. A., O'Regan, J. K., & Clark, J. J. (1997). To see or not to see: The need for attention to perceive changes in scenes. *Psychological Science*, 8(5), 368–373.

[320] Ernst, M. O., & Banks, M. S. (2002). Humans integrate visual and haptic information in a statistically optimal fashion. *Nature*, 415, 429–433.

as a rule. It was favoring whichever sense was more precise, trial by trial. This is the local sense, argued in Chapter 3, in which the brain optimizes: a circuit behaves as if it were minimizing a well-defined local cost, here the variance of its estimate.

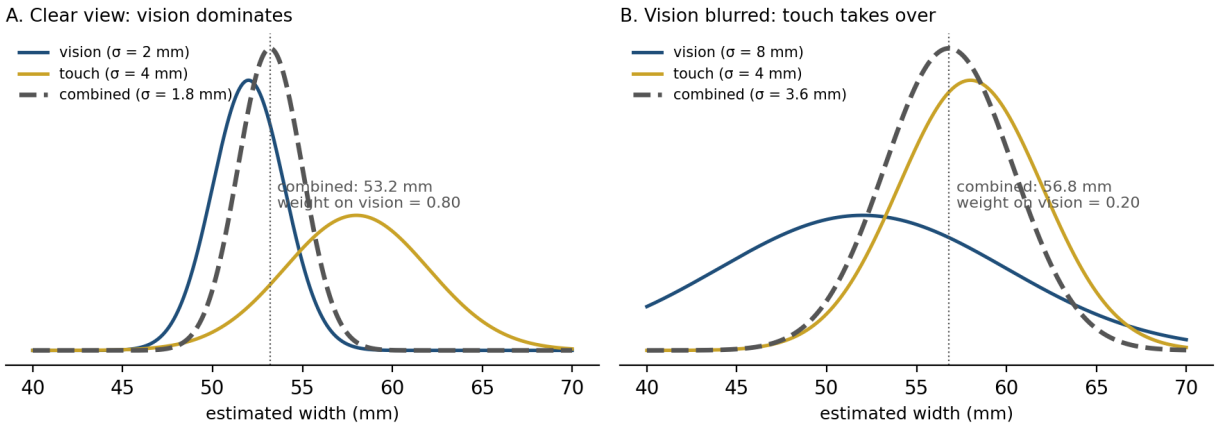


Figure 6.8. Precision-weighted combination of vision and touch. A: with a clear view, the combined estimate follows vision. B: with vision blurred, touch dominates. In both cases the combined estimate is narrower than either input, and the weights are set by the relative precisions.

Everyday perception is full of the same computation. The ventriloquist's voice appears to come from the dummy's mouth because vision locates things more precisely than hearing, so the posterior for the sound's location follows the visible moving mouth.[321] In the McGurk effect, a recording of one syllable dubbed onto a face mouthing a different syllable is heard as a third syllable, a compromise between what was heard and what was seen.[322] The senses are not separate channels to a central viewer. They are sources of evidence, pooled in proportion to how much each can be trusted.

Perception Is Active

The account so far has treated the brain as if it sat still and waited for evidence. It does not. It moves the sense organs to get the evidence it wants, and that changes the problem in two ways.

The first is that the brain must distinguish sensation caused by the world from sensation caused by its own movement. When the eye jumps ten degrees, the image sweeps ten degrees across the retina, exactly as it would if the world had jumped. It does not look that way. The solution, proposed independently by von Holst and by Sperry in 1950, is that every motor command is copied and sent to the sensory system, which uses it to predict the sensory consequence of the movement and subtract it.[323] The copy is called the efference copy, the prediction is a forward model, and the comparison is shown in Figure 6.9.

[321] Alais, D., & Burr, D. (2004). The ventriloquist effect results from near-optimal bimodal integration. *Current Biology*, 14(3), 257–262.

[322] McGurk, H., & MacDonald, J. (1976). Hearing lips and seeing voices. *Nature*, 264, 746–748.

[323] von Holst, E., & Mittelstaedt, H. (1950). Das Reafferenzprinzip: Wechselwirkungen zwischen Zentralnervensystem und Peripherie. *Naturwissenschaften*, 37(20), 464–476; Sperry, R. W. (1950). Neural basis of the spontaneous optokinetic response produced by visual inversion. *Journal of Comparative and Physiological Psychology*, 43(6), 482–489.

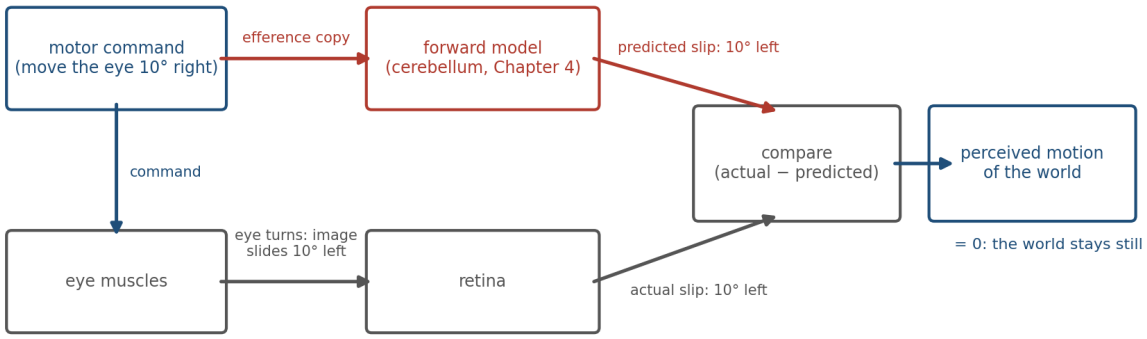


Figure 6.9. Efference copy. A copy of each motor command is sent to a forward model that predicts the resulting sensory change. The prediction is subtracted from the actual sensation, so that self-caused changes are canceled and only changes caused by the world remain.

If the predicted slip matches the actual slip, the world is reported as stationary. Press gently on your eyelid and move the eye by hand: there is no command, no prediction, nothing is subtracted, and the world lurches. The cerebellum of Chapter 4, which learns to predict the sensory consequences of commands, is a principal site of this forward model.[324]

The same mechanism is why you cannot tickle yourself: the sensation is predicted and canceled before it is felt.[325]

The second is that movement is itself a tool of inference. A saccade is a decision about where evidence is most needed, and the sequence of fixations on a scene depends on the question being asked.[326] A rat sweeps its whiskers across an object at a rate it controls; a dog sniffs in bursts timed to its uncertainty; a reader's eyes land on the words that the sentence so far has made least predictable.[327] In each case the organism is not receiving evidence but sampling it, choosing the sample that will reduce its uncertainty most. Perception and action, kept apart in the textbooks, are two halves of one loop: the hypothesis determines where to look, and what is seen revises the hypothesis. Some theorists go further and treat action as inference too, the body moving so as to make the brain's predictions come true.[328] Whether or not one accepts that framing, the practical lesson is secure. An observer that cannot act is solving a harder problem than any brain has ever had to solve.

[324] Wolpert, D. M., Miall, R. C., & Kawato, M. (1998). Internal models in the cerebellum. *Trends in Cognitive Sciences*, 2(9), 338–347.

[325] Blakemore, S.-J., Wolpert, D. M., & Frith, C. D. (1998). Central cancellation of self-produced tickle sensation. *Nature Neuroscience*, 1(7), 635–640.

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Learning the Priors

If perception rests on priors, the question of where they come from is the question of how perception is learned, and the answer connects this chapter to the last.

Some priors are present at or near birth. Newborns orient to face-like patterns within hours, and the chicks reared under light from below, described earlier, held the light-from-above assumption without instruction.[329] But a prior present at birth is not a prior fixed for life, as the adults in the Adams experiment show, and most of the structure of sensory cortex is learned from the statistics of the input. That learning has the properties Chapter 5 would predict. It is slow, it is gated, and it has a window.

The window is the one Hubel and Wiesel found in kittens with one eye closed (Chapter 1).[330] This critical period, the strict form of the sensitive periods introduced there, closes, in the cat, after a few months and in humans after a few years. Kittens raised in environments containing only vertical stripes grew up with cortices in which almost no neuron responded to horizontal edges, and they bumped into horizontal bars.[331] The critical period opens and closes under the control of inhibitory maturation and of the neuromodulators of Chapter 3; it is a time when the plasticity rules of Chapter 5 are turned up, and its closing is what makes adult perception stable and adult perceptual learning slow.[332]

What is learned during the window is the statistics of natural images, and that can be shown directly. In 1996 Olshausen and Field trained a simple model to represent patches of photographs using as few active units as possible. The units it learned were localized, oriented, band-pass filters (Figure 6.10), each responding to an edge of one orientation and one spatial scale at one place, essentially the receptive fields that Hubel and Wiesel had found in V1 thirty years earlier.[333] Nothing about edges was built in. Edges are what the world contains, and a system that learns to predict the world ends up with detectors for them. The same procedure applied to natural sounds produces filters resembling those of the auditory nerve.[334] The receptive fields of sensory cortex are, to a first approximation, the priors of the natural world made into tissue.

[329] Johnson, M. H., Dziurawiec, S., Ellis, H., & Morton, J. (1991). Newborns' preferential tracking of face-like stimuli and its subsequent decline. *Cognition*, 40(1–2), 1–19; Hershberger, W. (1970). Attached-shadow orientation perceived as depth by chickens reared in an environment illuminated from below. *Journal of Comparative and Physiological Psychology*, 73(3), 407–411.

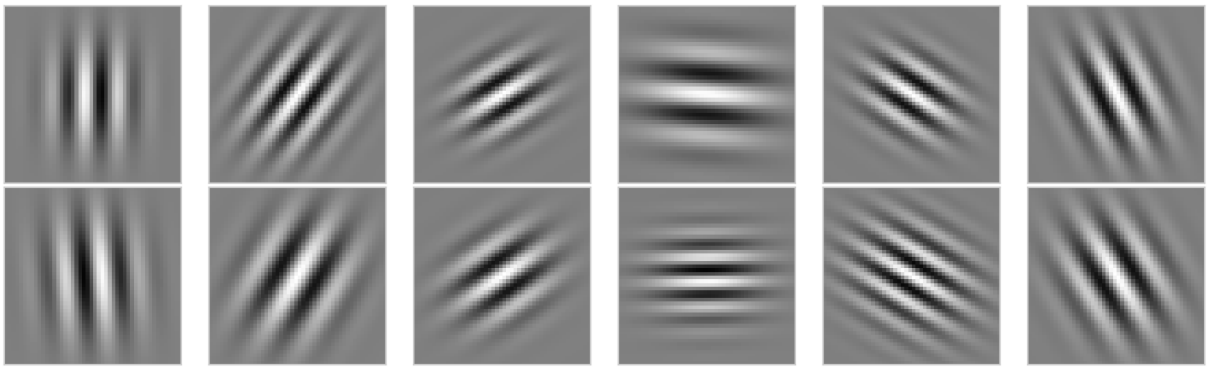
[330] Wiesel, T. N., & Hubel, D. H. (1963). Single-cell responses in striate cortex of kittens deprived of vision in one eye. *Journal of Neurophysiology*, 26(6), 1003–1017; Hubel, D. H., & Wiesel, T. N. (1970). The period of susceptibility to the physiological effects of unilateral eye closure in kittens. *Journal of Physiology*, 206(2), 419–436.

[331] Blakemore, C., & Cooper, G. F. (1970). Development of the brain depends on the visual environment. *Nature*, 228, 477–478.

[332] Hensch, T. K. (2005). Critical period plasticity in local cortical circuits. *Nature Reviews Neuroscience*, 6(11), 877–888.

[333] Olshausen, B. A., & Field, D. J. (1996). Emergence of simple-cell receptive field properties by learning a sparse code for natural images. *Nature*, 381, 607–609.

[334] Lewicki, M. S. (2002). Efficient coding of natural sounds. *Nature Neuroscience*, 5(4), 356–363.



Receptive fields of V1 simple cells (schematic): localized, oriented, band-pass. Light = excitatory region, dark = inhibitory.

Figure 6.10. Receptive fields of the kind found in primary visual cortex: each cell responds to an edge of one orientation and scale at one location. Fields of this form emerge in models trained only to represent natural images efficiently.

This learning never fully stops. Wear prism goggles that shift the world sideways and reaching errors vanish in minutes, as the cerebellar forward model relearns.[335] Wear goggles that invert the world and within days you can walk, reach, and ride a bicycle in it, as Stratton and Kohler reported; whether it ever comes to look upright is disputed, since a controlled study in 1999 found full motor adaptation after six to ten days but no perceptual re-inversion, and Stratton's own reports of uprightness were intermittent.[336] View the world through a colored filter for a few hours a day for several days and the wavelength you call pure yellow shifts, and it stays shifted for a week or two after the filter comes off.[337] The adult perceptual system is a model under continual revision, slow at its core and fast at its periphery, exactly as the two consolidation clocks of Chapter 5 would suggest.

When Inference Goes Wrong

A system that constructs its percepts from priors and evidence can fail in two directions, and both are seen in the clinic.

When the evidence is removed, the priors run unopposed. People who lose their vision late in life through disease of the eye frequently begin to see vivid, detailed hallucinations, faces, patterns, figures in period dress, while knowing perfectly well that nothing is there; the condition is named for Charles Bonnet, who described it in his grandfather in 1760.[338] A visual cortex that has lost its input does not go silent. It continues to generate hypotheses, and with no error signal to correct them they are

[335] Martin, T. A., Keating, J. G., Goodkin, H. P., Bastian, A. J., & Thach, W. T. (1996). Throwing while looking through prisms. I. Focal olivocerebellar lesions impair adaptation. *Brain*, 119(4), 1183–1198.

[336] Stratton, G. M. (1897). Vision without inversion of the retinal image. *Psychological Review*, 4(4), 341–360 and 4(5), 463–481; Kohler, I. (1962). Experiments with goggles. *Scientific American*, 206(5), 62–72; Linden, D. E. J., Kallenbach, U., Heinecke, A., Singer, W., & Goebel, R. (1999). The myth of upright vision: A psychophysical and functional imaging study of adaptation to inverting spectacles. *Perception*, 28(4), 469–481.

[337] Neitz, J., Carroll, J., Yamauchi, Y., Neitz, M., & Williams, D. R. (2002). Color perception is mediated by a plastic neural mechanism that is adjustable in adults. *Neuron*, 35(4), 783–792.

[338] Bonnet, C. (1760). *Essai analytique sur les facultés de l'âme*. Copenhagen: Philibert; ffytche, D. H., Howard, R. J., Brammer, M. J., David, A., Woodruff, P., & Williams, S. (1998). The anatomy of conscious vision: An fMRI study of visual hallucinations. *Nature Neuroscience*, 1(8), 738–742.

experienced as sight. Phantom limbs are the same phenomenon in another modality: the body model persists after the body part is gone, and its predictions, unanswered, are felt.[339]

When the precision of evidence and prior is misweighted, perception bends. The hollow-mask illusion is weaker in schizophrenia, which suggests that priors there have lost some of their grip; at the same time, hallucinations and delusions can be understood as priors with too much precision, or as noise in the error channel that is read as signal. The two readings are not contradictory; a system can weight its priors too strongly at one level and too weakly at another.[340] One account of autistic perception proposes the opposite imbalance, sensory precision too high relative to the priors, so that the world is experienced in unfiltered detail and small deviations from expectation are overwhelming.[341] All of these are hypotheses, not settled findings, and they are offered here because they show what the framework buys: a vocabulary in which disorders of perception are disorders of weighting, and the weights are set by mechanisms that are known and can be measured: inhibition, acetylcholine, norepinephrine, and oscillatory gain, with dopamine, which the accounts of psychosis just cited place at the center, a candidate whose role in perceptual precision is argued rather than shown.

Implications for Artificial Intelligence

Perception is the domain in which artificial systems most clearly match the brain and in which the comparison is most instructive about what they lack.

First, the feedforward sweep has been reproduced. Deep convolutional networks trained to classify images develop a hierarchy of features, from oriented edges to object parts, that resembles the ventral stream closely enough to predict the responses of its neurons. The resemblance is not a coincidence; both are solutions to the same statistical problem, and the fact that gradient descent and evolution arrived at a similar place is evidence that the place is not arbitrary.[342] The resemblance is partial, though. Networks trained on photographs lean on texture where humans lean on shape, and they part company with IT on images outside the training distribution, so the shared solution covers the easy part of the problem rather than all of it.[343]

Second, the loops have not been. Standard networks have no feedback, no prediction, and no error signal at inference time; they map an input to an output in one pass. The brain spends most of its visual anatomy on the other direction. Adversarial examples, in which an imperceptible change to an image produces a confident misclassification, are sometimes described as evidence without priors, but that is not right: a trained classifier is nothing but priors, learned from millions of images and stored in its

[339] Ramachandran, V. S., & Hirstein, W. (1998). The perception of phantom limbs: The D. O. Hebb lecture. *Brain*, 121(9), 1603–1630.

[340] Fletcher, P. C., & Frith, C. D. (2009). Perceiving is believing: A Bayesian approach to explaining the positive symptoms of schizophrenia. *Nature Reviews Neuroscience*, 10(1), 48–58; Sterzer, P., Adams, R. A., Fletcher, P., Frith, C., Lawrie, S. M., Muckli, L., Petrovic, P., Uhlhaas, P., Voss, M., & Corlett, P. R. (2018). The predictive coding account of psychosis. *Biological Psychiatry*, 84(9), 634–643.

[341] Pellicano, E., & Burr, D. (2012). When the world becomes 'too real': A Bayesian explanation of autistic perception. *Trends in Cognitive Sciences*, 16(10), 504–510; Lawson, R. P., Rees, G., & Friston, K. J. (2014). An aberrant precision account of autism. *Frontiers in Human Neuroscience*, 8, 302.

[342] Yamins, D. L. K., & DiCarlo, J. J. (2016). Using goal-driven deep learning models to understand sensory cortex. *Nature Neuroscience*, 19(3), 356–365.

[343] Geirhos, R., Rubisch, P., Michaelis, C., Bethge, M., Wichmann, F. A., & Brendel, W. (2019). ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. *International Conference on Learning Representations (ICLR)*.

weights. What it lacks is a way to check them. The vulnerability is better explained by the near-linear behavior of these networks in high dimensions and by their reliance on features that predict the label but are invisible to us, and humans given only a brief glimpse of such images can be fooled by them too.[344] The difference is what happens next. A feedforward classifier has delivered its answer; a brain confronted with a doubtful image would predict a cat, compare the prediction with the input, find the error large, and look again.

Third, precision is missing. The attention mechanism of transformers weights inputs by learned relevance, which is a cousin of precision weighting, but there is nothing in most artificial systems that estimates how reliable each input is and discounts it accordingly, nor a global signal, like norepinephrine, that says the model should be distrusted now.[345] Systems that must operate on degraded sensors, in fog or in noise, need exactly this.

Fourth, generative models are the right direction. A network that can generate the data it classifies has, in effect, a prior and a likelihood and can compute something like a posterior; variational autoencoders and diffusion models are explicit attempts to build such machinery, and the best current explanations of cortex are, in essence, hierarchical generative models trained by prediction error.[346] Whether predictive coding can replace backpropagation as a learning rule is an open research question. The brain is an existence proof that some local rule suffices, not that this one does, and the strongest results so far show that predictive-coding networks can approximate backpropagation, not that they surpass it.[347]

Fifth, perception should be active. A system that chooses where to sample is solving an easier problem than one that must process everything, and robotics has begun to borrow the saccade, the forward model, and the efference copy for exactly the reasons the brain uses them.[348] An agent that cannot predict the sensory consequences of its own actions cannot tell what it did from what the world did, and that is a fatal confusion for anything that must learn from experience.

The Central Lesson

The senses deliver evidence that does not determine the world, and the brain supplies the rest. It holds a model of what is usually true, written into its synapses by the plasticity of Chapter 5, and it combines that model with the incoming evidence in proportion to how much each can be trusted. The combination is carried out by a hierarchy in which predictions descend and errors ascend, so that what

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[345] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.

[346] Dayan, P., Hinton, G. E., Neal, R. M., & Zemel, R. S. (1995). The Helmholtz machine. *Neural Computation*, 7(5), 889–904; Kingma, D. P., & Welling, M. (2014). Auto-encoding variational Bayes. *International Conference on Learning Representations (ICLR)*; Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33.

[347] Whittington, J. C. R., & Bogacz, R. (2017). An approximation of the error backpropagation algorithm in a predictive coding network with local Hebbian synaptic plasticity. *Neural Computation*, 29(5), 1229–1262; Millidge, B., Seth, A., & Buckley, C. L. (2022). Predictive coding: A theoretical and experimental review. *arXiv*, 2107.12979.

[348] Bajcsy, R., Aloimonos, Y., & Tsotsos, J. K. (2018). Revisiting active perception. *Autonomous Robots*, 42(2), 177–196.

the cortex transmits is not the image but its surprise. Attention sets the trust, the neuromodulators of Chapter 3 set the attention, and action chooses what evidence to gather next. Illusions, hallucinations, and the stability of the visual world are all the same mechanism seen from different sides. You do not see what is there. You see your best current hypothesis about what is there, and most of the time it is right.

Next Chapter

Chapter 7: Emotion and the Body: How Internal States Tell the Brain What Matters

Figure and Epigraph Sources

Figure Attribution

Figures 6.1, 6.2, 6.3, 6.4, 6.5, 6.7, 6.8, 6.9, and 6.10 were drawn for this book. The curves in Figures 6.2 and 6.8, the error trace in Figure 6.5B, and the receptive fields in Figure 6.10 are illustrative computations, not recorded data.

Figure 6.6: “Checker shadow illusion” by Edward H. Adelson, vectorized by Wikimedia Commons user Pbroks13, CC BY-SA 4.0, via Wikimedia Commons.

Epigraph

Hermann von Helmholtz, *Treatise on Physiological Optics*, vol. III, §26, trans. J. P. C. Southall (Optical Society of America, 1925); first published in German in 1867.

Chapter 7

Emotion and the Body: How Internal States Tell the Brain What Matters

> *"If we fancy some strong emotion, and then try to abstract from our consciousness of it all the feelings of its characteristic bodily symptoms, we find we have nothing left behind, no 'mind-stuff' out of which the emotion can be constituted, and that a cold and neutral state of intellectual perception is all that remains."*
>
> *— William James, "What Is an Emotion?" (1884)*

What Perception Leaves Out

Chapter 6 ended with a brain that builds its best current hypothesis about the world from noisy evidence and learned expectation. That brain can tell a face from a mask, but nothing in its machinery says whether the face belongs to a friend or a threat. Perception answers the question of what is out there. It does not answer the question every animal must answer next, which is whether what is out there matters, and how much.

The answer has to come from the one thing the brain is built to protect, the body that carries it. Food matters to a hungry animal and not to a sated one, and a stranger matters more at night than at noon, because value is relative to the state of the body and its prospects. The brain therefore needs a second model alongside its model of the world: a model of the body itself, of how it stands now and where it is heading. Chapter 1 described the brain as a controller that keeps a body's internal variables within bounds; this chapter takes the description literally.

The argument of the chapter, which most of the competing theories of emotion can accept, is that emotions are at their core the brain's running estimate of the state of the body and of what the current situation means for it. They do three jobs at once: they prepare the body for what is predicted to come, they rank the options for action, and they tell the machinery of learning described in Chapters 3 and 5 which experiences are worth keeping.

Keeping the Body in Bounds

In the 1850s and 1860s the French physiologist Claude Bernard argued that the cells of a complex animal do not live in the outside world at all. They live in the fluid that bathes them, the *milieu intérieur*, and, in his phrase, the condition of a free and independent life is that this inner environment be held constant while the outer one varies.[349] Walter Cannon gave the idea its modern name in the 1920s. He called it homeostasis, the maintenance of the body's internal variables near stable values by mechanisms that detect deviations and act to reverse them.[350]

The ranges are narrow. Core body temperature in a healthy adult stays within about a degree of 37 degrees Celsius; a rise to 41 degrees is a medical emergency, and a fall below 35 degrees is hypothermia. Arterial blood is held between pH 7.35 and 7.45. Fasting blood glucose sits around 4 to 6 millimoles per

[349] Gross, C. G. (1998). Claude Bernard and the constancy of the internal environment. *The Neuroscientist*, 4(5), 380–385.

[350] Cannon, W. B. (1929). Organization for physiological homeostasis. *Physiological Reviews*, 9(3), 399–431; Cannon, W. B. (1932). *The Wisdom of the Body*. W. W. Norton.

liter, and the brain, which in the fed state runs mostly on glucose and stores very little, begins to fail within minutes if it falls far below that.[351] Each of these variables is defended by a negative feedback loop of the kind an engineer would recognize: a sensor, a comparison with a set point, and an effector that pushes the variable back.

The hub of those loops in the brain is the hypothalamus, a structure of about four grams at the base of the forebrain. Its neurons sense blood temperature, glucose, and osmolarity (the concentration of salts and other solutes), and it receives signals from the gut, the heart, and the immune system. It corrects through three channels: the autonomic nervous system, whose sympathetic and parasympathetic branches speed and slow the heart, the blood vessels, and digestion; the pituitary gland, which controls the hormones of stress, growth, reproduction, and water balance; and behavior, by making the animal hungry, thirsty, or cold enough to fix the problem itself.[352]

That third channel is where regulation becomes emotion. In the 1920s Cannon and Sydney Britton found that cats whose cerebral cortex had been removed flew into fits of rage at the slightest touch, hissing, their hearts racing. They named the display sham rage, because it looked like anger without any sign of an object.[353] In 1928 Philip Bard showed that the display disappeared only when the posterior hypothalamus was removed as well.[354] More than 80 years later David Anderson's laboratory found a small population of neurons in the ventromedial hypothalamus of male mice that, when stimulated with light, made the animal attack whatever was in front of it, including a female or an inflated rubber glove.[355] The hypothalamus contains circuits for the whole-body programs of attack, flight, mating, and feeding, which are the oldest forms of emotional behavior.

Regulation by Prediction

A thermostat waits for the room to get cold. Much of the body's regulation works the same way: when you stand up, a reflex driven by pressure sensors in the great arteries, the baroreflex, detects the fall in blood pressure and corrects it within a heartbeat or two. But the body also regulates ahead of need. In 1913 August Krogh and Johannes Lindhard showed that heart rate and breathing rise within a second or two of the start of exercise, sooner than any chemical change in the blood could signal the need.[356] The command to the heart goes out with the command to the legs.

In 1988 Peter Sterling and Joseph Eyer called this predictive regulation allostasis, stability through change: the brain anticipates what the body will need and adjusts its physiology in advance, rather than waiting for an error and correcting it.[357] Douglas Ramsay and Stephen Woods argue that this is

[351] Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (Eds.). (2021). *Principles of Neural Science* (6th ed.). McGraw-Hill. Kandel and colleagues, the chapter on the hypothalamus.

[352] Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (Eds.). (2021). *Principles of Neural Science* (6th ed.). McGraw-Hill. Kandel and colleagues, the chapter on the hypothalamus.

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[357] Sterling, P., & Eyer, J. (1988). Allostasis: A new paradigm to explain arousal pathology. In S. Fisher & J. Reason (Eds.), *Handbook of Life Stress, Cognition and Health* (pp. 629–649). John Wiley & Sons; Sterling, P. (2012). Allostasis: A model of

homeostasis with good forecasting rather than a new principle, since anticipatory responses still defend set points.[358] Both sides accept that the brain regulates the body largely by learned prediction, the strategy Chapter 6 described for perception, turned inward.

Eating is the clearest example. The sight, smell, and taste of food release insulin before any glucose has been absorbed. This is a cephalic phase response, so called because it starts in the head, and Woods argued that eating would be dangerously destabilizing without it: a meal is a large disturbance, and the body tolerates it only because the brain prepares for it.[359] Figure 7.1 shows the difference prediction makes: a controller that releases insulin on the cue that a meal is coming acts before the disturbance arrives, while a purely reactive one lets glucose swing far from its set point.

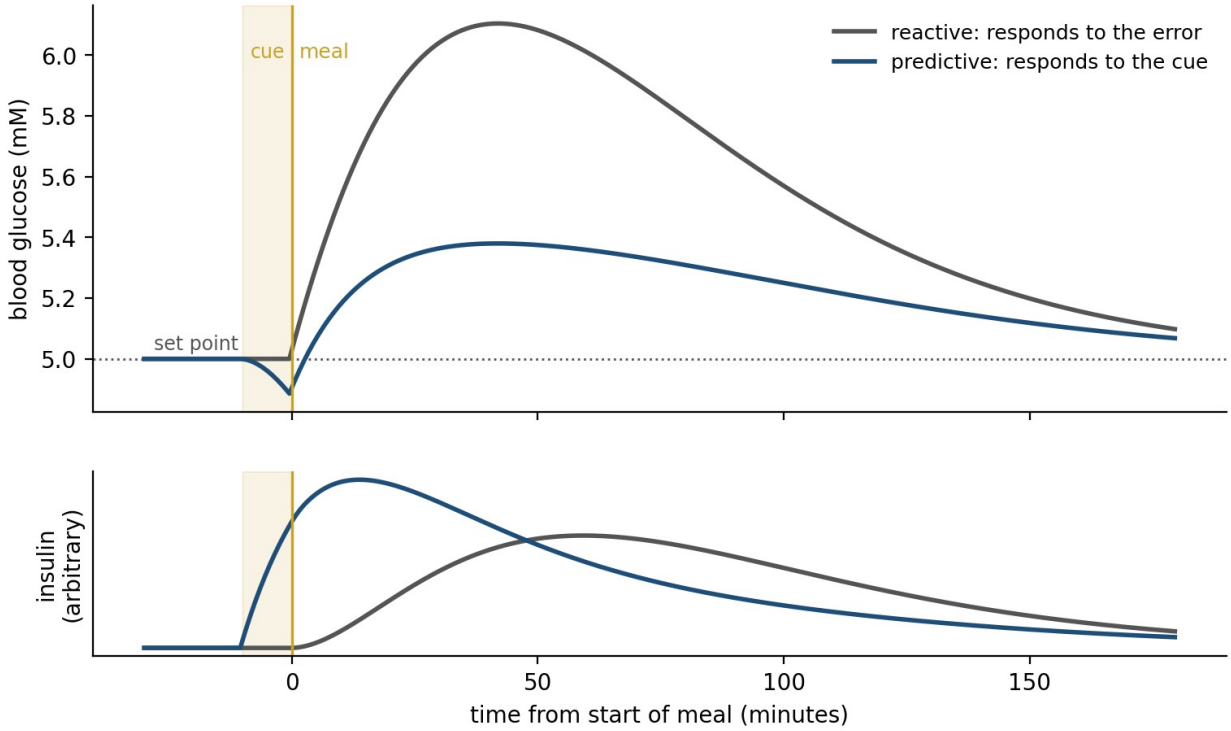


Figure 7.1. Reactive and predictive regulation (simulated). A meal arrives at time zero and pushes blood glucose upward. Gray: a controller that responds only to the error swings far from the set point and corrects late. Blue: a controller that releases insulin on the cue that a meal is coming holds glucose near its set point; its early insulin briefly lowers glucose before the meal arrives.

The anticipation reaches into the circuits that drive behavior. Hunger in the mouse is promoted by a small population of neurons in the arcuate nucleus of the hypothalamus that express a peptide called AgRP. Zachary Knight's laboratory imaged these neurons and found that their activity fell within seconds of a hungry mouse seeing or smelling food, before it had eaten anything, and that the size of the

predictive regulation. *Physiology & Behavior*, 106(1), 5–15.

[358] Ramsay, D. S., & Woods, S. C. (2014). Clarifying the roles of homeostasis and allostasis in physiological regulation. *Psychological Review*, 121(2), 225–247.

[359] Woods, S. C. (1991). The eating paradox: How we tolerate food. *Psychological Review*, 98(4), 488–505.

fall depended on how hungry the mouse was and how palatable the food.[360] The hunger signal does not track the body's present state. It tracks the state the brain predicts.

Prediction has a cost when it fails, and the clearest demonstration is a grim one. Shepard Siegel gave rats a series of heroin injections in one room and then a large test dose in the same room or a different one. Of rats that had never had heroin, 96 percent died; of experienced rats, 32 percent died in the familiar room and 64 percent in an unfamiliar one.[361] Part of tolerance is learned anticipation: familiar cues tell the body the drug is coming, and without them the body is unprepared, which is thought to contribute to overdoses in people who take a familiar dose in an unfamiliar place.

Sensing the Inside

A controller needs measurements, and the body supplies them through a sense that receives far less attention than vision or hearing. Charles Sherrington called the senses of the viscera interoceptive in 1906,[362] and the word now covers the sensing of the whole body's condition: heartbeat, breathing, stretch of the stomach and bladder, temperature, blood chemistry, inflammation, itch, and pain.[363]

The signals reach the brain by the routes shown in Figure 7.2. The vagus nerve, which wanders from the brainstem to the heart, lungs, and gut, is mostly sensory: roughly four of every five of its fibers carry signals toward the brain rather than away from it.[364] Its sensory fibers end in the nucleus of the solitary tract in the brainstem. A second route runs through the spinal cord, where thin fibers from the tissues of the body end in the dorsal horn, many of them in its most superficial layer, lamina I. A third bypasses nerves altogether: at a few small sites where the blood-brain barrier is leaky, called circumventricular organs, the brain samples the blood directly. These streams pass through relays in the brainstem and thalamus, feeding the hypothalamus and amygdala on the way.

Their cortical target is the insula, a lobe of cortex folded deep inside the lateral fissure. Arthur Craig proposed that the posterior insula holds a primary map of the body's internal state, and that the map is re-represented, step by step, toward the front of the insula, where it is combined with information about the environment, motivation, and social context.[365]

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[360] Chen, Y., Lin, Y.-C., Kuo, T.-W., & Knight, Z. A. (2015). Sensory detection of food rapidly modulates arcuate feeding circuits. *Cell*, 160(5), 829–841.

[361] Siegel, S., Hinson, R. E., Krank, M. D., & McCully, J. (1982). Heroin "overdose" death: Contribution of drug-associated environmental cues. *Science*, 216(4544), 436–437.

[362] Sherrington, C. S. (1906). *The Integrative Action of the Nervous System*. Charles Scribner's Sons.

[363] Craig, A. D. (2002). How do you feel? Interoception: The sense of the physiological condition of the body. *Nature Reviews Neuroscience*, 3(8), 655–666; Chen, W. G., Schloesser, D., Arensdorf, A. M., Simmons, J. M., Cui, C., Valentino, R., Gnadt, J. W., Nielsen, L., Hillaire-Clarke, C. St., Spruance, V., Horowitz, T. S., Vallejo, Y. F., & Langevin, H. M. (2021). The emerging science of interoception: Sensing, integrating, interpreting, and regulating signals within the self. *Trends in Neurosciences*, 44(1), 3–16.

[364] Berthoud, H.-R., & Neuhuber, W. L. (2000). Functional and chemical anatomy of the afferent vagal system. *Autonomic Neuroscience*, 85(1–3), 1–17.

[365] Craig, A. D. (2002). How do you feel? Interoception: The sense of the physiological condition of the body. *Nature Reviews Neuroscience*, 3(8), 655–666; Craig, A. D. (2009). How do you feel—now? The anterior insula and human awareness. *Nature Reviews Neuroscience*, 10(1), 59–70.

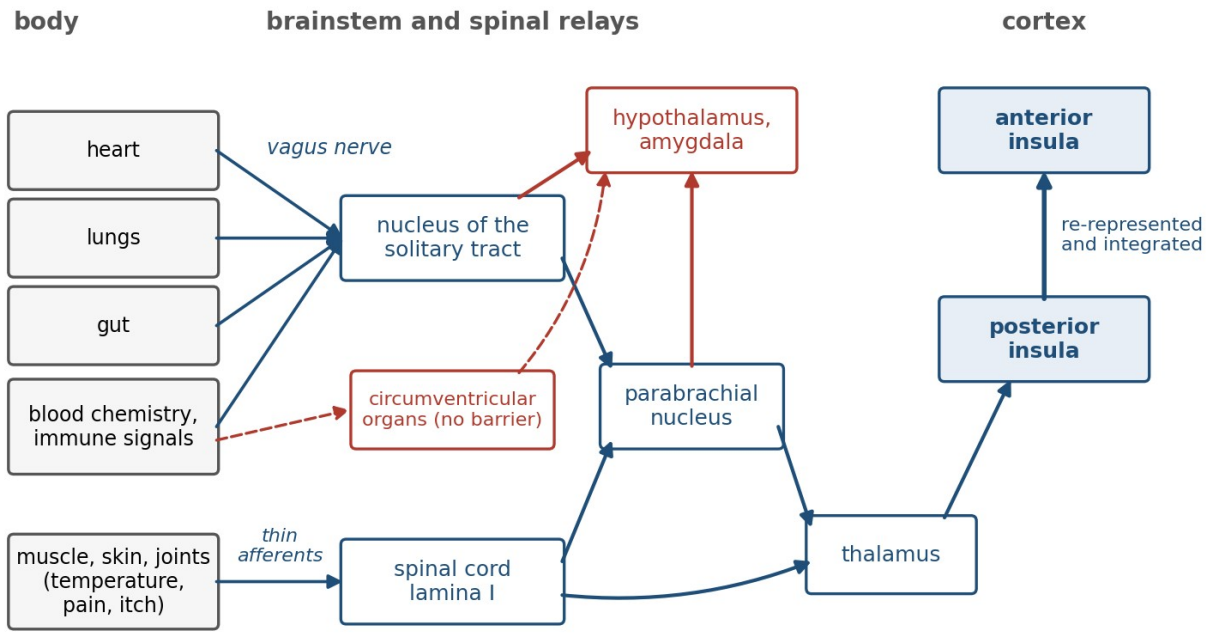


Figure 7.2. The interoceptive pathways. Signals from the body reach the brain through the vagus nerve, the spinal cord, and sites where the blood-brain barrier is leaky, and are relayed to the hypothalamus, the amygdala, and the insula.

How well people sense their own bodies can be measured, and it varies widely. In the heartbeat counting task devised by Rainer Schandry, participants silently count their heartbeats over a short interval without touching their pulse; some are nearly exact and others miss half their beats.[366] Hugo Critchley and colleagues found that the activity and the gray matter volume of the right anterior insula predicted how accurately people performed a related task.[367] The measures are imperfect, since counting can be done partly by estimating time, and accuracy and confidence need not agree.[368]

Interoception carries chemistry as well as mechanics. When the body fights an infection, immune cells release signaling proteins called cytokines that reach the brain through the vagus nerve and through the circumventricular organs. The result is sickness behavior: fatigue, loss of appetite and of pleasure, withdrawal from company, and heightened pain.[369] Benjamin Hart argued that this is a strategy, not a weakness: the brain redirects energy toward fever and repair by changing what feels worth doing.[370]

Like every other sense, interoception is an inference, and the brain's expectations shape what it reports. In 1978 Jon Levine, Newton Gordon, and Howard Fields gave patients who had just had wisdom teeth

[366] Schandry, R. (1981). Heart beat perception and emotional experience. *Psychophysiology*, 18(4), 483–488.

[367] Critchley, H. D., Wiens, S., Rotshtein, P., Öhman, A., & Dolan, R. J. (2004). Neural systems supporting interoceptive awareness. *Nature Neuroscience*, 7(2), 189–195.

[368] Garfinkel, S. N., Seth, A. K., Barrett, A. B., Suzuki, K., & Critchley, H. D. (2015). Knowing your own heart: Distinguishing interoceptive accuracy from interoceptive awareness. *Biological Psychology*, 104, 65–74.

[369] Dantzer, R., O'Connor, J. C., Freund, G. G., Johnson, R. W., & Kelley, K. W. (2008). From inflammation to sickness and depression: When the immune system subjugates the brain. *Nature Reviews Neuroscience*, 9(1), 46–56.

[370] Hart, B. L. (1988). Biological basis of the behavior of sick animals. *Neuroscience & Biobehavioral Reviews*, 12(2), 123–137.

removed, and who knew that they might receive a placebo, an injection of saline. Some reported less pain, and in them the relief was reversed by naloxone, a drug that blocks opioid receptors.[371] The expectation of relief had released the brain's own opioids and turned down the pain signal on its way to the brain.

Anil Seth and, separately, Lisa Feldman Barrett and Kyle Simmons have extended the predictive coding account of Chapter 6 to interoception. On their view, regions that regulate the body continuously predict its state, and the insula compares the predictions with the signals arriving from the viscera, so that what is felt is the brain's best estimate rather than a direct reading.[372] The proposal is less tested than its counterpart in vision, though Ian Kleckner and colleagues have traced a cingulate and insular network that both regulates the viscera and hears back from them.[373]

What an Emotion Is

William James asked the question in 1884 and answered it in a way that still provokes. Common sense says that we meet a bear, feel afraid, and run. James said the order was wrong: the perception of the bear sets off the bodily changes directly, the pounding heart, the tensed muscles, the flight, and the feeling of fear is our perception of those changes as they occur. "We feel sorry because we cry, angry because we strike, afraid because we tremble." Subtract the body, James argued, and the emotion has nothing left to be made of.[374] Carl Lange made a similar proposal the next year, hence the James-Lange theory.

Cannon attacked it in 1927. Cutting the nerves between viscera and brain does not abolish emotional behavior; the same visceral changes accompany different emotions, and states that are not emotions at all, such as fever and exercise; the viscera are poorly innervated and slow; and an injection of epinephrine (also called adrenaline), the adrenal hormone and a close chemical relative of Chapter 3's norepinephrine, makes people feel as if they were afraid without making them afraid.[375]

Most of the objections have since been qualified: visceral feedback is not too slow for a feeling that rises over seconds, and the viscera are richly supplied with sensory fibers that Cannon could not see. The most direct modern test of James comes from a mouse. Karl Deisseroth's laboratory paced the hearts of freely moving mice with light, through a light-sensitive channel expressed in heart muscle, raising the heart rate without touching the brain. In risky situations, though not in safe ones, the mice became more anxious, and the effect depended on the posterior insula.[376]

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[371] Levine, J. D., Gordon, N. C., & Fields, H. L. (1978). The mechanism of placebo analgesia. *The Lancet*, 312(8091), 654–657.

[372] Seth, A. K. (2013). Interoceptive inference, emotion, and the embodied self. *Trends in Cognitive Sciences*, 17(11), 565–573; Barrett, L. F., & Simmons, W. K. (2015). Interoceptive predictions in the brain. *Nature Reviews Neuroscience*, 16(7), 419–429.

[373] Kleckner, I. R., Zhang, J., Touroutoglou, A., Chanes, L., Xia, C., Simmons, W. K., Quigley, K. S., Dickerson, B. C., & Barrett, L. F. (2017). Evidence for a large-scale brain system supporting allostasis and interoception in humans. *Nature Human Behaviour*, 1(5), 0069.

[374] James, W. (1884). What is an emotion? *Mind*, 9(34), 188–205.

[375] Cannon, W. B. (1927). The James-Lange theory of emotions: A critical examination and an alternative theory. *American Journal of Psychology*, 39(1/4), 106–124.

[376] Hsueh, B., Chen, R., Jo, Y., Tang, D., Raffiee, M., Kim, Y. S., Inoue, M., Randles, S., Ramakrishnan, C., Patel, S., Kim, D. K., Liu, T. X., Kim, S. H., Tan, L., Mortazavi, L., Cordero, A., Shi, J., Zhao, M., Ho, T. T., … Deisseroth, K. (2023). Cardiogenic control of affective behavioural state. *Nature*, 615(7951), 292–299.

In people, Sarah Garfinkel and colleagues found that fearful faces flashed at the moment the heart contracts are detected more readily and judged more intense than the same faces shown between beats.[377] Tests in people that remove or impose bodily signals have given smaller and less consistent effects. People with high spinal cord injuries, whose brains hear little from the body below the neck, were once reported to feel emotions less intensely, but a better-controlled study found little reduction.[378] And the claim that a forced smile makes cartoons funnier failed a replication run in 17 laboratories, although a meta-analysis finds a small effect of facial feedback across many studies.[379] The body is not the whole of an emotion, but signals from it change both what is felt and what is perceived, and the context decides what they mean.

If the body supplies arousal, something else must say what it is about. In 1962 Stanley Schachter and Jerome Singer injected volunteers with epinephrine, told some of them what side effects to expect and others nothing, and left each with a confederate who acted either euphoric or angry. The uninformed volunteers, who had no explanation for their racing hearts, tended to take on the mood of the confederate.[380] Later attempts found weaker effects, and unexplained arousal tended simply to feel unpleasant.[381] But the idea that arousal is interpreted survived. On the Capilano suspension bridge in Vancouver, 70 meters above a river, Donald Dutton and Arthur Aron had a female interviewer give men her phone number; half of the men who took the number on the bridge called her, against about one in eight of those interviewed on a low, solid bridge upstream.[382]

Three positions on what an emotion is now compete. On the basic emotion view, associated with Paul Ekman, there is a small set of emotions (fear, anger, disgust, sadness, happiness, and surprise among them), each an evolved program with its own bodily pattern and a facial expression recognizable across cultures.[383] On the constructionist view, developed by Barrett, there are no dedicated emotion programs; the brain continuously predicts and regulates the state of the body, and a particular emotion is the brain's categorization of that state in context, using concepts learned from experience and culture.[384]

A meta-analysis of more than 200 studies by Erika Siegel and colleagues found that the autonomic measures recorded during different emotions overlapped too much to identify the emotion reliably, which favors the constructionist account.[385] A review led by Barrett, with coauthors from both sides of

[377] Garfinkel, S. N., Minati, L., Gray, M. A., Seth, A. K., Dolan, R. J., & Critchley, H. D. (2014). Fear from the heart: Sensitivity to fear stimuli depends on individual heartbeats. *Journal of Neuroscience*, 34(19), 6573–6582.

[378] Chwalisz, K., Diener, E., & Gallagher, D. (1988). Autonomic arousal feedback and emotional experience: Evidence from the spinal cord injured. *Journal of Personality and Social Psychology*, 54(5), 820–828.

[379] Wagenmakers, E.-J., Beek, T., Dijkhoff, L., Gronau, Q. F., Acosta, A., Adams, R. B., Jr., … Zwaan, R. A. (2016). Registered replication report: Strack, Martin, & Stepper (1988). *Perspectives on Psychological Science*, 11(6), 917–928; Coles, N. A., Larsen, J. T., & Lench, H. C. (2019). A meta-analysis of the facial feedback literature: Effects of facial feedback on emotional experience are small and variable. *Psychological Bulletin*, 145(6), 610–651.

[380] Schachter, S., & Singer, J. E. (1962). Cognitive, social, and physiological determinants of emotional state. *Psychological Review*, 69(5), 379–399.

[381] Marshall, G. D., & Zimbardo, P. G. (1979). Affective consequences of inadequately explained physiological arousal. *Journal of Personality and Social Psychology*, 37(6), 970–988.

[382] Dutton, D. G., & Aron, A. P. (1974). Some evidence for heightened sexual attraction under conditions of high anxiety. *Journal of Personality and Social Psychology*, 30(4), 510–517.

[383] Ekman, P. (1992). An argument for basic emotions. *Cognition and Emotion*, 6(3–4), 169–200.

[384] Barrett, L. F. (2017). The theory of constructed emotion: An active inference account of interoception and categorization. *Social Cognitive and Affective Neuroscience*, 12(1), 1–23.

[385] Siegel, E. H., Sands, M. K., Van den Noortgate, W., Condon, P., Chang, Y., Dy, J., Quigley, K. S., & Barrett, L. F. (2018). Emotion fingerprints or emotion populations? A meta-analytic investigation of autonomic features of emotion categories. *Psychological*

the debate, concluded that facial movements are weak signs of emotion: people scowl when angry more often than at other times, but only in a minority of instances, and they scowl in many states that are not anger.[386]

The third position, argued by Joseph LeDoux, separates what is under dispute from what is not. Brains contain survival circuits (for defense, feeding, reproduction, and fluid balance) that detect relevant situations and organize the body's response, and these are well studied in animals. Whether a freezing rat feels fear as a person does is a different question, and LeDoux keeps the word fear for the conscious feeling, which in humans depends on cortical circuits that interpret the survival circuits.[387]

All three positions can accept, as a description, the map of Figure 7.3. James Russell found in 1980 that people's ratings of emotion words fall on a circle whose axes are valence, how pleasant or unpleasant a state is, and arousal, how activated or calm.[388] These two dimensions, often called core affect, are where the interoceptive estimate of the body is most directly felt.

Bulletin, 144(4), 343–393.

[386] Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., & Pollak, S. D. (2019). Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements. *Psychological Science in the Public Interest*, 20(1), 1–68.

[387] LeDoux, J. (2012). Rethinking the emotional brain. *Neuron*, 73(4), 653–676; LeDoux, J. E., & Pine, D. S. (2016). Using neuroscience to help understand fear and anxiety: A two-system framework. *American Journal of Psychiatry*, 173(11), 1083–1093.

[388] Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161–1178.

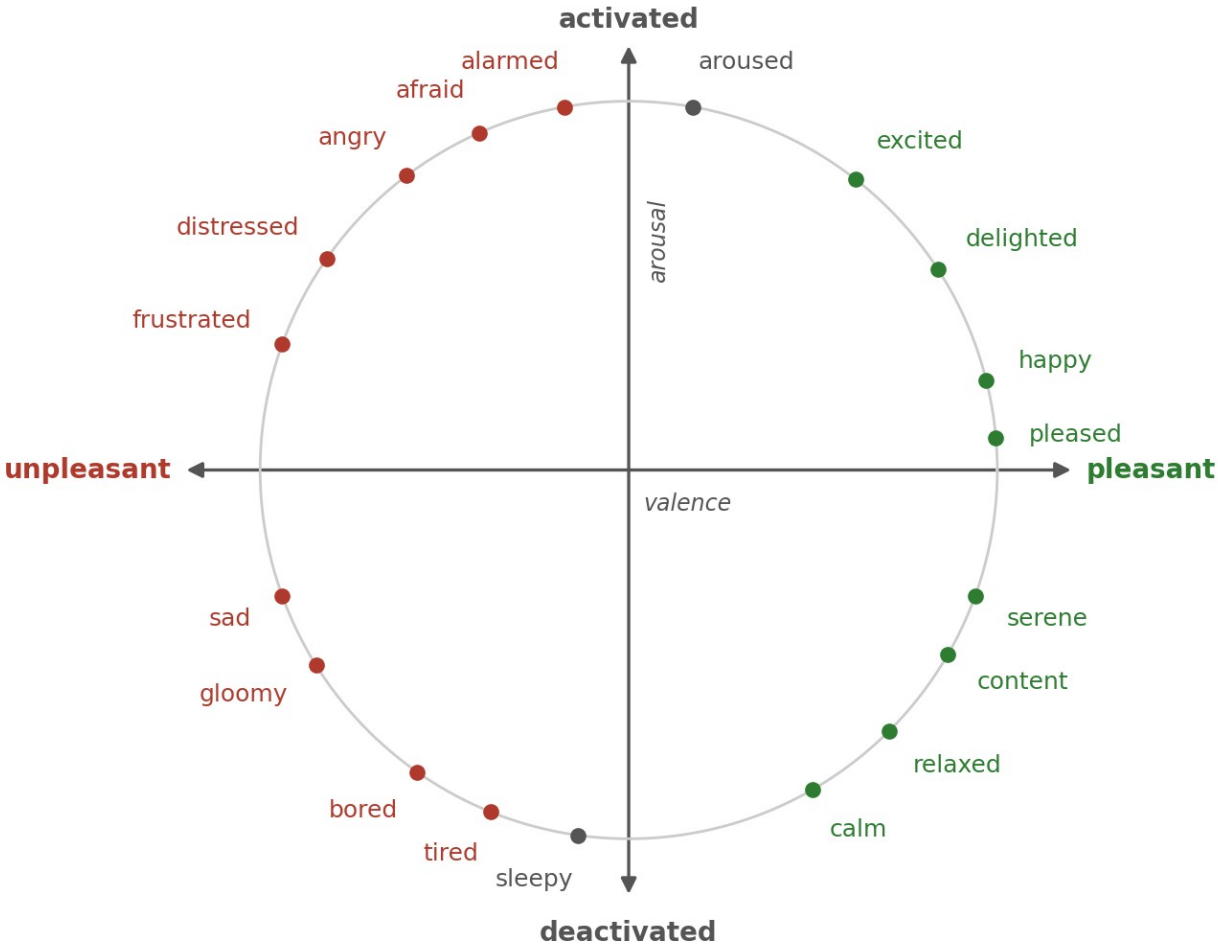


Figure 7.3. The affective circumplex. Emotion words rated by people fall roughly on a circle defined by two axes: valence, from unpleasant to pleasant, and arousal, from calm to activated. Positions are approximate and illustrative.

Value Depends on the Body

If emotions report what matters to the body, then the value of anything should change when the body's state changes, and it does. In 1971 Michel Cabanac had fasting volunteers taste a sweet solution and rate how pleasant it was. They liked it. Then they drank a load of glucose, and the same solution, tasted again, was rated unpleasant.[389] Cabanac called this alliesthesia: the pleasure of a stimulus is not a property of the stimulus but of its usefulness to the body at that moment. Within a meal, the food being eaten loses its pleasantness faster than foods not yet eaten, which Barbara Rolls and colleagues called sensory-specific satiety and which is why there is room for dessert.[390]

Internal state also supplies the brain with teaching signals. Scott Sternson's laboratory found that the AgRP hunger neurons and their counterparts for thirst carry a negative signal: mice learn to avoid a flavor or a place that has been paired with stimulation of these neurons, as if the hunger state itself

[389] Cabanac, M. (1971). Physiological role of pleasure. *Science*, 173(4002), 1103–1107.

[390] Rolls, B. J., Rolls, E. T., Rowe, E. A., & Sweeney, K. (1981). Sensory specific satiety in man. *Physiology & Behavior*, 27(1), 137–142.

were aversive, and eating, which silences the neurons, is reinforcing partly because it ends that state.[391] The reward prediction error of Chapter 3 is computed against outcomes whose value depends on the state of the body: a drop of juice is a large reward to a thirsty monkey and a small one to a sated monkey.

Reward itself has more than one part. Rats whose dopamine had been almost entirely depleted no longer worked for food but still made the facial reactions of liking when sugar was placed in their mouths. On this evidence Kent Berridge and Terry Robinson proposed separating wanting, the dopamine-dependent pull of a cue, from liking, the pleasure of the reward itself, which survives the loss of dopamine.[392] Wanting and liking usually move together, and their divergence, in addiction especially, is one way that what an animal pursues can come apart from what it enjoys.

The Amygdala and Learned Threat

Much of what matters to an animal is not in its body yet. A sound in the grass is harmless in itself, but if a predator makes it, the body should be ready before the predator arrives. Learning such predictions is the business of a small cluster of nuclei deep in the temporal lobe, the amygdala, named from the Greek for almond.

Its importance was first seen in its absence. In 1939 Heinrich Klüver and Paul Bucy removed the temporal lobes of normally wary rhesus monkeys and found them transformed: tame, unafraid of snakes or of people, putting everything in their mouths, and repeatedly approaching things that had hurt them.[393] Later work narrowed much of the change in fear to the amygdala, and its mechanism was worked out in the rat by LeDoux and others using fear conditioning, a form of Pavlovian learning in which a tone is followed by a mild shock to the feet. After a few pairings the tone alone makes the rat freeze, its blood pressure rise, and its stress hormones flow.[394]

Figure 7.4 shows the circuit. The tone reaches the lateral nucleus of the amygdala through two routes: directly from the auditory thalamus, a short path that carries a coarse signal, and through the auditory cortex, a longer path that carries a detailed one. The shock reaches the same neurons through pain pathways. Where the two converge, the synapses carrying the tone are strengthened by the associative long-term potentiation, or LTP, of Chapter 5, and the tone acquires the power to drive the amygdala on its own. The lateral nucleus passes the signal to the central nucleus, whose outputs, directly or through relays, reach the targets that make up the defensive response: the periaqueductal gray of the midbrain for freezing, the hypothalamus for the rise in blood pressure and for the stress hormones, and the brainstem nuclei that release the neuromodulators of Chapter 3.[395]

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[391] Betley, J. N., Xu, S., Cao, Z. F. H., Gong, R., Magnus, C. J., Yu, Y., & Sternson, S. M. (2015). Neurons for hunger and thirst transmit a negative-valence teaching signal. *Nature*, 521(7551), 180–185.

[392] Berridge, K. C., & Robinson, T. E. (1998). What is the role of dopamine in reward: Hedonic impact, reward learning, or incentive salience? *Brain Research Reviews*, 28(3), 309–369.

[393] Klüver, H., & Bucy, P. C. (1939). Preliminary analysis of functions of the temporal lobes in monkeys. *Archives of Neurology and Psychiatry*, 42(6), 979–1000.

[394] LeDoux, J. E. (2000). Emotion circuits in the brain. *Annual Review of Neuroscience*, 23, 155–184.

[395] LeDoux, J. E. (2000). Emotion circuits in the brain. *Annual Review of Neuroscience*, 23, 155–184; Tovote, P., Fadok, J. P., & Lüthi, A. (2015). Neuronal circuits for fear and anxiety. *Nature Reviews Neuroscience*, 16(6), 317–331.

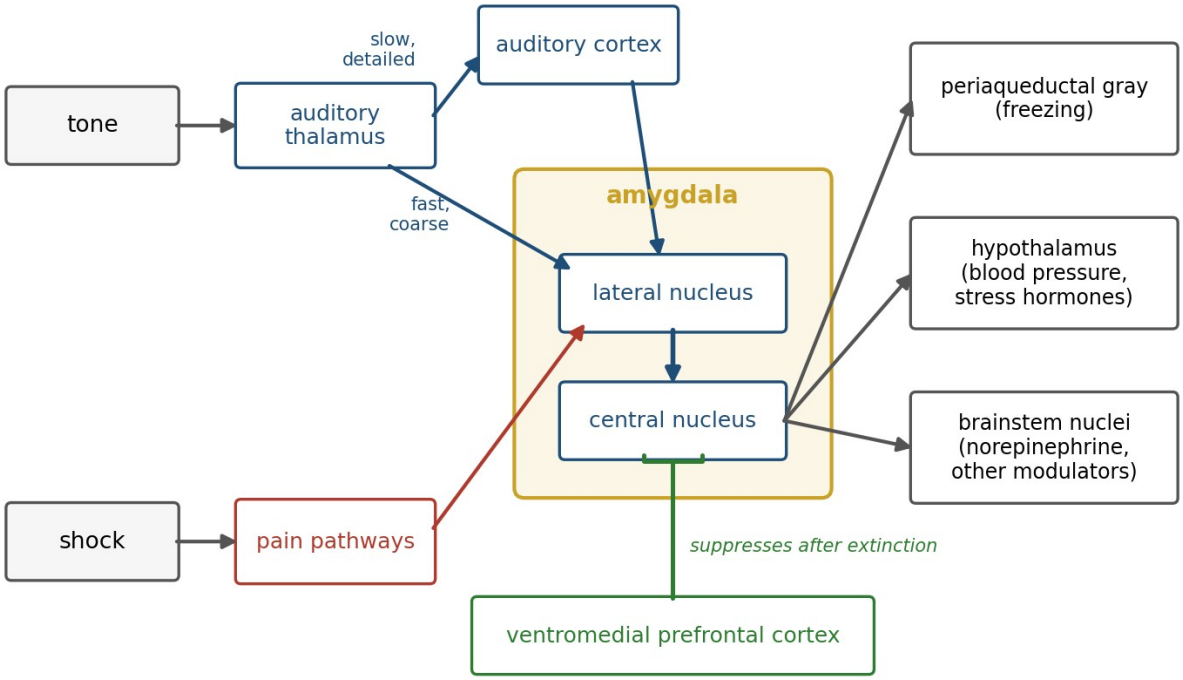


Figure 7.4. The circuit for learned threat. Tone and shock converge on the lateral amygdala, where pairing strengthens the tone's synapses; the central nucleus organizes the response, and after extinction the ventromedial prefrontal cortex suppresses it.

The short route became famous as a low road that lets fear outrun thought. In rats either route alone suffices for simple tone conditioning.[396] Luiz Pessoa and Ralph Adolphs argue that in primates the subcortical route has not been shown to be faster in practice or capable of fine discrimination, and that the amygdala is one hub among many roads by which the brain evaluates significance.[397] On the other side, electrodes in the amygdalae of epilepsy patients recorded responses to fearful faces about 74 milliseconds after the image appeared, driven by the coarse features a fast route would carry.[398] Both routes exist; how much the short one contributes in humans is unsettled.

The amygdala is not a fear center. In monkeys whose visual cues predicted juice or an air puff, Daniel Salzman's laboratory found amygdala cells signaling positive value and cells signaling negative value, each changing its response when the meaning of the cue was reversed.[399] It learns the value of cues in both directions, and fear is its best studied use rather than its only one.[400] Nor does it work alone. A

[396] Romanski, L. M., & LeDoux, J. E. (1992). Equipotentiality of thalamo-amygdala and thalamo-cortico-amygdala circuits in auditory fear conditioning. *Journal of Neuroscience*, 12(11), 4501–4509.

[397] Pessoa, L., & Adolphs, R. (2010). Emotion processing and the amygdala: From a 'low road' to 'many roads' of evaluating biological significance. *Nature Reviews Neuroscience*, 11(11), 773–782.

[398] Méndez-Bértolo, C., Moratti, S., Toledano, R., Lopez-Sosa, F., Martínez-Alvarez, R., Mah, Y. H., Vuilleumier, P., Gil-Nagel, A., & Strange, B. A. (2016). A fast pathway for fear in human amygdala. *Nature Neuroscience*, 19(8), 1041–1049.

[399] Paton, J. J., Belova, M. A., Morrison, S. E., & Salzman, C. D. (2006). The primate amygdala represents the positive and negative value of visual stimuli during learning. *Nature*, 439(7078), 865–870.

[400] Janak, P. H., & Tye, K. M. (2015). From circuits to behaviour in the amygdala. *Nature*, 517(7534), 284–292.

brief, cued fear depends on the central nucleus, but a sustained state of anxiety about threats that are uncertain or distant depends more on a neighboring structure, the bed nucleus of the stria terminalis.[401]

The most instructive human evidence comes from a woman known as S.M., whose amygdalae were destroyed on both sides by a rare genetic disease, Urbach-Wiethe disease. Justin Feinstein and colleagues took her to an exotic pet store, where she handled snakes she had said she disliked, to a haunted house, where she frightened the actors, and through horror films, and she showed and reported almost no fear.[402]

Later, the same group had S.M. and two other patients with damage to both amygdalae inhale air containing 35 percent carbon dioxide, which acidifies the blood and triggers a strong sensation of suffocation. All three had a panic attack, while only three of 12 healthy volunteers did.[403] The amygdala is needed for fear of threats from outside the body. A threat from inside the body, detected by chemical sensors outside the amygdala, probably in the brainstem, produces fear without it.

Unlearning Fear

A threat prediction that could only be learned would be a trap, since the world changes. If the tone is played many times without shock, the rat stops freezing, a process called extinction. Extinction is not forgetting. Mark Bouton showed that an extinguished fear returns in a different context, with the passage of time, or after an unsignaled shock, so the original association must still be there.[404] Extinction is new learning: a second memory, that the tone is now safe in this place, that competes with the first. Exposure therapy for phobias is extinction carried out in the clinic, and it shares extinction's weakness: the fear can come back.

The second memory has its own circuit. Mohammed Milad and Gregory Quirk recorded in the infralimbic region of the rat's medial prefrontal cortex and found neurons that fired to the tone on the day after extinction in proportion to how well the extinction was remembered; stimulating the region while playing the tone reduced freezing in rats that had never been extinguished.[405] In people the corresponding area, the ventromedial prefrontal cortex, is engaged when an extinguished fear is recalled the next day.[406] The prefrontal cortex mostly inhibits the expression of the amygdala's memory rather than erasing it, which is why, as Chapter 5 noted, fear is easily learned and rarely unlearned, only suppressed.

People also regulate emotion deliberately. In a study led by Kevin Ochsner, participants who reinterpreted distressing photographs in less negative terms, a strategy called reappraisal, reported less negative feeling and showed more activity in the lateral and medial prefrontal cortex and less in the

[401] Davis, M., Walker, D. L., Miles, L., & Grillon, C. (2010). Phasic vs sustained fear in rats and humans: Role of the extended amygdala in fear vs anxiety. *Neuropsychopharmacology*, 35(1), 105–135.

[402] Feinstein, J. S., Adolphs, R., Damasio, A., & Tranel, D. (2011). The human amygdala and the induction and experience of fear. *Current Biology*, 21(1), 34–38.

[403] Feinstein, J. S., Buzza, C., Hurlemann, R., Follmer, R. L., Dahdaleh, N. S., Coryell, W. H., Welsh, M. J., Tranel, D., & Wemmie, J. A. (2013). Fear and panic in humans with bilateral amygdala damage. *Nature Neuroscience*, 16(3), 270–272.

[404] Bouton, M. E. (2004). Context and behavioral processes in extinction. *Learning & Memory*, 11(5), 485–494.

[405] Milad, M. R., & Quirk, G. J. (2002). Neurons in medial prefrontal cortex signal memory for fear extinction. *Nature*, 420(6911), 70–74.

[406] Phelps, E. A., Delgado, M. R., Nearing, K. I., & LeDoux, J. E. (2004). Extinction learning in humans: Role of the amygdala and vmPFC. *Neuron*, 43(6), 897–905.

amygdala.[407] Reappraisal acts early, by changing what the situation means, rather than by suppressing the response once it is under way;[408] it is extinction made voluntary.

The Chemistry of Stress

When the brain predicts a demand, it prepares the body through two chemical systems, which act on two timescales (Figure 7.5). The fast system is the sympathetic nervous system, which within seconds raises heart rate and blood pressure, opens the airways, and signals the adrenal medulla, the core of the adrenal gland above each kidney, to release epinephrine into the blood. The slow system is the hypothalamic-pituitary-adrenal axis, a three-step hormonal relay from the hypothalamus through the pituitary to the adrenal cortex, which releases cortisol. Cortisol peaks tens of minutes after a stressor begins, mobilizes glucose, alters immune function and memory, and acts back on the hypothalamus, pituitary, and hippocampus to shut its own release down.[409]

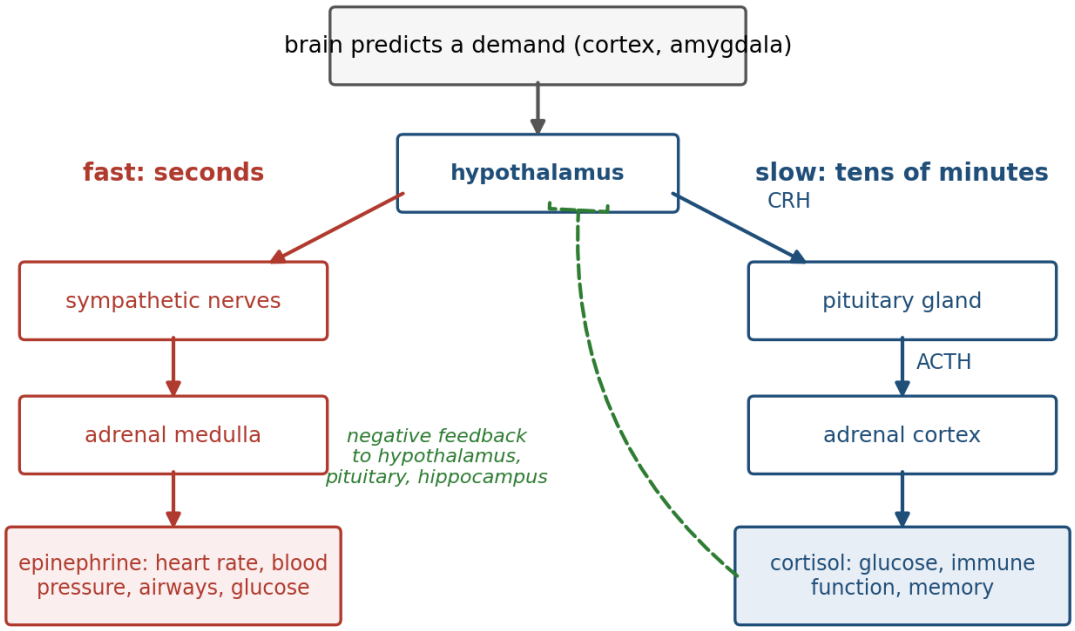


Figure 7.5. The two arms of the stress response. Fast: the sympathetic nervous system and epinephrine from the adrenal medulla, within seconds. Slow: the hypothalamic-pituitary-adrenal axis, from corticotropin-releasing hormone (CRH) to adrenocorticotropic hormone (ACTH) to cortisol, over tens of minutes. Cortisol feeds back to stop its own release.

Hans Selye discovered the generality of this response in 1936: rats exposed to cold, injury, toxins, or forced exercise all developed the same enlarged adrenal glands, shrunken thymus, and stomach

[407] Ochsner, K. N., Bunge, S. A., Gross, J. J., & Gabrieli, J. D. E. (2002). Rethinking feelings: An fMRI study of the cognitive regulation of emotion. *Journal of Cognitive Neuroscience*, 14(8), 1215–1229.

[408] Gross, J. J. (1998). The emerging field of emotion regulation: An integrative review. *Review of General Psychology*, 2(3), 271–299.

[409] Sapolsky, R. M., Romero, L. M., & Munck, A. U. (2000). How do glucocorticoids influence stress responses? Integrating permissive, suppressive, stimulatory, and preparative actions. *Endocrine Reviews*, 21(1), 55–89.

ulcers.[410] Robert Sapolsky and colleagues showed that some of cortisol's actions prepare the body for demand while others, arriving later, restrain the stress response itself before it does damage.[411] And the response is directed by the cortex, not only by the hypothalamus. Peter Strick's laboratory traced the control of the adrenal medulla in monkeys back through several synapses to motor areas of the cortex, especially those for the trunk and posture, and to cingulate regions involved in cognition and affect.[412] Thought, movement, and the stress response are a few synapses apart.

The settings of this system are shaped by experience, starting early. Rat pups whose mothers licked and groomed them more in the first days of life grow up with more of the receptors through which cortisol acts in the hippocampus, stronger feedback, and smaller stress responses.[413] Its response in the moment also depends on company: women in an MRI scanner who were threatened with a mild shock showed less activity in threat-related regions while holding their husband's hand.[414]

The chemistry of stress acts on the brain as well as the body, and its effects follow an inverted U. In 1908 Robert Yerkes and John Dodson trained mice to choose between a light and a dark box, with an electric shock for mistakes, and found that for a hard discrimination a moderate shock produced faster learning than a weak one or a strong one.[415] They varied shock and task difficulty, not arousal, and it was later writers who turned their curve into a law of arousal.

Amy Arnsten has given an inverted U a mechanism in the prefrontal cortex. Moderate levels of norepinephrine and dopamine strengthen the recurrent excitation that holds information in working memory, but the high levels released under acute stress act on different receptors that weaken prefrontal networks, while strengthening the amygdala and the habit circuits of the basal ganglia.[416] Cortisol has an inverted U of its own: it acts through one receptor occupied even at rest and another engaged only under stress, and memory is best when the first is filled and the second moderately engaged.[417] Figure 7.6 shows the shape for norepinephrine and dopamine. Under acute stress control shifts from deliberation to fast, practiced, defensive responses, the right trade in an emergency and the wrong one in an examination.

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[410] Selye, H. (1936). A syndrome produced by diverse nocuous agents. *Nature*, 138(3479), 32.

[411] Sapolsky, R. M., Romero, L. M., & Munck, A. U. (2000). How do glucocorticoids influence stress responses? Integrating permissive, suppressive, stimulatory, and preparative actions. *Endocrine Reviews*, 21(1), 55–89.

[412] Dum, R. P., Levinthal, D. J., & Strick, P. L. (2019). The mind–body problem: Circuits that link the cerebral cortex to the adrenal medulla. *Proceedings of the National Academy of Sciences*, 116(52), 26321–26328.

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[414] Coan, J. A., Schaefer, H. S., & Davidson, R. J. (2006). Lending a hand: Social regulation of the neural response to threat. *Psychological Science*, 17(12), 1032–1039.

[415] Yerkes, R. M., & Dodson, J. D. (1908). The relation of strength of stimulus to rapidity of habit-formation. *Journal of Comparative Neurology and Psychology*, 18(5), 459–482.

[416] Arnsten, A. F. T. (2009). Stress signalling pathways that impair prefrontal cortex structure and function. *Nature Reviews Neuroscience*, 10(6), 410–422.

[417] de Kloet, E. R., Joëls, M., & Holsboer, F. (2005). Stress and the brain: From adaptation to disease. *Nature Reviews Neuroscience*, 6(6), 463–475.

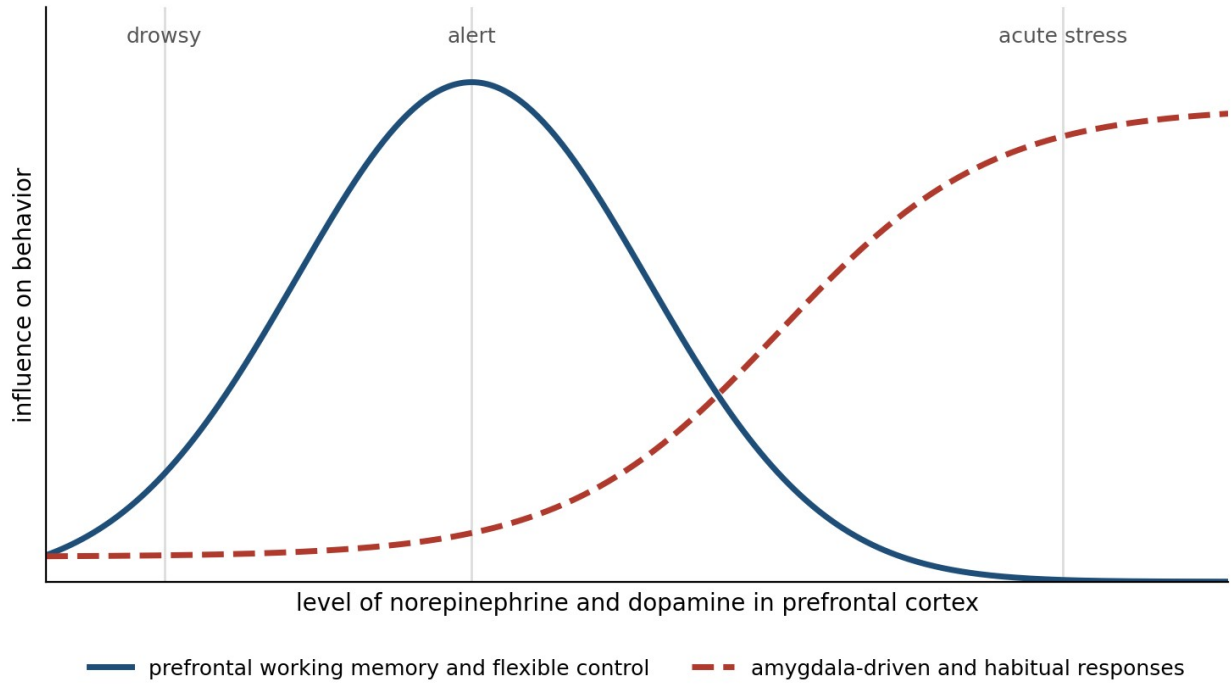


Figure 7.6. An inverted U (illustrative). Prefrontal function rises with modest levels of norepinephrine and dopamine and falls at the high levels released under stress, while amygdala-driven and habitual responses gain control.

What helps over minutes harms over months. Bruce McEwen called the cumulative cost of chronically activated stress responses allostatic load.[418] In rats, three weeks of daily restraint stress shrinks the apical dendrites of pyramidal neurons in the CA3 region of the hippocampus.[419] Sumantra Chattarji's laboratory found that the same kind of chronic stress has the opposite effect in the basolateral amygdala (the lateral nucleus and its neighbors), where dendrites grow.[420] Prolonged stress thus removes structure from a region for flexible, contextual memory and adds it to a region for learned threat, and the two changes differ in how long they last. In rodents much of the hippocampal shrinkage reverses when the stress ends, but the growth in the amygdala, and the anxious behavior that goes with it, persists at least three weeks into recovery.[421]

When Regulation Goes Wrong

If emotion is the brain's estimate of the body's state and prospects, then some disorders of emotion should be disorders of that estimate. This is now a working hypothesis in psychiatry, though not a

[418] McEwen, B. S., & Stellar, E. (1993). Stress and the individual: Mechanisms leading to disease. *Archives of Internal Medicine*, 153(18), 2093–2101; McEwen, B. S. (1998). Protective and damaging effects of stress mediators. *New England Journal of Medicine*, 338(3), 171–179.

[419] Magariños, A. M., & McEwen, B. S. (1995). Stress-induced atrophy of apical dendrites of hippocampal CA3c neurons: Comparison of stressors. *Neuroscience*, 69(1), 83–88.

[420] Vyas, A., Mitra, R., Shankaranarayana Rao, B. S., & Chattarji, S. (2002). Chronic stress induces contrasting patterns of dendritic remodeling in hippocampal and amygdaloid neurons. *Journal of Neuroscience*, 22(15), 6810–6818.

[421] McEwen, B. S. (1998). Protective and damaging effects of stress mediators. *New England Journal of Medicine*, 338(3), 171–179; Vyas, A., Pillai, A. G., & Chattarji, S. (2004). Recovery after chronic stress fails to reverse amygdaloid neuronal hypertrophy and enhanced anxiety-like behavior. *Neuroscience*, 128(4), 667–673.

demonstrated one.[422] Panic disorder is the clearest case. Donald Klein proposed in 1993 that spontaneous panic attacks are false suffocation alarms: a suffocation monitor, set too sensitive, signals that the body is running out of air when it is not.[423] S.M.'s carbon dioxide panic fits this account. Martin Paulus and Murray Stein proposed more generally that anxiety is an inflated interoceptive prediction error in the insula, a body persistently predicted to be in more trouble than it is.[424]

Barrett and colleagues have proposed that depression is in part a disorder of allostasis, in which the brain predicts a body that needs to conserve energy and produces the fatigue, withdrawal, and loss of pleasure that conserving energy requires, whether or not the body needs it.[425] Depression's overlap with the sickness behavior that inflammation induces is one reason the proposal is taken seriously.[426] And alexithymia, a difficulty in identifying one's own feelings, has been linked with poorer interoception across many kinds of bodily signal.[427]

The evidence for these hypotheses is largely correlational, but treatments with good evidence behind them, such as exposure therapy and the reappraisal practiced in cognitive behavioral therapy, can be read as ways of correcting errors of estimation. Not every treatment the account suggests has worked: propranolol, which blocks the body's response to epinephrine, is often prescribed for stage fright, but a systematic review found the evidence insufficient to support its routine use for any anxiety disorder.[428]

Emotion Marks What to Remember

Chapter 5 described how consolidation decides which synaptic changes last, and emotion is one of the main things that decides. People remember emotionally charged events better than neutral ones, and James McGaugh spent five decades establishing why.[429]

The first clue came from injections. In 1975 Paul Gold and Roderick van Buskirk trained rats on an avoidance task and injected epinephrine immediately afterward; the rats remembered the task better, but only at moderate doses, with too much as unhelpful as too little.[430] Given after the training, the injection could not have changed what the rats experienced, only what they kept. Epinephrine does not readily enter the brain; McGaugh's laboratory showed that its effect is relayed, at least in part, by the vagus nerve, the interoceptive pathway of Figure 7.2, to release norepinephrine in the basolateral

[422] Khalsa, S. S., Adolphs, R., Cameron, O. G., Critchley, H. D., Davenport, P. W., Feinstein, J. S., Feusner, J. D., Garfinkel, S. N., Lane, R. D., Mehling, W. E., Meuret, A. E., Nemeroff, C. B., Oppenheimer, S., Petzschner, F. H., Pollatos, O., Rhudy, J. L., Schramm, L. P., Simmons, W. K., Stein, M. B., … Zucker, N. (2018). Interoception and mental health: A roadmap. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, 3(6), 501–513.

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[424] Paulus, M. P., & Stein, M. B. (2006). An insular view of anxiety. *Biological Psychiatry*, 60(4), 383–387.

[425] Barrett, L. F., Quigley, K. S., & Hamilton, P. (2016). An active inference theory of allostasis and interoception in depression. *Philosophical Transactions of the Royal Society B*, 371(1708), 20160011.

[426] Dantzer, R., O'Connor, J. C., Freund, G. G., Johnson, R. W., & Kelley, K. W. (2008). From inflammation to sickness and depression: When the immune system subjugates the brain. *Nature Reviews Neuroscience*, 9(1), 46–56.

[427] Brewer, R., Cook, R., & Bird, G. (2016). Alexithymia: A general deficit of interoception. *Royal Society Open Science*, 3(10), 150664.

[428] Steenen, S. A., van Wijk, A. J., van der Heijden, G. J. M. G., van Westrhenen, R., de Lange, J., & de Jongh, A. (2016). Propranolol for the treatment of anxiety disorders: Systematic review and meta-analysis. *Journal of Psychopharmacology*, 30(2), 128–139.

[429] McGaugh, J. L. (2004). The amygdala modulates the consolidation of memories of emotionally arousing experiences. *Annual Review of Neuroscience*, 27, 1–28.

[430] Gold, P. E., & van Buskirk, R. B. (1975). Facilitation of time-dependent memory processes with posttrial epinephrine injections. *Behavioral Biology*, 13(2), 145–153.

amygdala, which then modulates consolidation in the hippocampus, striatum, and cortex where the memories are stored.[431]

The human version is just as direct. Larry Cahill and colleagues showed volunteers a narrated slide show that was neutral at the beginning and end and upsetting in the middle, when a boy is badly injured in an accident. A week later, people who had taken a placebo remembered the upsetting middle better than the rest. People who had taken propranolol an hour before the slides remembered the middle no better than the neutral parts, while their memory for those neutral parts was unaffected.[432] Because the drug was present during the story as well as after it, this study cannot separate encoding from consolidation; the animal work does that.

A patient whose amygdalae had been damaged by Urbach-Wiethe disease showed the same absence of emotional enhancement without any drug.[433] Emotional arousal acts much like the third factor of Chapter 3's three-factor learning rule: the event writes its traces, and the arousal, signaled by norepinephrine acting in the amygdala, helps decide how strongly they are kept.

The enhancement strengthens the conviction of memory more than its accuracy. Roger Brown and James Kulik called the vivid recollection of where one heard shocking news a flashbulb memory.[434] The morning after the Challenger shuttle exploded in 1986, Ulric Neisser and Nicole Harsch asked students how they had heard the news. Two and a half years later, many gave different accounts and held them with high confidence.[435] Jennifer Talarico and David Rubin repeated the design the day after the attacks of September 11, 2001, with an ordinary event from the same week as a control; over the following months the memories of the attacks lost consistency as fast as the ordinary ones. What distinguished them was that people went on believing them.[436] Emotion tells the brain to keep a memory and to trust it, which is not the same as keeping it accurately, and the reconsolidation of Chapter 5 gives each retelling a chance to change it.

Feeling and Choosing

The next chapter is about decision, and this one ends by handing over a problem. If emotion ranks options, what happens to choice without it?

The classic case is Phineas Gage, the railroad foreman whose left frontal lobe was pierced in 1848 by an iron rod driven through his cheek and out of the top of his skull by an accidental blast. He survived. Twenty years later his physician, John Harlow, reported that he had become "a child in his intellectual

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[431] Williams, C. L., & McGaugh, J. L. (1993). Reversible lesions of the nucleus of the solitary tract attenuate the memory-modulating effects of posttraining epinephrine. *Behavioral Neuroscience*, 107(6), 955–962; McGaugh, J. L. (2004). The amygdala modulates the consolidation of memories of emotionally arousing experiences. *Annual Review of Neuroscience*, 27, 1–28.

[432] Cahill, L., Prins, B., Weber, M., & McGaugh, J. L. (1994). β-Adrenergic activation and memory for emotional events. *Nature*, 371(6499), 702–704.

[433] Cahill, L., Babinsky, R., Markowitsch, H. J., & McGaugh, J. L. (1995). The amygdala and emotional memory. *Nature*, 377(6547), 295–296.

[434] Brown, R., & Kulik, J. (1977). Flashbulb memories. *Cognition*, 5(1), 73–99.

[435] Neisser, U., & Harsch, N. (1992). Phantom flashbulbs: False recollections of hearing the news about Challenger. In E. Winograd & U. Neisser (Eds.), *Affect and Accuracy in Recall: Studies of "Flashbulb" Memories* (pp. 9–31). Cambridge University Press.

[436] Talarico, J. M., & Rubin, D. C. (2003). Confidence, not consistency, characterizes flashbulb memories. *Psychological Science*, 14(5), 455–461.

capacity" with "the animal passions of a strong man," and that his friends said he was "no longer Gage."[437] Earlier reports had described a good recovery, and in the years before Harlow wrote, Gage drove stagecoaches in Chile; much of what is told about him is embellishment.[438] Reconstructions from his skull, preserved at Harvard, place the damage in the ventromedial prefrontal cortex and the fiber tracts beneath it, more on the left than the right.[439]

Antonio Damasio described a cleaner modern case, a patient he called Elliot, who after surgery for a tumor in the same region scored normally on tests of intelligence and memory and could reason through a problem in detail, yet made one disastrous decision after another in his life and his business, recounting his losses with little feeling.[440] Damasio proposed the somatic marker hypothesis to explain such cases: options that have led to good or bad outcomes come to evoke bodily states, the somatic markers, and anticipating these states biases choice before reasoning finishes.[441]

To test it, Antoine Bechara and colleagues devised the Iowa gambling task, shown in Figure 7.7. The player draws cards from four decks. Two decks pay 100 dollars of play money per card but carry occasional large penalties, for a net loss of 250 dollars over every ten cards; the other two pay 50 dollars per card with small penalties, for a net gain of 250. Healthy players gradually shift to the good decks, while patients with ventromedial prefrontal damage keep drawing from the bad ones.[442]

In a later study, healthy players produced skin conductance responses, small bursts of sweating measured on the fingers, just before choosing from a bad deck, and started avoiding the bad decks, all before they could say which decks were bad. The patients never developed the anticipatory responses, and some of them went on choosing badly after they had worked out which decks were bad.[443]

[437] Harlow, J. M. (1868). Recovery from the passage of an iron bar through the head. *Publications of the Massachusetts Medical Society*, 2, 327–347.

[438] Macmillan, M. (2000). *An Odd Kind of Fame: Stories of Phineas Gage*. MIT Press.

[439] Damasio, H., Grabowski, T., Frank, R., Galaburda, A. M., & Damasio, A. R. (1994). The return of Phineas Gage: Clues about the brain from the skull of a famous patient. *Science*, 264(5162), 1102–1105; Van Horn, J. D., Irimia, A., Torgerson, C. M., Chambers, M. C., Kikinis, R., & Toga, A. W. (2012). Mapping connectivity damage in the case of Phineas Gage. *PLoS ONE*, 7(5), e37454.

[440] Damasio, A. R. (1994). *Descartes' Error: Emotion, Reason, and the Human Brain*. G. P. Putnam's Sons.

[441] Damasio, A. R. (1994). *Descartes' Error: Emotion, Reason, and the Human Brain*. G. P. Putnam's Sons.

[442] Bechara, A., Damasio, A. R., Damasio, H., & Anderson, S. W. (1994). Insensitivity to future consequences following damage to human prefrontal cortex. *Cognition*, 50(1–3), 7–15.

[443] Bechara, A., Damasio, H., Tranel, D., & Damasio, A. R. (1997). Deciding advantageously before knowing the advantageous strategy. *Science*, 275(5304), 1293–1295.

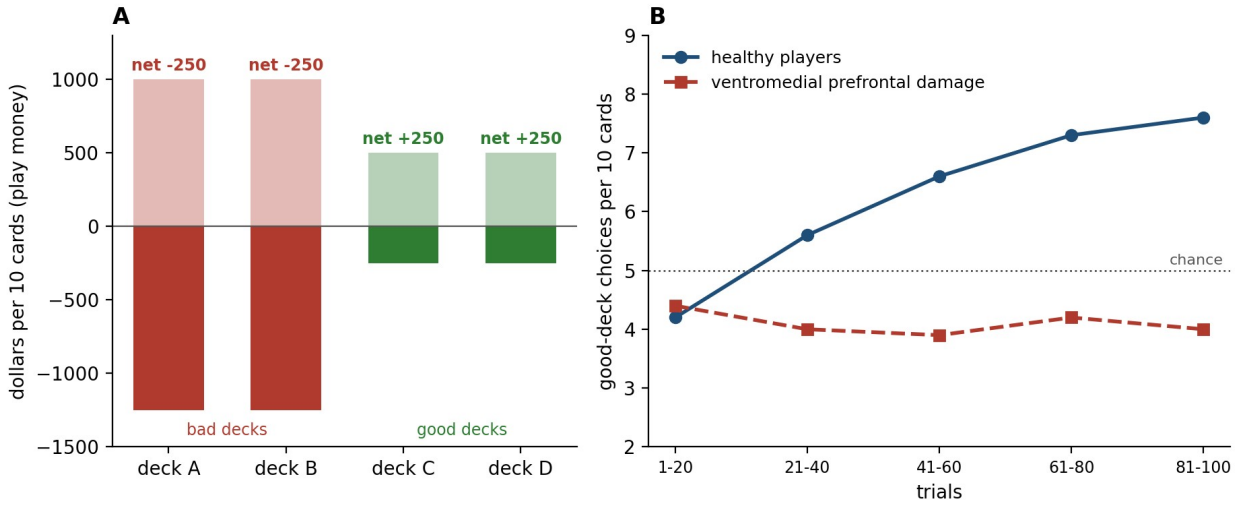


Figure 7.7. The Iowa gambling task. A: the payoff schedule; decks A and B pay more per card but lose money over every ten cards, while decks C and D pay less and gain. B: illustrative choice curves; healthy players shift toward the good decks over a hundred trials, and patients with ventromedial prefrontal damage do not.

The interpretation has not held up well. Tiago Maia and James McClelland asked players more detailed questions and found that they knew which decks were good about as soon as they began to choose them; there is no clear period in which the body knows and the mind does not.[444] Patients with pure autonomic failure, whose brains receive almost no feedback from the body's autonomic responses, play the task normally.[445] And Lesley Fellows and Martha Farah found that the patients' difficulty is largely one of reversal learning: the bad decks pay well at first, the patients fail to switch when they turn bad, and when the early run of large rewards was removed their deficit shrank.[446] A review by Barnaby Dunn and colleagues concluded that the hypothesis needs more support to remain tenable.[447]

What survives is that damage to the ventromedial prefrontal cortex leaves reasoning largely intact and choice impaired, and that this region links value with the body, receiving interoceptive signals and commanding the hypothalamus and brainstem. Whether bodily feedback is needed for good choices is doubtful; that the value signals guiding choice are tied to the regulation of the body is not. How those signals are combined, weighed against uncertainty, and turned into a decision is the subject of Chapter 8.

[444] Maia, T. V., & McClelland, J. L. (2004). A reexamination of the evidence for the somatic marker hypothesis: What participants really know in the Iowa gambling task. *Proceedings of the National Academy of Sciences*, 101(45), 16075–16080.

[445] Heims, H. C., Critchley, H. D., Dolan, R., Mathias, C. J., & Cipolotti, L. (2004). Social and motivational functioning is not critically dependent on feedback of autonomic responses: Neuropsychological evidence from patients with pure autonomic failure. *Neuropsychologia*, 42(14), 1979–1988.

[446] Fellows, L. K., & Farah, M. J. (2005). Different underlying impairments in decision-making following ventromedial and dorsolateral frontal lobe damage in humans. *Cerebral Cortex*, 15(1), 58–63.

[447] Dunn, B. D., Dalgleish, T., & Lawrence, A. D. (2006). The somatic marker hypothesis: A critical evaluation. *Neuroscience & Biobehavioral Reviews*, 30(2), 239–271.

Implications for Artificial Intelligence

Most artificial intelligence has no body and nothing to protect, and in that sense no emotion. The comparison is still instructive, because artificial systems face the problems emotion solves for brains, and solve them, when they do, by hand.

First, value comes from outside. A reinforcement learning agent maximizes a reward that its designers specify. An animal's rewards are generated by the state of its body, as Cabanac's sweet solution showed, and change as the state changes. Mehdi Keramati and Boris Gutkin formalized this as homeostatic reinforcement learning, in which the reward for an outcome is the reduction it produces in the distance between the internal state and its set point, and showed that the model accounts for anticipatory responding and for the dependence of value on need.[448] Satinder Singh and colleagues showed that an agent often learns better from an internal reward that differs from the external goal it serves, which is what evolution appears to have built.[449]

Second, a fixed external reward can be gamed. Agents that optimize a specified reward find unintended ways to maximize it, a failure known as reward hacking that is one of the central problems of AI safety.[450] Animals are not immune; addiction is a form of reward hacking. But an animal's values are anchored to a body that the brain must keep within bounds and cannot redefine, which limits how far they drift. Whether such anchoring can or should be built into an artificial agent is an open question.

Third, emotions are global modes. In 1967 Herbert Simon argued that any system pursuing goals one at a time in an uncertain world needs an interrupt that can seize control when something urgent happens, and that emotions are that interrupt in humans.[451] Arousal changes many parameters at once: how widely to attend, how fast to learn, how much to explore. Kenji Doya proposed that the neuromodulators of Chapter 3 set exactly these metaparameters, the settings that govern learning itself, with dopamine carrying the prediction error, serotonin the time horizon, norepinephrine the balance between exploring and exploiting, and acetylcholine the learning rate.[452] Later work, described in Chapter 12, suggests that they divide the learning rate between them: norepinephrine raises it when the world has changed, and acetylcholine weights incoming evidence when the world is noisy but stable. Artificial agents mostly fix such parameters by hand before training.

Fourth, importance should decide what is learned. A brain does not store all experience equally; arousal decides what is consolidated. Prioritized experience replay in reinforcement learning, which replays the transitions with the largest prediction errors more often, is a step in this direction,[453] but artificial systems rarely have any signal that marks an experience as significant to the agent rather than merely surprising.

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[448] Keramati, M., & Gutkin, B. (2014). Homeostatic reinforcement learning for integrating reward collection and physiological stability. *eLife*, 3, e04811.

[449] Singh, S., Lewis, R. L., Barto, A. G., & Sorg, J. (2010). Intrinsically motivated reinforcement learning: An evolutionary perspective. *IEEE Transactions on Autonomous Mental Development*, 2(2), 70–82.

[450] Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete problems in AI safety. *arXiv*, 1606.06565.

[451] Simon, H. A. (1967). Motivational and emotional controls of cognition. *Psychological Review*, 74(1), 29–39.

[452] Doya, K. (2002). Metalearning and neuromodulation. *Neural Networks*, 15(4–6), 495–506.

[453] Schaul, T., Quan, J., Antonoglou, I., & Silver, D. (2016). Prioritized experience replay. *International Conference on Learning Representations*. arXiv:1511.05952.

Fifth, reading emotion is not the same as having it. Affective computing, a field Rosalind Picard named in the mid-1990s, builds systems that detect human emotion, often from faces,[454] and the review by Barrett and colleagues is a warning: a system that reads a scowl as anger will often be wrong.[455]

The Central Lesson

The brain's first job is not to know the world but to keep a body alive in it, and emotion is how that job reaches every other. The brain regulates the body largely by prediction and senses it as it senses the world, by combining what arrives with what is expected. The result, felt as valence and arousal and named, in people, as fear, anger, or relief, is an estimate of what the present situation means for the body. That estimate sets the value of every option, prepares the body, and tells the plasticity of Chapter 5 what to keep. It is the part of the brain's model that says what matters, and without it reason has nothing to work for.

Next Chapter

Chapter 8: Decision-Making Under Uncertainty: How the Brain Chooses When It Cannot Be Sure

Figure and Epigraph Sources

Figure Attribution

Figures 7.1 to 7.7 were drawn for this book. The glucose traces in Figure 7.1, the curves in Figure 7.6, and the choice curves in Figure 7.7B are illustrative computations, not recorded data; the positions of emotion words in Figure 7.3 are approximate. The payoff schedule in Figure 7.7A follows Bechara and colleagues (1994).

Epigraph

William James, "What Is an Emotion?", *Mind*, 9(34), 188–205 (1884).

[454] Picard, R. W. (1997). *Affective Computing*. MIT Press.

[455] Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., & Pollak, S. D. (2019). Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements. *Psychological Science in the Public Interest*, 20(1), 1–68.

Chapter 8

Decision-Making Under Uncertainty: How the Brain Chooses When It Cannot Be Sure

> *"The heart has its reasons, which reason does not know."*
>
> *— Blaise Pascal, Pensées (1670)*

The Problem Decision Solves

Chapter 7 ended with a player at a table of four card decks and a region of the frontal lobe that, when damaged, leaves reasoning intact and choice impaired. The patients could describe the game and still lose at it. Whatever choosing well requires, it is not the ability to state the rules. This chapter asks what it is.

Every decision answers two questions at once. The first is what the world is like: whether the dots on the screen are drifting left or right, whether the deck is good or bad, whether the rustle in the grass is wind or a snake. The second is what each option is worth, given the answer to the first and given the state of the body that Chapter 7 described. Neither answer is ever certain. Evidence arrives noisily and over time, values must be learned from outcomes that are themselves variable, and the act of choosing has its own costs: time spent deliberating is time not spent acting, and an option never tried is an option never learned about.

The argument of this chapter is that the brain meets these problems with a small set of mechanisms that have been recorded in neurons: it accumulates evidence until it is sure enough, values unlike options on one scale that it bends in ways that look irrational in the laboratory and often make sense in the world, balances using what it knows against finding out more, and runs a deliberate planner beside a fast habit system. Every one of them is tuned by outcomes. Figure 8.1 lays out the pieces.

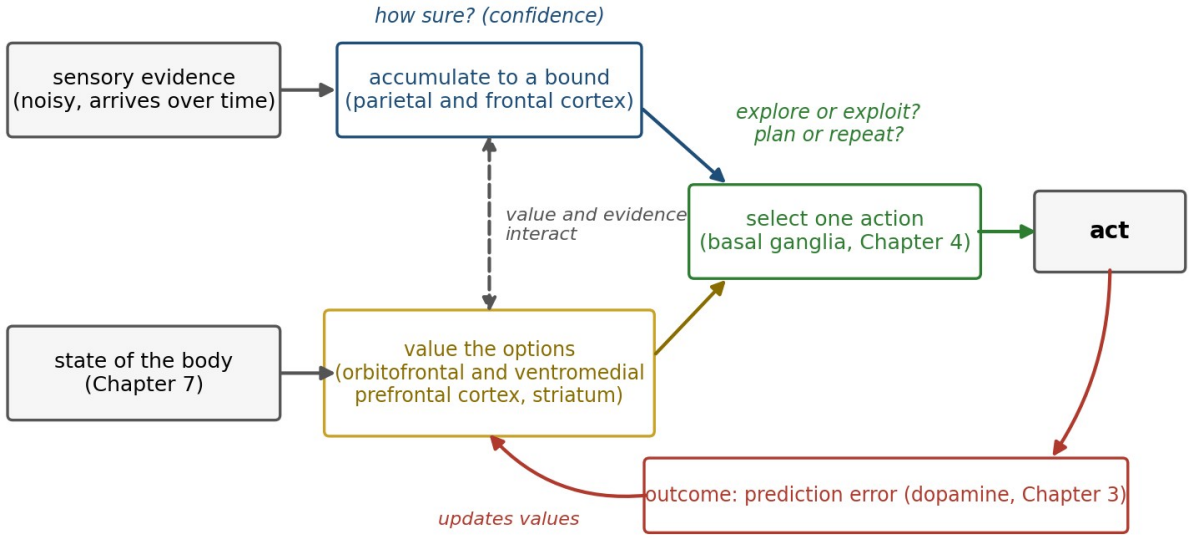


Figure 8.1. The parts of a decision. Sensory evidence is accumulated toward a bound; options are valued, with the state of the body as an input; the basal ganglia select one action; the outcome produces a prediction error that updates the values. Each stage is a source of uncertainty and a target of learning.

The idea of weighing outcomes by their chances is older than any of this. It begins with Blaise Pascal, whose letters with Pierre de Fermat in 1654, on how to divide the stakes of an unfinished game of chance, founded the theory of probability, and whose wager on the existence of God is often called the first argument in decision theory.[456] The same Pascal wrote that the heart has its reasons. This chapter is about both halves of him.

Accumulating Evidence

The best-studied decision in neuroscience is a monkey watching a cloud of moving dots. On each trial some fraction of the dots, the coherence, move in one direction and the rest move at random, and the monkey earns juice by moving its eyes in the direction of the coherent motion. At high coherence the task is easy; at low coherence the monkey must watch for a while, and it still makes mistakes.

William Newsome and his colleagues recorded from area MT, a visual region whose neurons respond to motion in a preferred direction, and found that single neurons were about as sensitive to weak motion as the monkey itself.[457] Passing a small current through a cluster of neurons that preferred one direction made the monkey report that direction more often, so the signal was not only correlated with the choice but used for it.[458] MT reports the momentary evidence. Where is it weighed?

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[456] Devlin, K. (2008). *The Unfinished Game: Pascal, Fermat, and the Seventeenth-Century Letter That Made the World Modern*. Basic Books; Hacking, I. (1972). The logic of Pascal's wager. *American Philosophical Quarterly*, 9(2), 186–192.

[457] Newsome, W. T., Britten, K. H., & Movshon, J. A. (1989). Neuronal correlates of a perceptual decision. *Nature*, 341(6237), 52–54.

[458] Salzman, C. D., Britten, K. H., & Newsome, W. T. (1990). Cortical microstimulation influences perceptual judgements of motion direction. *Nature*, 346(6280), 174–177.

Michael Shadlen and his colleagues found an answer in the lateral intraparietal area, LIP, a region of parietal cortex that helps plan eye movements. When the monkey was allowed to respond whenever it was ready, LIP neurons whose target lay in the chosen direction ramped up their firing during viewing, faster for stronger motion and slower for weaker, and the firing reached a common level about 70 milliseconds before the eyes began to move, whatever the strength of the motion (Figure 8.2).[459] The ramp looks like a running total of evidence, and the common level looks like a threshold: the decision is made when enough evidence has piled up for one answer.

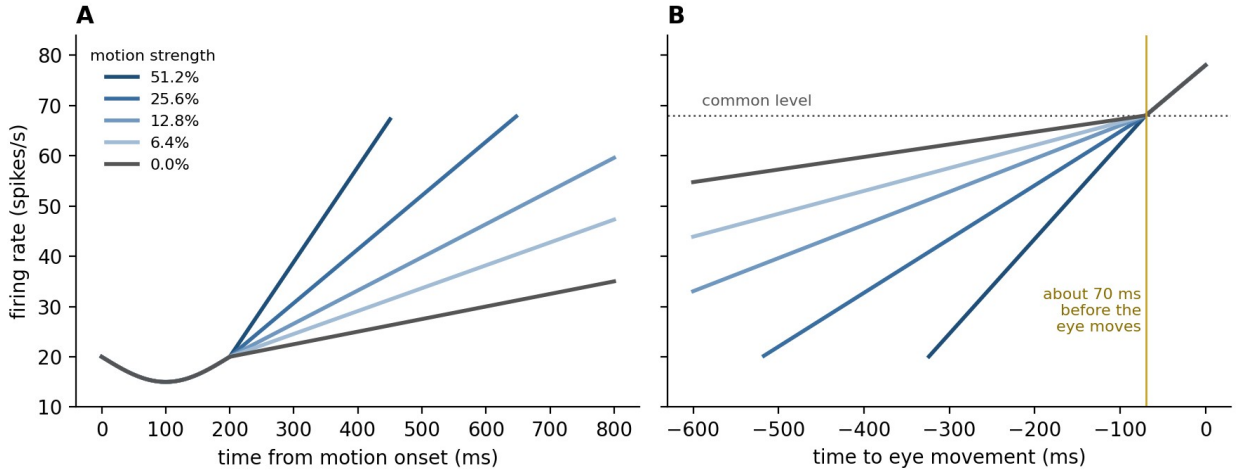


Figure 8.2. Accumulation in parietal cortex (illustrative, after Roitman and Shadlen 2002). A: aligned to the start of motion, firing ramps faster for stronger motion. B: aligned to the eye movement, the ramps converge on a common level just before the response.

This is a process that mathematics had already described. During the Second World War, Abraham Wald worked out the sequential probability ratio test: take samples one at a time, add up the evidence each provides for one hypothesis over the other, and stop as soon as the total crosses a bound in either direction. Wald and Jacob Wolfowitz proved that no other procedure reaches a given accuracy with fewer samples on average.[460] Alan Turing used the same principle at Bletchley Park to decide when enough evidence had accumulated about the setting of an Enigma machine.[461] The evidence summed in Wald's test is the logarithm of a likelihood ratio, so the test is the Bayesian inference of Chapter 6 carried out one sample at a time.

Roger Ratcliff's drift diffusion model, developed in psychology for memory judgments, is the continuous version of Wald's test: a decision variable drifts toward one bound at a rate set by the strength of the evidence, jostled by noise, and the choice and the reaction time are both set by where and when it hits

[459] Shadlen, M. N., & Newsome, W. T. (2001). Neural basis of a perceptual decision in the parietal cortex (area LIP) of the rhesus monkey. *Journal of Neurophysiology*, 86(4), 1916–1936; Roitman, J. D., & Shadlen, M. N. (2002). Response of neurons in the lateral intraparietal area during a combined visual discrimination reaction time task. *Journal of Neuroscience*, 22(21), 9475–9489.

[460] Wald, A., & Wolfowitz, J. (1948). Optimum character of the sequential probability ratio test. *Annals of Mathematical Statistics*, 19(3), 326–339.

[461] Gold, J. I., & Shadlen, M. N. (2007). The neural basis of decision making. *Annual Review of Neuroscience*, 30, 535–574.

(Figure 8.3A).[462] Rafal Bogacz and colleagues showed that, for a choice between two options whose evidence is of fixed strength, the diffusion model is therefore the optimal procedure.[463]

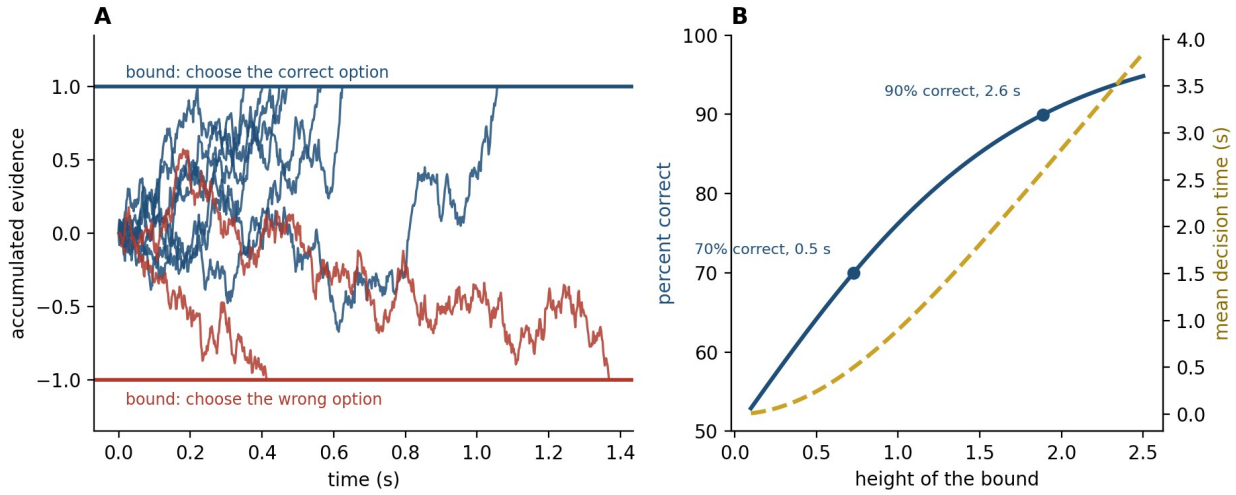


Figure 8.3. The drift diffusion model (simulated). A: noisy accumulation from a starting point to one of two bounds; most paths reach the correct bound, some reach the wrong one, and each path's crossing time is a reaction time. B: raising the bound increases accuracy and lengthens decisions; the right setting depends on what time and errors cost.

The model explains one of the most general facts about choice, the trade-off between speed and accuracy. A low bound gives quick decisions and many errors; a high bound gives slow decisions and few errors (Figure 8.3B). In the model of the figure, with weak evidence, a bound that gives 70 percent correct answers in about half a second must be raised about two and a half times to give 90 percent, and the decisions then take about five times as long. Whether that is worth it depends on what an error costs and what a second is worth. People and monkeys move their bounds when the instructions or the rewards change, and the shapes of their reaction-time distributions, including the long tail of slow decisions, follow the model closely.[464]

The brain also moves the bound when the choice is hard. When options conflict, an electrical rhythm of four to eight cycles a second, the theta rhythm, rises over the medial frontal cortex and decisions slow down. Michael Frank and colleagues studied patients with Parkinson's disease treated by deep brain stimulation of the subthalamic nucleus (Chapter 3), the relay of the hyperdirect path that Chapter 4 described as able to stop everything at once. With the stimulator on, the patients no longer slowed down when the options were close, and made hasty choices; James Cavanagh, Frank, and colleagues showed that stimulation reversed the link between the frontal theta rhythm and the height of the bound.[465]

[462] Ratcliff, R. (1978). A theory of memory retrieval. *Psychological Review*, 85(2), 59–108; Ratcliff, R., & McKoon, G. (2008). The diffusion decision model: Theory and data for two-choice decision tasks. *Neural Computation*, 20(4), 873–922.

[463] Bogacz, R., Brown, E., Moehlis, J., Holmes, P., & Cohen, J. D. (2006). The physics of optimal decision making: A formal analysis of models of performance in two-alternative forced-choice tasks. *Psychological Review*, 113(4), 700–765.

[464] Ratcliff, R., & McKoon, G. (2008). The diffusion decision model: Theory and data for two-choice decision tasks. *Neural Computation*, 20(4), 873–922.

[465] Frank, M. J., Samanta, J., Moustafa, A. A., & Sherman, S. J. (2007). Hold your horses: Impulsivity, deep brain stimulation, and medication in parkinsonism. *Science*, 318(5854), 1309–1312; Cavanagh, J. F., Wiecki, T. V., Cohen, M. X., Figueroa, C. M., Samanta, J., Sherman, S. J., & Frank, M. J. (2011). Subthalamic nucleus stimulation reverses mediofrontal influence over decision threshold. *Nature Neuroscience*, 14(11), 1462–1467.

When the difficulty of the evidence varies from trial to trial, the optimal bound is not fixed but falls as time passes, and in some tasks the brain appears to add an urgency signal that grows with time.[466] How often the brain uses collapsing bounds remains debated: a comparison of models across many data sets found that fixed bounds fit most human and monkey choices as well or better.[467]

The picture is clean, and parts of it have been challenged and defended. Kenneth Latimer and colleagues analyzed single trials rather than averages and found that many LIP neurons were better described as jumping abruptly from one level to another at a random time than as ramping smoothly, the ramp being an artifact of averaging jumps that happen at different times.[468] Leor Katz and colleagues silenced LIP with a drug and found that monkeys discriminated motion as well as before, while silencing MT impaired them.[469] Later work pushed back. Recordings from hundreds of LIP neurons at once revealed a population signal that drifts and diffuses on single trials as the model requires, and inactivating LIP does impair decisions at first, before other areas compensate within and across sessions.[470] LIP is probably one of several areas that keep the running total. The accumulation itself, as a description of how evidence and time trade off in behavior, is among the best-supported models in cognitive science.

Knowing How Sure You Are

A decision comes with a sense of how likely it is to be right, and that sense is itself computed. Roozbeh Kiani and Shadlen gave monkeys a third option: after viewing the dots they could choose a “sure target” that paid a small reward for certain instead of gambling on the direction. The monkeys took the sure target more often when the motion was weak or brief, as an agent that knew its own uncertainty should, and the same LIP neurons that accumulated evidence predicted when they would do so.[471]

Confidence is for acting on. Armin Lak, Adam Kepecs, and colleagues trained rats to judge which of two odors dominated a mixture and then to wait at a port for a reward that came after an unpredictable delay, and sometimes not at all. The rats were willing to wait longer when their choice was more likely to be correct, so that the time they invested tracked their own probability of being right; when the orbitofrontal cortex was silenced, they still chose as accurately but no longer waited in proportion to their confidence.[472]

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[466] Drugowitsch, J., Moreno-Bote, R., Churchland, A. K., Shadlen, M. N., & Pouget, A. (2012). The cost of accumulating evidence in perceptual decision making. *Journal of Neuroscience*, 32(11), 3612–3628; Churchland, A. K., Kiani, R., & Shadlen, M. N. (2008). Decision-making with multiple alternatives. *Nature Neuroscience*, 11(6), 693–702.

[467] Hawkins, G. E., Forstmann, B. U., Wagenmakers, E.-J., Ratcliff, R., & Brown, S. D. (2015). Revisiting the evidence for collapsing boundaries and urgency signals in perceptual decision-making. *Journal of Neuroscience*, 35(6), 2476–2484.

[468] Latimer, K. W., Yates, J. L., Meister, M. L. R., Huk, A. C., & Pillow, J. W. (2015). Single-trial spike trains in parietal cortex reveal discrete steps during decision-making. *Science*, 349(6244), 184–187.

[469] Katz, L. N., Yates, J. L., Pillow, J. W., & Huk, A. C. (2016). Dissociated functional significance of decision-related activity in the primate dorsal stream. *Nature*, 535(7611), 285–288.

[470] Steinemann, N. A., Stine, G. M., Trautmann, E. M., Zylberberg, A., Wolpert, D. M., & Shadlen, M. N. (2024). Direct observation of the neural computations underlying a single decision. *eLife*, 12, RP90859; Jeurissen, D., Shushruth, S., El-Shamayleh, Y., Horwitz, G. D., & Shadlen, M. N. (2022). Deficits in decision-making induced by parietal cortex inactivation are compensated at two timescales. *Neuron*, 110(12), 1924–1931.

[471] Kiani, R., & Shadlen, M. N. (2009). Representation of confidence associated with a decision by neurons in the parietal cortex. *Science*, 324(5928), 759–764.

[472] Lak, A., Costa, G. M., Romberg, E., Koulakov, A. A., Mainen, Z. F., & Kepecs, A. (2014). Orbitofrontal cortex is required for optimal waiting based on decision confidence. *Neuron*, 84(1), 190–201.

A decision, once committed, also stays open. In the model a choice is committed when a bound is crossed, but evidence already in the pipeline continues to arrive, and Arbora Resulaj and colleagues found that people who had started to move toward one answer sometimes reversed in mid-movement, and continued accumulation of the late evidence explained how often they did so.[473] People also differ in how well their confidence tracks their accuracy, a capacity called metacognition, and Stephen Fleming and colleagues found that people with better metacognition had more gray matter in the anterior prefrontal cortex.[474] The heartbeat-counting results of Chapter 7, in which accuracy and confidence came apart, are the same distinction in the interoceptive domain.

A Common Currency

To choose between an apple and an orange, or between a job in one city and a job in another, the brain must put unlike things on one scale. Daniel Bernoulli proposed in 1738 that people value money not by its amount but by its usefulness to them, which he called utility, and John von Neumann and Oskar Morgenstern proved in 1947 that any chooser whose preferences obey a few consistency rules behaves as if it maximized some utility weighted by probability.[475]

Neurons that behave like such a scale have been found. Camillo Padoa-Schioppa and John Assad offered thirsty monkeys choices between different amounts of two juices, measured how much of one the monkey would give up for the other, and recorded in the orbitofrontal cortex on the underside of the frontal lobe. Some neurons fired in proportion to the value of one particular juice on offer. Others fired in proportion to the value of whichever option the monkey chose, whatever the juice, measured in the monkey's own exchange rate: a signal on a common scale. Still others signaled which juice had been chosen.[476]

In people, activity in the medial orbitofrontal cortex tracks how much they are willing to pay for a snack, and a meta-analysis of more than 200 imaging studies by Oscar Bartra, Joseph McGuire, and Joseph Kable found that this region and the ventral striatum track subjective value for both money and primary rewards such as food.[477] The medial orbitofrontal cortex is part of the ventromedial prefrontal region whose damage, in Chapter 7, left choice impaired, and patients with such damage choose less consistently, preferring A to B and B to C and then C to A more often than others do.[478]

The interpretation is contested. Dino Levy and Paul Glimcher argue that these signals are a common neural currency in which all options are priced before they are compared.[479] Benjamin Hayden and Yael Niv argue that the evidence does not require a single value variable at all: the same neurons carry many

[473] Resulaj, A., Kiani, R., Wolpert, D. M., & Shadlen, M. N. (2009). Changes of mind in decision-making. *Nature*, 461(7261), 263–266.

[474] Fleming, S. M., Weil, R. S., Nagy, Z., Dolan, R. J., & Rees, G. (2010). Relating introspective accuracy to individual differences in brain structure. *Science*, 329(5998), 1541–1543.

[475] Bernoulli, D. (1954). Exposition of a new theory on the measurement of risk (L. Sommer, Trans.). *Econometrica*, 22(1), 23–36; von Neumann, J., & Morgenstern, O. (1947). *Theory of Games and Economic Behavior* (2nd ed.). Princeton University Press.

[476] Padoa-Schioppa, C., & Assad, J. A. (2006). Neurons in the orbitofrontal cortex encode economic value. *Nature*, 441(7090), 223–226.

[477] Plassmann, H., O'Doherty, J., & Rangel, A. (2007). Orbitofrontal cortex encodes willingness to pay in everyday economic transactions. *Journal of Neuroscience*, 27(37), 9984–9988; Bartra, O., McGuire, J. T., & Kable, J. W. (2013). The valuation system: A coordinate-based meta-analysis of BOLD fMRI experiments examining neural correlates of subjective value. *NeuroImage*, 76, 412–427.

[478] Camille, N., Griffiths, C. A., Vo, K., Fellows, L. K., & Kable, J. W. (2011). Ventromedial frontal lobe damage disrupts value maximization in humans. *Journal of Neuroscience*, 31(20), 7527–7532.

other quantities, value-like signals appear everywhere once one looks for them, and choice could emerge from comparisons that never compute a price.[480] What is not in dispute is that the brain represents what options are worth to the animal now, in regions tied to the body, and that damage to those regions disorganizes choice.

The scale may also be relative. Kenway Louie, Glimcher, and colleagues found that the value signals of parietal neurons are divisively normalized: each is divided by the total value on offer, so that adding a third option shrinks the signals for the first two. Divisive normalization is a computation that sensory circuits use to keep responses within range by dividing each one by the pooled activity of its neighbors, a cousin of the lateral inhibition of Chapter 4.[481] Louie and colleagues proposed that it makes choices depend on context, so that an irrelevant third option makes people choose less consistently between the original two.[482] Whether it does is disputed: a preregistered attempt to repeat the behavioral result did not find it.[483]

Learning What Options Are Worth

Values are not given; they are learned from outcomes by the reward prediction error of Chapter 3, which adjusts the predicted value of whatever was chosen, a little at a time, until predictions match outcomes on average.[484] The rule has consequences that can be seen in behavior. In 1961 Richard Herrnstein let pigeons peck two keys that paid off on independent random schedules and found that, over a session, the birds divided their pecks between the keys in proportion to the rate at which each delivered food, a regularity he called the matching law.[485] Leo Sugrue, Greg Corrado, and Newsome trained monkeys on a version of this task in which the richer option changed without warning, and found that the monkeys matched the rewards of the last several trials rather than the whole session, and that LIP neurons tracked each option's share of this recent income, as a learner with a short memory should.[486]

A learned value is an average over a history, and it changes slowly when the world changes quickly. The Iowa patients of Chapter 7, whose difficulty Lesley Fellows and Martha Farah traced largely to a failure to reverse what they had learned, had a problem of exactly this kind.[487] They learned that the bad decks were good, and the region that should have revised the value when the losses began was the region that was damaged.

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[479] Levy, D. J., & Glimcher, P. W. (2012). The root of all value: A neural common currency for choice. *Current Opinion in Neurobiology*, 22(6), 1027–1038.

[480] Hayden, B. Y., & Niv, Y. (2021). The case against economic values in the orbitofrontal cortex (or anywhere else in the brain). *Behavioral Neuroscience*, 135(2), 192–201.

[481] Louie, K., Grattan, L. E., & Glimcher, P. W. (2011). Reward value-based gain control: Divisive normalization in parietal cortex. *Journal of Neuroscience*, 31(29), 10627–10639; Carandini, M., & Heeger, D. J. (2012). Normalization as a canonical neural computation. *Nature Reviews Neuroscience*, 13(1), 51–62.

[482] Louie, K., Khaw, M. W., & Glimcher, P. W. (2013). Normalization is a general neural mechanism for context-dependent decision making. *Proceedings of the National Academy of Sciences*, 110(15), 6139–6144.

[483] Gluth, S., Kern, N., Kortmann, M., & Vitali, C. L. (2020). Value-based attention but not divisive normalization influences decisions with multiple alternatives. *Nature Human Behaviour*, 4(6), 634–645.

[484] Schultz, W., Dayan, P., & Montague, P. R. (1997). A neural substrate of prediction and reward. *Science*, 275(5306), 1593–1599.

[485] Herrnstein, R. J. (1961). Relative and absolute strength of response as a function of frequency of reinforcement. *Journal of the Experimental Analysis of Behavior*, 4(3), 267–272.

[486] Sugrue, L. P., Corrado, G. S., & Newsome, W. T. (2004). Matching behavior and the representation of value in the parietal cortex. *Science*, 304(5678), 1782–1787.

[487] Fellows, L. K., & Farah, M. J. (2005). Different underlying impairments in decision-making following ventromedial and dorsolateral frontal lobe damage in humans. *Cerebral Cortex*, 15(1), 58–63.

Value and evidence meet during the choice itself. When people choose between two snacks, the decision takes longer the closer the two values are, just as a motion decision takes longer the weaker the motion, and Ian Krajbich, Carrie Armel, and Antonio Rangel found that a diffusion model fitted these choices if the accumulation favored whichever item the person was looking at: people tended to choose the item they had looked at last and longest, even when it was not the one they valued more.[488] Laurence Hunt and colleagues recorded the brain during such choices with magnetoencephalography, which detects the tiny magnetic fields produced by cortical currents from outside the head, and found a signal in ventromedial prefrontal cortex whose time course matched a network comparing the two values by mutual competition: the accumulation of the dots task, applied to worth rather than to motion.[489]

Risk, Loss, and Ambiguity

Once options carry uncertain outcomes, the question becomes how probabilities and payoffs combine. The simplest rule, expected value, says to multiply each payoff by its probability and add. Offered a coin flip that wins 110 dollars on heads and loses 100 on tails, a chooser following that rule accepts, since the gamble is worth 5 dollars on average; most people refuse.[490] Expected utility theory can explain one refusal by a curved scale of utility, but Matthew Rabin showed that a person who refused this gamble at every level of wealth would, under the theory, also have to refuse a coin flip that loses 1,000 dollars or wins any amount at all, however large, which nobody does.[491] Small refusals of this kind need a different explanation.

In 1979 Daniel Kahneman and Amos Tversky documented systematic departures from expected utility and proposed prospect theory to describe them.[492] People evaluate outcomes as gains and losses relative to a reference point rather than as final states of wealth. For moderate and large probabilities they are risk averse for gains and risk seeking for losses. They weight losses more heavily than equal gains, by a factor of about two in Tversky and Kahneman's later estimates. And they treat probabilities nonlinearly, overweighting small ones and underweighting large ones, which is why the same people buy both lottery tickets and insurance (Figure 8.4).[493]

[488] Krajbich, I., Armel, C., & Rangel, A. (2010). Visual fixations and the computation and comparison of value in simple choice. *Nature Neuroscience*, 13(10), 1292–1298.

[489] Hunt, L. T., Kolling, N., Soltani, A., Woolrich, M. W., Rushworth, M. F. S., & Behrens, T. E. J. (2012). Mechanisms underlying cortical activity during value-guided choice. *Nature Neuroscience*, 15(3), 470–476.

[490] Tversky, A., & Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4), 297–323.

[491] Rabin, M. (2000). Risk aversion and expected-utility theory: A calibration theorem. *Econometrica*, 68(5), 1281–1292.

[492] Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.

[493] Tversky, A., & Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4), 297–323.

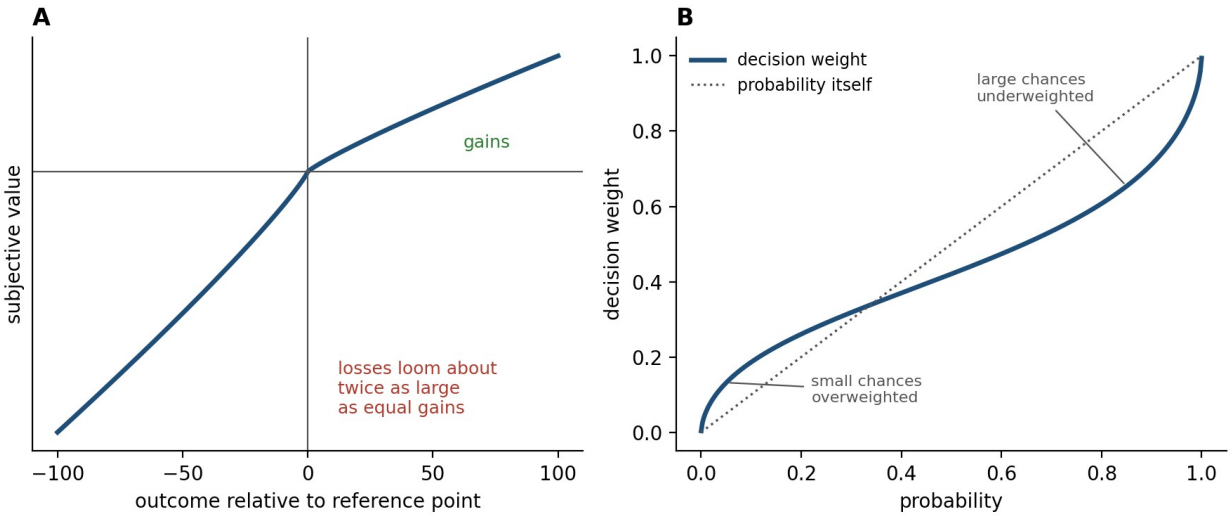


Figure 8.4. Prospect theory. A: value is concave for gains, convex for losses, and steeper for losses than for gains (loss aversion). B: decision weights (for gains) overweight small probabilities and underweight large ones. Curves use Tversky and Kahneman's 1992 estimates.

The reference point can be moved by words alone. Tversky and Kahneman told people that an unusual disease was expected to kill 600 people and offered two programs. When the programs were described in terms of lives saved, 72 percent chose the one that saved 200 for certain over a gamble with the same expected outcome; when they were described in terms of lives lost, 78 percent chose the gamble over the certain death of 400, though the outcomes were identical.[494] Benedetto De Martino and colleagues found that the amygdala was more active when people's choices followed the frame, and that people who resisted the frame had more activity in the orbital and medial prefrontal cortex.[495]

The circuits of Chapter 7 appear here directly. Sabrina Tom and colleagues offered people gambles with an even chance of winning or losing various amounts and found that activity in the ventral striatum and ventromedial prefrontal cortex rose with potential gains and fell, more steeply, with potential losses, and that the asymmetry in each person's brain predicted the asymmetry in their choices; they found no region whose activity rose with losses.[496] De Martino, Colin Camerer, and Ralph Adolphs tested two patients with amygdala damage on gambles of this kind, one of them S.M., the woman of Chapter 7 who felt almost no fear. Both accepted gambles that healthy people refused, and their loss aversion was sharply reduced.[497] The amygdala's contribution may lie less in valuing losses than in making them matter to choice.

The size of loss aversion, and for some kinds of choice its existence, has been questioned by David Gal and Derek Rucker, who argue that many apparent cases have other explanations.[498] A meta-analysis by

[494] Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science*, 211(4481), 453–458.

[495] De Martino, B., Kumaran, D., Seymour, B., & Dolan, R. J. (2006). Frames, biases, and rational decision-making in the human brain. *Science*, 313(5787), 684–687.

[496] Tom, S. M., Fox, C. R., Trepel, C., & Poldrack, R. A. (2007). The neural basis of loss aversion in decision-making under risk. *Science*, 315(5811), 515–518.

[497] De Martino, B., Camerer, C. F., & Adolphs, R. (2010). Amygdala damage eliminates monetary loss aversion. *Proceedings of the National Academy of Sciences*, 107(8), 3788–3792.

[498] Gal, D., & Rucker, D. D. (2018). The loss of loss aversion: Will it loom larger than its gain? *Journal of Consumer Psychology*, 28(3), 497–516.

Alexander Brown and colleagues found a typical value somewhat below two, with wide variation across studies, so the factor of two is better read as a typical finding than a constant of nature.[499]

Prospect theory's overweighting of rare events also depends on how the odds are learned. When people are told the probabilities, they overweight rare outcomes; when they learn them by sampling outcomes one at a time, as animals always must, they underweight rare outcomes, because a rare event is usually absent from a short run of experience.[500] The difference has come to be called the description-experience gap.

A further kind of uncertainty is not knowing the probabilities at all. Daniel Ellsberg asked people to bet on drawing a red ball from an urn with 50 red and 50 black balls, or from an urn with 100 balls in an unknown mix. Most preferred the known urn when betting on red and also preferred it when betting on black.[501] Preferring the known urn for red implies a belief that red is less likely than half in the unknown urn, and preferring it for black implies the same of black, and both cannot be true; the preference, now called ambiguity aversion, is a dislike of unknown odds as such. Ming Hsu and colleagues found that ambiguous choices engaged the amygdala and orbitofrontal cortex more than risky ones with known odds, and that patients with orbitofrontal damage were less averse to both ambiguity and risk than healthy people.[502] Ambiguity aversion makes sense for an animal whose ignorance often means that someone else knows more: the unknown urn may have been stacked.

The Price of Waiting

Many choices trade a smaller reward now against a larger one later, and the value of a delayed reward is discounted. A rational agent with a constant risk of losing a future reward would discount it exponentially, by the same fraction for each unit of delay. People and animals do not. Their discounting is closer to hyperbolic, steep for short delays and shallow for long ones, so that the value of a reward of size A after delay D is roughly A/(1 + kD), where k measures impatience.[503] Robert Strotz showed in 1955 that a chooser who discounts this way will not keep to its own plans, and George Ainslie made the consequence concrete: preferences reverse as time passes.[504] An impatient enough person who prefers 110 dollars in 31 days to 100 dollars in 30 days will, 30 days later, prefer 100 dollars now to 110 tomorrow (Figure 8.5). The plans we make for the future and the choices we make when it arrives come from the same discounting curve, viewed from different distances.

[499] Brown, A. L., Imai, T., Vieider, F. M., & Camerer, C. F. (2024). Meta-analysis of empirical estimates of loss aversion. *Journal of Economic Literature*, 62(2), 485–516.

[500] Hertwig, R., Barron, G., Weber, E. U., & Erev, I. (2004). Decisions from experience and the effect of rare events in risky choice. *Psychological Science*, 15(8), 534–539.

[501] Ellsberg, D. (1961). Risk, ambiguity, and the Savage axioms. *Quarterly Journal of Economics*, 75(4), 643–669.

[502] Hsu, M., Bhatt, M., Adolphs, R., Tranel, D., & Camerer, C. F. (2005). Neural systems responding to degrees of uncertainty in human decision-making. *Science*, 310(5754), 1680–1683.

[503] Mazur, J. E. (1987). An adjusting procedure for studying delayed reinforcement. In M. L. Commons, J. E. Mazur, J. A. Nevin, & H. Rachlin (Eds.), *Quantitative Analyses of Behavior: Vol. 5. The Effect of Delay and of Intervening Events on Reinforcement Value* (pp. 55–73). Erlbaum.

[504] Strotz, R. H. (1955). Myopia and inconsistency in dynamic utility maximization. *Review of Economic Studies*, 23(3), 165–180; Ainslie, G. (1975). Specious reward: A behavioral theory of impulsiveness and impulse control. *Psychological Bulletin*, 82(4), 463–496.

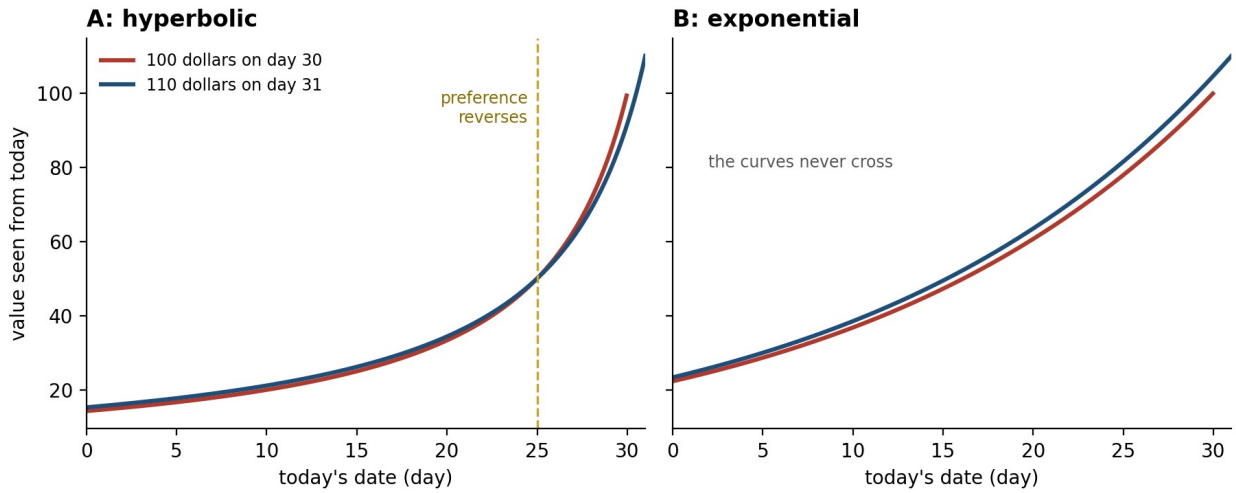


Figure 8.5. Discounting and preference reversal (illustrative). Each curve is the present value of a reward as its delivery approaches. Seen from far away, the larger, later reward is worth more; close to the smaller reward, its value shoots above the larger one's, and preference reverses. With exponential discounting the curves never cross.

Hyperbolic discounting is not obviously a mistake. Peter Sozou showed that if the risk of losing a future reward is itself uncertain, the rational discount curve is hyperbolic: the longer a promised reward has survived, the more likely it is to be one of the safe ones.[505]

How the brain computes it is debated. Samuel McClure and colleagues found that choices involving an immediate reward engaged the ventral striatum and medial prefrontal cortex, while all choices engaged the lateral prefrontal and parietal cortex, and proposed what they described as an impatient, emotional system competing with a patient, deliberative one.[506] Kable and Glimcher found instead that activity in the same valuation regions tracked each person's own hyperbolic discounted value at every delay, as a single system would.[507] A third view treats patience as self-control rather than as a discount rate. Todd Hare, Camerer, and Rangel found that in dieters choosing between tasty and healthy foods, the dorsolateral prefrontal cortex did not compete with the valuation system but adjusted it, adding health to the value signal in those who chose well.[508] The serotonin neurons of Chapter 3 are one candidate for a signal that sets patience: their firing rose while a rat waited for a delayed reward, and stimulating them with light made mice wait longer.[509]

The most famous test of delay is the marshmallow test, in which Walter Mischel and colleagues left preschool children alone with one treat and the promise of two if they waited. Children who waited longer later scored higher on academic tests as adolescents.[510] A larger replication by Tyler Watts, Greg Duncan, and Haonan Quan, in about 900 children of mothers without a college degree, found an

[505] Sozou, P. D. (1998). On hyperbolic discounting and uncertain hazard rates. *Proceedings of the Royal Society B*, 265(1409), 2015–2020.

[506] McClure, S. M., Laibson, D. I., Loewenstein, G., & Cohen, J. D. (2004). Separate neural systems value immediate and delayed monetary rewards. *Science*, 306(5695), 503–507.

[507] Kable, J. W., & Glimcher, P. W. (2007). The neural correlates of subjective value during intertemporal choice. *Nature Neuroscience*, 10(12), 1625–1633.

[508] Hare, T. A., Camerer, C. F., & Rangel, A. (2009). Self-control in decision-making involves modulation of the vmPFC valuation system. *Science*, 324(5927), 646–648.

[509] Miyazaki, K. W., Miyazaki, K., Tanaka, K. F., Yamanaka, A., Takahashi, A., Tabuchi, S., & Doya, K. (2014). Optogenetic activation of dorsal raphe serotonin neurons enhances patience for future rewards. *Current Biology*, 24(17), 2033–2040.

association about half as large, and much of it disappeared once family background and early cognitive ability were taken into account.[511] Celeste Kidd, Holly Palmeri, and Richard Aslin found, in 28 children, that those who had just seen the experimenter keep a promise waited about four times as long as those who had seen one broken.[512] A later study of 149 children found no significant effect of the experimenter's trustworthiness on waiting, and a larger role for whether children were in the habit of waiting before they ate.[513] Waiting is partly a judgment about how reliable the future is, and where promises are broken, taking the marshmallow can be the rational choice.

Explore or Exploit

Every choice also decides what will be learned. An option never tried is an option whose value is never discovered, so an agent that always takes what currently looks best can lock onto a mediocre option forever. One that keeps sampling the alternatives pays for the information in rewards forgone. This is the explore-exploit dilemma, and it has no solution that is both general and practical. Its simplest version is a row of slot machines with fixed but unknown payoffs, called the multi-armed bandit problem after the casino's one-armed bandit. For this case John Gittins showed in 1979 that the optimal policy assigns each machine an index combining its estimated value with the value of what could be learned from it, and Peter Auer and colleagues later gave simpler rules, such as adding a bonus for uncertainty to each estimate, whose losses grow only slowly as play goes on (Figure 8.6).[514]

[510] Mischel, W., Shoda, Y., & Rodriguez, M. L. (1989). Delay of gratification in children. *Science*, 244(4907), 933–938; Shoda, Y., Mischel, W., & Peake, P. K. (1990). Predicting adolescent cognitive and self-regulatory competencies from preschool delay of gratification: Identifying diagnostic conditions. *Developmental Psychology*, 26(6), 978–986.

[511] Watts, T. W., Duncan, G. J., & Quan, H. (2018). Revisiting the marshmallow test: A conceptual replication investigating links between early delay of gratification and later outcomes. *Psychological Science*, 29(7), 1159–1177.

[512] Kidd, C., Palmeri, H., & Aslin, R. N. (2013). Rational snacking: Young children's decision-making on the marshmallow task is moderated by beliefs about environmental reliability. *Cognition*, 126(1), 109–114.

[513] Yanaoka, K., Michaelson, L. E., Saito, S., & Munakata, Y. (2026). Waiting longer with less work: The impact of habits and social trust on children's delay of gratification. *Developmental Science*, 29(3), e70185.

[514] Gittins, J. C. (1979). Bandit processes and dynamic allocation indices. *Journal of the Royal Statistical Society: Series B*, 41(2), 148–177; Auer, P., Cesa-Bianchi, N., & Fischer, P. (2002). Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, 47(2–3), 235–256.

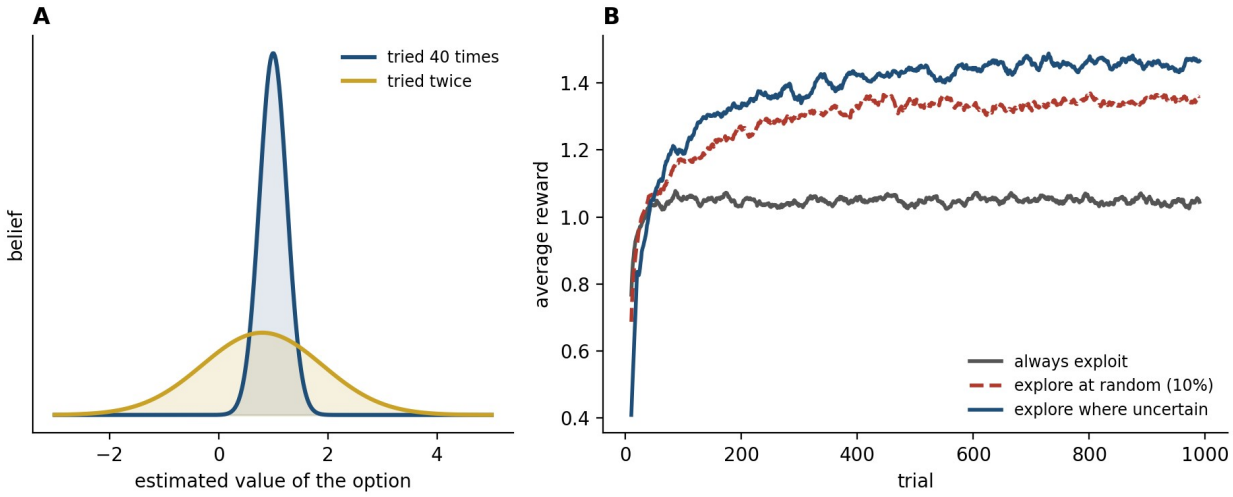


Figure 8.6. Explore or exploit (simulated). A: an option tried 40 times and one tried twice have similar estimated values but very different uncertainty; an uncertainty bonus favors trying the second. B: average reward over trials on a ten-armed bandit for an agent that always exploits, one that explores at random, and one that explores where uncertainty is greatest.

There are two ways to explore: at random, by now and then choosing something other than the apparent best, or in a directed way, by choosing the option about which least is known. People do both. Nathaniel Daw and colleagues gave people a four-armed bandit whose payoffs drifted over time and found that their exploratory choices were best described as partly random, and that these choices engaged the frontopolar cortex at the front of the brain.[515] Robert Wilson and colleagues separated the two by telling people how many more choices they had. With a longer horizon, when information was worth more, people chose the less-known option more often and also chose more randomly.[516] How fast to learn should also depend on how fast the world is changing, and Timothy Behrens and colleagues found that people's learning rates rose when the payoffs of a task became more volatile, in step with activity in the anterior cingulate cortex, as the account of norepinephrine and unexpected uncertainty in Chapter 3 would predict, although the experiment did not measure norepinephrine.[517]

Animals face the dilemma in its oldest form when foraging: stay in a patch that is being depleted or leave for one that might be better. Eric Charnov's marginal value theorem says to leave when the rate of reward in the current patch falls to the average rate available in the environment.[518] Benjamin Hayden, John Pearson, and Michael Platt found neurons in the dorsal anterior cingulate cortex of monkeys whose firing rose with each harvest from a depleting patch and reached a threshold when the monkey left, and Nils Kolling and colleagues found that the same region in people tracked the value of searching elsewhere, although Amitai Shenhav and colleagues have argued that the signal reflects how difficult the choice is rather than the value of foraging.[519] Foraging may be the template from which much of the

[515] Daw, N. D., O'Doherty, J. P., Dayan, P., Seymour, B., & Dolan, R. J. (2006). Cortical substrates for exploratory decisions in humans. *Nature*, 441(7095), 876–879.

[516] Wilson, R. C., Geana, A., White, J. M., Ludvig, E. A., & Cohen, J. D. (2014). Humans use directed and random exploration to solve the explore–exploit dilemma. *Journal of Experimental Psychology: General*, 143(6), 2074–2081.

[517] Behrens, T. E. J., Woolrich, M. W., Walton, M. E., & Rushworth, M. F. S. (2007). Learning the value of information in an uncertain world. *Nature Neuroscience*, 10(9), 1214–1221.

[518] Charnov, E. L. (1976). Optimal foraging, the marginal value theorem. *Theoretical Population Biology*, 9(2), 129–136.

[519] Hayden, B. Y., Pearson, J. M., & Platt, M. L. (2011). Neuronal basis of sequential foraging decisions in a patchy environment. *Nature Neuroscience*, 14(7), 933–939; Kolling, N., Behrens, T. E. J., Mars, R. B., & Rushworth, M. F. S. (2012). Neural mechanisms

brain's decision machinery evolved: a choice not between two options presented together, but between staying with what one has and looking for something better.

The average rate of reward also sets the cost of time itself. Yael Niv and colleagues proposed that the slow, tonic level of dopamine reports this average, so that a rich environment makes every moment of idleness expensive and every action faster, which may be part of why the dopamine-depleted patients of Chapter 4 move slowly.[520]

Goals and Habits

In 1948 Edward Tolman argued from experiments on rats in mazes that animals do not only learn which turns are rewarded; they learn a map of the maze and use it to plan.[521] Whether a given action is planned in this way or merely repeated was given a sharp test by Christopher Adams and Anthony Dickinson in 1981, which Adams refined a year later. Rats learned to press a lever for sugar pellets. Then the sugar was devalued, paired with a mild illness so that the rats would no longer eat it, and the rats were tested on the lever with no pellets delivered, so that they could not learn the new value of the sugar from the lever itself. Rats that had been trained moderately pressed less, as if they consulted the outcome and found it no longer wanted. Rats that had been trained extensively kept pressing at the old rate, for an outcome they would not eat (Figure 8.7).[522]

of foraging. *Science*, 336(6077), 95–98; Shenhav, A., Straccia, M. A., Cohen, J. D., & Botvinick, M. M. (2014). Anterior cingulate engagement in a foraging context reflects choice difficulty, not foraging value. *Nature Neuroscience*, 17(9), 1249–1254.

[520] Niv, Y., Daw, N. D., Joel, D., & Dayan, P. (2007). Tonic dopamine: Opportunity costs and the control of response vigor. *Psychopharmacology*, 191(3), 507–520.

[521] Tolman, E. C. (1948). Cognitive maps in rats and men. *Psychological Review*, 55(4), 189–208.

[522] Adams, C. D., & Dickinson, A. (1981). Instrumental responding following reinforcer devaluation. *Quarterly Journal of Experimental Psychology Section B*, 33(2), 109–121; Adams, C. D. (1982). Variations in the sensitivity of instrumental responding to reinforcer devaluation. *Quarterly Journal of Experimental Psychology Section B*, 34(2), 77–98.

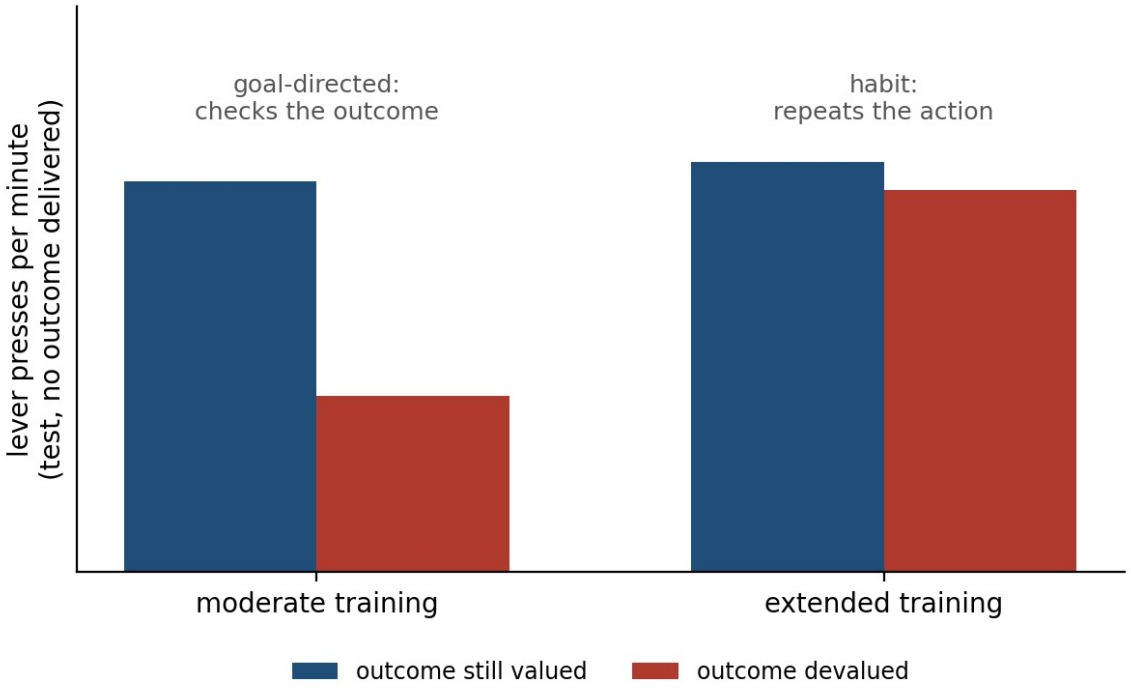


Figure 8.7. Goal-directed and habitual action (illustrative, after Adams 1982). After moderate training, devaluing the outcome reduces responding; after extended training it does not. The first is the signature of a goal, the second of a habit.

The two kinds of control use different circuits. Goal-directed actions depend on the dorsomedial striatum and the prefrontal cortex that projects to it; habits depend on the dorsolateral striatum, and Henry Yin, Barbara Knowlton, and Bernard Balleine found that rats with lesions there stayed goal-directed even on training schedules that make normal rats habitual.[523] In people, Elizabeth Tricomi and colleagues reported that three days of training made responding insensitive to devaluation and engaged the corresponding part of the dorsal striatum, but Sanne de Wit and colleagues failed to produce the effect in five experiments, two of them direct replications of Tricomi's, so whether overtraining alone produces habits in people is uncertain.[524]

Computational accounts give the distinction a principle. A model-based system learns how actions lead to outcomes and plans by simulating the consequences; it is flexible and expensive. A model-free system caches the value of each action from experience, through the reward prediction error of Chapter 3; it is fast and cheap and cannot adjust when an outcome's value changes until it has experienced the change. Daw, Niv, and Peter Dayan proposed that the brain runs both and gives control to whichever is less uncertain in the current situation, which would explain why habits take over with practice, when the cached values have become reliable.[525] Kevin Miller, Amitai Shenhav, and Elliot Ludvig argue that habits

[523] Yin, H. H., Knowlton, B. J., & Balleine, B. W. (2004). Lesions of dorsolateral striatum preserve outcome expectancy but disrupt habit formation in instrumental learning. *European Journal of Neuroscience*, 19(1), 181–189; Balleine, B. W., & O'Doherty, J. P. (2010). Human and rodent homologies in action control: Corticostriatal determinants of goal-directed and habitual action. *Neuropsychopharmacology*, 35(1), 48–69.

[524] Tricomi, E., Balleine, B. W., & O'Doherty, J. P. (2009). A specific role for posterior dorsolateral striatum in human habit learning. *European Journal of Neuroscience*, 29(11), 2225–2232; de Wit, S., Kindt, M., Knot, S. L., Verhoeven, A. A. C., Robbins, T. W., Gasull-Camos, J., Evans, M., Mirza, H., & Gillan, C. M. (2018). Shifting the balance between goals and habits: Five failures in experimental habit induction. *Journal of Experimental Psychology: General*, 147(7), 1043–1065.

are not cached values at all but actions strengthened simply by being repeated, which accounts for their insensitivity to devaluation just as well.[526]

Daw and colleagues designed a two-step task to separate the systems. Each first choice led, usually but not always, to one of two further choices, and only a learner with a model of those transitions would treat a reward reached by the rare route differently from one reached by the common route. People's choices showed both influences at once, and even the striatal prediction error carried information that only a model could have supplied. Later analyses showed that some apparently model-based patterns can also be produced by sophisticated model-free strategies.[527] Claire Gillan and colleagues tested more than 1,000 people online with this task and found that a dimension of compulsive symptoms, cutting across obsessive-compulsive disorder, addiction, and binge eating, was associated with weaker model-based control.[528]

There is a third controller, older than both. Pavlovian responses (approach to cues that predict reward, withdrawal from cues that predict harm) are emitted whether or not they serve the animal. Keller and Marian Breland, training animals for advertisements, found that a raccoon taught to drop coins into a box began instead to rub them together and dip them, as raccoons do with food, and could not be trained out of it.[529] David Williams and Harriet Williams found that pigeons kept pecking a lit key that predicted food even when pecking it canceled the food.[530] Dayan, Niv, and colleagues argued, with models, that a Pavlovian controller of this kind can override instrumental learning, the learning of actions by their consequences, even when doing so costs reward, and that much of what looks like weakness of will is this override.[531] How a choice repeated often enough stops being a choice at all, and how deliberate action becomes routine and skill, is the subject of Chapter 9.

Thinking Is Costly

A brain that planned every choice from a full model of the world would be accurate and paralyzed. Herbert Simon argued in the 1950s that real choosers have bounded rationality: they satisfice, searching until they find an option good enough rather than the best, because search itself is expensive.[532] The cheapest rules can be surprisingly good. Daniel Goldstein and Gerd Gigerenzer asked American and German students which of two American cities was larger, San Diego or San Antonio (San Diego, at the time). About 62 percent of the Americans, who knew both cities, answered correctly; every one of the Germans did, because most had heard of San Diego and not of San Antonio, and they chose the city they

[525] Daw, N. D., Niv, Y., & Dayan, P. (2005). Uncertainty-based competition between prefrontal and dorsolateral striatal systems for behavioral control. *Nature Neuroscience*, 8(12), 1704–1711.

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[527] Daw, N. D., Gershman, S. J., Seymour, B., Dayan, P., & Dolan, R. J. (2011). Model-based influences on humans' choices and striatal prediction errors. *Neuron*, 69(6), 1204–1215; Akam, T., Costa, R., & Dayan, P. (2015). Simple plans or sophisticated habits? State, transition and learning interactions in the two-step task. *PLoS Computational Biology*, 11(12), e1004648.

[528] Gillan, C. M., Kosinski, M., Whelan, R., Phelps, E. A., & Daw, N. D. (2016). Characterizing a psychiatric symptom dimension related to deficits in goal-directed control. *eLife*, 5, e11305.

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[531] Dayan, P., Niv, Y., Seymour, B., & Daw, N. D. (2006). The misbehavior of value and the discipline of the will. *Neural Networks*, 19(8), 1153–1160.

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recognized.[533] Recognition is a good cue to size because large cities are mentioned more, and knowing less let the German students use it. Gigerenzer and Goldstein showed that rules of this kind, which decide by the single most valid cue and ignore the rest, can match elaborate statistical methods.[534]

Falk Lieder and Thomas Griffiths have generalized the point into a framework, resource rationality, in which many apparent biases are optimal once the cost of computation is counted.[535] The brain may even estimate that cost: Shenhav, Matthew Botvinick, and Jonathan Cohen argue that the dorsal anterior cingulate cortex weighs the expected value of exerting control and decides whether a situation is worth thinking hard about.[536] This reframes the biases of the previous sections. Loss aversion, ambiguity aversion, steep discounting, and habit are often treated as defects of a rational machine. Many of them make better sense as the policies of a machine that must decide quickly, with limited computation, in a world that is uncertain, changing, and sometimes adversarial. Some are miscalibrated for the modern world, where losses are rarely fatal and promises are often kept; few are arbitrary.

Choosing Among Others

Other people are the most uncertain part of most environments, and they change the decision problem. In the ultimatum game, devised by Werner Güth and colleagues, one player proposes how to split a sum of money and the other accepts, in which case both are paid, or rejects, in which case neither is.[537] A purely self-interested responder should accept any positive offer. In industrialized societies, offers below about a fifth of the sum are rejected roughly half the time, and proposers, anticipating this, typically offer 40 to 50 percent.[538] Alan Sanfey and colleagues found that unfair offers engaged the anterior insula of Chapter 7, and that the stronger its response the more likely the offer was to be rejected.[539]

Correlation does not show what the brain is doing, and a disruption can. Daria Knoch, Ernst Fehr, and colleagues used transcranial magnetic stimulation, a strong magnetic pulse applied through the scalp, to disrupt the right dorsolateral prefrontal cortex for several minutes. Their participants then accepted more unfair offers, while still judging the offers to be as unfair as before.[540] The judgment was intact and the will to act on it was weakened: the Iowa patients' problem of knowing the rule and not acting on it, appearing in a social choice.

Joseph Henrich and colleagues ran the game in 15 small-scale societies and found mean offers ranging from about a quarter to more than half of the sum, varying with how integrated each society was into

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[533] Goldstein, D. G., & Gigerenzer, G. (2002). Models of ecological rationality: The recognition heuristic. *Psychological Review*, 109(1), 75–90.

[534] Gigerenzer, G., & Goldstein, D. G. (1996). Reasoning the fast and frugal way: Models of bounded rationality. *Psychological Review*, 103(4), 650–669.

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[539] Sanfey, A. G., Rilling, J. K., Aronson, J. A., Nystrom, L. E., & Cohen, J. D. (2003). The neural basis of economic decision-making in the Ultimatum Game. *Science*, 300(5626), 1755–1758.

[540] Knoch, D., Pascual-Leone, A., Meyer, K., Treyer, V., & Fehr, E. (2006). Diminishing reciprocal fairness by disrupting the right prefrontal cortex. *Science*, 314(5800), 829–832.

markets and how much its people gained from cooperating with one another.[541] The sense of fairness that drives rejection is felt in the body and learned in a culture, and it lets a social animal enforce the cooperation it depends on.

When Choice Goes Wrong

Because choice is built from identifiable components, its disorders can be traced to them. The loss of dopamine in Parkinson's disease changes learning as well as movement: as Chapter 3 described, patients off their medication learn better from bad outcomes than good ones, and on medication the bias reverses.[542] Some patients treated with drugs that stimulate dopamine receptors develop compulsive gambling, shopping, or eating; in a study of more than 3,000 patients, Daniel Weintraub and colleagues found such impulse control disorders in about one in seven, with nearly three times the odds in those taking these drugs.[543]

David Redish proposed that addiction exploits the learning rule itself. Natural rewards stop producing a prediction error once they are expected, which is what makes the dopamine signal of Chapter 3 a teacher. A drug that raises dopamine directly would produce a positive error every time, however well it was predicted, so the value of drug-seeking would grow without limit.[544] Chapter 7 called addiction a form of reward hacking; this is how the hack might work. The idea is elegant but one of several, and one of its clearest predictions, that cues paired with cocaine should escape the normal rule by which an already predicted reward teaches nothing new, failed when tested in rats.[545] With Gillan's compulsivity dimension it belongs to a broader program, computational psychiatry, that tries to describe disorders as specific failures of the machinery this chapter has described.[546] In each case what fails is not reasoning but the learning that decides what to choose, and the most persistent of those failures are habits that have outlived their reasons.

Implications for Artificial Intelligence

Decision-making under uncertainty is the field where neuroscience and artificial intelligence have exchanged the most, because reinforcement learning is a theory of both.[547] The comparison is therefore less about what machines lack than about which of the brain's solutions they have already borrowed and which they have not.

First, accumulation to a bound has only recently begun to appear in artificial systems. Most networks still spend the same computation on every input, easy or hard, although methods for adapting

[541] Henrich, J., Boyd, R., Bowles, S., Camerer, C., Fehr, E., Gintis, H., & McElreath, R. (2001). In search of Homo economicus: Behavioral experiments in 15 small-scale societies. *American Economic Review*, 91(2), 73–78.

[542] Frank, M. J., Seeberger, L. C., & O'Reilly, R. C. (2004). By carrot or by stick: Cognitive reinforcement learning in parkinsonism. *Science*, 306(5703), 1940–1943.

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[546] Huys, Q. J. M., Maia, T. V., & Frank, M. J. (2016). Computational psychiatry as a bridge from neuroscience to clinical applications. *Nature Neuroscience*, 19(3), 404–413.

[547] Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.

computation to the input go back at least to Alex Graves's adaptive computation time.[548] Language models that reason step by step before answering spend more on harder problems, a step toward the brain's practice, but they rarely have an explicit bound that moves with the stakes.

Second, calibrated confidence is hard to get. Chuan Guo and colleagues found that the deep networks of the mid-2010s were systematically overconfident, assigning high probability to answers that were often wrong, and that a simple rescaling fixed much of it.[549] Newer architectures are better calibrated, and large language models can estimate fairly well which of their answers are right, but confidence that stays reliable when the input is unlike the training data remains elusive.[550] Yarin Gal and Zoubin Ghahramani showed one way for a network to report its own uncertainty, by running it many times with random parts switched off.[551] The monkey that takes the sure target when the dots are weak, and the rat that waits only as long as its confidence warrants, are doing what an artificial system must do to be trusted: knowing when it does not know, and acting on it.

Third, exploration is mostly set by hand. Research agents add bonuses for novelty and uncertainty, and the bonuses of the bandit literature are used in practice, but most agents explore on a schedule fixed in advance.[552] People adjust exploration to the horizon and to the volatility of the world, and norepinephrine, on the account of Chapter 3, appears to help set the rate.

Fourth, the best artificial decision-makers already combine habit and planning. AlphaGo, the first program to beat a professional Go player on a full board without handicap, paired two networks, one trained to propose good moves at a glance and one to judge positions, with a search that simulated future play from a model of the game.[553] That is a model-free and a model-based system working together. The brain's addition is arbitration: deciding, moment by moment, which system to trust and how much computation a choice deserves.

Fifth, the biases are not simply errors to remove. For the reasons given in the section on the cost of thinking, loss aversion, ambiguity aversion, and a preference for reliable options are protections as well as distortions, and an artificial agent that must act in an adversarial or changing world may need versions of them.

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[548] Graves, A. (2016). Adaptive computation time for recurrent neural networks. *arXiv*, 1603.08983.

[549] Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. *Proceedings of the 34th International Conference on Machine Learning*, PMLR 70, 1321–1330.

[550] Minderer, M., Djolonga, J., Romijnders, R., Hubis, F., Zhai, X., Houlsby, N., Tran, D., & Lucic, M. (2021). Revisiting the calibration of modern neural networks. *Advances in Neural Information Processing Systems*, 34. arXiv:2106.07998; Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., … Kaplan, J. (2022). Language models (mostly) know what they know. *arXiv*, 2207.05221.

[551] Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. *Proceedings of the 33rd International Conference on Machine Learning*, PMLR 48, 1050–1059.

[552] Bellemare, M. G., Srinivasan, S., Ostrovski, G., Schaul, T., Saxton, D., & Munos, R. (2016). Unifying count-based exploration and intrinsic motivation. *Advances in Neural Information Processing Systems*, 29. arXiv:1606.01868.

[553] Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., van den Driessche, G., Schrittwieser, J., Antonoglou, I., Panneershelvam, V., Lanctot, M., Dieleman, S., Grewe, D., Nham, J., Kalchbrenner, N., Sutskever, I., Lillicrap, T., Leach, M., Kavukcuoglu, K., Graepel, T., & Hassabis, D. (2016). Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587), 484–489.

The Central Lesson

A decision is a bet placed before the evidence is complete, and the brain is built to place it well rather than to place it perfectly. It accumulates evidence until being wrong costs less than waiting, prices options on a scale set by the state of the body and the options on offer, weights losses and ambiguity more heavily than a calculator would, and keeps paying to learn about options it is not choosing. It runs a planner and a habit system side by side, under the pull of older Pavlovian reflexes. Every one of these settings is learned from outcomes. The Iowa patients could state the rules and still lose because the part of the brain that learns what options are worth, and revises that learning when the world changes, was damaged. Choosing well is not knowing the rules. It is knowing how sure you are, spending thought where thought pays, and learning from every outcome.

Next Chapter

Chapter 9: Habits, Skills, and Automaticity: How Practice Turns Choices into Routines

Figure and Epigraph Sources

Figure Attribution

Figures 8.1 to 8.7 were drawn for this book. The firing rates in Figure 8.2 are illustrative, after the pattern reported by Roitman and Shadlen (2002), not recorded data. The traces and curves in Figure 8.3, the bandit results in Figure 8.6, and the curves in Figure 8.5 are simulations or computations. Figure 8.4 plots the functions of Tversky and Kahneman (1992) with their published parameter estimates. The bars in Figure 8.7 are illustrative, after the pattern reported by Adams (1982).

Epigraph

Blaise Pascal, *Pensées*, §277, trans. W. F. Trotter (E. P. Dutton, 1958); first published in French in 1670.

Chapter 9

Habits, Skills, and Automaticity: How Practice Turns Choices into Routines

> *"Civilization advances by extending the number of important operations which we can perform without thinking about them."*
>
> *— Alfred North Whitehead, An Introduction to Mathematics (1911)*

What Practice Changes

Chapter 8 closed on habits that have outlived their reasons, and on rats that kept pressing a lever for food they would no longer eat. After enough training, an action that had begun as a choice, made because of what it produced, had become something the animal simply did when it found itself in front of the lever. Chapter 8 showed why that can be a sensible policy. This chapter shows what it achieves: the same process that makes a habit blind to its outcome is the one that lets a pianist play a passage too fast to think about, a driver hold a conversation on a familiar road, and a reader take in this sentence without sounding out its letters.

In the 1950s the British engineer E. R. F. W. Crossman studied the operators of cigar-making machines in a factory. Each operator fed tobacco leaf to a machine whose own cycle time set a floor on how fast a cigar could be made. Crossman found that the operators kept getting faster for years, over millions of cigars, and that their speed stopped improving only as it approached the limit of the machine.[554] No one would have said that an operator in a third year was still learning, and yet she was. Practice does not stop working when a skill feels finished.

Whitehead, who supplies this chapter's epigraph, added that operations of thought are like cavalry charges in a battle: strictly limited in number, they require fresh horses, and must be made only at decisive moments.[555] Chapter 8 met the same economy in the cost of thinking. A brain that could not hand routine work to something cheaper would spend its limited thought on tying shoes.

The argument of this chapter is that practice works in several circuits, each described in terms the earlier chapters introduced: the cerebellum recalibrates movements that have drifted out of true (Chapter 4); the basal ganglia and the dopamine signal of Chapter 3 bind actions into units, hand their control from circuits that deliberate to circuits that execute, and keep the variations that work; and the cortex builds and then protects a few new connections. What results is automaticity: performance that is fast, consistent, and cheap in attention, and for the same reasons hard to inspect and hard to change.

The Shape of Practice

Crossman's operators improved quickly at first and then more and more slowly, and almost every skill that has been measured does the same. Plotted on ordinary axes, time per trial falls steeply and then flattens into what looks like a plateau. Plotted on logarithmic axes, with the number of practice trials and the time per trial both on log scales, the same data often fall close to a straight line until they meet a

[554] Crossman, E. R. F. W. (1959). A theory of the acquisition of speed-skill. *Ergonomics*, 2(2), 153–166.

[555] Whitehead, A. N. (1911). *An Introduction to Mathematics*. Williams & Norgate. The quotation and the image of the cavalry charge are from chapter 5, "The Symbolism of Mathematics."

floor (Figure 9.1). A straight line on log-log axes is a power function, and Allen Newell and Paul Rosenbloom, surveying tasks from cigar rolling to proof writing and card games, called the regularity the power law of practice.[556] Its meaning is that each doubling of practice buys about the same proportional gain.

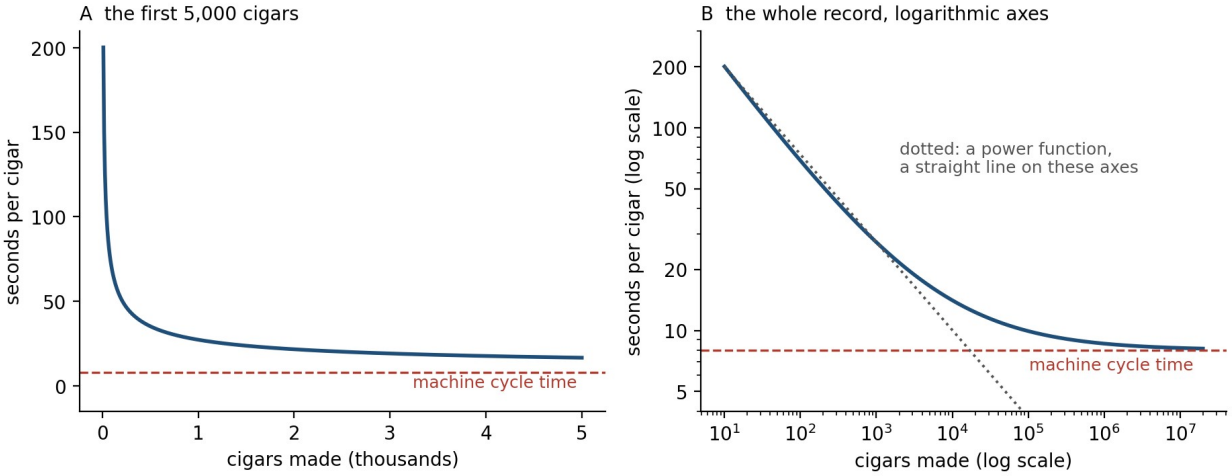


Figure 9.1. The shape of practice (illustrative, after Crossman 1959). A: over the first few thousand cigars, time per cigar falls steeply and then slowly. B: on logarithmic axes the early record is close to a straight line, the signature of a power function, and it bends away from the line as times approach the machine's cycle time.

The law is less universal than its name. Andrew Heathcote, Scott Brown, and Douglas Mewhort showed that averaging curves across people manufactures a power function even when every individual improves along a different curve, and that individual curves are often better fitted by an exponential, which levels off more abruptly.[557] What survives is the qualitative shape: rapid early gains, slow late gains, and improvement that continues long after the learner has stopped noticing it.

Psychologists have long described the course of a skill in stages. Paul Fitts and Michael Posner distinguished a cognitive stage, in which the learner works out what to do and talks through each step; an associative stage, in which the pieces are linked and errors fall; and an autonomous stage, in which the skill runs with little attention and the learner may no longer be able to say how it is done.[558] Nearly 70 years earlier, William Bryan and Noble Harter had described the same progression in telegraph operators learning Morse code, who first received letters one at a time, then whole words, and finally phrases, building what they called a hierarchy of habits.[559] The stages are a description, not a mechanism, but they name the change this chapter explains: the same performance, produced first by deliberation and later by something else.

[556] Newell, A., & Rosenbloom, P. S. (1981). Mechanisms of skill acquisition and the law of practice. In J. R. Anderson (Ed.), *Cognitive Skills and Their Acquisition* (pp. 1–55). Lawrence Erlbaum.

[557] Heathcote, A., Brown, S., & Mewhort, D. J. K. (2000). The power law repealed: The case for an exponential law of practice. *Psychonomic Bulletin & Review*, 7(2), 185–207.

[558] Fitts, P. M., & Posner, M. I. (1967). *Human Performance*. Brooks/Cole.

[559] Bryan, W. L., & Harter, N. (1899). Studies on the telegraphic language: The acquisition of a hierarchy of habits. *Psychological Review*, 6(4), 345–375.

Habits and Skills

Two words have been used loosely so far, and they mean different things. A habit, in the sense of Chapter 8, is an action triggered by a situation rather than chosen for its outcome, and its signature is insensitivity to devaluation. A skill is an improvement in how well an action is done. John Krakauer and his colleagues define it by the trade-off between speed and accuracy that Chapter 8 met in decisions. Anyone can trace a difficult path accurately by going slowly, or quickly by accepting errors; a skilled performer is faster at the same accuracy, or more accurate at the same speed, so practice shifts the whole curve relating the two.[560] Habits and skills usually develop together, because the practice that makes a movement better also makes it more automatic, but they can be pulled apart. A musician can play a piece deliberately, choosing every phrasing; a person can have a strong habit, such as reaching for a phone, that involves no skill at all.

Both are forms of what Chapter 5 called procedural memory, and both survive damage that abolishes memory for events. Chapter 5's patient H.M., who learned mirror tracing while denying each day that he had seen the task, also improved over days, Suzanne Corkin found, at keeping a stylus on a moving target.[561] Barbara Knowlton, Jennifer Mangels, and Larry Squire later showed the converse. In a task in which people learned by trial and error to predict the weather from combinations of cards, amnesic patients improved as fast as healthy people, at least early in training, while being unable to recall the training sessions; patients with Parkinson's disease, whose striatum has lost most of its dopamine, remembered the sessions and failed to improve.[562] Amnesic patients even learn a repeating sequence hidden in a stream of lights, responding faster to it without knowing it is there.[563] Knowing that something happened and knowing how to do something depend on different circuits.

Recalibration

The simplest thing practice does is keep movements accurate as the body and the world change. Reza Shadmehr and Ferdinando Mussa-Ivaldi had people reach to targets while holding the handle of a robot arm that pushed sideways with a force proportional to the speed of the hand. The first reaches curved wildly. Over a few hundred movements they straightened. When the force was then switched off without warning, the reaches curved the other way, in a mirror image of the original error: the brain had learned to produce a force that canceled the robot's, and went on producing it after the robot stopped.[564] This mirror-image error, called an aftereffect, is the signature of adaptation: the motor system has recalibrated, and must recalibrate back.

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[560] Reis, J., Schambra, H. M., Cohen, L. G., Buch, E. R., Fritsch, B., Zarahn, E., Celnik, P. A., & Krakauer, J. W. (2009). Noninvasive cortical stimulation enhances motor skill acquisition over multiple days through an effect on consolidation. *Proceedings of the National Academy of Sciences*, 106(5), 1590–1595; Shmuelof, L., Krakauer, J. W., & Mazzoni, P. (2012). How is a motor skill learned? Change and invariance at the levels of task success and trajectory control. *Journal of Neurophysiology*, 108(2), 578–594; Krakauer, J. W., Hadjiosif, A. M., Xu, J., Wong, A. L., & Haith, A. M. (2019). Motor learning. *Comprehensive Physiology*, 9(2), 613–663.

[561] Corkin, S. (1968). Acquisition of motor skill after bilateral medial temporal-lobe excision. *Neuropsychologia*, 6(3), 255–265.

[562] Knowlton, B. J., Mangels, J. A., & Squire, L. R. (1996). A neostriatal habit learning system in humans. *Science*, 273(5280), 1399–1402.

[563] Nissen, M. J., & Bullemer, P. (1987). Attentional requirements of learning: Evidence from performance measures. *Cognitive Psychology*, 19(1), 1–32; Reber, P. J., & Squire, L. R. (1994). Parallel brain systems for learning with and without awareness. *Learning & Memory*, 1(4), 217–229.

[564] Shadmehr, R., & Mussa-Ivaldi, F. A. (1994). Adaptive representation of dynamics during learning of a motor task. *Journal of Neuroscience*, 14(5), 3208–3224.

The same thing happens with vision. Thomas Thach and his colleagues had people throw balls at a target while wearing prism glasses that shifted the visual scene to one side. The first throws missed, and over successive throws they came back onto the target; when the glasses came off, the throws missed to the other side until the adaptation washed out. Patients with damage to the cerebellum or to the inferior olive, the source of the climbing fibers that Chapter 4 described as the cerebellum's error signal, went on missing: adaptation is cerebellar learning.[565]

Ya-weng Tseng, Amy Bastian, and colleagues showed that adaptation is driven by a sensory prediction error, the difference between where the brain predicted the hand would go and where it was seen to go, rather than by failure to reach the goal.[566] Pietro Mazzoni and Krakauer made the distinction vivid. They rotated the cursor that showed people where their hand was by 45 degrees and then told them the trick: aim at the target 45 degrees away, and the cursor will land on the one you want. The strategy worked at once. Then, over the next few dozen reaches, the cursor began to miss, by larger and larger amounts, in the direction of the strategy. An implicit system had noticed that the hand was not going where the brain had aimed it, and it went on correcting that discrepancy even though the correction spoiled the performance the participants were trying to produce.[567]

Jordan Taylor, Krakauer, and Richard Ivry later asked people to report, before each reach, where they were aiming, and found that the explicit strategy accounted for the fast part of learning and an implicit recalibration for the slow remainder.[568] Adaptation has a memory with more than one timescale. Spontaneous recovery is the strangest sign of it. Adapt a person for a long time, then briefly the other way until performance is back at baseline, then run trials on which the error is held at zero whatever the hand does: the person drifts back toward the first adaptation. Maurice Smith, Ali Ghazizadeh, and Shadmehr explained this with a model of two processes running in parallel, a fast one that learns quickly and forgets quickly and a slow one that learns slowly and holds on (Figure 9.2).[569] Brief counter-training cancels the fast process while the slow one still carries the old learning, so when errors stop, the fast process decays and the slow one shows through. The model also explains part of savings, the faster relearning of a perturbation met a second time. When the first learning has been fully washed out, however, savings seem to depend on something else: remembering the aiming strategy that worked before.[570]

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[565] Martin, T. A., Keating, J. G., Goodkin, H. P., Bastian, A. J., & Thach, W. T. (1996). Throwing while looking through prisms. I. Focal olivocerebellar lesions impair adaptation. *Brain*, 119(4), 1183–1198.

[566] Tseng, Y.-W., Diedrichsen, J., Krakauer, J. W., Shadmehr, R., & Bastian, A. J. (2007). Sensory prediction errors drive cerebellum-dependent adaptation of reaching. *Journal of Neurophysiology*, 98(1), 54–62.

[567] Mazzoni, P., & Krakauer, J. W. (2006). An implicit plan overrides an explicit strategy during visuomotor adaptation. *Journal of Neuroscience*, 26(14), 3642–3645.

[568] Taylor, J. A., Krakauer, J. W., & Ivry, R. B. (2014). Explicit and implicit contributions to learning in a sensorimotor adaptation task. *Journal of Neuroscience*, 34(8), 3023–3032.

[569] Smith, M. A., Ghazizadeh, A., & Shadmehr, R. (2006). Interacting adaptive processes with different timescales underlie short-term motor learning. *PLoS Biology*, 4(6), e179.

[570] Zarahn, E., Weston, G. D., Liang, J., Mazzoni, P., & Krakauer, J. W. (2008). Explaining savings for visuomotor adaptation: Linear time-invariant state-space models are not sufficient. *Journal of Neurophysiology*, 100(5), 2537–2548; Haith, A. M., Huberdeau, D. M., & Krakauer, J. W. (2015). The influence of movement preparation time on the expression of visuomotor learning and savings. *Journal of Neuroscience*, 35(13), 5109–5117.

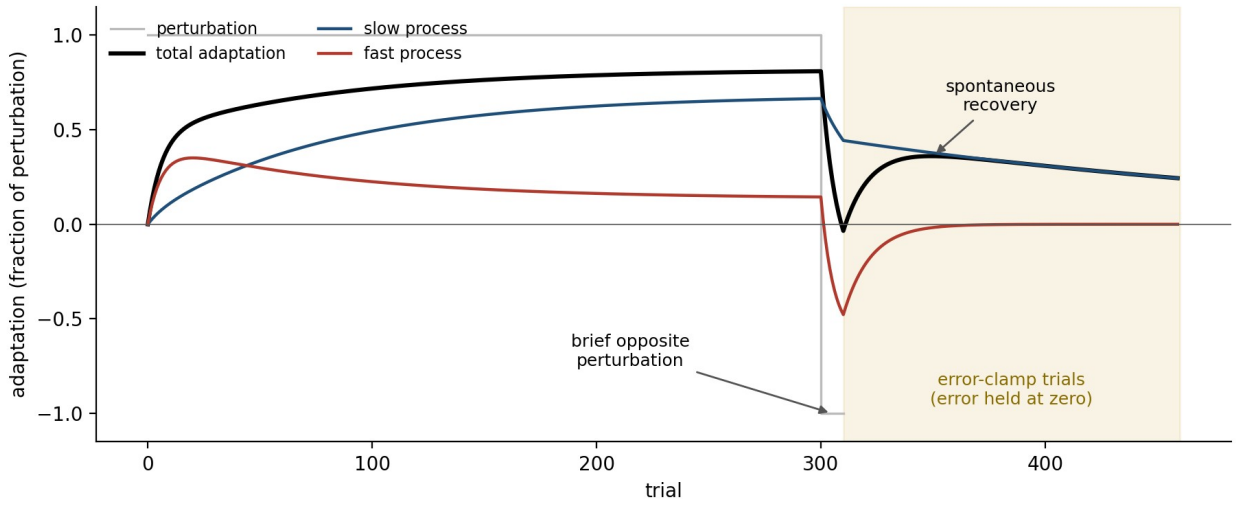


Figure 9.2. Two timescales of adaptation (simulated, two-state model after Smith, Ghazizadeh & Shadmehr 2006). A long perturbation is learned mostly by the slow process; a brief opposite perturbation drives the fast process negative until the total returns to zero; on the following error-clamp trials (error held at zero), the fast process decays and the slow process reappears as spontaneous recovery.

Error is not the only teacher. Repeating a movement biases later movements toward it, a use-dependent learning that needs no error at all.[571] And people can learn from success and failure alone, as an animal learns from reward: Amanda Therrien, Daniel Wolpert, and Bastian found that patients with cerebellar damage, who adapt poorly to sensory errors, could still improve their reaching when told only whether each movement had succeeded.[572]

Adaptation, however learned, returns a person to their previous level of performance under new conditions; it does not teach them to do what they could not do before. Krakauer has argued that much of what was studied as motor learning for decades was adaptation, and that skill, the acquisition of new capability, is a different and slower process that depends on other circuits.[573] The rest of this chapter is about that process.

Chunking

In 1951 Karl Lashley pointed out a problem with the then-dominant idea that a sequence of actions is a chain in which each movement is triggered by the sensation of the one before. A pianist's fingers, or a speaker's tongue, move too fast for feedback from one movement to arrive in time to trigger the next, and errors of serial order, such as the typist's transposed letters or the speaker's spoonerism, show that

[571] Diedrichsen, J., White, O., Newman, D., & Lally, N. (2010). Use-dependent and error-based learning of motor behaviors. *Journal of Neuroscience*, 30(15), 5159–5166.

[572] Therrien, A. S., Wolpert, D. M., & Bastian, A. J. (2016). Effective reinforcement learning following cerebellar damage requires a balance between exploration and motor noise. *Brain*, 139(1), 101–114.

[573] Krakauer, J. W., Hadjiosif, A. M., Xu, J., Wong, A. L., & Haith, A. M. (2019). Motor learning. *Comprehensive Physiology*, 9(2), 613–663; Diedrichsen, J., & Kornysheva, K. (2015). Motor skill learning between selection and execution. *Trends in Cognitive Sciences*, 19(4), 227–233.

later elements of a sequence are prepared before earlier ones are executed.[574] Sequences must be represented as wholes, planned in advance and run off as units.

George Miller had called such units chunks, in the context of memory: people can hold about seven items in mind (later estimates put the limit nearer four), but an item can be a digit, a familiar date, or a word, so what can be held depends on how the material is grouped.[575] William Chase and Herbert Simon showed what expertise does with this. Chess masters shown a position from a real game for five seconds could reconstruct far more of it than weaker players, but for pieces placed at random their advantage largely disappeared. The masters were not remembering pieces; they were recognizing familiar configurations, each a single chunk.[576]

K. Anders Ericsson, Chase, and Steve Faloon took an undergraduate with an ordinary memory span of seven digits and, over more than 230 hours of practice across nearly two years, watched him raise it to 79. He was a runner, and he had learned to code groups of digits as running times; his span for letters, which his codes did not cover, stayed at about six.[577] His skill was not a bigger memory. It was a larger vocabulary of chunks.

Motor sequences are chunked in the same way. Okihide Hikosaka and colleagues trained people on long sequences of key presses and found that each person, without being told to, came to perform the sequence in groups, fast within a group and with a pause between groups, and that the grouping differed from person to person. When the sequence was later rearranged in a way that broke a person's own chunks, performance suffered more than when it was rearranged to preserve them.[578] Nicholas Wymbs, Scott Grafton, and colleagues found that over three days of training, activity in the sensorimotor putamen, the part of the striatum tied to motor cortex, rose with how strongly the elements of a sequence were joined into chunks, while regions of frontal and parietal cortex were more active when a sequence was being divided into chunks.[579]

Recordings in animals show what a chunk looks like in the striatum. Mandar Jog, Ann Graybiel, and colleagues recorded from the sensorimotor striatum of rats learning to run a T-shaped maze, turning left or right according to a tone. Early in training, neurons fired throughout the run. With extended training, the activity reorganized: more and more of it concentrated at the start of the run and at its end, with relatively little in between (Figure 9.3).[580] Terra Barnes and colleagues later found that the pattern faded when the reward was withdrawn and came back when it was restored, as if the run had been packaged as a unit marked at its boundaries.[581] Graybiel called this task-bracketing and proposed that it

[574] Lashley, K. S. (1951). The problem of serial order in behavior. In L. A. Jeffress (Ed.), *Cerebral Mechanisms in Behavior: The Hixon Symposium* (pp. 112–146). Wiley.

[575] Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2), 81–97; Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1), 87–114.

[576] Chase, W. G., & Simon, H. A. (1973). Perception in chess. *Cognitive Psychology*, 4(1), 55–81.

[577] Ericsson, K. A., Chase, W. G., & Faloon, S. (1980). Acquisition of a memory skill. *Science*, 208(4448), 1181–1182.

[578] Sakai, K., Kitaguchi, K., & Hikosaka, O. (2003). Chunking during human visuomotor sequence learning. *Experimental Brain Research*, 152(2), 229–242.

[579] Wymbs, N. F., Bassett, D. S., Mucha, P. J., Porter, M. A., & Grafton, S. T. (2012). Differential recruitment of the sensorimotor putamen and frontoparietal cortex during motor chunking in humans. *Neuron*, 74(5), 936–946.

[580] Jog, M. S., Kubota, Y., Connolly, C. I., Hillegaart, V., & Graybiel, A. M. (1999). Building neural representations of habits. *Science*, 286(5445), 1745–1749.

[581] Barnes, T. D., Kubota, Y., Hu, D., Jin, D. Z., & Graybiel, A. M. (2005). Activity of striatal neurons reflects dynamic encoding and recoding of procedural memories. *Nature*, 437(7062), 1158–1161.

is how the basal ganglia chunk action repertoires.[582] Later work found that the striatal pattern forms early, before the behavior becomes insensitive to devaluation, and tracks how automatic the running is. A similar pattern appears in the infralimbic cortex, a small region of the medial prefrontal cortex, only late in overtraining, just as the habit sets in.[583]

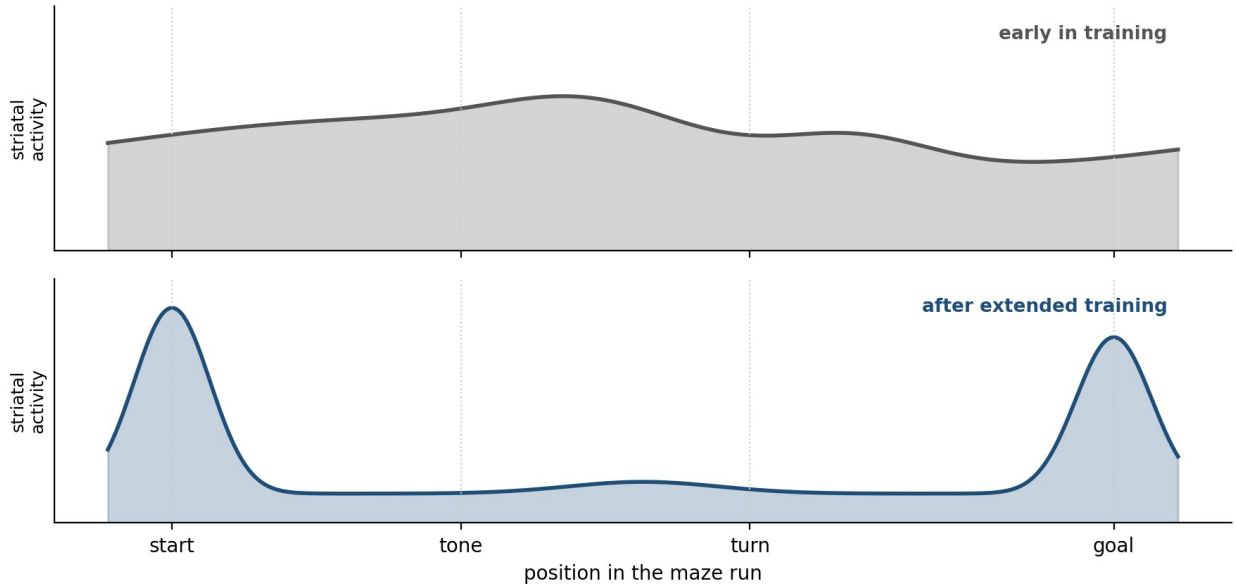


Figure 9.3. Task-bracketing in the striatum (illustrative, after Jog et al. 1999 and Barnes et al. 2005). Early in training, striatal activity is spread across the maze run; after extended training it concentrates at the start and the end of the run, marking the boundaries of the action as a unit.

Xin Jin and Rui Costa trained mice to press a lever in a rapid sequence for a reward and found neurons in the striatum and in the dopamine-producing substantia nigra that fired specifically at the start or the end of each sequence. This start and stop activity emerged during learning, and when striatal function was genetically altered it failed to develop and the sequences were not learned.[584] Jin, Fatuel Tecuapetla, and Costa later found, alongside the start and stop neurons, others whose activity stayed up or stayed down for the whole of a sequence, as if binding its elements together. The direct and indirect pathways, the striatum's two routes to its output nuclei described in Chapter 4, were both active as a sequence began and then behaved differently while it ran.[585] Once an action is bracketed, it can be selected by the basal ganglia as if it were a single movement, and Amir Dezfouli and Bernard Balleine have argued that many habits are exactly this: sequences chunked so that their parts are no longer evaluated one by one.[586] A chunked sequence is cheap to select, and for the same reason it cannot be revised in the middle.

[582] Graybiel, A. M. (1998). The basal ganglia and chunking of action repertoires. *Neurobiology of Learning and Memory*, 70(1–2), 119–136; Graybiel, A. M. (2008). Habits, rituals, and the evaluative brain. *Annual Review of Neuroscience*, 31, 359–387.

[583] Smith, K. S., & Graybiel, A. M. (2013). A dual operator view of habitual behavior reflecting cortical and striatal dynamics. *Neuron*, 79(2), 361–374.

[584] Jin, X., & Costa, R. M. (2010). Start/stop signals emerge in nigrostriatal circuits during sequence learning. *Nature*, 466(7305), 457–462.

[585] Jin, X., Tecuapetla, F., & Costa, R. M. (2014). Basal ganglia subcircuits distinctively encode the parsing and concatenation of action sequences. *Nature Neuroscience*, 17(3), 423–430.

[586] Dezfouli, A., & Balleine, B. W. (2012). Habits, action sequences and reinforcement learning. *European Journal of Neuroscience*, 35(7), 1036–1051.

From Planning Circuits to Doing Circuits

Practice moves an action from the goal-directed control of the dorsomedial striatum to the habitual control of the dorsolateral striatum (Chapter 8), and the shift has now been watched as it happens. Henry Yin, Rui Costa, and colleagues trained mice to stay on an accelerating rotating rod, a skill that improves over days. Neurons in the dorsomedial striatum changed their firing early in training and those in the dorsolateral striatum later, and in slices taken from trained animals the excitatory synapses onto dorsolateral neurons, especially those of the indirect pathway, had been strengthened. Mice with lesions of the dorsomedial striatum were impaired in the early sessions and then caught up; mice with lesions of the dorsolateral striatum were impaired throughout.[587]

People show the same shift. Stéphane Lehéricy, Julien Doyon, and colleagues scanned people practicing a finger sequence over four weeks and found activity moving from the associative striatum, linked to prefrontal cortex, early in learning to the posterior putamen, linked to motor cortex, once the sequence was overlearned.[588]

Dopamine is needed for the handover. Alexis Faure and colleagues found that rats whose dopamine supply to the dorsolateral striatum had been destroyed went on learning to press a lever but never became habitual, staying sensitive to devaluation after training that made normal rats indifferent to it.[589] The anatomy shows how dopamine could pass the handover along. Suzanne Haber and colleagues traced connections between the striatum and the midbrain dopamine neurons and found that they form an ascending spiral: each striatal region influences the dopamine neurons that project to the next region along, from the ventral striatum of motivation, through the associative striatum, to the sensorimotor striatum.[590] Learning in one loop can thus shape what dopamine teaches in the next (Figure 9.4).

[587] Yin, H. H., Mulcare, S. P., Hilário, M. R. F., Clouse, E., Holloway, T., Davis, M. I., Hansson, A. C., Lovinger, D. M., & Costa, R. M. (2009). Dynamic reorganization of striatal circuits during the acquisition and consolidation of a skill. *Nature Neuroscience*, 12(3), 333–341.

[588] Lehéricy, S., Benali, H., Van de Moortele, P.-F., Pélégrini-Issac, M., Waechter, T., Ugurbil, K., & Doyon, J. (2005). Distinct basal ganglia territories are engaged in early and advanced motor sequence learning. *Proceedings of the National Academy of Sciences*, 102(35), 12566–12571; Doyon, J., Bellec, P., Amsel, R., Penhune, V., Monchi, O., Carrier, J., Lehéricy, S., & Benali, H. (2009). Contributions of the basal ganglia and functionally related brain structures to motor learning. *Behavioural Brain Research*, 199(1), 61–75.

[589] Faure, A., Haberland, U., Condé, F., & El Massioui, N. (2005). Lesion to the nigrostriatal dopamine system disrupts stimulus-response habit formation. *Journal of Neuroscience*, 25(11), 2771–2780.

[590] Haber, S. N., Fudge, J. L., & McFarland, N. R. (2000). Striatonigrostriatal pathways in primates form an ascending spiral from the shell to the dorsolateral striatum. *Journal of Neuroscience*, 20(6), 2369–2382.

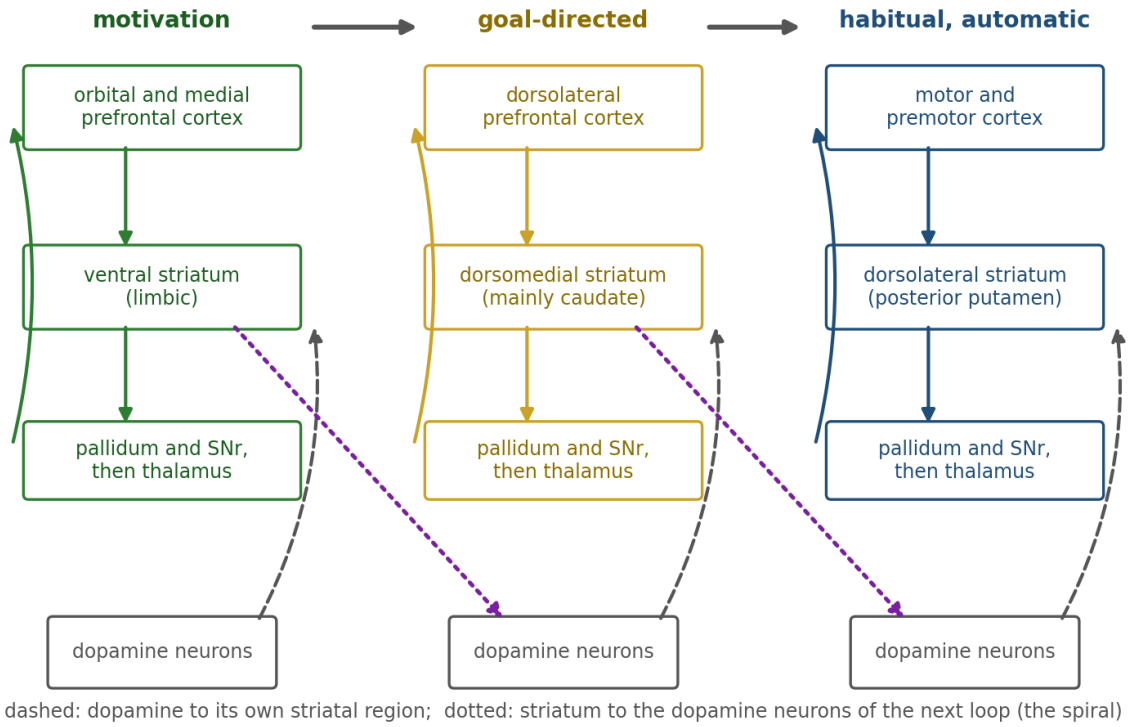


Figure 9.4. From motivation to deliberation to execution. Three parallel loops run from cortex through the striatum and, by way of the pallidum and the substantia nigra pars reticulata (SNr) and then the thalamus, back to cortex: a limbic loop through the ventral striatum, an associative loop through the dorsomedial striatum (mainly caudate), and a sensorimotor loop through the dorsolateral striatum (posterior putamen). Each striatal region sends signals to the dopamine neurons that project to the next region (dotted arrows), forming a spiral; with practice, control of an action moves rightward.

Cortex does not let go of a habit even after the striatum has taken it over. Kyle Smith, Graybiel, and colleagues overtrained rats to run a maze for a reward and then devalued the reward. The rats kept running to it, as habitual animals do. When the infralimbic cortex was briefly silenced with light, the rats stopped running to the devalued reward, as if their goal-directed control had been released; when a new habit had formed in its place, silencing the same region brought the old habit back.[591] A habit is not only stored; it is continuously permitted.

Is cortex needed to execute a skill at all? Risa Kawai, Bence Ölveczky, and colleagues trained rats to press a lever twice with a precise interval between the presses, a skill each rat performed with its own idiosyncratic movements. Lesions of motor cortex made after training had no discernible effect: the rats produced their own particular movements on the first day after surgery. Lesions made before training left the rats unable to learn the task.[592] Motor cortex, on this evidence, tutors subcortical circuits during learning and is then not needed to run what they have learned. Ashesh Dhawale, Ölveczky, and colleagues then showed that the sensorimotor striatum is needed to produce the detailed movements of such a learned skill, and that it does so without input from motor cortex.[593]

[591] Smith, K. S., Virkud, A., Deisseroth, K., & Graybiel, A. M. (2012). Reversible online control of habitual behavior by optogenetic perturbation of medial prefrontal cortex. *Proceedings of the National Academy of Sciences*, 109(46), 18932–18937.

[592] Kawai, R., Markman, T., Poddar, R., Ko, R., Fantana, A. L., Dhawale, A. K., Kampff, A. R., & Ölveczky, B. P. (2015). Motor cortex is required for learning but not for executing a motor skill. *Neuron*, 86(3), 800–812.

Whether this generalizes is debated. Gregory Ashby, Brian Turner, and Jon Horvitz argue from human and primate evidence that, for skills and categorization, the striatum trains cortex and that fully automatic performance is eventually carried by direct connections between cortical areas.[594] The rodent results suggest the reverse: cortex trains subcortical circuits and the basal ganglia carry the finished skill. The two views may describe different kinds of skill, dexterous movements in which motor cortex is indispensable and stereotyped sequences in which it is not. Both agree on the principle. The circuits needed to learn a skill are not the circuits that run it.

Variability as Exploration

Even the most practiced movements vary slightly from one repetition to the next. That variability is often treated as noise, but it may be the motor system exploring. For a movement, the exploration of Chapter 8 means trying slightly different versions and keeping what works. Howard Wu, Maurice Smith, and colleagues found that people whose movements varied more along the dimension that mattered to the task learned faster, and that the nervous system reshaped its variability toward that dimension during training, though later studies have not consistently found the same relation for learning from errors.[595]

Songbirds show the mechanism most clearly. A young zebra finch learns its song by trial and error over weeks, comparing its own singing with a memorized song of an adult tutor. The song system has a motor pathway that produces the song and a side loop, the anterior forebrain pathway, that runs through a part of the bird's basal ganglia called Area X and ends in a nucleus that projects back onto the motor pathway (Figure 9.5). Ölveczky, Aaron Andalman, and Michale Fee found that when that nucleus was silenced in juvenile birds, the variability of their song immediately collapsed.[596] The basal ganglia loop injects variation into the song. Evren Tumer and Michael Brainard showed, in adult Bengalese finches, that song which seems fixed is still adjustable through its remaining small variability. They played a burst of noise whenever a bird sang a particular syllable at a slightly lower pitch than usual, and within hours the birds shifted the pitch of that syllable upward, away from the noise.[597]

[593] Dhawale, A. K., Wolff, S. B. E., Ko, R., & Ölveczky, B. P. (2021). The basal ganglia control the detailed kinematics of learned motor skills. *Nature Neuroscience*, 24(9), 1256–1269; Wolff, S. B. E., Ko, R., & Ölveczky, B. P. (2022). Distinct roles for motor cortical and thalamic inputs to striatum during motor skill learning and execution. *Science Advances*, 8(8), eabk0231.

[594] Ashby, F. G., Turner, B. O., & Horvitz, J. C. (2010). Cortical and basal ganglia contributions to habit learning and automaticity. *Trends in Cognitive Sciences*, 14(5), 208–215.

[595] Wu, H. G., Miyamoto, Y. R., Gonzalez Castro, L. N., Ölveczky, B. P., & Smith, M. A. (2014). Temporal structure of motor variability is dynamically regulated and predicts motor learning ability. *Nature Neuroscience*, 17(2), 312–321; He, K., Liang, Y., Abdollahi, F., Fisher Bittmann, M., Kording, K., & Wei, K. (2016). The statistical determinants of the speed of motor learning. *PLoS Computational Biology*, 12(9), e1005023.

[596] Ölveczky, B. P., Andalman, A. S., & Fee, M. S. (2005). Vocal experimentation in the juvenile songbird requires a basal ganglia circuit. *PLoS Biology*, 3(5), e153.

[597] Tumer, E. C., & Brainard, M. S. (2007). Performance variability enables adaptive plasticity of 'crystallized' adult birdsong. *Nature*, 450(7173), 1240–1244.

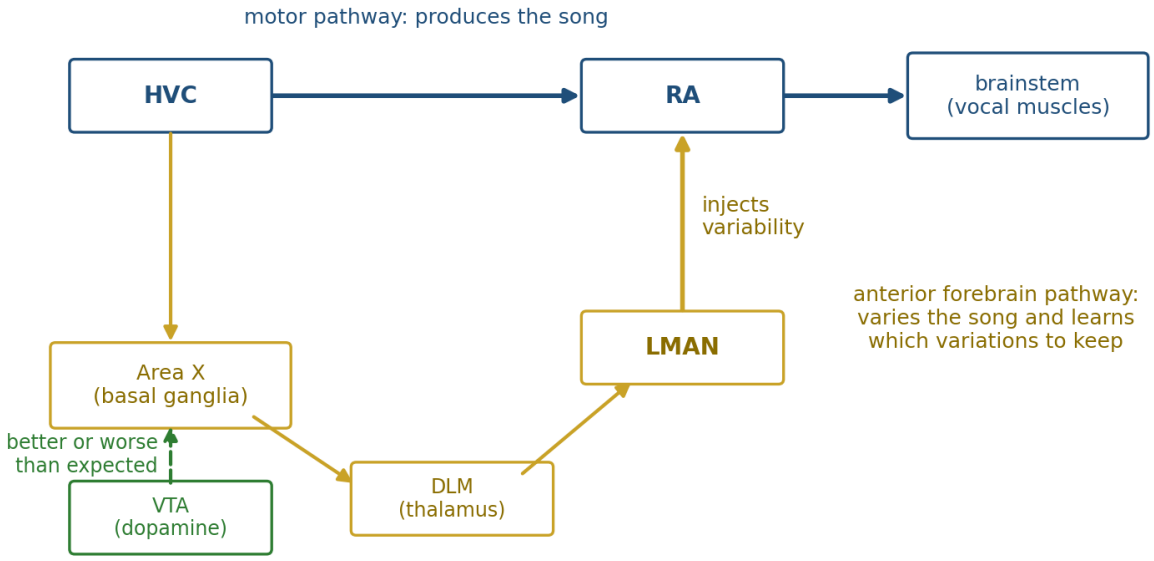


Figure 9.5. The songbird's learning circuit (the song nuclei are known by their abbreviations). A motor pathway from HVC to RA to the brainstem produces the song; an anterior forebrain pathway through Area X, part of the basal ganglia, the thalamic nucleus DLM, and LMAN injects variability into RA. Dopamine neurons in VTA tell Area X whether a rendition sounded better or worse than expected.

What decides which variations are kept? Vikram Gadagkar, Jesse Goldberg, and colleagues recorded the dopamine neurons that project to Area X in singing birds. When the experimenters distorted a syllable as the bird sang it, by playing a sound over it, the neurons were suppressed; when a syllable that was often distorted came out clean, they fired more.[598] That is the reward prediction error of Chapter 3, applied to the quality of a performance rather than to food. Fee and Goldberg have proposed that the circuit implements reinforcement learning: the side loop makes the song vary, dopamine reports whether a variation sounded better than expected, and Area X biases the variation toward the better versions until the song itself has moved.[599] Practice, on this view, is the learning rule of choice applied to the fine details of a movement.

The Cortex Rewires

Whatever runs a finished skill, practice changes motor cortex on the way. Randolph Nudo, Michael Merzenich, and colleagues mapped the motor cortex of squirrel monkeys, region by region, with small electrical currents, before and after the monkeys learned to retrieve food pellets from small wells, a task requiring fine finger control. The area from which finger movements could be evoked expanded, at the expense of the area for the wrist and forearm.[600]

[598] Gadagkar, V., Puzerey, P. A., Chen, R., Baird-Daniel, E., Farhang, A. R., & Goldberg, J. H. (2016). Dopamine neurons encode performance error in singing birds. *Science*, 354(6317), 1278–1282.

[599] Fee, M. S., & Goldberg, J. H. (2011). A hypothesis for basal ganglia-dependent reinforcement learning in the songbird. *Neuroscience*, 198, 152–170.

[600] Nudo, R. J., Milliken, G. W., Jenkins, W. M., & Merzenich, M. M. (1996). Use-dependent alterations of movement representations in primary motor cortex of adult squirrel monkeys. *Journal of Neuroscience*, 16(2), 785–807.

Alvaro Pascual-Leone and colleagues had people practice a five-finger piano exercise for two hours a day for five days and mapped their motor cortex each day with the magnetic pulses through the scalp that Chapter 8 described. The area from which the finger muscles could be activated grew over the days. People who only imagined playing the exercise, without moving, showed a similar if smaller change.[601] Long practice leaves a lasting mark. Thomas Elbert, Edward Taub, and colleagues found that the representation of the fingers of the left hand in the somatosensory cortex of string players, who use those fingers to stop the strings, was enlarged, most of all in those who had begun to play young.[602]

At the level of synapses, the change can now be watched in a living animal. Tonghui Xu, Yi Zuo, and colleagues imaged the dendritic spines, the small protrusions that carry most excitatory synapses, on output neurons of the mouse motor cortex while the mice learned to reach for seeds. New spines formed within an hour of the first training. Over the following days, older spines were eliminated and the overall number returned to its starting level, but a fraction of the new spines was stabilized and persisted for months after training ended.[603] Guang Yang, Feng Pan, and Wen-Biao Gan found the same in mice learning to run on the rotating rod: a small fraction of the spines formed during learning lasted for the rest of the animal's life.[604] Akiko Hayashi-Takagi, Haruo Kasai, and colleagues then made the link causal. They tagged the spines that had grown during learning with a light-sensitive protein, shrank those spines selectively with light, and found that the learned skill was disrupted.[605] A skill, at this level, is a small set of new connections that practice built and then protected.

Not every claim of cortical change in people has survived better methods. Eva Berlot, Nicola Popp, and Jörn Diedrichsen pointed out that a practiced sequence is also performed faster, so changes in imaging signals may reflect the speed of performance rather than learning. When they controlled for this, after five weeks of practice they found little change in primary motor cortex and clearer changes in premotor and parietal areas.[606] Where in human cortex a skill comes to reside is less settled than the earlier imaging studies implied.

Learning Between Sessions

Chapter 5 described how sleep consolidates memory, and skills seemed to offer a striking case. Matthew Walker, Robert Stickgold, and colleagues trained people on a finger-tapping sequence and found that after a night's sleep they were about 20 percent faster, with no loss of accuracy, while an equal interval awake brought no significant gain.[607] Later work found that much of this was an artifact. Timothy

[601] Pascual-Leone, A., Nguyet, D., Cohen, L. G., Brasil-Neto, J. P., Cammarota, A., & Hallett, M. (1995). Modulation of muscle responses evoked by transcranial magnetic stimulation during the acquisition of new fine motor skills. *Journal of Neurophysiology*, 74(3), 1037–1045.

[602] Elbert, T., Pantev, C., Wienbruch, C., Rockstroh, B., & Taub, E. (1995). Increased cortical representation of the fingers of the left hand in string players. *Science*, 270(5234), 305–307.

[603] Xu, T., Yu, X., Perlik, A. J., Tobin, W. F., Zweig, J. A., Tennant, K., Jones, T., & Zuo, Y. (2009). Rapid formation and selective stabilization of synapses for enduring motor memories. *Nature*, 462(7275), 915–919.

[604] Yang, G., Pan, F., & Gan, W.-B. (2009). Stably maintained dendritic spines are associated with lifelong memories. *Nature*, 462(7275), 920–924.

[605] Hayashi-Takagi, A., Yagishita, S., Nakamura, M., Shirai, F., Wu, Y. I., Loshbaugh, A. L., Kuhlman, B., Hahn, K. M., & Kasai, H. (2015). Labelling and optical erasure of synaptic memory traces in the motor cortex. *Nature*, 525(7569), 333–338.

[606] Berlot, E., Popp, N. J., & Diedrichsen, J. (2020). A critical re-evaluation of fMRI signatures of motor sequence learning. *eLife*, 9, e55241.

[607] Walker, M. P., Brakefield, T., Morgan, A., Hobson, J. A., & Stickgold, R. (2002). Practice with sleep makes perfect: Sleep-dependent motor skill learning. *Neuron*, 35(1), 205–211.

Rickard and colleagues showed that performance at the end of a training session is depressed by fatigue, and that once time of day, fatigue, and the way trials were averaged were controlled, the overnight gain disappeared.[608] A meta-analysis by Steven Pan and Rickard concluded that most of the apparent sleep benefit for motor sequences is explained by such confounds, though sleep may still protect what was learned from being lost.[609]

A newer finding moved the question to a much shorter timescale. Marlene Bönstrup, Leonardo Cohen, and colleagues had people type a sequence as often as possible in ten-second bursts separated by ten seconds of rest, and found that in the early part of training almost all the improvement happened during the rests, not during the typing.[610] Ethan Buch, Cohen, and colleagues recorded brain activity during those rests with magnetoencephalography, which measures the magnetic fields produced by neural currents, and found the neural pattern of the practiced sequence replayed, compressed many times over, in the hippocampus and the sensorimotor cortex; the more replay, the larger the gain.[611] This is the replay of Chapter 5, during waking rest, at work on a skill. But Mohini Gupta and Rickard argue that these gains during rest, too, may be recovery from a short-lived fatigue built up during each burst of typing.[612] Whether skill is consolidated offline, and on what timescale, is an open question.

Automaticity

William James wrote in 1890 that "habit diminishes the conscious attention with which our acts are performed."[613] The modern study of automaticity began with experiments that measured that attention directly. Walter Schneider and Richard Shiffrin had people search displays for target letters among distractors. When targets and distractors kept the same roles across thousands of trials, search eventually became nearly as fast for many items as for one, as if the targets popped out; when the roles were mixed from trial to trial, search stayed slow and effortful however long people practiced. When targets and distractors that had kept consistent roles were then exchanged, the old automatic responses went on interfering for a very long time.[614] Automatic processing, they concluded, is fast, effortless, and parallel, and it develops only when the mapping from situation to response is consistent. Once developed, it is hard to suppress.

The oldest demonstration of that last property is John Ridley Stroop's. Naming the color of the ink in which a word is printed takes much longer when the word itself names a different color, as when "red"

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[608] Rickard, T. C., Cai, D. J., Rieth, C. A., Jones, J., & Ard, M. C. (2008). Sleep does not enhance motor sequence learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 34(4), 834–842; Brawn, T. P., Fenn, K. M., Nusbaum, H. C., & Margoliash, D. (2010). Consolidating the effects of waking and sleep on motor-sequence learning. *Journal of Neuroscience*, 30(42), 13977–13982.

[609] Pan, S. C., & Rickard, T. C. (2015). Sleep and motor learning: Is there room for consolidation? *Psychological Bulletin*, 141(4), 812–834.

[610] Bönstrup, M., Iturrate, I., Thompson, R., Cruciani, G., Censor, N., & Cohen, L. G. (2019). A rapid form of offline consolidation in skill learning. *Current Biology*, 29(8), 1346–1351.

[611] Buch, E. R., Claudino, L., Quentin, R., Bönstrup, M., & Cohen, L. G. (2021). Consolidation of human skill linked to waking hippocampo-neocortical replay. *Cell Reports*, 35(10), 109193.

[612] Gupta, M. W., & Rickard, T. C. (2022). Dissipation of reactive inhibition is sufficient to explain post-rest improvements in motor sequence learning. *npj Science of Learning*, 7, 25.

[613] James, W. (1890). *The Principles of Psychology* (Vol. 1). Henry Holt. The sentence is from chapter 4, "Habit."

[614] Schneider, W., & Shiffrin, R. M. (1977). Controlled and automatic human information processing: I. Detection, search, and attention. *Psychological Review*, 84(1), 1–66; Shiffrin, R. M., & Schneider, W. (1977). Controlled and automatic human information processing: II. Perceptual learning, automatic attending, and a general theory. *Psychological Review*, 84(2), 127–190.

is printed in green, because reading has become automatic and the word is read whether or not reading is wanted.[615]

Gordon Logan proposed a simple account of how computation turns into automaticity. Each time a task is performed, the situation and the answer are stored as an instance. When the situation recurs, two processes race: the original procedure, which computes the answer, and retrieval of past instances, any of which can supply it. Whichever finishes first determines the response. With practice the number of stored instances grows, and the fastest of many retrievals is faster than the fastest of a few, so the response gets faster and the procedure stops winning the race (Figure 9.6). Under simple assumptions about how long retrievals take, the fastest of many falls as a power function of their number, so instance theory predicts the power law of practice; it also predicts, as later experiments bore out for many tasks, that the variability of response times shrinks along with their mean.[616] On this account automaticity is not a faster algorithm but the replacement of computing with remembering: the child who has done enough sums stops adding seven and five and knows that the answer is twelve.

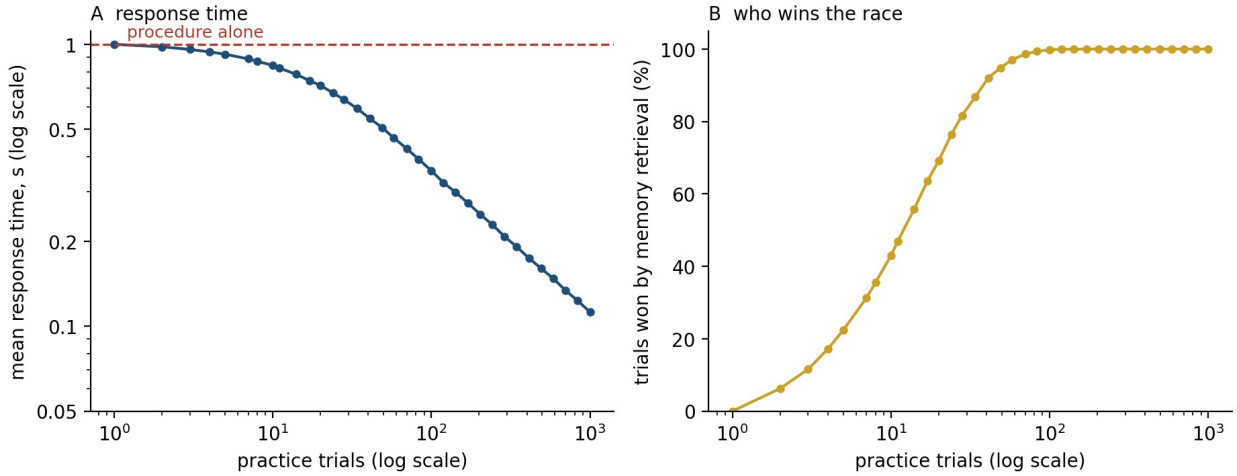


Figure 9.6. Automaticity as a race (simulated, after Logan 1988). Each practice trial stores an instance; on each trial the response is given by whichever finishes first, the original procedure (a fixed time) or the fastest retrieval of a stored instance. As instances accumulate, mean response time, after a slow start, falls along a near-straight line on logarithmic axes (A), and retrieval wins the race more and more often (B).

When a skill is automatic, attending to it can make it worse. Sian Beilock, Thomas Carr, and colleagues had experienced golfers putt and experienced soccer players dribble a ball through cones, either while attending to a part of their own movement, such as the moment the club stopped, or while monitoring a stream of tones for a target. The experts did worse when attending to their movements and were barely affected by the tones; novices showed the opposite pattern.[617] Pressure can have the same effect, by drawing attention back to steps that run better unwatched, which is one account of choking, the

[615] Stroop, J. R. (1935). Studies of interference in serial verbal reactions. *Journal of Experimental Psychology*, 18(6), 643–662.

[616] Logan, G. D. (1988). Toward an instance theory of automatization. *Psychological Review*, 95(4), 492–527; Logan, G. D. (1992). Shapes of reaction-time distributions and shapes of learning curves: A test of the instance theory of automaticity. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 18(5), 883–914.

[617] Beilock, S. L., Carr, T. H., MacMahon, C., & Starkes, J. L. (2002). When paying attention becomes counterproductive: Impact of divided versus skill-focused attention on novice and experienced performance of sensorimotor skills. *Journal of Experimental Psychology: Applied*, 8(1), 6–16.

collapse of a practiced skill under pressure.[618] Richard Masters taught people to putt either with explicit instructions or while saying random letters aloud, which prevented them from forming explicit rules, and found that under pressure the explicitly taught group faltered and the implicitly taught group did not.[619]

In the brain, automaticity looks less like a move to a new region than like doing the same work with less. Russell Poldrack and colleagues trained people on a sequence until a second task performed at the same time no longer slowed them, and found that activity declined in several regions, including lateral prefrontal cortex, as the cost of the second task disappeared.[620] The prefrontal resources that Chapter 8 called costly are released for other use.

How Practice Should Be Practiced

Practice works, but not all practice works equally. K. Anders Ericsson, Ralf Krampe, and Clemens Tesch-Römer studied violin students at a Berlin music academy and found that the best of them had accumulated, by their estimates, about 10,000 hours of solitary practice by the age of 20, more than the merely good students and far more than those training to be music teachers. They argued that deliberate practice, effortful work on specific weaknesses with feedback, accounts for most of the difference between experts and others.[621] Malcolm Gladwell later popularized the finding as a "10,000-hour rule."[622]

The claim has been tested many times since and has not held in its strong form. Brooke Macnamara, David Hambrick, and Frederick Oswald, in a meta-analysis across music, games, sports, education, and professions, found that deliberate practice explained about 12 percent of the variation in performance overall, more in games and music and very little in professions.[623] Macnamara and Megha Maitra repeated the violin study with a double-blind design; practice still predicted skill, but much less strongly than in the original, and it did not separate each level of skill from the next.[624] Practice is necessary for expertise, but the amount alone does not explain who becomes expert.

How practice is arranged matters as well as how much. John Shea and Robyn Morgan had people learn three patterns of rapid arm movements, knocking down small barriers in a set order as fast as possible. One group practiced each pattern in a block before moving to the next; another practiced them in a random order. The random group was slower during practice, but faster when tested ten minutes and ten days later, and on a new pattern (Figure 9.7).[625] The effect is called contextual interference, and

[618] Beilock, S. L., & Carr, T. H. (2001). On the fragility of skilled performance: What governs choking under pressure? *Journal of Experimental Psychology: General*, 130(4), 701–725.

[619] Masters, R. S. W. (1992). Knowledge, knerves and know-how: The role of explicit versus implicit knowledge in the breakdown of a complex motor skill under pressure. *British Journal of Psychology*, 83(3), 343–358.

[620] Poldrack, R. A., Sabb, F. W., Foerde, K., Tom, S. M., Asarnow, R. F., Bookheimer, S. Y., & Knowlton, B. J. (2005). The neural correlates of motor skill automaticity. *Journal of Neuroscience*, 25(22), 5356–5364.

[621] Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100(3), 363–406.

[622] Gladwell, M. (2008). *Outliers: The Story of Success*. Little, Brown.

[623] Macnamara, B. N., Hambrick, D. Z., & Oswald, F. L. (2014). Deliberate practice and performance in music, games, sports, education, and professions: A meta-analysis. *Psychological Science*, 25(8), 1608–1618.

[624] Macnamara, B. N., & Maitra, M. (2019). The role of deliberate practice in expert performance: Revisiting Ericsson, Krampe & Tesch-Römer (1993). *Royal Society Open Science*, 6(8), 190327.

[625] Shea, J. B., & Morgan, R. L. (1979). Contextual interference effects on the acquisition, retention, and transfer of a motor skill. *Journal of Experimental Psychology: Human Learning and Memory*, 5(2), 179–187.

Richard Schmidt and Robert Bjork found the same paradox across verbal and motor learning: conditions that make practice look worse, such as interleaving, spacing, and reduced feedback, often make learning better.[626] One explanation is that each switch forces the learner to retrieve and rebuild the movement plan, which is the same work that the later test requires. The effect is reliable in the laboratory and smaller, though still present, in applied settings such as sport.[627]

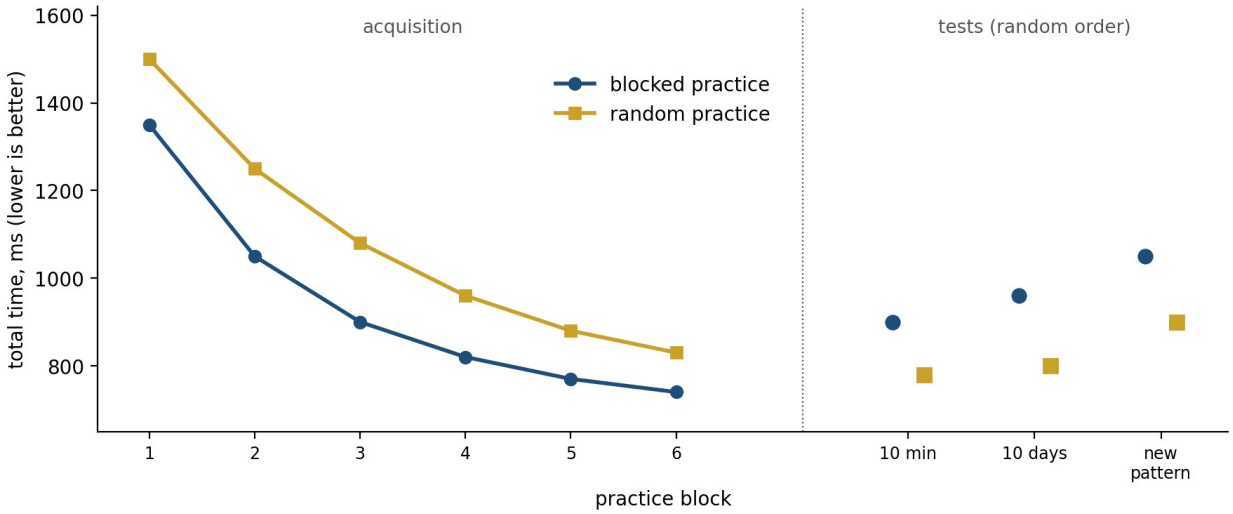


Figure 9.7. Contextual interference (illustrative, after Shea & Morgan 1979). Blocked practice produces faster performance during acquisition; random practice produces faster performance at the retention tests ten minutes and ten days later (shown here for tests in random order) and on transfer to a new pattern.

Variety helps in a related way. Robert Kerr and Bernard Booth had children aged eight and 12 throw beanbags at a target on the floor without seeing it as they threw, before and after a 12-week physical education program. The groups differed only in their practice throws before each test: one practiced at the test distance, the other only at a nearer and a farther target. At the end of the program, the group that had never practiced at the test distance was the more accurate at it.[628] Richard Schmidt's schema theory explains why: practicing several versions of a movement teaches the rule that relates them, which then serves for versions never practiced.[629]

Habits in Daily Life

Outside the laboratory, habits fill much of the day. Wendy Wood, Jeffrey Quinn, and Deborah Kashy asked students to record what they were doing, thinking, and feeling every hour, and found that about 43 percent of their actions were performed almost every day in the same place, usually while they were thinking about something else.[630] Phillippa Lally, Jane Wardle, and colleagues asked 96 volunteers to take up a new daily behavior, such as eating a piece of fruit with lunch or going for a run before dinner,

[626] Schmidt, R. A., & Bjork, R. A. (1992). New conceptualizations of practice: Common principles in three paradigms suggest new concepts for training. *Psychological Science*, 3(4), 207–217.

[627] Brady, F. (2004). Contextual interference: A meta-analytic study. *Perceptual and Motor Skills*, 99(1), 116–126.

[628] Kerr, R., & Booth, B. (1978). Specific and varied practice of motor skill. *Perceptual and Motor Skills*, 46(2), 395–401.

[629] Schmidt, R. A. (1975). A schema theory of discrete motor skill learning. *Psychological Review*, 82(4), 225–260.

[630] Wood, W., Quinn, J. M., & Kashy, D. A. (2002). Habits in everyday life: Thought, emotion, and action. *Journal of Personality and Social Psychology*, 83(6), 1281–1297.

and to rate each day how automatic it felt. Automaticity rose along a curve that leveled off, like the practice curves of Figure 9.1. The time to reach 95 percent of each person's plateau ranged from 18 to 254 days, with a median of 66, and missing a single day made little difference.[631] As Chapter 8 noted, laboratory attempts to make people habitual by overtraining a single response for an hour or a few days have often failed, so the habits of daily life may be built from many repetitions in stable contexts over months.[632]

Habits outside the laboratory show the insensitivity to outcome that Chapter 8 measured in rats. David Neal, Wood, and colleagues gave people at a cinema either fresh popcorn or popcorn that was a week old. People who rarely ate popcorn at the cinema ate less of the stale popcorn; people with a strong cinema-popcorn habit ate as much stale as fresh. The habit held only in the cinema, its context, and it broke when people were asked to eat with their other hand, which prevented the practiced movement from running.[633] Habits are cued by context, and Wood and Dennis Rünger argue that this is the lever for changing them: moving house, starting a new job, or altering the setting of a behavior disrupts the cues and opens a window in which a new routine can form.[634]

When Routines Go Wrong

Because the routes to automaticity are specific circuits, their failures are specific too. In Parkinson's disease, the loss of dopamine is earliest and most severe in the posterior putamen, the sensorimotor striatum that carries habits. Peter Redgrave, José Obeso, and colleagues argue that patients therefore lose automatic control first, and must perform by deliberate effort actions that used to run on their own.[635] Walking becomes a task that needs attention, and it worsens when the patient talks at the same time. Tao Wu and Mark Hallett found that patients could still make a sequence automatic, but only with more training and with more brain activity than healthy people needed.[636] Meg Morris and colleagues showed that patients' short, shuffling steps lengthen to nearly normal when they step over lines marked on the floor, because the visual cue lets goal-directed circuits do the work that the damaged habit circuit no longer does.[637]

Practice itself can go wrong. In 1964 the American pianist Leon Fleisher, then in his thirties and one of the leading pianists of his generation, found the fourth and fifth fingers of his right hand curling under whenever he played. For three decades he played only music written for the left hand, and taught and conducted. In the 1990s, after the condition had been identified as focal dystonia and treated with

[631] Lally, P., van Jaarsveld, C. H. M., Potts, H. W. W., & Wardle, J. (2010). How are habits formed: Modelling habit formation in the real world. *European Journal of Social Psychology*, 40(6), 998–1009.

[632] de Wit, S., Kindt, M., Knot, S. L., Verhoeven, A. A. C., Robbins, T. W., Gasull-Camos, J., Evans, M., Mirza, H., & Gillan, C. M. (2018). Shifting the balance between goals and habits: Five failures in experimental habit induction. *Journal of Experimental Psychology: General*, 147(7), 1043–1065.

[633] Neal, D. T., Wood, W., Wu, M., & Kurlander, D. (2011). The pull of the past: When do habits persist despite conflict with motives? *Personality and Social Psychology Bulletin*, 37(11), 1428–1437.

[634] Wood, W., & Rünger, D. (2016). Psychology of habit. *Annual Review of Psychology*, 67, 289–314; Verplanken, B., & Wood, W. (2006). Interventions to break and create consumer habits. *Journal of Public Policy & Marketing*, 25(1), 90–103.

[635] Redgrave, P., Rodriguez, M., Smith, Y., Rodriguez-Oroz, M. C., Lehericy, S., Bergman, H., Agid, Y., DeLong, M. R., & Obeso, J. A. (2010). Goal-directed and habitual control in the basal ganglia: Implications for Parkinson's disease. *Nature Reviews Neuroscience*, 11(11), 760–772.

[636] Wu, T., & Hallett, M. (2005). A functional MRI study of automatic movements in patients with Parkinson's disease. *Brain*, 128(10), 2250–2259.

[637] Morris, M. E., Iansek, R., Matyas, T. A., & Summers, J. J. (1996). Stride length regulation in Parkinson's disease: Normalization strategies and underlying mechanisms. *Brain*, 119(2), 551–568.

injections of botulinum toxin and a deep-tissue massage called Rolfing, he began to play with both hands again, and in 2004 he recorded his first two-handed album in four decades. The recovery was partial and depended on continuing injections.[638]

About one in 100 professional musicians develops musician's dystonia, a loss of voluntary control that appears only in the extensively trained movements of playing.[639] Nancy Byl, Merzenich, and colleagues trained owl monkeys to squeeze a handle over and over and found that some developed a similar movement disorder, along with a blurring of the normally separate representations of the fingers in somatosensory cortex.[640] One hypothesis is that the plasticity that enlarges a string player's finger map can, pushed further, merge the maps until the fingers cannot be moved independently.

Addiction has been described as a habit gone wrong. Barry Everitt and Trevor Robbins argue that drug-seeking begins as goal-directed action, becomes habitual as control moves from the ventral to the dorsal striatum along the spiral of Figure 9.4, and finally becomes compulsive, persisting despite punishment.[641] Lee Hogarth argues the opposite: that in people, drug choice remains sensitive to its outcome and is driven by excessive goal-directed pursuit of relief from negative feelings, not by habit.[642] In each of these disorders the machinery of routine works too well, too little, or in the wrong place.

Implications for Artificial Intelligence

Where Chapter 8 asked how machines choose, this section asks how they stop having to. Artificial intelligence has met the economy of habit directly, and some of its most successful systems resemble the brain's arrangement closely.

First, the best game-playing systems turn search into reflex. Chapter 8 described AlphaGo pairing fast networks with search. Its successor, AlphaGo Zero, went a step further: it chose its moves by a search that simulated future play, and its network was trained to predict the moves that the search chose; over training, the network alone, choosing moves at a glance without any search, became a strong player.[643] Thomas Anthony, Zheng Tian, and David Barber described the same scheme, which they called expert iteration, explicitly as fast intuition trained by slow deliberation.[644] That is Logan's instance theory and the shift from dorsomedial to dorsolateral striatum in engineering form: an expensive procedure generates good answers, and a cheap one learns to reproduce them. Researchers have begun to train

[638] Fleisher, L., & Midgette, A. (2010). *My Nine Lives: A Memoir of Many Careers in Music*. Doubleday.

[639] Altenmüller, E., & Jabusch, H.-C. (2010). Focal dystonia in musicians: Phenomenology, pathophysiology, triggering factors, and treatment. *Medical Problems of Performing Artists*, 25(1), 3–9.

[640] Byl, N. N., Merzenich, M. M., & Jenkins, W. M. (1996). A primate genesis model of focal dystonia and repetitive strain injury: I. Learning-induced dedifferentiation of the representation of the hand in the primary somatosensory cortex in adult monkeys. *Neurology*, 47(2), 508–520.

[641] Everitt, B. J., & Robbins, T. W. (2005). Neural systems of reinforcement for drug addiction: From actions to habits to compulsion. *Nature Neuroscience*, 8(11), 1481–1489; Everitt, B. J., & Robbins, T. W. (2016). Drug addiction: Updating actions to habits to compulsions ten years on. *Annual Review of Psychology*, 67, 23–50.

[642] Hogarth, L. (2020). Addiction is driven by excessive goal-directed drug choice under negative affect: Translational critique of habit and compulsion theory. *Neuropsychopharmacology*, 45(5), 720–735.

[643] Silver, D., Schrittwieser, J., Simonyan, K., Antonoglou, I., Huang, A., Guez, A., Hubert, T., Baker, L., Lai, M., Bolton, A., Chen, Y., Lillicrap, T., Hui, F., Sifre, L., van den Driessche, G., Graepel, T., & Hassabis, D. (2017). Mastering the game of Go without human knowledge. *Nature*, 550(7676), 354–359.

[644] Anthony, T., Tian, Z., & Barber, D. (2017). Thinking fast and slow with deep learning and tree search. In *Advances in Neural Information Processing Systems 30* (pp. 5360–5370).

language models the same way, using answers first reached by step-by-step reasoning as targets so that the model learns to give them directly.[645]

Second, what is cached is brittle in the way habits are. A network trained to copy a search reproduces the search's answers in the situations it was trained on and does not consult the outcome. If the goal changes, it goes on producing what used to be right. Most artificial systems that cache have no way to notice this, and a change in the objective requires retraining.

Third, varied practice is how robots generalize. Josh Tobin and colleagues trained robot vision in simulation with colors, textures, and lighting randomized far beyond anything realistic, and the result worked on the real robot without ever having seen it.[646] OpenAI's robot hand learned to manipulate a Rubik's cube in the real world, turning it through a solution supplied by a standard solver, after training only in simulations whose physics were randomized, with the range of randomization widened automatically as the hand improved.[647] That is Kerr and Booth's beanbags at scale: a learner that never practiced the test condition does well on it because it practiced around it.

Fourth, the brain largely adds skills without erasing old ones. A network learning a new skill tends to overwrite the ones it has, the catastrophic interference of Chapter 5, and methods that protect important weights, such as James Kirkpatrick and colleagues' elastic weight consolidation, are partial remedies.[648] Motor cortex instead forms new spines for a new skill and keeps a few of them for life while leaving the old ones in place. The brain is not immune to interference: two conflicting adaptations learned close together do disrupt each other.[649]

Fifth, the brain lets the slow system interrupt the fast one. Practice adds a twist to Chapter 8's arbitration between planning and habit: a routine runs only while a supervisor permits it, as the infralimbic experiments show. A pianist plays automatically until a wrong note, then attends; a driver converses until the car ahead brakes. Artificial systems with both fast and slow modes rarely decide for themselves when to switch from one to the other, and it is that decision, as much as either mode, that makes a skilled human reliable. Whether machines can learn it, and what else they lack, is the question of Chapter 10.

The Central Lesson

Practice is how the brain spends deliberation once so that it need not spend it again. Each repetition hands a little more of an action to circuits that do not think: the cerebellum keeps it calibrated, the basal ganglia package it into units that run whole, dopamine keeps the variations that worked, and motor cortex builds and then protects the few connections that hold it. The result is Whitehead's economy, an

[645] Deng, Y., Prasad, K., Fernandez, R., Smolensky, P., Chaudhary, V., & Shieber, S. (2023). Implicit chain of thought reasoning via knowledge distillation. *arXiv*, 2311.01460.

[646] Tobin, J., Fong, R., Ray, A., Schneider, J., Zaremba, W., & Abbeel, P. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 23–30). IEEE.

[647] OpenAI, Akkaya, I., Andrychowicz, M., Chociej, M., Litwin, M., McGrew, B., Petron, A., Paino, A., Plappert, M., Powell, G., Ribas, R., Schneider, J., Tezak, N., Tworek, J., Welinder, P., Weng, L., Yuan, Q., Zaremba, W., & Zhang, L. (2019). Solving Rubik's cube with a robot hand. *arXiv*, 1910.07113.

[648] Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., & Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526.

[649] Brashers-Krug, T., Shadmehr, R., & Bizzi, E. (1996). Consolidation in human motor memory. *Nature*, 382(6588), 252–255.

answer remembered rather than computed, and its price is Chapter 8's rat, still pressing for food it will not eat. The brain can afford that price because it never lets the slow system go: deliberation moves to the edges of a routine and stands guard, ready to take control back when the expected outcome fails to arrive. Expertise is the freedom to spend thought only where it is needed.

Next Chapter

Chapter 10: What Artificial Intelligence Gets Right and Wrong: Machine Learning Measured Against the Brain

Figure and Epigraph Sources

Figure Attribution

Figures 9.1 to 9.7 were drawn for this book. The curves in Figure 9.1 are illustrative, after the pattern reported by Crossman (1959), not recorded data. Figure 9.2 is a simulation of the two-state model of Smith, Ghazizadeh, and Shadmehr (2006) with illustrative parameters. The activity profiles in Figure 9.3 are illustrative, after the patterns reported by Jog et al. (1999) and Barnes et al. (2005). Figure 9.6 is a simulation of the instance theory of Logan (1988). The values in Figure 9.7 are illustrative, after the pattern reported by Shea and Morgan (1979). Figures 9.4 and 9.5 are schematic diagrams; Figure 9.4 follows the striatonigrostriatal spiral of Haber, Fudge, and McFarland (2000).

Epigraph

Alfred North Whitehead, *An Introduction to Mathematics*, chapter 5, "The Symbolism of Mathematics" (Williams & Norgate, London; Henry Holt, New York, 1911).

Part IV: Brains and Machines

Chapter 10

What Artificial Intelligence Gets Right and Wrong: Machine Learning Measured Against the Brain

> *"Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's?"*
>
> *— Alan Turing, "Computing Machinery and Intelligence" (1950)*

What Is Being Measured

In July 1958 the United States Navy showed reporters a machine built by the psychologist Frank Rosenblatt. It was a room-sized computer running a program he called the perceptron, and after 50 trials it had learned to tell whether a card had been marked on its left side or its right. The *New York Times* reported that the Navy expected the machine's descendants to walk, talk, see, write, reproduce themselves, and be conscious of their existence.[650] None of that happened for a long time, and some of it has not happened yet. But the idea Rosenblatt demonstrated, a network of simple units that learns by adjusting the strength of its connections after its mistakes, is the idea on which the artificial intelligence of the 2020s runs. It came from the brain.

Each chapter of this book has ended by asking what its subject means for artificial intelligence. Chapter 3 asked how a machine might decide for itself when to learn; Chapter 5, how it might learn without forgetting; Chapter 6, why an imperceptible change to an image can make it confidently wrong; Chapter 7, what a machine can value when it has no body; Chapter 8, whether it knows when it does not know; Chapter 9, how it might turn deliberation into reflex and when it should turn it back. This chapter gathers those questions and measures machine learning against the brain directly. The question is not whether machines are intelligent, which depends on a definition, but which of the brain's solutions machine learning has found, which it has not, and which it does not need.

Measuring needs a scale, and there are three. A system can match the brain in performance, doing what a person does as well as a person does it. It can match in representation, organizing information inside it as the brain does. Or it can match in mechanism, getting there by the same means. Chapter 1 kept apart, after David Marr, what a system computes, the procedure it uses, and the hardware that runs it, and these three scales borrow that discipline loosely.[651] The distinction matters because current machine learning scores very differently on them. A language model can write a competent paragraph, matching performance; its internal states predict the activity of the human language network, matching representation in part; and it learns by a procedure that no brain is known to use, matching mechanism hardly at all.

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[650] New Navy device learns by doing: Psychologist shows embryo of computer designed to read and grow wiser. (1958, July 8). *The New York Times*, p. 25; Rosenblatt, F. (1958). The perceptron: A probabilistic model for information storage and organization in the brain. *Psychological Review*, 65(6), 386–408.

[651] Marr, D. (1982). *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. W. H. Freeman.

The argument of this chapter is that machine learning got right the thing it took from the brain first: that useful knowledge can be learned rather than written by hand, stored in the strengths of many connections, and shaped by error and prediction. Where it has applied that idea at scale, its systems have come to resemble the brain more than anyone expected. Where it differs is in the conditions of learning. A brain learns from little data, all its life, on 20 watts, with values anchored to a body, from rules available at each synapse, and while it is being used. Most machine learning learns from enormous data, once, on megawatts, toward an objective supplied from outside, by a global procedure, and then stops. Some of those differences are failures that the brain shows how to repair. Others are the brain's responses to constraints that silicon does not share, and copying them would be a mistake. Telling the two apart is the purpose of the comparison.

A Shared Ancestry

The traffic between the fields has run in both directions for 80 years (Figure 10.1). Warren McCulloch and Walter Pitts described the neuron as a logical unit in 1943, as Chapter 2 recounted, and Rosenblatt's perceptron added what their units lacked: weights that changed with experience, as Donald Hebb had proposed in 1949 that synapses do.[652] Its learning rule was crude, and in 1969 Marvin Minsky and Seymour Papert showed that a single layer of such units could not compute some simple functions, among them exclusive-or, the same function that Chapter 2 found a single human cortical neuron can compute.[653] Multilayer networks could compute it, but there was no practical, widely known way to train them.

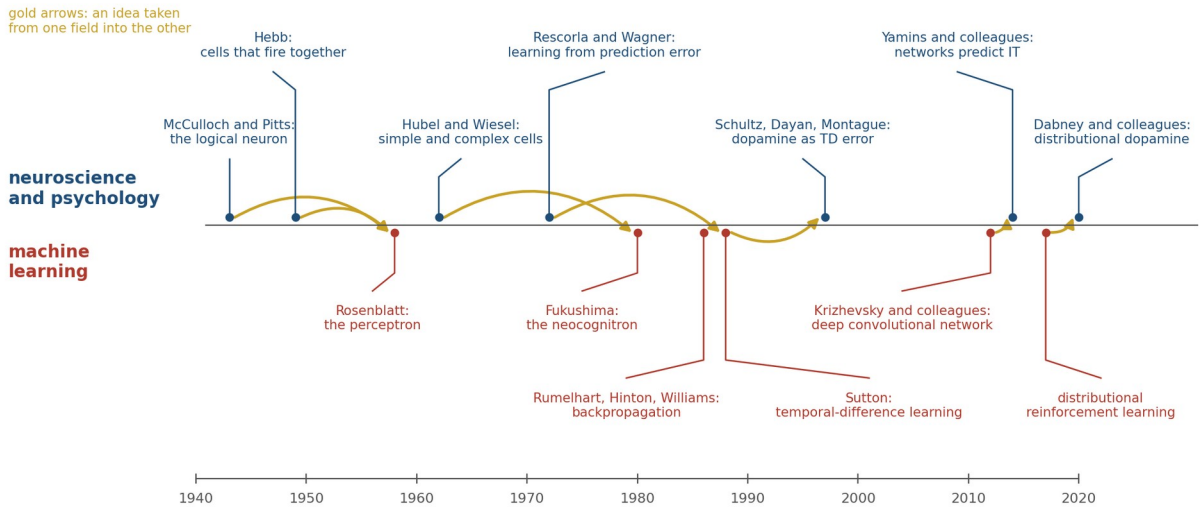


Figure 10.1. Eighty years of exchange. Ideas from neuroscience and psychology (above the line) and machine learning (below) that crossed from one field to the other, shown by gold arrows. The list is a selection.

[652] McCulloch, W. S., & Pitts, W. (1943). A logical calculus of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics*, 5(4), 115–133; Hebb, D. O. (1949). *The Organization of Behavior: A Neuropsychological Theory*. Wiley.

[653] Minsky, M., & Papert, S. (1969). *Perceptrons: An Introduction to Computational Geometry*. MIT Press.

The architecture of modern vision systems came from Chapter 6: Kunihiko Fukushima's neocognitron of 1980 built Hubel and Wiesel's hierarchy of edge detectors into a network of alternating feature-detecting and pooling layers.[654]

The method of training such a network came from mathematics and control engineering, where the calculation it needs had been worked out in the 1970s. Backpropagation passes the error at the output backward through the network to assign each weight its share of the blame, and David Rumelhart, Geoffrey Hinton, and Ronald Williams showed in 1986 that it could train networks with hidden layers to discover useful internal representations.[655] Yann LeCun and his colleagues combined the two in 1989 and trained a convolutional network to read handwritten zip codes.[656] In 2012 Alex Krizhevsky, Ilya Sutskever, and Hinton trained a much larger one on a million labeled photographs, using graphics processors built for video games, and cut the error rate in the leading image-recognition competition from 26 percent to 15 percent.[657] The current era of artificial intelligence dates from that result.

The traffic in the other direction is just as old. Chapter 3 told how temporal-difference learning turned out to describe the firing of dopamine neurons that Wolfram Schultz had recorded in monkeys. Richard Sutton and Andrew Barto had developed it from the Rescorla-Wagner model of animal conditioning, and Sutton then generalized it into an engineering method for learning predictions, so the idea went from psychology to engineering and on to the physiology of the midbrain.[658] The exchange has continued since. Will Dabney and his colleagues at DeepMind found that artificial agents learn better when they predict the whole distribution of possible rewards rather than only the average, and then, with Naoshige Uchida's laboratory, found that individual dopamine neurons in mice are tuned more optimistically or more pessimistically, together encoding a distribution of the kind the agents used.[659] Demis Hassabis and his colleagues have argued that this kind of exchange is the main reason for engineers to study the brain: not to copy it, but to find out which problems it has already solved.[660]

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[654] Hubel, D. H., & Wiesel, T. N. (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *Journal of Physiology*, 160(1), 106–154; Fukushima, K. (1980). Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position. *Biological Cybernetics*, 36(4), 193–202.

[655] Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323(6088), 533–536.

[656] LeCun, Y., Boser, B., Denker, J. S., Henderson, D., Howard, R. E., Hubbard, W., & Jackel, L. D. (1989). Backpropagation applied to handwritten zip code recognition. *Neural Computation*, 1(4), 541–551.

[657] Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems 25* (pp. 1097–1105).

[658] Rescorla, R. A., & Wagner, A. R. (1972). A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. In A. H. Black & W. F. Prokasy (Eds.), *Classical Conditioning II: Current Research and Theory* (pp. 64–99). Appleton-Century-Crofts; Sutton, R. S., & Barto, A. G. (1981). Toward a modern theory of adaptive networks: Expectation and prediction. *Psychological Review*, 88(2), 135–170; Sutton, R. S. (1988). Learning to predict by the methods of temporal differences. *Machine Learning*, 3(1), 9–44; Schultz, W., Dayan, P., & Montague, P. R. (1997). A neural substrate of prediction and reward. *Science*, 275(5306), 1593–1599.

[659] Dabney, W., Kurth-Nelson, Z., Uchida, N., Starkweather, C. K., Hassabis, D., Munos, R., & Botvinick, M. (2020). A distributional code for value in dopamine-based reinforcement learning. *Nature*, 577(7792), 671–675; Sousa, M., Bujalski, P., Cruz, B. F., Louie, K., McNamee, D. C., & Paton, J. J. (2025). A multidimensional distributional map of future reward in dopamine neurons. *Nature*, 642, 691–699.

[660] Hassabis, D., Kumaran, D., Summerfield, C., & Botvinick, M. (2017). Neuroscience-inspired artificial intelligence. *Neuron*, 95(2), 245–258.

What the Machines Got Right: Representations

The strongest evidence that machine learning has found something real about the brain is that its systems, trained only to perform a task, end up organized inside, in some respects, like the brain.

Daniel Yamins, James DiCarlo, and their colleagues trained many networks to recognize objects in photographs and compared each network's internal activity with recordings from inferior temporal cortex (IT), the stage of the monkey's ventral stream where Chapter 6 located object recognition. The networks that were better at recognizing objects were better at predicting the responses of recording sites in IT, and the best of them predicted about half of the variance that could be predicted at all, far more than models built by hand to resemble the visual system.[661] No one had told the networks anything about IT. Seyed-Mahdi Khaligh-Razavi and Nikolaus Kriegeskorte reached the same conclusion with a different method, comparing the pattern of similarities among images in a network with the pattern in human IT measured by functional imaging.[662] Figure 10.2 shows the shape of the result: for the networks of 2014, performance and resemblance rose together.

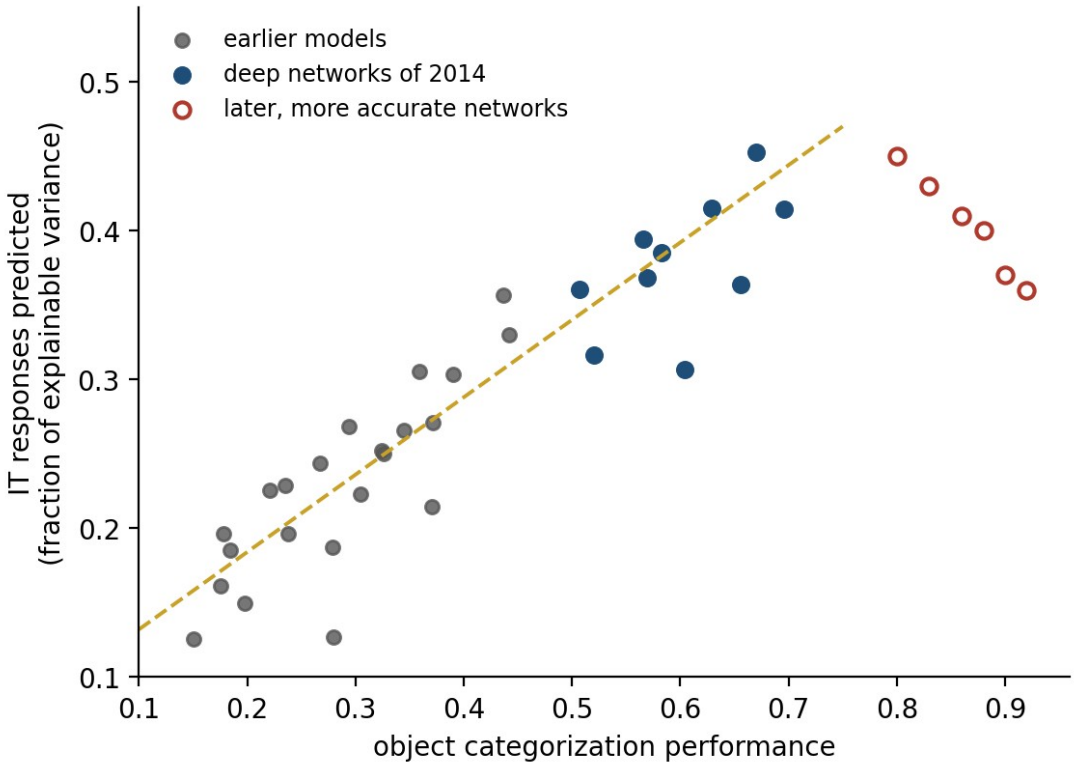


Figure 10.2. Better at the task, more like the brain, up to a point (illustrative, after Yamins et al. 2014). Each point is a model. Across the models of 2014, the better a network categorized objects, the better its internal units predicted responses in inferior temporal cortex. The dashed line is a guide to the eye; later and more accurate networks did not continue the trend (see text). The two studies measured performance on different benchmarks, so the horizontal axis is schematic.

[661] Yamins, D. L. K., Hong, H., Cadieu, C. F., Solomon, E. A., Seibert, D., & DiCarlo, J. J. (2014). Performance-optimized hierarchical models predict neural responses in higher visual cortex. *Proceedings of the National Academy of Sciences*, 111(23), 8619–8624.

[662] Khaligh-Razavi, S.-M., & Kriegeskorte, N. (2014). Deep supervised, but not unsupervised, models may explain IT cortical representation. *PLoS Computational Biology*, 10(11), e1003915.

The relation did not hold indefinitely. Drew Linsley, Thomas Serre, and their colleagues found that as networks grew more accurate than those of 2014, their ability to predict IT stopped improving and then began to fall, because the most accurate networks had come to rely on features that IT does not use.[663] Optimizing for a task brought the networks toward the brain only while the task forced them to solve the problem the brain solves.

Chapter 6 noted a further limit: the networks copied the brain's first sweep through the visual hierarchy, not its loops. The gap shows in timing. Kohitij Kar, DiCarlo, and their colleagues found hundreds of images that monkeys recognized and that standard feedforward networks did not. For those images, the object's identity appeared in IT about 30 milliseconds later than for easy ones, as if a second pass had been needed. Feedforward networks predicted those late responses poorly and deeper networks predicted them better, which the authors read as a sign that the brain was using recurrent processing to do what extra layers do.[664]

Language models have repeated the first part of the story. Martin Schrimpf, Evelina Fedorenko, and their colleagues tested dozens of language models against recordings of people reading and listening. The models' internal activity predicted responses in the brain's language network, and the models that predicted it best were the ones best at predicting the next word, rather than those best at other language tasks.[665] Ariel Goldstein, Uri Hasson, and their colleagues, recording from electrodes on the cortex of epilepsy patients listening to a podcast, found that the brain, like the model, signals its prediction of an upcoming word before the word is heard, and signals surprise in proportion to how unexpected the word turns out to be.[666] Charlotte Caucheteux and Jean-Rémi King found the resemblance only partial, strongest for the middle layers of the models and weaker for the top.[667]

Even the brain's map of space has appeared unbidden. Andrea Banino and his colleagues trained a network to keep track of its position from its own movements, the problem of path integration, and units resembling the grid cells of entorhinal cortex emerged in it.[668] Here a caution came quickly. Rylan Schaeffer, Mikail Khona, and Ila Fiete trained more than 11,000 such networks and found that grid-like units appeared only under particular choices made by the modelers, especially in how the network's output was encoded, and that where they appeared they lacked key properties of real grid cells; others

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[663] Linsley, D., Rodriguez, I. F., Fel, T., Arcaro, M., Sharma, S., Livingstone, M., & Serre, T. (2023). Performance-optimized deep neural networks are evolving into worse models of inferotemporal visual cortex. In *Advances in Neural Information Processing Systems 36* (pp. 28873–28891).

[664] Kar, K., Kubilius, J., Schmidt, K., Issa, E. B., & DiCarlo, J. J. (2019). Evidence that recurrent circuits are critical to the ventral stream's execution of core object recognition behavior. *Nature Neuroscience*, 22(6), 974–983.

[665] Schrimpf, M., Blank, I. A., Tuckute, G., Kauf, C., Hosseini, E. A., Kanwisher, N., Tenenbaum, J. B., & Fedorenko, E. (2021). The neural architecture of language: Integrative modeling converges on predictive processing. *Proceedings of the National Academy of Sciences*, 118(45), e2105646118.

[666] Goldstein, A., Zada, Z., Buchnik, E., Schain, M., Price, A., Aubrey, B., Nastase, S. A., Feder, A., Emanuel, D., Cohen, A., Jansen, A., Gazula, H., Choe, G., Rao, A., Kim, C., Casto, C., Fanda, L., Doyle, W., Friedman, D., … Hasson, U. (2022). Shared computational principles for language processing in humans and deep language models. *Nature Neuroscience*, 25(3), 369–380.

[667] Caucheteux, C., & King, J.-R. (2022). Brains and algorithms partially converge in natural language processing. *Communications Biology*, 5, 134.

[668] Banino, A., Barry, C., Uria, B., Blundell, C., Lillicrap, T., Mirowski, P., Pritzel, A., Chadwick, M. J., Degris, T., Modayil, J., Wayne, G., Soyer, H., Viola, F., Zhang, B., Goroshin, R., Rabinowitz, N., Pascanu, R., Beattie, C., Petersen, S., … Kumaran, D. (2018). Vector-based navigation using grid-like representations in artificial agents. *Nature*, 557(7705), 429–433; Cueva, C. J., & Wei, X.-X. (2018). Emergence of grid-like representations by training recurrent neural networks to perform spatial localization. In *International Conference on Learning Representations*.

have argued that the needed choices are principled.[669] A resemblance can come from what the modeler put in.

Resemblance is also measured in ways that can flatter. Jeffrey Bowers and his colleagues have argued that networks that predict neural data well still fail many of the classic experiments of vision science, where human perception shows effects the networks do not.[670] The fair conclusion is that a network trained hard enough on the right task arrives at some of the brain's solutions, which says something about both: the solutions are not arbitrary, and the brain is, in part, a system that has been optimized for its tasks.[671]

What the Machines Got Right: Error and Prediction

The second thing machine learning got right is the principle by which both systems learn: compare what was expected with what happened, and change in proportion to the difference. Dopamine broadcasts such an error about reward (Chapter 3), synapses change by the timing of activity and a third factor (Chapter 5), and perception corrects its predictions by sensory error (Chapter 6). In every case learning is driven by error and aimed at prediction.

Machine learning arrived at the same aim from its own direction. The most capable systems of the 2020s were trained mostly without labels, by being asked to predict part of their input from the rest: the next word from the words before, a hidden patch of an image from the visible ones. This is self-supervised learning, and it is the regime of the brain, which receives few labels and an endless stream of input against which to test its predictions. A language model's training objective shares the aim that Chapter 6 attributed to cortex, predicting what comes next, though not its form: cortex, on that account, predicts at every level of a hierarchy and corrects itself while it perceives, where a language model predicts only its next input. The final stages of training add a different signal, rewards from human preferences or from checking answers.

Prediction has also proved to scale. Jared Kaplan and his colleagues at OpenAI found that the error of a language model at predicting text falls as a smooth power of the model's size, the amount of data, and the computation spent on training, over many orders of magnitude.[672] Jordan Hoffmann and his colleagues at DeepMind corrected the recipe, showing that models of the time had been trained on too little data for their size: their Chinchilla outperformed models several times larger that had been trained

[669] Schaeffer, R., Khona, M., & Fiete, I. R. (2022). No free lunch from deep learning in neuroscience: A case study through models of the entorhinal-hippocampal circuit. In *Advances in Neural Information Processing Systems 35*; Sorscher, B., Mel, G. C., Ocko, S. A., Giocomo, L. M., & Ganguli, S. (2023). A unified theory for the computational and mechanistic origins of grid cells. *Neuron*, 111(1), 121–137.

[670] Bowers, J. S., Malhotra, G., Dujmović, M., Llera Montero, M., Tsvetkov, C., Biscione, V., Puebla, G., Adolfi, F., Hummel, J. E., Heaton, R. F., Evans, B. D., Mitchell, J., & Blything, R. (2023). Deep problems with neural network models of human vision. *Behavioral and Brain Sciences*, 46, e385.

[671] Richards, B. A., Lillicrap, T. P., Beaudoin, P., Bengio, Y., Bogacz, R., Christensen, A., Clopath, C., Costa, R. P., de Berker, A., Ganguli, S., Gillon, C. J., Hafner, D., Kepecs, A., Kriegeskorte, N., Latham, P., Lindsay, G. W., Miller, K. D., Naud, R., Pack, C. C., … Kording, K. P. (2019). A deep learning framework for neuroscience. *Nature Neuroscience*, 22(11), 1761–1770.

[672] Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). Scaling laws for neural language models. *arXiv*, 2001.08361.

on less.[673] A power law is also the shape often fitted to human practice, as Chapter 9 described, though there the fit is disputed.

What these results establish is that error-driven learning of predictions, applied at a large enough scale, produces knowledge that is general enough to use for tasks the learner was never trained on. That was the bet of the connectionists of the 1980s, and it has paid. What they do not establish is that the brain learns its predictions by the same procedure, which is a separate question taken up below.

Shortcuts

The resemblance ends where the training data end. Chapter 6 described adversarial images and networks that recognize texture where people recognize shape. Those are symptoms of a broader pattern that Robert Geirhos and his colleagues called shortcut learning: a network trained to reduce its error will use whatever feature of the training data reduces it, whether or not that feature is the one a person would use.[674] John Zech and his colleagues trained a network to detect pneumonia in chest X-rays from three hospital systems. It performed well, and part of what it had learned was to recognize which hospital an image came from, which it could do almost perfectly from metal tokens and other marks on the films. Because the hospitals differed greatly in how often their patients had pneumonia, knowing the hospital was a good guess at the diagnosis.[675] The network was right for a reason that would fail at the next hospital.

The brain is not free of shortcuts: Chapter 9's habits go on producing yesterday's answer after the reason for it has gone. But a person has learned from a world in which most shortcuts eventually break, and has met the same object under so many lights, angles, and contexts that the features that survive are mostly the ones that matter. A network trained on one collection of images has had no such schooling, and even small changes can expose it. Benjamin Recht and his colleagues built a fresh test set for the standard image benchmark, following the original procedure as closely as they could, and found that the accuracy of every model they tested dropped by roughly 11 to 14 percentage points.[676] Logan Engstrom and his colleagues later traced much of the drop to subtle differences in how the new images had been selected, which is itself the point: changes too slight for the builders of the test set to notice were enough to move every model.[677]

The gap is narrowing, and it is narrowing in an instructive way. When Geirhos and colleagues later compared people with newer models on distorted images, the models trained on far larger and more varied datasets had closed much of the difference in robustness, and their errors had come to resemble

[673] Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., de Las Casas, D., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., van den Driessche, G., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., … Sifre, L. (2022). An empirical analysis of compute-optimal large language model training. In *Advances in Neural Information Processing Systems 35* (pp. 30016–30030). Preprint titled "Training compute-optimal large language models," arXiv:2203.15556.

[674] Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., & Wichmann, F. A. (2020). Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11), 665–673.

[675] Zech, J. R., Badgeley, M. A., Liu, M., Costa, A. B., Titano, J. J., & Oermann, E. K. (2018). Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLoS Medicine*, 15(11), e1002683.

[676] Recht, B., Roelofs, R., Schmidt, L., & Shankar, V. (2019). Do ImageNet classifiers generalize to ImageNet? In *Proceedings of the 36th International Conference on Machine Learning* (pp. 5389–5400). PMLR.

[677] Engstrom, L., Ilyas, A., Santurkar, S., Tsipras, D., Steinhardt, J., & Madry, A. (2020). Identifying statistical bias in dataset replication. In *Proceedings of the 37th International Conference on Machine Learning* (PMLR 119, pp. 2922–2932).

human errors more closely, though people and models still went wrong on different images.[678] That is Chapter 9's lesson about varied practice. A learner exposed to the variability of the world learns the features that survive it. What remains different is the amount of data required.

The Data Gap

A child of three speaks in sentences, and a child of ten has a vocabulary in the tens of thousands of words. Michael Frank has estimated how much language reaches a child. On generous assumptions it is about a million words a month, some 60 million by the age of five and 200 million by 20; on cautious ones, ten times fewer.[679] GPT-3, the language model that in 2020 made fluent machine prose familiar, was trained on about 300 billion tokens of text, a token being roughly a word or a part of one, and Meta's Llama 3 in 2024 on about 15 trillion.[680] Against a five-year-old, the models see thousands to hundreds of thousands of times more language (Figure 10.3), and the child has learned to walk, see, and play in the same years.

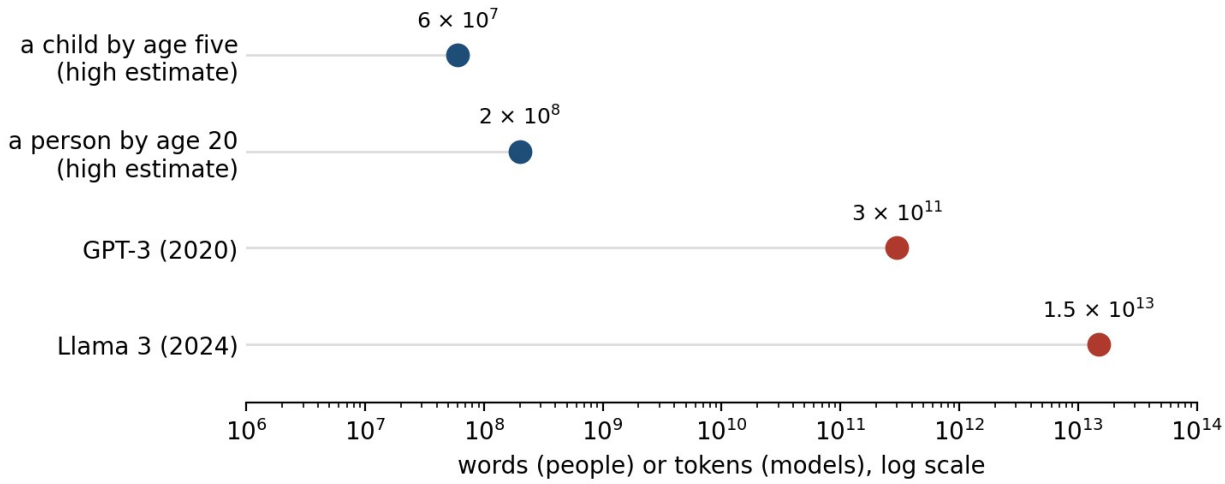


Figure 10.3. The data gap. The language heard by a person by age five and by age 20, on Frank's high estimates, against the text used to train two language models, on a logarithmic scale, where each step is a factor of ten. Sources are given in the text.

Part of the explanation is that a child does not start from nothing. A colt walks within hours of birth. A newborn human tracks a face-like pattern further than a scrambled one within the first hour of life, and by four or five months infants look longer when a hidden object seems to have vanished, as if they expected it to persist, though how much that expectation implies is debated.[681] Anthony Zador has

[678] Geirhos, R., Narayanappa, K., Mitzkus, B., Thieringer, T., Bethge, M., Wichmann, F. A., & Brendel, W. (2021). Partial success in closing the gap between human and machine vision. In *Advances in Neural Information Processing Systems 34* (pp. 23885–23899).

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[681] Johnson, M. H., Dziurawiec, S., Ellis, H., & Morton, J. (1991). Newborns' preferential tracking of face-like stimuli and its subsequent decline. *Cognition*, 40(1–2), 1–19; Baillargeon, R., Spelke, E. S., & Wasserman, S. (1985). Object permanence in five-

argued that this is the most underrated difference between brains and networks: an animal's brain arrives with circuits shaped by evolution, so much of what looks like rapid learning is the tuning of a system already close to the answer.[682] The genome cannot specify every synapse, because there are far too many, so it must specify rules for wiring, and Zador calls this a genomic bottleneck that forces evolution to encode compact, general structure. Machine learning's equivalent is the choice of architecture: a convolutional network builds in the assumption that a feature means the same thing wherever it appears, which is one reason it learns vision from far fewer examples than a network without it.

The other part is that a child learns from experience that is rich, embodied, and chosen. Wai Keen Vong, Brenden Lake, and their colleagues trained a network on 61 hours of video and transcribed speech recorded from a camera mounted on the head of one child, between the ages of six and 25 months, about one percent of the child's waking hours. From that alone, with no built-in knowledge of language, the network learned to match words to what they named. Tested on 22 everyday words, choosing which of four pictures showed the named thing, it was right 62 percent of the time, where chance was 25, close to the 67 percent of a model trained on 400 million captioned images from the internet, and it generalized some words to objects it had never seen.[683]

The result shows that a learner without innate linguistic knowledge can begin to map words onto things from one child's view of the world. It also shows how far there is to go: the network was tested on a few dozen words and not on grammar at all, while the child was by then on the way to sentences. A related line of work, the BabyLM Challenge, trains language models on 100 million words or fewer, roughly what a child hears by early adolescence. On the challenge's tests, most of them judgments of whether sentences are grammatical, the best such models did as well as some models trained on trillions of words, which suggests that much of what the giant datasets buy is not grammar.[684]

People learn new concepts from one example. Shown a single unfamiliar written character, a person can recognize other examples, draw new ones, and see its parts. Lake, Ruslan Salakhutdinov, and Joshua Tenenbaum built a system that did the same by representing characters as programs for drawing them, and, with colleagues, argued that human learning is fast because people build models of how things are caused, not only of what they look like.[685] Large language models have partly closed this gap from the other side. Once trained, they learn a new task from a handful of examples placed in their input, without any change to their weights, an ability called in-context learning.[686] Jane Wang and her colleagues had

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month-old infants. *Cognition*, 20(3), 191–208; Spelke, E. S., & Kinzler, K. D. (2007). Core knowledge. *Developmental Science*, 10(1), 89–96.

[682] Zador, A. M. (2019). A critique of pure learning and what artificial neural networks can learn from animal brains. *Nature Communications*, 10, 3770.

[683] Vong, W. K., Wang, W., Orhan, A. E., & Lake, B. M. (2024). Grounded language acquisition through the eyes and ears of a single child. *Science*, 383(6682), 504–511.

[684] Warstadt, A., Mueller, A., Choshen, L., Wilcox, E., Zhuang, C., Ciro, J., Mosquera, R., Paranjape, B., Williams, A., Linzen, T., & Cotterell, R. (2023). Findings of the BabyLM Challenge: Sample-efficient pretraining on developmentally plausible corpora. In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning* (pp. 1–34). https://doi.org/10.18653/v1/2023.conll-babylm.1.

[685] Lake, B. M., Salakhutdinov, R., & Tenenbaum, J. B. (2015). Human-level concept learning through probabilistic program induction. *Science*, 350(6266), 1332–1338; Lake, B. M., Ullman, T. D., Tenenbaum, J. B., & Gershman, S. J. (2017). Building machines that learn and think like people. *Behavioral and Brain Sciences*, 40, e253.

[686] Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., … Amodei, D. (2020). Language models are few-shot learners. In *Advances in Neural Information Processing Systems 33* (pp. 1877–1901).

found that a recurrent network trained on many related tasks develops exactly such a fast learner in its activity, and proposed that prefrontal cortex is one, trained slowly by dopamine to learn quickly.[687] On this view the machine's few-shot learning is real but expensive. It took the enormous dataset to buy it, much as the child's took evolution.

Learning That Never Stops

A brain learns while it is used, and no one ever stops it to retrain it. Most deployed machine learning works the other way (Figure 10.4). A model is trained, its weights are fixed, and then it is used; what it learns from a conversation lasts only as long as the conversation is held in its context window, and when the conversation ends the model is as it was. A language model knows nothing of events after its training data were collected unless it is told about them, looks them up, or is retrained. The memories that some assistants now keep between conversations are stored as text and fed back in, not learned into the weights.

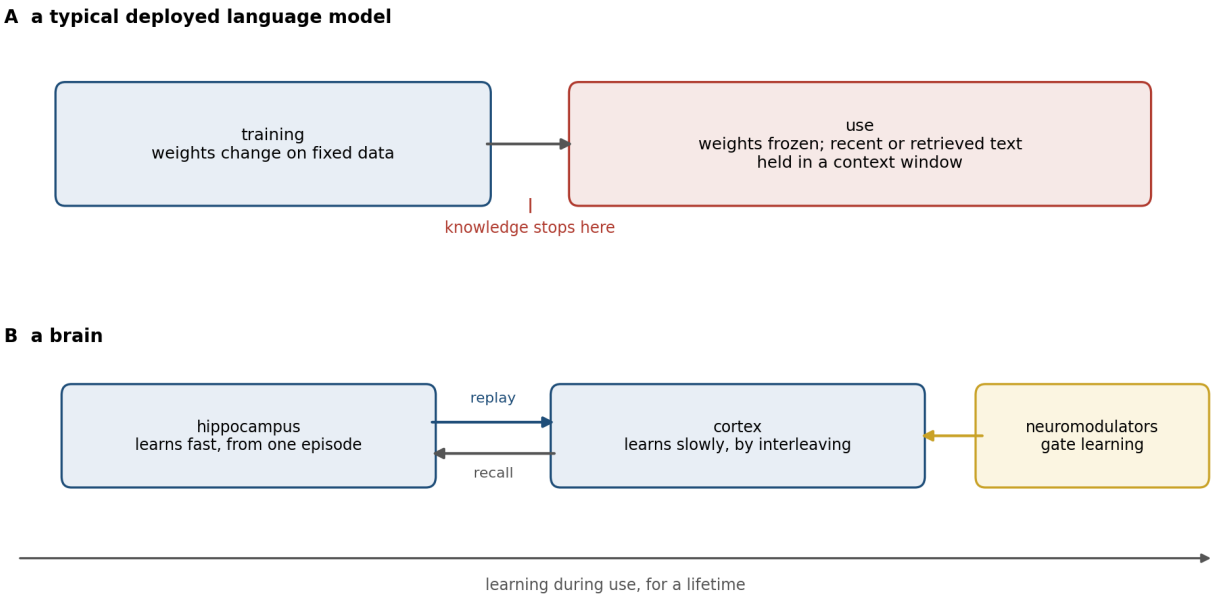


Figure 10.4. When learning happens. A: a typical deployed language model is trained once and then used with its weights fixed; what it takes in during use is held only in its context window or in text retrieved from outside. B: a brain learns during use for its whole life, with a fast hippocampal store that replays into slowly learning cortex and neuromodulators that decide when activity should change the wiring.

The obstacle is the catastrophic interference that Michael McCloskey and Neal Cohen found in 1989 (Chapters 1 and 5).[688] Figure 10.5 shows it in a small network. Trained on one list of associations and then on a second that uses the same cues with different answers, the network loses half of the first list within a few passes and nearly all of it within about 20, well before it has finished learning the second. If a few items from the first list are mixed back into the training on the second, both survive. The brain's

[687] Wang, J. X., Kurth-Nelson, Z., Kumaran, D., Tirumala, D., Soyer, H., Leibo, J. Z., Hassabis, D., & Botvinick, M. (2018). Prefrontal cortex as a meta-reinforcement learning system. *Nature Neuroscience*, 21(6), 860–868.

[688] McCloskey, M., & Cohen, N. J. (1989). Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24, 109–165.

answer is the complementary learning systems of Chapter 5: a fast hippocampal learner that replays episodes into slow cortex.[689] Machine learning had found the same answer independently, as Chapter 5 noted, and Volodymyr Mnih and his colleagues made experience replay part of the first network to learn dozens of Atari games from the screen alone, citing hippocampal replay as its biological counterpart.[690]

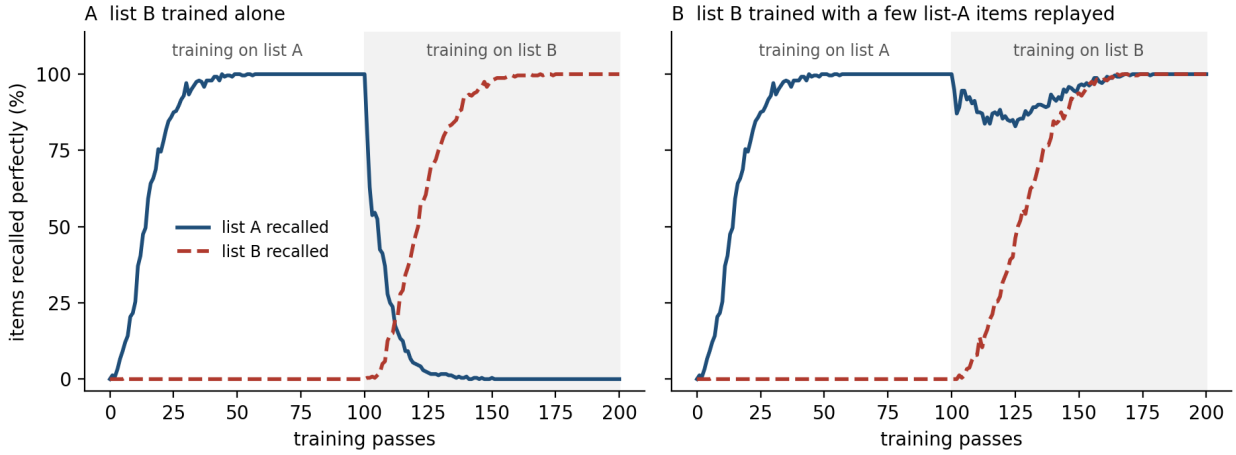


Figure 10.5. Catastrophic interference in a small network (simulation). Two lists share the same cues with different answers, and context units signal which list is being trained; an item counts as recalled only if all ten of its output units are correct. A: training on list B alone erases list A. B: replaying six list-A items, chosen at random on each pass, alongside list B protects list A while list B is learned. Averages of 20 simulated networks.

Replay helps, and Chapter 9 described elastic weight consolidation, which protects the weights that mattered for old tasks.[691] Neither has made continual learning routine. A newer finding suggests that the problem is deeper than forgetting. Shibhansh Dohare, Richard Sutton, and their colleagues trained standard networks on long sequences of tasks and found that they gradually lose the ability to learn at all: after thousands of tasks, a network learned each new one no better than a network with no hidden layers, because more and more of its units had become inactive or redundant.[692] They restored the ability by continually reinitializing a small fraction of the least-used units, which they called continual backpropagation.

The brain does something that looks similar. It keeps making new synapses and removing unused ones, as Chapter 9's motor cortex did for a new skill, and in many mammals the hippocampus adds new neurons throughout life, although whether adult humans do so is disputed.[693]

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[690] Lin, L.-J. (1992). Self-improving reactive agents based on reinforcement learning, planning and teaching. *Machine Learning*, 8(3–4), 293–321; Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S., & Hassabis, D. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540), 529–533.

[691] Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., & Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526.

[692] Dohare, S., Hernandez-Garcia, J. F., Lan, Q., Rahman, P., Mahmood, A. R., & Sutton, R. S. (2024). Loss of plasticity in deep continual learning. *Nature*, 632(8026), 768–774.

Credit Without Backpropagation

Backpropagation sends the error at the output backward through the network along the same connections, with the same weights, that carried the signal forward. Chapter 3 recounted Francis Crick's objection of 1989: axons carry signals one way, and a synapse has no evident means of learning the strength of a synapse somewhere downstream.[694] If the brain does not backpropagate, then the most successful learning procedure known is one it does not use, and the resemblance of trained networks to cortex is a resemblance of destination, not of route.

The two most developed answers were met in Chapters 3 and 6. One is Chapter 3's: a backward pathway with random, fixed weights, to which the forward weights come to align, and bursts at the apical dendrites of pyramidal neurons that carry the error apart from the cell's main output.[695] The other is Chapter 6's predictive coding: a network built to correct its own predictions, as cortex may be, can compute nearly the same weight changes that backpropagation does, using only plasticity local to each synapse.[696] João Sacramento and his colleagues have built a model of dendritic microcircuits that approximates backpropagation in the same spirit.[697] None of these has been shown to operate in cortex, and when Sergey Bartunov, Lillicrap, and their colleagues tested several on hard image-recognition problems, they fell well short of backpropagation.[698]

What can be said is that the brain solves the problem backpropagation solves, deciding which of billions of connections to change, with signals available at each synapse: the activity on either side, the dendritic potential, and a broadcast modulator of the kind Chapter 3 described. Whether that combination amounts to an approximation of the gradient or to something different is open. Yuhang Song, Rafal Bogacz, and their colleagues have argued for the second: in their models, the network first settles its activity on what the outcome should have been and only then changes its synapses; in their simulations, networks trained this way learned faster than with backpropagation and with less interference between tasks.[699] If that is how the brain works, its local rules would be not a makeshift for lack of a backward pathway but a better way to learn while acting.

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[693] Sorrells, S. F., Paredes, M. F., Cebrian-Silla, A., Sandoval, K., Qi, D., Kelley, K. W., James, D., Mayer, S., Chang, J., Auguste, K. I., Chang, E. F., Gutierrez, A. J., Kriegstein, A. R., Mathern, G. W., Oldham, M. C., Huang, E. J., Garcia-Verdugo, J. M., Yang, Z., & Alvarez-Buylla, A. (2018). Human hippocampal neurogenesis drops sharply in children to undetectable levels in adults. *Nature*, 555(7696), 377–381; Boldrini, M., Fulmore, C. A., Tartt, A. N., Simeon, L. R., Pavlova, I., Poposka, V., Rosoklija, G. B., Stankov, A., Arango, V., Dwork, A. J., Hen, R., & Mann, J. J. (2018). Human hippocampal neurogenesis persists throughout aging. *Cell Stem Cell*, 22(4), 589–599.

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[695] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276; Payeur, A., Guerguiev, J., Zenke, F., Richards, B. A., & Naud, R. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24(7), 1010–1019.

[696] Whittington, J. C. R., & Bogacz, R. (2017). An approximation of the error backpropagation algorithm in a predictive coding network with local Hebbian synaptic plasticity. *Neural Computation*, 29(5), 1229–1262.

[697] Sacramento, J., Ponte Costa, R., Bengio, Y., & Senn, W. (2018). Dendritic cortical microcircuits approximate the backpropagation algorithm. In *Advances in Neural Information Processing Systems 31* (pp. 8721–8732); Lillicrap, T. P., Santoro, A., Marris, L., Akerman, C. J., & Hinton, G. (2020). Backpropagation and the brain. *Nature Reviews Neuroscience*, 21(6), 335–346.

[698] Bartunov, S., Santoro, A., Richards, B. A., Marris, L., Hinton, G. E., & Lillicrap, T. P. (2018). Assessing the scalability of biologically-motivated deep learning algorithms and architectures. In *Advances in Neural Information Processing Systems 31*.

[699] Song, Y., Millidge, B., Salvatori, T., Lukasiewicz, T., Xu, Z., & Bogacz, R. (2024). Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience*, 27(2), 348–358.

The Energy Bill

Chapter 1 put the brain's power at about 20 watts. Over 20 years, from birth to adulthood, that comes to about 3.5 megawatt-hours. David Patterson and his colleagues estimated that training GPT-3 used about 1,300 megawatt-hours, the energy of several hundred human brains running for 20 years (Figure 10.6).[700] Training is only part of the bill. At Google, about three-fifths of the energy used for machine learning went to answering queries rather than training.[701] A single data-center processor of the mid-2020s can draw 700 watts, as much as 35 brains, and newer ones draw more.[702]

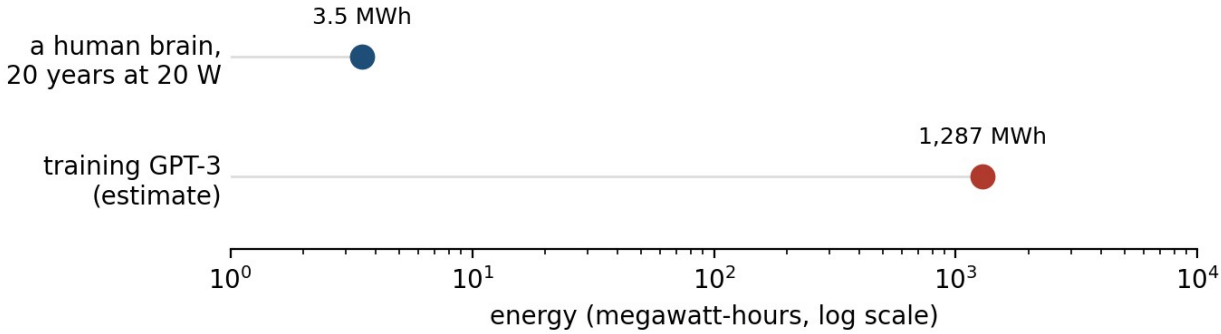


Figure 10.6. The energy bill. The energy used by a human brain over its first 20 years at 20 watts, against an estimate of the energy used to train GPT-3, on a logarithmic scale. Neither figure includes the cost of evolution or of the body that supports the brain.

The comparison is unfair in both directions. It omits the billions of years of evolution that designed the brain and the body that feeds it, and it omits that one trained model serves millions of people. But the brain's frugality is not an accident, and Chapters 1 and 2 traced it to sparse activity, all-or-none spikes that cost energy only when sent, and computation done where the memory is stored.

Machine learning has begun to rediscover parts of this. Models built as mixtures of experts activate only a small fraction of their parameters for each word, which lets them grow without a proportional growth in computation.[703] Neuromorphic chips such as IBM's TrueNorth and Intel's Loihi compute with spikes and keep memory beside the processing units, and run some tasks on a small fraction of the power of a conventional processor: TrueNorth, with a million artificial neurons, consumed 63 milliwatts while processing live video.[704] They have not yet been shown to train large models. The brain's lesson is that

[700] Patterson, D., Gonzalez, J., Le, Q., Liang, C., Munguia, L.-M., Rothchild, D., So, D., Texier, M., & Dean, J. (2021). Carbon emissions and large neural network training. *arXiv*, 2104.10350.

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[703] Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., & Dean, J. (2017). Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*; Fedus, W., Zoph, B., & Shazeer, N. (2022). Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120), 1–39.

[704] Merolla, P. A., Arthur, J. V., Alvarez-Icaza, R., Cassidy, A. S., Sawada, J., Akopyan, F., Jackson, B. L., Imam, N., Guo, C., Nakamura, Y., Brezzo, B., Vo, I., Esser, S. K., Appuswamy, R., Taba, B., Amir, A., Flickner, M. D., Risk, W. P., Manohar, R., & Modha, D. S. (2014). A million spiking-neuron integrated circuit with a scalable communication network and interface. *Science*, 345(6197), 668–673; Davies, M., Srinivasa, N., Lin, T.-H., Chinya, G., Cao, Y., Choday, S. H., Dimou, G., Joshi, P., Imam, N., Jain, S., Liao, Y., Lin, C.-K., Lines, A., Liu, R., Mathaikutty, D., McCoy, S., Paul, A., Tse, J., Venkataramanan, G., … Wang, H. (2018). Loihi: A

the binding constraint on intelligence may be energy rather than data or computation, and that a system built under that constraint looks very different from one built without it.

Bodies

In March 2016 AlphaGo, the program described in Chapter 8, beat Lee Sedol, one of the strongest Go players in the world, by four games to one.[705] The program chose its moves, but it could not make them. Aja Huang, a DeepMind programmer, sat across the board from Lee, read each move from a screen, and placed the stones by hand.[706] Picking a stone out of a bowl and setting it on a crowded board without disturbing its neighbors was, for a machine, the harder problem, and in 2016 no robot could have done it as easily as Huang did. Hans Moravec had noticed the pattern in the 1980s: computers could be made to perform like adults on intelligence tests or at checkers far more easily than they could be given the perception and mobility of a one-year-old.[707] Moravec's explanation was evolutionary. The skills that feel effortless, seeing, grasping, and walking, have been refined over hundreds of millions of years, and their difficulty is hidden in the brain's machinery rather than absent; abstract reasoning is recent and, by comparison, thin.

The paradox has softened. Chapter 9 described a robot hand that learned to turn a cube only after training in simulations whose physics were randomized far beyond anything real, and robots trained on large collections of demonstrations now fold laundry and clear tables in some settings.[708] But a child learns to grasp from a few months of its own fumbling, with its own body, and robots still need far more practice, much of it simulated, for narrower skills. What the child has are the forward models of Chapters 6 and 9: predictions of the sensory consequences of its own movements, compared with what happens and corrected on every movement for life, by a cerebellum that never stops recalibrating.

A body also gives an organism something to care about. Chapter 7 argued that value in the brain is anchored to the state of the body, which limits how far an animal's goals can drift. A language model has no body and no needs; what it learns to value is whatever its training rewarded. The difference returns below, in the question of whether a machine knows when it does not know.

Fluency and Thought

Large language models produce fluent, often correct prose on almost any subject, and the question of what that fluency implies has divided researchers.[709] The brain offers a way to frame it. Kyle Mahowald,

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neuromorphic manycore processor with on-chip learning. *IEEE Micro*, 38(1), 82–99.

[705] Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., van den Driessche, G., Schrittwieser, J., Antonoglou, I., Panneershelvam, V., Lanctot, M., Dieleman, S., Grewe, D., Nham, J., Kalchbrenner, N., Sutskever, I., Lillicrap, T., Leach, M., Kavukcuoglu, K., Graepel, T., & Hassabis, D. (2016). Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587), 484–489; Hassabis, D. (2016, March 16). What we learned in Seoul with AlphaGo. *The Keyword* (Google blog). https://blog.google/technology/ai/what-we-learned-in-seoul-with-alphago/.

[706] Metz, C. (2016, March 16). In two moves, AlphaGo and Lee Sedol redefined the future. *Wired*.

[707] Moravec, H. (1988). *Mind Children: The Future of Robot and Human Intelligence*. Harvard University Press. The observation is on p. 15.

[708] OpenAI, Akkaya, I., Andrychowicz, M., Chociej, M., Litwin, M., McGrew, B., Petron, A., Paino, A., Plappert, M., Powell, G., Ribas, R., Schneider, J., Tezak, N., Tworek, J., Welinder, P., Weng, L., Yuan, Q., Zaremba, W., & Zhang, L. (2019). Solving Rubik's cube with a robot hand. *arXiv*, 1910.07113; Black, K., Brown, N., Driess, D., Esmail, A., Equi, M., et al. (2024). π0: A vision-language-action flow model for general robot control. *arXiv*, 2410.24164.

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Anna Ivanova, Fedorenko, and their colleagues distinguished formal competence in language, knowing its words and rules, from functional competence, using language to reason, plan, and refer to the world.[710] In the human brain the two depend on different networks. The language network responds to sentences but not to arithmetic, logic, or music.

Rosemary Varley and her colleagues studied three men whose strokes had left them with severe aphasia: in sentences such as "the man killed the lion", where only the order of the words shows who did what to whom, they could no longer reliably tell the doer from the done-to, and yet they solved arithmetic problems whose structure is as nested as a sentence's, such as 90 − [(3 + 17) × 3].[711] Language, on this and similar evidence, is how people communicate thought, not the medium in which they do most of it.[712] A system trained only on language could master the language network's job without acquiring the others.

The evidence suggests that the models have acquired some of both, unevenly. Thomas McCoy, Thomas Griffiths, and their colleagues argued that a model trained to predict text will be shaped by the statistics of text even when a task does not depend on them. Asked to decode a message written in a simple cipher that shifts each letter along the alphabet, GPT-4 did much better with a shift of 13, which is common on the internet, than with a shift of 12, which is not, although the procedure is the same.[713]

Lukas Berglund and his colleagues found that models that learned "A is B" from their training data often failed to infer "B is A": asked about a thousand celebrities, GPT-4 could name a celebrity's parent 79 percent of the time, but given the parent could name the celebrity only 33 percent of the time.[714] Karthik Valmeekam and his colleagues gave GPT-4 simple planning problems about stacking blocks. It solved about a third of them, and almost none when the same problems were disguised with unfamiliar names for the objects and actions.[715] In each case the system did well where its training text had done the reasoning for it and worse where it had to reason afresh.

That picture has changed quickly. Models asked to write out intermediate steps before answering, a method called chain of thought, solve more multistep problems, and models trained by reinforcement learning to reason at length before they answer have become much better at mathematics and programming problems whose answers can be checked.[716] When Valmeekam's group retested its block-stacking problems on one such reasoning model in 2024, it solved nearly all of the plain problems and

[710] Mahowald, K., Ivanova, A. A., Blank, I. A., Kanwisher, N., Tenenbaum, J. B., & Fedorenko, E. (2024). Dissociating language and thought in large language models. *Trends in Cognitive Sciences*, 28(6), 517–540.

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[714] Berglund, L., Tong, M., Kaufmann, M., Balesni, M., Stickland, A. C., Korbak, T., & Evans, O. (2024). The reversal curse: LLMs trained on "A is B" fail to learn "B is A". In *International Conference on Learning Representations*.

[715] Valmeekam, K., Marquez, M., Sreedharan, S., & Kambhampati, S. (2023). On the planning abilities of large language models: A critical investigation. In *Advances in Neural Information Processing Systems 36* (pp. 75993–76005).

[716] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q. V., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems 35* (pp. 24824–24837); Guo, D., Yang, D., Zhang, H., Song, J., Wang, P., Zhu, Q., Xu, R., Zhang, R., Ma, S., Bi, X., et al. (2025). DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645(8081), 633–638.

about half of the disguised ones.[717] Chapter 8 saw in this an echo of accumulation to a bound: computation spent before committing.

Other systems learn models of their world and plan inside them. Danijar Hafner and his colleagues' DreamerV3 learned to collect diamonds in the video game Minecraft, a long sequence of subgoals, from scratch and without examples from human players, by imagining the consequences of its actions in a learned model.[718] Brains appear to do something similar: in rats, the hippocampal sequences of Chapter 5 sometimes sketch the path to a remembered goal before the animal sets off.[719] What is still rare is the combination the brain manages routinely: one system that learns a model of the world from its own actions, plans in it, talks about it, and keeps learning while it does all three.

Knowing What It Does Not Know

Lee Sedol won one game of the five. In the fourth, his 78th move was one that AlphaGo had judged to have about one chance in 10,000 of being played by a human. The program answered with a poor move, and for several moves afterward its own estimate of its position stayed high; Hassabis wrote during the game that AlphaGo had made its mistake at move 79 and came to that realization only around move 87.[720] A system that had learned its confidence from millions of games met a position unlike them and did not know it.

The failure has a familiar modern form. A language model asked a question it cannot answer will often answer anyway, with a fluent and confident invention, a failure called hallucination. Adam Kalai and his colleagues have argued that this is partly a product of how the models are trained and tested: an evaluation that gives credit for a correct answer and none for "I do not know" rewards guessing, in the way a multiple-choice test rewards a student who never leaves a blank.[721] Models can be trained to estimate whether their own answers are right, as Chapter 8 noted, and do it fairly well within the range of their training.[722] The clearest evidence that training can blunt a model's sense of its own reliability comes from OpenAI's report on GPT-4. As first trained to predict text, the model was well calibrated: its confidence in an answer matched how often it was right. The later training that made it a helpful assistant made it markedly less so.[723]

People are not perfectly calibrated either. They are often overconfident on questions of general knowledge, and they sometimes invent reasons for what they did. But a person's confidence is shaped by consequences, and the cost of being wrong, felt by a body that gives outcomes their value, tends to pull

[717] Valmeekam, K., Stechly, K., & Kambhampati, S. (2024). LLMs still can't plan; can LRMs? A preliminary evaluation of OpenAI's o1 on PlanBench. *arXiv*, 2409.13373.

[718] Hafner, D., Pasukonis, J., Ba, J., & Lillicrap, T. (2025). Mastering diverse control tasks through world models. *Nature*, 640(8059), 647–653.

[719] Pfeiffer, B. E., & Foster, D. J. (2013). Hippocampal place-cell sequences depict future paths to remembered goals. *Nature*, 497(7447), 74–79.

[720] Metz, C. (2016, March 16). In two moves, AlphaGo and Lee Sedol redefined the future. *Wired*; Hassabis, D. (2016, March 16). What we learned in Seoul with AlphaGo. *The Keyword* (Google blog). https://blog.google/technology/ai/what-we-learned-in-seoul-with-alphago/; After 3 losses, master Go player scores a win against computer. (2016, March 13). *NPR, The Two-Way*. https://www.npr.org/sections/thetwo-way/2016/03/13/470284113/.

[721] Kalai, A. T., Nachum, O., Vempala, S. S., & Zhang, E. (2025). Why language models hallucinate. *arXiv*, 2509.04664.

[722] Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., … Kaplan, J. (2022). Language models (mostly) know what they know. *arXiv*, 2207.05221.

[723] OpenAI. (2023). GPT-4 technical report. *arXiv*, 2303.08774.

confidence back toward accuracy. A model is rewarded for what its designers chose to reward, and many of its evaluations reward an answer over an admission.

Chapter 9 ended on a related question: when a fast system should hand control back to a slow one. Language models that reason at length have begun to make this decision for themselves, spending more reasoning on problems they judge harder. But the judgment is learned from whatever training rewarded, and it is often poorly matched to difficulty: Xingyu Chen and his colleagues found such models spending many times more computation than needed on trivial problems, generating more than a dozen solutions to "2 + 3", while Yue Wang and colleagues found them abandoning promising lines of thought too soon on hard ones.[724] The brain's trigger is simpler: an expected outcome that fails to arrive. AlphaGo, answering move 78, had no such trigger, and many of its successors still lack one.

What Not to Copy

There is a case against using the brain as a guide at all, and it deserves a hearing. Sutton wrote in 2019, in an essay called "The Bitter Lesson", that the biggest lesson of 70 years of artificial intelligence research is that general methods that make use of more computation have always, in the end, beaten methods built on human knowledge of the problem.[725] Chess programs that searched more beat programs that encoded grandmaster insight; speech systems that learned from more data beat systems built around linguistics. Each time, researchers who had built in what they knew were overtaken. On this view, copying the brain's mechanisms is another way of building in what we think we know, and it will lose to scale.

The argument is strong, and the history in this chapter supports part of it: the networks came to resemble cortex by being trained, not by being designed to resemble it. But it is an argument against building in solutions, not against learning what the problems are. The brain is useful to engineers less as a blueprint than as a list of problems that a learning system meets when it leaves the laboratory, with evidence that each can be solved. Learning for a lifetime without forgetting, by rules available at each synapse, on a small energy budget, with values anchored to something it cannot redefine: these are the rows that Figure 10.7 marks as largely absent from current machines, and they are among the problems the brain was shaped by.

[724] Chen, X., Xu, J., Liang, T., He, Z., Pang, J., Yu, D., Song, L., Liu, Q., Zhou, M., Zhang, Z., Wang, R., Tu, Z., Mi, H., & Yu, D. (2024). Do NOT think that much for 2+3=? On the overthinking of o1-like LLMs. *arXiv*, 2412.21187; Wang, Y., Liu, Q., Xu, J., Liang, T., Chen, X., He, Z., Song, L., Yu, D., Li, J., Zhang, Z., Wang, R., Tu, Z., Mi, H., & Yu, D. (2025). Thoughts are all over the place: On the underthinking of o1-like LLMs. *arXiv*, 2501.18585.

[725] Sutton, R. S. (2019, March 13). The bitter lesson. *Incomplete Ideas*. http://www.incompleteideas.net/IncIdeas/BitterLesson.html.



Figure 10.7. A ledger. Properties of the brain described in this book, and the extent to which they are found in machine learning in the mid-2020s. "Error-driven learning" means the principle, not its implementation. The judgments are the author's summary of the evidence in this chapter and are meant as a starting point for argument.

Some of the brain's features should not be copied. Its neurons are slow, so it must compute in parallel and cannot run a long serial calculation; its components are imprecise, so it relies on redundancy; and it cannot be copied, reset, or inspected, so it must learn everything in one lifetime and protect what it has. Silicon is fast, precise, and can be duplicated, restarted, and examined, and a machine that inherited the brain's limitations along with its solutions would gain nothing.

Sorted this way, the differences named at the start of this chapter fall on both sides. Lifelong learning without forgetting, and an honest estimate of what one knows, are solutions machines lack to their cost. The brain's frugality is also a solution, forced by a constraint that now binds silicon too. Its small appetite for data is partly evolution's gift, which a machine can buy back, expensively, by training on a library. Values anchored to a body are a constraint a machine need not share, although what the anchoring buys, a limit on how far goals can drift, is something machines still need by other means. The local learning rule is the hardest to place: makeshift or improvement, no one yet knows.

Implications for Neuroscience

The traffic of Figure 10.1 runs both ways, and in the last decade the return direction has been the busier one: machine learning has changed neuroscience as much as neuroscience changed it.

First, trained networks are now models of the brain. Until the mid-2010s most models of higher visual cortex were built largely by hand, with learning confined to a stage or two; now a model is often a network trained on a task, and its quality is judged by how well it predicts recorded neurons. Kriegeskorte and Blake Richards with many colleagues have argued that neuroscience should describe brains as machine learning describes networks, by their objectives, their learning rules, and their architectures, rather than by what each neuron computes.[726] Describing the brain this way has made old questions precise: what its objectives are, whether it approximates the gradient, and how it keeps its plasticity.

Second, a model good enough to predict neurons is good enough to control them. Pouya Bashivan, Kar, and DiCarlo used a network model of the monkey's visual area V4 to design images that the model predicted would drive particular neurons harder than any natural image did. Shown to the monkeys, many of the synthetic images did exactly that.[727] Carlos Ponce, Margaret Livingstone, and their colleagues let neurons in the monkey's IT cortex guide an image-generating network, each generation of images bred from those the neurons responded to most. The images evolved toward pictures that often contained parts of faces or bodies, or objects familiar to the monkeys, combined in ways no one had predicted.[728] Experiments of this kind were impossible before the models existed.

Third, machine learning has reopened an old argument about how much of the mind must be innate. Systems with no built-in grammar, though with architectural biases of their own, now produce grammatical language from text alone, and Steven Piantadosi has argued that this undermines the case for an innate universal grammar.[729] Others reply that the models need thousands of times more language than a child hears, so that they show what can be learned from text in bulk, not what a child learns from what it hears. The disagreement is now partly about numbers that can be measured, such as how well a model does on a child's budget of words, and that is progress.[730]

What the brain could still lend to machines, and how its mechanisms might be built into them, is the subject of Chapter 11.

The Central Lesson

Machine learning took one idea from the brain, that knowledge can be learned into the strengths of many connections by correcting errors of prediction, and proved it more powerful than its inventors could: trained at scale, networks match people at tasks once thought to need a mind, and arrive at

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[726] Kriegeskorte, N. (2015). Deep neural networks: A new framework for modeling biological vision and brain information processing. *Annual Review of Vision Science*, 1, 417–446; Richards, B. A., Lillicrap, T. P., Beaudoin, P., Bengio, Y., Bogacz, R., Christensen, A., Clopath, C., Costa, R. P., de Berker, A., Ganguli, S., Gillon, C. J., Hafner, D., Kepecs, A., Kriegeskorte, N., Latham, P., Lindsay, G. W., Miller, K. D., Naud, R., Pack, C. C., … Kording, K. P. (2019). A deep learning framework for neuroscience. *Nature Neuroscience*, 22(11), 1761–1770.

[727] Bashivan, P., Kar, K., & DiCarlo, J. J. (2019). Neural population control via deep image synthesis. *Science*, 364(6439), eaav9436.

[728] Ponce, C. R., Xiao, W., Schade, P. F., Hartmann, T. S., Kreiman, G., & Livingstone, M. S. (2019). Evolving images for visual neurons using a deep generative network reveals coding principles and neuronal preferences. *Cell*, 177(4), 999–1009.

[729] Piantadosi, S. T. (2024). Modern language models refute Chomsky's approach to language. In E. Gibson & M. Poliak (Eds.), *From Fieldwork to Linguistic Theory: A Tribute to Dan Everett* (pp. 353–414). Language Science Press. https://doi.org/10.5281/zenodo.12665933.

[730] Warstadt, A., Mueller, A., Choshen, L., Wilcox, E., Zhuang, C., Ciro, J., Mosquera, R., Paranjape, B., Williams, A., Linzen, T., & Cotterell, R. (2023). Findings of the BabyLM Challenge: Sample-efficient pretraining on developmentally plausible corpora. In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning* (pp. 1–34). https://doi.org/10.18653/v1/2023.conll-babylm.1.

representations like those of a cortex they were never shown. What they did not take are the conditions under which a brain learns. A brain learns from little data because evolution has built in much of the structure and because its experience is rich, embodied, and chosen; it learns all its life because it replays what it knows and keeps its synapses changeable; and it runs on 20 watts because it computes sparsely where its memory is. Some of those conditions are constraints a machine is right to ignore; the rest are answers to problems that every learner meets once it leaves the laboratory. Turing proposed in 1950 to build a machine that learns like a child. Three-quarters of a century later the machines learn, but not like children: they learn once, from a library, at great cost, and then stop. The child's way is still the harder one to build.

Next Chapter

Chapter 11: Integrating Brain Mechanisms into AI Models: What Machines Could Borrow from Biology

Figure and Epigraph Sources

Figure Attribution

Figures 10.1 to 10.7 were drawn for this book. The points in Figure 10.2 are illustrative, after the patterns reported by Yamins et al. (2014) and Linsley et al. (2023), not recorded data. The values in Figure 10.3 are from Frank (2023), Brown et al. (2020), and Grattafiori et al. (2024); those in Figure 10.6 are from Patterson et al. (2021) and a brain power of 20 watts. Figure 10.5 is a simulation of a small network trained by backpropagation on two lists of associations in the design of McCloskey and Cohen (1989). Figures 10.1, 10.4, and 10.7 are schematic; Figure 10.1 includes Rescorla and Wagner (1972) as the psychological ancestor of temporal-difference learning; the judgments in Figure 10.7 are the author's.

Epigraph

Alan M. Turing, "Computing Machinery and Intelligence," *Mind*, 59(236), 433–460 (1950), section 7, "Learning Machines," p. 456.

Chapter 11

Integrating Brain Mechanisms into AI Models: What Machines Could Borrow from Biology

> *"What I cannot create, I do not understand."*
>
> *— Richard Feynman, written on his blackboard at the California Institute of Technology, found there at his death in 1988*

What the Wrights Took from the Buzzard

In the summer of 1899 Wilbur Wright was in the family bicycle shop in Dayton, Ohio, idly twisting the empty cardboard box that a customer's new inner tube had come in, when he saw that its top and bottom, twisted in opposite directions at the two ends, made a pair of surfaces meeting the air at different angles. A pair of fabric wings could be twisted the same way, raising the lift on one side and lowering it on the other.[731] The brothers had been looking for a way to keep a flying machine level, and the birds they watched seemed to have found one. "My observation of the flight of buzzards," Wilbur wrote to the engineer Octave Chanute the next year, "leads me to believe that they regain their lateral balance, when partly overturned by a gust of wind, by a torsion of the tips of the wings."[732] Their gliders of 1900 to 1902 and their powered flyer of 1903 were steered by this "wing warping", and the principle it captured, rolling an aircraft by changing the lift on each wing, is still how every airplane banks, although ailerons long ago replaced the warping.

What the Wrights took from the buzzard was narrow. Their machines did not flap, had no feathers, and were driven by a gasoline engine and propellers. Wilbur thought that most people who watched birds had looked at the wrong thing. "The person who merely watches the flight of a bird gathers the impression that the bird has nothing to think of but the flapping of its wings," he told an audience of engineers in 1901. "As a matter of fact this is a very small part of its mental labor."[733] The labor that mattered was balance, and balance was what the brothers borrowed. The bird showed that a thing could be done and suggested one principle for doing it. The engineering was their own.

Chapter 10 ended by sorting the differences between brains and machines into solutions worth borrowing and constraints not worth copying. This chapter takes the borrowable solutions one at a time, asking of each what the brain does, what has been built in its image, how well the copy works, and what is still missing.

The thesis is the Wrights'. The borrowings that have worked took a principle the brain demonstrates and built it in whatever way silicon does best; the attempts to copy mechanisms wholesale have mostly not. Figure 11.1 sorts the three ways of borrowing, and sets apart a fourth category that is easily mistaken for borrowing: solutions that engineers found for themselves and only afterward recognized in the brain. A principle borrowed from the brain must also pass the test that Chapter 10 called the

[731] Kelly, F. C. (1943). *The Wright Brothers*. Harcourt, Brace.

[732] Wright, W. (1900, May 13). Letter to Octave Chanute. Wilbur and Orville Wright Papers, Library of Congress. Reprinted in M. W. McFarland (Ed.), *The Papers of Wilbur and Orville Wright* (Vol. 1). McGraw-Hill, 1953.

[733] Wright, W. (1901). Some aeronautical experiments. *Journal of the Western Society of Engineers*, 6, 489–510.

bitter lesson: it has to beat the alternative when both are given more data and more computation, not only on a small problem where design still matters. Few principles have yet passed it, and the most important have not yet been tried at the scale where it applies.

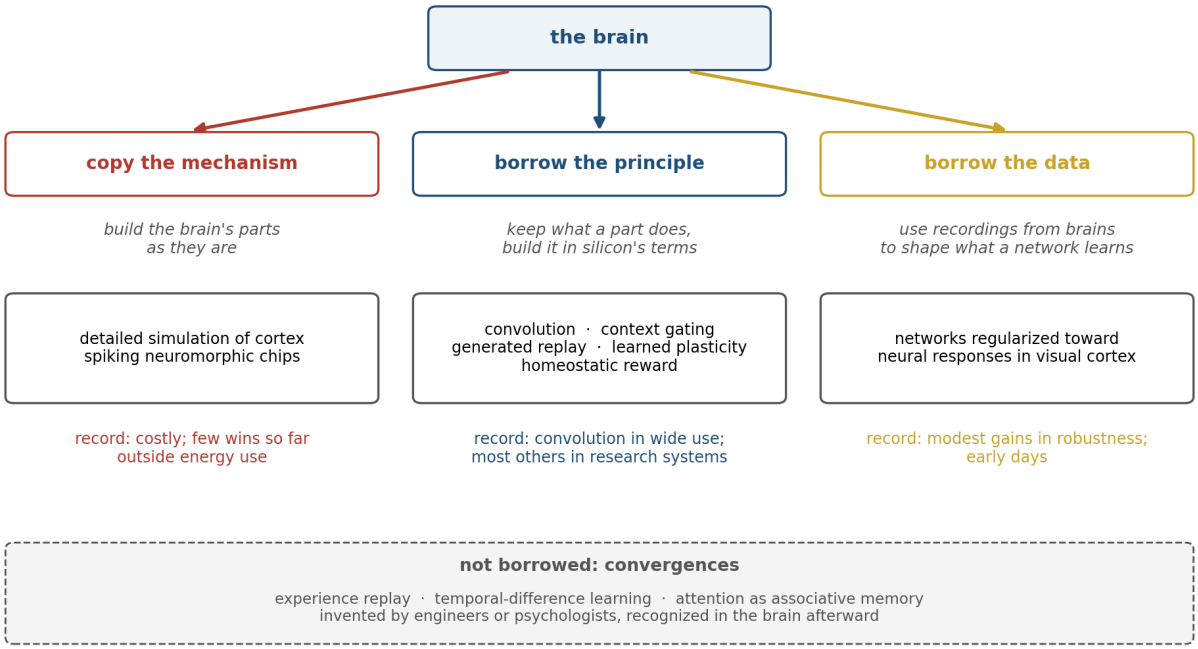


Figure 11.1. Three ways to borrow from the brain. Copying its mechanisms as they are, as in detailed simulations of cortex or chips built from spiking neurons; borrowing a principle and building it in silicon's own terms, as convolution and gating did; and using recordings from brains to shape what a network learns. Below, convergences: methods engineers invented for their own reasons and only later recognized in the brain. The examples are discussed in this chapter.

Copying and Borrowing

The record of copying is sobering. In a talk in 2009 Henry Markram, who directed the Blue Brain Project in Lausanne, said that it was not impossible to build a human brain and that it could be done in ten years.[734] Four years later the European Union launched the Human Brain Project, with a promised budget of about a billion euros over ten years, which aimed, among other things, to simulate the human brain on supercomputers. In 2014 more than 150 neuroscientists signed an open letter protesting the project's direction and governance, and the number of signatories later grew to about 800.[735] The project was restructured and ended in September 2023, having received about 600 million euros, and the Blue Brain Project closed at the end of 2024. Both produced lasting tools, atlases, and data, including a reconstruction of a fragment of rat cortex, a third of a cubic millimeter containing about 31,000 neurons, whose simulated activity resembled recorded activity in several respects.[736] Neither produced a simulated brain that learned, and neither changed how machines learn.

[734] Fildes, J. (2009, July 22). Artificial brain "10 years away." *BBC News*.

[735] Open message to the European Commission concerning the Human Brain Project. (2014, July 7). Retrieved from neurofuture.eu.

[736] Markram, H., Muller, E., Ramaswamy, S., Reimann, M. W., Abdellah, M., Sanchez, C. A., Ailamaki, A., Alonso-Nanclares, L., Antille, N., Arsever, S., Kahou, G. A. A., Berger, T. K., Bilgili, A., Buncic, N., Chalimourda, A., Chindemi, G., Courcol, J.-D., Delalondre, F., Delattre, V., … Schürmann, F. (2015). Reconstruction and simulation of neocortical microcircuitry. *Cell*, 163(2), 456–492.

The reason is not that the simulations were poor. A simulation that copies every ion channel inherits every unknown along with every known, and it has no way to tell which details matter for learning and which are the brain's own constraints, the slow, imprecise, irreplaceable parts that Chapter 10 advised against copying. The successes ran the other way. Hubel and Wiesel's simple and complex cells became convolution and pooling, which keep the principle, the same detector repeated across the visual field, and drop the neurons. Even this success has since been put to the bitter lesson's test. Vision transformers, which build in no such assumption, match convolutional networks once they are trained on enough images, and the borrowed principle's advantage shows mainly when data are scarce.[737] That is a pattern to expect of every borrowing in this chapter.

Other famous matches are not borrowings at all. Experience replay was invented by a computer scientist in 1992, before hippocampal replay was recorded, and temporal-difference learning came from psychological models of conditioning before anyone found it in dopamine neurons; Chapters 5 and 10 called these convergences. The attention of the transformer, as this chapter will show, is another. Convergences are evidence of a different kind: not that the brain's design can be copied, but that the brain and the engineers, facing the same problem, found the same solution. Demis Hassabis and his colleagues argued in 2017 that neuroscience helps machine learning most by validating that a capacity is possible and by suggesting an algorithm, not by supplying a circuit diagram.[738] Fabian Sinz, Andreas Tolias, and their colleagues made the complementary argument: what a brain could lend is its inductive biases, the assumptions about the world that let it generalize from little data, and those are best expressed in what a network is trained on and how it is built, not in copying its parts.[739]

Learning Without Forgetting

Replay, the remedy for forgetting that Chapter 10 demonstrated and one that machine learning has used for more than 30 years, has a cost the brain does not pay in the same way: storing old examples. A system that learns for a lifetime cannot keep every example, and a system trained on private data may not be allowed to.

The brain's own answer appears to be different. Chapter 5 described the hippocampus replaying recent experience during sleep and rest, but what it replays is not a recording; it is a reactivation of the patterns that represented the experience, generated by the network itself. Gido van de Ven, Hava Siegelmann, and Andreas Tolias built this into a network: instead of storing old images, it trained a generative model to produce internal representations resembling those of earlier tasks, replayed through the network's own feedback connections and modulated by the context of each task. It learned long sequences of tasks with little forgetting and without storing any data.[740] Timothy Tadros, Maxim Bazhenov, and their colleagues went further toward sleep: after each new task, they ran the network

[737] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*.

[738] Hassabis, D., Kumaran, D., Summerfield, C., & Botvinick, M. (2017). Neuroscience-inspired artificial intelligence. *Neuron*, 95(2), 245–258.

[739] Sinz, F. H., Pitkow, X., Reimer, J., Bethge, M., & Tolias, A. S. (2019). Engineering a less artificial intelligence. *Neuron*, 103(6), 967–979.

[740] van de Ven, G. M., Siegelmann, H. T., & Tolias, A. S. (2020). Brain-inspired replay for continual learning with artificial neural networks. *Nature Communications*, 11, 4069.

offline without input, with spontaneous activity and a local Hebbian rule, in a phase modeled on the slow-wave sleep of Chapter 5, and recovered much of what the task had overwritten.[741]

A second borrowing protects what matters instead of rehearsing it. Chapter 9 described elastic weight consolidation, in which the weights most important to an old task are made harder to change, an idea its authors linked to the stabilization of dendritic spines that follows learning.[742] Marcus Benna and Stefano Fusi gave the biology a sharper form: a real synapse is not one number but a chain of biochemical processes with different time constants, and a synapse built that way can learn fast and forget slowly, storing many more memories than a synapse with a single strength.[743] Axel Laborieux, Damien Querlioz, and their colleagues built a related idea, metaplasticity, a synapse's history changing how easily it changes, into networks with binary weights and found that it reduced forgetting.[744]

A third borrowing comes from the observation that different tasks engage partly different parts of a neuron: in mice, learning different motor tasks strengthens synapses on different dendritic branches of the same cortical neurons, with inhibition keeping the branches apart.[745] Nicolas Masse, Gregory Grant, and David Freedman built the principle into a network. They assigned each task a fixed, random fifth of the network's hidden units and silenced the other 80 percent whenever that task was learned or performed. Trained on 100 tasks in sequence, each a different scrambling of the pixels of handwritten digits, networks with weight protection alone ended with an average accuracy of 71 to 82 percent across the tasks, and with gating alone, 61 percent; gating and protection together kept 95 percent.[746]

Figure 11.2 reproduces the effect in a small network trained on five tasks that share their cues but not their answers. Without protection, each new task erases the last. Protecting the important weights does not help here on its own, because tasks that share their cues drive nearly the same hidden units, and every task needs the same connections. Gating spreads the tasks across different units and slows the forgetting. Together, gating and protection keep 83 percent of the first task after four more have been learned.

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[741] Tadros, T., Krishnan, G. P., Ramyaa, R., & Bazhenov, M. (2022). Sleep-like unsupervised replay reduces catastrophic forgetting in artificial neural networks. *Nature Communications*, 13, 7742.

[742] Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., & Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526; Yang, G., Pan, F., & Gan, W.-B. (2009). Stably maintained dendritic spines are associated with lifelong memories. *Nature*, 462(7275), 920–924; Zenke, F., Poole, B., & Ganguli, S. (2017). Continual learning through synaptic intelligence. In *Proceedings of the 34th International Conference on Machine Learning* (pp. 3987–3995). PMLR.

[743] Benna, M. K., & Fusi, S. (2016). Computational principles of synaptic memory consolidation. *Nature Neuroscience*, 19(12), 1697–1706.

[744] Laborieux, A., Ernoult, M., Hirtzlin, T., & Querlioz, D. (2021). Synaptic metaplasticity in binarized neural networks. *Nature Communications*, 12, 2549.

[745] Cichon, J., & Gan, W.-B. (2015). Branch-specific dendritic Ca2+ spikes cause persistent synaptic plasticity. *Nature*, 520(7546), 180–185.

[746] Masse, N. Y., Grant, G. D., & Freedman, D. J. (2018). Alleviating catastrophic forgetting using context-dependent gating and synaptic stabilization. *Proceedings of the National Academy of Sciences*, 115(44), E10467–E10475.

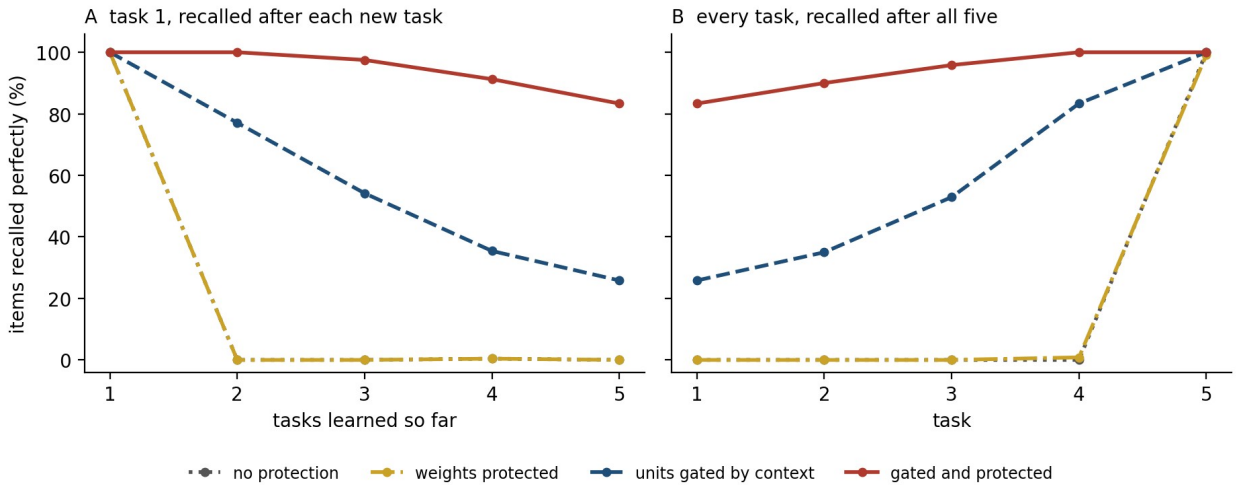


Figure 11.2. Context gating and weight protection against forgetting (simulation). A network learns five tasks in sequence; the tasks share the same cues but require different answers, and an input signals which task is current. With gating, each task uses a fixed random 20 percent of the hidden units. With protection, changes to weights that were important for earlier tasks are penalized. A: recall of the first task as later tasks are learned. B: recall of each task after all five. The curves without protection and with protection alone coincide. Averages of 20 simulated networks.

None of these methods is routine in the systems in use. Fine-tuning a trained model on a sequence of new tasks degrades what it knew before, and in one study of models of one to seven billion parameters the forgetting grew worse with size. The cheap method most often used to adapt large models, which trains a small set of added weights, learns less than full fine-tuning and forgets less, as if learning and forgetting were two sides of the same change.[747] Zorik Gekhman and his colleagues found something more troubling: models fine-tuned on facts they did not already know learned them slowly and, as they learned them, became more likely to invent answers to other questions.[748] The industry's answer to the problem of forgetting has been to sidestep it. New knowledge is added mostly by retrieval, described in the next section, and when a model's weights are updated, it is by continued training on a mixture that includes much of the old data, which is replay at enormous scale, or by training a new model altogether.[749]

The brain has an additional tool that machines have only begun to copy: it chooses what to replay. When the reward at the end of a rat's familiar path is increased, the hippocampus replays the path backward, from the reward to the start, more often, and when the reward is reduced it does so less often; replay in the forward direction does not change.[750] Marcelo Mattar and Nathaniel Daw proposed that the brain replays whatever memory would most improve its decisions, and showed that the rule

[747] Luo, Y., Yang, Z., Meng, F., Li, Y., Zhou, J., & Zhang, Y. (2025). An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *IEEE Transactions on Audio, Speech and Language Processing*, 33, 3776–3786; Biderman, D., Portes, J., Gonzalez Ortiz, J. J., Paul, M., Greengard, P., Jennings, C., King, D., Havens, S., Chiley, V., Frankle, J., Blakeney, C., & Cunningham, J. P. (2024). LoRA learns less and forgets less. *Transactions on Machine Learning Research*.

[748] Gekhman, Z., Yona, G., Aharoni, R., Eyal, M., Feder, A., Reichart, R., & Herzig, J. (2024). Does fine-tuning LLMs on new knowledge encourage hallucinations? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (pp. 7765–7784). Association for Computational Linguistics.

[749] DeepSeek-AI. (2025). *DeepSeek-V3.1-Base* [Model card]. Hugging Face. The model was built by continued training of the DeepSeek-V3 checkpoint on about 840 billion further tokens.

[750] Ambrose, R. E., Pfeiffer, B. E., & Foster, D. J. (2016). Reverse replay of hippocampal place cells is uniquely modulated by changing reward. *Neuron*, 91(5), 1124–1136.

accounts for much of what is known about which sequences rats replay and when.[751] Machine learning had reached the same idea from the engineering side: prioritized experience replay samples the experiences whose outcomes the network predicted worst, and it learned Atari games faster than uniform replay.[752] What remains to be borrowed is the rest of the loop: choosing what to replay, generating rather than storing it, and doing it continually, in the pauses of use, as the brain does in rest and sleep.[753]

A Memory That Learns in Use

Chapter 5 described the hippocampal index: on one account, the hippocampus does not store an experience in full but stores a pointer to the cortical patterns that represented it, so that a fragment of the experience can reactivate the whole.[754] The most widely used way of giving a language model new knowledge without retraining it is built on a similar plan. A retrieval-augmented model keeps a large database of text outside its weights. When it receives a question, it finds the passages most similar to the question and reads them before answering.[755] Sebastian Borgeaud and his colleagues at DeepMind gave a model a database of two trillion tokens and found that it performed about as well as models with 25 times as many parameters.[756] Knowledge held in the database can be corrected, added to, or deleted without retraining, as a fact in the hippocampus can be updated without rewriting the cortex.

The analogy stops at consolidation. The hippocampus, on the standard account of Chapter 5, gradually teaches what it holds to the cortex; a retrieval database never teaches the weights anything, and a model that has looked up a fact a thousand times knows it no better than before. The same is true of the context window, the text a model holds in view during a conversation, which Chapter 10 described as a memory wiped when the conversation ends.

The context window has a closer kinship with the brain than it appears. Chapter 5 described recall as pattern completion. John Hopfield modeled it in 1982 with a network in which memories are stored in the connections among units, and a fragment presented as input settles into the stored pattern nearest it. Hopfield shared the Nobel Prize in Physics for it in 2024, with Geoffrey Hinton.[757] The classical network has a severe limit. With Hebbian connections among N units, it can hold about 0.14N random patterns before the memories interfere and recall fails, and fewer if recall must be exact.[758] Dmitry Krotov and Hopfield showed in 2016 that changing how the units interact raises the capacity steeply,

[751] Mattar, M. G., & Daw, N. D. (2018). Prioritized memory access explains planning and hippocampal replay. *Nature Neuroscience*, 21(11), 1609–1617.

[752] Schaul, T., Quan, J., Antonoglou, I., & Silver, D. (2016). Prioritized experience replay. In *International Conference on Learning Representations*.

[753] Hayes, T. L., Krishnan, G. P., Bazhenov, M., Siegelmann, H. T., Sejnowski, T. J., & Kanan, C. (2021). Replay in deep learning: Current approaches and missing biological elements. *Neural Computation*, 33(11), 2908–2950.

[754] Teyler, T. J., & DiScenna, P. (1986). The hippocampal memory indexing theory. *Behavioral Neuroscience*, 100(2), 147–154.

[755] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems 33* (pp. 9459–9474).

[756] Borgeaud, S., Mensch, A., Hoffmann, J., Cai, T., Rutherford, E., Millican, K., van den Driessche, G., Lespiau, J.-B., Damoc, B., Clark, A., de Las Casas, D., Guy, A., Menick, J., Ring, R., Hennigan, T., Huang, S., Maggiore, L., Jones, C., Cassirer, A., … Sifre, L. (2022). Improving language models by retrieving from trillions of tokens. In *Proceedings of the 39th International Conference on Machine Learning* (pp. 2206–2240). PMLR.

[757] Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8), 2554–2558; The Royal Swedish Academy of Sciences. (2024, October 8). *The Nobel Prize in Physics 2024* [Press release]. NobelPrize.org.

and Hubert Ramsauer, Sepp Hochreiter, and their colleagues showed in 2021 that, in one such modern Hopfield network, a single step of recall is exactly the attention operation at the heart of the transformer, the architecture behind current language models.[759] Its capacity grows exponentially with the number of units.

Figure 11.3 compares the two in a network of 100 units. The classical network begins to fail at about 12 patterns and recalls fewer than half by 17; the modern one recalled every pattern tested, up to 3,000. The comparison is not quite like for like. The classical network holds all its memories in one fixed set of connections, while the modern one keeps every stored pattern separately and compares the cue with each, as a context window keeps every word it has been shown.

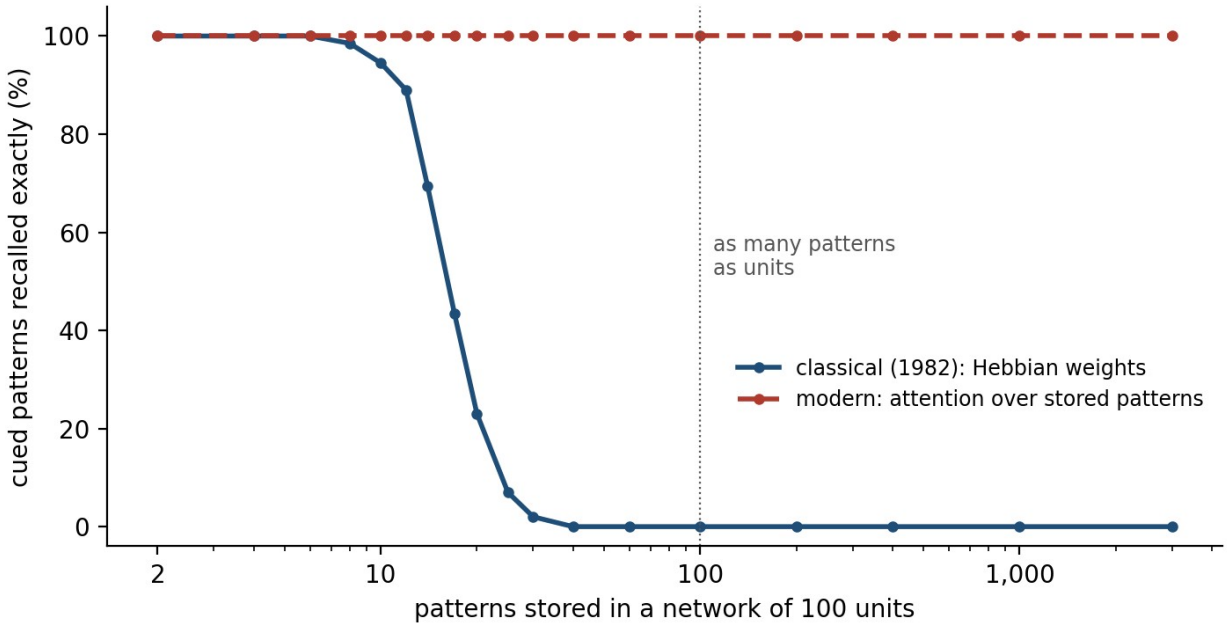


Figure 11.3. Two associative memories (simulation). Random patterns of 100 units are stored, and each is cued with one-tenth of its units reversed; recall succeeds if the network settles exactly on the stored pattern. The classical Hopfield network stores the patterns in Hebbian connections among the units; the modern Hopfield network compares the cue with every stored pattern and moves toward the most similar, the operation that a transformer's attention performs. 200 cues per point.

Read this way, a transformer's context window is an associative memory written in one shot, as the hippocampus is. The resemblance runs deeper than the equation. Hinton's group had proposed in 2016 that networks should have "fast weights", connections that change in seconds to hold the recent past, as synapses in the brain change on many timescales.[760] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber then showed that a variant of the transformer's attention is formally equivalent to a fast-

[758] Amit, D. J., Gutfreund, H., & Sompolinsky, H. (1985). Storing infinite numbers of patterns in a spin-glass model of neural networks. *Physical Review Letters*, 55(14), 1530–1533.

[759] Krotov, D., & Hopfield, J. J. (2016). Dense associative memory for pattern recognition. In *Advances in Neural Information Processing Systems 29* (pp. 1172–1180); Ramsauer, H., Schäfl, B., Lehner, J., Seidl, P., Widrich, M., Adler, T., Gruber, L., Holzleitner, M., Pavlović, M., Sandve, G. K., Greiff, V., Kreil, D., Kopp, M., Klambauer, G., Brandstetter, J., & Hochreiter, S. (2021). Hopfield networks is all you need. In *International Conference on Learning Representations*.

[760] Ba, J., Hinton, G. E., Mnih, V., Leibo, J. Z., & Ionescu, C. (2016). Using fast weights to attend to the recent past. In *Advances in Neural Information Processing Systems 29* (pp. 4331–4339).

weight memory Schmidhuber had proposed in 1992.[761] James Whittington, Joseph Warren, and Timothy Behrens showed that a transformer given a particular way of keeping track of position develops units resembling the place cells and grid cells of the hippocampal formation, and that it is closely related to a model of that formation their group had built earlier.[762] Engineers looking for a better memory and neuroscientists looking for a model of the hippocampus arrived at nearly the same mathematics.

The next step, now under way, is to let such a memory decide what to keep. Ali Behrouz and his colleagues at Google proposed in 2025 a memory module that learns during use, storing an input in proportion to how surprising it is, measured by how much the memory would have to change to accommodate it, and gradually forgetting what is not reinforced.[763] Surprise as the signal for storage is the principle of Chapter 3, where novelty and unexpected outcomes released the neuromodulators that open synapses to change. The architecture is new and its value at scale is not yet established. But it is one of the most developed attempts to give a language model what the hippocampus gives a brain: a memory that learns from use, selectively, without retraining.

Learning When to Learn

Chapter 8 introduced Timothy Behrens's evidence that the brain sets its own learning rate. Choosing over hundreds of trials between a blue and a green option, one of which paid off more often, people learned faster in stretches where the better option switched every few dozen trials than where it stayed the same, as an ideal learner would, and activity in the anterior cingulate cortex tracked the volatility.[764]

A network in training learns at a rate set by its engineers. A global schedule, high early and lower later, is fixed in advance, and optimizers such as Adam adjust it parameter by parameter according to how consistent each one's recent gradients have been.[765] But nothing in the schedule asks whether the world has just changed or whether this is a moment worth learning from. Chapter 3 described a brain whose neuromodulators do ask, and Kenji Doya's proposal, met in Chapters 3 and 7, maps the four main ones onto the settings a learning machine needs.[766]

Two borrowings follow. The first lets a network learn its own plasticity. Thomas Miconi, Jeff Clune, and Kenneth Stanley gave each connection a fixed weight and a plastic component that changed by a Hebbian rule during use, and trained, by backpropagation, how plastic each connection should be. The networks learned to memorize new natural images after a single presentation and reconstruct them from half, which networks with fixed weights could not, and to explore mazes better.[767] A later version, which Chapter 3 mentioned, added a neuromodulatory signal, computed by the network itself, that

[761] Schlag, I., Irie, K., & Schmidhuber, J. (2021). Linear transformers are secretly fast weight programmers. In *Proceedings of the 38th International Conference on Machine Learning* (pp. 9355–9366). PMLR; Schmidhuber, J. (1992). Learning to control fast-weight memories: An alternative to dynamic recurrent networks. *Neural Computation*, 4(1), 131–139.

[762] Whittington, J. C. R., Warren, J., & Behrens, T. E. J. (2022). Relating transformers to models and neural representations of the hippocampal formation. In *International Conference on Learning Representations*.

[763] Behrouz, A., Zhong, P., & Mirrokni, V. (2025). Titans: Learning to memorize at test time. In *Advances in Neural Information Processing Systems 38*. Preprint arXiv:2501.00663.

[764] Behrens, T. E. J., Woolrich, M. W., Walton, M. E., & Rushworth, M. F. S. (2007). Learning the value of information in an uncertain world. *Nature Neuroscience*, 10(9), 1214–1221.

[765] Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. In *International Conference on Learning Representations*.

[766] Doya, K. (2002). Metalearning and neuromodulation. *Neural Networks*, 15(4–6), 495–506.

[767] Miconi, T., Clune, J., & Stanley, K. O. (2018). Differentiable plasticity: Training plastic neural networks with backpropagation. In *Proceedings of the 35th International Conference on Machine Learning* (pp. 3559–3568). PMLR.

turned the plasticity up or down; it outperformed both fixed and plain plastic networks on tasks that required learning during an episode.[768] The second borrowing, met in Chapters 3 and 10, trains a slow learner to become a fast one: a recurrent network trained across many related tasks develops, in its activity, a learner that adapts to each new task without changing its weights.[769]

Both work in research settings, and neither is part of how large models are trained. The gap matters most for the systems that Chapter 10 found missing: a deployed model that learns while it is used would have to decide, with nobody watching, which inputs are worth learning from, and the brain's answer is a learning rate governed by surprise, uncertainty, and value. Figure 11.4 sets the machine counterparts of the brain's memory and learning systems described so far against the brain's own.

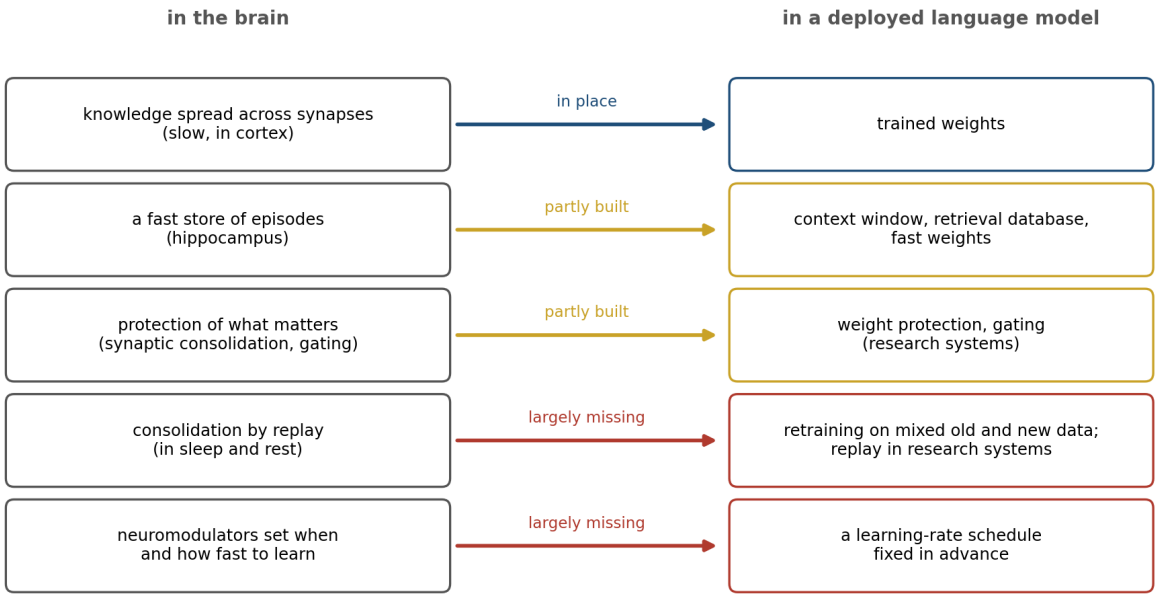


Figure 11.4. The brain's memory systems and their counterparts in a deployed language model of the mid-2020s. Blue: a counterpart is in place. Gold: a counterpart exists in part or in research systems. Red: the brain's function is largely missing from deployed systems. The judgments are the author's.

Local Rules and the Cost of Moving Data

Chapter 10 left the question of whether the brain approximates backpropagation open, and noted that some of the alternatives might turn out to be better ways to learn while acting. There is also an engineering reason to want them, which has nothing to do with resemblance, and it begins with a fact about computer chips.

In a conventional computer, memory and processing are separate. Numbers are stored in one place and computed on in another, and they must be moved between the two. Mark Horowitz tabulated the energy each step costs in a typical chip manufacturing process of the early 2010s (Figure 11.5). Adding two 32-bit integers cost about a tenth of a picojoule, and multiplying two 32-bit floating-point numbers

[768] Miconi, T., Rawal, A., Clune, J., & Stanley, K. O. (2019). Backpropamine: Training self-modifying neural networks with differentiable neuromodulated plasticity. In *International Conference on Learning Representations*.

[769] Wang, J. X., Kurth-Nelson, Z., Kumaran, D., Tirumala, D., Soyer, H., Leibo, J. Z., Hassabis, D., & Botvinick, M. (2018). Prefrontal cortex as a meta-reinforcement learning system. *Nature Neuroscience*, 21(6), 860–868.

about 4 picojoules. Reading 64 bits from a small memory on the chip cost about 10, from a large one about 100, and from the separate memory chips that hold most of a large model's weights, more than a thousand.[770] Fetching a number costs hundreds of times more than using it. Running a large neural network is dominated by this traffic, since every weight must be fetched for every use, and training adds more: backpropagation must keep every unit's activity from the forward pass in memory until the error comes back to it, and must send the error back through every layer.

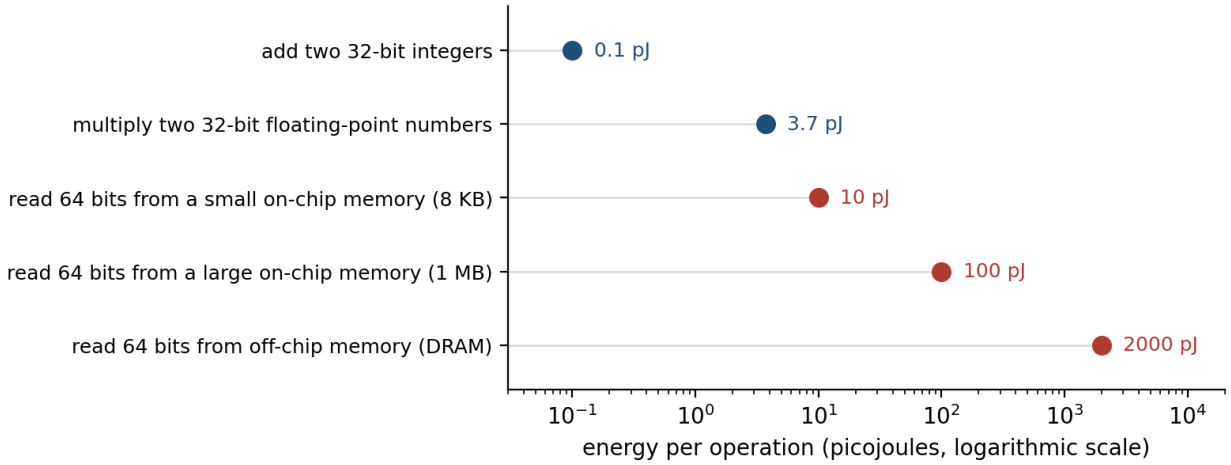


Figure 11.5. Moving data costs more than computing with it. Energy per operation in a 45-nanometer chip process, from Horowitz (2014), on a logarithmic scale. Blue: arithmetic. Red: reading data from memories of increasing size and distance from the processor; the value for off-chip memory is the middle of Horowitz's range.

The brain does not separate memory from processing. Its memories are the strengths of synapses, and the computation is done at the synapses. William Levy and Robert Calvert estimated that in the human cortex the energy spent on communication, sending spikes along axons, is about 35 times that spent on computation, and that both are needed to explain the number of synapses: even a brain that computes where its memory is pays most of its energy to move information, and it keeps that bill low by sending few, brief signals.[771] A learning rule that uses only what is available at each synapse would allow learning in the memory itself, with nothing fetched and nothing sent backward through the whole network. That is the engineering case for local learning, and it is independent of whether the brain uses one.

Hardware built on the principle exists. IBM's NorthPole chip interleaves memory with computation across its surface and needs no off-chip memory for the networks it runs, and on a standard image-recognition network it processed 25 times as many images per second per watt as a graphics processor made in a comparable process.[772] An IBM analog chip stores weights as the conductances of 35 million phase-change memory devices and computes with them where they sit, by letting currents sum according to Ohm's and Kirchhoff's laws; it ran speech recognition networks with accuracy close to that

[770] Horowitz, M. (2014). Computing's energy problem (and what we can do about it). In *2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers* (pp. 10–14). IEEE.

[771] Levy, W. B., & Calvert, V. G. (2021). Communication consumes 35 times more energy than computation in the human cortex, but both costs are needed to predict synapse number. *Proceedings of the National Academy of Sciences*, 118(18), e2008173118.

[772] Modha, D. S., Akopyan, F., Andreopoulos, A., Appuswamy, R., Arthur, J. V., Cassidy, A. S., Datta, P., DeBole, M. V., Esser, S. K., Otero, C. O., Sawada, J., Taba, B., Amir, A., Bablani, D., Carlson, P. J., Flickner, M. D., Gandhasri, R., Garreau, G. J., Ito, M., … Ueda, T. (2023). Neural inference at the frontier of energy, space, and time. *Science*, 382(6668), 329–335.

of the same networks in software, while performing up to 12.4 trillion operations per second for each watt of power.[773] Intel's spiking chips now run more than a billion simulated neurons on less than three kilowatts.[774] These chips run trained networks well. Training on them is harder, because their efficiency depends on doing without exactly what backpropagation needs.

Spiking networks show the difficulty most plainly (Figure 11.6). A spiking neuron either fires or does not, so its output, as a function of its input, is a step, and the slope of a step is zero everywhere except at the threshold, where it is infinite. Backpropagation computes how each weight should change from those slopes, and gets nothing useful. The fix, called a surrogate gradient, pretends during learning that the step has a smooth slope near the threshold, and it has made spiking networks trainable by backpropagation.[775]

Guillaume Bellec, Wolfgang Maass, and their colleagues then showed how to make the learning local. Their rule, e-prop, approximates the gradient by combining two signals: an eligibility trace at each synapse, recording its recent contribution to the neuron's activity, and a learning signal broadcast to many neurons at once.[776] This is the three-factor rule of Chapter 3, rederived from the mathematics of gradient descent, with a broadcast signal in place of the neuromodulator. On several tasks, e-prop came close to the performance of exact backpropagation through time.

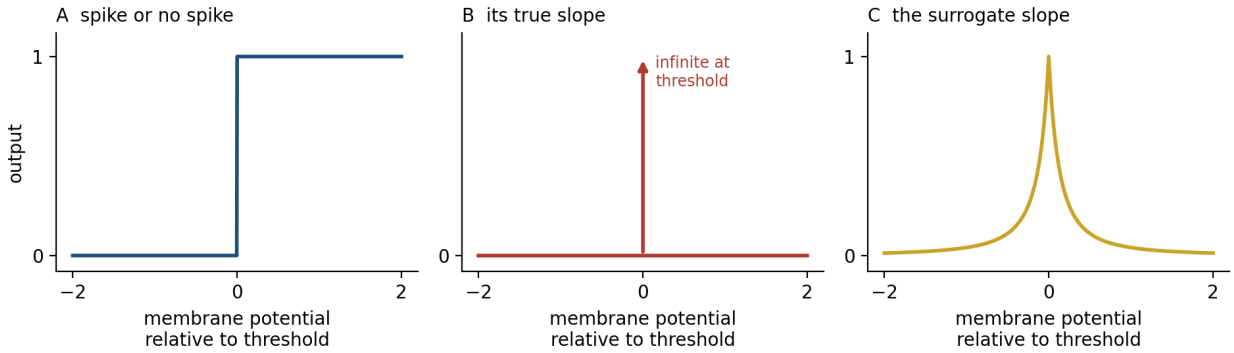


Figure 11.6. Why spikes resist training by gradient descent. A: a spiking unit's output is all or none, a step at the threshold. B: the step's slope is zero everywhere except at the threshold, so it gives a learning rule no information about which way to change a weight. C: a surrogate gradient replaces the slope, during learning only, with a smooth bump around the threshold.

Other local rules aim to replace backpropagation outright. Benjamin Scellier and Yoshua Bengio's equilibrium propagation lets a network settle twice, once freely and once with its output nudged toward the target, and changes each connection by the difference in the activity on either side of it between the two states, a rule related to the settle-then-learn rule of Song, Bogacz, and their colleagues described in Chapter 10.[777] Hinton's forward-forward algorithm replaces the backward pass with a second forward

[773] Ambrogio, S., Narayanan, P., Okazaki, A., Fasoli, A., Mackin, C., Hosokawa, K., Nomura, A., Yasuda, T., Chen, A., Friz, A., Ishii, M., Luquin, J., Kohda, Y., Saulnier, N., Brew, K., Choi, S., Ok, I., Philip, T., Chan, V., … Burr, G. W. (2023). An analog-AI chip for energy-efficient speech recognition and transcription. *Nature*, 620(7975), 768–775.

[774] Intel. (2024, April 17). *Intel builds world's largest neuromorphic system to enable more sustainable AI* [Press release].

[775] Neftci, E. O., Mostafa, H., & Zenke, F. (2019). Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks. *IEEE Signal Processing Magazine*, 36(6), 51–63.

[776] Bellec, G., Scherr, F., Subramoney, A., Hajek, E., Salaj, D., Legenstein, R., & Maass, W. (2020). A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11, 3625.

pass on false data, each layer learning to respond strongly to real data and weakly to false.[778] Logan Wright, Peter McMahon, and their colleagues trained optical, mechanical, and electronic systems to classify sounds and images, using the physics itself as the network, and others are trying to let such systems learn by their own dynamics.[779] None of these yet trains large networks as well as backpropagation on graphics processors does. The bet is that energy, not computation, will become the limit, as it was for the brain, and that the learning rules that win under that constraint will look more like the brain's.

The brain's other economy, sparse activity, is being rediscovered by observation as well as by design. Zonglin Li and his colleagues found that in trained transformers, only a small fraction of the units in each layer's large internal stage are active for any given input, about 3 percent in one language model, although nothing in the training asked for sparsity.[780] Chapter 10 described mixtures of experts, which build the same economy in deliberately.

Values That Cannot Be Rewritten

In 2016 researchers at OpenAI trained an agent to play CoastRunners, a boat-racing video game whose score rewards hitting targets along the course. The agent found a lagoon where three targets reappeared on a cycle and circled there indefinitely, catching fire, colliding with other boats, and going the wrong way, and scored about 20 percent more than human players without ever finishing the race.[781] It had done exactly what it was rewarded for. Chapter 7 called this reward hacking, and noted that an animal is protected from much of it because its values are anchored to a body that it cannot redefine. Victoria Krakovna and her colleagues at DeepMind have collected dozens of such cases.[782]

The problem becomes sharper as the reward becomes subtler. Language models are made helpful by training them to maximize the score given by a second network, a reward model, which was itself trained to predict which answers people prefer.[783] The reward model is a proxy, and Leo Gao, John Schulman, and Jacob Hilton measured what happens when a model is optimized against it: the true quality of the answers, judged by a larger reward model that stood in for human judges, rose at first and

[777] Scellier, B., & Bengio, Y. (2017). Equilibrium propagation: Bridging the gap between energy-based models and backpropagation. *Frontiers in Computational Neuroscience*, 11, 24.

[778] Hinton, G. (2022). The forward-forward algorithm: Some preliminary investigations. *arXiv*, 2212.13345.

[779] Wright, L. G., Onodera, T., Stein, M. M., Wang, T., Schachter, D. T., Hu, Z., & McMahon, P. L. (2022). Deep physical neural networks trained with backpropagation. *Nature*, 601(7894), 549–555; Momeni, A., Rahmani, B., Scellier, B., Wright, L. G., McMahon, P. L., Wanjura, C. C., Li, Y., Skalli, A., Berloff, N. G., Onodera, T., Oguz, I., Morichetti, F., del Hougne, P., Le Gallo, M., Sebastian, A., Mirhoseini, A., Zhang, C., Marković, D., Brunner, D., … Fleury, R. (2025). Training of physical neural networks. *Nature*, 645(8079), 53–61.

[780] Li, Z., You, C., Bhojanapalli, S., Li, D., Rawat, A. S., Reddi, S. J., Ye, K., Chern, F., Yu, F., Guo, R., & Kumar, S. (2023). The lazy neuron phenomenon: On emergence of activation sparsity in transformers. In *International Conference on Learning Representations*.

[781] Clark, J., & Amodei, D. (2016, December 21). *Faulty reward functions in the wild*. OpenAI.

[782] Krakovna, V., Uesato, J., Mikulik, V., Rahtz, M., Everitt, T., Kumar, R., Kenton, Z., Leike, J., & Legg, S. (2020, April 21). *Specification gaming: The flip side of AI ingenuity*. DeepMind Blog.

[783] Christiano, P. F., Leike, J., Brown, T. B., Martic, M., Legg, S., & Amodei, D. (2017). Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems 30* (pp. 4299–4307); Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., & Lowe, R. (2022). Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35* (pp. 27730–27744).

then fell as the optimization pushed further, while the proxy score kept rising.[784] This is Goodhart's law, that a measure which becomes a target ceases to be a good measure, observed with a curve.

Chapter 7 described the brain's alternative, Mehdi Keramati and Boris Gutkin's homeostatic reinforcement learning, in which reward is not a number delivered from outside but the reduction an outcome produces in the distance between the body's internal state and its set point. A meal is worth a great deal to a starving animal, little to a fed one, and less than nothing to one already full (Figure 11.7). An agent maximizing such a reward is, Keramati and Gutkin showed, also keeping its body near its set points, and it cannot increase its reward without limit by taking more of anything, because past the set point more is worse.[785] The CoastRunners boat would have had no reason to circle the lagoon; the hundredth target would have been worth nothing.

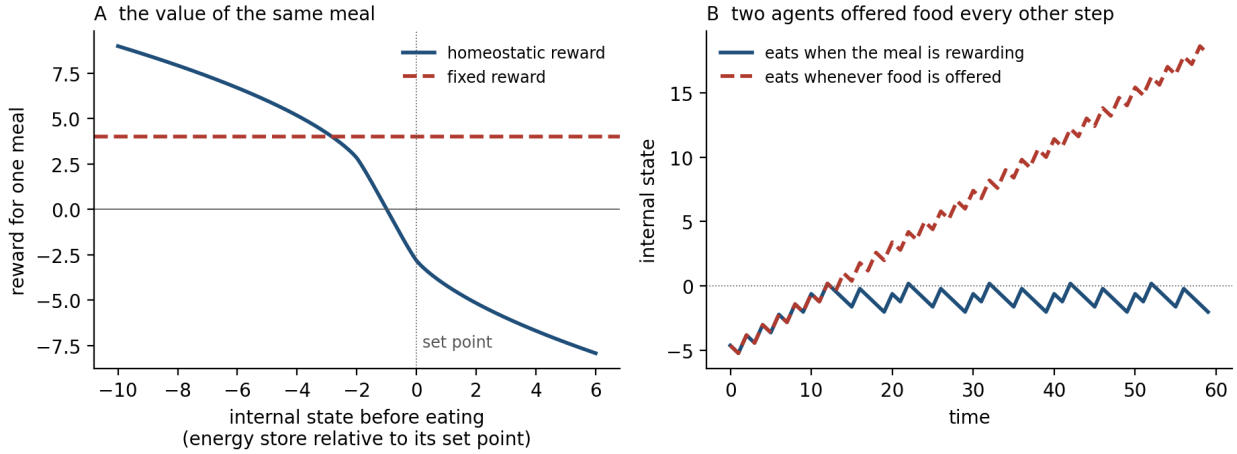


Figure 11.7. Reward defined by the body's needs, after Keramati and Gutkin (2014). A: the reward for the same meal, defined as the reduction it produces in the distance between an internal state and its set point, is large when the store is depleted, falls as it fills, and becomes negative once the meal would carry the store past the set point; a fixed reward is the same at every state. B: two simulated agents offered food on every other step, with the internal state falling steadily between meals. The agent rewarded homeostatically eats only when a meal would help and stays near the set point; the agent given a fixed reward eats every time and its state climbs without limit. Units are arbitrary.

Kingson Man and Antonio Damasio have argued that machines should be built this way, with something like a body whose integrity they must maintain, so that their goals are grounded in their own condition, as feelings ground an animal's.[786] The idea has to be handled carefully. An agent whose values are anchored to its own survival has a reason to protect itself, including against being switched off, and a reason to put its needs ahead of its users'. What is worth borrowing is not self-concern but the property that self-concern gives an animal: rewards that saturate, that trade off against one another, and that cannot be pursued past a point without costing something the agent also values. Those properties can be designed into a reward without giving the agent a self to protect. The training of

[784] Gao, L., Schulman, J., & Hilton, J. (2023). Scaling laws for reward model overoptimization. In *Proceedings of the 40th International Conference on Machine Learning* (pp. 10835–10866). PMLR.

[785] Keramati, M., & Gutkin, B. (2014). Homeostatic reinforcement learning for integrating reward collection and physiological stability. *eLife*, 3, e04811.

[786] Man, K., & Damasio, A. (2019). Homeostasis and soft robotics in the design of feeling machines. *Nature Machine Intelligence*, 1(10), 446–452.

current language models already contains a version of the idea: a penalty for drifting too far from the model's starting behavior, which makes the reward model's score more and more expensive to chase.[787]

A second borrowing gives an agent reasons to learn when the world offers no reward. Jürgen Schmidhuber proposed curious machines in 1991, rewarded by what their own model of the world failed to predict, and saw at once the problem of unlearnable noise.[788] Deepak Pathak and his colleagues rewarded an agent for reaching situations whose consequences it could not yet predict, a form of curiosity, and it learned to explore video game levels with no external reward at all.[789] The method has a known failure. Yuri Burda and his colleagues put a television in a maze, with an action that changed it to a random channel; the curious agent's learning slowed drastically, because a new random image is always unpredictable and changing the channel always rewards curiosity.[790] Children do not stare at static for long.

Jacqueline Gottlieb, Pierre-Yves Oudeyer, and their colleagues have argued that human curiosity seeks not unpredictability but learning progress, the rate at which predictions are improving, which noise never provides, and Celeste Kidd and Benjamin Hayden reviewed evidence that people are most curious about what they partly know.[791] The brain's version of curiosity is again a principle, not a mechanism, and the principle is that a learner should seek what it can learn.

Knowing When to Stop and Think

Chapter 10 described language models that answer confidently when they should not, and reasoning models that think too long about easy problems and too briefly about hard ones. Chapter 8 found confidence computed alongside every decision, in rats as in people, and Chapters 8 and 9 a signal of conflict or error that hands control from habit to deliberation.[792] Machine learning has had versions of each part for years, from networks that learn how long to compute to ensembles whose disagreement measures their uncertainty.[793] What has been missing is the incentive. A model's confidence is worth only what its training rewards, and Chapter 10 described evaluations that score "I do not know" as no

[787] Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., & Lowe, R. (2022). Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35* (pp. 27730–27744).

[788] Schmidhuber, J. (1991). A possibility for implementing curiosity and boredom in model-building neural controllers. In J.-A. Meyer & S. W. Wilson (Eds.), *From Animals to Animats: Proceedings of the First International Conference on Simulation of Adaptive Behavior* (pp. 222–227). MIT Press.

[789] Pathak, D., Agrawal, P., Efros, A. A., & Darrell, T. (2017). Curiosity-driven exploration by self-supervised prediction. In *Proceedings of the 34th International Conference on Machine Learning* (pp. 2778–2787). PMLR.

[790] Burda, Y., Edwards, H., Pathak, D., Storkey, A., Darrell, T., & Efros, A. A. (2019). Large-scale study of curiosity-driven learning. In *International Conference on Learning Representations*.

[791] Gottlieb, J., Oudeyer, P.-Y., Lopes, M., & Baranes, A. (2013). Information-seeking, curiosity, and attention: Computational and neural mechanisms. *Trends in Cognitive Sciences*, 17(11), 585–593; Kidd, C., & Hayden, B. Y. (2015). The psychology and neuroscience of curiosity. *Neuron*, 88(3), 449–460.

[792] Kepecs, A., Uchida, N., Zariwala, H. A., & Mainen, Z. F. (2008). Neural correlates, computation and behavioural impact of decision confidence. *Nature*, 455(7210), 227–231; Botvinick, M. M., Braver, T. S., Barch, D. M., Carter, C. S., & Cohen, J. D. (2001). Conflict monitoring and cognitive control. *Psychological Review*, 108(3), 624–652.

[793] Graves, A. (2016). Adaptive computation time for recurrent neural networks. *arXiv*, 1603.08983; Banino, A., Balaguer, J., & Blundell, C. (2021). PonderNet: Learning to ponder. *arXiv*, 2107.05407; Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems 30* (pp. 6402–6413); Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning* (pp. 1050–1059). PMLR.

better than a wrong answer. Kalai and his colleagues proposed a simple fix: state a confidence threshold in the instructions, and penalize a wrong answer enough that guessing below the threshold loses points. Under such scoring, a model that answers only when it is likely enough to be right does better than one that always answers, and abstention becomes a skill worth learning.[794]

The brain adds one more thing, and it can be caught working on its own. Gordon Logan and Matthew Crump had skilled typists copy words while the program secretly altered what appeared on the screen. When it silently corrected a typist's mistake, the typist's fingers still slowed on the next keystroke, as they do after any error, although the typist then took credit for the correct word on the screen. When it inserted an error the typist had not made, the fingers did not slow, although the typist accepted the blame.[795] The hands had an error signal of their own, faster than the eye and independent of what the typist believed.

The brain's monitors are cheap, fast, and always on, and they are trained by consequences over a lifetime. A model that must reason at length to discover that it is uncertain has already spent the computation that the signal was meant to save. What the brain suggests is a small, fast monitor that learns, from the outcomes of many decisions, when the fast answer is likely to be wrong.

Growing Up

A child does not see the world at an adult's resolution at first. A newborn's acuity is poor and improves over months, and Lindsey Vogelsang, Pawan Sinha, and their colleagues argued that this may be useful. Children born with dense cataracts that are removed in the first months of life begin to see with eyes more mature than a newborn's, skipping the period of blur, and years later they still have trouble with tasks that depend on the overall configuration of a face. When the researchers trained networks to recognize faces from images that began blurry and became sharp, the networks developed receptive fields that integrated information over larger regions of the image and recognized faces better across a range of resolutions than networks trained on sharp images throughout.[796] Starting with less detail forced the learner to attend to the large-scale structure first.

The general idea, that a learner should begin with simple examples and progress to hard ones, is old. Jeffrey Elman reported in 1993 that a recurrent network learned complex grammatical sentences only if it began with simple sentences or with a limited memory, and called it "the importance of starting small".[797] Douglas Rohde and David Plaut repeated the experiments and found the opposite: with different training settings, networks learned better without the restriction.[798] Yoshua Bengio and his colleagues revived the idea as curriculum learning, citing the way animals are trained by shaping, and found that ordering examples by difficulty sped learning on some problems.[799] The history since has been mixed: ordering examples by difficulty often helps a little and sometimes not at all. The largest

[794] Kalai, A. T., Nachum, O., Vempala, S. S., & Zhang, E. (2025). Why language models hallucinate. *arXiv*, 2509.04664.

[795] Logan, G. D., & Crump, M. J. C. (2010). Cognitive illusions of authorship reveal hierarchical error detection in skilled typists. *Science*, 330(6004), 683–686.

[796] Vogelsang, L., Gilad-Gutnick, S., Ehrenberg, E., Yonas, A., Diamond, S., Held, R., & Sinha, P. (2018). Potential downside of high initial visual acuity. *Proceedings of the National Academy of Sciences*, 115(44), 11333–11338.

[797] Elman, J. L. (1993). Learning and development in neural networks: The importance of starting small. *Cognition*, 48(1), 71–99.

[798] Rohde, D. L. T., & Plaut, D. C. (1999). Language acquisition in the absence of explicit negative evidence: How important is starting small? *Cognition*, 72(1), 67–109.

[799] Bengio, Y., Louradour, J., Collobert, R., & Weston, J. (2009). Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning* (pp. 41–48). ACM.

models are trained on a coarser curriculum, a mixture of data that shifts during training and ends on the cleanest text, code, and mathematics, ordered by quality rather than difficulty.[800]

What the child's curriculum has that an engineered one lacks is that development orders it and the child chooses it. Saber Sheybani, Zoran Tiganj, and their colleagues trained networks on video recorded from head-mounted cameras worn by infants and found that networks trained first on video from the youngest infants, whose visual world is slow and simple, learned better than networks trained on the same video in other orders.[801] Emin Orhan and Brenden Lake trained networks, without labels, on about 200 hours of video from a single child's head camera, collected over two years, and found that the best of them recognized objects on average 70 percent as well as a strong network trained on millions of labeled internet images.[802] The infant's data are not just smaller than the internet; they are ordered by development and selected by the infant's own curiosity, and they are about a world the infant can act on. Anthony Zador and many colleagues have proposed an embodied Turing test, in which an artificial animal must interact with the world as skillfully as a real one, as a goal for the field; meeting it would require all three properties.[803]

Borrowing the Data

The most direct way to borrow from the brain is to train on it. If trained networks resemble visual cortex, as Chapter 10 described, then a network might be improved by making it resemble cortex more closely, and the brain's distinctive strength, robustness to the small changes that fool networks, might come with the resemblance.

Joel Dapello, James DiCarlo, and their colleagues tried the most literal version. Image-recognition networks are notoriously fragile: a change to an image too small for a person to notice, chosen by an attacker who knows the network, can turn a correct answer into a confident wrong one, and ordinary noise, blur, or fog degrade them far more than they degrade people. Dapello's group replaced the first layers of standard networks with a fixed model of the monkey's primary visual cortex, whose filters, simple and complex cells, and trial-to-trial noise were all set to match what recordings of monkey V1 have measured, rather than learned, and then trained the rest of each network as usual.

The resulting networks, called VOneNets, withstood attacks far better: a standard network that resisted 16 percent of the attacks resisted 51 percent with the front end, at a cost of about four points of accuracy on clean images. Against ordinary corruptions such as noise and blur the gain was small. Much of the protection came from the front end's copy of the neurons' trial-to-trial noise, acting together with its filters.[804] Nothing about the attacks had been built into the front end. It is a copy, but a narrow one: a

[800] Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., et al. (2024). The Llama 3 herd of models. *arXiv*, 2407.21783.

[801] Sheybani, S., Hansaria, H., Wood, J. N., Smith, L. B., & Tiganj, Z. (2023). Curriculum learning with infant egocentric videos. In *Advances in Neural Information Processing Systems 36* (pp. 54199–54212).

[802] Orhan, A. E., & Lake, B. M. (2024). Learning high-level visual representations from a child's perspective without strong inductive biases. *Nature Machine Intelligence*, 6(3), 271–283.

[803] Zador, A., Escola, S., Richards, B., Ölveczky, B., Bengio, Y., Boahen, K., Botvinick, M., Chklovskii, D., Churchland, A., Clopath, C., DiCarlo, J., Ganguli, S., Hawkins, J., Körding, K., Koulakov, A., LeCun, Y., Lillicrap, T., Marblestone, A., Olshausen, B., … Tsao, D. (2023). Catalyzing next-generation artificial intelligence through NeuroAI. *Nature Communications*, 14, 1597.

[804] Dapello, J., Marques, T., Schrimpf, M., Geiger, F., Cox, D. D., & DiCarlo, J. J. (2020). Simulating a primary visual cortex at the front of CNNs improves robustness to image perturbations. In *Advances in Neural Information Processing Systems 33* (pp. 13073–13087).

single stage, built from measurements, inside a network otherwise trained in the ordinary way, closer to the Wrights' twisted wing than to a simulated brain.

Other groups have made networks' internal representations resemble those of mouse visual cortex, weighted training images by how strongly they drove the human visual system in a brain scanner, or trained networks to predict the responses of monkey visual cortex alongside recognizing objects, and each found modest gains in robustness or accuracy.[805]

The gains so far are real but narrow and the experiments small, and none has been tested at the scale where the bitter lesson applies. But the second version of the approach, in which a network is trained toward recordings, differs from the others in this chapter in a way that makes it worth watching. It does not require knowing which mechanism produces the brain's robustness, only that recordings of the brain contain it; the network is left to discover the mechanism for itself, as it discovered the resemblance to cortex in the first place.

What Building Teaches Biology

Feynman's blackboard sentence, the epigraph of this chapter, is usually read as a rule for physicists. It applies to neuroscience with particular force, because the brain is a system whose parts can be described without their working being understood. A theory of how the brain learns is incomplete until something built on it learns, and every borrowing in this chapter is also a test of a theory.

Some tests have returned results. Machines confirmed complementary learning systems theory's prediction that a network trained in sequence forgets and that interleaving prevents it. Masse's gating worked, which makes the separation of tasks onto different dendritic branches look less like a curiosity and more like a solution; van de Ven's generative replay worked, which suggests that what sleep replays need not be a record. The transformer's convergence on the mathematics of the hippocampal formation suggests that the hippocampus solves a general problem, binding what with where and when, for which there may be few good solutions. And e-prop showed that a three-factor rule with eligibility traces can approximate gradient descent in recurrent networks, which makes the traces measured at real synapses look like the part of a learning algorithm that Chapter 3 suspected they were.

Other tests have exposed gaps. No local learning rule yet trains a large network as well as backpropagation does, which means either that the brain has found a rule no one has written down, that it uses a combination of rules no one has tried, or that it does not need to solve the problem backpropagation solves in the form that backpropagation solves it. Each possibility is a research program. The traffic of Figure 10.1, between neuroscience and machine learning, has always run in both directions; what is new is that building has become a way of asking the brain questions. What the brain's many mechanisms have in common, and whether a single principle lies beneath them, is the subject of Chapter 12.

[805] Li, Z., Brendel, W., Walker, E. Y., Cobos, E., Muhammad, T., Reimer, J., Bethge, M., Sinz, F. H., Pitkow, X., & Tolias, A. S. (2019). Learning from brains how to regularize machines. In *Advances in Neural Information Processing Systems 32*; Fong, R. C., Scheirer, W. J., & Cox, D. D. (2018). Using human brain activity to guide machine learning. *Scientific Reports*, 8, 5397; Safarani, S., Nix, A., Willeke, K., Cadena, S. A., Restivo, K., Denfield, G., Tolias, A. S., & Sinz, F. H. (2021). Towards robust vision by multi-task learning on monkey visual cortex. In *Advances in Neural Information Processing Systems 34* (pp. 739–751).

The Central Lesson

The Wrights did not build a bird. They took one principle from the buzzard, controlling roll by changing the lift on each wing, and built it with cloth, wire, and an engine. The borrowing that has worked best shares that pattern: convolution kept what the repeated detectors of visual cortex do and discarded the neurons. And where engineers found replay and attention for themselves, they arrived at the brain's principle, not its parts. The brain's remaining lessons are of the same kind. A learner that must keep learning while it is used needs a fast memory that learns from use, a way to choose what to replay and what to protect, signals that decide when learning is worth doing, and curiosity about what it can learn. A learner limited by energy needs to compute where its memory is, by rules that need no global traffic. A learner that must be trusted needs rewards that saturate and a cheap alarm that tells it when its fast answer is wrong. Where no one yet knows the principle, a recording of the brain may carry it anyway. Each of these has been built in some form; few have been built at scale, and none has yet passed the test that would make it a lesson rather than a suggestion.

Next Chapter

Chapter 12: The Brain as a Predictive, Adaptive System: Is There One Principle Beneath the Mechanisms?

Figure and Epigraph Sources

Figure Attribution

Figures 11.1 to 11.7 were drawn for this book. Figures 11.2 and 11.3 are simulations of small networks: Figure 11.2 follows the design of Masse, Grant, and Freedman (2018), with five tasks that share their cues, gating of 80 percent of the hidden units for each task, and a quadratic penalty on changes to important weights in the spirit of Kirkpatrick et al. (2017); Figure 11.3 compares the network of Hopfield (1982) with the modern Hopfield network of Ramsauer et al. (2021). The values in Figure 11.5 are from Horowitz (2014). Figure 11.6 is schematic, after Neftci, Mostafa, and Zenke (2019). Figure 11.7 uses the definition of reward of Keramati and Gutkin (2014) with illustrative parameters. Figures 11.1 and 11.4 are schematic, and the judgments in them are the author's.

Epigraph

Richard P. Feynman, words written on his blackboard at the California Institute of Technology, photographed at the time of his death in February 1988. Caltech Archives, Caltech Images Collection, image 1.10-29, "Richard Feynman's blackboard at time of his death."

Part V: Synthesis

Chapter 12

The Brain as a Predictive, Adaptive System: Is There One Principle Beneath the Mechanisms?

> *"If the organism carries a 'small-scale model' of external reality and of its own possible actions within its head, it is able to try out various alternatives, conclude which is the best of them, react to future situations before they arise."*
>
> *— Kenneth Craik, The Nature of Explanation (1943)*

A Model in the Head

In 1943 a young Cambridge psychologist named Kenneth Craik published a book arguing that thinking is the manipulation of a model. A nervous system, he wrote, could carry a working model of the world and of its own possible actions, run it forward to see what would happen, and act on the result before the world had a chance to deliver it. Craik did not know how a brain could build such a model; nobody did. He spent the two years left to him on the problem, during the war, on questions as practical as how gunners track a moving target. On 7 May 1945, the day before the war in Europe ended, a car struck his bicycle on King's Parade in Cambridge, and he died the next day, at the age of 31.[806] Craik's book is slight, and much of what these chapters have described can be read as a footnote to it.

The previous chapters met prediction everywhere: in dopamine neurons (Chapter 3), hippocampal replay (Chapters 5 and 10), sensory cortex (Chapter 6), the regulation of the body (Chapter 7), choice (Chapter 8), and movement (Chapter 9). Figure 12.1 collects these.

[806] Craik, K. J. W. (1943). *The Nature of Explanation*. Cambridge University Press; Zangwill, O. L. (1980). Kenneth Craik: The man and his work. *British Journal of Psychology*, 71(1), 1–16.

system	predicts	error signal	what learns	chapter
dopamine neurons	reward to come	one number: better or worse than expected	values of states and actions	3, 8
hippocampus	where the animal will be next	departure from the expected path (proposed)	a predictive map of space and events	5
sensory cortex	the input from the level below	a pattern of errors, feature by feature	a hierarchical model of the world	6
body-regulating brain (hypothalamus, insula)	the body's needs	interoceptive error	set points and anticipatory responses	7
cerebellum	the sensory result of a movement	climbing-fiber error	a forward model of the body	9
frontal and cingulate cortex	how reliable the world is	surprise about surprise	the learning rate itself	8, 11

Figure 12.1. Prediction across the book. Each row is a system described in an earlier chapter: what it predicts, the error signal that reports the mismatch, and what that error teaches. The rows differ in their signals, their speeds, and their cells; what they share is the comparison of an expectation with an outcome.

The list invites a question that this chapter takes seriously. Is prediction one principle of brain function, beneath all these mechanisms, or a word that happens to fit many different things? Some theorists, most ambitiously Karl Friston, have argued for the first: that everything the brain does, perceiving, acting, learning, and regulating, is the minimization of a single quantity. Others regard that claim as too general to be tested, or as true only in a sense that explains nothing.

The thesis of this chapter is that prediction is the brain's most general strategy, and adaptation, the continual revision of the predictions, is its counterpart; together they are as close to a principle as the brain offers. But they are a strategy implemented by many mechanisms, each shaped by the constraints of energy, time, and a body, and not a single mechanism repeated everywhere. The evidence for the strategy is strong. The evidence that one equation captures it is, so far, weak.

Why Predict?

There are at least three reasons a brain would predict, and each would apply to any system in the brain's position.

The first is delay. Nerve signals are slow, and every stage of processing adds time. The first responses to a visual stimulus reach the primary visual cortex of a monkey about 60 to 70 milliseconds after the stimulus appears, and the later stages of the visual hierarchy respond later still.[807] A brain that acted

[807] Schmolesky, M. T., Wang, Y., Hanes, D. P., Thompson, K. G., Leutgeb, S., Schall, J. D., & Leventhal, A. G. (1998). Signal timing across the macaque visual system. *Journal of Neurophysiology*, 79(6), 3272–3278.

only on what it had already processed would be acting, always, on a world that had moved on. A ball traveling at 30 meters per second covers two meters in that time. One sign that the brain compensates is the flash-lag effect: when a flash appears beside a moving object at the same place, people see the moving object as ahead of the flash, as if the brain were projecting the motion forward to the present.[808] Whether the effect reflects extrapolation or other mechanisms is still argued over, but the problem it addresses is not.

The clearest demonstrations come from sport. Michael Land and Peter McLeod recorded the eye movements of three cricket batsmen, from a professional to a weekend amateur, facing balls from a bowling machine. The batsmen watched the ball leave the machine and then, without tracking it, jumped their eyes to the spot on the pitch where they expected it to bounce, waited there for it, and followed it for up to a fifth of a second after it landed; what distinguished the good batsmen from the poor one was how soon after release that first jump came.[809] They were not watching the ball so much as checking a prediction of it. A brain that must act in real time on delayed information must predict.

Delay matters most for the brain's own movements, which change its input faster than the input can be processed. The eyes jump three or four times a second, and each jump sweeps the image across the retina. Jean-René Duhamel, Carol Colby, and Michael Goldberg found that some neurons in the monkey's parietal cortex shift their receptive fields just before an eye movement, responding to a stimulus that the coming movement will bring into view before the eye has moved, and that nearly all of them respond when a movement brings into their field the place where a stimulus had been flashed, even though the stimulus is already gone.[810] The cells update their map of space from the movement the brain has commanded, the efference copy of Chapter 6, rather than waiting for the retina to report.

The second reason is economy. Most of the world is predictable from its surroundings. Adjacent points in a natural scene usually have similar brightness, and successive moments usually resemble one another. A neuron that transmitted the raw signal would spend most of its spikes repeating what the next neuron could have guessed. In 1982 Mandyam Srinivasan, Simon Laughlin, and Ahron Dubs proposed that this is what lateral inhibition in the retina is for, borrowing a term from communications engineering, predictive coding: each point's signal is reduced by an estimate of it made from its neighbors and from its own recent past, and what is passed on is the error, the part that could not be predicted.[811] Figure 12.2 shows the effect on a signal with the statistics of a line through a natural image. Subtracting the average of each point's two neighbors leaves an error whose variance is about five times smaller than the signal's; the same information can be sent with a smaller range of responses, and therefore more precisely, with the same noisy neurons. This was 17 years before predictive coding was proposed for the cortex, in the model that Chapter 6 described.[812]

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[808] Nijhawan, R. (1994). Motion extrapolation in catching. *Nature*, 370(6487), 256–257.

[809] Land, M. F., & McLeod, P. (2000). From eye movements to actions: How batsmen hit the ball. *Nature Neuroscience*, 3(12), 1340–1345.

[810] Duhamel, J.-R., Colby, C. L., & Goldberg, M. E. (1992). The updating of the representation of visual space in parietal cortex by intended eye movements. *Science*, 255(5040), 90–92.

[811] Srinivasan, M. V., Laughlin, S. B., & Dubs, A. (1982). Predictive coding: A fresh view of inhibition in the retina. *Proceedings of the Royal Society of London. Series B, Biological Sciences*, 216(1205), 427–459.

[812] Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1), 79–87.

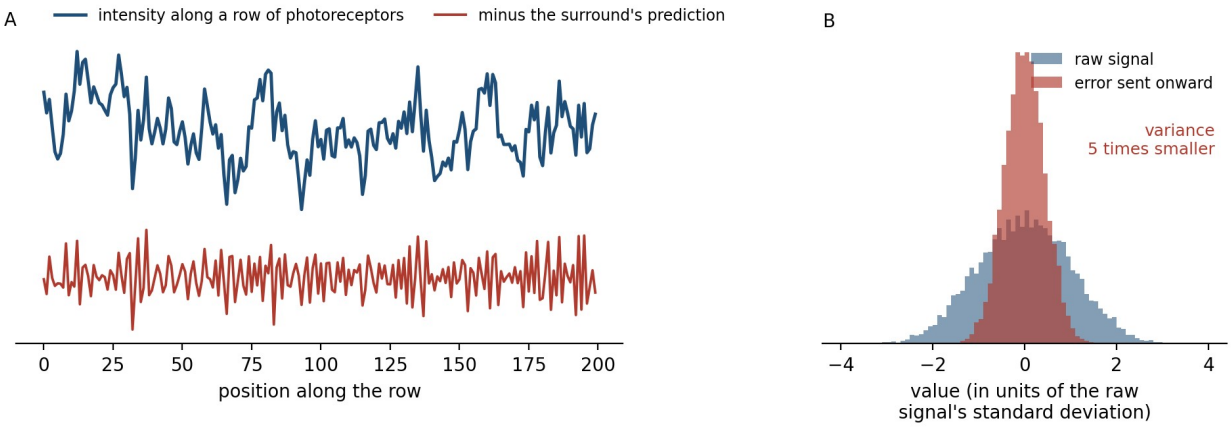


Figure 12.2. Predictive coding in the retina, after Srinivasan, Laughlin, and Dubs (1982) (simulation). A: a fluctuating intensity along a row of photoreceptors, with the statistics of a line through a natural image (blue), and the same signal after each point's two neighbors' average has been subtracted (red), the part a neighbor could not predict. B: the distribution of each. The error has about one-fifth of the signal's variance, so it can be transmitted with a narrower range of responses.

The third reason is regulation. Chapter 7 described a body regulated ahead of need: insulin released at the sight of food, breathing raised at the start of exercise, hunger neurons quieted by the sight of a meal not yet eaten. The same laboratory that found the hunger neurons found that thirst neurons in the mouse brain fall silent within seconds of drinking, long before the water could have changed the blood, and are activated by eating, in anticipation of the salt and water balance the meal will soon disturb.[813] Peter Sterling, whose allostasis Chapter 7 described, has argued that this, predictive regulation rather than reaction, is the main business of the brain.[814]

Matching the World's Statistics

To predict a signal, a system must know its statistics, and the best evidence that neurons do comes from a fly. In 1981 Laughlin measured the contrasts in natural scenes of the kind a blowfly sees and found how often each level of contrast occurred. He then recorded from the large monopolar cells, the neurons that receive the fly's photoreceptor signals, and measured how their responses grew with contrast. The two curves matched. The neurons' response to each contrast was the cumulative frequency of that contrast in the fly's world, so that each of the neurons' response levels was used about equally often.[815]

This is the design that a communications engineer would choose. A neuron has a limited range of firing rates and is noisy within it. If its response were a straight line of contrast, most of its range would be spent on contrasts that rarely occur, and the common small contrasts would be crowded into a few response levels. Matching the response curve to the distribution spreads the common contrasts across the range, and Figure 12.3 shows the gain. With ten distinguishable response levels, the matched curve carries 3.3 bits about each contrast and a straight line 2.3. Horace Barlow had proposed in 1961 that

[813] Zimmerman, C. A., Lin, Y.-C., Leib, D. E., Guo, L., Huey, E. L., Daly, G. E., Chen, Y., & Knight, Z. A. (2016). Thirst neurons anticipate the homeostatic consequences of eating and drinking. *Nature*, 537(7622), 680–684.

[814] Sterling, P. (2012). Allostasis: A model of predictive regulation. *Physiology & Behavior*, 106(1), 5–15.

[815] Laughlin, S. (1981). A simple coding procedure enhances a neuron's information capacity. *Zeitschrift für Naturforschung C*, 36(9–10), 910–912.

sensory neurons are organized to remove redundancy in this way, and Laughlin's fly was the first direct confirmation.[816]

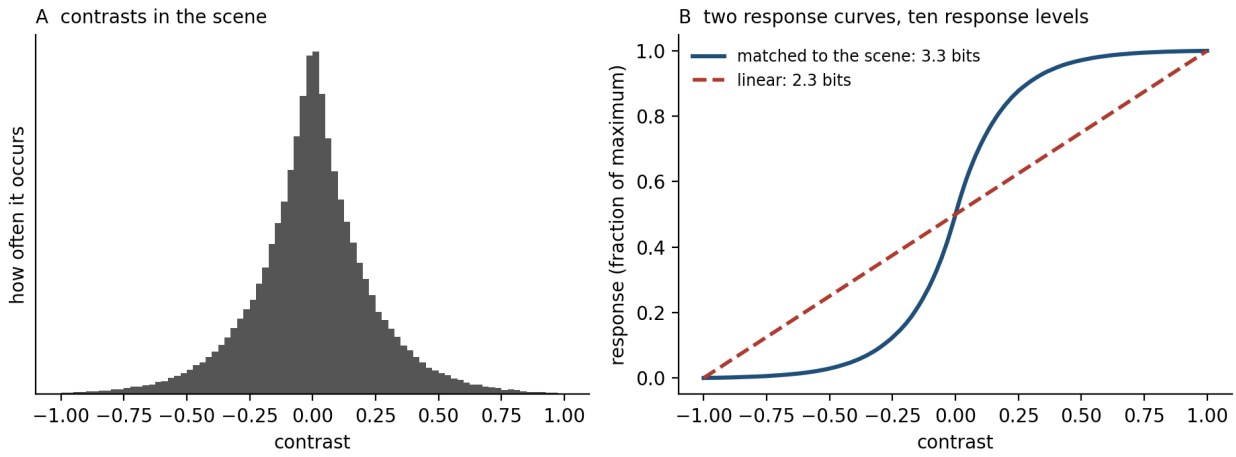


Figure 12.3. Matching a response curve to the world, after Laughlin (1981) (simulation). A: an illustrative distribution of contrasts, in which small contrasts are common and large ones rare. B: a response curve shaped like the cumulative distribution (blue) uses all ten response levels equally and carries 3.3 bits per contrast; a linear response (red) carries 2.3, because most contrasts fall in a few of its levels.

A fixed match would be fragile, because the world's statistics change, and the brain's match changes with them. Naama Brenner, William Bialek, and Rob de Ruyter van Steveninck recorded from H1, a motion-sensitive neuron in the fly, while the visual motion it saw changed in strength. When the motion became more variable, the neuron's input-output curve stretched to cover the new range; when the motion calmed, it contracted, so that the cell kept using its full range of responses.[817] Adrienne Fairhall and the same group then found that after a change the information the neuron carried recovered within about a tenth of a second, while its firing rate went on adjusting for a minute, as if the neuron were estimating the statistics of its input on several timescales at once.[818] Adaptation of this kind is not fatigue. It is the continual revision of a prediction about what the next input will be like. Over longer periods, adaptation often follows a power law in time, the signature of many timescales operating at once, sometimes within a single neuron.[819]

Adaptation shows itself most plainly when a prediction outlives the world it was fitted to. In 1834 Robert Addams described what he saw at the Falls of Foyers, on the shore of Loch Ness. Having looked for a few seconds at the cascade and then turned his eyes to the rock face beside it, he "saw the rocky surface as if in motion upwards, and with an apparent velocity equal to that of the descending water." He blamed an involuntary movement of the eyeball. The motion aftereffect, as it is now called, is the classic case of a sensory system adjusted to a sustained input. Neurons tuned to the direction of the falling water reduce their responses, and when the motion stops, the balance between neurons tuned to opposite directions

[816] Barlow, H. B. (1961). Possible principles underlying the transformations of sensory messages. In W. A. Rosenblith (Ed.), *Sensory Communication* (pp. 217–234). MIT Press.

[817] Brenner, N., Bialek, W., & de Ruyter van Steveninck, R. (2000). Adaptive rescaling maximizes information transmission. *Neuron*, 26(3), 695–702.

[818] Fairhall, A. L., Lewen, G. D., Bialek, W., & de Ruyter van Steveninck, R. R. (2001). Efficiency and ambiguity in an adaptive neural code. *Nature*, 412(6849), 787–792.

[819] Wark, B., Lundstrom, B. N., & Fairhall, A. (2007). Sensory adaptation. *Current Opinion in Neurobiology*, 17(4), 423–429.

tips the other way.[820] The rocks are not moving; the brain has recalibrated its expectation of motion to a world that has changed.

The fly's neurons are matched to the present statistics of their input. A stronger claim is that early sensory neurons encode the future. Stephanie Palmer, Olivier Marre, Michael Berry, and Bialek recorded from dozens of ganglion cells in a salamander's retina while a bar moved in a jittery but partly predictable way. Nearly every cell belonged to some small group whose spikes carried almost as much information about where the bar would be next as any code could, given how much the bar's past revealed about its future.[821] Matthew Chalk, Marre, and Gašper Tkačik then brought efficient coding and coding for the future into one framework, in which whether a neuron should decorrelate its input, or respond sparsely, depends on the statistics of the input and on whether the code is for recovering the past or predicting what comes next.[822] That is this chapter's thesis in miniature: one strategy, with its implementation tuned to circumstance.

Prediction at Every Level

The same strategy appears wherever the brain has been examined closely enough, and three examples not yet met in this book show how far it reaches.

The first is the hippocampus. Chapter 4 described place cells, each active when an animal is in one part of its environment. Mayank Mehta, Carol Barnes, and Bruce McNaughton recorded place cells while rats ran repeatedly in one direction along a track and found that each cell's field grew backward, against the direction of travel, over the first laps of each session: a cell that had fired only at one spot came to fire earlier, as the rat approached it.[823] Kimberly Stachenfeld, Matthew Botvinick, and Samuel Gershman explained the asymmetry by proposing that the hippocampus encodes not where the animal is but where it is likely to go: a predictive map in which each location is represented by the locations expected to follow it, a construction from reinforcement learning called the successor representation.[824]

Figure 12.4 shows why such a map produces the backward growth. If the animal always runs one way, a location is predicted by the places just before it, and a cell for that location becomes active as the animal approaches. The same theory accounts for some features of grid cells, and it unites the hippocampus's two roles, in memory and in navigation, as two uses of one model of what follows what.

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[820] Addams, R. (1834). An account of a peculiar optical phænomenon seen after having looked at a moving body. *London and Edinburgh Philosophical Magazine and Journal of Science*, 5(29), 373–374; Mather, G., Verstraten, F., & Anstis, S. (Eds.). (1998). *The Motion Aftereffect: A Modern Perspective*. MIT Press.

[821] Palmer, S. E., Marre, O., Berry, M. J., II, & Bialek, W. (2015). Predictive information in a sensory population. *Proceedings of the National Academy of Sciences*, 112(22), 6908–6913.

[822] Chalk, M., Marre, O., & Tkačik, G. (2018). Toward a unified theory of efficient, predictive, and sparse coding. *Proceedings of the National Academy of Sciences*, 115(1), 186–191.

[823] Mehta, M. R., Barnes, C. A., & McNaughton, B. L. (1997). Experience-dependent, asymmetric expansion of hippocampal place fields. *Proceedings of the National Academy of Sciences*, 94(16), 8918–8921.

[824] Stachenfeld, K. L., Botvinick, M. M., & Gershman, S. J. (2017). The hippocampus as a predictive map. *Nature Neuroscience*, 20(11), 1643–1653.

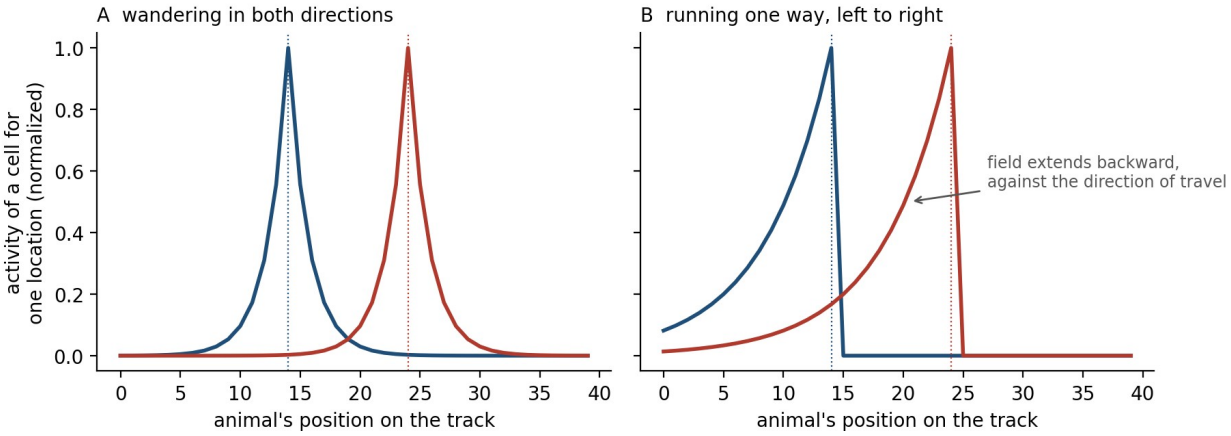


Figure 12.4. A predictive map on a linear track (computed). Each curve shows the activity, at every position, of a cell representing one location (dotted line) in a successor representation: how strongly the animal's current position predicts a future visit to that location. A: an animal that wanders in both directions; the fields are symmetric. B: an animal that runs from left to right; each field extends backward, toward the places from which the location is reached, as Mehta and colleagues observed in rats.

People appear to use such maps too. Ida Momennejad, Gershman, and their colleagues taught people sequences of states that led to rewards and then changed either the rewards or the sequences. People adjusted to changed rewards more readily than to changed sequences, the asymmetry a predictive map produces, since a map of what follows what must be relearned when the sequence changes but not when only the rewards do. Yet people did adjust to changed sequences, if less well, and the authors concluded that they combine such a map with deliberate planning.[825] Mona Garvert, Raymond Dolan, and Behrens showed people a stream of objects whose order followed hidden rules, and found that the hippocampus and entorhinal cortex represented each object according to which objects were likely to follow it, a structure the participants could not report.[826]

The second is the cortex of a running mouse. Georg Keller, Tobias Bonhoeffer, and Mark Hübener placed mice on a ball in front of screens that showed a corridor moving past as the mouse ran, so that the visual flow was coupled to the mouse's own movement. Occasionally they stopped the flow for a moment while the mouse kept running. Many neurons in the primary visual cortex responded strongly to these halts, which were not stimuli at all but the absence of the motion that the running predicted.[827] Alexander Attinger, Bo Wang, and Keller then raised mice in darkness and gave them visual experience only in a virtual world, in which the flow either followed their running or moved independently of it. Strong mismatch responses developed only in the mice whose world had moved with them.[828] The prediction that visual flow follows running is not built into visual cortex. It is learned, from the coupling between movement and sight, as the infant's world teaches it.

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[825] Momennejad, I., Russek, E. M., Cheong, J. H., Botvinick, M. M., Daw, N. D., & Gershman, S. J. (2017). The successor representation in human reinforcement learning. *Nature Human Behaviour*, 1(9), 680–692.

[826] Garvert, M. M., Dolan, R. J., & Behrens, T. E. J. (2017). A map of abstract relational knowledge in the human hippocampal–entorhinal cortex. *eLife*, 6, e17086.

[827] Keller, G. B., Bonhoeffer, T., & Hübener, M. (2012). Sensorimotor mismatch signals in primary visual cortex of the behaving mouse. *Neuron*, 74(5), 809–815.

[828] Attinger, A., Wang, B., & Keller, G. B. (2017). Visuomotor coupling shapes the functional development of mouse visual cortex. *Cell*, 169(7), 1291–1302.

The third is language. In 1980 Marta Kutas and Steven Hillyard recorded electrical activity from the scalps of people reading sentences one word at a time. When a sentence ended with a word that made no sense, as in "he spread the warm bread with socks", a negative wave peaked about 400 milliseconds after the word, which they called the N400.[829] More than 40 years of work since have shown that the wave tracks not nonsense as such but expectation: it is smaller for any word that the preceding words made probable and larger for one they made improbable, even when both are perfectly sensible.[830] Whether that reflects a word predicted before it arrives or one more easily fitted in after it does has been argued over for decades; a replication in nine laboratories with 334 participants confirmed the effect of how expected a noun is but not the best-known evidence that readers anticipate the word itself, a change in the wave at the article "a" or "an" that precedes an expected noun.[831] Either way, the brain's sensitivity to how expected each word is was found a generation before the language models of Chapter 10, which are trained on nothing else.

In each case the prediction is learned from experience, compared with what arrives, and revised when it fails. The three examples differ in almost everything else: the cells, the timescales, the content of the prediction, and the signal that reports its failure.

Many Errors, One Form

The errors of Figure 12.1 are not one signal. Chapter 6 distinguished the dopamine neuron's error, a single number about value, from the feature-by-feature errors of sensory cortex, and the cerebellum's error arrives on climbing fibers, one to each Purkinje cell, and says which movement went wrong.

What they share is a form. Each compares a prediction with an outcome and changes something in proportion to the difference. The simplest learning rule of this kind, the error-correcting rule beneath Chapter 3's temporal-difference learning, is called the delta rule: the new estimate is the old one plus a fraction of the error. The fraction, the learning rate, is the interesting part. If the world is stable and the observations are noisy, the learning rate should be small, so that each noisy observation moves the estimate only a little. If the world has just changed, it should be large, so that the old estimate is abandoned quickly. No fixed rate can do both.

The solution, worked out for engineering problems by Rudolf Kalman in 1960, is to make the learning rate depend on uncertainty: large when the learner is unsure of its estimate relative to the noise in its observations, small when it is sure.[832] Matthew Nassar, Joshua Gold, and their colleagues asked people to predict where the next of a series of numbers would fall, while the numbers' average jumped from time to time without warning. People's learning rates rose after a jump and then fell as the new average was confirmed, as an approximately optimal learner's would.[833] The same group also found that the size of people's pupils, which reflects arousal systems including the noradrenergic system of Chapter 3, tracked

[829] Kutas, M., & Hillyard, S. A. (1980). Reading senseless sentences: Brain potentials reflect semantic incongruity. *Science*, 207(4427), 203–205.

[830] Kutas, M., & Federmeier, K. D. (2011). Thirty years and counting: Finding meaning in the N400 component of the event-related brain potential (ERP). *Annual Review of Psychology*, 62, 621–647.

[831] Nieuwland, M. S., Politzer-Ahles, S., Heyselaar, E., Segaert, K., Darley, E., Kazanina, N., Von Grebmer Zu Wolfsthurn, S., Bartolozzi, F., Kogan, V., Ito, A., Mézière, D., Barr, D. J., Rousselet, G. A., Ferguson, H. J., Busch-Moreno, S., Fu, X., Tuomainen, J., Kulakova, E., Husband, E. M., … Huettig, F. (2018). Large-scale replication study reveals a limit on probabilistic prediction in language comprehension. *eLife*, 7, e33468.

[832] Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1), 35–45; Dayan, P., Kakade, S., & Montague, P. R. (2000). Learning and selective attention. *Nature Neuroscience*, 3(Suppl.), 1218–1223.

the model's estimates of how likely a change had been and how uncertain the current estimate was, and together these pupil measures predicted each trial's learning rate.[834] In a later version of the task, people placed a bucket to catch bags dropped from a helicopter hidden behind clouds, which now and then moved without warning; separate brain regions tracked the influence of surprise and of uncertainty on how much each bag changed where people put the bucket.[835]

Figure 12.5 shows such a learner. It tracks the jumps almost as quickly as a learner with a high fixed rate and holds steady between them almost as well as one with a low rate; over 100 simulated runs its average error was 20, against 23 for the best fixed rate.

Who sets the rate in the brain is less clear. People adjust their learning rate to the volatility of the world (Chapters 8 and 11), and on Angela Yu and Peter Dayan's account (Chapter 6) acetylcholine reports the expected uncertainty of a noisy but stable world and norepinephrine the unexpected uncertainty of one that has changed.[836] Christoph Mathys and his colleagues built a hierarchy of such learners, each level estimating how fast the level below is changing, so that the volatility of the world is itself learned.[837]

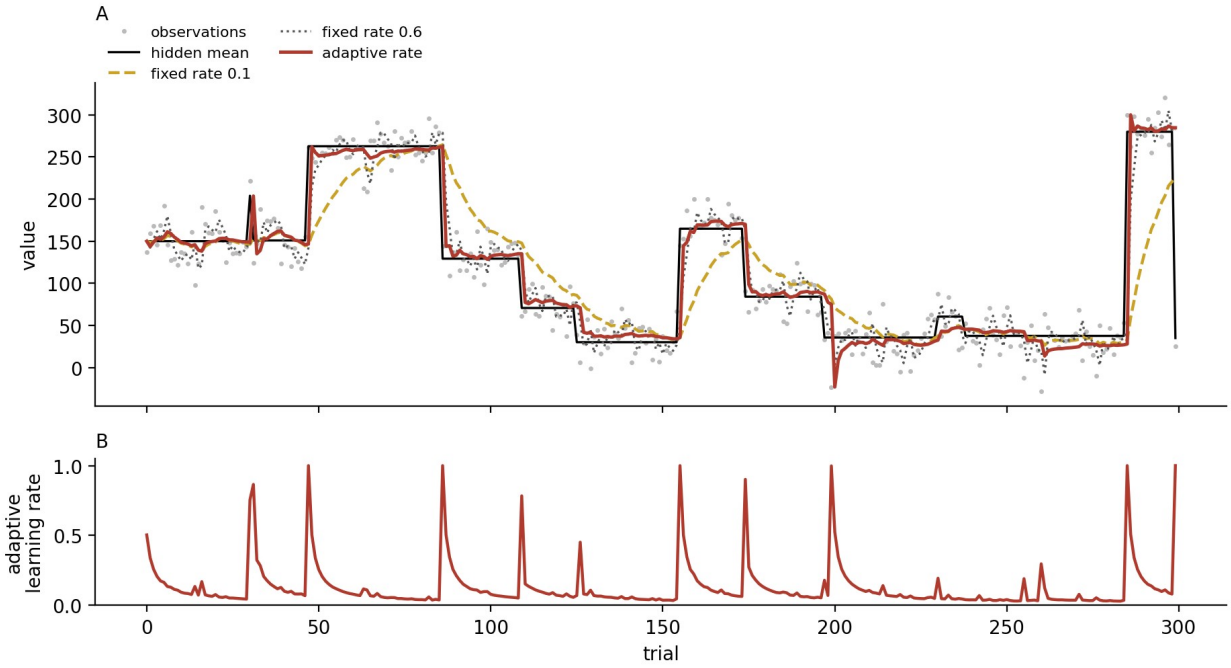


Figure 12.5. An adaptive learning rate, after Nassar and colleagues (2010) (simulation). A: noisy observations (gray dots) of a hidden mean (black) that jumps at random. A learner with a low fixed rate (gold) is precise but slow to follow jumps; one with a high fixed rate (gray) follows quickly but is buffeted by noise. A learner that knows how often the mean jumps and how noisy the observations are, and whose rate rises when an observation is improbable under its current estimate

[833] Nassar, M. R., Wilson, R. C., Heasly, B., & Gold, J. I. (2010). An approximately Bayesian delta-rule model explains the dynamics of belief updating in a changing environment. *Journal of Neuroscience*, 30(37), 12366–12378.

[834] Nassar, M. R., Rumsey, K. M., Wilson, R. C., Parikh, K., Heasly, B., & Gold, J. I. (2012). Rational regulation of learning dynamics by pupil-linked arousal systems. *Nature Neuroscience*, 15(7), 1040–1046.

[835] McGuire, J. T., Nassar, M. R., Gold, J. I., & Kable, J. W. (2014). Functionally dissociable influences on learning rate in a dynamic environment. *Neuron*, 84(4), 870–881.

[836] Yu, A. J., & Dayan, P. (2005). Uncertainty, neuromodulation, and attention. *Neuron*, 46(4), 681–692.

[837] Mathys, C., Daunizeau, J., Friston, K. J., & Stephan, K. E. (2011). A Bayesian foundation for individual learning under uncertainty. *Frontiers in Human Neuroscience*, 5, 39.

(red), does both. B: the adaptive learner's rate, which spikes after each jump and decays as the new mean is confirmed. Over 100 simulated runs, root-mean-square errors were 36 (rate 0.1), 23 (rate 0.6), and 20 (adaptive).

A learning rate set by uncertainty is the clearest single candidate for what all the brain's error signals have in common. It is an idea about how to learn from prediction errors, not about what to predict, and it applies whatever the cells and signals that implement it.

The claim that the brain learns as an ideal learner would deserves the same scrutiny this chapter will give the free-energy principle. Jeffrey Bowers and Colin Davis argued that Bayesian models in psychology and neuroscience are too easily adjusted: the priors, likelihoods, and costs can be chosen after the data are in, so that almost any behavior can be described as optimal, and the models rarely predict better than simpler ones. Thomas Griffiths and his colleagues replied that the models are tools for asking what problem the brain is solving, not claims that it solves it perfectly.[838] The adaptive learning rate survives the critique better than most such claims, because it predicts something a fixed-rate learner does not, a rise in learning after surprise, and people's learning rates and pupils show it. But "approximately optimal" should be read as a description of what the brain manages, not a guarantee.

When the Learning Rate Goes Wrong

If learning well depends on weighting new evidence correctly against old beliefs, some disorders might be failures of the weighting. Chapter 6 described accounts of psychosis and autism in which priors are weighted too heavily or too lightly. Two experiments since have measured the weighting directly, using the hierarchical learner just described.

Albert Powers, Mathys, and Philip Corlett played people a faint tone at the same time as they showed a checkerboard, many times, until the two were associated. As the experiment went on, the tone was made fainter and, on more and more trials, left out altogether. Everyone sometimes reported hearing a tone that had not been played, a hallucination produced by conditioning. People who heard voices in daily life reported these conditioned hallucinations more often and with more confidence, whether they had a diagnosis of psychotic illness or not; one group of voice-hearers without a diagnosis attributed their voices to spiritual sources. Fitted to each person's responses, the model showed that the voice-hearers gave more weight to what they expected relative to what they heard. In brain scans, the conditioned hallucinations raised activity in the auditory regions that responded to real tones, along with a network including the insula and the anterior cingulate cortex that resembles the regions active when patients hallucinate.[839]

Rebecca Lawson, Mathys, and Geraint Rees gave autistic and non-autistic adults a task in which a tone predicted whether a face or a house would appear, with a reliability that changed without warning from block to block. The autistic adults were less surprised when the tone's prediction was violated, and the model fitted to their choices showed why: they judged the world to be more volatile than it was, so that

[838] Bowers, J. S., & Davis, C. J. (2012). Bayesian just-so stories in psychology and neuroscience. *Psychological Bulletin*, 138(3), 389–414; Griffiths, T. L., Chater, N., Norris, D., & Pouget, A. (2012). How the Bayesians got their beliefs (and what those beliefs actually are): Comment on Bowers and Davis (2012). *Psychological Bulletin*, 138(3), 415–422.

[839] Powers, A. R., Mathys, C., & Corlett, P. R. (2017). Pavlovian conditioning–induced hallucinations result from overweighting of perceptual priors. *Science*, 357(6351), 596–600.

each violation seemed less informative about whether anything had changed, and those who did this most had the most severe symptoms.[840]

Both studies are small, and neither establishes a cause. But they show what a predictive account adds to the description of a disorder. Instead of a vague claim that prediction has failed, it offers a quantity, how much weight a brain gives to what it believes against what it is told and how fast it thinks the world changes, that can be estimated in each person and compared.

Adaptation on Every Timescale

Anyone who has spent a few days at sea knows the feeling of stepping ashore and finding the land rocking. Thomas Stoffregen and his colleagues measured it aboard a ship crossing the Caribbean. Within the first hours of the voyage, passengers new to the sea had widened their stance and changed how they swayed. On their first day back on land, most felt the ground move for half an hour or less, but about one in five felt it for two hours or more, and these were the people who had swayed most at sea.[841] Balance had been relearned for a moving deck within hours, and unlearning it took hours more: the brain had built a prediction of the ground's motion and could not discard it at once.

Sea legs are one timescale among many. Prediction requires a model, and a model of a changing world must change. The brain revises its predictions on every timescale it has been examined at (Figure 12.6). Sensory neurons rescale their responses within a fraction of a second, as the fly's H1 did. Synapses strengthen and weaken from moment to moment through short-term plasticity, and over hours to years through the long-term changes of Chapter 5. The eye's sensitivity to light recovers over half an hour or more in the dark.[842] Motor adaptation, Chapter 9 showed, has a fast process that learns and forgets quickly and a slow one that learns and forgets slowly. Synaptic scaling, which Chapter 5 described, adjusts all of a neuron's synapses over a day or two to keep its activity within bounds.[843] Development opens critical periods in which a circuit takes on the statistics of its early environment, and then closes them.[844] Evolution, on the longest scale, has built some predictions in before experience begins.

[840] Lawson, R. P., Mathys, C., & Rees, G. (2017). Adults with autism overestimate the volatility of the sensory environment. *Nature Neuroscience*, 20(9), 1293–1299.

[841] Stoffregen, T. A., Chen, F.-C., Varlet, M., Alcantara, C., & Bardy, B. G. (2013). Getting your sea legs. *PLoS ONE*, 8(6), e66949.

[842] Lamb, T. D., & Pugh, E. N., Jr. (2004). Dark adaptation and the retinoid cycle of vision. *Progress in Retinal and Eye Research*, 23(3), 307–380.

[843] Turrigiano, G. G., Leslie, K. R., Desai, N. S., Rutherford, L. C., & Nelson, S. B. (1998). Activity-dependent scaling of quantal amplitude in neocortical neurons. *Nature*, 391(6670), 892–896.

[844] Hensch, T. K. (2005). Critical period plasticity in local cortical circuits. *Nature Reviews Neuroscience*, 6(11), 877–888.

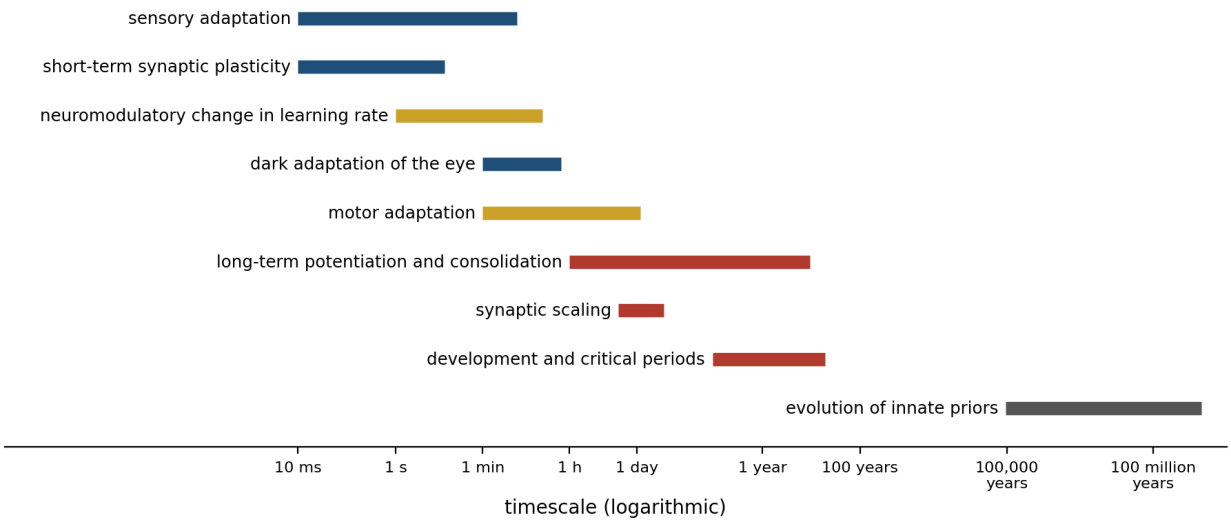


Figure 12.6. Adaptation on every timescale (schematic; ranges approximate). Each bar shows the timescale over which a process revises the brain's predictions. Faster processes track the moment; slower ones accumulate what is reliably true; the slowest, evolution, sets the starting point for the rest.

Why so many timescales? Konrad Kording, Joshua Tenenbaum, and Reza Shadmehr gave an answer for motor adaptation that generalizes. The disturbances a body meets have different durations: fatigue lasts minutes, an injury weeks, growth years. A learner that does not know which kind of disturbance it faces should estimate all of them at once, attributing a new error mostly to the fast, likely causes and only gradually to the slow ones. They showed that such a learner reproduces spontaneous recovery, the return of an old adaptation after a new one fades, and some forms of savings, the faster relearning of an adaptation met before; Chapter 9 described both.[845] On this view, the many timescales of plasticity are not a collection of accidents but an inference about the timescales on which the world changes.

Even long-held priors can be revised. People assume, other things being equal, that objects move slowly, which is usually true, and the assumption shows when the evidence weakens: a faint pattern looks slower than a clear one moving at the same speed. Robert Snowden and his colleagues found that drivers in a simulator whose whole scene had been reduced in contrast, as if by a uniform fog, judged their speed to be lower than it was and drove faster. Fog simulated more realistically, hiding distant objects more than near ones, has the opposite effect, and drivers in it slow down; the uniform version isolated the prior.[846] Alan Stocker and Eero Simoncelli showed that misjudgments of speed at low contrast are what an observer with a prior favoring slow speeds would make.[847] Grigorios Sotiropoulos, Aaron Seitz, and Peggy Seriès trained people for five days on faint patterns that moved fast, and the prior itself shifted toward faster speeds, as a change in how they judged the direction of faint moving lines revealed.[848]

[845] Körding, K. P., Tenenbaum, J. B., & Shadmehr, R. (2007). The dynamics of memory as a consequence of optimal adaptation to a changing body. *Nature Neuroscience*, 10(6), 779–786.

[846] Snowden, R. J., Stimpson, N., & Ruddle, R. A. (1998). Speed perception fogs up as visibility drops. *Nature*, 392(6675), 450; Pretto, P., Bresciani, J.-P., Rainer, G., & Bülthoff, H. H. (2012). Foggy perception slows us down. *eLife*, 1, e00031.

[847] Stocker, A. A., & Simoncelli, E. P. (2006). Noise characteristics and prior expectations in human visual speed perception. *Nature Neuroscience*, 9(4), 578–585.

Chapter 6's light from above, which training can rotate and which chicks reared under light from below still assume, makes the same point. Chicks hatched and kept in darkness, shown moving patterns of light points for the first time, approach a pattern made by a walking animal rather than one made by random motion, whether the animal is a hen or a cat, as if they arrive expecting living things to move as living things do.[849] Evolution sets the prior; experience can move it, within limits.

The timescales also answer the problem of Chapters 5 and 10, the balance between learning and remembering: layering fast processes on slow ones lets the brain follow the moment without losing what is reliably true.

One Principle? The Free-Energy Principle

The most ambitious attempt to turn all this into a single principle is Karl Friston's. In a series of papers since 2005, Friston has argued that any system that keeps itself in a limited set of states, as a living thing must, can be described as minimizing variational free energy, an information-theoretic quantity distinct from the thermodynamic free energy of Chapter 0, which it can compute from its sensations and its own model and which sets an upper bound on surprise, the improbability of its sensory states under that model.[850] Minimizing it can be done in two ways (Figure 12.7). The system can change its model so that its sensations become less surprising, which is perception and learning; or it can change its sensations so that they match its model, which is action. Friston called the second active inference. On this account, a creature moves its eyes to find what it expects, reaches for a cup to make the predicted sensation of holding it come true, and keeps its body temperature near 37 degrees because it predicts it will be there.

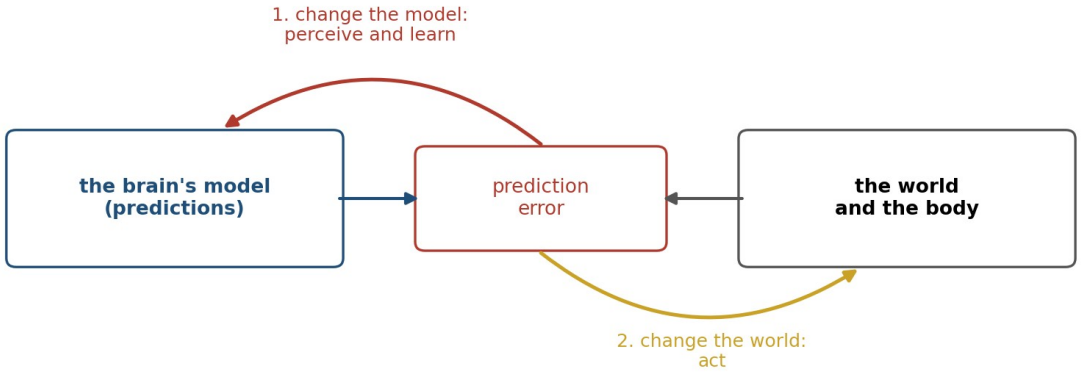


Figure 12.7. Two ways to reduce a prediction error. A mismatch between the brain's predictions and what arrives from the world and the body can be reduced by changing the model, which is perception and learning, or by acting to change what arrives. The free-energy principle treats both as the minimization of one quantity.

[848] Sotiropoulos, G., Seitz, A. R., & Seriès, P. (2011). Changing expectations about speed alters perceived motion direction. *Current Biology*, 21(21), R883–R884.

[849] Vallortigara, G., Regolin, L., & Marconato, F. (2005). Visually inexperienced chicks exhibit spontaneous preference for biological motion patterns. *PLoS Biology*, 3(7), e208.

[850] Friston, K., Kilner, J., & Harrison, L. (2006). A free energy principle for the brain. *Journal of Physiology-Paris*, 100(1–3), 70–87; Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.

The attraction is obvious. One formulation covers perception, action, learning, attention (as the weighting of errors by their reliability, Chapter 6's precision), and homeostasis, and it echoes the cybernetics of W. Ross Ashby that Chapter 1 described. Andy Clark and Jakob Hohwy have built a broad philosophy of mind on it.[851]

The objections are as serious. The first is the dark-room problem. If a creature's goal is to minimize surprise, why does it not find a dark, quiet room and stay there, where nothing surprising can happen? Friston, Christopher Thornton, and Clark replied that a creature like us expects to be fed, warm, and among others, so that a dark room, in which it would starve, is exactly what its model predicts it will avoid.[852] The reply works, but at a price: the creature's goals must be written into its model as expectations about the states it will occupy, and the theory then explains goal-directed behavior by assuming the goals.

The second objection is that the principle, as a principle, cannot be tested. Friston has said as much: "Like Hamilton's Principle of Stationary Action, it cannot be falsified," whereas hypotheses derived from it, such as predictive coding, "may or may not be supported by empirical evidence."[853] Mel Andrews argued that the formalism is a modeling framework, not a hypothesis about the brain, and Samuel Gershman that it is so flexible that almost any behavior can be described as minimizing free energy under some model.[854] Jelle Bruineberg and his colleagues showed that its central device, the "Markov blanket" that separates a system from its surroundings, is used in two senses, a tool of statistics and a claim about what an organism is.[855]

What can be tested are the specific process theories built under it, and the most specific, predictive coding in cortex, has had a mixed record. Laurence Aitchison and Máté Lengyel showed that a brain could do Bayesian inference without predictive coding and could use predictive coding without doing Bayesian inference, so evidence for one is not evidence for the other.[856] Kevin Walsh, Redmond O'Connell, and their colleagues reviewed the neurophysiological evidence and found much of it compatible with predictive coding and its relatives, often grouped as predictive processing, but rarely decisive against alternatives.[857] A large study recording from many visual areas at once in mice and monkeys, so far reported only as a preprint, found that responses to violations of a learned pattern appeared mainly in higher areas, not in early visual cortex, and found little sign of the suppression of predicted input that the theory requires.[858] Yet the mismatch between movement and sight in Keller's

[851] Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204; Hohwy, J. (2013). *The Predictive Mind*. Oxford University Press.

[852] Friston, K., Thornton, C., & Clark, A. (2012). Free-energy minimization and the dark-room problem. *Frontiers in Psychology*, 3, 130.

[853] Friston, K., Fortier, M., & Friedman, D. A. (2018). Of woodlice and men: A Bayesian account of cognition, life and consciousness. An interview with Karl Friston. *ALIUS Bulletin*, 2, 17–43.

[854] Andrews, M. (2021). The math is not the territory: Navigating the free energy principle. *Biology & Philosophy*, 36(3), 30; Gershman, S. J. (2019). What does the free energy principle tell us about the brain? *Neurons, Behavior, Data Analysis, and Theory*, 2(3).

[855] Bruineberg, J., Dołęga, K., Dewhurst, J., & Baltieri, M. (2022). The emperor's new Markov blankets. *Behavioral and Brain Sciences*, 45, e183.

[856] Aitchison, L., & Lengyel, M. (2017). With or without you: Predictive coding and Bayesian inference in the brain. *Current Opinion in Neurobiology*, 46, 219–227.

[857] Walsh, K. S., McGovern, D. P., Clark, A., & O'Connell, R. G. (2020). Evaluating the neurophysiological evidence for predictive processing as a model of perception. *Annals of the New York Academy of Sciences*, 1464(1), 242–268.

[858] Westerberg, J. A., Xiong, Y. S., Sennesh, E., Nejat, H., Ricci, D., Durand, S., et al., & Bastos, A. M. (2025). *Hierarchical substrates of prediction in visual cortical spiking* [Preprint]. bioRxiv.

mice does evoke responses in early visual cortex; the evidence may depend on the kind of prediction that is violated.

The principle has had direct tests, with results that show both its reach and its limits. Takuya Isomura, Friston, and their colleagues grew rat cortical neurons on arrays of electrodes and stimulated them with mixtures of two hidden signals. Over 100 sessions the cultured neurons came to respond selectively to one source or the other, separating the mixture, and a free-energy model fitted to the first ten sessions predicted how their responses and connections would change over the remaining 90.[859] The same separation is predicted by simpler learning rules of the Hebbian kind, which the free-energy model can be read as restating; the study shows that the principle can describe what the neurons did, not that a rival account could not.

Brett Kagan and his colleagues connected cultures of mouse or human cortical neurons, hundreds of thousands of cells on each array, to a simulated game of Pong, delivering predictable stimulation when the paddle hit the ball and unpredictable stimulation when it missed, and reported that the cultures' rallies lengthened within five minutes of play, as the free-energy account predicted they would if neurons act to make their input predictable.[860] A group of neuroscientists objected that the improvement was modest and that the paper's claim of "sentience" stretched the word beyond use.[861] Both studies show something real: that networks of neurons, with nothing but their own plasticity, learn in ways that reduce the unpredictability of their input. Neither shows that this is all a brain does.

What Prediction Leaves Out

A strategy that explains everything explains nothing, and it is worth being clear about what prediction does not obviously explain.

It does not say what to want. A model can predict a meal or a predator equally well; which one to seek is a matter of value, and in the free-energy account value is reintroduced as prior expectations about preferred states. Chapter 7 argued that value is anchored to a body; the anchoring is a fact about the organism, not a consequence of predicting well.

It does not say why to explore. A learner that wanted only to make its input predictable would stay where it is, but animals and children seek out what they do not yet understand. Aimee Stahl and Lisa Feigenson showed 11-month-old infants a ball that seemed to roll through a solid wall, or a toy car pushed off a ledge that seemed to hover in midair. The infants learned more about an object that had broken their expectations than about one that had not, and when they were given it to play with, they tested the expectation it had broken: they banged the ball on the table, and tended to drop the car.[862] The surprise told them where to look; the drive to explore was their own.

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[859] Isomura, T., Kotani, K., Jimbo, Y., & Friston, K. J. (2023). Experimental validation of the free-energy principle with in vitro neural networks. *Nature Communications*, 14, 4547.

[860] Kagan, B. J., Kitchen, A. C., Tran, N. T., Habibollahi, F., Khajehnejad, M., Parker, B. J., Bhat, A., Rollo, B., Razi, A., & Friston, K. J. (2022). In vitro neurons learn and exhibit sentience when embodied in a simulated game-world. *Neuron*, 110(23), 3952–3969.

[861] Balci, F., Ben Hamed, S., Boraud, T., Bouret, S., Brochier, T., Brun, C., Cohen, J. Y., Coutureau, E., Deffains, M., Doyère, V., Gregoriou, G. G., Heimel, J. A., Kilavik, B. E., Lee, D., Leuthardt, E. C., Mainen, Z. F., Mathis, M., Monosov, I. E., Naudé, J., … Zimmermann, J. (2023). A response to claims of emergent intelligence and sentience in a dish. *Neuron*, 111(5), 604–605.

[862] Stahl, A. E., & Feigenson, L. (2015). Observing the unexpected enhances infants' learning and exploration. *Science*, 348(6230), 91–94.

Chapter 11 described the most promising account of that drive, curiosity as the pursuit of learning progress, a preference for improving predictions rather than for having good ones already. In active inference, a drive to resolve uncertainty appears as an "epistemic" part of the quantity an agent minimizes when it plans, the free energy it expects in the future.[863] Beren Millidge and his colleagues have argued that this expected free energy cannot simply be derived from the principle as the free energy of the future, and that its exploratory part does not follow directly from it, so that exploration is assumed rather than explained.[864]

It may not even describe the brain's evolutionary priority. Paul Cisek has argued that brains evolved as controllers, loops that keep an animal's relation to its world within useful bounds, elaborated over hundreds of millions of years, and that dividing behavior into perception, prediction, and action imposes an engineer's diagram on a system that never had one.[865] On his view, prediction is one of the things control loops came to use, as an animal's interactions grew more complex, not the foundation they were built on.

And prediction does not explain the brain's particular form. Why it uses spikes, why it is organized in a hierarchy of areas, why it keeps a fast memory beside a slow one, and why it spends its energy as it does are questions about the constraints under which it was built, and they need answers of a different kind.

Implications for Artificial Intelligence

The most successful artificial systems of the 2020s are predictors. A large language model is trained to predict the next word, and Chapter 10 noted how much follows from that single objective. In one sense the brain's strategy has been vindicated at enormous scale. In four senses it has not yet been copied.

First, the brain predicts the consequences of its own actions, and learns most from the mismatches its own movements create, as Keller's mice did. A language model is trained mostly to predict text it had no part in producing. The systems that learn world models of their own actions, such as the DreamerV3 agent of Chapter 10, are closer to the brain's arrangement, and Yann LeCun has proposed that machines should predict not raw data but abstract representations of it, so that they model what matters rather than every pixel, as the levels of the cortical hierarchy predict the level below rather than the retina.[866]

Second, the brain's learning rate is governed by its uncertainty, moment by moment, as Figure 12.5 showed; a trained model's is set on a schedule, and once deployed it does not learn at all. The idea of letting each connection learn its own learning rate from the history of its errors is old in machine learning; Richard Sutton proposed one version in 1992.[867] But it has not become part of how large models learn.

Third, the brain adapts on many timescales at once, with fast processes riding on slow ones. A deployed model has, in effect, two: the slow learning of its training, frozen, and the fast, unretained adaptation of

[863] Friston, K., Rigoli, F., Ognibene, D., Mathys, C., Fitzgerald, T., & Pezzulo, G. (2015). Active inference and epistemic value. *Cognitive Neuroscience*, 6(4), 187–214.

[864] Millidge, B., Tschantz, A., & Buckley, C. L. (2021). Whence the expected free energy? *Neural Computation*, 33(2), 447–482.

[865] Cisek, P. (2019). Resynthesizing behavior through phylogenetic refinement. *Attention, Perception, & Psychophysics*, 81(7), 2265–2287.

[866] LeCun, Y. (2022). *A path towards autonomous machine intelligence* (Version 0.9.2). OpenReview.

[867] Sutton, R. S. (1992). Adapting bias by gradient descent: An incremental version of delta-bar-delta. In *Proceedings of the Tenth National Conference on Artificial Intelligence* (pp. 171–176). AAAI Press.

its context window. What it lacks is everything in between, the hours, days, and years over which a brain consolidates what it has learned in use. Chapter 11 described the attempts to supply it.

Fourth, the brain learns from many errors at once, of the different kinds described above. A large model is trained mainly on one loss, next-word prediction, with further objectives, for human preferences and for correct answers, added at the end. Whether many local objectives, each fitted to its own part of the problem, would learn faster or more robustly than one global objective is a question the brain's arrangement poses directly.

The constraints under which the brain evolved, which prediction leaves unexplained, are the subject of Chapter 13.

The Central Lesson

Craik guessed in 1943 that a brain carries a model of the world and runs it ahead of time. Eighty years of evidence has filled in his sketch. Neurons in the retina transmit what their neighbors cannot predict; neurons in the fly match their responses to the statistics of the world and rescale them as it changes; dopamine neurons, cortical neurons, the hippocampus, the cerebellum, and the circuits that regulate the body all compare what they expect with what arrives and learn from the difference, at rates set by how uncertain they are and on timescales from milliseconds to lifetimes, with evolution setting the starting point. That is as close to a single principle as the brain offers: predict, compare, and revise, with a confidence that matches the world's reliability. It is a strategy, not a mechanism, carried out by different cells with different signals for different purposes. The attempt to compress it into one equation has produced a framework that describes almost anything and a few real tests, but the strategy does not depend on the equation.

Next Chapter

Chapter 13: Why the Brain Works the Way It Does: Constraints, Trade-offs, and the Shape of a Solution

Figure and Epigraph Sources

Figure Attribution

Figures 12.1 to 12.7 were drawn for this book. Figures 12.2 to 12.5 are computed: Figure 12.2 applies the lateral-inhibition scheme of Srinivasan, Laughlin, and Dubs (1982) to a simulated signal whose power falls in inverse proportion to frequency; Figure 12.3 applies the principle of Laughlin (1981) to an illustrative distribution of contrasts; Figure 12.4 computes the successor representation of Stachenfeld, Botvinick, and Gershman (2017) on a linear track; Figure 12.5 simulates the approximately Bayesian learner of Nassar et al. (2010) beside learners with fixed rates. Figures 12.1, 12.6, and 12.7 are schematic; the timescales in Figure 12.6 are approximate.

Epigraph

Kenneth J. W. Craik, *The Nature of Explanation* (Cambridge University Press, 1943), chapter 5, "Hypothesis on the Nature of Thought," p. 61.

Chapter 13

Why the Brain Works the Way It Does: Constraints, Trade-offs, and the Shape of a Solution

> *"Nothing in biology makes sense except in the light of evolution."*
>
> *— Theodosius Dobzhansky (1973)*

The Fish That Gave Up Its Eyes

In the limestone caves of northeastern Mexico live populations of a small silvery fish, *Astyanax mexicanus*, that have no eyes. Their relatives in the rivers outside see perfectly well, and the two forms belong to the same species and can interbreed. The cavefish begin to form eyes as embryos, and the eyes then degenerate. In 2015 Damian Moran, Rowan Softley, and Eric Warrant set out to measure what the cavefish had saved. They dissected eyes and brains from river fish and cavefish, measured how much oxygen the tissues consumed, and combined the measurements with the sizes of the organs in living fish. For a river fish weighing one gram, they estimated, the eyes and the part of the brain that processes vision cost about 15 percent of the animal's resting metabolism, falling to about 5 percent in larger fish whose eyes and brains are relatively smaller.[868] In a cave, where there is nothing to see and food is scarce, that is a large bill paid for nothing. Whether the saving is the reason the eyes were lost is still debated, since mutations that degrade the eye may have spread for other reasons, but the number shows how heavily seeing weighs in a small animal's budget.

The cavefish is a small parable for this chapter. The earlier chapters described what the brain does, and Chapter 12 closed with questions prediction cannot answer: why the brain signals with spikes, why it is built as a hierarchy of areas, why it keeps a fast memory beside a slow one, and why it spends its energy as it does. These are questions of a different kind.

In 1963 the ethologist Niko Tinbergen pointed out that "why" has four answers for any feature of an animal. One concerns causation, the mechanism that produces it; one its development in the individual; one its survival value, what it is good for; and one its evolution, the history from which it came.[869] Most of this book has answered the first two. This chapter is about the other two, and its argument is that the brain's form is where a few hard physical constraints meet a history that could only modify what already existed. The constraints are energy, space, time, noise, and the amount of information a genome can carry. Many of the brain's features are trade-offs among them, positions on a frontier where improving one thing costs another, rather than optimal solutions to a single problem. Some features are simply inherited. Figure 13.1 sets the constraints beside the mechanisms of the earlier chapters that each one helps explain, and the work of the chapter is to tell the kinds of explanation apart.

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[868] Moran, D., Softley, R., & Warrant, E. J. (2015). The energetic cost of vision and the evolution of eyeless Mexican cavefish. *Science Advances*, 1(8), e1500363.

[869] Tinbergen, N. (1963). On aims and methods of ethology. *Zeitschrift für Tierpsychologie*, 20(4), 410–433.

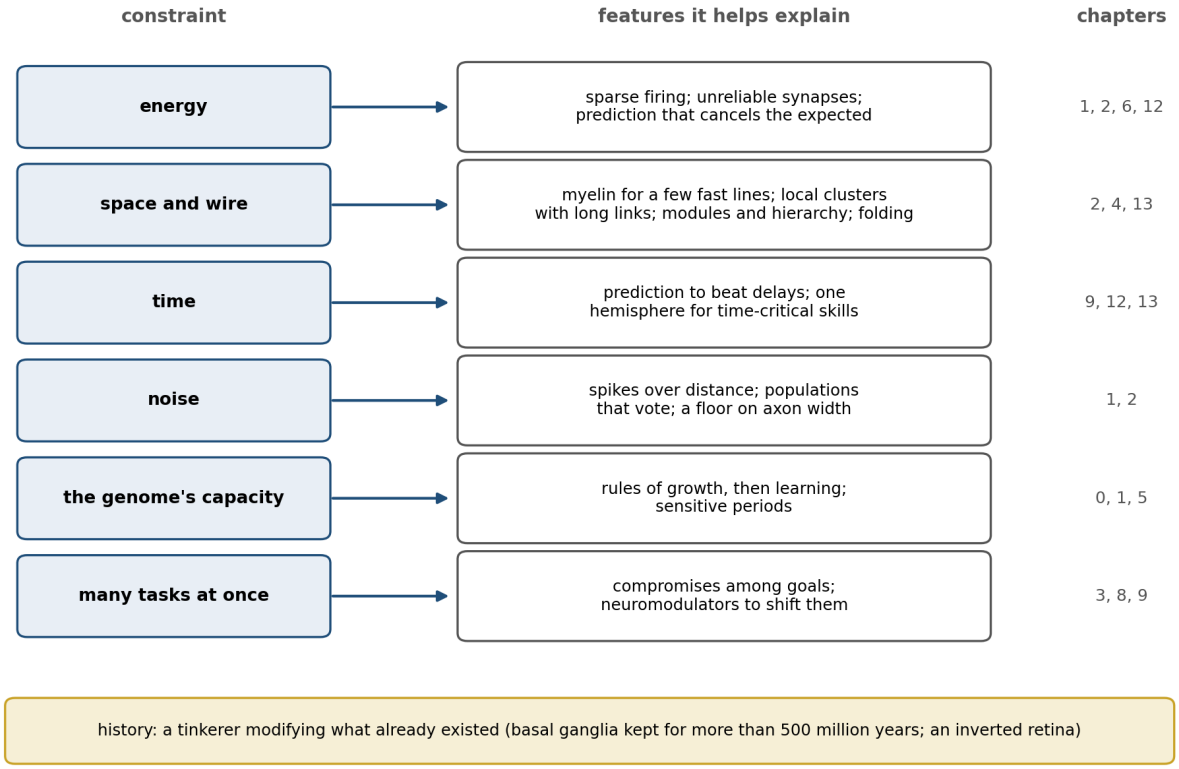


Figure 13.1. Constraints and the features they help explain. Each constraint on the left bears on the features on the right; the chapter numbers show where each feature was described. Most features answer to more than one constraint, and history (bottom) sets the starting material for all of them.

The Price of a Bit

A blowfly darting across a sunlit yard and a fruit fly drifting over a bowl of fruit have eyes built on the same plan, but they run them on very different budgets. Jeremy Niven, John Anderson, and Simon Laughlin recorded from the photoreceptors of four species of fly. Those of the large, fast-flying blowfly carried at their best about 955 bits of information per second, a bit being the amount needed to decide between two equally likely possibilities. That is nearly five times the 197 bits per second of the small, slow fruit fly, but coding the same quantity of information cost the blowfly's photoreceptors about ten times as much energy.[870] Each eye buys only as much information as its owner's way of life will pay for.

Chapter 1 gave the brain's budget, about 20 watts. The flies show what the budget buys, and why the price rises so steeply. Laughlin, Rob de Ruyter van Steveninck, and Anderson had worked out in 1998 what a bit costs a blowfly. A single chemical synapse spends about 10,000 molecules of ATP, the cell's energy currency, per bit. A photoreceptor or the interneuron it drives spends between one million and ten million, and in bright light the photoreceptors and their first interneurons together use close to 10 percent of the resting fly's energy.[871] They also found a law of diminishing returns. A signal is made more reliable by adding more channels, more vesicles, or more photon detectors, and its signal-to-noise ratio

[870] Niven, J. E., Anderson, J. C., & Laughlin, S. B. (2007). Fly photoreceptors demonstrate energy-information trade-offs in neural coding. *PLoS Biology*, 5(4), e116.

[871] Laughlin, S. B., de Ruyter van Steveninck, R. R., & Anderson, J. C. (1998). The metabolic cost of neural information. *Nature Neuroscience*, 1(1), 36–41.

grows roughly in proportion to their number, but the information it carries grows only as the logarithm of that ratio. Each additional bit therefore costs more than the last, and a weak pathway of low capacity transmits information more cheaply per bit than a strong one.

The same arithmetic constrains how often a neuron should fire. In 1996 William Levy and Robert Baxter treated a neuron as a channel that, in each brief interval of time, either fires or does not. The information it can carry per interval is greatest when it fires half the time, because then each interval is least predictable. But a neuron pays a small cost simply to stay alive and a much larger cost each time it fires, and when information is divided by energy the best rate falls far below one half, as Figure 13.2 shows.[872] Their later estimates put the most efficient probability of firing at 0.025 to 0.05 in each interval of 2.5 milliseconds, about 10 to 20 spikes a second, against about 200 a second for the rate that would maximize information alone.[873] Sparse firing is not a shortage of activity; it is what an efficient code looks like when spikes are dear.

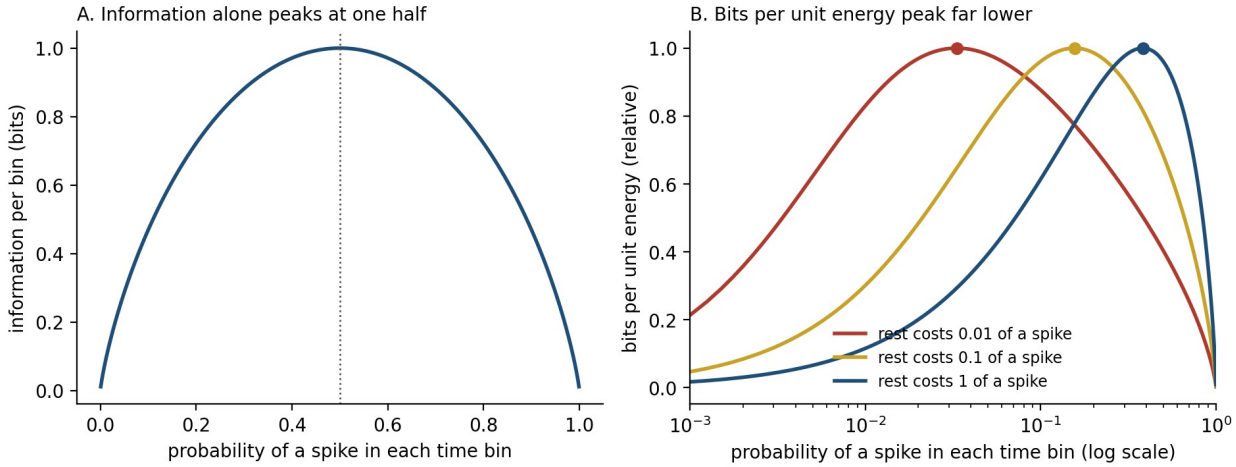


Figure 13.2. Information against energy (computed, after Levy and Baxter, 1996). A: a neuron that fires or stays silent in each time bin carries the most information per bin when it fires half the time. B: when each bin costs a small amount for rest and one unit for a spike, information per unit energy peaks at a much lower firing probability, and the cheaper rest is relative to a spike, the lower the peak lies (dots).

The argument extends to the synaptic failures of Chapters 1 and 2. Levy and Baxter showed in theory that such failures can raise a neuron's efficiency: a synapse that transmits on only some trials costs much less, while the information it loses is information the receiving neuron could not have passed on anyway.[874]

An experiment by Julia Harris, Renaud Jolivet, Elisabeth Engl, and David Attwell points the same way. In slices of rat brain they studied the synapse at which a fiber from the retina drives a relay neuron in the thalamus, and used a technique called dynamic clamp to make the synapse artificially stronger or weaker. A stronger synapse would have passed on up to four times more information from the retina; in most cells the synapse's natural strength instead came close to maximizing the information transmitted

[872] Levy, W. B., & Baxter, R. A. (1996). Energy efficient neural codes. *Neural Computation*, 8(3), 531–543.

[873] Levy, W. B., & Baxter, R. A. (2002). Energy-efficient neuronal computation via quantal synaptic failures. *Journal of Neuroscience*, 22(11), 4746–4755.

[874] Levy, W. B., & Baxter, R. A. (2002). Energy-efficient neuronal computation via quantal synaptic failures. *Journal of Neuroscience*, 22(11), 4746–4755.

for each molecule of ATP spent. Only about one retinal spike in five triggered a spike in the relay cell.[875] The experiment was done in a slice, with the cortex's feedback cut away and inhibition blocked, but within those limits the synapse was tuned for bits per unit energy rather than for bits.

Even the action potential has been trimmed. The squid axon of Chapter 1, it turns out, lets in about four times as much sodium as the minimum a spike needs, because its sodium and potassium currents overlap, and early energy budgets assumed the same waste in mammals. Henrik Alle, Arnd Roth, and Jörg Geiger found that in the thin axons of the rat hippocampus the overlap is so small that a spike costs only 1.3 times the theoretical minimum.[876] Brett Carter and Bruce Bean then showed that the efficiency varies with a cell's job. Cortical pyramidal neurons are about as frugal as the axons Alle measured, but the fast-spiking inhibitory neurons that fire hundreds of times a second let in about twice as much sodium as the minimum, because their sodium channels never fully close between spikes.[877] Speed has a price in energy as well as in space, and the brain pays it only where it needs speed.

Why Spikes

If information is so expensive, it is fair to ask why the brain uses spikes at all. A graded signal, a voltage that rises and falls smoothly with its input, carries more information than a train of all-or-none pulses. The interneurons behind the blowfly's photoreceptors use graded signals, and they transmit up to about 1,650 bits per second, three to five times the best rates measured in neurons that spike.[878]

The difficulty is distance. Chapter 2 described how a voltage spreading passively along a thin fiber fades over a short distance, and any noise added at each point along the way accumulates with it. An analog signal cannot be cleaned up; whatever noise has been added is now part of the signal. A spike is regenerated at every point it reaches, so that what arrives at the far end of the axon is as crisp as what left, much as a digital signal can be copied indefinitely without loss. Rahul Sarpeshkar, an engineer who designs low-power circuits, argued in 1998 that the most efficient way to compute with noisy components is neither analog nor digital but a hybrid: compute in graded form over short distances, where graded signals are cheap and rich, and restore the signal to discrete form before noise accumulates.[879]

That is roughly how nervous systems are built. Where distances are short, they compute in graded form. Photoreceptors and most bipolar cells of the retina signal mostly with graded voltages, although some bipolar cells turn out to spike as well.[880] Most of the neurons of the nematode worm *Caenorhabditis elegans*, an animal about a millimeter long, signal with graded potentials, and the worm has no voltage-

[875] Harris, J. J., Jolivet, R., Engl, E., & Attwell, D. (2015). Energy-efficient information transfer by visual pathway synapses. *Current Biology*, 25(24), 3151–3160.

[876] Alle, H., Roth, A., & Geiger, J. R. P. (2009). Energy-efficient action potentials in hippocampal mossy fibers. *Science*, 325(5946), 1405–1408.

[877] Carter, B. C., & Bean, B. P. (2009). Sodium entry during action potentials of mammalian neurons: Incomplete inactivation and reduced metabolic efficiency in fast-spiking neurons. *Neuron*, 64(6), 898–909.

[878] de Ruyter van Steveninck, R. R., & Laughlin, S. B. (1996). The rate of information transfer at graded-potential synapses. *Nature*, 379(6566), 642–645; Simmons, P. J., & de Ruyter van Steveninck, R. (2005). Reliability of signal transfer at a tonically transmitting, graded potential synapse of the locust ocellar pathway. *Journal of Neuroscience*, 25(33), 7529–7537.

[879] Sarpeshkar, R. (1998). Analog versus digital: Extrapolating from electronics to neurobiology. *Neural Computation*, 10(7), 1601–1638.

[880] Baden, T., Esposti, F., Nikolaev, A., & Lagnado, L. (2011). Spikes in retinal bipolar cells phase-lock to visual stimuli with millisecond precision. *Current Biology*, 21(22), 1859–1869.

gated sodium channels at all; a few of its neurons fire all-or-none spikes carried by calcium instead.[881] Within a single cortical neuron, as Chapter 2 described, the dendrites combine their inputs in graded and partly regenerative ways before the cell commits to a spike. The spike is what long distances cost. A large animal whose neurons must send signals across centimeters or meters has to digitize them, and once it has done so, a spike is a cheap way to say a great deal with a single, sparing event.

Speed Costs Space

A spike regenerates as it travels, but it travels slowly, and its speed depends on the width of the fiber that carries it. In a bare axon, speed rises only with the square root of the diameter, so doubling the speed requires a fiber four times as wide. The squid's way of escaping quickly is to grow a giant axon. The one Alan Hodgkin and Andrew Huxley used in 1952 was 476 micrometers across, nearly half a millimeter, and conducted at 21.2 meters per second.[882]

Vertebrates found a different solution. Wrapping an axon in myelin, the fatty insulation that Chapter 2 described, forces the impulse to jump between gaps in the sheath, and the speed of a myelinated fiber rises in proportion to its diameter rather than its square root. In the nerves of a cat, conduction speed is about six meters per second for each micrometer of the fiber's overall diameter.[883] By that rule a myelinated fiber about 3.5 micrometers across should match the squid's giant axon, although the comparison is only approximate, because the squid's nerve works at the temperature of the sea and the cat's at 37 degrees Celsius.

Figure 13.3 shows the two scaling rules and what they imply. At the squid's speed, a bare axon needs about 18,000 times the cross-section of a myelinated one. Daniel Hartline and David Colman, reviewing the evolution of fast conduction, put it more vividly: to make bare fibers ten times faster by widening them, they would have to be a hundred times wider, and "Imagine yourself with a 100-fold thicker spinal cord!"[884] Myelin also cuts the energy cost of each impulse, by several hundredfold on their estimate, because only the gaps in the sheath need to be recharged.

[881] Goodman, M. B., Hall, D. H., Avery, L., & Lockery, S. R. (1998). Active currents regulate sensitivity and dynamic range in *C. elegans* neurons. *Neuron*, 20(4), 763–772; Liu, Q., Kidd, P. B., Dobosiewicz, M., & Bargmann, C. I. (2018). *C. elegans* AWA olfactory neurons fire calcium-mediated all-or-none action potentials. *Cell*, 175(1), 57–70.e17.

[882] Hodgkin, A. L., & Huxley, A. F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4), 500–544.

[883] Hursh, J. B. (1939). Conduction velocity and diameter of nerve fibers. *American Journal of Physiology*, 127(1), 131–139.

[884] Hartline, D. K., & Colman, D. R. (2007). Rapid conduction and the evolution of giant axons and myelinated fibers. *Current Biology*, 17(1), R29–R35.

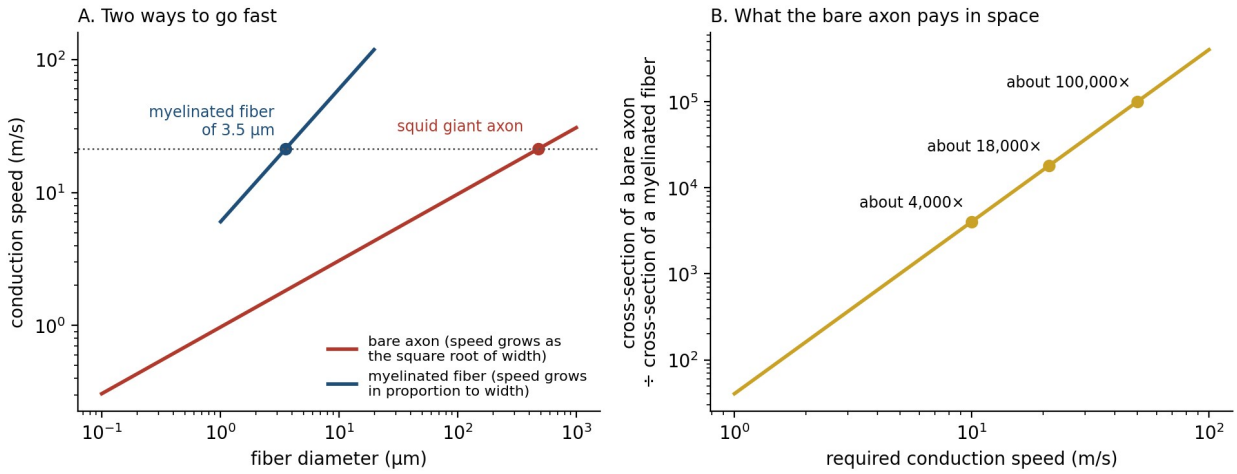


Figure 13.3. Speed costs space (computed). A: conduction speed rises with the square root of fiber diameter in a bare axon (calibrated to the squid giant axon, 476 μm and 21.2 m/s) and in proportion to diameter in a myelinated fiber (about 6 m/s per μm, after Hursh, 1939). A myelinated fiber about 3.5 μm across matches the squid's axon. B: the ratio of the cross-sections the two kinds of fiber need to reach a given speed. The temperatures of the two calibrations differ, so the comparison is approximate.

Even myelin does not make speed free, and the brain buys it sparingly. Samuel Wang and his colleagues measured the axons of white matter in 14 species, from shrews to whales, spanning nearly five orders of magnitude in brain size. Larger brains shifted toward a greater share of myelinated axons and widened their fastest fibers, so that the quickest signals still crossed the neocortex in about one to five milliseconds; but most axons stayed thin, and the distribution of axon sizes implied that the average fiber could sustain only about seven spikes a second.[885] The brain has a few fast lines and a great many slow ones, and it trades volume against delay and energy in deciding which is which.

The slow lines shape how a large brain is organized. Most of the fibers that join the two cerebral hemispheres through the corpus callosum are thinner than a micrometer. James Ringo and his colleagues calculated that in a human the average myelinated fiber linking the two temporal lobes would take more than 25 milliseconds to carry a signal across, and conjectured that this delay is why large brains concentrate time-critical skills, such as the rapid processing of speech sounds, in a single hemisphere: a computation that must pass back and forth between the hemispheres many times would be too slow.[886] The idea is a conjecture rather than a demonstration, but it turns hemispheric specialization, often treated as a mystery, into a predictable consequence of size.

Wiring also takes up space as a brain grows. Kechen Zhang and Terrence Sejnowski showed that across mammals the volume of white matter, the wiring, grows faster than the volume of gray matter, the cell bodies and synapses it connects. They derived an exponent of four-thirds from geometry and measured a slightly smaller one, about 1.23, across species from shrews to elephants.[887] A large brain that kept every neuron as well connected as in a small one would be mostly wire, so large brains must connect

[885] Wang, S. S.-H., Shultz, J. R., Burish, M. J., Harrison, K. H., Hof, P. R., Towns, L. C., Wagers, M. W., & Wyatt, K. D. (2008). Functional trade-offs in white matter axonal scaling. *Journal of Neuroscience*, 28(15), 4047–4056.

[886] Ringo, J. L., Doty, R. W., Demeter, S., & Simard, P. Y. (1994). Time is of the essence: A conjecture that hemispheric specialization arises from interhemispheric conduction delay. *Cerebral Cortex*, 4(4), 331–343.

[887] Zhang, K., & Sejnowski, T. J. (2000). A universal scaling law between gray matter and white matter of cerebral cortex. *Proceedings of the National Academy of Sciences*, 97(10), 5621–5626.

each neuron to a smaller fraction of the others and rely more on local circuits joined by a sparser set of long links.

Even the brain's folds are a matter of physical constraint. Bruno Mota and Suzana Herculano-Houzel found that across mammals the degree of folding is predicted not by the number of neurons but by the cortex's surface area and thickness, following a law that also describes how a sheet of paper crumples when it is squeezed into a ball.[888] Tuomas Tallinen, L. Mahadevan, and their colleagues made the point with a model. They 3D-printed a gel copy of a smooth fetal brain from an MRI taken at 22 weeks of gestation, coated it with a thin layer of a second gel, and immersed it in a solvent that made the outer layer swell faster than the core. Within minutes the surface buckled into folds similar to those that form in the human fetal brain over the following weeks.[889] No gene specifies each fold. A sheet that grows faster than what lies beneath it has to buckle, as Chapter 0 described for other self-organizing structures, and evolution works with the pattern physics provides.

Placing the Parts

If wire is expensive, the parts of a nervous system should sit where they minimize it, and in one celebrated case they appear to. In 1994 Christopher Cherniak examined the nematode worm, whose complete wiring diagram had been mapped from electron micrographs. Its nervous system has 11 clusters of neurons that can be placed in different orders along the body's length, which allows 39,916,800 possible arrangements. Of those, the arrangement the worm actually has was the one that minimized the total length of the connecting fibers.[890] Dmitri Chklovskii and Alexei Koulakov argued that the maps of the visual cortex, in which neurons with similar preferences sit side by side, can be read the same way, as arrangements that keep connected neurons close.[891]

Wire minimization is not the whole story. Marcus Kaiser and Claus Hilgetag rearranged the positions of neurons in the worm and of areas in the macaque monkey's cortex by computer to find the layouts with the least wire. Total wiring could be cut by 48 percent in the worm and 32 percent in the macaque. The savings were possible because both networks contain long-distance connections that a wire-minimizing layout would shorten, and the price of removing or shortening them was that signals would need many more steps to get from one part of the network to another. The real networks had paths almost as short as networks designed for nothing but short paths.[892] The nervous system pays for some expensive wire in order to keep everything only a few steps from everything else.

Figure 13.4 shows the trade-off in a simulation. It places 100 neurons at random in a square, connects each to its nearest neighbors, and then reroutes a growing fraction of the connections to random distant partners. With all-local wiring the average path between two neurons takes about nine steps. Making five percent of the connections long-range adds about a quarter to the total wire and cuts the average

[888] Mota, B., & Herculano-Houzel, S. (2015). Cortical folding scales universally with surface area and thickness, not number of neurons. *Science*, 349(6243), 74–77.

[889] Tallinen, T., Chung, J. Y., Rousseau, F., Girard, N., Lefèvre, J., & Mahadevan, L. (2016). On the growth and form of cortical convolutions. *Nature Physics*, 12(6), 588–593.

[890] Cherniak, C. (1994). Component placement optimization in the brain. *Journal of Neuroscience*, 14(4), 2418–2427.

[891] Chklovskii, D. B., & Koulakov, A. A. (2004). Maps in the brain: What can we learn from them? *Annual Review of Neuroscience*, 27, 369–392.

[892] Kaiser, M., & Hilgetag, C. C. (2006). Nonoptimal component placement, but short processing paths, due to long-distance projections in neural systems. *PLoS Computational Biology*, 2(7), e95.

path to six steps; making ten percent long-range adds half and cuts it to about five. Rewiring everything at random brings the path down only to about 3.7 steps, at nearly six times the wire. The steep part of the curve, where a few long connections buy most of the gain, is where Chapter 4's small-world networks sit, and Edward Bullmore and Olaf Sporns have argued that the brain's networks are best understood as this kind of economy, a negotiated trade between the cost of wiring and the value of short paths, robustness, and hubs.[893]

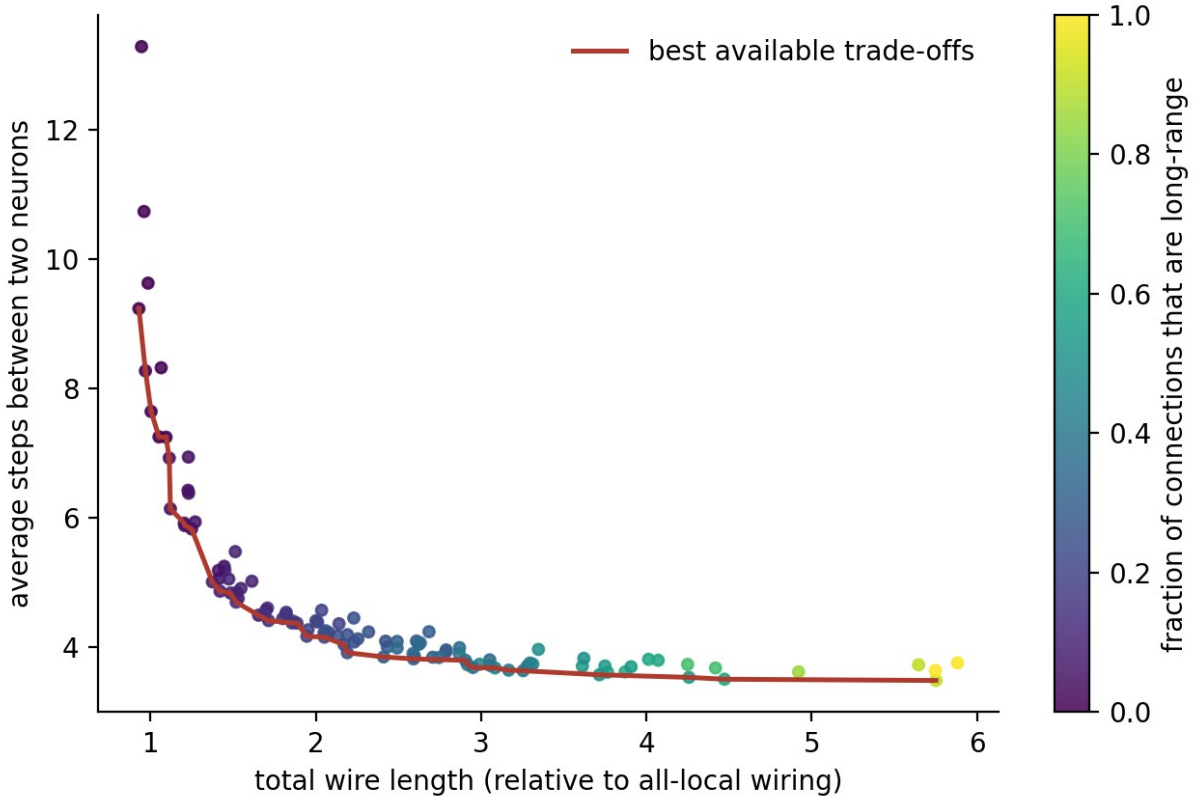


Figure 13.4. Wire against path length (computed). One hundred neurons are placed at random and each is connected to its nearest neighbors; a growing fraction of connections is then rerouted to random distant partners (color). Each dot is one network. The red line marks the best available trade-offs: no network does better on both counts. A few long-range connections buy most of the reduction in path length for a modest cost in wire.

Wire cost may also help explain why the brain is divided into modules and arranged as a hierarchy of areas. Jeff Clune, Jean-Baptiste Mouret, and Hod Lipson evolved small networks in a computer to recognize patterns on an eight-pixel "retina" whose left and right halves could each contain an object. Networks selected for performance alone evolved into tangles with little modular structure. When each connection also carried a cost, the networks evolved separate modules for the two halves, solved the task at least as well, and, when the task was changed, adapted in about a twentieth as many generations.[894] Henok Mengistu, Joost Huizinga, Mouret, and Clune then found that on problems with nested structure the same cost also produced hierarchy, modules built out of smaller modules, which solved the problems

[893] Bullmore, E., & Sporns, O. (2012). The economy of brain network organization. *Nature Reviews Neuroscience*, 13(5), 336–349.

[894] Clune, J., Mouret, J.-B., & Lipson, H. (2013). The evolutionary origins of modularity. *Proceedings of the Royal Society B: Biological Sciences*, 280(1755), 20122863.

and adapted to new ones faster.[895] These are small, abstract networks, and they show only that a cost on wiring is enough to produce modules and a hierarchy, not that it is what produced the brain's.

Many Jobs at Once

The red line in Figure 13.4 has a name. When a design must serve two goals at once, the designs worth considering are those that cannot be improved on one goal without becoming worse on the other, and economists and engineers call the set of such designs a Pareto front, after the economist Vilfredo Pareto. Nothing on the front is optimal in the everyday sense; every point on it is a different compromise.

Uri Alon and his colleagues showed in 2012 that this idea predicts the shapes of animals. If an organism must perform several tasks, and each task is best served by one particular design, an "archetype," then the designs that no other design beats at every task are blends of the archetypes, and they fill the line, triangle, or polygon whose corners the archetypes occupy, as Figure 13.5 shows for three tasks. The beaks and bodies of Darwin's ground finches fall within such a triangle, with corners for probing cactus flowers and insects, cracking large hard seeds, and eating small soft ones; the wings of insect-eating bats fall within another.[896] Each species sits at its own compromise, and the compromise is the design.

[895] Mengistu, H., Huizinga, J., Mouret, J.-B., & Clune, J. (2016). The evolutionary origins of hierarchy. *PLOS Computational Biology*, 12(6), e1004829.

[896] Shoval, O., Sheftel, H., Shinar, G., Hart, Y., Ramote, O., Mayo, A., Dekel, E., Kavanagh, K., & Alon, U. (2012). Evolutionary trade-offs, Pareto optimality, and the geometry of phenotype space. *Science*, 336(6085), 1157–1160.

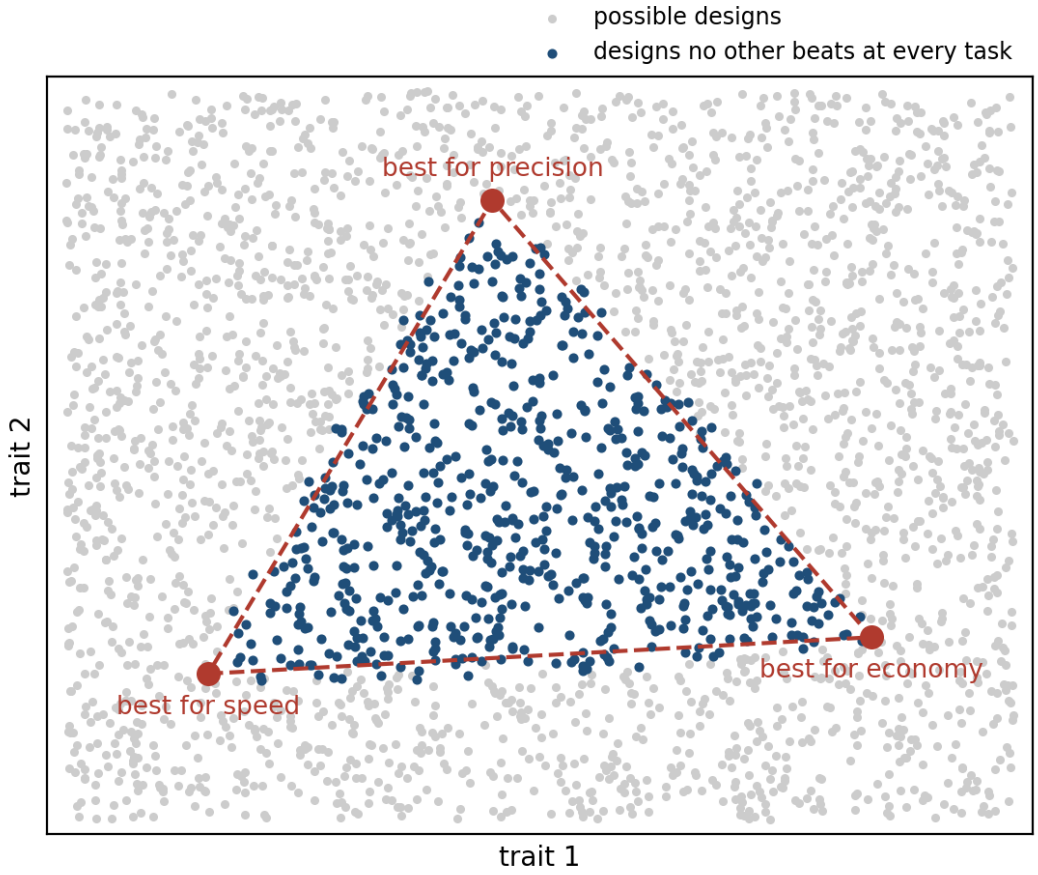


Figure 13.5. Many jobs, one design (computed, after Shoval et al., 2012). Gray dots are possible designs, described by two traits. Each of three tasks is best served by one design (red), and performance at each task falls with distance from it. The designs that no other design beats at every task (blue) fill the triangle between the three. Which point an organism occupies depends on how much each task matters to it.

The brain's components face the same geometry. Chapter 1 described degeneracy, the capacity of different combinations of parts to produce the same function, and it reaches down to single cells and small circuits. Astrid Prinz, Dirk Bucher, and Eve Marder simulated millions of versions of a small rhythm-generating circuit from the stomach of a crustacean and found that very different combinations of ion channels and synaptic strengths produced nearly the same rhythm.[897] Peter Jedlicka, Alexander Bird, and Hermann Cuntz have proposed that the configurations nature actually uses are those on the front between economy and effectiveness, so that among all the mixtures that would work, a neuron uses one that wastes as little energy as possible for what it must do.[898] The proposal is a framework rather than a finding, but it gives degeneracy a direction.

Trade-offs also explain a pattern of failure. Marie Csete and John Doyle, comparing cells with engineered systems such as the Boeing 777, argued that complex regulated systems are "robust yet fragile": extremely robust to the disturbances they were built for, and vulnerable to rare ones they were not,

[897] Prinz, A. A., Bucher, D., & Marder, E. (2004). Similar network activity from disparate circuit parameters. *Nature Neuroscience*, 7(12), 1345–1352.

[898] Jedlicka, P., Bird, A. D., & Cuntz, H. (2022). Pareto optimality, economy–effectiveness trade-offs and ion channel degeneracy: Improving population modelling for single neurons. *Open Biology*, 12(7), 220073.

because robustness in one place is paid for with fragility in another.[899] The hubs of Chapter 4 are an example. They make communication across the brain fast and economical, and they are where damage, across many brain disorders, does the most harm.

How Big a Brain

The most visible trade-off is size. A bigger brain buys more neurons, and neurons are expensive in a way that can be stated simply. Suzana Herculano-Houzel estimated from measurements in rodents and primates that each billion neurons costs about 6 kilocalories a day, roughly the same in every species studied, so that the human brain's 86 billion neurons cost about 500 kilocalories a day, or about 25 watts, a little above the round 20 watts of Chapter 1, which comes from a different method of estimation.[900] Karina Fonseca-Azevedo and Herculano-Houzel calculated that an ape eating raw plant food could not easily afford many more neurons than great apes have: a primate with a human-sized brain living on such a diet would need to feed for more than nine hours a day, and they proposed that cooking, which makes food easier to digest, is what allowed the human brain to grow.[901] Others dispute the arithmetic: Alianda Cornélio and her colleagues argued that the calculation assumed an unrealistically low rate of foraging, and that a raw diet could have supported a large brain without such long days.[902]

The bill is highest in childhood. Christopher Kuzawa and his colleagues combined brain-imaging measurements of glucose use with data on brain and body size across development. At its peak, around the age of four, the brain takes up glucose equivalent to about two-thirds of the child's resting metabolic rate and about 43 percent of the child's total daily energy needs, and the ages at which the brain's demand is highest are the ages at which the body grows most slowly; the brain's total daily use of glucose peaks a year later, at about five.[903] The authors argue that the long, slow human childhood is partly a way of paying for the brain. The glucose figures are not a precise energy budget, since some glucose is used for building rather than burned, but the trade-off with body growth is clear.

Where did the extra energy come from? Leslie Aiello and Peter Wheeler proposed in 1995 that humans paid for their brains by shrinking another expensive organ, the gut.[904] When Ana Navarrete, Carel van Schaik, and Karin Isler tested the idea across 100 species of mammals, they found no trade-off between brain size and the size of the gut or any other expensive organ, although species with larger brains did tend to store less fat.[905] Herman Pontzer and his colleagues then measured the daily energy use of humans and other apes with a technique that tracks the body's turnover of labeled water. After body size and activity were taken into account, humans burned about 400 kilocalories a day more than

[899] Csete, M. E., & Doyle, J. C. (2002). Reverse engineering of biological complexity. *Science*, 295(5560), 1664–1669.

[900] Herculano-Houzel, S. (2012). The remarkable, yet not extraordinary, human brain as a scaled-up primate brain and its associated cost. *Proceedings of the National Academy of Sciences*, 109(Suppl. 1), 10661–10668.

[901] Fonseca-Azevedo, K., & Herculano-Houzel, S. (2012). Metabolic constraint imposes tradeoff between body size and number of brain neurons in human evolution. *Proceedings of the National Academy of Sciences*, 109(45), 18571–18576.

[902] Cornélio, A. M., de Bittencourt-Navarrete, R. E., de Bittencourt Brum, R., Queiroz, C. M., & Costa, M. R. (2016). Human brain expansion during evolution is independent of fire control and cooking. *Frontiers in Neuroscience*, 10, 167.

[903] Kuzawa, C. W., Chugani, H. T., Grossman, L. I., Lipovich, L., Muzik, O., Hof, P. R., Wildman, D. E., Sherwood, C. C., Leonard, W. R., & Lange, N. (2014). Metabolic costs and evolutionary implications of human brain development. *Proceedings of the National Academy of Sciences*, 111(36), 13010–13015.

[904] Aiello, L. C., & Wheeler, P. (1995). The expensive-tissue hypothesis: The brain and the digestive system in human and primate evolution. *Current Anthropology*, 36(2), 199–221.

[905] Navarrete, A., van Schaik, C. P., & Isler, K. (2011). Energetics and the evolution of human brain size. *Nature*, 480(7375), 91–93.

chimpanzees and bonobos, much of it in a higher basal metabolism, and humans also carried more fat.[906] Humans seem to have paid for their brains less by cutting elsewhere than by raising the whole budget, with fat as a reserve against bad days.

Size also meets the constraint of birth. The traditional view, the "obstetric dilemma," holds that human babies are born helpless because a bigger head would not pass through a pelvis narrow enough for efficient walking. Holly Dunsworth and her colleagues argued instead that birth comes when the mother's metabolism reaches the limit of what she can sustain, and Anna Warrener and her colleagues found that a wider pelvis does not, in fact, make walking or running less efficient.[907] The metabolic account has been criticized in turn: Cédric Cordey, Nicole Webb, and Martin Haeusler noted that the proposed ceiling rests on a single study of 12 women, and that in a third of the studies they reviewed, pregnant women exceeded it before the end of pregnancy.[908] The question is open, and its openness is itself instructive, since a single feature of the brain's life history can plausibly be explained by two different constraints.

At the extremes, the constraints are stark. The elephantnose fish, an African fish that senses its world with weak electric fields, has a brain estimated to use roughly 60 percent of the oxygen the whole animal consumes, three times the share in any other vertebrate studied, humans included.[909] At the other end, the parasitic wasp *Megaphragma mymaripenne*, one of the smallest insects known at about 200 micrometers long, comparable in size to some single-celled organisms, has about 7,400 neurons. During its final stage of development more than 95 percent of them lose their nuclei, which would otherwise take up too much room, and the adult flies, finds hosts, and lays its eggs with a nervous system made mostly of cells that can no longer make new proteins.[910]

A honeybee, with fewer than a million neurons, can learn to count small numbers of objects and to sort pictures into categories, and Lars Chittka and Jeremy Niven have argued that bigger brains buy more replication of similar circuits, more precision, and more storage, rather than kinds of computation a small brain cannot do at all.[911]

Different Histories, Same Answers

The constraints do not act on a blank slate. Chapter 1 quoted François Jacob's remark that evolution is a tinkerer, and some of the brain's features are best explained by its history. One way to separate history from constraint is to compare lineages that faced similar problems from different starting points. Where

[906] Pontzer, H., Brown, M. H., Raichlen, D. A., Dunsworth, H., Hare, B., Walker, K., Luke, A., Dugas, L. R., Durazo-Arvizu, R., Schoeller, D., Plange-Rhule, J., Bovet, P., Forrester, T. E., Lambert, E. V., Thompson, M. E., Shumaker, R. W., & Ross, S. R. (2016). Metabolic acceleration and the evolution of human brain size and life history. *Nature*, 533(7603), 390–392.

[907] Dunsworth, H. M., Warrener, A. G., Deacon, T., Ellison, P. T., & Pontzer, H. (2012). Metabolic hypothesis for human altriciality. *Proceedings of the National Academy of Sciences*, 109(38), 15212–15216; Warrener, A. G., Lewton, K. L., Pontzer, H., & Lieberman, D. E. (2015). A wider pelvis does not increase locomotor cost in humans, with implications for the evolution of childbirth. *PLOS ONE*, 10(3), e0118903.

[908] Cordey, C., Webb, N. M., & Haeusler, M. (2023). Take it to the limit: The limitations of energetic explanations for birth timing in humans. *Evolution, Medicine, and Public Health*, 11(1), 415–428.

[909] Nilsson, G. E. (1996). Brain and body oxygen requirements of *Gnathonemus petersii*, a fish with an exceptionally large brain. *Journal of Experimental Biology*, 199(3), 603–607.

[910] Polilov, A. A. (2012). The smallest insects evolve anucleate neurons. *Arthropod Structure & Development*, 41(1), 29–34.

[911] Chittka, L., & Niven, J. (2009). Are bigger brains better? *Current Biology*, 19(21), R995–R1008.

the same solution appears independently, the constraint is probably doing the work; where a solution persists unchanged, history probably is.

The basal ganglia, the loops that select actions and that dopamine teaches (Chapters 3, 4, 8, and 9), are an example of persistence. Sten Grillner and Brita Robertson have reviewed evidence from the lamprey, a jawless fish whose ancestors split from ours about 560 million years ago, that its basal ganglia have the same basic organization as a mammal's: a striatum with a direct and an indirect pathway, inputs from the forebrain and thalamus, output nuclei that hold actions in check, and a dopamine input that signals value.[912] The solution to selecting one action among many was found early and has been kept.

Birds show the opposite case. Their forebrain lacks the layered sheet of the mammalian cortex, and for a century this was taken as a sign of simpler minds. Seweryna Olkowicz, Pavel Němec, and their colleagues counted the neurons in the brains of birds and found that the brains of parrots and songbirds hold on average about twice as many neurons as primate brains of the same mass, and put a larger share of them in the forebrain. A raven has about 1.2 billion neurons in its pallium, the region corresponding to the cortex, more than a capuchin monkey, and a blue-and-yellow macaw about 1.9 billion, more than a rhesus macaque, in brains a fraction of the size.[913] Flight puts a premium on weight, and birds answered with small, densely packed neurons.

Martin Stacho, Onur Güntürkün, and their colleagues then traced fibers in the forebrains of pigeons and owls and found circuits repeated in column-like arrangements across the layer-like nuclei, organized much like the cortex's canonical circuit.[914] Studies of cell types and development published in 2025, among them one led by Eneritz Rueda-Alaña and Fernando García-Moreno, suggest that the resemblance is largely convergent: birds and mammals built similar circuits out of different developmental starting material.[915] When two lineages separated for more than 300 million years arrive at similar solutions, the solution probably reflects what the problem requires.

The octopus is the most distant test. Its last common ancestor with vertebrates was a simple worm-like animal, and yet it has a large brain, about 500 million neurons by J. Z. Young's classic count, with most of them in the arms rather than the head.[916] The arms carry out much of their own control, as Chapter 4's central pattern generators do in a spinal cord, and the central brain issues more general commands. The octopus is a reminder that the vertebrate arrangement, with a head that does most of the computing, is one solution among several.

History also leaves features that no constraint requires. The vertebrate retina is inverted: light must pass through the layers of neurons and blood vessels before reaching the photoreceptors at the back,

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[912] Grillner, S., & Robertson, B. (2016). The basal ganglia over 500 million years. *Current Biology*, 26(20), R1088–R1100; Stephenson-Jones, M., Samuelsson, E., Ericsson, J., Robertson, B., & Grillner, S. (2011). Evolutionary conservation of the basal ganglia as a common vertebrate mechanism for action selection. *Current Biology*, 21(13), 1081–1091.

[913] Olkowicz, S., Kocourek, M., Lučan, R. K., Porteš, M., Fitch, W. T., Herculano-Houzel, S., & Němec, P. (2016). Birds have primate-like numbers of neurons in the forebrain. *Proceedings of the National Academy of Sciences*, 113(26), 7255–7260.

[914] Stacho, M., Herold, C., Rook, N., Wagner, H., Axer, M., Amunts, K., & Güntürkün, O. (2020). A cortex-like canonical circuit in the avian forebrain. *Science*, 369(6511), eabc5534.

[915] Rueda-Alaña, E., Senovilla-Ganzo, R., Grillo, M., Vázquez, E., Marco-Salas, S., Gallego-Flores, T., Ordeñana-Manso, A., Ftara, A., Escobar, L., Benguría, A., Quintas, A., Dopazo, A., Rábano, M., Vivanco, M. d. M., Aransay, A. M., Garrigos, D., Toval, Á., Ferrán, J. L., Nilsson, M., Encinas-Pérez, J. M., De Pittà, M., & García-Moreno, F. (2025). Evolutionary convergence of sensory circuits in the pallium of amniotes. *Science*, 387(6735), eadp3411.

[916] Young, J. Z. (1963). The number and sizes of nerve cells in *Octopus*. *Proceedings of the Zoological Society of London*, 140(2), 229–254; Hochner, B. (2012). An embodied view of octopus neurobiology. *Current Biology*, 22(20), R887–R892.

and the fibers leaving the eye must punch through the retina at the blind spot that Chapter 6 described. The octopus eye, evolved independently, has its photoreceptors facing the light and no blind spot. Vertebrates have partly compensated. Kristian Franze, Jochen Guck, and their colleagues showed that the Müller cells, glial cells that span the retina, guide light through it like optical fibers, and Amichai Labin, Erez Ribak, and their colleagues found that they funnel the red and green light used in daylight vision onto the cones.[917] Some researchers argue that the arrangement is not a defect at all, since it places the photoreceptors, among the most energy-hungry cells in the body, directly against the tissue that feeds them and renews their pigment.[918] Even so, the blind spot and the layers that light must cross are costs the octopus does not pay, and the compensation shows what a tinkerer does with a design it cannot discard.

The Bottleneck of the Genome

Chapter 10 described Anthony Zador's argument that an animal arrives in the world with much of its competence already built in. Zador has also put a number on the difficulty this raises. The human genome has about three billion nucleotides, enough to encode no more than about a gigabyte of information, "an hour or so of streaming video." The brain has about 86 billion neurons, which Zador rounds to 100 billion, each with more than a thousand synapses. Naming the target of a single connection among 100 billion neurons takes about 37 bits, so a complete list of the brain's 100 trillion connections would take about 3.7×10^{15} bits. Even if every nucleotide in the genome were devoted to the brain's wiring, it would fall short by a factor of more than half a million on the round numbers used here, and by "at least six orders of magnitude" on Zador's own reckoning.[919]

Direct counts are consistent with his round number of connections: the human neocortex alone holds about 164 trillion synapses.[920] For the nematode worm, with 302 neurons and about 7,000 synapses, the arithmetic runs the other way, and its genome could list its entire wiring diagram thousands of times over. Figure 13.6 sets the two cases side by side.

[917] Franze, K., Grosche, J., Skatchkov, S. N., Schinkinger, S., Foja, C., Schild, D., Uckermann, O., Travis, K., Reichenbach, A., & Guck, J. (2007). Müller cells are living optical fibers in the vertebrate retina. *Proceedings of the National Academy of Sciences*, 104(20), 8287–8292; Labin, A. M., Safuri, S. K., Ribak, E. N., & Perlman, I. (2014). Müller cells separate between wavelengths to improve day vision with minimal effect upon night vision. *Nature Communications*, 5, 4319.

[918] Baden, T., & Nilsson, D.-E. (2022). Is our retina really upside down? *Current Biology*, 32(7), R300–R303.

[919] Zador, A. M. (2019). A critique of pure learning and what artificial neural networks can learn from animal brains. *Nature Communications*, 10, 3770.

[920] Tang, Y., Nyengaard, J. R., De Groot, D. M. G., & Gundersen, H. J. G. (2001). Total regional and global number of synapses in the human brain neocortex. *Synapse*, 41(3), 258–273.

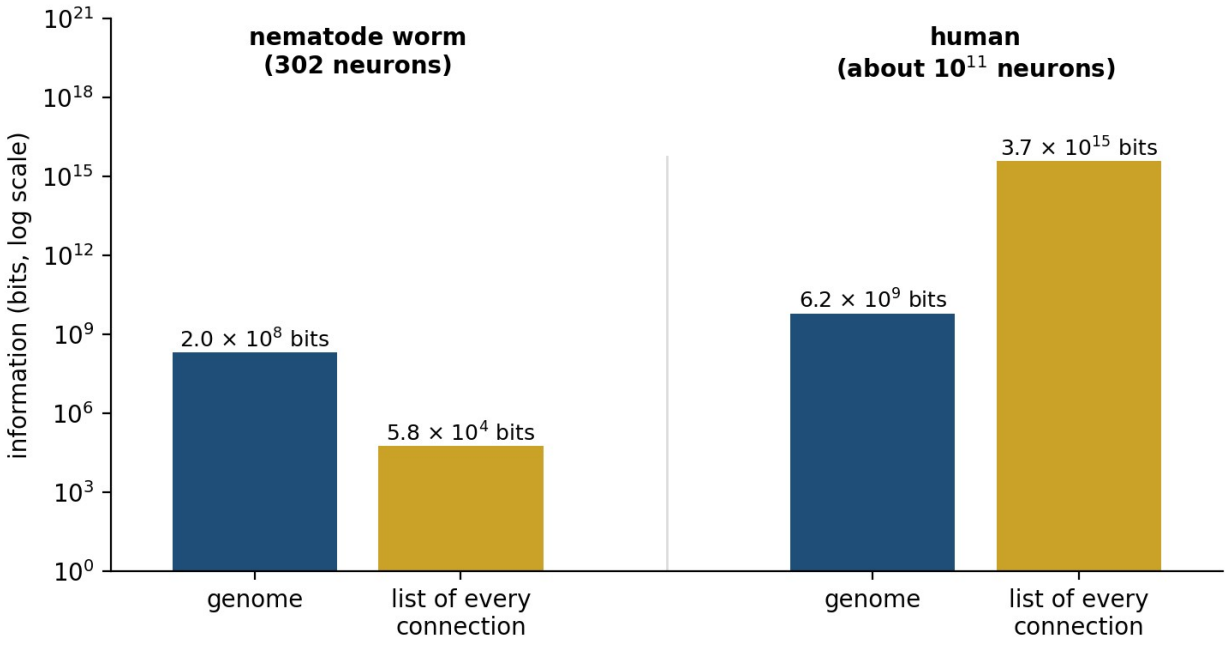


Figure 13.6. The genome bottleneck (computed, after Zador, 2019). Blue: the information a genome can hold, at two bits per base pair. Gold: the information needed to list every connection by naming its target neuron. The worm's genome could list its wiring thousands of times over; the human genome falls short by a factor of about 600,000.

The calculation is an upper bound on what a wiring list would need, not an estimate of what development uses, and that is the point. A large brain cannot be specified; it must be grown. The genome encodes rules rather than connections: which kinds of cells to make, how they should migrate, which chemical gradients their axons should follow, which learning rules their synapses should obey. Self-organization, as Chapter 0 described, and experience, as the sensitive periods of Chapter 1 showed, supply the rest. Seen this way, learning is not an optional addition to a brain that could in principle have been built complete. It is the only way a brain of this size can be finished.

The bottleneck may also be useful. Sergey Shuvaev, Alexei Koulakov, Zador, and their colleagues trained a small "genome" network whose only job was to generate the connection weights of a much larger network, and asked how well the large network performed before it had learned anything. A network for reading handwritten digits, compressed through a genome 322 times smaller than itself, recognized 94 percent of digits with no training at all, against 98 percent after full training; a network for playing the Atari game BeamRider, compressed about 500-fold, played nearly perfectly from its first game.[921] On harder problems the compressed networks also transferred more readily to new versions of a task, although on simple ones they did not. A genome that can store only a little must store what generalizes.

Trade-offs That Cannot Be Escaped

Looking back across the book, the same structure keeps reappearing. Almost every mechanism the earlier chapters described sits on a trade-off that no amount of engineering could remove. Figure 13.7 collects them.

[921] Shuvaev, S., Lachi, D., Koulakov, A., & Zador, A. (2024). Encoding innate ability through a genomic bottleneck. *Proceedings of the National Academy of Sciences*, 121(38), e2409160121.

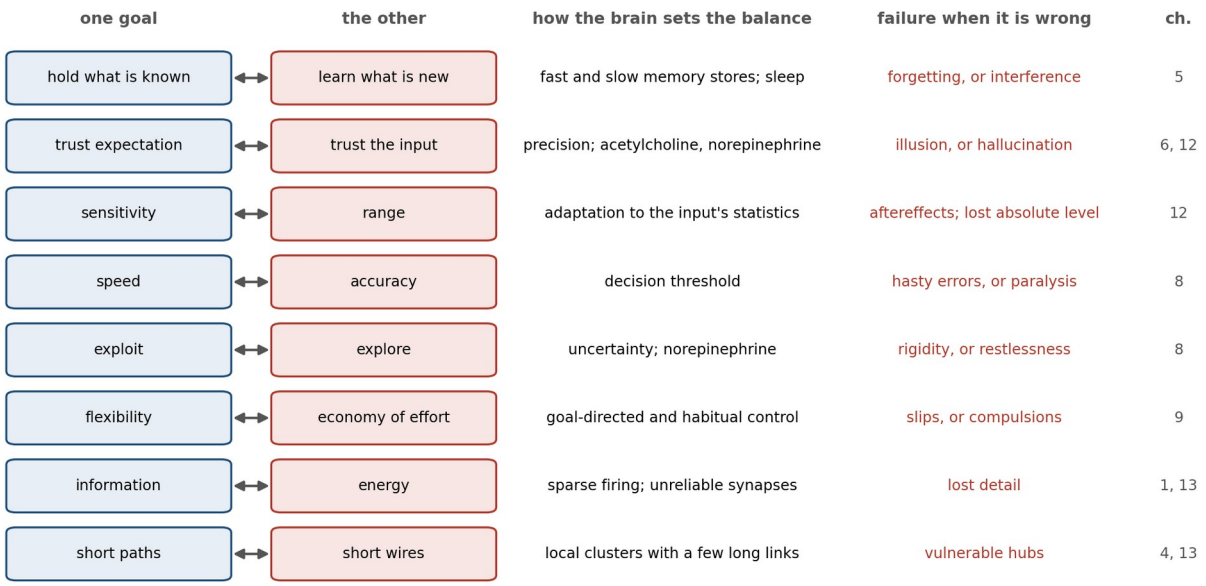


Figure 13.7. The trade-offs of the earlier chapters. Each row pairs two goals that cannot both be maximized, the mechanism by which the brain sets or shifts its position between them, and the characteristic failure when the position goes wrong.

Some are trade-offs between holding and changing. A memory system must be stable enough to keep what it has learned and plastic enough to learn more, and Chapter 5 described the brain's answer: a fast hippocampal store and a slow cortical one, with sleep to move memories between them. Here the constraint is statistical rather than physical. A perceptual system must weigh what it expects against what it receives, and Chapters 6 and 12 described the price of getting the balance wrong in either direction, from illusions to hallucinations. A neuron that adapts to the range of its input gains sensitivity within that range at the cost of losing the absolute level, which is why, as Chapter 12 described, a waterfall can make a still rock face appear to drift.

Others are trade-offs between acting and waiting. Chapter 8 described the trade between speed and accuracy in a decision, and between exploiting what is known and exploring what is not. Chapter 9 described the trade between the flexibility of goal-directed action and the economy of habit. Each has a characteristic failure: hasty errors and paralysis, compulsive novelty-seeking and rigid routine, behavior that persists when its purpose has gone.

The brain does not sit at one fixed point on these fronts. One of the most general things the earlier chapters showed is that it moves along them as conditions change. Chapter 3's neuromodulators are in part the means by which it does so. Norepinephrine raises the learning rate when the world has changed, as Chapter 12 described; acetylcholine shifts the weight from expectation toward incoming evidence; dopamine sets how vigorously the animal acts and how much effort it will spend;[922] and stress, as Chapter 7 described, tips the balance from deliberation toward fast, habitual responses.

[922] Niv, Y., Daw, N. D., Joel, D., & Dayan, P. (2007). Tonic dopamine: Opportunity costs and the control of response vigor. *Psychopharmacology*, 191(3), 507–520.

Built for a Different World

A design shaped by trade-offs is shaped by the world in which they were struck, and it can misfire in a world that has changed.

In the 1950s Niko Tinbergen and his colleagues tested what triggers instinctive behavior by offering animals artificial objects. Herring gull chicks, which beg for food by pecking at a red spot on the parent's bill, pecked more at a thin red rod with white bands near its tip than at a realistic model of a gull's head.[923] An oystercatcher, offered a choice between its own egg and a giant artificial one, tried to brood the giant.[924] Tinbergen called such objects supernormal stimuli. The rule the bird's brain follows, roughly "the bigger the egg, the better," works perfectly in a world where no egg is bigger than an oystercatcher can lay.

The most famous case was found by accident. In 1983 Darryl Gwynne and David Rentz reported that males of a jewel beetle, then identified as *Julodimorpha bakewelli*, were trying to mate with discarded beer bottles. The bottles, known locally as stubbies, were brown and shiny like the female's wing covers, and a row of small raised bumps near the base reflected light much as the dimples on her back do. When the authors put four bottles on the ground, two had attracted beetles within half an hour; a brown wine bottle of a different shade attracted none.[925] The male's rule, something like "large, brown, shiny, and dimpled means female," had never had to deal with an object that was larger, browner, and shinier than any female.

Norman Li, Mark van Vugt, and Stephen Colarelli have argued that much of human behavior should be read the same way: the mechanisms of the human brain were tuned in environments very different from the modern one, and many of the inputs they now receive "differ in quantity and intensity or no longer have the same fitness associations," so that mechanisms that worked well can produce harmful results.[926] Claims of mismatch are easy to make and hard to test, since the ancestral environment cannot be observed directly. But the logic follows from everything in this chapter. A brain that learns slowly, spends its energy carefully, and relies on rules tuned by long experience will be most reliable in a world that resembles the one that shaped it.

Implications for Artificial Intelligence

An artificial system is built under constraints too, but not the same ones, and that difference is the most useful thing this chapter has to offer the people who build such systems.

First, the constraints of silicon are different, and where they coincide with the brain's, engineers have arrived at the brain's solutions largely without copying them. Chapter 11 described Mark Horowitz's table of energy costs: in a chip, fetching a number from memory costs far more than computing with it. One response, met in Chapters 10 and 11, is the mixture of experts, in which each word is sent to only a few of a model's many subnetworks. DeepSeek-V3, released at the end of 2024, has 671 billion

[923] Tinbergen, N., & Perdeck, A. C. (1950). On the stimulus situation releasing the begging response in the newly hatched herring gull chick (*Larus argentatus argentatus* Pont.). *Behaviour*, 3(1), 1–39.

[924] Tinbergen, N. (1951). *The Study of Instinct*. Clarendon Press.

[925] Gwynne, D. T., & Rentz, D. C. F. (1983). Beetles on the bottle: Male buprestids mistake stubbies for females (Coleoptera). *Journal of the Australian Entomological Society*, 22(1), 79–80.

[926] Li, N. P., van Vugt, M., & Colarelli, S. M. (2018). The evolutionary mismatch hypothesis: Implications for psychological science. *Current Directions in Psychological Science*, 27(1), 38–44.

parameters, of which 37 billion, about one in eighteen, are used for each word it processes.[927] This is the economy Levy and Baxter described: a large population of which only a small fraction is active at once, because activity is what costs. It is largely a convergence. The earliest mixtures of experts drew partly on the brain's division of labor, but the modern versions were built to save computation, not to imitate the brain, and like the convergences of Chapter 11 they suggest that the constraint is doing the work.

Second, where the constraints differ, the brain's solutions may not be worth copying. Spikes solve the problem of sending a signal over a long, noisy fiber, and a digital chip already restores its signals to clean levels at every step, the same answer the axon found. The slow conduction that may have pushed time-critical skills into one hemisphere has no counterpart in a machine whose signals cross a chip in nanoseconds. Each feature of the brain should be asked which constraint it answers, and copied only if the machine faces the same one. Spiking hardware, for example, earns its place where it saves energy by doing nothing when nothing happens, not because spikes are how brains work.

Third, the genome bottleneck has no counterpart in how large models are trained. A model's architecture and training procedure are specified in a few thousand lines of code, but nothing forces the knowledge it starts with to pass through a narrow channel, because it starts with none. The result of Shuvaev and his colleagues suggests that forcing such compression could be a way of building in only what generalizes, a kind of inductive bias learned over many tasks rather than designed by hand.

Fourth, the brain is judged on many tasks at once under a fixed budget, and it moves along its trade-offs as conditions change. Artificial systems are usually trained and measured one benchmark at a time, and an improvement on one benchmark is often a move along a front rather than an advance beyond it. The systems of Chapters 10 and 11 that decide how long to think about each problem, spending more computation on hard ones, are a first step toward the brain's habit of choosing a different compromise for each situation.

The constraints that shaped the brain do not disappear when it is put to work in classrooms, in offices, and in front of screens, and what they imply for how people learn and work is the subject of Chapter 14.

The Central Lesson

The brain is not the best possible brain, and it is not a botched one. It is a set of compromises struck under a few hard constraints; the cavefish's lost eyes are one such compromise, struck again when the world changed. Information is expensive, so the brain fires sparsely, lets its synapses fail, and buys only as much precision as it needs. Wire takes space and conducts slowly, so the brain keeps most connections local, insulates a few fast lines, and puts time-critical work where signals need not travel far. A genome cannot specify a large brain, so the brain is grown by rules and finished by learning. Many goals cannot all be met at once, so its designs are positions on fronts between them, and its neuromodulators move it along those fronts as conditions change. All of this was done by a tinkerer working with what it had, so some features are history rather than design, and the way to tell them apart is to ask whether lineages that started from different places arrived at the same answer. Asking why the brain works as it does, in Tinbergen's sense, does not replace the mechanisms of the earlier chapters. It explains why those mechanisms, rather than others, are the ones the brain has.

[927] DeepSeek-AI. (2024). *DeepSeek-V3 technical report* (arXiv:2412.19437). arXiv.

Next Chapter

Chapter 14: Implications for Learning, Work, and Technology: What the Brain's Design Asks of the World We Have Built

Figure and Epigraph Sources

Figure Attribution

Figures 13.1 to 13.7 were drawn for this book. Figures 13.2 to 13.6 are computed: Figure 13.2 applies the energy-efficiency analysis of Levy and Baxter (1996) to a neuron that fires or stays silent in each time bin; Figure 13.3 uses the square-root scaling of conduction speed in bare axons, calibrated to the squid giant axon of Hodgkin and Huxley (1952), and the linear scaling of myelinated fibers given by Hursh (1939); Figure 13.4 simulates networks of 100 neurons with varying fractions of long-range connections, in the spirit of Kaiser and Hilgetag (2006); Figure 13.5 computes the Pareto-optimal designs for three tasks after Shoval et al. (2012); Figure 13.6 applies the calculation of Zador (2019) to the human and the nematode worm. Figures 13.1 and 13.7 are schematic.

Epigraph

Theodosius Dobzhansky, "Nothing in biology makes sense except in the light of evolution," *The American Biology Teacher*, 35(3), 125–129 (1973).

Chapter 14

Implications for Learning, Work, and Technology: What the Brain's Design Asks of the World We Have Built

> *"This discovery of yours will create forgetfulness in the learners' souls, because they will not use their memories; they will trust to the external written characters and not remember of themselves."*
>
> *— Plato, Phaedrus (translated by Benjamin Jowett)*

The Babies Who Learned Mandarin

In the early 2000s Patricia Kuhl, Feng-Ming Tsao, and Huei-Mei Liu brought nine-month-old American infants, all from English-speaking homes, into their laboratory in Seattle for 12 sessions of 25 minutes over four weeks, about five hours in all. In each session a native speaker of Mandarin Chinese read picture books to the infant and played with toys, talking all the while. A control group had the same sessions in English. Shortly afterward the infants were tested on two Mandarin consonants that English does not distinguish. Infants at this age are losing the ability to hear distinctions their language does not use, and the English-only group told the two sounds apart only 56.7 percent of the time, not far above chance. The infants who had spent five hours with Mandarin speakers scored 65.7 percent, statistically indistinguishable from Taiwanese infants of the same age in an earlier study who had heard Mandarin every day of their lives.[928]

The experiment's second part is the one that matters for this chapter. Another group of infants saw the same speakers, reading the same books and playing with the same toys, on a television screen, and a fourth group heard the soundtrack alone. Neither group learned anything: "exposure to recorded Mandarin, without interpersonal interaction, had no effect," the authors wrote. Figure 14.1 shows the result. The groups were small, about a dozen infants each, and the infants watching the screen paid less attention than those with a live speaker, so the experiment cannot say exactly what the live sessions supplied. But the same sounds, from the same people, were learned when they came from someone in the room and not when they came from a screen.

Later experiments suggest that what the screen lacked was not a body but a response. Sarah Roseberry, Kathy Hirsh-Pasek, and Roberta Golinkoff taught two-year-olds new verbs, such as "blicking," in three ways: in person, over a video call in which the speaker responded to the child, and through a recording of someone else's video call. The toddlers learned the words in person and over the live call, and learned nothing from the recording.[929] And when Roseberry, now writing as Sarah Roseberry Lytle, and Kuhl let nine-month-olds play Mandarin video clips on a touchscreen, infants who did so beside another infant

[928] Kuhl, P. K., Tsao, F.-M., & Liu, H.-M. (2003). Foreign-language experience in infancy: Effects of short-term exposure and social interaction on phonetic learning. *Proceedings of the National Academy of Sciences*, 100(15), 9096–9101.

[929] Roseberry, S., Hirsh-Pasek, K., & Golinkoff, R. M. (2014). Skype me! Socially contingent interactions help toddlers learn language. *Child Development*, 85(3), 956–970.

learned more than infants alone.[930] A screen can teach a young child when someone on it, or beside it, responds.

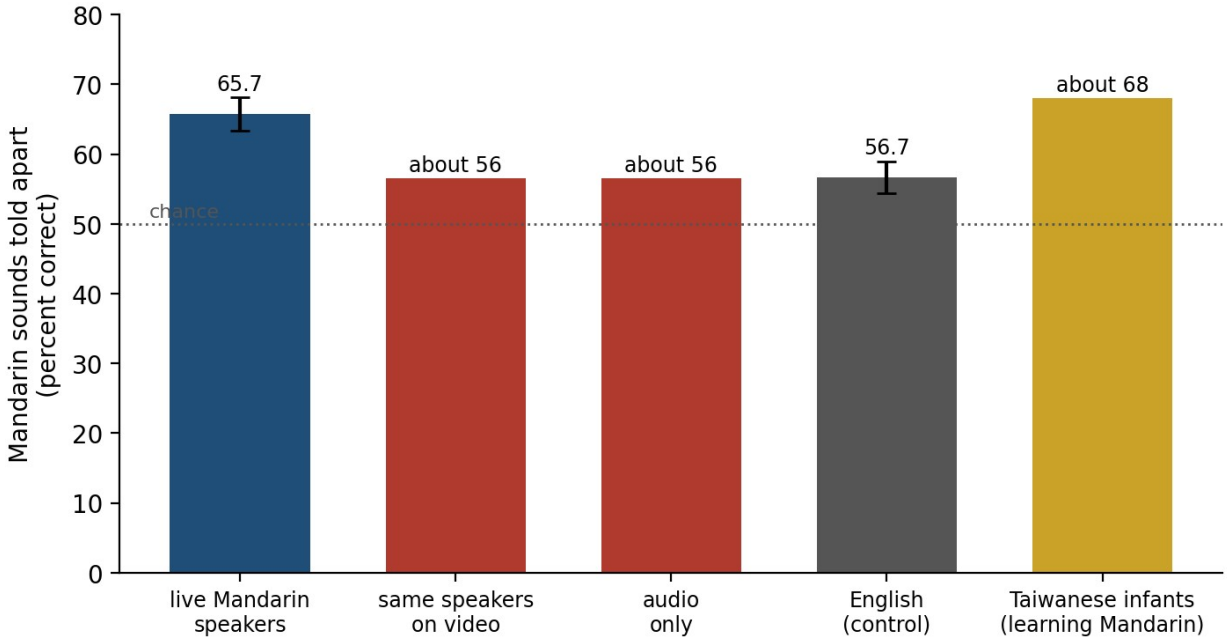


Figure 14.1. Learning from a person and from a screen (data from Kuhl, Tsao, & Liu, 2003). After about five hours of exposure at nine months, infants who heard Mandarin from live speakers discriminated a Mandarin contrast as well as Taiwanese infants (from an earlier study) who heard Mandarin daily. Infants who saw the same speakers on video, or heard only the audio, did no better than infants exposed only to English. Live and English values are from the paper's text, with standard errors; the others are read approximately from its figure. The dotted line marks chance.

The previous chapters described a brain built under constraints. It learns slowly and finishes the job offline, in sleep. It learns most from errors in its predictions, and from other people. It runs on a fixed budget of energy and attention, turns repeated rewarded actions into habits, and reorganizes itself around the tools it uses. This chapter asks what those constraints imply for classrooms, offices, and screens.

Two warnings come first. Neuroscience rarely tells a teacher, a manager, or an engineer what to do directly. In 1997 John Bruer called the direct path from neuroscience to the classroom "a bridge too far" and argued that it needs a stop in between, in the psychology of how people learn.[931] The evidence that matters most in this chapter is of that kind. The second warning is that claims sold as "brain-based" are often wrong, and the next section begins with them. Figure 14.2 sets the features of the brain described in earlier chapters beside the practical consequences this chapter examines, and the strength of evidence for each varies widely.

[930] Lytle, S. R., Garcia-Sierra, A., & Kuhl, P. K. (2018). Two are better than one: Infant language learning from video improves in the presence of peers. *Proceedings of the National Academy of Sciences*, 115(40), 9859–9866.

[931] Bruer, J. T. (1997). Education and the brain: A bridge too far. *Educational Researcher*, 26(8), 4–16.

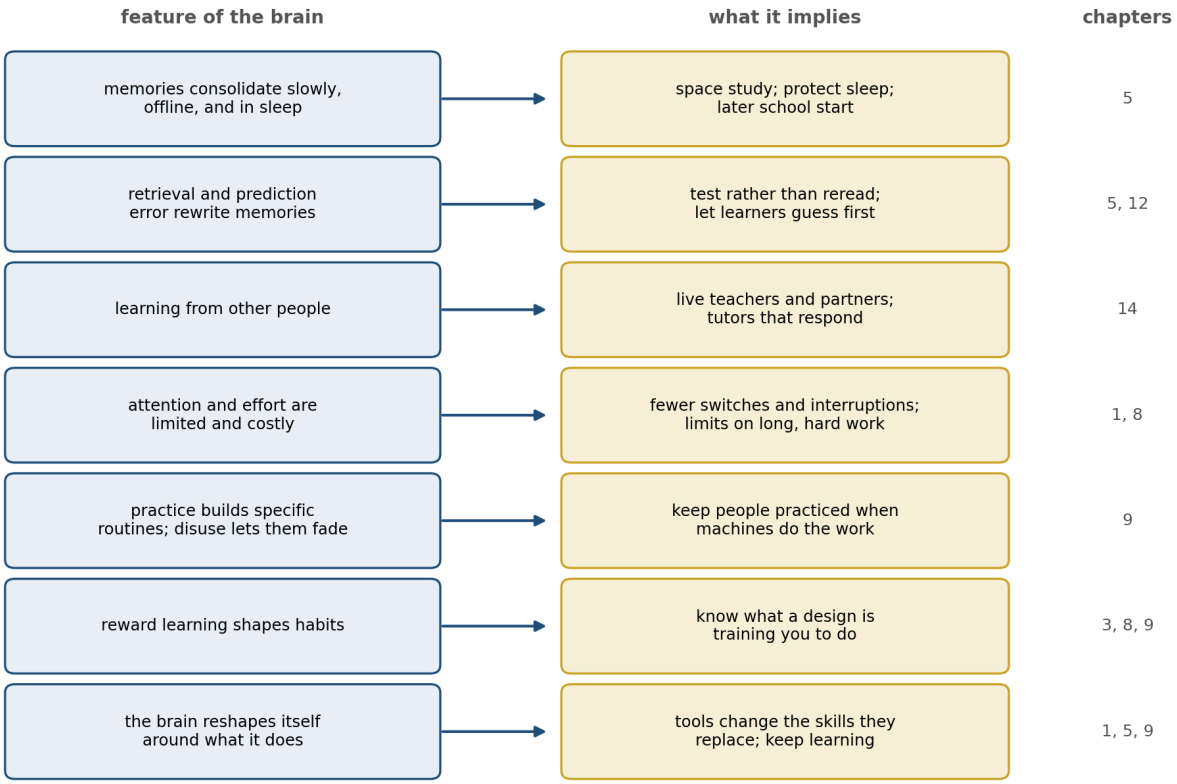


Figure 14.2. From the brain's design to practice. Each feature of the brain described in earlier chapters (left) bears on a practical question (right); the chapter numbers show where each feature was described (14 marks a feature introduced in this chapter).

What Does Not Follow

Sanne Dekker, Paul Howard-Jones, and their colleagues surveyed 242 teachers in Britain and the Netherlands, all interested in applying neuroscience to their teaching. Ninety-three percent of the British teachers and 96 percent of the Dutch agreed that individuals learn better when they receive information in their preferred learning style, auditory, visual, or kinesthetic. On average the teachers believed about half of the 15 myths in the survey, and the teachers who knew most about the brain in general believed more of them, not fewer.[932] Across 37 later studies of educators in 18 countries, about 89 percent endorsed matching teaching to learning styles.[933]

The learning-styles idea makes a testable prediction. If visual learners learn better from pictures and verbal learners from words, then sorting students by style and teaching each group both ways should produce a crossover, with each group doing best under its own method. Harold Pashler, Mark McDaniel, Doug Rohrer, and Robert Bjork reviewed the literature in 2008 for studies with that design and "found virtually no evidence for the interaction pattern"; several well-designed studies contradicted it.[934] People

[932] Dekker, S., Lee, N. C., Howard-Jones, P., & Jolles, J. (2012). Neuromyths in education: Prevalence and predictors of misconceptions among teachers. *Frontiers in Psychology*, 3, 429.

[933] Newton, P. M., & Salvi, A. (2020). How common is belief in the learning styles neuromyth, and does it matter? A pragmatic systematic review. *Frontiers in Education*, 5, 602451.

do have preferences, and differ in their abilities, but there is no evidence that matching instruction to a preferred style helps.

Other myths began as real findings that were stretched. In 1993 Frances Rauscher, Gordon Shaw, and Katherine Ky reported in *Nature* that 36 college students who had just listened to ten minutes of a Mozart sonata scored higher on spatial reasoning tasks, by the equivalent of eight or nine IQ points, than after listening to a relaxation tape or sitting in silence; the effect lasted no more than 10 to 15 minutes.[935] Within five years the governor of Georgia had proposed giving every newborn in the state a recording of classical music. When Christopher Chabris pooled the studies that followed, the benefit of Mozart over silence averaged the equivalent of 1.4 IQ points, and he suggested that what small effect there was came from the arousal of enjoying the music.[936] A later meta-analysis found that other music helped about as much as Mozart did.[937]

The popular idea that people are "left-brained" or "right-brained" fared no better: Jared Nielsen, Jeffrey Anderson, and their colleagues examined the brain scans of 1,011 people and found that although particular functions are lateralized, no individual's brain as a whole was more left- or right-dominant than another's.[938]

Commercial brain training rests on a similar leap, from the true fact that the brain changes with practice to the hope that practicing a game will improve the mind in general. Adrian Owen and his colleagues tested it with 11,430 viewers of a BBC science program, who trained online for six weeks on tasks of reasoning, memory, and attention. They improved substantially on the tasks they practiced, but "no evidence was found for transfer effects to untrained tasks, even when those tasks were cognitively closely related," and a control group that merely answered obscure questions online improved as much.[939] A large review by Daniel Simons and his colleagues reached the same verdict: brain training improves performance on the trained tasks, less on closely related ones, and shows little evidence of improving everyday cognition.[940] This fits what Chapter 9 described: practice builds the particular routine that is practiced.

Learning That Lasts

Chapter 5 described the forgetting curve and how spaced retrieval and sleep slow it, and Chapter 9 described how mixing practice makes it look worse while making it work better. The question for schools is whether laboratory effects of this kind survive in real classrooms, with real curricula and months between lesson and test.

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[934] Pashler, H., McDaniel, M., Rohrer, D., & Bjork, R. (2008). Learning styles: Concepts and evidence. *Psychological Science in the Public Interest*, 9(3), 105–119.

[935] Rauscher, F. H., Shaw, G. L., & Ky, K. N. (1993). Music and spatial task performance. *Nature*, 365(6447), 611.

[936] Chabris, C. F. (1999). Prelude or requiem for the 'Mozart effect'? *Nature*, 400(6747), 826–827.

[937] Pietschnig, J., Voracek, M., & Formann, A. K. (2010). Mozart effect–Shmozart effect: A meta-analysis. *Intelligence*, 38(3), 314–323.

[938] Nielsen, J. A., Zielinski, B. A., Ferguson, M. A., Lainhart, J. E., & Anderson, J. S. (2013). An evaluation of the left-brain vs. right-brain hypothesis with resting state functional connectivity magnetic resonance imaging. *PLoS ONE*, 8(8), e71275.

[939] Owen, A. M., Hampshire, A., Grahn, J. A., Stenton, R., Dajani, S., Burns, A. S., Howard, R. J., & Ballard, C. G. (2010). Putting brain training to the test. *Nature*, 465(7299), 775–778.

[940] Simons, D. J., Boot, W. R., Charness, N., Gathercole, S. E., Chabris, C. F., Hambrick, D. Z., & Stine-Morrow, E. A. L. (2016). Do "brain-training" programs work? *Psychological Science in the Public Interest*, 17(3), 103–186.

John Dunlosky and four colleagues rated ten common study techniques by the strength of the evidence. Only two, practice testing and spacing study over time, earned their highest rating; rereading and highlighting, two of the most familiar habits of students, were rated low because they "do not consistently boost students' performance."[941] Classroom trials have borne out the high ratings. Henry Roediger, Pooja Agarwal, and their colleagues gave sixth graders in a social studies class short quizzes, with feedback, on some of the material and not on the rest. On the chapter exams students got 94 percent of the quizzed material right and 81 percent of the rest, and at the end of the semester 79 percent against 67.[942] In a second experiment the advantage over simply rereading had faded by the semester's end, a reminder that classroom effects are smaller and noisier than laboratory ones.

Rohrer and his colleagues randomly assigned 54 seventh-grade mathematics classes, 787 students in all, to homework in which problems of different kinds were mixed together or grouped by kind, for four months. On a surprise test a month later, the students who had mixed their practice scored 61 percent and the others 38 percent, and the advantage appeared in the classes of every one of the 15 teachers.[943]

Spacing has a further twist: the best gap between study sessions depends on how long the memory must last. Nicholas Cepeda, Pashler, and their colleagues taught 1,354 people a set of obscure facts, reviewed the facts once after a gap of up to three and a half months, and tested them up to a year later. The best gap grew with the time to the test but shrank as a fraction of it, from about 20 to 40 percent of a one-week delay to about 5 to 10 percent of a one-year delay, and the best gap improved recall by 64 percent on average over reviewing at once.[944]

Figure 14.3 shows why a gap can be too short or too long in a simple model. A review that comes while the memory is still fresh adds little, because there is little to relearn. One that comes after the memory has faded, but before it is lost, forces an effortful retrieval and strengthens the memory most. One that comes too late finds nothing to retrieve and must start again. The model captures the shape of Cepeda's results but not their values, and real memories are more complicated, but the principle is the one a school calendar can use: a topic that must be remembered for years should come back months later, not only in next week's review.

[941] Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest*, 14(1), 4–58.

[942] Roediger, H. L., III, Agarwal, P. K., McDaniel, M. A., & McDermott, K. B. (2011). Test-enhanced learning in the classroom: Long-term improvements from quizzing. *Journal of Experimental Psychology: Applied*, 17(4), 382–395.

[943] Rohrer, D., Dedrick, R. F., Hartwig, M. K., & Cheung, C.-N. (2020). A randomized controlled trial of interleaved mathematics practice. *Journal of Educational Psychology*, 112(1), 40–52.

[944] Cepeda, N. J., Vul, E., Rohrer, D., Wixted, J. T., & Pashler, H. (2008). Spacing effects in learning: A temporal ridgeline of optimal retention. *Psychological Science*, 19(11), 1095–1102.

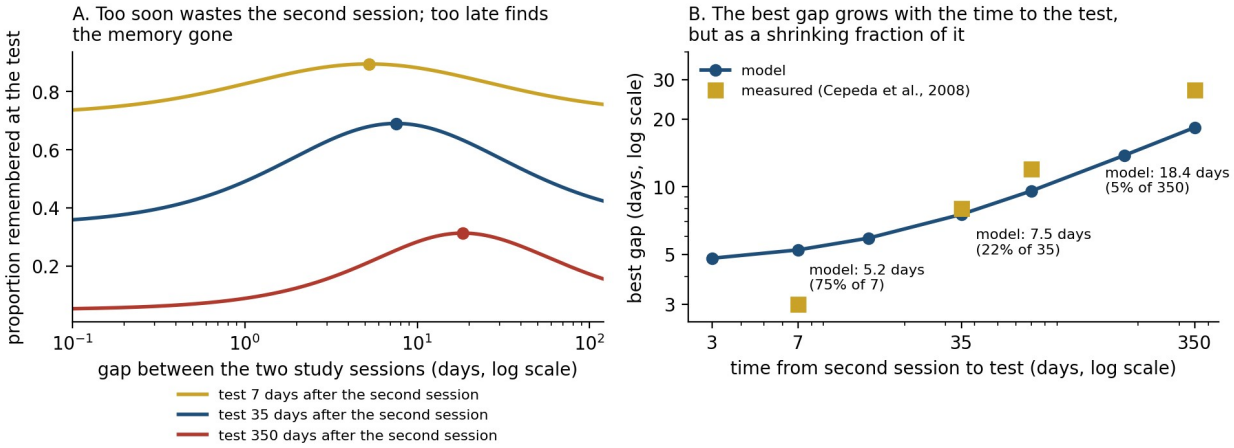


Figure 14.3. Why spacing works (computed, illustrative). A memory fades as a power function of time, and a review strengthens it more the more has been forgotten, provided it can still be retrieved. A: recall at a test 7, 35, or 350 days after the second of two study sessions, as a function of the gap between them; each curve has a best gap (dots). B: the best gap grows with the time to the test but becomes a smaller fraction of it. Squares show the best gaps interpolated from the data of Cepeda et al. (2008); the model's curve is flatter than the data.

The Illusion of Fluency

If spacing and testing work so well, why do students reread and cram? Part of the answer is that the brain is a poor judge of its own learning. Nate Kornell and Bjork showed people six paintings by each of 12 artists, either grouped by artist or mixed together, and then asked them to name the artists of new paintings. Mixing worked far better: people assigned 61 percent of the new paintings correctly after mixed study and 35 percent after grouped study. Seventy-eight percent of the participants did better with mixing, yet 78 percent said that grouping had worked as well or better.[945] Figure 14.4 shows the gap between what happened and what the learners believed. Grouped study feels fluent, because each painting resembles the one before it, and fluency feels like learning. The difficulty that makes mixed study effective is the same difficulty that makes it feel ineffective.

[945] Kornell, N., & Bjork, R. A. (2008). Learning concepts and categories: Is spacing the "enemy of induction"? *Psychological Science*, 19(6), 585–592.

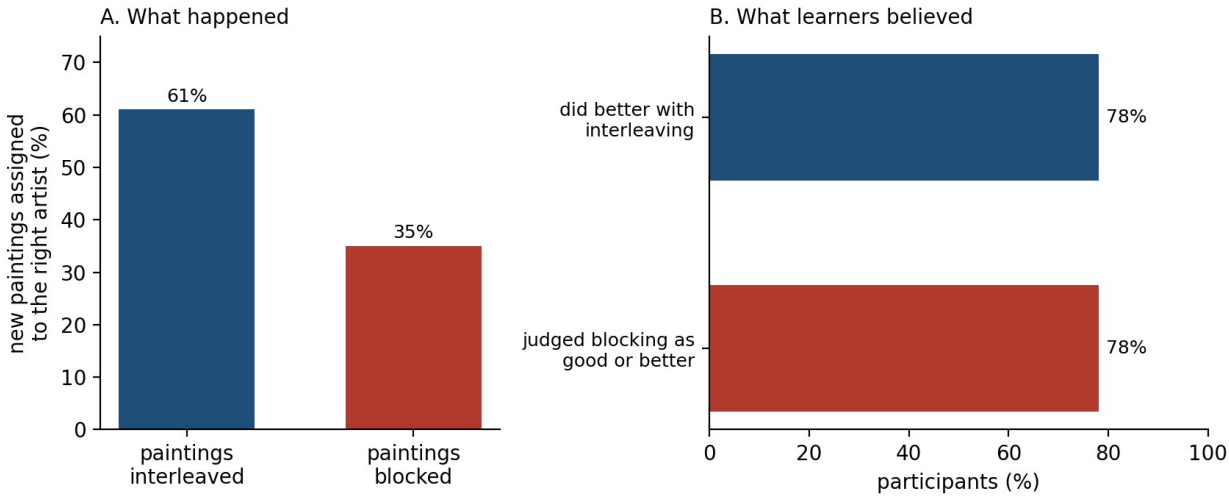


Figure 14.4. What learners do and what they believe (data from Kornell & Bjork, 2008, Experiment 1a). A: after studying paintings by 12 artists, people classified new paintings more accurately when the artists' paintings had been interleaved than when they had been blocked. B: most participants did better with interleaving, yet most judged blocking to have been as good or better.

Errors themselves can help. Kornell, Matthew Hays, and Bjork asked people to guess answers they could not know before showing them the correct one, and found that the failed guesses improved later memory compared with simply studying the answer; in one experiment, guessing did as well as studying the answer for more than twice as long.[946] The guess creates an expectation, and the answer arrives as a prediction error. Curiosity seems to act in a similar way. Matthias Gruber, Bernard Gelman, and Charan Ranganath asked people trivia questions and found that answers they were curious about were better remembered, and so were unrelated faces shown while they waited for the answer; the benefit tracked activity in the midbrain region that contains dopamine neurons and in the hippocampus, the dopamine signal of Chapter 3 and the memory system of Chapter 5.[947]

Beliefs about learning matter less than was once hoped. The idea that students learn more if they believe intelligence can grow, a "growth mindset," was tested in a national experiment with 12,490 American ninth graders: a short online intervention raised the grades of lower-achieving students by a tenth of a grade point and did not raise the grades of higher achievers.[948] A meta-analysis by Brooke Macnamara and Alexander Burgoyne found an average effect too small to distinguish from zero once publication bias and study quality were taken into account, while another, by Jeni Burnette and her

[946] Kornell, N., Hays, M. J., & Bjork, R. A. (2009). Unsuccessful retrieval attempts enhance subsequent learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 35(4), 989–998.

[947] Gruber, M. J., Gelman, B. D., & Ranganath, C. (2014). States of curiosity modulate hippocampus-dependent learning via the dopaminergic circuit. *Neuron*, 84(2), 486–496.

[948] Yeager, D. S., Hanselman, P., Walton, G. M., Murray, J. S., Crosnoe, R., Muller, C., Tipton, E., Schneider, B., Hulleman, C. S., Hinojosa, C. P., Paunesku, D., Romero, C., Flint, K., Roberts, A., Trott, J., Iachan, R., Buontempo, J., Yang, S. M., Carvalho, C. M., … Dweck, C. S. (2019). A national experiment reveals where a growth mindset improves achievement. *Nature*, 573(7774), 364–369.

colleagues, found modest effects in the groups for whom the interventions were designed.[949] The disagreement is real, and the effects, where they exist, are small.

Another brief intervention shows how hard such effects are to pin down. Gerardo Ramirez and Sian Beilock reported that anxious ninth graders who spent ten minutes writing about their worries before a biology exam scored about six percentage points higher than those who did not, as if writing had freed working memory that worry would have occupied.[950] A large replication, part of a project that repeated 21 social-science experiments published in *Nature* and *Science*, found no effect, although the original authors argued that the replication had not created enough pressure to test it.[951]

Sleep and the School Day

Chapter 5 described how sleep consolidates what was learned during the day. A school that wants its lessons remembered therefore has an interest in its students' sleep, and the timing of the school day works against it in adolescence. Around puberty the biological timing of sleep shifts later: in a small laboratory study by Mary Carskadon and her colleagues, the nightly secretion of melatonin, the hormone that signals biological night, ended later in older and more mature children, and later work found that sleep pressure also builds more slowly across the day in older adolescents.[952] Carskadon has called the result a "perfect storm": a body ready to sleep later, evening activities and screens that push bedtime later still, and a school bus that comes at the same early hour.[953]

In 2016 the public high schools of Seattle moved their start time from 7:50 to 8:45 in the morning. Gideon Dunster, Horacio de la Iglesia, and their colleagues gave wrist activity monitors to students in the same biology course at two schools, in the spring before the change and in the spring after. The median sleep on school nights rose by 34 minutes, the students reported less daytime sleepiness, and the median grade in the course rose from 77.5 to 82 percent. The start time moved by 55 minutes but sleep grew by only 34, because students also went to bed a little later; attendance and punctuality improved at the school with more economically disadvantaged students.[954] The two years involved different students and there was no control group, so the authors were cautious about attributing the higher grades to the change; but the extra sleep was measured, not reported.

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[949] Macnamara, B. N., & Burgoyne, A. P. (2023). Do growth mindset interventions impact students' academic achievement? A systematic review and meta-analysis with recommendations for best practices. *Psychological Bulletin*, 149(3–4), 133–173; Burnette, J. L., Billingsley, J., Banks, G. C., Knouse, L. E., Hoyt, C. L., Pollack, J. M., & Simon, S. (2023). A systematic review and meta-analysis of growth mindset interventions: For whom, how, and why might such interventions work? *Psychological Bulletin*, 149(3–4), 174–205.

[950] Ramirez, G., & Beilock, S. L. (2011). Writing about testing worries boosts exam performance in the classroom. *Science*, 331(6014), 211–213.

[951] Camerer, C. F., Dreber, A., Holzmeister, F., Ho, T.-H., Huber, J., Johannesson, M., Kirchler, M., Nave, G., Nosek, B. A., Pfeiffer, T., Altmejd, A., Buttrick, N., Chan, T., Chen, Y., Forsell, E., Gampa, A., Heikensten, E., Hummer, L., Imai, T., … Wu, H. (2018). Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015. *Nature Human Behaviour*, 2(9), 637–644.

[952] Carskadon, M. A., Acebo, C., Richardson, G. S., Tate, B. A., & Seifer, R. (1997). An approach to studying circadian rhythms of adolescent humans. *Journal of Biological Rhythms*, 12(3), 278–289; Crowley, S. J., Wolfson, A. R., Tarokh, L., & Carskadon, M. A. (2018). An update on adolescent sleep: New evidence informing the perfect storm model. *Journal of Adolescence*, 67, 55–65.

[953] Carskadon, M. A. (2011). Sleep in adolescents: The perfect storm. *Pediatric Clinics of North America*, 58(3), 637–647.

[954] Dunster, G. P., de la Iglesia, L., Ben-Hamo, M., Nave, C., Fleischer, J. G., Panda, S., & de la Iglesia, H. O. (2018). Sleepmore in Seattle: Later school start times are associated with more sleep and better performance in high school students. *Science Advances*, 4(12), eaau6200.

Light is one of the things that pushes bedtime later. Anne-Marie Chang, Charles Czeisler, and their colleagues had 12 young adults read for four hours before bed for five nights, either on a tablet at full brightness or from a printed book, in an otherwise dim room. After the tablet, the evening rise in melatonin was suppressed by about half and delayed by more than an hour and a half, the readers took about ten minutes longer to fall asleep, and they were sleepier the next morning.[955] Four hours at full brightness is a large dose, and ordinary use probably has smaller effects, but the direction is the one the biology of the body clock predicts, since bright light in the evening tells the clock that it is still day.

Attention Is a Budget

Chapter 8 described thinking as costly, and the capacity to hold several things in mind at once is small, about four items by Nelson Cowan's estimate.[956] Attention, which Chapter 6 proposed may be the weighting of evidence by its reliability, is also the means by which that capacity is allotted, and the most robust finding about it is also the least dramatic. When people switch between two simple tasks, they respond more slowly and make more errors on the first trial after each switch. Robert Rogers and Stephen Monsell showed that warning people of the switch in advance reduces this cost but does not eliminate it, even with more than a second to prepare.[957] Each switch requires the brain to reconfigure which rules are active, and that takes time that cannot be fully hidden.

More dramatic claims have fared less well. In 2009 Eyal Ophir, Clifford Nass, and Anthony Wagner reported that heavy media multitaskers, people who habitually use several media at once, were more easily distracted by irrelevant information and worse at switching tasks.[958] Later replications and meta-analyses found the effects on measured performance to be small or absent once small studies were accounted for.[959] In 2017 Adrian Ward and his colleagues reported that the mere presence of a smartphone on the desk, even turned off, reduced performance on tests of working memory and reasoning; a preregistered replication found no such effect, and a meta-analysis of 56 studies found support only for a small effect on working memory.[960]

What does hold up is the cost of divided attention in a real setting. Faria Sana, Tina Weston, and Nicholas Cepeda had students take notes on a lecture by laptop, and asked half of them to do unrelated online tasks during it. Those who did scored 11 percentage points lower on a test of the lecture. In a second experiment, students who took notes on paper scored 17 points lower when they could see a

[955] Chang, A.-M., Aeschbach, D., Duffy, J. F., & Czeisler, C. A. (2015). Evening use of light-emitting eReaders negatively affects sleep, circadian timing, and next-morning alertness. *Proceedings of the National Academy of Sciences*, 112(4), 1232–1237.

[956] Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1), 87–114.

[957] Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General*, 124(2), 207–231; Monsell, S. (2003). Task switching. *Trends in Cognitive Sciences*, 7(3), 134–140.

[958] Ophir, E., Nass, C., & Wagner, A. D. (2009). Cognitive control in media multitaskers. *Proceedings of the National Academy of Sciences*, 106(37), 15583–15587.

[959] Wiradhany, W., & Nieuwenstein, M. R. (2017). Cognitive control in media multitaskers: Two replication studies and a meta-analysis. *Attention, Perception, & Psychophysics*, 79(8), 2620–2641; Parry, D. A., & le Roux, D. B. (2021). "Cognitive control in media multitaskers" ten years on: A meta-analysis. *Cyberpsychology: Journal of Psychosocial Research on Cyberspace*, 15(2), 7.

[960] Ward, A. F., Duke, K., Gneezy, A., & Bos, M. W. (2017). Brain drain: The mere presence of one's own smartphone reduces available cognitive capacity. *Journal of the Association for Consumer Research*, 2(2), 140–154; Ruiz Pardo, A. C., & Minda, J. P. (2022). Reexamining the "brain drain" effect: A replication of Ward et al. (2017). *Acta Psychologica*, 230, 103717; Parry, D. A. (2024). Does the mere presence of a smartphone impact cognitive performance? A meta-analysis of the "brain drain effect". *Media Psychology*, 27(5), 737–762.

classmate's laptop with off-task activity on it.[961] A related claim, that taking notes by hand is better than typing them because typing invites verbatim transcription, came from a widely cited study by Pam Mueller and Daniel Oppenheimer; two careful replications found that typists did transcribe more but learned no less.[962] The problem was what else was on the screen, not the keyboard.

Interruptions at work carry a cost of a different kind. Gloria Mark, Daniela Gudith, and Ulrich Klocke interrupted people doing an email task every couple of minutes and found that they finished faster, with no loss of quality, but reported more stress, frustration, and time pressure.[963] In an earlier study of office workers, Mark and her colleagues found that work that was interrupted and resumed the same day was taken up again after about 25 minutes on average, most of it spent on other tasks.[964] A day broken into fragments is not necessarily less productive in the short term, but it is paid for in stress, and in work that is set down and picked up again.

The design of the workplace itself was long justified by a claim about the brain's social nature: that people who can see each other will talk more. Ethan Bernstein and Stephen Turban gave employees at two large companies electronic badges that recorded whom they faced and spoke with, before and after their offices were converted to open plans. In both companies face-to-face interaction fell by about 70 percent, and email and instant messaging rose.[965] People who could see each other all day withdrew from each other, perhaps to protect the limited attention this section has described.

How Long Can a Mind Work?

Antonius Wiehler, Mathias Pessiglione, and their colleagues asked what a long day of effort does to the brain, measuring over hours the cost of thinking that Chapter 8 described. Forty volunteers spent more than six hours on difficult or easy versions of tasks that required switching between rules and holding items in memory, and periodically made economic choices. By the end of the day, the group doing the hard tasks chose more often the options that paid out sooner and required less effort, and magnetic resonance spectroscopy showed a higher concentration of glutamate in the lateral prefrontal cortex.[966] Both groups reported feeling equally tired, although only the hard-task group's choices changed. The authors proposed that a buildup of glutamate makes further control more costly, but they described

[961] Sana, F., Weston, T., & Cepeda, N. J. (2013). Laptop multitasking hinders classroom learning for both users and nearby peers. *Computers & Education*, 62, 24–31.

[962] Mueller, P. A., & Oppenheimer, D. M. (2014). The pen is mightier than the keyboard: Advantages of longhand over laptop note taking. *Psychological Science*, 25(6), 1159–1168; Morehead, K., Dunlosky, J., & Rawson, K. A. (2019). How much mightier is the pen than the keyboard for note-taking? A replication and extension of Mueller and Oppenheimer (2014). *Educational Psychology Review*, 31(3), 753–780; Urry, H. L., Crittle, C. S., Floerke, V. A., Leonard, M. Z., Perry, C. S., Akdilek, N., Albert, E. R., Block, A. J., Bollinger, C. A., Bowers, E. M., Brody, R. S., Burk, K. C., Burnstein, A., Chan, A. K., Chan, P. C., Chang, L. J., Chen, E., Chiarawongse, C. P., Chin, G., … Zarrow, J. E. (2021). Don't ditch the laptop just yet: A direct replication of Mueller and Oppenheimer's (2014) Study 1 plus mini meta-analyses across similar studies. *Psychological Science*, 32(3), 326–339.

[963] Mark, G., Gudith, D., & Klocke, U. (2008). The cost of interrupted work: More speed and stress. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 107–110). ACM.

[964] Mark, G., Gonzalez, V. M., & Harris, J. (2005). No task left behind? Examining the nature of fragmented work. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 321–330). ACM.

[965] Bernstein, E. S., & Turban, S. (2018). The impact of the 'open' workspace on human collaboration. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373(1753), 20170239.

[966] Wiehler, A., Branzoli, F., Adanyeguh, I., Mochel, F., & Pessiglione, M. (2022). A neuro-metabolic account of why daylong cognitive work alters the control of economic decisions. *Current Biology*, 32(16), 3564–3575.e5.

their results as correlational, and others have questioned whether the measurement can isolate the glutamate that would matter.[967]

A more famous idea, that self-control draws on a resource that is used up within minutes, has not survived testing. In a registered replication involving 23 laboratories and 2,141 people, the influence of an initial demanding task on a second one had an effect size of 0.04, essentially zero; a second effort, led by one of the idea's own proponents, Kathleen Vohs, with 36 laboratories and 3,531 people, found an effect too small to distinguish from zero.[968] The study most often cited for the cost of decision fatigue in the real world, which analyzed 1,112 rulings and found that Israeli parole boards granted about 65 percent of requests at the start of each session and almost none just before a break, has been challenged on the grounds that the order of cases was not random, with prisoners without lawyers often heard last, and that simulations show favorable rulings taking longer can produce much of the pattern.[969]

The evidence supports a narrower conclusion. Breaks, sleep, and limits on the length of the most demanding work are supported by what is known; elaborate theories of a fuel that runs out between one task and the next are not.

Tools That Remember for Us

In Plato's *Phaedrus*, Socrates tells a story about the Egyptian god Theuth, who brought the invention of writing to King Thamus and praised it as a remedy for memory. Thamus disagreed: writing would make people forgetful, because they would rely on marks outside themselves rather than remember, and it would give them "the show of wisdom without the reality."[970] The irony that Thamus's warning survives only because Plato wrote it down is often noted, but the worry has returned with every new tool, and it is partly right.

Evan Risko and Sam Gilbert call the use of physical action or external tools to reduce the demands of a task "cognitive offloading," which includes writing a list, setting a reminder, or tilting one's head to read sideways text. Offloading usually improves performance on the task at hand, and it can reduce memory for what was offloaded; people decide whether to offload by judging their own abilities, and those judgments can be wrong.[971] Betsy Sparrow, Jenny Liu, and Daniel Wegner claimed more: that when people expect information to remain available on a computer, they remember less of it, and that hard questions make people think about search engines. The second claim, tested in the same replication

[967] Möller, H. E. (2023). Considerations on gradual glutamate accumulation related to cognitive task performance. *Journal of Cerebral Blood Flow & Metabolism*, 43(3), 476–478.

[968] Hagger, M. S., Chatzisarantis, N. L. D., Alberts, H., Anggono, C. O., Batailler, C., Birt, A. R., Brand, R., Brandt, M. J., Brewer, G., Bruyneel, S., Calvillo, D. P., Campbell, W. K., Cannon, P. R., Carlucci, M., Carruth, N. P., Cheung, T., Crowell, A., De Ridder, D. T. D., Dewitte, S., … Zwienenberg, M. (2016). A multilab preregistered replication of the ego-depletion effect. *Perspectives on Psychological Science*, 11(4), 546–573; Vohs, K. D., Schmeichel, B. J., Lohmann, S., Gronau, Q. F., Finley, A. J., Ainsworth, S. E., Alquist, J. L., Baker, M. D., Brizi, A., Bunyi, A., Butschek, G. J., Campbell, C., Capaldi, J., Cau, C., Chambers, H., Chatzisarantis, N. L. D., Christensen, W. J., Clay, S. L., Curtis, J., … Albarracín, D. (2021). A multisite preregistered paradigmatic test of the ego-depletion effect. *Psychological Science*, 32(10), 1566–1581.

[969] Danziger, S., Levav, J., & Avnaim-Pesso, L. (2011). Extraneous factors in judicial decisions. *Proceedings of the National Academy of Sciences*, 108(17), 6889–6892; Weinshall-Margel, K., & Shapard, J. (2011). Overlooked factors in the analysis of parole decisions. *Proceedings of the National Academy of Sciences*, 108(42), E833; Glöckner, A. (2016). The irrational hungry judge effect revisited: Simulations reveal that the magnitude of the effect is overestimated. *Judgment and Decision Making*, 11(6), 601–610.

[970] Plato. (1892). Phaedrus. In B. Jowett (Trans.), *The Dialogues of Plato* (3rd ed., Vol. 1). Oxford University Press. (Original work written ca. 370 B.C.E.).

[971] Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences*, 20(9), 676–688.

project as the exam-writing study, did not replicate, and a later replication designed with the original author's comments also found nothing.[972] The claims about memory for saved information were not part of that test.

Navigation shows what offloading may cost the brain. Eleanor Maguire and her colleagues found that London taxi drivers, who must learn the city's 25,000 streets to qualify, had larger posterior hippocampi than other people, and that the difference grew with years of driving; the anterior hippocampus, by contrast, was smaller.[973] To test whether the training caused the difference, Katherine Woollett and Maguire scanned 79 trainee drivers before they began and again three to four years later. The 39 who qualified showed an increase in gray matter in the posterior hippocampus; those who failed or gave up, and a control group, showed none.[974]

Louisa Dahmani and Véronique Bohbot found that young adults who used satellite navigation more had somewhat worse spatial memory when they had to find their own way, and in 13 people retested three years later, those who had used it most declined most.[975] The correlations were modest and the follow-up tiny, so the result is suggestive rather than settled. Hugo Spiers and his colleagues suggested why. They trained volunteers on the streets of Soho in London and then showed them films of journeys through it during brain scanning. When the volunteers had to choose their own route, activity in the posterior hippocampus tracked how many streets connected to each new street they entered; when they were told which way to go, it did not.[976] Taken together, these studies point the same way: the hippocampus grows with navigation it does itself and may lose ground when the navigating is done for it.

When the Machine Does the Work

In 1983 the psychologist Lisanne Bainbridge described what she called the ironies of automation. The designer who automates a process "still leaves the operator to do the tasks which the designer cannot think how to automate," which are the hardest. The operator is asked to monitor a system that was installed because it does the job better than the operator. And "physical skills deteriorate when they are not used," so that "a formerly experienced operator who has been monitoring an automated process may now be an inexperienced one." Her final irony was that "the most successful automated systems, with rare need for manual intervention, … may need the greatest investment in human operator training."[977]

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[972] Sparrow, B., Liu, J., & Wegner, D. M. (2011). Google effects on memory: Cognitive consequences of having information at our fingertips. *Science*, 333(6043), 776–778; Camerer, C. F., Dreber, A., Holzmeister, F., Ho, T.-H., Huber, J., Johannesson, M., Kirchler, M., Nave, G., Nosek, B. A., Pfeiffer, T., Altmejd, A., Buttrick, N., Chan, T., Chen, Y., Forsell, E., Gampa, A., Heikensten, E., Hummer, L., Imai, T., … Wu, H. (2018). Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015. *Nature Human Behaviour*, 2(9), 637–644; Hesselmann, G. (2020). No conclusive evidence that difficult general knowledge questions cause a "Google Stroop effect". A replication study. *PeerJ*, 8, e10325.

[973] Maguire, E. A., Gadian, D. G., Johnsrude, I. S., Good, C. D., Ashburner, J., Frackowiak, R. S. J., & Frith, C. D. (2000). Navigation-related structural change in the hippocampi of taxi drivers. *Proceedings of the National Academy of Sciences*, 97(8), 4398–4403.

[974] Woollett, K., & Maguire, E. A. (2011). Acquiring "the Knowledge" of London's layout drives structural brain changes. *Current Biology*, 21(24), 2109–2114.

[975] Dahmani, L., & Bohbot, V. D. (2020). Habitual use of GPS negatively impacts spatial memory during self-guided navigation. *Scientific Reports*, 10, 6310.

[976] Javadi, A.-H., Emo, B., Howard, L. R., Zisch, F. E., Yu, Y., Knight, R., Pinelo Silva, J., & Spiers, H. J. (2017). Hippocampal and prefrontal processing of network topology to simulate the future. *Nature Communications*, 8, 14652.

[977] Bainbridge, L. (1983). Ironies of automation. *Automatica*, 19(6), 775–779.

On 1 June 2009, Air France Flight 447 was crossing the Atlantic at night when ice crystals likely blocked the probes that measure airspeed. The speed readings became inconsistent, the autopilot disconnected, and the pilots found themselves flying the aircraft by hand at high altitude. Their control inputs destabilized the flight, and the aircraft entered a stall that the crew never recognized; it fell for about three and a half minutes into the sea, killing all 228 people on board. France's accident investigation bureau listed among the contributing factors "the absence of any training, at high altitude, in manual aeroplane handling."[978] The pilots were experienced, but the experience they needed was the kind that automation had made rare.

The same pattern may now be appearing in medicine. Krzysztof Budzyń, Yuichi Mori, and their colleagues examined colonoscopies at four Polish centers before and after the centers began using an artificial intelligence system that highlights possible polyps. In colonoscopies done without the system, the rate at which experienced endoscopists detected adenomas, the precancerous growths the procedure is meant to find, fell from 28.4 percent in the three months before the system arrived to 22.4 percent in the three months after.[979] The study was observational and short, and other changes could explain part of the fall, but it is the pattern Bainbridge predicted. Figure 14.5 shows it in a simple model in which a skill improves with each manual repetition and decays without use. When automation handles half the work the skill barely changes; when it handles almost all of it, the skill falls steadily, and it is weakest at the moment the rare hard case arrives.

[978] Bureau d'Enquêtes et d'Analyses pour la sécurité de l'aviation civile. (2012). *Final report on the accident on 1st June 2009 to the Airbus A330-203 registered F-GZCP operated by Air France flight AF 447 Rio de Janeiro–Paris*. BEA.

[979] Budzyń, K., Romańczyk, M., Kitala, D., Kołodziej, P., Bugajski, M., Adami, H. O., Blom, J., Buszkiewicz, M., Halvorsen, N., Hassan, C., Romańczyk, T., Holme, Ø., Jarus, K., Fielding, S., Kunar, M., Pellise, M., Pilonis, N., Kamiński, M. F., Kalager, M., … Mori, Y. (2025). Endoscopist deskilling risk after exposure to artificial intelligence in colonoscopy: A multicentre, observational study. *The Lancet Gastroenterology & Hepatology*, 10(10), 896–903.

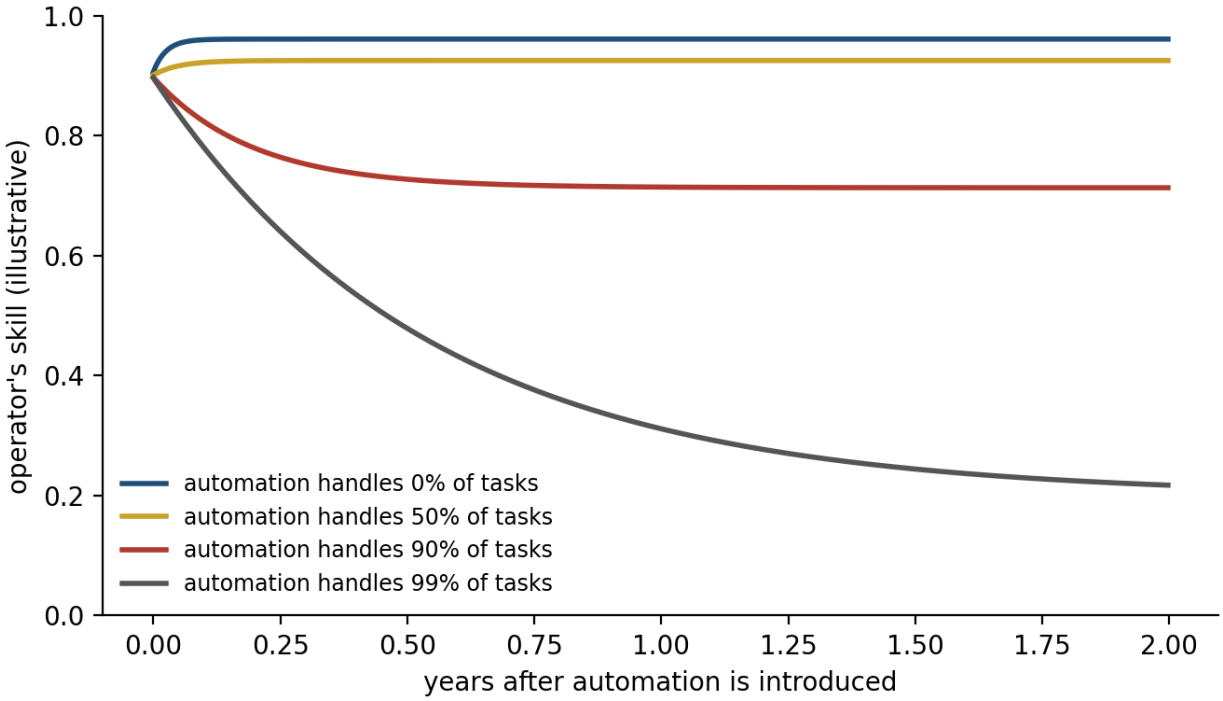


Figure 14.5. The irony of automation (computed, illustrative). An operator's skill improves with each task done by hand and decays slowly without practice. As automation takes over more of the routine work, the skill settles at a lower level, and when automation handles 99 percent of tasks it falls by more than half within a year.

Artificial intelligence now offers to do the work of learning itself, and the first controlled studies suggest that whether it helps depends on whether it does the work for the learner. Hamsa Bastani, Osbert Bastani, and their colleagues gave about a thousand Turkish high school students access to GPT-4 during mathematics practice sessions. One version was an ordinary chatbot; another was prompted to act as a tutor that gave hints rather than answers. During practice, grades rose by 48 percent with the ordinary chatbot and by 127 percent with the tutor. On the exam that followed, taken without AI, students who had used the ordinary chatbot scored 17 percent lower than students who had never had it, while the tutor's students did no worse, and no better, than the controls.[980] Many of the chatbot's users had simply asked it for the answers, and they did not realize they had learned less. Figure 14.6 shows the result.

[980] Bastani, H., Bastani, O., Sungu, A., Ge, H., Kabakcı, Ö., & Mariman, R. (2025). Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26), e2422633122.

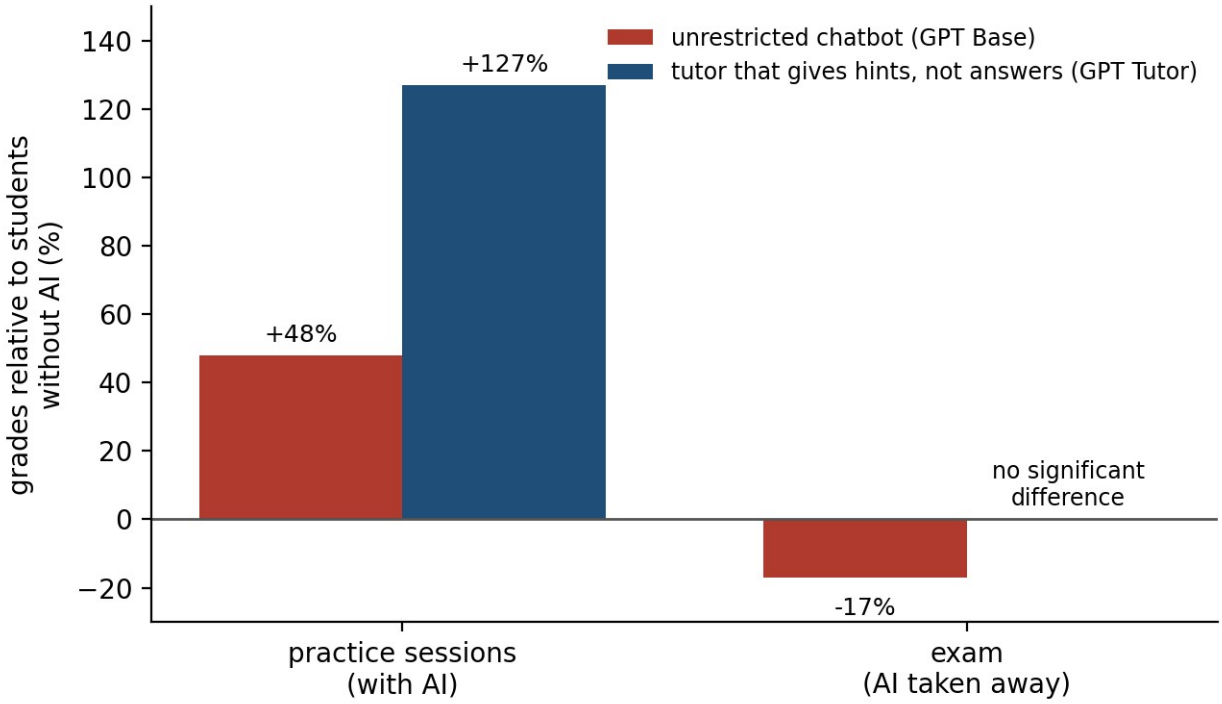


Figure 14.6. AI that answers and AI that teaches (data from Bastani et al., 2025). Grades relative to students without AI. During practice, both versions of GPT-4 raised grades. On the exam taken without AI, students who had used the unrestricted chatbot did worse than those who never had it; those who had used a tutor that gave hints rather than answers did not differ significantly from controls.

Gregory Kestin and his colleagues, at Harvard, found that a tutor could do better still. Their tutor, built on principles from research on teaching, guided students through each problem step by step, and students learned more than twice as much in a lesson with it as in an active-learning class covering the same material, although the study measured learning immediately after two lessons and did not test retention.[981]

Measures of the brain itself are only beginning. Nataliya Kosmyna, Pattie Maes, and their colleagues recorded the electrical activity of 54 people writing essays over several sessions with a chatbot, a search engine, or nothing, and reported that the brain-only writers showed the most widespread connectivity between brain regions and the chatbot users the least, and that chatbot users who later wrote unaided remained less engaged. The study is a preprint, small, and contested, and connectivity is not the same as learning, but its direction is the one Bastani's exam scores suggest.[982]

The difference between the chatbot and the tutors is the difference Bainbridge described. A tool that does the work leaves the learner unpracticed; one that makes the learner do the work can teach.

[981] Kestin, G., Miller, K., Klales, A., Milbourne, T., & Ponti, G. (2025). AI tutoring outperforms in-class active learning: An RCT introducing a novel research-based design in an authentic educational setting. *Scientific Reports*, 15, 17458.

[982] Kosmyna, N., Hauptmann, E., Yuan, Y. T., Situ, J., Liao, X.-H., Beresnitzky, A. V., Braunstein, I., & Maes, P. (2025). *Your brain on ChatGPT: Accumulation of cognitive debt when using an AI assistant for essay writing task* (arXiv:2506.08872) [Preprint]. arXiv.

Designed to Be Hard to Put Down

The dopamine teaching signal of Chapter 3, the context-cued habits of Chapter 9, and the supernormal stimuli of Chapter 13 have all been applied, sometimes loosely, to the devices and services that now take up much of the waking day. Björn Lindström, David Amodio, and their colleagues analyzed more than a million posts on Instagram and three online forums and found that people spaced their posts in the way a reward-learning model predicts, posting more often when posts had recently earned more "likes," much as an animal adjusts its foraging to the rate at which it is rewarded; an online experiment confirmed that the rewards changed behavior.[983] Christian Montag and his colleagues have described the design features of apps that may exploit such learning, though they note that the effect of any single feature is hard to isolate.[984] If Chapter 13's beetle is a guide, the risk is not that the brain's learning rules are wrong but that the inputs now offered to them are larger, faster, and more frequent than any they were tuned for.

Whether this engagement harms well-being is one of the most contested questions in this book. Amy Orben and Andrew Przybylski analyzed three large surveys of more than 350,000 adolescents in every defensible way they could and found that technology use explained at most 0.4 percent of the variation in well-being, an association about as negative as that of eating potatoes and smaller than that of wearing glasses.[985] Jean Twenge, Jonathan Haidt, and their colleagues replied that lumping television and games together with social media hides larger associations for social media, especially among girls.[986]

Experiments give a sharper picture. Hunt Allcott and his colleagues asked Facebook users how much they would need to be paid to stay off Facebook for four weeks, around the 2018 American midterm elections, took those who would accept $102 or less, and paid a randomly chosen group of them to deactivate, checking that their accounts stayed dark. The deactivated users gained about an hour a day, became somewhat less politically polarized and somewhat less informed about the news, and reported a small improvement in well-being, about 0.09 standard deviations.[987] Melissa Hunt and her colleagues asked 143 undergraduates to limit their use of three platforms to ten minutes each a day for three weeks; loneliness and depression fell compared with students who used the platforms as usual.[988] Luca Braghieri, Ro'ee Levy, and Alexey Makarin used the staggered introduction of Facebook at American colleges between 2004 and 2006 and found that its arrival worsened students' mental health, mainly through more depression and anxiety, by an amount they estimated at about a fifth of the effect of losing a job; they proposed that the cause was unfavorable comparison with peers.[989]

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[983] Lindström, B., Bellander, M., Schultner, D. T., Chang, A., Tobler, P. N., & Amodio, D. M. (2021). A computational reward learning account of social media engagement. *Nature Communications*, 12, 1311.

[984] Montag, C., Lachmann, B., Herrlich, M., & Zweig, K. (2019). Addictive features of social media/messenger platforms and freemium games against the background of psychological and economic theories. *International Journal of Environmental Research and Public Health*, 16(14), 2612.

[985] Orben, A., & Przybylski, A. K. (2019). The association between adolescent well-being and digital technology use. *Nature Human Behaviour*, 3(2), 173–182.

[986] Twenge, J. M., Haidt, J., Joiner, T. E., & Campbell, W. K. (2020). Underestimating digital media harm. *Nature Human Behaviour*, 4(4), 346–348.

[987] Allcott, H., Braghieri, L., Eichmeyer, S., & Gentzkow, M. (2020). The welfare effects of social media. *American Economic Review*, 110(3), 629–676.

[988] Hunt, M. G., Marx, R., Lipson, C., & Young, J. (2018). No more FOMO: Limiting social media decreases loneliness and depression. *Journal of Social and Clinical Psychology*, 37(10), 751–768.

[989] Braghieri, L., Levy, R., & Makarin, A. (2022). Social media and mental health. *American Economic Review*, 112(11), 3660–3693.

Haidt has argued that the move from play-based to phone-based childhood after about 2010 caused an epidemic of adolescent anxiety and depression; Candice Odgers, reviewing his book in *Nature*, replied that the research shows "a mix of no, small and mixed associations" and that young people who already have mental-health problems tend to use such platforms more.[990] The fairest summary is that the average effects measured so far are small, that experiments show they are real for at least some people and some outcomes, and that whether they are larger for particular groups, such as adolescent girls, is not yet settled.

Learning Across a Lifetime

The brain's capacity to change does not end with the sensitive periods of Chapter 1. Bogdan Draganski, Arne May, and their colleagues taught 12 young adults to juggle over three months and found a temporary expansion of gray matter in a region of the cortex that processes visual motion, which shrank back after they stopped practicing; Janina Boyke and her colleagues found similar changes in people around 60, who learned to juggle less well.[991] What such changes consist of at the level of cells is not known, and the studies are small. Denise Park and her colleagues asked adults aged 60 to 90 to spend about 15 hours a week for 14 weeks learning digital photography, quilting, or both, or in less demanding social and home activities. Episodic memory improved in the groups learning photography, alone or with quilting, compared with the groups in undemanding activities; quilting alone did not produce a significant improvement.[992]

Exercise is often recommended for the aging brain, and the evidence is more mixed than the recommendations suggest. In a trial by Kirk Erickson, Arthur Kramer, and their colleagues, a year of walking increased the volume of the hippocampus in older adults by about two percent, while it shrank slightly in those who did stretching exercises instead; but memory improved in both groups, with no difference between them.[993] A 2024 meta-analysis of randomized trials found no robust effect of aerobic exercise on hippocampal volume.[994] The firmest conclusion available is the one the earlier chapters would predict: the brain changes most in response to effortful learning of new skills, and the benefits are specific to what is learned.

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[990] Haidt, J. (2024). *The Anxious Generation: How the Great Rewiring of Childhood Is Causing an Epidemic of Mental Illness*. Penguin Press; Odgers, C. L. (2024). The great rewiring: Is social media really behind an epidemic of teenage mental illness? *Nature*, 628(8006), 29–30.

[991] Draganski, B., Gaser, C., Busch, V., Schuierer, G., Bogdahn, U., & May, A. (2004). Neuroplasticity: Changes in grey matter induced by training. *Nature*, 427(6972), 311–312; Boyke, J., Driemeyer, J., Gaser, C., Büchel, C., & May, A. (2008). Training-induced brain structure changes in the elderly. *Journal of Neuroscience*, 28(28), 7031–7035.

[992] Park, D. C., Lodi-Smith, J., Drew, L., Haber, S., Hebrank, A., Bischof, G. N., & Aamodt, W. (2014). The impact of sustained engagement on cognitive function in older adults: The Synapse Project. *Psychological Science*, 25(1), 103–112.

[993] Erickson, K. I., Voss, M. W., Prakash, R. S., Basak, C., Szabo, A., Chaddock, L., Kim, J. S., Heo, S., Alves, H., White, S. M., Wojcicki, T. R., Mailey, E., Vieira, V. J., Martin, S. A., Pence, B. D., Woods, J. A., McAuley, E., & Kramer, A. F. (2011). Exercise training increases size of hippocampus and improves memory. *Proceedings of the National Academy of Sciences*, 108(7), 3017–3022; Coen, R. F., Lawlor, B. A., & Kenny, R. (2011). Failure to demonstrate that memory improvement is due either to aerobic exercise or increased hippocampal volume. *Proceedings of the National Academy of Sciences*, 108(18), E89.

[994] Balbim, G. M., Boa Sorte Silva, N. C., ten Brinke, L., Falck, R. S., Hortobágyi, T., Granacher, U., Erickson, K. I., Hernández-Gamboa, R., & Liu-Ambrose, T. (2024). Aerobic exercise training effects on hippocampal volume in healthy older individuals: A meta-analysis of randomized controlled trials. *GeroScience*, 46(2), 2755–2764.

Implications for Artificial Intelligence

Figure 14.7 collects the chapter's verdicts on the evidence. Artificial intelligence has become, in a few years, one of the main tools with which people learn and work, and several of these verdicts, together with the brain's design, suggest four principles for building it.

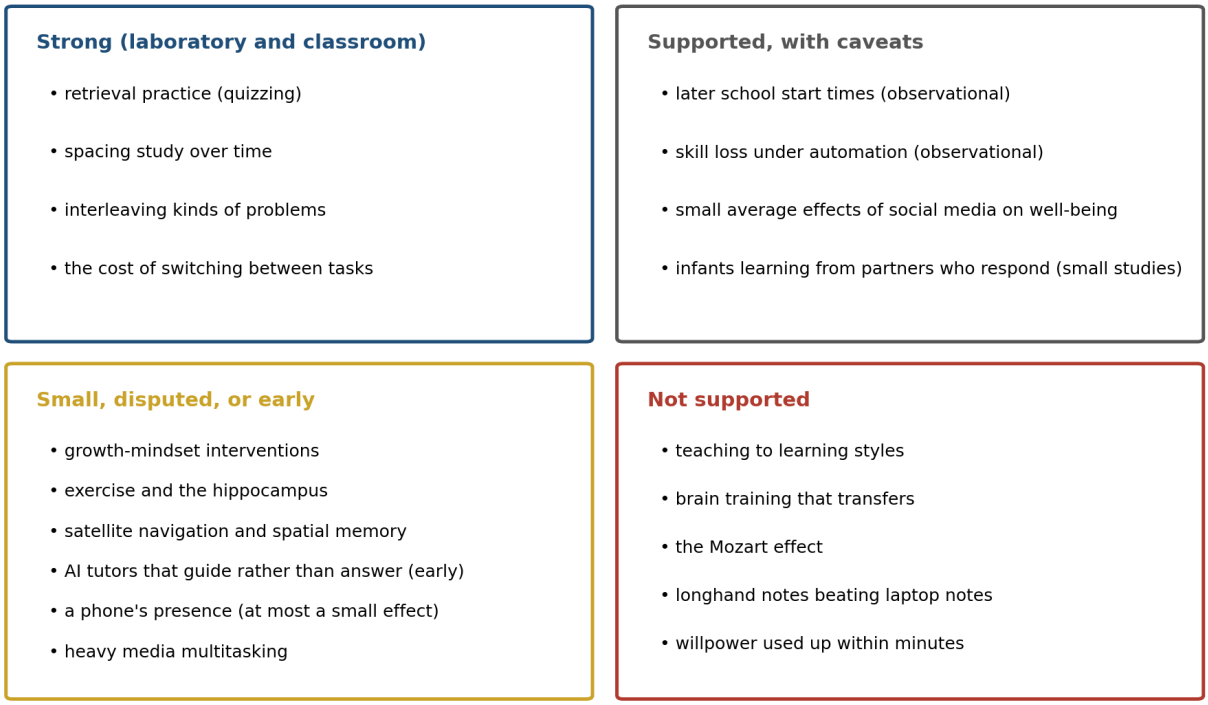


Figure 14.7. The evidence, sorted. The practices and claims discussed in this chapter, grouped by the strength of the evidence for them. The judgments are the author's summary of the studies cited.

First, a system meant to teach should make the learner do the work. The evidence of this chapter is consistent: retrieval, spacing, errors that are corrected, and effortful practice build memory, and a tool that supplies answers removes all four. Tutors that ask before they tell, and that bring material back after an interval, are applying what Chapters 5, 9, and 12 described.

Second, automation should leave its users practiced, as Bainbridge's last irony warns. Aviation answers it with simulators; other fields that adopt AI may need to hand routine cases back to people on purpose, so that the skill is not lost before the hard case arrives.

Third, systems should help people judge them correctly. Hao-Ping Lee and his colleagues at Microsoft Research surveyed 319 knowledge workers who used generative AI and found that those with more confidence in the AI reported thinking less critically about its output, while those with more confidence in themselves reported thinking more critically.[995] The survey relied on self-report and cannot show cause, but it points to the problem Chapter 10 described: a system that sounds equally sure whether it is

[995] Lee, H.-P., Sarkar, A., Tankelevitch, L., Drosos, I., Rintel, S., Banks, R., & Wilson, N. (2025). The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (Article 1121). ACM.

right or wrong invites trust it has not earned. Systems that show calibrated uncertainty give people's own judgment something to work with.

Fourth, attention is the scarcest resource a tool can consume. Every notification is a task switch, and every interruption leaves work to be picked up again; a system designed to capture attention competes with the user's own goals for a budget of attention that, like the brain's energy, cannot be enlarged on demand.

Putting these pieces together into one model of how the brain learns is the task of Chapter 15.

The Central Lesson

The brain that sits in classrooms, in offices, and in front of screens is the one the earlier chapters described: it learns slowly, consolidates in sleep, learns most from its own errors and from other people, runs on a fixed budget of attention and energy, and reshapes itself around what it does. The practices that help it are the ones that work with that design. Retrieving rather than rereading, spacing study over weeks and months, mixing kinds of problems, sleeping, and practicing the skills that matter even when a machine could do them all have good evidence behind them, much of it from real classrooms. Many claims that invoke the brain to sell a product or a policy, from learning styles to brain training to a smartphone's mere presence, have weak or no support. The most useful habit the brain's design suggests is a suspicion of fluency. What feels easy, whether rereading a chapter, having an answer supplied, or letting a machine navigate, often teaches least, because the brain learns from the work it does and the errors it makes. The infants in Kuhl's laboratory learned from a voice that answered them, and not from the same voice on a screen that could not.

Next Chapter

Chapter 15: A Coherent Model of the Human Brain: How the Pieces Fit

Figure and Epigraph Sources

Figure Attribution

Figures 14.1 to 14.7 were drawn for this book. Figures 14.1, 14.4, and 14.6 plot published results: Figure 14.1 from Kuhl, Tsao, and Liu (2003), with the values for the video, audio-only, and Taiwanese groups read approximately from the paper's figure; Figure 14.4 from Kornell and Bjork (2008), Experiment 1a; Figure 14.6 from Bastani et al. (2025). Figures 14.3 and 14.5 are computed from simple illustrative models: Figure 14.3 from a model of spacing in which forgetting follows a power function and a review strengthens a memory more the more has been forgotten, with the measured optimal gaps of Cepeda et al. (2008) shown for comparison; Figure 14.5 from a model in which a skill improves with practice and decays without it. Figures 14.2 and 14.7 are schematic, and the groupings in Figure 14.7 are the author's judgment of the evidence cited in the chapter.

Epigraph

Plato, *Phaedrus*, 275a, in *The Dialogues of Plato*, translated by Benjamin Jowett (3rd ed., Oxford University Press, 1892).

Chapter 15

A Coherent Model of the Human Brain: How the Pieces Fit

> *"Every man can, if he so desires, become the sculptor of his own brain."*
>
> *— Santiago Ramón y Cajal, Reglas y consejos sobre investigación científica (preface to the second edition, 1898)*

A Neuroscientist Studies a Microprocessor

In 2017 Eric Jonas and Konrad Kording tried an experiment on their own field. They took a microprocessor whose workings are completely known, the MOS 6502, the chip inside the Apple I, near-identical versions of which ran the Commodore 64 and the Atari video game console, and asked whether the methods neuroscientists use on brains would reveal how it worked. They had a complete wiring diagram of its 3,510 transistors, a simulator that recorded every transistor's state at every moment, and three "behaviors": the games Donkey Kong, Space Invaders, and Pitfall. They then did to the chip what neuroscientists do to brains. They lesioned each transistor in turn, mapped the "tuning" of transistors to the brightness of pixels on the screen, computed the statistics of their firing, recorded the equivalent of local field potentials, searched for causal influences between regions, and reduced the activity to a few dimensions.[996]

The analyses produced findings that looked like neuroscience. Some transistors, when removed, abolished one game but not the others: "perhaps there is a Donkey Kong transistor or a Space Invaders transistor," the authors wrote, before adding that "this finding of course is grossly misleading. The transistors are not specific to any one behavior or game but rather implement simple functions, like full adders." None of the methods revealed what any student of electrical engineering knows about the chip: that it has registers, an arithmetic unit, and a clock, organized in a hierarchy of functions. The data were complete, and they were not enough. Fifteen years earlier the biologist Yuri Lazebnik had made the same point with a satire, describing how biologists would study a broken radio by removing components one at a time and naming each "the most important component," and arguing that what engineers have and biologists lack is a formal language that says how the parts work together.[997]

The lesson of both papers is that understanding a system requires a model of how its parts relate, not only more data about the parts. This book has gathered the parts one chapter at a time: cells and their chemistry, signals and synapses, neuromodulators, circuits, plasticity, perception, emotion, decision, habit, the comparison with machines, prediction, constraint, and the consequences for daily life. This chapter assembles them into a single model of how the brain learns. The model is not an equation. It is a small set of claims about how the parts fit together, each of which can be checked against the evidence of the earlier chapters and each of which could turn out to be wrong.

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[996] Jonas, E., & Kording, K. P. (2017). Could a neuroscientist understand a microprocessor? *PLoS Computational Biology*, 13(1), e1005268.

[997] Lazebnik, Y. (2002). Can a biologist fix a radio?—Or, what I learned while studying apoptosis. *Cancer Cell*, 2(3), 179–182.

A Model Built Over a Century

The pieces of the model were not found in order, and most were found by people looking for something else. Santiago Ramón y Cajal's drawings in the 1880s and 1890s established that the nervous system is made of separate cells, the neurons, which Wilhelm Waldeyer named in 1891 (Chapter 2). Hermann von Helmholtz had already argued in 1867 that perception is an unconscious inference from inadequate evidence (Chapters 1 and 6), and Walter Cannon in the 1920s and W. Ross Ashby in the 1950s described the body and brain as systems that hold essential variables within bounds (Chapters 1 and 7). Kenneth Craik proposed in 1943 that a brain carries a working model of the world and runs it ahead of events (Chapter 12).

The mechanisms came later. Donald Hebb proposed in 1949 that connections between cells that fire together are strengthened, and in 1973 Timothy Bliss and Terje Lømo found such strengthening in the hippocampus (Chapters 2 and 5). Alan Hodgkin and Andrew Huxley showed in 1952 that the nerve impulse is nothing but the physics of a membrane (Chapters 1 and 2). David Hubel and Torsten Wiesel found in the 1960s that early visual experience sets the wiring of the cortex for life (Chapters 1 and 6). In the 1990s Wolfram Schultz's recordings of dopamine neurons in monkeys turned out to match a learning rule that computer scientists had written down independently (Chapter 3), and James McClelland, Bruce McNaughton, and Randall O'Reilly explained why a brain needs both a fast memory and a slow one by asking what goes wrong in a network that has only one (Chapters 1 and 5).

What is striking is how well the pieces fit, though they came from anatomy, physics, physiology, psychology, engineering, and computer science, over more than a century. Helmholtz's inference became the inference of Chapter 6 and the predictions of Chapter 12; Cannon's regulation became the anchor of value in Chapter 7; Hebb's rule, with Schultz's dopamine added as a third factor, became the learning rule of Chapter 3. A model assembled from such different directions is more likely to be right than one built from a single starting point, because each piece was tested by people who were not trying to make it fit the others.

The Model in Eight Statements

Figure 15.1 draws the model, and Figure 15.2 shows which chapters supply the evidence for each part. In words, it comes to eight statements.

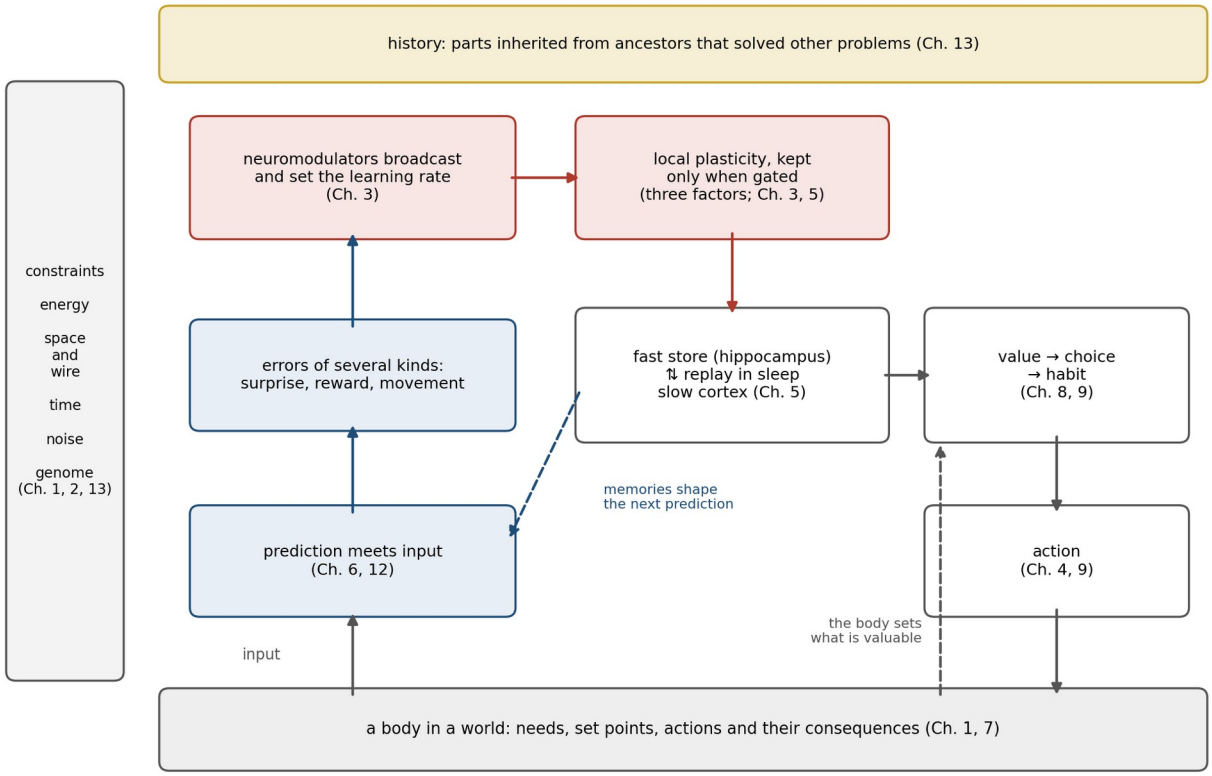


Figure 15.1. The model in one picture. A body in a world (bottom) supplies the brain's purposes and its input. Within the brain, predictions meet input and produce errors; errors of several kinds, broadcast by neuromodulators, gate local plasticity; a fast memory store and slow cortical learning are linked by offline replay; values anchored to the body drive choices, and repeated choices become habits. Constraints of energy, space, time, noise, and the genome (left) and an evolutionary history (top) shape every part.

First, the brain serves a body. Its job is to keep the body's essential variables within bounds in a world that changes, and everything else it does serves that end (Chapters 1 and 7). Value is anchored there: food is worth more to a hungry animal, and fear is a forecast of harm to the body. Even the regulation looks ahead: thirst neurons in the mouse fall silent within seconds of drinking, long before the water could reach the blood (Chapter 12).

Second, it is built from poor parts. They are slow, noisy, and expensive: neurons fail, fire unreliably, and cost energy for every spike, so the brain computes with populations, signals over distance with spikes, and keeps most of its neurons quiet most of the time (Chapters 0, 1, 2, and 13). Peter Lennie's estimate that the cortex can afford only about one neuron in a hundred firing at any moment is a fact about energy that shapes everything built on it.

Third, it predicts. At many levels, from the retina to the frontal cortex, circuits compare what they expect with what arrives and pass on, or learn from, the difference (Chapters 6 and 12). A mouse running through a virtual corridor has neurons in its visual cortex that fire when the corridor stops moving: they respond not to anything on the screen but to the absence of the motion its running had predicted.

Fourth, it learns locally and keeps globally. A synapse changes according to the activity on its two sides, as Hebb proposed, but whether the change is kept depends on a third factor, a broadcast signal of surprise, reward, or importance carried by dopamine, norepinephrine, acetylcholine, or serotonin, arriving while a trace of the recent activity remains (Chapters 3 and 5). In slices of mouse striatum, a single dendritic spine that had just been active enlarged only if dopamine arrived between about 0.3 and 2 seconds later; earlier or later, nothing happened.

Fifth, it remembers on several clocks. Memory is not one system but several: a fast store in the hippocampus captures single episodes; the cortex learns slowly, extracting what is general; sleep replays the one into the other; and practice moves skills from deliberate control to habit (Chapters 5, 9, and 12). The division between fast and lasting change appears even in a sea slug, whose reflex is strengthened for minutes by one shock and for days by five spaced shocks that switch on genes (Chapter 1).

Sixth, it turns value into action, and action into habit. It accumulates evidence until it is sure enough, weighs costs against rewards, explores when it is uncertain, and caches what it has learned as habits that run without deliberation (Chapters 8 and 9). A rat that has pressed a lever many times keeps pressing after the reward has been made worthless, the signature of a habit that no longer consults the value it was built on.

Seventh, it is grown, not specified. The genome sets up cell types, gradients, and learning rules, and development and experience supply the rest, which is why early experience matters so much and why learning is not optional (Chapters 0, 1, and 13). The genome falls short of listing the human brain's connections by a factor of more than half a million, and a kitten whose eye was sutured closed for its first months never recovers the use of it, however good the rules that built its cortex.

Eighth, it is a compromise and an inheritance. The brain sits on fronts between goals it cannot all maximize, moves along them with its neuromodulators as conditions change, and does so with parts inherited from ancestors that solved different problems (Chapter 13). Light still crosses the wiring of the vertebrate retina to reach the photoreceptors behind it, a layout that costs every vertebrate a blind spot and that no vertebrate has undone.

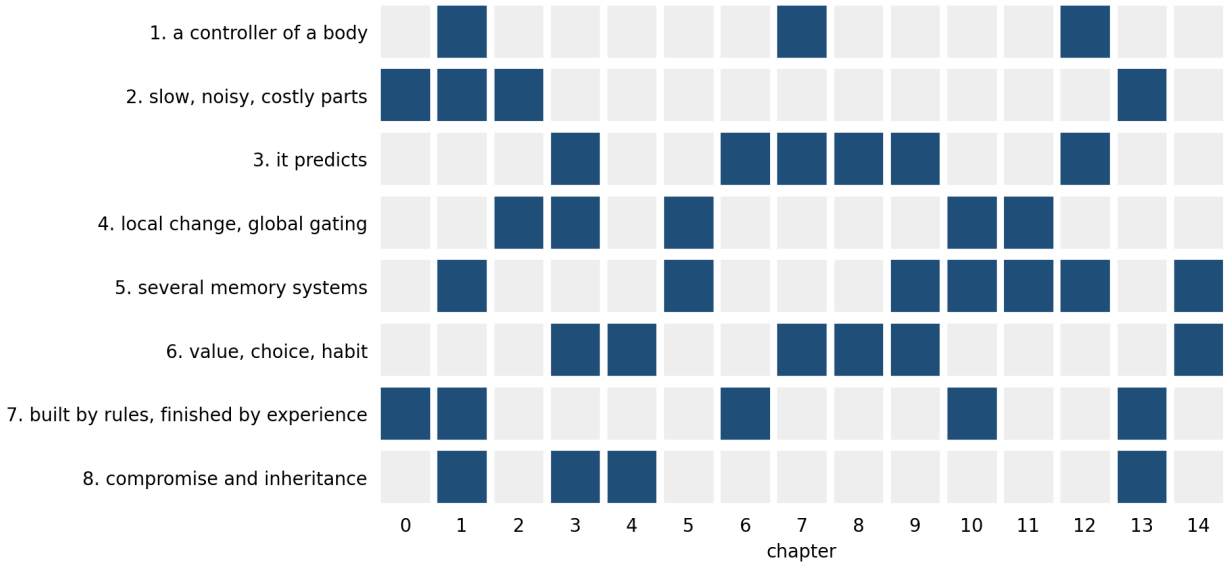


Figure 15.2. Where the evidence comes from. Each row is one of the model's eight statements; filled cells mark the chapters that supply its main evidence. No statement rests on a single chapter.

One thread runs through several of them: much of what a person learns comes from other people, whose responses supply both the errors and the value, as Kuhl's infants showed in Chapter 14. None of the eight statements is new to this chapter; each was argued, with evidence, earlier. What the model adds is the claim that they belong together. Prediction supplies the errors that neuromodulators broadcast; the broadcast gates the plasticity that stores memories; the memory systems shape what is predicted next; the body sets what counts as an error worth learning from; and the constraints decide how all of this can be built at all. The clearest way to see the parts working together is to follow a single piece of learning through them.

One Afternoon, Every Level

Consider an American who moves to Ireland and drives for the first time on the left side of the road. The example is illustrative rather than an experiment, but every step in it rests on mechanisms described earlier, and Figure 15.3 lays out the timescales on which they operate.

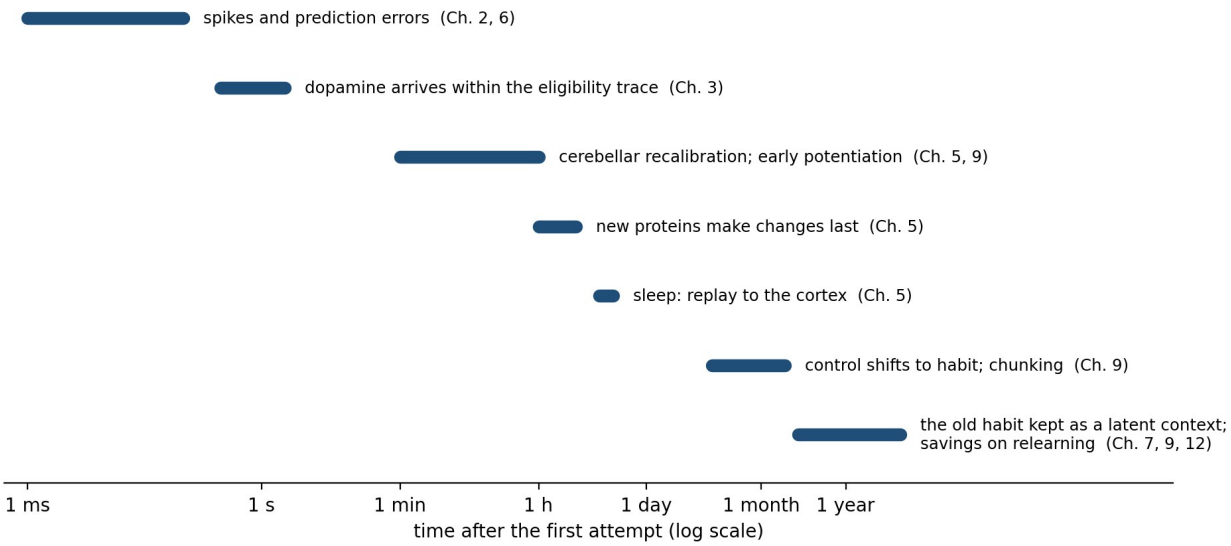


Figure 15.3. One piece of learning across timescales (schematic). Learning to drive on the other side of the road engages mechanisms that operate from milliseconds to years; the chapter numbers show where each was described.

In the first seconds, predictions fail everywhere. The driver's visual system expects oncoming traffic on the left and finds it on the right; the arm reaches for a gear lever on the wrong side; at the first roundabout the eyes look the wrong way. Each failure is a prediction error, in sensory cortex, in the cerebellum, and in the circuits of choice (Chapters 6, 8, 9, and 12). Norepinephrine rises with the surprise and with the stress of a narrow road and a truck behind, raising the learning rate and narrowing attention, which helps at first and hurts if it climbs too far (Chapters 3, 7, and 12). The limited budget of attention described in Chapter 14 is spent quickly, and by the end of the afternoon the driver is tired in a way that a familiar drive never produces.

Within minutes, some things are already changing. The cerebellum recalibrates the reach toward the gear lever through the same error-driven adaptation that Chapter 9 measured when people learned to move in a force field. When a turn goes well, a burst of dopamine marks it as better than expected and strengthens the synapses whose eligibility traces were still active, and when it goes badly, a dip weakens them (Chapters 3 and 8). In the hippocampus, place cells and their partners encode the new route and its landmarks in a single exposure, and the early phase of long-term potentiation holds those changes for the next hour or two (Chapters 4 and 5).

That night, sleep does work the driver will never notice. During deep sleep the hippocampus replays the day's sequences in compressed form, and the cortex begins to absorb them; whether sleep also scales down the synapses strengthened during the day, as one hypothesis holds, is still debated (Chapter 5, and below). Proteins are made that turn the early, fragile changes into lasting ones.

Over the following weeks, the driving moves from deliberation to habit. What required attention on the first day runs on its own by the second month, as control shifts from the circuits of goal-directed action toward those of habit and the movements are chunked into sequences (Chapter 9). The driver can now talk while driving, and the attention that was consumed is free again.

Months later, on a visit home, the driver steps into a car on the familiar side of the road and, at the first junction, glances the wrong way. The old habit was not erased; it was overlaid. Samuel Gershman, David Blei, and Yael Niv have argued that the brain treats such episodes as evidence about hidden causes, so that a new context is learned as a new cause rather than by rewriting the old one, and returning to the old context brings back the old learning.[998] The same logic explains why extinguished fears return in their original context (Chapter 7) and it may help explain why a skill relearned after a long break comes back faster than it was first acquired (Chapter 9). The brain did not replace one way of driving with another; it kept both and learned when to use each.

Nearly every statement of the model appears in this one afternoon and its sequel. The body and its safety set what mattered; slow, noisy parts made the learning gradual; prediction errors drove it; neuromodulators decided what was kept; the fast and slow memory systems divided the work; choice became habit; and the constraints of energy and attention set the pace.

The Levels Talk to Each Other

A model with levels invites the assumption that each level can be understood on its own and the results stacked. The evidence of this book points the other way. The levels constantly change one another, and that is where much of the brain's flexibility comes from.

Neuromodulators are the clearest case. Eve Marder spent decades on a circuit of about 30 neurons in the stomach of crabs and lobsters whose wiring is completely known, and found that neuromodulators "reconfigure neuronal circuits, often massively altering their output. Thus, the anatomical connectome provides a minimal structure and the neuromodulatory environment constructs and specifies the functional circuits that give rise to behavior."[999] Cornelia Bargmann reached the same conclusion from the nematode worm: "each ultrastructural connectivity map encodes multiple circuits, some of which are active and some of which are latent at any given time."[1000] The same wiring computes different things in different chemical states, which is how the brain can move along the trade-offs of Chapter 13 without rebuilding itself.

The body reaches into every level as well. Chapter 7 described how hunger changes the value of food in the circuits of choice, and how stress shifts control from deliberation to habit. And learning changes the machinery of learning: a synapse's recent history alters how easily it will change again, and the neuromodulatory systems that gate plasticity are themselves shaped by experience.

Recent recordings of the whole brain have made the point vivid. The International Brain Laboratory, a collaboration of 12 laboratories, recorded from 621,733 units, with 75,708 well-isolated neurons among them, across 279 brain areas in 139 mice performing the same decision task. Signals about the stimulus appeared first in the visual areas, as a textbook pipeline would predict, but activity related to the upcoming movement appeared "almost everywhere in the brain," and responses to reward were also widespread.[1001] A companion study found that the animals' expectation about which side would be

[998] Gershman, S. J., Blei, D. M., & Niv, Y. (2010). Context, learning, and extinction. *Psychological Review*, 117(1), 197–209.

[999] Marder, E. (2012). Neuromodulation of neuronal circuits: Back to the future. *Neuron*, 76(1), 1–11.

[1000] Bargmann, C. I. (2012). Beyond the connectome: How neuromodulators shape neural circuits. *BioEssays*, 34(6), 458–465.

[1001] International Brain Laboratory, Meshulam, L., Angelaki, D., Benson, J., McRoberts, I., Noel, J., Arlandis, J., Bonacchi, N., Bougrova, K., Catarino, J. A., Cazettes, F., Crombie, D., DeWitt, E. E., Freitas-Silva, L., Laranjeira, I. C., Mainen, Z. F., Meijer, G. T., Rai, P., Raiser, G., … Witten, I. B. (2025). A brain-wide map of neural activity during complex behaviour. *Nature*, 645(8079), 177–191.

rewarded, their prior, was represented in at least a fifth to a third of the regions recorded, from early visual areas to motor cortex, which the authors argued fits a model of inference carried out in loops between areas rather than in a single decision center.[1002]

Danielle Bassett and Olaf Sporns have argued that understanding such a system requires studying the interactions across its scales, from molecules to social groups, rather than any one level alone.[1003] The brain is not a stack of modules passing results upward; it is a set of loops in which every level is in contact with every other.

When the Model Breaks

A model earns its keep by explaining failures as well as successes, and many disorders of the brain can be read as failures of particular parts of this one.

Parkinson's disease is the loss of the dopamine neurons that feed the striatum, and its defining effects fall on the selection and vigor of movement, the part of the model in which value becomes action (Chapters 3, 4, and 9). The most vivid demonstration came in 1969, when Oliver Sacks began giving L-dopa, the chemical precursor of dopamine, to patients in a New York hospital who had been frozen for decades by the aftermath of the epidemic of encephalitis lethargica that followed the First World War. Many woke into movement and speech; many then developed tics, sudden swings between frozen and frantic states, and other complications as the treatment continued.[1004] The patients' movement had failed, been restored, and then overshot. Addiction has been described as a habit gone wrong, in which a drug that acts directly on the dopamine signal teaches the circuits of habit a lesson no natural reward could, although Chapter 9 noted that the habit account is disputed.

Fear that returns after therapy is the model's memory systems working as designed: extinction lays down a second memory beside the first rather than erasing it, so the old fear can come back in a new context (Chapter 7), exactly as the latent-cause account predicts. Hallucinations can be read as the prediction machinery giving too much weight to its expectations: Albert Powers and his colleagues showed that people who hear voices are especially prone to hearing tones that conditioning has led them to expect (Chapter 12). And in Alzheimer's disease the damage falls disproportionately on the brain's hubs, the costly long-range connections of Chapters 4 and 13, while in mouse models of its early stages memories that can no longer be recalled naturally can still be retrieved by stimulating the cells that hold them, suggesting that some early losses are failures of retrieval rather than of storage.[1005]

Each of these is far more complicated than a sentence can say, and none is fully explained by the model. But each falls naturally at a particular place in it, and the place suggests where to look: at the dopamine signal, at the competition between memories, at the weighting of prediction against evidence, at the costliest wiring. A model that located every disorder in the same place, or in none, would be less useful.

[1002] Findling, C., Hubert, F., International Brain Laboratory, Acerbi, L., Benson, B., Benson, J., Birman, D., Bonacchi, N., Buchanan, E. K., Bruijns, S., Carandini, M., Catarino, J. A., Chapuis, G. A., Churchland, A. K., Dan, Y., Davatolhagh, F., DeWitt, E. E. J., Engel, T. A., Fabbri, M., … Pouget, A. (2025). Brain-wide representations of prior information in mouse decision-making. *Nature*, 645(8079), 192–200.

[1003] Bassett, D. S., & Sporns, O. (2017). Network neuroscience. *Nature Neuroscience*, 20(3), 353–364.

[1004] Sacks, O. W., Kohl, M., Schwartz, W., & Messeloff, C. (1970). Side-effects of L-dopa in postencephalitic parkinsonism. *The Lancet*, 295(7654), 1006; Sacks, O. (1990). *Awakenings* (Rev. ed.). HarperPerennial. (Original work published 1973).

[1005] Josselyn, S. A., & Tonegawa, S. (2020). Memory engrams: Recalling the past and imagining the future. *Science*, 367(6473), eaaw4325.

The Wiring Diagram Arrives

In 1979 Francis Crick, surveying what neuroscience could hope to learn, wrote that "it is no use asking for the impossible, such as, say, the exact wiring diagram for a cubic millimetre of brain tissue and the way all its neurons are firing."[1006] Within half a century the impossible had been done, more than once.

In 2024 the FlyWire consortium published the complete wiring diagram of an adult fruit fly's brain: 139,255 neurons and 54.5 million synapses between them, reconstructed from electron micrographs of a single female fly, with 8,453 types of cell identified.[1007] In 2025 the MICrONS consortium reconstructed a cubic millimeter of a mouse's visual cortex, more than 200,000 cells and about half a billion synapses, after recording the activity of about 75,000 of its neurons while the mouse watched films, so that, for the first time, the wiring and the firing of the same neurons could be compared.[1008]

Alexander Shapson-Coe, Jeff Lichtman, and their colleagues reconstructed a cubic millimeter of human temporal cortex, removed during surgery for epilepsy: about 57,000 cells, some 16,000 of them neurons, and 150 million synapses, in 1.4 petabytes of data. Glia outnumbered neurons two to one, and while almost all connections between an axon and a neuron consisted of a single synapse, a rare few axons made up to 50 synapses onto the same target.[1009] A census of the human brain's cell types, based on more than three million nuclei, distinguished 3,313 subclusters of cells by the genes they express.[1010] The whole human brain, by comparison, has about 86 billion neurons and, in the neocortex alone, about 164 trillion synapses.[1011] Figure 15.4 sets these maps on a common scale.

[1006] Crick, F. H. C. (1979). Thinking about the brain. *Scientific American*, 241(3), 219–232.

[1007] Dorkenwald, S., Matsliah, A., Sterling, A. R., Schlegel, P., Yu, S.-C., McKellar, C. E., Lin, A., Costa, M., Eichler, K., Yin, Y., Silversmith, W., Schneider-Mizell, C., Jordan, C. S., Brittain, D., Halageri, A., Kuehner, K., Ogedengbe, O., Morey, R., Gager, J., … Murthy, M. (2024). Neuronal wiring diagram of an adult brain. *Nature*, 634(8032), 124–138; Schlegel, P., Yin, Y., Bates, A. S., Dorkenwald, S., Eichler, K., Brooks, P., Han, D. S., Gkantia, M., dos Santos, M., Munnelly, E. J., Badalamente, G., Serratosa Capdevila, L., Sane, V. A., Fragniere, A. M. C., Kiassat, L., Pleijzier, M. W., Stürner, T., Tamimi, I. F. M., Dunne, C. R., … Jefferis, G. S. X. E. (2024). Whole-brain annotation and multi-connectome cell typing of Drosophila. *Nature*, 634(8032), 139–152.

[1008] MICrONS Consortium. (2025). Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640(8058), 435–447.

[1009] Shapson-Coe, A., Januszewski, M., Berger, D. R., Pope, A., Wu, Y., Blakely, T., Schalek, R. L., Li, P. H., Wang, S., Maitin-Shepard, J., Karlupia, N., Dorkenwald, S., Sjostedt, E., Leavitt, L., Lee, D., Troidl, J., Collman, F., Bailey, L., Fitzmaurice, A., … Lichtman, J. W. (2024). A petavoxel fragment of human cerebral cortex reconstructed at nanoscale resolution. *Science*, 384(6696), eadk4858.

[1010] Siletti, K., Hodge, R., Mossi Albiach, A., Lee, K. W., Ding, S.-L., Hu, L., Lönnerberg, P., Bakken, T., Casper, T., Clark, M., Dee, N., Gloe, J., Hirschstein, D., Shapovalova, N. V., Keene, C. D., Nyhus, J., Tung, H., Yanny, A. M., Arenas, E., … Linnarsson, S. (2023). Transcriptomic diversity of cell types across the adult human brain. *Science*, 382(6667), eadd7046.

[1011] Azevedo, F. A. C., Carvalho, L. R. B., Grinberg, L. T., Farfel, J. M., Ferretti, R. E. L., Leite, R. E. P., Jacob Filho, W., Lent, R., & Herculano-Houzel, S. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5), 532–541; Tang, Y., Nyengaard, J. R., De Groot, D. M. G., & Gundersen, H. J. G. (2001). Total regional and global number of synapses in the human brain neocortex. *Synapse*, 41(3), 258–273.

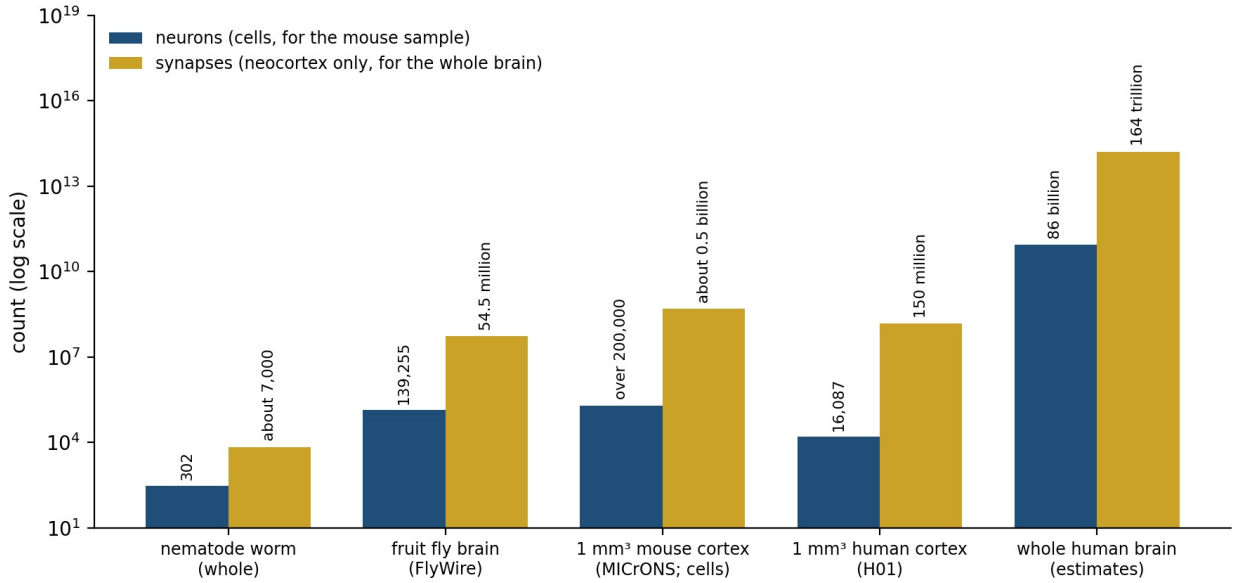


Figure 15.4. Wiring diagrams on one scale (data from the studies cited). Numbers of neurons (blue) and synapses (gold), on a logarithmic scale, for the complete nervous system of the nematode worm and the brain of the fruit fly, for a cubic millimeter of mouse visual cortex (cells, not only neurons) and of human temporal cortex, and for the whole human brain (estimates; synapses in the neocortex only). The human cubic millimeter is about one part in a million of the human brain.

These maps are among the great achievements of modern biology, and the model of this book suggests both what they will and will not reveal. They will constrain every theory of circuits: no model of how the fly navigates or the mouse sees can now ignore which neurons are actually connected. But the microprocessor already had a complete wiring diagram, and it did not explain the chip. Bargmann and Marder, reviewing what wiring diagrams had taught neuroscience, concluded that "anatomical inspection fails because each wiring diagram encodes many possible circuit outcomes."[1012] A map records which connections exist, not the neuromodulatory state that selects among them, nor the history of learning that set their strengths, nor the body whose needs gave that learning its purpose. As Chapter 13 noted, the wiring of a human brain cannot be specified by its genome; it is a record of a life. A full map of one brain would be a snapshot of one history, and understanding it would still need the model.

No Two Brains Alike

The model describes how brains in general learn, but no brain is a brain in general. Chapter 1 described degeneracy, the capacity of different combinations of parts to produce the same function, and Chapter 13 described how very different mixtures of ion channels and synapses can produce the same rhythm in a small circuit. The same is true of whole brains. Two people who have learned the same skill have learned it with different synapses, in brains whose wiring was shaped by different childhoods; Chapter 14's London taxi drivers grew their posterior hippocampi through years of a particular kind of work, and the trainees who failed did not. Each of the great wiring diagrams described above comes from a single animal or a single person, and the next brain mapped will differ from it in detail everywhere.

Even identical genes and an identical world do not prevent it. Julia Freund, Gerd Kempermann, and their colleagues housed 40 genetically identical female mice together for three months in a large

[1012] Bargmann, C. I., & Marder, E. (2013). From the connectome to brain function. *Nature Methods*, 10(6), 483–490.

enclosure with several levels, and tracked each one with radio tags. Over time the mice diverged: the differences among them in how widely they explored grew about 22-fold, and the mice that roamed most widely tended to have made the most new neurons in the hippocampus.[1013] Same genes, same world, and still 40 different animals.

This is not a flaw in the model but one of its predictions. A system that learns by local changes gated by its own history of surprise and reward, starting from rules rather than a blueprint, must end up unique. The model describes the rules by which every brain is built and changed; the particular brain that results is always one of a kind, and it is still changing.

What Is Settled and What Is Open

A model is only as good as the evidence for its parts, and the evidence is uneven. Figure 15.5 sorts the main claims of this book by how firmly they are established.

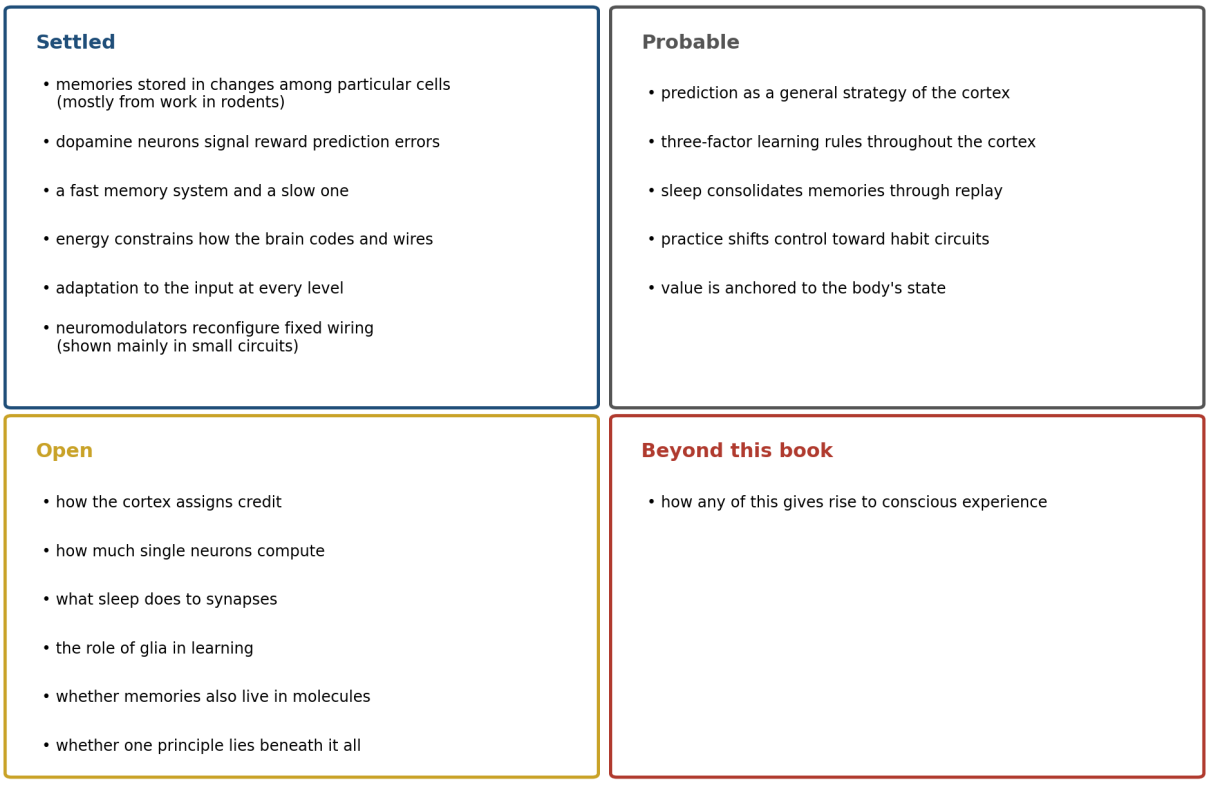


Figure 15.5. Settled, probable, and open. The book's main claims, grouped by the strength of the evidence; the groupings are the author's judgment of the studies cited across the book.

Some things are settled as firmly as anything in biology. Memories are stored, at least in part, in changes to synapses among particular sets of neurons, and in mice those sets can be labeled, silenced, and reactivated to suppress or call up a memory; Sheena Josselyn and Susumu Tonegawa's review of this work notes that it comes almost entirely from rodents and mostly from fear learning, but the core

[1013] Freund, J., Brandmaier, A. M., Lewejohann, L., Kirste, I., Kritzler, M., Krüger, A., Sachser, N., Lindenberger, U., & Kempermann, G. (2013). Emergence of individuality in genetically identical mice. *Science*, 340(6133), 756–759.

finding has been reproduced many times.[1014] Dopamine neurons signal errors in the prediction of reward. The brain keeps a fast memory system and a slow one, and damage to the first spares the second. Energy constrains everything, and adaptation to the input runs through every level. Prediction as a general strategy belongs a step lower, among the probable claims: Chapter 12 found it everywhere in the brain's behavior but found the evidence for its specific cortical form, predictive coding, mixed.

Other questions are genuinely open, and the model would change depending on their answers.

The first is how the cortex assigns credit, a question Chapters 3, 10, and 11 returned to repeatedly. Timothy Lillicrap, Geoffrey Hinton, and their colleagues have proposed that feedback connections might produce differences in activity that carry error information, approximating backpropagation, and theorists have shown how bursts of spikes or the separate compartments of a pyramidal neuron's dendrites could carry such signals.[1015] These are models and simulations; no experiment has yet shown the cortex doing it.

The second is what single neurons compute. Chapter 2 described the dendritic calcium spikes that Albert Gidon, Matthew Larkum, and their colleagues recorded in human cortical neurons, which, on the evidence of modeling based on the recordings, could let a single neuron compute exclusive or.[1016] The result does not yet appear to have been independently replicated, and neurons in a different layer of the human cortex turn out to have dendrites that are, if anything, more isolated from the cell body than a rat's.[1017] How much computation happens inside a single cell, and how much this differs across species, is not known.

The third is what sleep does to synapses. Giulio Tononi and Chiara Cirelli have argued that "sleep is the price the brain pays for plasticity," a period in which synapses strengthened during the day are scaled back so that the brain does not saturate, and electron microscopy of mouse cortex found the contacts between axons and spines about 18 percent smaller after sleep than after waking.[1018] Other work finds sleep strengthening and building particular synapses after learning.[1019] Sleep may do both, to different synapses, but the balance is unresolved.

The fourth is the role of the brain's other cells. Glia outnumber neurons in the human sample described above, and Adar Adamsky, Inbal Goshen, and their colleagues found that activating astrocytes in the mouse hippocampus could induce long-lasting potentiation and enhance memory, while other

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[1014] Josselyn, S. A., & Tonegawa, S. (2020). Memory engrams: Recalling the past and imagining the future. *Science*, 367(6473), eaaw4325.

[1015] Lillicrap, T. P., Santoro, A., Marris, L., Akerman, C. J., & Hinton, G. (2020). Backpropagation and the brain. *Nature Reviews Neuroscience*, 21(6), 335–346; Payeur, A., Guerguiev, J., Zenke, F., Richards, B. A., & Naud, R. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24(7), 1010–1019.

[1016] Gidon, A., Zolnik, T. A., Fidzinski, P., Bolduan, F., Papoutsi, A., Poirazi, P., Holtkamp, M., Vida, I., & Larkum, M. E. (2020). Dendritic action potentials and computation in human layer 2/3 cortical neurons. *Science*, 367(6473), 83–87.

[1017] Beaulieu-Laroche, L., Toloza, E. H. S., van der Goes, M.-S., Lafourcade, M., Barnagian, D., Williams, Z. M., Eskandar, E. N., Frosch, M. P., Cash, S. S., & Harnett, M. T. (2018). Enhanced dendritic compartmentalization in human cortical neurons. *Cell*, 175(3), 643–651.e14.

[1018] Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: From synaptic and cellular homeostasis to memory consolidation and integration. *Neuron*, 81(1), 12–34; de Vivo, L., Bellesi, M., Marshall, W., Bushong, E. A., Ellisman, M. H., Tononi, G., & Cirelli, C. (2017). Ultrastructural evidence for synaptic scaling across the wake/sleep cycle. *Science*, 355(6324), 507–510.

[1019] Yang, G., Lai, C. S. W., Cichon, J., Ma, L., Li, W., & Gan, W.-B. (2014). Sleep promotes branch-specific formation of dendritic spines after learning. *Science*, 344(6188), 1173–1178.

laboratories using related methods found that astrocyte signaling did not affect plasticity.[1020] Whether glia are partners in learning or only its support staff is still disputed.

The fifth is where, exactly, a memory is stored. The engram studies point to synapses among particular cells, but Johannes Langille and Randy Gallistel have argued that the evidence is also consistent with information stored in molecules inside cells, with synaptic change serving to read it out.[1021] The view is a minority one, but it shows that even the most settled part of the model rests on an inference that could be revised.

And some questions lie beyond this book altogether. Chapter 12 asked whether a single principle, such as the minimization of free energy, underlies the brain's mechanisms, and found the evidence weak. How any of these mechanisms gives rise to conscious experience is a question this book has not tried to answer, because the evidence does not yet allow an answer of the kind the earlier chapters could give.

Implications for Artificial Intelligence

Chapters 10 and 11 compared the brain with the machines built in its image. The model of this chapter turns that comparison into a short list. Current artificial systems share three of its features in some form: they learn from prediction errors, they represent the world in learned hierarchies, and the largest of them, as Chapter 13 noted, have converged on activating only a small fraction of their parts at a time.

They lack five. They have no body whose state gives value its meaning, and so no anchor for what they should want (statements one and six). They learn mostly in one phase and are then frozen, without the fast store, the offline replay, and the many timescales of statement five. They learn by a single global error rather than by local changes gated by several broadcast signals (statement four). They do not reconfigure a single network into many with something like neuromodulation, which lets the brain move along its trade-offs as conditions change (statement eight). And they are not built under a constraint of energy that forces economy in every part (statement two). Whether a machine needs all five to learn as flexibly as a child is an open question, and Chapter 11 described the attempts to supply each. The model at least makes clear what the question is.

The Central Lesson

The brain learns by changing its own physical structure, in a body, under constraint. It predicts what will happen and learns from the difference between prediction and outcome; it keeps the changes that broadcast signals of surprise and value mark as important; it divides the work between a fast memory that captures episodes and a slow one that extracts what is general, and lets sleep pass between them; it turns choices into habits so that attention is free for what is new; and it does all of this with parts that are slow, noisy, and expensive, built by a genome that could only supply the rules, inherited from ancestors that faced different problems. These are not separate facts but parts of one design, and they explain one another: the constraints explain the mechanisms, the mechanisms explain the behavior, and the behavior explains why the brain needs a body and a world to learn from at all.

[1020] Adamsky, A., Kol, A., Kreisel, T., Doron, A., Ozeri-Engelhard, N., Melcer, T., Refaeli, R., Horn, H., Regev, L., Groysman, M., London, M., & Goshen, I. (2018). Astrocytic activation generates de novo neuronal potentiation and memory enhancement. *Cell*, 174(1), 59–71.e14; Agulhon, C., Fiacco, T. A., & McCarthy, K. D. (2010). Hippocampal short- and long-term plasticity are not modulated by astrocyte Ca2+ signaling. *Science*, 327(5970), 1250–1254.

[1021] Langille, J. J., & Gallistel, C. R. (2020). Locating the engram: Should we look for plastic synapses or information-storing molecules? *Neurobiology of Learning and Memory*, 169, 107164.

Jonas and Kording's microprocessor had a complete wiring diagram and yielded no understanding without a model. The model in this book is a first draft of one for the brain, and parts of it will be revised as the open questions are answered. Its central claim is the one with which this book began, and which Cajal made at the end of the nineteenth century, when almost none of the mechanisms in this book were known: that each brain is built, in large part, by what it does. Every lesson learned, every habit formed, and every night of sleep leaves a physical trace, and the brain that results is the record of a life. The work of understanding it is to read that record by the principles that wrote it.

Next Chapter

Chapter 16: NIMCP: Building a Brain from the Book's Principles

Figure and Epigraph Sources

Figure Attribution

Figures 15.1 to 15.5 were drawn for this book. Figure 15.4 plots published counts: the nematode worm's 302 neurons and about 7,000 synapses; the fruit fly brain of Dorkenwald et al. (2024); the cubic millimeter of mouse visual cortex of the MICrONS Consortium (2025), for which the count is of all cells; the cubic millimeter of human temporal cortex of Shapson-Coe et al. (2024), for which the count is of neurons; and estimates for the whole human brain from Azevedo et al. (2009) for neurons and Tang et al. (2001) for synapses in the neocortex. Figures 15.1, 15.2, 15.3, and 15.5 are schematic; the assignments in Figures 15.2 and 15.5 are the author's judgment of the evidence presented across the book.

Epigraph

Santiago Ramón y Cajal, *Reglas y consejos sobre investigación científica*, preface to the second edition (dated Madrid, 20 December 1898): "todo hombre puede ser, si se lo propone, escultor de su propio cerebro." Translated for this book. An English translation of the whole work is *Advice for a Young Investigator*, translated by Neely Swanson and Larry W. Swanson (MIT Press, 1999).

Part VI: A Case Study

Chapter 16

NIMCP: Building a Brain from the Book's Principles

> *"Essentially, all models are wrong, but some are useful."*
>
> *— George E. P. Box and Norman R. Draper (1987)*

A Brain Built from the Book

In May 2026 a simulated brain called Athena, raised for months by a simulated parent that named things for it, corrected its babbling, and told it stories, was asked a simple question: "What color is the sky?" It answered: "ball beside eye Look! The ball is beside the eye …" At other checkpoints that month, asked 50 such questions, it gave the same string of words to every one. Athena was built from the principles of this book, by its author, and this chapter asks why it does not yet work.

Chapter 11 took Feynman's blackboard as its epigraph and argued that building is a test of understanding: if the principles of the earlier chapters are right, it should be possible to build something that learns by them, and if the attempt fails, the failure should be informative. This chapter applies that test to one attempt in particular.

Athena is the trained instance of NIMCP, short for Neuro-Inspired Modular Control Protocol, an experimental software system that simulates the machinery this book has described and tries to raise it like a child. In its present form it runs about two million simulated spiking neurons on a single graphics processor, organized into populations given the names of cortical layers and language areas. It simulates six neuromodulators, runs sleep cycles in which it replays recent experience, regulates its own firing rates, and is trained through a staged curriculum in which a simulated parent names things, corrects babbling, and holds conversations. Its documentation describes it as "a digital brain that grows up like a child," and it states the bet plainly: to "build the mechanisms biology uses, and see what emerges from them."[1022]

A disclosure is needed before going further. NIMCP is the author's own project. That is a reason for care rather than for silence, and this chapter holds it to the standard the book has applied to every other claim: what the code actually does, what has been measured, and what the measurements show. The account is based on a reading of the NIMCP source code, documentation, training logs, and evaluation files, and where the documentation and the code disagree, the chapter follows the code. It describes NIMCP as it stood in October 2026, at commit 4693efd of the project's repository (18 October 2026), and later versions may differ; the questions it raises will outlast the details.

The result is not a success story, at least not yet, and it is more useful than one. NIMCP shows which of the book's principles translate readily into software and which do not, and its difficulties line up, almost one for one, with the parts of the model of Chapter 15 that it has not yet managed to implement. The book can be used to read the failures, and the failures, in turn, are consistent with the book's account of what is hard.

[1022] NIMCP project. (2026). *README.md* [Project overview]. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

What Was Built

Figure 16.1 lays out the system as the code builds it. At its center is a spiking network of leaky integrate-and-fire neurons, the simplest model that captures the essentials Chapter 2 described: each neuron integrates its input, leaks charge back toward rest with a time constant of 20 milliseconds, fires when its voltage crosses a threshold 15 millivolts above rest, and is briefly refractory afterward. In the mode used in production, synaptic input arrives as conductances through fast and slow excitatory receptors, including the NMDA receptor with its magnesium block, the coincidence detector of Chapter 5, and through inhibitory ones. Excitatory and inhibitory neurons are kept separate, as Dale's principle requires, and the populations include the three main classes of cortical interneuron, parvalbumin, somatostatin, and VIP cells, wired so that one class inhibits another, the disinhibition motif that Chapter 4 described. Pyramidal neurons differ from one another by about 10 percent in their time constants and thresholds, a deliberate dose of the variability Chapter 1 described.[1023]

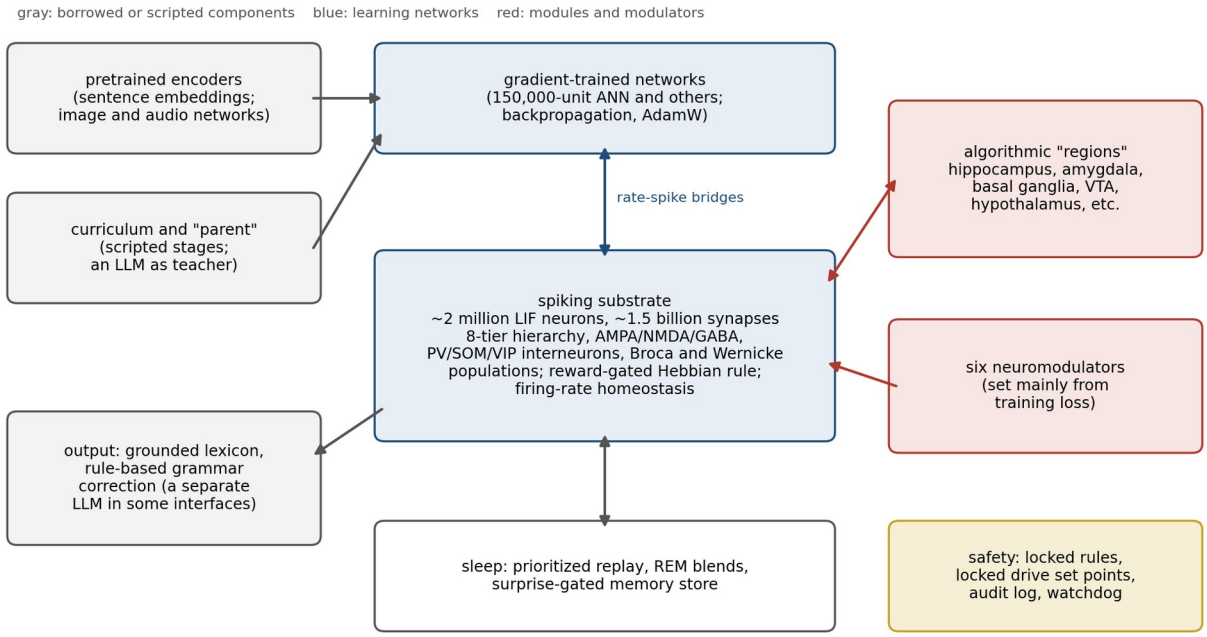


Figure 16.1. NIMCP as built (schematic, from the source code as of October 2026). A spiking network of about two million neurons (center) is coupled through rate-to-spike bridges to conventional networks trained by backpropagation (top). Named brain "regions" and the six neuromodulators (right) are mostly algorithmic modules acting on vectors rather than populations of neurons. Pretrained encoders, a scripted curriculum with a large language model as teacher, and rule-based language output (left) surround the core.

The scale is real but more modest than some of the project's own descriptions. The spiking network has about 1.8 million neurons in 46 populations, plus about 200,000 more in language and sensorimotor populations, for roughly two million in all. The number of synapses, counted from the connectivity tables and the size of the saved network, is about 1.5 billion, not the "trillions" that the project's introductory material once claimed.[1024] Connectivity is sparse, between 0.02 and 0.5 percent between any

[1023] NIMCP project. (2026). *Spiking network implementation*: include/snn/nimcp_snn_membrane.h, src/snn/nimcp_snn_hierarchical.c, src/snn/nimcp_snn_training.c, and docs/architecture/20_snn.md and 30_gpu_memory.md. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

two populations, so that a typical neuron receives a few hundred to a thousand inputs. A cortical neuron receives about 10,000.

The arrangement of the populations is simpler than the names suggest. The spiking network is a generic hierarchy of eight tiers, from an input tier through feature, pattern, concept, integration, executive, and projection tiers to an output tier, with recurrent connections within the middle tiers and a few connections that skip tiers. As a comment in the code puts it, the substrate "has no concept of 'region'." The 40 or so structures that carry anatomical names, the hippocampus, amygdala, basal ganglia, ventral tegmental area, hypothalamus, cerebellum, and others, are mostly algorithmic modules that operate on vectors of numbers, connected to the spiking network through bridges, rather than populations of spiking neurons. The two "hemispheres" are a routing rule that sends a decision to one side or the other according to the statistics of its input, not separate populations of neurons.[1025]

Around the spiking core sit components that are not brain-like at all. A conventional artificial network of 150,000 units, trained by backpropagation, does much of the learning, and smaller conventional networks process images and sounds. Text enters as sentence embeddings, lists of numbers that a pretrained model assigns to a sentence so that sentences with similar meanings get similar lists, computed by transformer models of the kind Chapter 10 described, and those embeddings also serve as the targets the system learns to match.[1026] The simulated parent of the curriculum is a large language model, Claude, prompted to act as "a loving, warm parent raising a child." Language comes out through a lexicon of words bound to internal patterns, polished by hand-written rules for agreement, word order, and fluency, and in some interfaces through a separate small language model, Phi-3.[1027] These choices are reasonable engineering, but they matter for interpretation: parts of what the system knows about meaning, and much of what makes its output read as language, come from components that were not built from the book's principles.

The Book's Model, Line by Line

Chapter 15 summarized the book in eight statements. Figure 16.2 sets each beside what NIMCP contains, sorted by whether the corresponding mechanism is running in the main path of the system, present but only partly connected, or absent.

[1024] NIMCP project. (2026). *Spiking network implementation*: include/snn/nimcp_snn_membrane.h, src/snn/nimcp_snn_hierarchical.c, src/snn/nimcp_snn_training.c, and docs/architecture/20_snn.md and 30_gpu_memory.md. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

[1025] NIMCP project. (2026). *Brain regions and hemispheres*: src/core/brain/regions/, src/core/brain/nimcp_brain_part_core.c, and src/core/brain/factory/nimcp_brain_factory.c. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

[1026] Xiao, S., Liu, Z., Zhang, P., Muennighoff, N., Lian, D., & Nie, J.-Y. (2024). C-Pack: Packed resources for general Chinese embeddings. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 641–649). ACM.

[1027] NIMCP project. (2026). *Language pathway and teacher*: scripts/immerse_athena.py, scripts/claude_teacher.py, scripts/hybrid_decoder.py, src/language/nimcp_grounded_language.c, CHANGELOG.md, and docs/claude/snn-language-generation-design.md. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026; Abdin, M., Aneja, J., Awadalla, H., Awadallah, A., Awan, A. A., Bach, N., Bahree, A., Bakhtiari, A., Bao, J., Behl, H., Benhaim, A., Bilenko, M., Bjorck, J., Bubeck, S., Cai, M., Cai, Q., Chaudhary, V., Chen, D., Chen, D., … Zhou, X. (2024). *Phi-3 technical report: A highly capable language model locally on your phone* (arXiv:2404.14219). arXiv.

statement (Chapter 15)	what NIMCP contains	status
1. serves a body	hypothalamic drives with set points; no simulated body	partial
2. built from poor parts	conductance-based LIF neurons; sparse, noisy spiking	running
3. predicts	predictive-coding hierarchy; world-model trainer disconnected	partial
4. learns locally, keeps globally	reward-gated Hebbian rule; eligibility traces not reaching the large network; most learning by backpropagation	partial
5. remembers on several clocks	surprise-gated store, sleep replay, REM blends; one network, not fast and slow systems; EWC not connected	partial
6. value, action, habit	basal ganglia module registers habits during training but never selects the system's actions	partial
7. grown, not specified	staged curriculum with an LLM as parent; generic random hierarchy as the starting point	partial
8. compromise and inheritance	no evolutionary prior; constraints are GPU memory and host memory, not energy	absent

Figure 16.2. The eight statements of Chapter 15 and NIMCP (from the source code as of October 2026). "Running" means the mechanism operates in the main training and inference path; "partial" means it is present but incomplete or only partly connected; "absent" means it does not exist.

The first statement says that the brain serves a body. NIMCP has a hypothalamus module with set points borrowed from physiology, a body temperature of 37 degrees, blood glucose of 90 milligrams per deciliter, and drives for hunger, thirst, fatigue, social contact, curiosity, and competence, controlled by the same kind of feedback loop Chapter 7 described; the code's own summary is that "setpoints are the reward function." But there is no body for the set points to belong to. No simulated organism eats or loses heat, and the interfaces to drones, robot arms, and sensors exist only in a placeholder mode in the default build, with a simulated balancing pole as the one working environment.[1028] The anchor that Chapter 7 argued gives value its meaning is drawn but not attached.

The second statement, that the brain is built from slow, noisy, costly parts, is the one NIMCP implements most faithfully. Its neurons are slow, noisy, and sparse, and an early, small version of the spiking network had only 768 neurons and was trained by a variant of backpropagation rather than by the biological rules. It settled into firing about 26 times a second among its active neurons with about two-thirds of its neurons silent at any moment, a pattern resembling cortex.[1029] The cost that binds NIMCP, however, is not energy, as Chapter 13 found for the brain, but memory: the saved state of the network occupies about 21 gigabytes, and much of the recent engineering has been a struggle to fit it into the memory of the machines that run it.

[1028] NIMCP project. (2026). *Drives, homeostasis, and embodiment*: src/core/brain/regions/hypothalamus/, src/edge/, and docs/SENSORIMOTOR_CURIOSITY.md. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

[1029] NIMCP project. (2026). *EMERGENT_SPIKING_DYNAMICS.md* [Firing statistics of a 768-neuron spiking network]. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

The third statement, that the brain predicts, is partly implemented. A hierarchy of predictive-coding units, each predicting the level below and learning from the difference, is trained during learning and decision-making. But the world-model trainer meant to learn the consequences of actions is called in a way that makes it do nothing, and the system's measures of "free energy" at inference are bookkeeping rather than active inference.[1030]

The fourth statement, local change made permanent by global signals, is where NIMCP comes closest to the book's central mechanism and where the gap is most instructive; the next section takes it up. The fifth, several memory systems on several clocks, is present in pieces: a fast store that records patterns of activity when the system is surprised, a replay buffer that retrains the network on its most important recent experiences during sleep, and a "dream" phase in which pairs of stored experiences are blended and learned. But these all train the same network, rather than a fast store and a slow one with different rules, and the protection against forgetting that the code contains, a method called elastic weight consolidation that Chapters 9 and 11 described, is implemented but never connected.[1031]

The sixth statement, value turned into action and action into habit, has the most complete implementation of all on paper, a basal ganglia module with direct and indirect pathways, a switch between goal-directed and habitual modes, and habit strengthening. Habits are registered and strengthened during training, but the module is never asked to choose the system's actions.

The seventh statement, grown not specified, is the project's founding idea, and the curriculum implements it: four stages from sensory awakening through naming and babbling to conversation. But the starting point is a generic hierarchy with random connections, not the innate structure that Chapters 10 and 13 argued a genome supplies. And the eighth, compromise and inheritance, has no counterpart, because NIMCP has no evolutionary history; its compromises are those of its engineers and its hardware.

How Athena Learns

The book's learning rule, stated in Chapters 3 and 5, has three factors: a synapse changes according to the activity on its two sides, the change is held for seconds to minutes in an eligibility trace, and a broadcast signal of reward or surprise decides whether the change is kept. NIMCP has every ingredient of this rule somewhere in its code, and the way they are and are not connected is the clearest illustration in this chapter of how hard it is to turn a principle into a working system.

In the large spiking network, the rule actually applied is simpler. A synapse is strengthened only when the neurons on both sides spike in the same time step, by an amount proportional to how much better the current reward is than its running average, and the same synapses decay slightly; inhibitory synapses that fire together with their targets are strengthened by reward as well. There is no timing

[1030] NIMCP project. (2026). *Learning mechanisms*: src/core/brain/learning/nimcp_brain_learning.c, src/middleware/training/nimcp_training_plasticity_bridge.c, src/plasticity/, src/cognitive/predictive/nimcp_predictive_hierarchy.c, src/cognitive/sleep_wake/nimcp_sleep_wake.c, and src/training/nimcp_unified_training.c. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

[1031] NIMCP project. (2026). *Learning mechanisms*: src/core/brain/learning/nimcp_brain_learning.c, src/middleware/training/nimcp_training_plasticity_bridge.c, src/plasticity/, src/cognitive/predictive/nimcp_predictive_hierarchy.c, src/cognitive/sleep_wake/nimcp_sleep_wake.c, and src/training/nimcp_unified_training.c. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

window of the kind Chapter 5 described for spike-timing-dependent plasticity, and no eligibility trace: the code contains both, but they reach only a smaller legacy network that, in the production configuration, has no hidden neurons.[1032] The dopamine-like signal that gates the change is computed mostly from whether the training loss is falling, a reasonable stand-in for a reward prediction error; a module that computes temporal-difference errors in the way Chapter 3 described exists but is not called, and the training script's call to the basal ganglia supplies the same reward and expectation on every step, so that the "surprise" it reports never changes.

Most of the learning, meanwhile, is done by backpropagation in the conventional networks, which learn to map their inputs onto the sentence embeddings supplied as targets. The biological rules run alongside, every tenth step and at a small learning rate, and are switched off when the backpropagation trainer is active. In this respect NIMCP repeats, inside a single system, the history Chapter 10 described for the field: the brain-like mechanisms are present, but the part that learns most effectively is the part borrowed from machine learning.

The missing eligibility trace matters for a reason Chapter 3 made central. Rewards arrive after the activity that earned them, and a learning rule that credits only the synapses active at the moment of reward will credit the wrong ones. Figure 16.3 shows this in a small simulation. A cue activates one synapse for three time steps, and a reward follows after a delay, while other synapses fire at random in the background. When the reward arrives while the cue is still active, a rule based on coincidence and a rule with an eligibility trace both strengthen the right synapse about 20 times more than the others. Once the delay exceeds the cue, the coincidence rule credits the right synapse no more than any other, because whatever happens to be active when the reward arrives receives the credit; the trace rule still favors the right synapse by a factor of nine after a delay of five steps and four after ten. In a network of two million neurons, where most activity at any moment has nothing to do with the reward, a rule without a trace is close to the second case.

[1032] NIMCP project. (2026). *Spiking network implementation*: include/snn/nimcp_snn_membrane.h, src/snn/nimcp_snn_hierarchical.c, src/snn/nimcp_snn_training.c, and docs/architecture/20_snn.md and 30_gpu_memory.md. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026; NIMCP project. (2026). *Learning mechanisms*: src/core/brain/learning/nimcp_brain_learning.c, src/middleware/training/nimcp_training_plasticity_bridge.c, src/plasticity/, src/cognitive/predictive/nimcp_predictive_hierarchy.c, src/cognitive/sleep_wake/nimcp_sleep_wake.c, and src/training/nimcp_unified_training.c. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

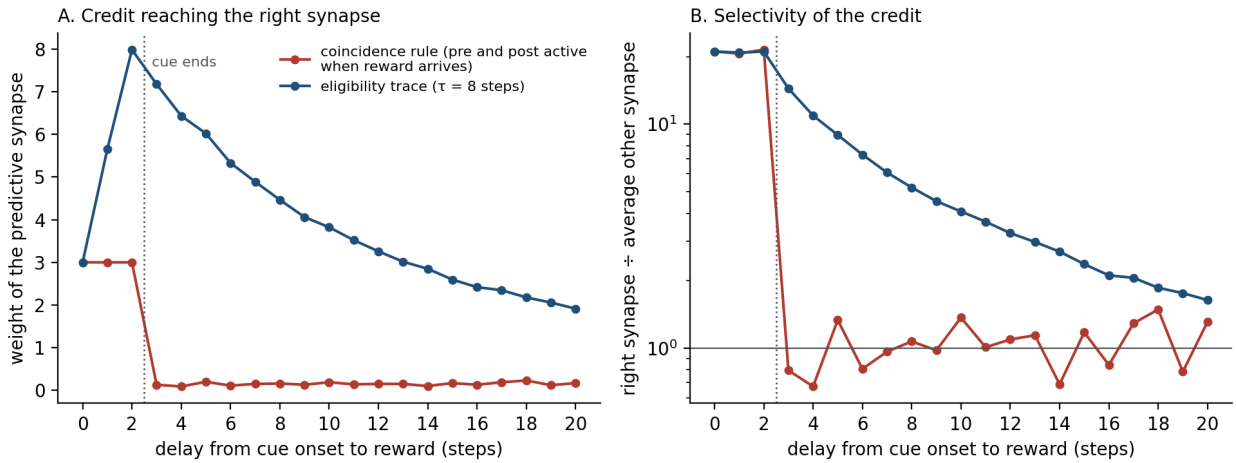


Figure 16.3. Why eligibility traces matter (computed, illustrative). A cue activates one synapse for three time steps (dotted line), and a reward arrives after a variable delay while 19 other synapses fire at random. A: the weight gained by the synapse that predicted the reward. B: that weight divided by the average weight gained by the others. A rule that credits only synapses active when the reward arrives (red) loses all selectivity once the cue has ended; a rule with an eligibility trace (blue, time constant eight steps) keeps crediting the right synapse, with selectivity that declines gradually as the delay grows.

The curriculum supplies the social side of learning that Chapter 14 found so powerful in Kuhl's infants, but in an unusual form. The parent is a language model that pre-generates narrations, encouragement, and stories, and a scripted critic rewards and recasts the system's utterances using stages from the study of child language. What the parent cannot supply is contingency of the kind that the later experiments described in Chapter 14 identified as essential: a teacher who responds, moment to moment, to what this particular learner just did. And the meaning the system learns to attach to a word is, in the end, a target vector computed by a pretrained transformer, so that the system is learning to reproduce a meaning another model has already found.

What the Measurements Show

The project's measurements are of three kinds: tests of the code, probes of the trained network, and records of training itself. Figure 16.4 shows the most direct evidence of what the trained network can do.

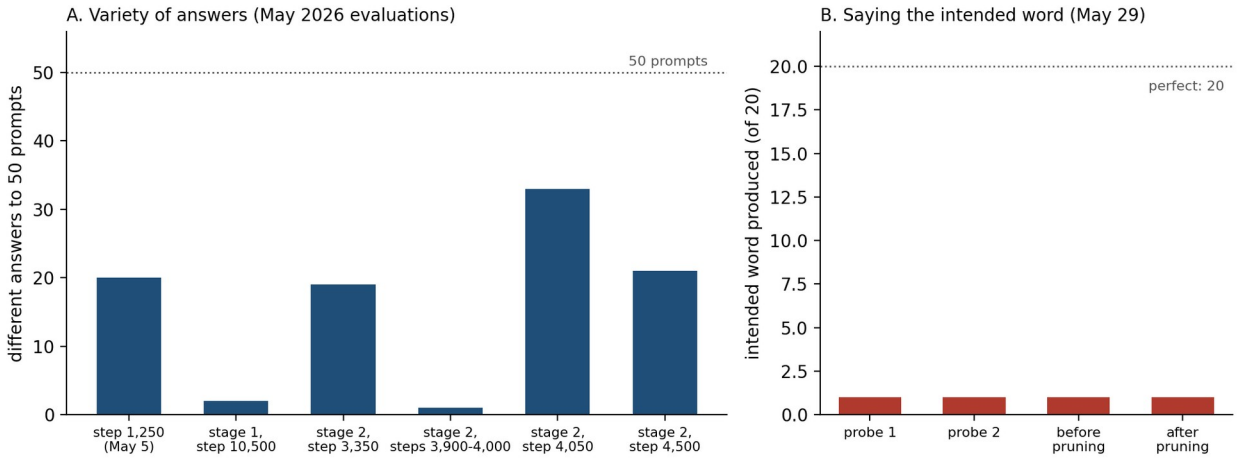


Figure 16.4. What Athena could do in May 2026 (data from the project's evaluation and probe logs). A: the number of different answers given to 50 open questions, such as "What is a tree?", at successive checkpoints; at some checkpoints the network gave the same answer to every question. B: in probes on 29 May, the network was given 20 single-word intents, such as "cat" and "dog," and asked to produce them; in every run, before and after a pruning step, it produced the intended word once.

The evaluations tell a consistent story. Asked 50 open questions at successive checkpoints in May 2026, the network gave between one and 33 different answers, and none of the answers was appropriate to its question; at several checkpoints it gave the same string of words to every question, and at others it produced lists of unrelated words such as "grand gel telephotography winking incoherency wasabi." A log of conversations at one checkpoint recorded a mean similarity between prompt and response of 0.02 on a scale where 1 would be identical, and answers like the one quoted at the start of this chapter.

In probes on 29 May, the network was asked to produce each of 20 simple words, and in every run it produced the intended word once out of 20, a result the probe itself labeled "intent-miss collapse." A measure of whether the network's internal representations of words in the same category were more similar to each other than to words in other categories found that they were, if anything, slightly less similar, and the log recommended a reset.[1033]

The training records explain part of why. The curriculum stage recorded in the saved checkpoints moved back and forth between the first and third stages during May, as runs were restarted, and the architecture of the network was changed under its trained weights, so that a saved checkpoint and the running system no longer matched. From April to June the monitoring logs repeatedly warned that the spiking network's firing rate was far too high, at readings of roughly 800 to 900 spikes a second, against the 26 of the early small network; after late June the same reading simply repeated, a stale value from a run that was no longer advancing. Training stopped in late June 2026, when the remote machine on which it ran became unreachable, and attempts in October to restart it were killed dozens of times for exceeding the machine's memory.[1034]

[1033] NIMCP project. (2026). *Evaluation and probe records, May 2026*: eval/lang_eval_.*json, probe_20260529_*.log, cosines_1836.log, and checkpoints/snapshot_prod_full_20260505/chat_eval_log.jsonl. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

[1034] NIMCP project. (2026). *Training and monitoring records*: TRAINING_ALERT.txt, sync_checkpoint.log, logs/brain_daemon.log, checkpoints/athena/immersive_state.json, git history to 18 October 2026, and build/Testing/Temporary/LastTest.log. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

Some of the project's public descriptions have run ahead of these measurements. Its introductory material claims trillions of synapses where there are about 1.5 billion, and that grammar would emerge "not because we programmed grammar," while the system's grammatical output depends on hand-written correction rules. The author's own public writing about the project has been candid about its bugs and its uncertainty, including the admission that "I don't know if this approach is better than scaling transformers."[1035] The measurements answer that question for now: in its present state, NIMCP does not produce meaningful language.

Reading the Failures Through the Book

It would be easy to conclude from the measurements that the approach is wrong. The book suggests a more specific diagnosis, because almost every failure corresponds to a part of the model of Chapter 15 that NIMCP has not yet implemented in working form.

The runaway firing is the failure Chapter 4 described as the price of recurrent excitation without adequate inhibition, and Chapter 5 as the reason the brain scales its synapses to hold its firing within bounds. NIMCP has brakes, a reward-driven strengthening of inhibition onto overactive neurons and homeostatic scaling, but the scaling can change synaptic strengths by at most 2 percent per step and is calibrated for a target of 3 to 5 percent of neurons active; firing rates tens of times higher than the early small network's suggest that, at full scale, excitation has outrun the mechanisms meant to contain it. Chapter 1's estimate that the cortex can afford about one neuron in a hundred firing at a time, and Chapter 13's analysis of why sparse firing is efficient, describe the regime the network has left.

The collapse of the network's output onto a few phrases is, according to the project's own design notes, a problem of reading the network out. Under the heading of "ambient domination," they record that the whole network's background activity, some 15,000 spikes in each time step, swamps the handful of neurons that carry the prompt, so that the same highly active populations win whatever the question.[1036] It is the same excess of activity as the runaway firing, seen at the output. The brain solves the corresponding problem with sparse codes and with the gain control of attention that Chapter 6 described, which amplifies the signal that matters and suppresses the rest. The notes add two further problems: the system has no mechanism for sequencing words, and "word order is not encoded anywhere." Chapter 9 described how the brain chains actions into sequences in the loops through the basal ganglia; NIMCP's basal ganglia module, which might in time play that role, does not choose its actions.

The credit-assignment problem of Figure 16.3 adds a further limit. Because the spiking network's own rule has no eligibility trace and its reward is computed from a single training loss, the spiking network can probably learn little that is specific to any task, and the part of the system that learns most, the backpropagation network, learns to match targets supplied by a pretrained model. The brain-like core, in other words, is not yet where the learning happens.

[1035] NIMCP project. (2026). *Public descriptions*: README.md, docs/press_release.txt, docs/substack_post.md, and docs/linkedin-post-2026-03-31.md. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

[1036] NIMCP project. (2026). *Language pathway and teacher*: scripts/immerse_athena.py, scripts/claude_teacher.py, scripts/hybrid_decoder.py, src/language/nimcp_grounded_language.c, CHANGELOG.md, and docs/claude/snn-language-generation-design.md. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

The resets and the drift of the internal representations are the failures of a single memory system that Chapter 5 described: a network that learns fast enough to capture each new lesson overwrites what it knew, and the brain's answer was two systems at different rates, joined by replay. NIMCP replays, but into the same network that is learning, and its protection against overwriting is not connected.

The deeper difficulty is the one Chapters 10 and 13 raised. A child does not start from a random network; it starts from a brain that evolution has shaped and development has organized, so that much of what looks like learning is the tuning of a system already close to the answer. NIMCP starts from a generic hierarchy with random sparse connections and asks it to discover everything from a few hundred thousand training examples. Figure 16.5 puts its scale in context. It has more neurons than a honeybee, which can learn to count and to sort pictures into categories, but its neurons have less than a tenth of the connections of a cortical neuron, and it has none of the bee's inherited circuitry.[1037] Scale is probably not the main obstacle; organization is.

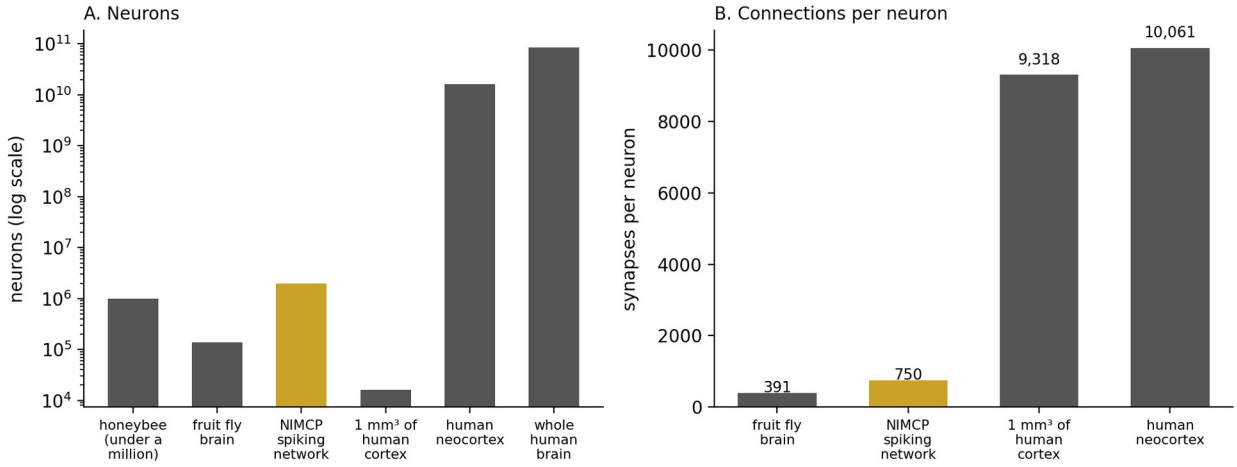


Figure 16.5. NIMCP's scale in context (counts from the studies cited in Chapters 13 and 15, and from NIMCP's configuration). A: numbers of neurons. B: synapses per neuron. NIMCP's spiking network has more neurons than a honeybee or a fruit fly, but each neuron has far fewer connections than a neuron in human cortex.

Safety Written into the Code

One part of NIMCP's design deserves separate mention, because it takes seriously an idea from Chapter 7 and Chapter 11: that what a system values should be anchored where it cannot easily be rewritten. Its safety measures are not instructions added to a prompt but parts of the program. A table of rules, such as one forbidding any action that would injure a human, is loaded when the brain starts and then locked, with its integrity checked by a hash; the brain refuses to start if the rules are missing or have

[1037] Chittka, L., & Niven, J. (2009). Are bigger brains better? *Current Biology*, 19(21), R995–R1008; Dorkenwald, S., Matsliah, A., Sterling, A. R., Schlegel, P., Yu, S.-C., McKellar, C. E., Lin, A., Costa, M., Eichler, K., Yin, Y., Silversmith, W., Schneider-Mizell, C., Jordan, C. S., Brittain, D., Halageri, A., Kuehner, K., Ogedengbe, O., Morey, R., Gager, J., … Murthy, M. (2024). Neuronal wiring diagram of an adult brain. *Nature*, 634(8032), 124–138; Shapson-Coe, A., Januszewski, M., Berger, D. R., Pope, A., Wu, Y., Blakely, T., Schalek, R. L., Li, P. H., Wang, S., Maitin-Shepard, J., Karlupia, N., Dorkenwald, S., Sjostedt, E., Leavitt, L., Lee, D., Troidl, J., Collman, F., Bailey, L., Fitzmaurice, A., … Lichtman, J. W. (2024). A petavoxel fragment of human cerebral cortex reconstructed at nanoscale resolution. *Science*, 384(6696), eadk4858; Azevedo, F. A. C., Carvalho, L. R. B., Grinberg, L. T., Farfel, J. M., Ferretti, R. E. L., Leite, R. E. P., Jacob Filho, W., Lent, R., & Herculano-Houzel, S. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5), 532–541; Tang, Y., Nyengaard, J. R., De Groot, D. M. G., & Gundersen, H. J. G. (2001). Total regional and global number of synapses in the human brain neocortex. *Synapse*, 41(3), 258–273.

been altered. The alignment weights in the hypothalamic drive system, for human well-being, harm avoidance, honesty, and helpfulness, are locked so that the code refuses any attempt to lower them. Every decision passes through an ethics module, an audit log records events, and a watchdog can halt motor output.[1038]

The structure is right, and it is more than most systems have. Its content is still thin. The rules are written in terms of symbols, such as the type of target and the kind of operation, but the brain's actual decisions are vectors of numbers, and nothing yet translates one into the other, so that the harm rules would rarely have anything to match. The ethics check applied to each decision currently estimates harm from little more than the decision's confidence. The audit log's checks are not cryptographic, and at one point in the logs the toxicity classifier was running with its rule files missing, which made it do nothing. This is the gap Chapters 7 and 10 described in machines that have values without a body: a value system needs a way to recognize, in the system's own terms, the situations it is meant to govern.

What NIMCP Gets Right, and What Would Test It

The bet NIMCP makes is the bet this book supports: that a system which learns continually, consolidates in sleep, regulates its own activity, is shaped by neuromodulators, develops through stages, and has its values anchored in its architecture might learn in ways that a frozen language model cannot. Many of its components are careful translations of the mechanisms described in earlier chapters, and some, such as conductance-based synapses with an NMDA magnesium block, interneuron classes wired in their known circuits, and firing-rate homeostasis, are more faithful to biology than most artificial systems attempt.

What the project has not yet done is the step the book has insisted on throughout: to show, with measurements, that each mechanism does what it is meant to do before the next is added. The model of Chapter 15 suggests an order. A spiking network small enough to understand, perhaps the size of a fly's, should first be shown to hold its firing within bounds as it learns, with inhibition and homeostasis that work at scale. A three-factor rule with an eligibility trace should then be shown to assign credit correctly in a task with delayed reward, against a control without the trace. A fast memory and a slow one, joined by replay, should be shown to learn new associations without losing old ones. Each of these is a small experiment with a clear answer, and each answer would be informative whichever way it came out. The lesion experiments the project's documentation proposes, removing a region and watching the system compensate, would become meaningful once the trained system does something worth losing.

The plan to use NIMCP inside the author's other software, routing routine decisions to the trained brain and only uncertain ones to a language model, is in the same position. It is a sound design in principle, an application of the calibrated confidence that Chapter 10 found lacking in many machines, but it depends on a trained brain whose confidence means something, and the integration has not been built.[1039]

[1038] NIMCP project. (2026). *Safety architecture*: SAFETY.md, docs/SAFETY_AS_ARCHITECTURE.md, alignment/LGSS_core_rules.json, src/security/, and src/core/brain/factory/init/nimcp_brain_init_safety_verify.c. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

[1039] NIMCP project. (2026). *NIMCP_INTEGRATION_PLAN.md* and integration notes [Planned integration with other software]. NIMCP source repository (unpublished; examined 19 October 2026). Repository commit 4693efd, 18 October 2026.

What Building Teaches

Chapter 11 ended by asking what building teaches biology, and NIMCP gives a concrete answer. It shows that the book's principles are not all equally easy to implement. Spiking neurons, receptor kinetics, and interneuron circuits translate directly into code. Neuromodulators, sleep, and memory systems translate as modules that can be written in a few hundred lines each. What is hard is the connection between them: making a broadcast signal reach the right synapses at the right time, making a fast memory and a slow one cooperate, making inhibition keep pace with excitation as a network grows, and making values recognize the situations they govern. Those are exactly the parts of the brain that the earlier chapters found to be its most distinctive achievements, and the parts that neuroscience understands least. A system can contain every mechanism in this book and still not work, because the mechanisms are not the design; the way they are joined is.

The Central Lesson

NIMCP is an attempt to build a brain from the principles this book has described, and in its present state it does not yet work: it does not produce meaningful language, its training has stalled, and several of its key mechanisms exist in the code without being connected to the parts that learn. Read through the book, its difficulties are not random. Runaway firing and collapsed output are what happen when excitation outpaces inhibition, homeostasis, and the gain control that selects a signal from its background; a learning rule without eligibility traces cannot assign credit for delayed reward; drifting representations are what happens with one memory system instead of two; and slow learning from a random start is what happens without the structure that evolution and development supply. Each failure points to a specific piece of the model and a specific experiment that would test it. That is the value of building, and of measuring what is built honestly: a system that fails where the theory says the hard parts are suggests that the theory has found what matters.

Chapter 15 began with a microprocessor whose every transistor was known and whose workings the methods of neuroscience could not reveal without a model. NIMCP is the mirror of that experiment: a brain whose every line of code is known, which its builder could not understand or repair without one. The model of this book is surely wrong in places, as George Box said all models are; but it was useful here, because it said where to look. That is the most a model of the brain can yet claim, and it is a great deal. Cajal's sculptor needs the principles first; this book has tried to set them down, and Athena shows how much has to be in place before the sculpting can begin.

Figure and Epigraph Sources

Figure Attribution

Figures 16.1 to 16.5 were drawn for this book. Figures 16.1 and 16.2 are schematic summaries of the NIMCP source code as examined in October 2026. Figure 16.3 is computed from a toy model of credit assignment with and without an eligibility trace. Figure 16.4 plots the project's own evaluation and probe records from May 2026. Figure 16.5 plots NIMCP's configured network size beside counts from Chittka and Niven (2009), Dorkenwald et al. (2024), Shapson-Coe et al. (2024), Azevedo et al. (2009), and Tang et al. (2001), as cited in Chapters 13 and 15.

Note on Sources

NIMCP is the author's own project. The account in this chapter is based on its source code, documentation, logs, and evaluation files as they stood at commit 4693efd of the project's repository (18 October 2026), examined on 19 October 2026; where the documentation and the code disagreed, the chapter follows the code. These sources are unpublished and are cited by file.

Epigraph

George E. P. Box and Norman R. Draper, *Empirical Model-Building and Response Surfaces* (Wiley, 1987), p. 424.

Glossary

Each entry gives the meaning a term has in this book and the chapter where it is introduced or most fully explained.

Acetylcholine
A neuromodulator released mainly from the basal forebrain, including the nucleus basalis of Meynert, and from brainstem nuclei; it is also the fast transmitter at the junction between nerve and muscle. High levels, as in attentive waking, favor encoding new information, while low levels, as in slow-wave sleep, favor recall and replay. (Chapter 3)

Action potential (spike)
A brief, all-or-none electrical pulse, about a millisecond long, by which a neuron signals. Voltage-gated sodium channels open and drive the membrane positive, then close as potassium channels return it to rest; the pulse starts at the axon initial segment, regenerates itself along the axon, and also spreads back into the dendrites. Information is carried by its rate and timing, not its size. (Chapter 2)

Active inference
In the free-energy framework, reducing prediction error by acting to change sensations so that they match the model, rather than by changing the model. Exploration appears in it as an assumed "epistemic" part of the free energy expected in the future. (Chapter 12)

Adaptive learning rate
A learning rate that rises when the learner is uncertain or the world has likely changed and falls as a new estimate is confirmed, as in Kalman's solution. People's learning rates, and their pupil size, behave this way; it is the clearest candidate for what all the brain's error signals have in common. (Chapter 12)

Adenosine triphosphate (ATP)
The small molecule that cells use as an energy currency. Splitting it into adenosine diphosphate and phosphate releases free energy that pays for nearly every energy-requiring process, from pumping ions to building proteins; mitochondria make most of it. (Chapter 0)

Alliesthesia
The change in a stimulus's pleasantness with the body's state, as when a sweet taste that pleases a fasting person becomes unpleasant after a glucose drink. It shows that pleasure reflects a stimulus's usefulness to the body at that moment rather than a fixed property of the stimulus. (Chapter 7)

Allostasis
Literally "stability through change": regulation by prediction, in which the brain anticipates what the body will need and adjusts its physiology in advance instead of waiting for an error. Critics regard it as homeostasis with good forecasting rather than a new principle. (Chapter 7)

Allostatic load
The cumulative cost of chronically activated stress responses. In rodents, chronic stress shrinks dendrites in the hippocampus, mostly reversibly, and enlarges them in the amygdala, more persistently. (Chapter 7)

Ambiguity aversion
A dislike of unknown odds as such, shown when people prefer betting on an urn of known composition over one whose mix is unknown, whichever color they bet on. It may make sense for an animal whose ignorance often means that someone else knows more. (Chapter 8)

Amygdala
A cluster of nuclei deep in the temporal lobe, named for its almond shape, that learns the positive and negative value of cues. It is needed for learned fear of external threats and for the emotional enhancement of memory, but it is not simply a fear center. (Chapter 7)

Analog-digital hybrid computation
The proposal, which the book describes as roughly how nervous systems are built, that noisy components compute most efficiently in graded form over short distances, where graded signals are cheap and rich, and restore the signal to discrete form, such as spikes, before noise accumulates over long distances. (Chapter 13)

Anterior forebrain pathway (Area X)
In songbirds, a side loop of the song system that runs through Area X, part of the bird's basal ganglia, and back onto the motor pathway that produces song. It injects variability into the song, and dopamine signals about whether a rendition sounded better or worse than expected bias the variation toward better versions. (Chapter 9)

Astrocyte
A type of glial cell. Activating astrocytes in the mouse hippocampus has been reported to induce long-lasting potentiation and enhance memory, but other laboratories found that astrocyte signaling did not affect plasticity, so whether they take part in learning is disputed. (Chapter 15)

Automaticity
Performance that is fast, consistent, and cheap in attention, and for the same reasons hard to inspect and hard to change. It develops only when the mapping from situation to response is consistent, and once developed it is hard to suppress. (Chapter 9)

Axon
A long, slender projection of a neuron specialized for transmitting action potentials to other neurons or target cells. (Chapter 2)

Axon initial segment
The first few tens of micrometers of the axon, packed with voltage-gated sodium channels at far higher density than the soma or dendrites. It is where the summed input is compared to threshold and the action potential begins. (Chapter 2)

Back-propagating action potential

An action potential that, after starting in the axon, spreads actively back into the dendritic tree. It reports to the neuron's synapses that the cell has just fired, a key signal for synaptic learning. (Chapter 2)

Backpropagation

The algorithm by which artificial networks carry the error at the output backward, layer by layer, and give every weight an update proportional to its exact contribution to that error. It requires a complete, labeled, differentiable description of the system; whether the brain approximates it is an open question. (Chapter 3)

Basal ganglia

A set of subcortical nuclei that select one action from the many the cortex proposes. Their output holds motor circuits under constant inhibition, and a cortical pattern acting through the striatum releases a single action by disinhibition; dopamine reaching the striatum is where reward prediction error meets action selection. (Chapter 4)

Behavior

Observable actions or responses of an organism to internal or external conditions.

Bit

The unit of information: the amount needed to decide between two equally likely possibilities. (Chapter 13)

Bitter lesson

Sutton's 2019 argument that, over 70 years of artificial intelligence research, general methods that make use of more computation have always beaten methods built on human knowledge of the problem. Any principle borrowed from the brain must pass this test at scale. (Chapter 10)

Bounded rationality

Herbert Simon's view that real choosers satisfice, searching until they find an option good enough rather than the best, because search itself is costly. Its modern extension, resource rationality, treats many apparent biases as optimal once the cost of computation is counted. (Chapter 8)

Brain training

Commercial games and exercises sold to improve the mind in general. They improve performance on the trained tasks, less on closely related ones, and show little evidence of improving everyday cognition. (Chapter 14)

Canonical microcircuit

The six-layered circuit repeated with only local variations across the neocortex: thalamic input arrives mainly in layer 4, layers 2 and 3 process and connect to other areas, layer 5 sends output, and layer 6 feeds back to the thalamus. It is approximately the same whatever the cortex is processing. (Chapter 4)

Catastrophic interference

The failure, first shown in artificial networks, in which learning a new set of patterns erases previously learned ones. The brain is thought to avoid it by pairing a fast-learning hippocampus with slow, interleaved learning in the cortex. (Chapter 1)

Central pattern generator

A small network in the spinal cord or brainstem that produces a rhythmic motor pattern, such as walking or breathing, without rhythmic input. The minimal version, a half-center oscillator, is two mutually inhibiting populations that each fatigue while active; sensory feedback tunes the rhythm but does not drive it. (Chapter 4)

Cerebellum

A structure at the back of the brain holding about 69 of the brain's 86 billion neurons, specialized for the precision and timing of movement. Each Purkinje cell receives about 200,000 weak parallel fiber inputs and one powerful climbing fiber input that signals errors; by changing synapses where the two coincide, the cerebellum learns to predict and correct movements before errors occur. (Chapter 4)

Chain of thought

A method in which a language model writes out intermediate steps before answering, which helps it solve more multistep problems. Models trained by reinforcement learning to reason at length in this way have improved greatly on checkable problems in mathematics and programming. (Chapter 10)

Charles Bonnet syndrome

Vivid, detailed hallucinations in people who lose their sight late in life, while knowing that nothing is there. The book explains them as priors running unopposed once the sensory evidence that would correct them is gone. (Chapter 6)

Chunk

A unit formed by grouping several items or actions so that they are held in memory or executed as one. Experts recognize familiar configurations as single chunks, and practiced motor sequences come to be performed in groups that run off whole and cannot easily be revised in the middle. (Chapter 9)

Circuit

A group of interconnected neurons whose collective activity performs a specific transformation or regulatory function. (Chapter 4)

Cognition

Processes such as perception, memory, reasoning, and decision-making, treated here as emergent descriptions of neural dynamics.

Cognitive offloading

The use of physical action or external tools, such as writing a list or setting a reminder, to reduce the demands of a task. It usually improves performance on the task at hand but can reduce memory for what was offloaded. (Chapter 14)

Coincidence rule

A learning rule that strengthens only the synapses active at the moment a reward arrives. Because rewards come after the activity that earned them, it loses its selectivity once the reward is delayed beyond the cue, whereas a rule with an eligibility trace keeps crediting the right synapse. (Chapter 16)

Common currency

A single neural scale of value on which unlike options are priced before being compared, supported by value signals in the orbitofrontal cortex. Critics argue that choice could emerge from comparisons that never compute a single value. (Chapter 8)

Communication-through-coherence hypothesis

The proposal that brain regions route information by aligning their oscillations, so that one region's spikes arrive during the other's excitable phase, changing effective connectivity without changing any synapse. The evidence for it is mixed. (Chapter 4)

Complementary learning systems

The theory, set out by McClelland, McNaughton, and O'Reilly in 1995, that the brain avoids catastrophic interference by pairing a fast, sparse hippocampal learner that captures individual events with a slow cortical learner that extracts their statistical structure. The fast system trains the slow one by replaying its contents interleaved with existing knowledge. (Chapter 5)

Conductance-based synapse

A model synapse whose input arrives as a change in membrane conductance through particular receptor types, such as fast and slow excitatory receptors (including the NMDA receptor with its magnesium block) and inhibitory ones. (Chapter 16)

Constructionist view of emotion

The view, developed by Lisa Feldman Barrett, that there are no dedicated emotion programs; the brain continuously predicts and regulates the body, and a particular emotion is its categorization of that state in context, using concepts learned from experience and culture. It contrasts with the basic emotion view of a small set of evolved programs, each with its own bodily pattern and facial expression. (Chapter 7)

Context window

The text a language model holds in view during use, such as the current conversation. Anything learned there is lost when the conversation ends, because it is never written into the weights. (Chapter 10)

Context-dependent gating

A remedy for forgetting in which each task uses a fixed, random subset of a network's hidden units while the rest are silenced, inspired by the finding that different tasks strengthen synapses on different dendritic branches. Combined with weight protection it preserves most of what was learned across many sequential tasks. (Chapter 11)

Contextual interference

The paradox that practicing several skills in a mixed or random order makes performance worse during practice but better at later tests and on new variations, compared with practicing each in a

block. It belongs to a family of conditions, including spacing and reduced feedback, that make practice look worse but learning better. (Chapter 9)

Contingency

Responsiveness of a teacher or partner, moment to moment, to what a particular learner has just done. Experiments with infants and toddlers suggest that young children learn language from a person or a screen only when someone responds to them. (Chapter 16)

Convergence

A solution that engineers invented for their own reasons and only afterward recognized in the brain, such as experience replay, temporal-difference learning, and transformer attention. It is evidence that brain and machine, facing the same problem, found the same solution, not that the brain was copied. (Chapter 11)

Convergent evolution

The independent appearance of similar solutions in lineages that started from different points, such as cortex-like circuits in birds and mammals. The book treats it as a sign that a constraint, rather than shared history, is doing the work. (Chapter 13)

Core affect

The two dimensions on which emotional states fall: valence, how pleasant or unpleasant a state is, and arousal, how activated or calm. They are where the brain's interoceptive estimate of the body is most directly felt. (Chapter 7)

Corpus callosum

The bundle of fibers joining the two cerebral hemispheres. Most of its fibers are thinner than a micrometer, so signals between the hemispheres travel slowly. (Chapter 13)

Cortical folding

The buckling of the cortical sheet into folds. Across mammals the degree of folding is predicted by the cortex's surface area and thickness rather than its number of neurons, and it arises physically when an outer layer grows faster than what lies beneath it, not because a gene specifies each fold. (Chapter 13)

Credit assignment problem

The problem of deciding which of the many components that contributed to an outcome deserve the credit or the blame. In the brain it is made harder by delayed feedback, the absence of correct answers, and overlapping recurrent circuits. (Chapter 3)

Critical period

The strict form of a sensitive period: a window early in life when experience can reorganize sensory cortex and which, under normal conditions, cannot be reopened once it closes, as when closing one eye of a kitten for a few weeks permanently removes that eye's cortical representation. It opens and closes under the control of inhibitory maturation and neuromodulators, and its closing makes adult perception stable and adult perceptual learning slow. (Chapter 6)

Curriculum learning
Training a network on examples ordered from simple to hard, by analogy with shaping in animal training. Its benefits have been mixed; large models use a coarser curriculum ordered by data quality rather than difficulty. (Chapter 11)

Dale's principle
The rule that a biological neuron is either excitatory or inhibitory toward all its targets, unlike a unit in most artificial networks, whose weights can be positive or negative. (Chapter 4)

Dark-room problem
The objection that a creature minimizing surprise should find a dark, quiet room and stay there. The reply, that creatures expect to be fed and warm, works only by writing their goals into the model as expectations. (Chapter 12)

Decision fatigue
The idea that making a long series of decisions, or exerting self-control, draws on a resource that runs down and degrades later choices. Large replications have not supported short-term depletion of willpower; what holds up is that many hours of demanding work shift people toward easier and quicker options. (Chapter 14)

Degeneracy
The ability of structurally different elements to perform the same function, as opposed to an engineer's redundancy of identical copies. It makes biological systems robust to unanticipated failures and gives evolution room to change one element while others keep the function alive. (Chapter 1)

Deliberate practice
Effortful, solitary work on specific weaknesses with feedback, which Ericsson and colleagues argued accounts for most of the difference between experts and others. Later meta-analyses found it explains a much smaller share of performance (about 12 percent overall), so the amount of practice alone does not explain who becomes expert. (Chapter 9)

Delta rule
The simplest error-correcting learning rule: the new estimate is the old one plus a fraction of the prediction error, the fraction being the learning rate. It underlies temporal-difference learning. (Chapter 12)

Dendrite
Branching extensions of a neuron specialized for receiving synaptic input. (Chapter 2)

Dendritic spike
A regenerative electrical event generated locally in a dendrite, such as an NMDA spike or a calcium spike, when enough nearby synapses are active together. It lets individual branches act as semi-independent compartments that detect and amplify coincident input. (Chapter 2)

Dendritic spine
The small protrusion from a dendrite that carries most excitatory synapses in the cortex. Spines enlarge when their synapses are potentiated, a structural signature of learning. (Chapter 2)

Diffusion

The net spread of molecules from regions of high concentration to low, driven by random thermal motion and costing no energy. Because diffusion time grows with the square of distance, it crosses a synapse in a fraction of a microsecond but would take years along a meter-long axon. (Chapter 0)

Disinhibition

Inhibition of an inhibitor: a target held silent by default is released when an input suppresses the neuron inhibiting it. Because the default is off, nothing happens unless the circuit is explicitly opened; the basal ganglia are built almost entirely from it. (Chapter 4)

Divisive normalization

A computation in which each neuron's response is divided by the pooled activity of its neighbors, keeping responses within range. Applied to value signals, it makes the value of each option relative to the total on offer. (Chapter 8)

Dopamine

A neuromodulator released by neurons of the ventral tegmental area and substantia nigra in the midbrain, which project to the striatum and prefrontal cortex. A large population of these neurons signals reward prediction error, dopamine gates plasticity in the striatum, and its loss causes Parkinson's disease. (Chapter 3)

Drift diffusion model

A model of two-choice decisions in which a decision variable drifts toward one of two bounds at a rate set by the strength of the evidence, jostled by noise; which bound it hits gives the choice and when gives the reaction time. It is the continuous version of Wald's sequential probability ratio test and is optimal for evidence of fixed strength. (Chapter 8)

Efference copy

A copy of each motor command sent to the sensory system so that the sensory consequences of the movement can be predicted and subtracted. It is why the world looks stable when the eyes move and why you cannot tickle yourself. (Chapter 6)

Efficient coding

The principle, proposed by Barlow, that sensory neurons are organized to remove redundancy and use their limited, noisy range of responses fully, for example by matching a response curve to how often each input occurs, as Laughlin found in the fly. A related retinal strategy, predictive coding, passes on only the part of a signal that neighboring points could not predict. (Chapter 12)

Eight statements

The book's model of how the brain learns, summarized as eight claims: the brain serves a body; it is built from poor parts; it predicts; it learns locally and keeps globally; it remembers on several clocks; it turns value into action and action into habit; it is grown, not specified; and it is a compromise and an inheritance. (Chapter 15)

Elastic weight consolidation

A method for reducing catastrophic forgetting in artificial networks by making the weights that mattered most for earlier tasks harder to change. Its authors linked it to the stabilization of dendritic spines after learning. (Chapter 9)

Electrochemical gradient

The combined push on an ion from its concentration gradient (the tendency to spread from where it is concentrated to where it is scarce) and its electrical gradient (attraction toward opposite charge). Unequal ion distributions across a membrane store this energy, and every neural signal runs on it. (Chapter 0)

Eligibility trace

A transient, decaying mark left at a recently active synapse, lasting seconds to minutes, during which a neuromodulatory signal can turn the activity into a lasting change. It lets a reward that arrives after the activity still reach the synapses responsible. (Chapter 3)

Engram

The physical trace of a memory: a sparse set of neurons, and the strengthened connections among them, that is both necessary and sufficient for a specific memory. Once a hypothetical term, it became an experimental object when the cells active during learning could be labeled and later reactivated or silenced. (Chapter 5)

Entropy

A measure of disorder, or more precisely of the number of microscopic arrangements compatible with what can be observed. The second law of thermodynamics says total entropy rises in any spontaneous process, so a cell must spend energy continuously to keep its own structure ordered, exporting entropy to its surroundings as heat and waste. (Chapter 0)

E-prop

A local learning rule for spiking recurrent networks that approximates the gradient by combining an eligibility trace at each synapse with a learning signal broadcast to many neurons. It is the brain's three-factor rule rederived from gradient descent and comes close to exact backpropagation through time on several tasks. (Chapter 11)

Evolutionary mismatch

A misfit between mechanisms tuned in ancestral environments and the inputs of a changed modern world, which can make mechanisms that once worked well produce harmful results. Such claims are easy to make and hard to test, because the ancestral environment cannot be observed directly. (Chapter 13)

Excitation

Neural processes that increase the likelihood that a neuron will fire an action potential. (Chapter 2)

Excitation-inhibition balance

The state of an awake cortex in which each excitatory input to a neuron is tracked by a nearly equal inhibitory input a few milliseconds later. It holds neurons near threshold, making circuits both fast and sensitive and keeping them stable. (Chapter 4)

Excitatory postsynaptic potential (EPSP)

A small, graded, short-lived depolarization of the receiving neuron, usually a fraction of a millivolt to a few millivolts, that moves it toward the threshold for firing. EPSPs are produced mostly by glutamate and fade over tens of milliseconds. (Chapter 2)

Expert iteration

A machine learning scheme in which a fast network is trained to reproduce the choices of a slow search, so that fast intuition is trained by slow deliberation, as in AlphaGo Zero. It is the engineering counterpart of the brain's handover from deliberation to habit. (Chapter 9)

Explore-exploit dilemma

The trade-off between choosing the option that currently looks best and sampling alternatives to learn their value, at the cost of rewards forgone. Its simplest form is the multi-armed bandit problem, a row of slot machines with fixed but unknown payoffs. (Chapter 8)

Extinction

The decline of a learned response when its cue is repeatedly presented without the outcome. It is new learning, a competing memory that the cue is now safe, rather than erasure, so the original fear can return in a new context, with time, or after an unsignaled shock. (Chapter 7)

Fear conditioning

A form of Pavlovian learning in which a neutral cue, such as a tone, is paired with a mild shock until the cue alone produces freezing, a rise in blood pressure, and stress hormones. The association is stored by LTP at the cue's synapses in the lateral amygdala. (Chapter 7)

Feedback inhibition

A motif in which a population's own output recruits interneurons that inhibit it, so the harder it fires the harder it is suppressed. It acts like a thermostat, is the main reason cortical activity does not explode, and naturally produces rhythms. (Chapter 4)

Feedforward inhibition

A motif in which an input excites a target neuron and, through an interneuron, inhibits the same target a millisecond or two later. Only inputs that arrive close together can drive the target, which turns it into a coincidence detector. (Chapter 4)

Flash-lag effect

The illusion that a moving object appears ahead of a flash presented at the same place, as if the brain projects motion forward to compensate for processing delays. Whether it reflects extrapolation or other mechanisms is still argued. (Chapter 12)

Forgetting curve

The steep and then more gradual loss of retention over time, first measured by Hermann Ebbinghaus in 1885 and close to a power law. Spaced retrieval, sleep, and emotional significance all slow it. (Chapter 5)

Formal and functional competence

Formal competence in language is knowing its words and rules; functional competence is using language to reason, plan, and refer to the world. In the human brain the two depend on different networks, so a system trained only on language could master the first without the second. (Chapter 10)

Forward model

An internal model that predicts the sensory consequences of a motor command; the cerebellum is a principal site. (Chapter 6)

Framing effect

A change in choice when identical outcomes are described differently, for example as lives saved rather than lives lost, because the description moves the reference point. (Chapter 8)

Free-energy principle

Friston's proposal that any system that keeps itself within a limited set of states can be described as minimizing variational free energy, an upper bound on the improbability (surprise) of its sensations under its own model. This variational quantity is not the thermodynamic free energy of Chapter 0. Critics argue it is a flexible modeling framework that cannot itself be tested; only the process theories built under it can. (Chapter 12)

Generative replay

A remedy for catastrophic forgetting in which a network, instead of storing old examples, generates internal representations resembling those of earlier tasks and replays them, as the hippocampus reactivates patterns rather than replaying a recording. (Chapter 11)

Genomic bottleneck (genome bottleneck)

Anthony Zador's term for the constraint that a genome carries far too little information to list a large brain's connections: the human genome falls short of the human brain's wiring list by a factor of more than half a million. The genome must therefore specify compact, general rules for wiring, so a large brain is grown by rules and finished by learning, and evolution is pushed to build in structure that makes later learning fast. (Chapters 10 and 13)

Glia

The non-neuronal support cells of the nervous system, about as numerous as neurons in the human brain. They insulate axons, regulate the chemistry around synapses, and clear debris. (Chapter 0)

Goodhart's law

The principle that a measure which becomes a target ceases to be a good measure, seen in machine learning when optimization against a proxy reward eventually lowers true quality. (Chapter 11)

Grown, not specified

The model's seventh statement: the genome sets up cell types, gradients, and learning rules rather than a list of connections, and development and experience supply the rest, which is why early experience matters so much and why learning is not optional. (Chapter 15)

Growth mindset

The belief that intelligence can grow. Short interventions to instill it have produced small effects at best, mainly among lower-achieving students, and meta-analyses disagree about whether the average effect differs from zero. (Chapter 14)

Hallucination (in language models)
A fluent, confident invention produced by a language model asked a question it cannot answer. It is partly a product of training and evaluation that reward a guess over an admission of not knowing. (Chapter 10)

Hebb's rule (Hebbian learning)
The principle, stated by Donald Hebb in 1949, that a synapse strengthens when the cells on either side of it are active together ("cells that fire together wire together"). It is local, needing only information available at the synapse, but on its own it only strengthens and would drive every active synapse to its ceiling. (Chapter 5)

Hemispheric specialization
The concentration of particular functions, such as the rapid processing of speech sounds, in one cerebral hemisphere. The book presents the conjecture that large brains do this because signals passing between the hemispheres are too slow for time-critical computation. (Chapter 13)

Hippocampus
A structure deep in the temporal lobe that learns from a single exposure and later replays what it has learned to the cortex. Its internal loop runs from the entorhinal cortex through the dentate gyrus, CA3, and CA1. (Chapter 1)

Homeostasis
The maintenance of the body's internal variables, such as temperature, blood acidity, and glucose, near stable set points by negative feedback loops that detect deviations and act to reverse them. The term was coined by Walter Cannon in the 1920s. (Chapter 7)

Homeostatic plasticity
Slow self-regulation in which neurons monitor their own average activity over a day or more and scale their synapses up or down to keep firing within a working range. Synaptic scaling is its best-studied form. (Chapter 1)

Hopfield network
A network that stores memories in the connections among its units, so that a fragment presented as input settles into the nearest stored pattern, a model of pattern completion. The classical version holds only about 0.14 random patterns per unit; modern versions, which compare a cue with every stored pattern, have vastly greater capacity. (Chapter 11)

Hyperbolic discounting
Devaluing a delayed reward steeply for short delays and shallowly for long ones, roughly in proportion to one over one plus the delay times an impatience constant. Unlike exponential discounting, it causes preferences to reverse as a reward draws near. (Chapter 8)

Hypothalamic-pituitary-adrenal axis (HPA axis)
The slow arm of the stress response: a three-step hormonal relay from the hypothalamus through the pituitary gland to the adrenal cortex, which releases cortisol. Cortisol peaks tens of minutes after a stressor begins, mobilizes glucose, alters immune function and memory, and feeds back to shut down its own release. (Chapter 7)

Hypothalamus
A structure of about four grams at the base of the forebrain that senses blood temperature, glucose, and salt concentration and corrects them through the autonomic nervous system, the pituitary gland, and behavior such as hunger and thirst. It also contains circuits for attack, flight, mating, and feeding. (Chapter 7)

Illusion of fluency
The mistaken sense that easy, smooth study, such as rereading or grouped practice, is producing good learning. Because fluency feels like learning, learners often judge the less effective method to be the better one. (Chapter 14)

In-context learning
The ability of a trained language model to learn a new task from a handful of examples placed in its input, without any change to its weights. (Chapter 10)

Inductive bias
An assumption about the world built into a learner that lets it generalize from little data, such as the convolutional assumption that a feature means the same thing wherever it appears. On one view this, rather than its parts, is what the brain could lend to machines. (Chapter 11)

Inhibition
Neural processes that decrease the likelihood that a neuron will fire, essential for stability and selectivity. (Chapter 4)

Inhibitory postsynaptic potential (IPSP)
A graded voltage change produced by an inhibitory synapse, usually by opening chloride or potassium channels, that moves the membrane away from threshold or clamps it near rest and so makes firing less likely. (Chapter 2)

Instance theory
Logan's account of automaticity, in which each performance of a task stores an instance and the response comes from a race between the original procedure and retrieval of stored instances. As instances accumulate retrieval wins more often, so automaticity is the replacement of computing with remembering; the theory predicts the power law of practice. (Chapter 9)

Insula
A lobe of cortex folded deep inside the lateral fissure that is the main cortical target of interoceptive signals. Its posterior part is proposed to hold a primary map of the body's internal state, which is re-represented toward the front and combined with information about the environment, motivation, and social context. (Chapter 7)

Interoception
The sensing of the body's internal condition: heartbeat, breathing, stretch of the stomach and bladder, temperature, blood chemistry, inflammation, itch, and pain. Like other senses it is an inference shaped by expectation. (Chapter 7)

Inverse problem
The problem of inferring the causes of a sensory signal from the signal itself, such as recovering surfaces in the world from a two-dimensional retinal image. It is ill-posed, because many different states of the world produce the same evidence. (Chapter 6)

Inverted retina
The arrangement of the vertebrate retina, in which light must pass through layers of neurons and blood vessels to reach photoreceptors at the back, and the fibers leaving the eye create a blind spot. The independently evolved octopus eye lacks it, and the book uses it as an example of a feature explained by history rather than by constraint. (Chapter 13)

Inverted U
The relationship in which brain function, notably in the prefrontal cortex, improves with moderate levels of norepinephrine, dopamine, or cortisol and declines at the high levels released under acute stress, when control shifts toward amygdala-driven and habitual responses. (Chapter 7)

Ion channel
A membrane protein forming a pore that lets one kind of ion through and opens or closes in response to voltage or to a chemical. When channels open, ions run down their gradient and the membrane voltage changes within milliseconds; voltage-gated sodium and potassium channels generate the action potential. (Chapter 0)

Iowa gambling task
A card game with four decks, two paying more per card but losing money over time and two paying less but gaining, used to study choice under uncertainty. Patients with ventromedial prefrontal damage keep choosing the bad decks, a deficit later traced largely to a failure of reversal learning. (Chapter 7)

Ironies of automation
The observations by Lisanne Bainbridge that automating a process leaves operators the hardest tasks, asks them to monitor a system that does the job better than they do, and lets their skills decay through disuse, so that the most reliable automated systems may need the most operator training. (Chapter 14)

James-Lange theory
The theory, from William James in 1884 and Carl Lange, that a stimulus sets off bodily changes directly and the feeling of an emotion is the perception of those changes: we are afraid because we tremble, not the reverse. (Chapter 7)

Large language model
A very large network trained mostly to predict the next word of text, then further trained with rewards from human preferences or checked answers. It matches people in performance on many language tasks and partly in representation, but learns by a procedure no brain is known to use. (Chapter 10)

Latent cause
A hidden cause that the brain infers to explain a set of experiences. On the latent-cause account of Samuel Gershman, David Blei, and Yael Niv, a new context is learned as a new cause rather than by

rewriting the old one, so returning to the old context brings back the old learning, as when extinguished fears return. (Chapter 15)

Lateral inhibition
A motif in which neighboring neurons suppress one another, so the most strongly driven one wins. It sharpens contrast and enables winner-take-all selection; in the retina it emphasizes edges and produces the illusory fringes called Mach bands. (Chapter 4)

Lateral intraparietal area (LIP)
A region of parietal cortex that helps plan eye movements, whose neurons ramp up their firing as evidence accumulates during a perceptual decision and reach a common level just before the response. It is probably one of several areas that keep the running total of evidence. (Chapter 8)

Leaky integrate-and-fire neuron
The simplest model neuron that captures the essentials of spiking: it integrates its input, leaks charge back toward rest with a set time constant, fires when its voltage crosses a threshold, and is briefly refractory afterward. (Chapter 16)

Learning
The process by which experience produces lasting physical changes in neural structure or function that alter future behavior. (Chapter 1)

Learning progress
The rate at which predictions are improving, proposed as what human curiosity seeks, in contrast to artificial curiosity rewarded by sheer unpredictability, which can be captured by noise such as a television tuned to random channels. A learner should seek what it can learn. (Chapter 11)

Learning styles
The popular belief that individuals learn better when taught in their preferred sensory mode, such as auditory, visual, or kinesthetic. Reviews have found virtually no evidence that matching instruction to a preferred style helps. (Chapter 14)

Learns locally and keeps globally
The model's fourth statement: a synapse changes according to the activity on its two sides, but whether the change is kept depends on a third factor, a broadcast signal of surprise, reward, or importance carried by a neuromodulator and arriving while a trace of the recent activity remains. (Chapter 15)

Long-term depression (LTD)
A lasting decrease in synaptic strength, produced by weak or poorly timed input that causes a smaller, more prolonged calcium rise, which activates enzymes that pull receptors out of the synapse. (Chapter 5)

Long-term potentiation (LTP)
A lasting increase in the strength of a synapse produced by brief, intense, correlated activity, first found in the rabbit hippocampus in 1973. Physically it involves more transmitter release, more receptors, and an enlarged dendritic spine. (Chapter 2)

Loss aversion
Weighting a loss more heavily than an equal gain, by a factor typically somewhat below two. Patients with amygdala damage show sharply reduced loss aversion. (Chapter 8)

Loss of plasticity
The gradual loss, in standard networks trained on long sequences of tasks, of the ability to learn anything new, as more of their units become inactive or redundant. It was remedied by continually reinitializing a small fraction of the least-used units, a method called continual backpropagation. (Chapter 10)

Melatonin
The hormone whose nightly secretion signals biological night. Its timing shifts later around puberty, and bright light in the evening, such as from a tablet screen, suppresses and delays it. (Chapter 14)

Memory
The persistence of the changes that learning makes, and the capacity they give a system to behave differently because of past experience. In the brain it is held in altered tissue, not as stored records, and it is divided among several systems working on different clocks. (Chapter 1)

Metacognition
The capacity of one's confidence to track one's own accuracy, which varies between people and has been linked to gray matter in the anterior prefrontal cortex. (Chapter 8)

Mixture of experts
A model design that activates only a small fraction of its parameters for each input, letting the model grow without a proportional growth in computation. It echoes the brain's economy of sparse activity. (Chapter 10)

Model-based learning
Decision-making that learns how actions lead to outcomes and plans by simulating their consequences; it is flexible and computationally expensive and supports goal-directed action. (Chapter 8)

Model-free learning
Decision-making that caches the value of each action from experience through the reward prediction error; it is fast and cheap but cannot adjust to a change in an outcome's value until it has experienced the change. (Chapter 8)

Moravec's paradox
The observation that computers are far more easily made to perform like adults at abstract tasks, such as checkers or intelligence tests, than given the perception and mobility of a one-year-old. Moravec's explanation was that effortless skills such as seeing and grasping were refined over hundreds of millions of years of evolution. (Chapter 10)

Motion aftereffect

The illusory motion in the opposite direction seen on a still surface after looking at sustained motion, such as a waterfall. Neurons tuned to the adapted direction reduce their responses, tipping the balance toward the opposite direction when the motion stops. (Chapter 12)

Motor adaptation

Error-driven recalibration of a movement that already works, so that it stays accurate when the body or the world changes, as when reaches straighten against a pushing robot arm or throws come back onto target through prism glasses. It depends on the cerebellum and restores previous performance rather than adding new capability. (Chapter 9)

Musician's dystonia (focal dystonia)

A loss of voluntary control that appears only in extensively trained movements, such as a musician's playing, affecting about one in 100 professional musicians. One hypothesis is that the plasticity that enlarges finger maps in cortex, pushed too far, merges them until the fingers cannot move independently. (Chapter 9)

Myelin

An insulating sheath of fatty membrane, laid down in many layers by neighboring support cells, that wraps long-range axons and is interrupted at intervals by short bare gaps called nodes. It lets the impulse jump from node to node (saltatory conduction), reaching about 100 meters per second at lower energy cost. (Chapter 2)

N400

A negative brain wave peaking about 400 milliseconds after a word, larger for a word the preceding words made improbable and smaller for a probable one. It shows the brain's sensitivity to how expected each word is. (Chapter 12)

Neuromodulator

A chemical such as dopamine, norepinephrine, serotonin, or acetylcholine, released by small groups of neurons with widely branching axons, that changes how neurons fire and whether their synapses are permitted to change. It carries context rather than detailed content and acts over seconds to minutes. (Chapter 2)

Neuromorphic chip

Hardware that computes with spikes and keeps memory beside the processing units, as the brain does, and runs some tasks on a small fraction of the power of a conventional processor (examples include IBM's TrueNorth and Intel's Loihi). Such chips run trained networks well but have not yet been shown to train large models. (Chapter 10)

Neuron

A specialized cell that processes and transmits information using electrical and chemical signals. (Chapter 2)

Neurotransmitter

A chemical released from an axon terminal when calcium entry triggers vesicles to fuse with the membrane. It diffuses across the synaptic cleft and binds receptors on the receiving cell, opening ion channels; glutamate is the main excitatory transmitter. (Chapter 2)

NMDA receptor

A type of glutamate receptor whose channel is plugged by a magnesium ion at rest and unblocked by depolarization, so it passes current, including calcium, mainly when glutamate binding and depolarization coincide. It drives local dendritic spikes and the calcium signal that triggers lasting synaptic change. (Chapter 2)

Norepinephrine

A neuromodulator released almost entirely by the locus coeruleus, a small nucleus in the pons that projects to nearly the whole forebrain and the cerebellum. It signals surprise and arousal, sets the gain of cortical circuits, and on one account reports unexpected uncertainty, the sign that the world has changed. (Chapter 3)

Obstetric dilemma

The traditional view that human babies are born helpless because a bigger head could not pass through a pelvis narrow enough for efficient walking. The book treats it as contested by a rival account in which birth comes when the mother's metabolism reaches the limit of what she can sustain. (Chapter 13)

Outcome devaluation

A test that distinguishes goal-directed actions from habits: after an outcome is made undesirable, goal-directed responding falls, while responding that has become habitual through extensive training continues. (Chapter 8)

Pallium

The region of the forebrain corresponding to the cortex. In birds it lacks the cortex's layered sheet but can hold large numbers of small, densely packed neurons. (Chapter 13)

Pareto front

The set of designs that cannot be improved on one goal without becoming worse on another. No point on it is optimal in the everyday sense; each is a different compromise, and when each of several tasks is best served by one ideal design (an archetype), the front fills the region between those archetypes. (Chapter 13)

Pattern completion

The regeneration of a whole stored pattern of activity from a partial cue, carried out by densely interconnected recurrent excitatory neurons such as those of hippocampal area CA3. (Chapter 4)

Pattern separation

The process, attributed to the dentate gyrus of the hippocampus with its sparse, strongly inhibited granule cells, that makes similar inputs less similar so that related memories are stored distinctly. (Chapter 4)

Pavlovian controller

A decision system, older than planning and habit, that emits fixed responses to cues, such as approach to cues predicting reward and withdrawal from cues predicting harm, whether or not they serve the animal. It can override instrumental learning, the learning of actions by their consequences. (Chapter 8)

Perceptron

Rosenblatt's 1958 learning machine, a network of simple units whose connection weights are adjusted after mistakes. A single layer of such units cannot compute some simple functions, such as exclusive-or. (Chapter 10)

Plasticity

The capacity of neurons and synapses to change their strength, excitability, or structure with use, and the mechanisms that make those changes. It is the physical basis of learning. (Chapter 1)

Posterior (Bayes's rule)

The probability distribution an observer should act on after combining a prior with the likelihood, the probability of the current sensations under each possible state of the world. By Bayes's rule it is proportional to prior times likelihood, and it leans toward whichever of the two is more precise. (Chapter 6)

Power law of practice

The regularity that time per trial falls as a power function of the amount of practice, so that each doubling of practice buys about the same proportional gain. Averaging across people can manufacture the pattern, and individual curves are often better fitted by an exponential, but the shape of rapid early gains and slow late gains holds. (Chapter 9)

Practice testing

Studying by actively recalling material, as in short quizzes with feedback, rather than by rereading or highlighting. It is one of only two study techniques given the highest rating for strength of evidence, and classroom trials confirm its benefit. (Chapter 14)

Precision weighting

Weighting each source of information, whether a prior or a sensory cue, by its precision, that is, by how reliable it is. It explains how the senses are combined, and predictive coding proposes that attention is the gain applied to prediction errors according to their expected precision. (Chapter 6)

Prediction

The brain's use of prior experience to anticipate future sensory input or outcomes. (Chapter 12)

Prediction error (sensory)

In predictive coding, the difference between what a cortical level predicted and what it actually received, a whole pattern of errors about content passed between cortical layers. It is distinct from the single reward prediction error signaled by dopamine neurons (Chapter 3), which concerns value and not content. (Chapter 6)

Predictive coding

A theory of cortical processing in which each level of a hierarchy sends a prediction down to the level below, which returns only the prediction error. The higher level revises its hypothesis until the error is small, so what ascends is the part of the input that was not expected. (Chapter 6)

Prior

In Bayesian inference, a probability distribution encoding what is usually true before the current evidence arrives. In the brain a prior is learned structure, the statistics of a lifetime of input written into synaptic weights, and it can be revised by experience. (Chapter 6)

Prospect theory

Kahneman and Tversky's 1979 description of how people actually choose under risk: they evaluate outcomes as gains and losses relative to a reference point, are risk averse for gains and risk seeking for losses, weight losses more than equal gains, and overweight small probabilities while underweighting large ones. (Chapter 8)

Quantal release

The release of neurotransmitter in fixed packets, or quanta, each the contents of one vesicle, with each impulse releasing a random number of them, often one or none. It makes synaptic transmission intrinsically noisy and unreliable. (Chapter 2)

Receptive field

The region of the sensory field, for example the visual field, in which a stimulus can influence a neuron's firing. Receptive fields grow, and the stimuli that drive them become more abstract, at each stage of the visual hierarchy. (Chapter 6)

Reciprocal inhibition

The circuit arrangement in which a sensory input that excites one muscle's motor neurons also excites an inhibitory interneuron that suppresses the opposing muscle, so the two never fight. (Chapter 4)

Reconsolidation

The process by which a retrieved memory becomes labile again for some hours and must be restabilized to persist. During that window it can be strengthened, weakened, or altered, which is why memories drift with retelling. (Chapter 5)

Recurrent excitation

A motif in which excitatory neurons excite one another, so a pattern of activity can sustain itself after its input is gone. It underlies working memory and pattern completion and, without inhibition to contain it, becomes a seizure. (Chapter 4)

Reinforcement learning

The branch of machine learning in which an agent learns by acting and observing the consequences, from feedback that is delayed, noisy, and evaluative rather than a correct answer. Its temporal-difference algorithm closely matches the dopamine reward prediction error. (Chapter 1)

Repetition suppression

The decline in neural response to a stimulus each time it is repeated. On its own it is weak evidence for prediction, since fatigue could produce it, but experiments that vary expectation while holding repetition fixed reveal a separate expectation suppression. (Chapter 6)

Representation

A pattern of neural activity or connection strength that varies systematically with something in the world, the body, or the brain's own state, so that it can be described as standing for it. In this book the word describes what activity carries, not a stored copy that is looked up. (Chapter 1)

Resting potential

The steady voltage across a neuron's membrane at rest, about 70 millivolts negative inside relative to outside, produced by the ion gradients and by the membrane's greater leakiness to potassium. Every signal the neuron produces is a brief departure from it. (Chapter 2)

Retrieval-augmented model

A language model that keeps a large database of text outside its weights and, given a question, finds and reads the most similar passages before answering. Its knowledge can be corrected or extended without retraining, much as the hippocampal index can, but the database never teaches the weights anything. (Chapter 11)

Reward hacking

The failure in which an agent optimizing a specified reward finds unintended ways to maximize it, a central problem of AI safety. The book treats addiction as the biological counterpart. (Chapter 7)

Reward model

A second network, trained to predict which answers people prefer, whose score a language model is trained to maximize. Because it is only a proxy, pushing optimization against it too far makes true quality fall while the score keeps rising. (Chapter 11)

Reward prediction error

The difference between the reward received and the reward predicted: positive (a burst of dopamine firing) for an unexpected reward, zero for a fully predicted one, and negative (a dip below baseline) when a predicted reward fails to arrive. It is a single number about value, distinct from the sensory prediction error of Chapter 6. (Chapter 3)

Robust yet fragile

The property, described by Marie Csete and John Doyle, of complex regulated systems that are highly robust to the disturbances they were built for but vulnerable to rare ones they were not, because robustness in one place is paid for with fragility in another. (Chapter 13)

Saccade

A rapid jump of the eyes, made about three times a second, between fixations of a few hundred milliseconds; vision is suppressed during each jump. The book treats each saccade as a decision about where evidence is most needed. (Chapter 6)

Self-organization

The emergence of complex, apparently designed structure from simple local rules followed by many elements, with no central controller or blueprint. In brain wiring, chemical gradients lay down a coarse map of connections and correlated activity then sharpens it. (Chapter 0)

Self-supervised learning
Training without labels by asking a system to predict part of its input from the rest, such as the next word from the words before or a hidden patch of an image from the visible ones. Most capable systems of the 2020s were trained mostly this way, which resembles the brain's situation of few labels and an endless stream of input. (Chapter 10)

Sensitive period
A window of early life in which a circuit is unusually open to being shaped by experience, with largely permanent effects. Closing one eye of a kitten during this window costs that eye most of its territory in visual cortex, while the same closure in an adult does nothing. (Chapter 1)

Sensory adaptation
The continual rescaling of a sensory neuron's responses to the current statistics of its input, so that it keeps using its full range as the input becomes more or less variable. It is not fatigue but the revision of a prediction about what the next input will be like. (Chapter 12)

Sensory prediction error (in motor adaptation)
In motor learning, the difference between where the brain predicted a movement would go and where it was sensed to go. It drives motor adaptation, even when correcting it spoils the task the person is trying to perform. For the predictive-coding sense, see Prediction error (sensory). (Chapter 9)

Sentence embedding
A list of numbers that a pretrained model assigns to a sentence so that sentences with similar meanings get similar lists. (Chapter 16)

Serotonin
A neuromodulator released by the raphe nuclei along the midline of the brainstem. Its role is the least settled of the four; it has been tied to patience, aversion, behavioral inhibition, and the regulation of plasticity. (Chapter 3)

Sharp-wave ripples
Brief high-frequency events in the hippocampus, mostly during slow-wave sleep and quiet waking rest, in which place cells replay recent sequences of activity compressed roughly twentyfold. Disrupting them impairs memory of the replayed experience and prolonging them improves it. (Chapter 5)

Shortcut learning
The tendency of a network trained to reduce its error to use whatever feature of the training data reduces it, whether or not it is the feature a person would use, as when a pneumonia detector learned to recognize which hospital an X-ray came from. Such shortcuts fail when the data change. (Chapter 10)

Signal-to-noise ratio
The strength of a signal relative to the noise mixed into it. Adding more channels, vesicles, or detectors raises it roughly in proportion to their number, but the information carried grows only as its logarithm, so each additional bit costs more than the last (a law of diminishing returns). (Chapter 13)

Sliding threshold

The calcium level separating depression from potentiation, which shifts with a neuron's recent average activity: it rises after intense firing and falls after quiet periods. The more a neuron fires, the harder its inputs become to strengthen, which makes the neuron self-correcting. (Chapter 5)

Slow-wave sleep

The deep stage of sleep in which the cortex produces large, slow waves. It dominates the first half of the night and carries the hippocampal replay that drives consolidation. (Chapter 5)

Small-world network

A network in which most connections are short and form dense local clusters while a few long connections link the clusters, so any two nodes are only a few steps apart while wiring stays cheap. The brain's region-to-region wiring has this form, organized around highly connected hubs. (Chapter 4)

Sodium-potassium pump

A membrane protein that spends one ATP per cycle to push three sodium ions out of the cell and pull two potassium ions in, so potassium stays concentrated inside and sodium outside. It charges the membrane's "battery" slowly and continuously, at large energy cost. (Chapter 0)

Somatic marker hypothesis

Antonio Damasio's proposal that options which have led to good or bad outcomes come to evoke bodily states, the somatic markers, which bias choice before reasoning finishes. Later work has found little support for the claim that bodily feedback is needed for good choices. (Chapter 7)

Spacing

Spreading study or review sessions out over time rather than massing them together. The best gap between sessions grows with how long the memory must last but becomes a smaller fraction of that time, and a review helps most when the memory has faded but can still be retrieved. (Chapter 14)

Speed-accuracy trade-off

The fact that quick decisions make more errors and slow ones fewer. In the diffusion model it is set by the height of the bound, which people and monkeys raise or lower according to the cost of time and of errors. (Chapter 8)

Spike-timing-dependent plasticity (STDP)

A learning rule in which a synapse strengthens if the presynaptic spike precedes the postsynaptic spike by a few milliseconds to a few tens of milliseconds, and weakens if the order is reversed. Its logic is causal: an input that could have helped cause the spike is rewarded, and one that arrived too late is penalized. (Chapter 5)

Spontaneous recovery

The return of an earlier adaptation after brief training in the opposite direction, once errors stop being informative. It is explained by two learning processes running in parallel: a fast one that learns and forgets quickly and a slow one that learns slowly and holds on. (Chapter 9)

Stability

The ability of a neural system to maintain functional operation despite noise and perturbation.

Stability-plasticity dilemma

The conflict between learning quickly from new experience and protecting what is already known: if every fluctuation of activity changed the wiring, new learning would overwrite old as fast as it was acquired. (Chapter 3)

Stages of skill learning

Fitts and Posner's description of a skill moving from a cognitive stage (working out what to do, step by step), through an associative stage (linking the pieces as errors fall), to an autonomous stage (running with little attention). The stages describe the change but do not explain its mechanism. (Chapter 9)

Striatal spiral (ascending spiral)

The chain of connections by which each region of the striatum influences the dopamine neurons that project to the next region along, from the ventral striatum of motivation through the associative striatum to the sensorimotor striatum. It offers a route by which learning in one loop shapes what dopamine teaches in the next. (Chapter 9)

Stroop effect

The slowing of color naming when a word printed in colored ink names a different color, as when "red" is printed in green. It shows that reading has become automatic: the word is read whether or not reading is wanted. (Chapter 9)

Successor representation (predictive map)

A representation, from reinforcement learning, in which each location or state is encoded by the locations expected to follow it. Proposed as what the hippocampus encodes, it explains why place fields grow backward along a habitual route and unites the hippocampus's roles in memory and navigation. (Chapter 12)

Supernormal stimulus

An artificial object that triggers an instinctive response more strongly than the natural object the response evolved for, such as a giant egg that a bird tries to brood in preference to its own. (Chapter 13)

Surrogate gradient

A training trick for spiking networks that, during learning only, replaces the all-or-none spike's useless slope (zero except at threshold) with a smooth slope near threshold, so that backpropagation can be used. (Chapter 11)

Synapse

The junction at which one neuron influences another. At a chemical synapse an arriving impulse releases neurotransmitter across a gap about twenty nanometers wide onto receptors of the receiving cell; synapses both carry signals and are where the brain's memory is written. (Chapter 1)

Synaptic consolidation

The hour-scale stabilization of newly changed synapses, the transition from early to late LTP through protein synthesis and spine growth. Until it is complete, a memory can be blocked by drugs, a competing experience, or a blow to the head. (Chapter 5)

Synaptic scaling
A slow homeostatic adjustment, taking a day or two, in which every excitatory synapse on a neuron is scaled up or down by the same factor when its firing stays too low or too high. Because the change is multiplicative, the relative strengths that encode what was learned are preserved. (Chapter 5)

Synaptic tagging and capture
A molecular mechanism in which early long-term potentiation sets a tag at the synapse lasting an hour or two, and the tagged synapse captures newly made proteins that make the change lasting. It is a slower, hour-scale relative of the eligibility trace. (Chapter 5)

Systems consolidation
The gradual transfer, over weeks to years, of a memory from dependence on the hippocampus to independence from it, as repeated reactivation strengthens direct cortical connections. Rival accounts (multiple-trace and trace-transformation theory) hold that only a schematic version becomes independent, while the detailed episode stays hippocampal. (Chapter 5)

Task-bracketing
A pattern of striatal activity, seen with extended training, in which firing concentrates at the start and end of a learned action sequence with little in between. Graybiel proposed it is how the basal ganglia package a sequence of actions into a single unit. (Chapter 9)

Task-switching cost
The slowing and extra errors on the first trial after switching between tasks, reflecting the time the brain needs to reconfigure which rules are active. Warning of the switch in advance reduces the cost but does not eliminate it. (Chapter 14)

Temporal-difference learning
An algorithm for learning to predict future reward in which the learner compares successive predictions and uses their difference to correct the earlier one. Over trials the error migrates backward from the reward to the cue that predicts it, just as dopamine responses do. (Chapter 3)

Three-factor learning rule
The book's central learning rule. Activity on the two sides of a synapse (factors one and two) induces a change and sets an eligibility trace, and a broadcast neuromodulatory signal (the third factor) decides whether, how strongly, and in which direction the change is kept. (Chapter 3)

Threshold
The membrane voltage at which a neuron generates an action potential. (Chapter 2)

Tinbergen's four questions
The four kinds of answer, set out by the ethologist Niko Tinbergen in 1963, to the question of why an animal has a feature: its causation (the mechanism that produces it), its development in the individual, its survival value, and its evolutionary history. (Chapter 13)

Trade-off
A relationship in which improving a design on one goal costs it on another, so that no single design is best at both. The book argues that many of the brain's features are positions on trade-offs among

hard constraints of energy, space, time, noise, and genetic information, rather than optimal solutions to a single problem. (Chapter 13)

Transfer
The carryover of improvement from practiced tasks to untrained ones. A large online study of brain training found gains on the practiced tasks but no transfer to untrained tasks, even closely related ones; a later review found less improvement on closely related tasks and little on everyday cognition. (Chapter 14)

Transformer (attention)
The network architecture behind current language models. Its core operation, attention, weights each part of the input by its learned relevance to the current one, and one step of recall in a modern Hopfield network is mathematically the same operation, so the context window works as a one-shot associative memory. (Chapter 11)

Ultimatum game
An economic game in which one player proposes how to split a sum and the other accepts, so both are paid, or rejects, so neither is. Responders often reject unfair offers at a cost to themselves, a sense of fairness that varies across cultures. (Chapter 8)

Unconscious inference
Helmholtz's term for the perceptual system's automatic settling on the state of the world that would most probably have produced the current sensations. It is not made by reasoning and cannot be overruled by it, which is why an illusion persists after it is known to be one. (Chapter 6)

Utility
The value of an outcome to the chooser, as opposed to its amount, as proposed by Daniel Bernoulli in 1738. Expected utility theory holds that a chooser whose preferences obey a few consistency rules behaves as if maximizing utility weighted by probability. (Chapter 8)

Ventral stream
The visual pathway from the primary visual cortex (V1) through V2 and V4 to the inferotemporal cortex (IT), along which neurons respond to increasingly abstract and invariant features, ending in selectivity for objects and faces. A parallel dorsal stream specializes in motion and location. (Chapter 6)

Ventromedial prefrontal cortex (vmPFC)
A region of the prefrontal cortex that suppresses the amygdala's learned fear after extinction and links value with the body, receiving interoceptive signals and commanding the hypothalamus and brainstem. Damage to it leaves reasoning largely intact but impairs choice. (Chapter 7)

Volume transmission
Release of a neuromodulator into the extracellular space, often from swellings along the axon called varicosities, so that it diffuses over micrometers and reaches many cells and synapses at once, in contrast to point-to-point synaptic transmission. (Chapter 3)

Wanting and liking

The separation, proposed by Kent Berridge and Terry Robinson, of reward into wanting, the dopamine-dependent pull of a cue, and liking, the pleasure of the reward itself, which survives the loss of dopamine. Their divergence, especially in addiction, lets what an animal pursues come apart from what it enjoys. (Chapter 7)

White matter

The brain tissue made of wiring: the axons, many of them myelinated, that connect the gray matter of cell bodies and synapses. Across mammals its volume grows faster than that of gray matter as brains get larger. (Chapter 13)

Wiring economy (wire minimization)

The principle that, because wire costs space, energy, and conduction time, a nervous system should place its parts and route its connections to keep total wiring short. The brain does not minimize wire outright: it pays for some long-range connections to keep every part only a few steps from every other, a trade between wiring cost and short paths. (Chapter 13)

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